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Eberhard Freitag Reinhardt Kiehl

Etale Cohomology and the Weil Conjecture



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Etale Cohomology and the Weil Conjecture

With an Historical Introduction by J.A. Dieudonné



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Preface

Some years ago a conference on l-adic cohomology in Oberwolfach was held with the aim of reaching an understanding of Deligne's proof of the Weil conjectures. For the convenience of the speakers the present authors – who were also the organisers of that meeting – prepared short notes containing the central definitions and ideas of the proofs. The unexpected interest for these notes and the various suggestions to publish them encouraged us to work somewhat more on them and fill out the gaps. Our aim was to develop the theory in as self-contained and as short a manner as possible. We intended especially to provide a complete introduction to etale and l-adic cohomology theory including the monodromy theory of Lefschetz pencils. Of course, all the central ideas are due to the people who created the theory, especially Grothendieck and Deligne. The main references are the SGA-notes [64–69].

With the kind permission of Professor J.A. Dieudonné we have included in the book that finally resulted his excellent notes on the history of the Weil conjectures, as a second introduction.

Our original notes were written in German. However, we finally followed the recommendation made variously to publish the book in English.

We had the good fortune that Professor W. Waterhouse and his wife Betty agreed to translate our manuscript. We want to thank them very warmly for their willing involvement in such a tedious task.

We are very grateful to the staff of Springer-Verlag for their careful work.

Heidelberg and Mannheim
October 1987

Eberhard Freitag
Reinhardt Kiehl

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On the History of the Weil Conjectures*

By J. A. Dieudonné

The story of the "Weil conjectures" is a marvelous example of mathematical imagination, and one of the most striking instances exhibiting the fundamental unity of mathematics. The essential ideas which led to their proof are due to six men: E. Artin, F.K. Schmidt, H. Hasse, A. Weil, A. Grothendieck, and P. Deligne, over a period of fifty years (1923–1973).

I. The Number of Solutions of Congruences

Appropriately enough, the story, as with so many problems in number theory, begins with Gauss. In his works on the laws of reciprocity, he introduced what are now called Gauss sums (the best known being $\sum_{x=0}^{p-1} \exp(2\pi i x^2/p)$ for prime p); in order to compute them, he was led to evaluate, by elementary arguments, the number of solutions of congruences of one of the types

$$\begin{aligned} &ax^3 - by^3 \equiv 1 \pmod{p}, \qquad ax^4 - by^4 \equiv 1 \pmod{p}, \\ &y^2 \equiv ax^4 - b \pmod{p} \end{aligned} \tag{1}$$

where a, b are fixed integers not divisible by p, solutions (x, y) are identified when they are congruent mod p (so that the congruences (1) are considered as equations in the prime field $\mathbf{F}_p$), and p *varies* in an infinite set of primes; one looks for *asymptotic* expressions (in terms of simple functions of p) of the number of solutions. A little later, Jacobi observed that conversely, by using the elementary properties of Gauss sums, good estimates of the number of solutions could be obtained in more general cases, where elementary methods became unwieldy. After him, not much work was done on this topic, until Hardy and Littlewood, in their study of the Waring problem, and in order to derive the properties of their "Singular series", were confronted with the asymptotic evaluation of the number of solutions of the congruence

$$x_1^k + \ldots + x_r^k \equiv 0 \pmod{p} \tag{2}$$

when the prime p tends to $+\infty$. They used Jacobi's method for that purpose;

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more generally, it was shown simultaneously by Hua-Vandiver and A. Weil in 1949 that this method gave estimates of the number N of solutions of equations

$$a_0 x_0^{k_0} + \ldots + a_r x_r^{k_r} = 0 \quad (a_0 a_1 \ldots a_r \neq 0) \tag{3}$$

in any *finite field* $\mathbf{F}_q$ with $q = p^m$ elements; the result is

$$N = q^r + O(q^{(r+1)/2}). \tag{4}$$

Similar theorems had been obtained by Davenport (1931) and Mordell (1933) for equations of type $y^m = P_n(x)$ in $\mathbf{F}_p$, where P_n is a polynomial of degree n; for some small values of m and n, they had estimates of the form $N = p + O(p^{\phi(m,n)})$ with $1/2 < \phi(m, n) < 1$.

II. Enter the Zêta Functions

Let us first recall the classical properties of the Riemann zêta function: it is defined for $\mathscr{R}s > 1$ by the series $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, and satisfies the Euler relation

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1} \tag{5}$$

where the product is extended over all primes. Riemann showed that ζ can be extended to a meromorphic function in the complex plane with a single pole at $s = 1$, and that if

$$\xi(s) = \tfrac{1}{2} s(s-1)\, \pi^{-s/2}\, \Gamma(s/2)\, \zeta(s)$$

then ξ is an entire function satisfying the functional equation $\xi(1-s) = \xi(s)$. Furthermore he advanced the (still unproved) *Riemann hypothesis* that all the zeroes of ξ are on the line $\mathscr{R}s = 1/2$.

A little later, Dedekind generalized Riemann's theory to an arbitrary field K of algebraic numbers (finite extension of $\mathbf{Q}$), by defining $\zeta_K(s) = \sum_{\mathfrak{a}} (N\mathfrak{a})^{-s}$, where $\mathfrak{a}$ runs through all ideals of the ring $\mathfrak{o}$ of algebraic integers in K, and the *norm* $N\mathfrak{a}$ is the number of elements of the ring $\mathfrak{o}/\mathfrak{a}$. He generalized the Euler relation to

$$\zeta_K(s) = \prod_{\mathfrak{p}} (1 - (N\mathfrak{p})^{-s})^{-1} \tag{6}$$

the product being extended to all prime ideals $\mathfrak{p}$ of $\mathfrak{o}$; many years later Hecke showed that ζ_K could be extended to a meromorphic function and satisfied a functional equation generalizing Riemann's.

It is immediate to see that, formally, relation (6) only uses two properties of the ring $\mathfrak{o}$: 1) it is a Dedekind ring: 2) the field $\mathfrak{o}/\mathfrak{p}$ is finite for all primes: indeed, if $\mathfrak{a} = \mathfrak{p}_1^{\nu_1} \ldots \mathfrak{p}_k^{\nu_k}$ is the decomposition into primes of the ideal $\mathfrak{a}$, $\mathfrak{o}/\mathfrak{a}$ is isomorphic to the direct product of the $\mathfrak{o}/\mathfrak{p}_j^{\nu_j}$, and for any prime ideal $\mathfrak{p}$, each $(\mathfrak{o}/\mathfrak{p})$-module $\mathfrak{p}^h/\mathfrak{p}^{h+1}$ is isomorphic to $\mathfrak{o}/\mathfrak{p}$; this shows that the norm is multiplica-

tive, from which (6) follows formally (the proof of the convergence of the infinite product needs simple estimates of the number of primes of given norm). In 1923, E. Artin observed that these properties are true for rings defined in the following way: start with a finite field $\mathbf{F}_q$, consider the field $K_0 = \mathbf{F}_q(T)$ of rational fractions, and a quadratic extension $K = K_0(v)$, where $v^2 = P(T)$, with P a polynomial having no multiple roots. Then the integral closure $\mathfrak{o}$ of $\mathbf{F}_q[T]$ in K satisfies properties 1) and 2), $\mathfrak{o}/\mathfrak{p}$ being a *finite extension* of $\mathbf{F}_q$ for each prime $\mathfrak{p}$; it is easy to prove that the series and products involved in the corresponding definition of ζ_K converge for $\mathscr{R}s > 1$. Furthermore, Artin noticed that the theory was essentially simpler than for the Dedekind functions, for his functions could be written $Z(q^{-s})$ where $Z(u)$ is a *rational function* with coefficients in $\mathbf{Q}$; the functional equation expressed the quotient $Z(1/qu)/Z(u)$ by a rational function with known zeroes and poles; he then conjectured that the zeroes of $Z(u)$ would all be on the circle $|u| = q^{1/2}$, and verified that conjecture for many polynomials P of low degree.

Now the primes $\mathfrak{p}$ such that $\mathfrak{o}/\mathfrak{p}$ is isomorphic to $\mathbf{F}_q$ itself (i.e. $N\mathfrak{p} = q$) correspond bijectively to the homomorphisms $\mathfrak{o} \to \mathbf{F}_q$; any such homomorphism sends (T, v) to $(a, b) \in (\mathbf{F}_q)^2$ such that $b^2 = P(a)$. In other words the number of solutions of the equation $y^2 - P(x) = 0$ in $\mathbf{F}_q^2$ is exactly the number N_1 of these primes; but from the Euler relation (6) it follows at once that

$$\log Z(u) = N_1 u + \dots$$

near $u = 0$, hence the knowledge of $Z(u)$ implies that of N_1, and Artin's "Riemann hypothesis" yields the evaluation

$$|N_1 - q| \leq c \cdot q^{1/2}, \tag{7}$$

thus tying his results with the previous work on Gauss' problem on congruences.

III. Enters Algebraic Geometry

When k is an arbitrary commutative field, it is tempting to visualize the set of solutions $(x_1, \dots, x_r) \in k^r$ of an equation $P(x_1, \dots, x_r) = 0$, with P an irreducible polynomial of $k[T_1, \dots, T_r]$ as an "affine algebraic hypersurface" ("curve" for $r = 2$, "surface" for $r = 3$) in "affine space" k^r, by an extension of the usual language of geometry. Furthermore, for any field $K \supset k$, one may also consider the solutions of $P(y_1, \dots, y_r) = 0$ for values of the y_j *in the larger field* K, hence an "algebraic hypersurface" V in the "affine space" K^r; the fact that the coefficients of P are in the subfield k is now expressed by saying that V is *defined over* k. Experience shows that the transfer of geometrical language and intuition to these "abstract" varieties is satisfactory only when K is *algebraically closed* (think of the "curve" $x_1^2 + x_2^2 + 1 = 0$ when $k = \mathbf{R}$). We will in fact restrict ourselves to the case in which $K = \bar{k}$, algebraic closure of k; furthermore, we shall only consider *nonsingular* hypersurfaces V in $\bar{k}^r$, i.e. such that at every point the "tangent hyperplane" to V is uniquely defined in the usual way (which means

that the partial derivatives of P do not all vanish simultaneously in V). For every point $x=(x_1, \ldots, x_r)\in V$, the coordinates x_j are in $\bar{k}$, hence there is a smallest finite extension $k(x)$ of k containing all the x_j, and $[k(x):k]=\deg(x)$ is defined as the *degree* of the point x. If $\mathfrak{m}$ is the kernel of the homomorphism $k[T_1, \ldots, T_r]\to\bar{k}$ sending each T_j to x_j, $\mathfrak{m}$ is a maximal ideal of the ring $k[T_1, \ldots, T_r]$ and $k[T_1, \ldots, T_r]/\mathfrak{m}$ is isomorphic to $k(x)$; one writes $k(\mathfrak{m})=k(x)$ and $\deg(\mathfrak{m})=\deg(x)$; it can be shown that every maximal ideal $\mathfrak{m}$ of $k[T_1, \ldots, T_r]$ containing $P(T_1, \ldots, T_r)$ corresponds to a number of points x of V equal to $\deg(\mathfrak{m})$.

When $k=\mathbf{F}_q$, let

$$Z_V(u)=\prod_{P\in\mathfrak{m}}(1-u^{\deg(\mathfrak{m})})^{-1}; \tag{8}$$

the function $Z(u)$ defined by E. Artin is then equal to $Z_C(u)$, where C is the "affine curve" $x_2^2-P(x_1)=0$ defined over $\mathbf{F}_q$. In general Z_V will be called the *zêta function of* V. The points of V in $(\mathbf{F}_{q^n})^r$ are those for which $\deg(x)$ divides n; their number is obviously $\leq q^{nr}$, hence the number of maximal ideals $\mathfrak{m}$ such that $P\in\mathfrak{m}$, corresponding to these points, is *a fortiori* $\leq q^{nr}$, and this shows that the product (8) converges for small u; furthermore, one may write for small u

$$uZ_V'(u)/Z_V(u)=\sum_{P\in\mathfrak{m}}\frac{\deg(\mathfrak{m})\,u^{\deg(\mathfrak{m})}}{1-u^{\deg(\mathfrak{m})}}=$$

$$=\sum_{\nu=1}^{\infty}\sum_{P\in\mathfrak{m}}\deg(\mathfrak{m})\,u^{\nu\deg(\mathfrak{m})}=\sum_{\nu=1}^{\infty}N_\nu u^\nu \tag{9}$$

where N_ν is the number of points of V in $(\mathbf{F}_{q^\nu})^r$.

These considerations can be generalized to other types of nonsingular varieties, not necessarily embedded in an "affine space" $\bar{k}^r$. Historically, the language of algebraic geometry in the theory of zêta functions was introduced in 1931 by F.K. Schmidt, who treated nonsingular *projective curves* over a finite field $\mathbf{F}_q$. He showed that the Dedekind-Weber theory of algebraic curves over $\mathbf{C}$ (including the definition of the genus and the Riemann-Roch theorem) could be extended to projective curves over an arbitrary algebraically closed field $\bar{k}$; this enabled him to prove that for a projective nonsingular curve C of genus g defined over $\mathbf{F}_q$, the zêta function could be written

$$Z_C(u)=P_{2g}(u)/(1-u)(1-qu) \tag{10}$$

where the numerator is a polynomial of degree $2g$ with integral coefficients, and that one had the functional equation

$$Z_C(1/qu)=(qu^2)^{1-g}Z_C(u). \tag{11}$$

The "Riemann hypothesis" for C was therefore that the zeroes of P_{2g} are on the circle $|u|=q^{1/2}$; this is easily seen to be equivalent to the inequalities

$$|N_\nu-q^\nu-1|\leq 2g\cdot q^{\nu/2}\qquad\text{for all }\nu\geq 1. \tag{12}$$

IV. Daydreaming About Algebraic Topology

Returning to the case of a hypersurface V in $(\bar{\mathbf{F}}_q)^r$, observe that the elements of $\bar{\mathbf{F}}_q$ belonging to $\mathbf{F}_{q^n}$ are those for which $t^{q^n}=t$. Consider then the map

$$\Phi\colon (x_1, \ldots, x_r)\mapsto(x_1^q, \ldots, x_r^q)$$

of $(\bar{\mathbf{F}}_q)^r$ onto itself. As the coefficients of P are in $\mathbf{F}_q$, hence satisfy $t^q=t$, one has

$$P(\Phi(x))=(P(x))^q,$$

hence Φ maps V onto itself; its restriction to V is called the *Frobenius morphism* of V. In 1936, Hasse observed that for a curve C, the number N_v is exactly the number of points $x\in C$ for which $\Phi^v(x)=x$, i.e. the *fixed points* of Φ^v. Now let us for a moment abandon the chronological sequence of events, and pretend that we are dealing with "classical" nonsingular algebraic varieties X in a complex projective space. Since the days of Picard and Poincaré, it has been realized that most properties of such varieties are closely linked to their *homological* properties. In their present form (essentially due to the work of Lefschetz and Hodge), for an irreducible nonsingular projective variety X of dimension d over C (hence of dimension $2d$ as a differentiable manifold), these properties center on the *cohomology algebra* $H^\cdot(X)=\bigoplus_i H^i(X)$ of X over a field K of *characteristic* 0; it is a graded algebra over K, having the following properties:

A) 1) Each $H^i(X)$ is a finite dimensional vector space over K, which is 0 except for $0\leq i\leq 2d$.

2) There is a natural isomorphism $H^{2d}(X)\xrightarrow{\sim} K$, and for each i, the multiplication in $H^\cdot(X)$ gives rise to a nondegenerate bilinear form $H^i(X)\times H^{2d-i}(X)\to H^{2d}(X)\xrightarrow{\sim} K$ (Poincaré duality) allowing one to identify $H^{2d-i}(X)$ to

$$H_i(X)=\mathrm{Hom}_K(H^i(X), K),$$

the *homology* over K in dimension i.

3) For two nonsingular varieties X, Y, there is a natural isomorphism of graded algebras

$$H^\cdot(X)\otimes H^\cdot(Y)\xrightarrow{\sim} H^\cdot(X\times Y)\quad\text{(Künneth formulas).}$$

B) Any morphism $f\colon X\to X$ defines naturally, for each i, a linear mapping $f^{(i)}\colon H^i(X)\to H^i(X)$, such that the $f^{(i)}$ for $0\leq i\leq 2d$ constitute a homomorphism $f^\cdot\colon H^\cdot(X)\to H^\cdot(X)$ of graded algebras. The *fixed points* of f are the projection on X of the intersections of the graph Γ of f and the diagonal Δ of $X\times X$; if Γ and Δ intersect *transversally* at each point (which means their tangent spaces have an intersection reduced to a point), the number of fixed points of f is given by the *Lefschetz trace formula*

$$N=\sum_{i=0}^{2d}(-1)^i\mathrm{Tr}(f^{(i)}). \tag{13}$$

C) If Y is a nonsingular subvariety of X of dimension $d-1$, there are natural linear mappings $H^i(X) \to H^i(Y)$ which are bijective for $i \leq d-2$ and injective for $i=d-1$.

D) Let $h \in H^2(X)$ correspond by Poincaré duality to the homology class in $H_{2d-2}(X)$ of a hyperplane section of X, and let L: $a \to ha$ be multiplication by h in $H^*(X)$; then L^{d-i}: $H^i(X) \to H^{2d-i}(X)$ is an isomorphism for $i \leq d$.

This implies, by an easy algebraic argument, that if the morphism f: $X \to X$ is such that $f^{(2)}(h)=q \cdot h$ where $q>0$ is a rational number, and if $g_i=q^{-i/2}f^{(i)}$ (considered as an endomorphism of $H^i(X) \otimes_K \bar{K}$), g_i is bijective, and g_i^{-1} is identified to ${}^t g_{2d-i}$ by Poincaré duality. This implies that if α_{ij} are the eigenvalues of $f^{(i)}$ in $\bar{K}$, the set of the elements $q^{i/2}/\alpha_{ij}$ is the same as the set of elements $\alpha_{2d-i,j}/q^{d-(i/2)}$.

E) In each $H^i(X)$ for $i \leq d$ there is a subspace $A^i(X)$ which is stable under $f^{(i)}$ for any morphism f: $X \to X$, and on each $A^i(X)$, considered as a vector space over the rationals, there is a nondegenerate *scalar product*, such that, if f satisfies the conditions of D), each g_i is a *unitary* mapping for that scalar product; this implies that all the eigenvalues of $f^{(i)}$ (elements of $\bar{\mathbf{Q}}$) have absolute values equal to $q^{i/2}$.

Returning now to the hypersurface V defined over $\mathbf{F}_q$, *suppose* that it could be possible to attach to it a graded algebra $H^*(V)$ having all the preceding properties, and that in addition $\Phi^{(2)}(h)=q \cdot h$ for the Frobenius morphism. It is easy to see that the graph of Φ^v intersects Δ transversally; therefore, if α_{ij} are the eigenvalues of $(\Phi^v)^{(i)}$, the number N_v would be given by the formula

$$N_v=\sum_i (-1)^i \sum_j \alpha_{ij}^v ; \tag{14}$$

hence one would have (for $d=r-1=\dim(V)$)

$$Z_V(u)=\frac{P_1(u)\,P_3(u) \ldots P_{2d-1}(u)}{P_0(u)\,P_2(u) \ldots P_{2d}(u)} \tag{15}$$

where $P_i(u)=\det(1-u \cdot \Phi^{(i)})$ is a polynomial with integral coefficients. In particular, $Z_V(u)$ should be a *rational* function; furthermore $Z_V(1/q^d u)$ should have the same zeroes and poles as $Z_V(u)$ with the exception of $u=0$, and one should have $|\alpha_{ij}|=q^{i/2}$. Finally, if the coefficients of the equation of V were the classes mod. p of integers, coefficients of the equation of a nonsingular hypersurface V_0 in $\bar{\mathbf{Q}}^r$, the degree of each P_i should be equal to the i-th Betti number of V_0.

These statements are the *Weil conjectures* for Z_V.

V. The "Substitutes" for Cohomology of Hasse and Weil

To understand how Weil could have been led to such bold conceptions, one must go back to the first ideas of Hasse for the proof of the "Riemann hypothe-

sis" for curves of *genus 1* over $\mathbf{F}_q$. In the classical theory of nonsingular curves over the complex field, to a curve C is associated its *jacobian* $J=J(C)$, which may be considered as the Pontrjagin dual of the homology group $H_1(C,\mathbf{Z})$; the duality being derived from the bilinear form $(\gamma,\omega)\to\int_\gamma\omega$ defined on cycles γ and holomorphic abelian differentials ω on the Riemann surface of C ("period" of ω over γ). If C has genus g, $J(C)$ is a complex torus $\mathbf{C}^g/\Delta$, where Δ is a discrete subgroup of rank $2g$, satisfying the classical Riemann bilinear conditions. One can give an algebraic definition of the group J, by considering the additive group G/G_l of classes of divisors of degree 0 on C, modulo linear equivalence: one associates to a divisor D of degree 0, which can be written $\partial\gamma$ for a 1-chain γ on the Riemann surface, the class $\phi(D)$ in $\mathbf{C}^g/\Delta$ of the vector $(\int_\gamma\omega_1,\ldots,\int_\gamma\omega_g)$ where the ω_j's form a basis of the space of holomorphic differentials; the Abel-Jacobi theorem shows that this mapping ϕ is surjective and has G_l as kernel. It is then possible to define on J a structure of *algebraic group* (a particular case of the algebraic groups over C known as *abelian varieties*) with the help of the (transcendental) Riemann bilinear conditions. Finally, if x_0 is a point of C,

$$x\mapsto\phi((x)-(x_0))$$

is a morphism of C into J, which is an isomorphism for $g=1$.

Hasse's first method to deal with a curve C of genus 1 defined over $\mathbf{F}_q$ consisted in "lifting" C back to a "classical" curve C_0 of genus 1 defined over the rational field $\mathbf{Q}$: if E is the field of rational functions on C, he showed that it was possible to determine C_0 such that, if ω_1,ω_2 are the periods of the elliptic functions corresponding to C_0 (so that the field E_0 of these functions is the field of rational functions on C_0), ω_1/ω_2 should generate an imaginary quadratic field K over $\mathbf{Q}$, and E would be the field of residues of the ring of integers of K modulo a suitably chosen prime ideal of that ring. Hasse could then use the classical results on "complex multiplications" of C_0 (i.e. endomorphisms of the jacobian $J(C_0)$) to determine the number of points of C of degree 1, and prove the "Riemann hypothesis" for C.

A little later, Hasse abandoned that method in favor of a more intrinsic one: as mentioned above, $J(C)$ may be defined algebraically as an "abstract" group, and the Frobenius morphism naturally defines an endomorphism of that group; Hasse proved that the numerator of the zêta function Z_C in (10) (in that case a polynomial of degree 2) was identical to the characteristic polynomial of that endomorphism. The main tool which he introduced for this purpose was an integer $\nu(\lambda)$ associated to any surjective endomorphism λ of $J(C)$: if E is the field of rational functions of C, λ defines a "comorphism" $R(\lambda)$, which is an isomorphism of E into itself, and $\nu(\lambda)$ is the degree $[E\colon R(\lambda)(E)]$, which is finite when λ is surjective. Hasse could then prove that for any positive integers a,b,

$$\nu(a\cdot 1+b\cdot\lambda)=a^2+\sigma(\lambda)\,ab+\nu(\lambda)b^2$$

for any surjective endomorphism λ of $J(C)$, and the positivity of this quadratic form gave him the proof of the "Riemann hypothesis".

The extension of these ideas to curves C of arbitrary genus g over $\mathbf{F}_q$ was far from obvious: the classical theory showed that $J(C)$ should now be an alge-

braic group of dimension g (instead of being isomorphic to C as an algebraic variety, as in Hasse's case), and until 1940 nobody had extended to algebraic geometry over a field of characteristic $p>0$ the theory of algebraic groups, and in particular the theory of abelian varieties. This was done single handed by A. Weil, who first had to develop, in his famous book *Foundations of algebraic geometry*, the basic properties of intersection numbers independently of any appeal to algebraic topology. He then could study the structure of the ring of endomorphisms of any abelian variety A; for any surjective endomorphism λ of A, he defined as Hasse the integer $v(\lambda)$, (where now E is the field of rational functions on A), and proved that this time

$$v(a\cdot 1+b\cdot\lambda)=a^{2g}+\sigma(\lambda)\,a^{2g-1}b+\ldots+v(\lambda)\,b^{2g}.$$

The invariant $\sigma(\lambda)$ turns out to be the "substitute" playing the role of $\mathrm{Tr}(f^{(1)})$ when λ is the endomorphism of $J(C)$ corresponding to a morphism f of C. A "substitute" for the Poincaré duality is found in a general duality for abelian varieties which can be defined algebraically (in the classical case it is defined using Pontrjagin duality); finally, if λ' is the "transpose" of an endomorphism λ for that duality, it can be proved that $\sigma(\lambda\lambda')>0$ for $\lambda\neq 0$, and this property (which serves as "substitute" for the positivity of the Hodge scalar products) enabled Weil to give a proof of the "Riemann hypothesis" for a curve of arbitrary genus.

In all this work, Weil had constantly in mind the classical theory of "correspondences" developed by Hurwitz: a correspondence on C may be thought of as a "multivalued" morphism, and more precisely as a curve Γ on the surface $C\times C$; better still, it is defined in general as a divisor (linear combination of curves) on $C\times C$. A correspondence Γ associates in a natural way (mimicking the set-theoretic one) to any divisor D on C (linear combination of points) another divisor $\Gamma(D)$, and this again defines an endomorphism of $J(C)$; conversely it turns out that any endomorphism of $J(C)$ may be obtained in that way. In the classical case, the Lefschetz formula (13) can be extended to give the intersection number of a correspondence Γ and of the "identity correspondence", i.e. the diagonal Δ of $C\times C$, and in fact had been obtained by Hurwitz in 1886, using the theory of abelian integrals; it was the fact that Weil was able to prove a similar formula by purely algebraic means which led him to formulate his general conjectures.

VI. Etale Cohomology and Deligne's Theorem

Using his result for curves, Weil was able to check that his conjectures were verified for hypersurfaces of type (3), as well as for other varieties such as the grassmannians. But for some time no suitable definition of a "good" cohomology was in sight. Beginning around 1953, Cartan and Serre had used the Leray cohomology with values in sheaves as a very efficient tool for complex manifolds, and Serre had shown how to translate this technique to algebraic varieties over an algebraically closed field of arbitrary characteristic p. But for $p>0$,

the cohomology groups thus defined obviously could not be used for a Lefschetz formula (13), where the left hand side must be an arbitrary integer, and not an element of a field of characteristic p. It was only after Grothendieck had built the general theory of schemes that, following a remark of Serre, he was able to generalize in a very original way both the concept of "topology" and the concept of "sheaf", and to attach to every variety (or scheme) X a cohomology algebra $H^{\cdot}(X_{\text{et}}, \mathbf{Q}_l)$ over an *l-adic field* $\mathbf{Q}_l$, where l is a prime different from the characteristic (the intervention of l-adic fields in these questions had already been noticed by Weil and Deuring).

The depth and complexity of the techniques involved in the definition of this "étale cohomology" $H^{\cdot}(X_{\text{et}})$ are such as to rule out the possibility of giving any further details on its definition. Let us only mention that Grothendieck (with the help of M. Artin (E. Artin's son) and J.L. Verdier) could prove that it satisfies conditions A), B), C)* above, and Deligne has recently shown that D) also holds for varieties over a finite field $\mathbf{F}_q$; however, no property similar to E) has yet been obtained for the étale cohomology (or other types of cohomology proposed more recently). Properties A), B), C) were at any rate enough to prove formula (15), as well as the functional equation

$$Z_V(1/q^d u) = \pm q^{n\chi/2} u^{\chi} Z_V(u) \tag{16}$$

where

$$\chi = \sum_{i=0}^{2d} (-1)^i \dim H^i(X_{\text{et}}, \mathbf{Q}_l). \tag{17}$$

However, it was not known until recently that the coefficients of the P_j in (15) were independent of the prime l. This has finally been proved by P. Deligne in 1973, as well as the last and most difficult of the Weil conjectures, namely that $|\alpha_{ij}| = q^{i/2}$.

Here again, it is impossible to describe in detail the extremely ingenious proof, which is quite different from those of Hasse and Weil, since it cannot rely on a "positivity" argument. One reduces the problem to the case $i = d$ (i.e. the "middle" cohomology space $H^d(X)$); to show that $|\alpha_{dj}| = q^{d/2}$ it is enough to prove that

$$q^{(d-1)/2} \leq |\alpha_{dj}| \leq q^{(d+1)/2} \tag{18}$$

for then one applies this result to the product X^k, and using Künneth formulas, one has

$$q^{(kd-1)/2} \leq |\alpha_{dj}^k| \leq q^{(kd+1)/2};$$

letting k tend to $+\infty$ gives the result. One may even suppose in (18) that d is even, for then one may take k to be even in the last step. The proof of (18) is then made by induction on the even number d; this is the deep and difficult part of the proof, which rests on the old "monodromy" technique of Picard and Lefschetz: it was essentially a topological technique in the classical case, but it also was transferred by Grothendieck and his school to étale cohomology.

* Before l-adic cohomology had been invented, Dwork proved, by an ingenious use of p-adic analytic functions, that the zêta function $Z_V(u)$ is *rational.*

As usual in mathematics, this breakthrough opens the way to a host of new problems; but as far as Gauss' initial problem is concerned, it is the end of the story, for Deligne's theorem implies, for the number of points of degree 1 of a non-singular projective hypersurface of dimension d the estimate

$$|N-(1+q+\ldots+q^d)| \leq b\, q^{d/2}$$

where even the constant b can be computed explicitly: it is the d-th Betti number of hypersurfaces over $\mathbf{C}$ having the same degree as V.

Introduction

The aim of this book is to develop Grothendieck's etale cohomology theory of algebraic varieties as far as necessary and then to present Deligne's proof of the Weil conjecture using this cohomology.

Here is what the Weil conjecture says.

Let X_0 be a smooth, absolutely irreducible, projective algebraic variety of dimension d over a finite field κ with q elements. Let $X = X_0 \otimes k$ be the extension of X_0 to the algebraic closure k of κ. We denote by κ_n the uniquely determined intermediate field with q^n elements. We then have the following *Diophantine Problem:* Determine the number A_n of points of X_0 with coordinates in κ_n.

For this purpose we form the formal power series

$$\sum_{n=1}^{\infty} A_n t^{n-1}.$$

There is then a uniquely determined series $f(t)$ such that

$$f(0) = 1$$

and

$$f'(t)/f(t) = \sum_{n=1}^{\infty} A_n t^{n-1}.$$

Just like the Riemann ζ-function, the Dirichlet series $f(q^{-s})$ has an Euler product expansion, namely

$$f(q^{-s}) = \prod (1 - (\mathcal{N}x)^{-s})^{-1}.$$

Here x runs through the closed points (not the geometric points) of the scheme X_0. The symbol $\mathcal{N}x$ denotes the absolute norm of x, that is, the number of elements in the residue class field of x. One also calls $\zeta_{X_0}(s) = f(q^{-s})$ the ζ-function of X_0.

Weil's conjectures assert:

(1) $f(t)$ is a rational function.

(2) $f(t)$ has a representation in the following form:

$$f(t) = f_1(t) \ldots f_{2d-1}(t) / f_0(t) f_2(t) \ldots f_{2d}(t).$$

Here $d = \dim(X)$, $f_0(t) = 1 - t$, $f_{2d}(t) = 1 - q^d t$, and each $f_j(t)$ is a polynomial with integer coefficients with $f_j(0) = 1$. Furthermore, the reciprocal roots λ of the polynomial $f_j(t)$ all have the same absolute value, $|\lambda| = q^{j/2}$.

(3) If X arises by "good" reduction of constants from a complex algebraic variety $\tilde{X}$, the degree of $f_j(t)$ is precisely the j-th Betti number of $\tilde{X}$ (interpreted as a complex analytic space).

(4) There is a functional equation for the Dirichlet series $\zeta_{X_0}(s)$.

Assertion (2) is an analogue of the *Riemann Hypothesis.* Assertions (2) and (3) yield, among other things, the precise asymptotic growth of the number of congruence solutions for $\tilde{X}$ modulo prime numbers p, when $\tilde{X}$ is defined over $\mathbb{Q}$.

The Weil conjecture has a series of profound consequences. We may mention estimates for exponential sums, such as the Kloosterman sums, and the proof (likewise due to Deligne) of the famous Ramanujan conjecture – or, more generally, the Ramanujan-Petersson conjecture on the growth behavior of Fourier coefficients of cusp forms for the elliptic modular group.

André Weil succeeded in proving his conjectures for curves using his algebraic theory of abelian varieties. He already recognized the connection between his conjectures and the question of finding a theory that would be analogous to singular cohomology and applicable to algebraic varieties over arbitrary base fields.

Let us provisionally introduce the idea of a *Weil cohomology.* Consider the category of smooth irreducible projective algebraic varieties Y over a fixed algebraically closed base field k. A Weil cohomology theory with coefficients in a fixed field L of characteristic 0 is a system $H^i(Y, L)$ $(i=0, 1, 2, \ldots)$ of contravariant functors on that category with values in the category of vector spaces over L. This system should satisfy the following provisional conditions.

(1) The vector spaces $H^i(Y, L)$ are finite-dimensional over L,

$$H^i(Y, L)=0 \quad \text{for } i>2\dim(Y)=2d,$$

and

$$\dim H^0(Y, L)=\dim H^{2d}(Y, L)=1.$$

(2) If Y arises by "good reduction of constants" from a complex algebraic variety $\tilde{Y}$ with Betti numbers $b_0, \ldots, b_{2d}$, then

$$\dim H^i(Y, L)=b_i\,.$$

(3) There is a *Lefschetz Fixed-Point Formula:* Let $\phi\colon Y\to Y$ be a morphism with finitely many fixed points of multiplicity 1, and let s_i be the trace of the linear mapping $H^i(\phi, L)$ induced by ϕ on $H^i(Y, L)$. Then the number A of fixed points of ϕ satisfies the fixed-point formula

$$A=\sum_{i=0}^{2\dim Y}(-1)^i s_i\,.$$

Now back to our original variety X_0. The number A_n is precisely the number of fixed points of the n-th power of the Frobenius mapping

$$\mathrm{Fr}\colon X\to X$$

$$\text{“}(x_1, \ldots, x_m)\mapsto(x_1^q, \ldots, x_m^q)\text{”}$$

on X. The fixed-point formula for the mapping Fr^n yields Grothendieck's formula

$$f(t)=\sum_{i=0}^{2d}\det(\mathrm{Id}-tH^i(\mathrm{Fr},L))^{(-1)^{i+1}}.$$

Thus $f(t)$ is rational! To prove the fundamental part (2) of the Weil conjectures, we have to prove that $\det(\mathrm{Id}-tH^i(\mathrm{Fr},L))$ is a polynomial with integer coefficients and that the eigenvalues λ of $H^i(\mathrm{Fr},L)$ have absolute value $|\lambda|=q^{i/2}$.

In the first chapter we shall develop Grothendieck's etale cohomology theory of algebraic varieties, going far enough to obtain the *comparison theorems* (§11) relating etale to singular cohomology and the *finiteness theorems* for the cohomology groups of projective algebraic varieties over an algebraically closed base field, both for $H^i(Y,\mathbb{Z}/n\mathbb{Z})$ with torsion coefficients (§8) and for $H^i(Y,L)$ with coefficients in the field L of l-adic numbers (§12).

Grothendieck's cohomology theory is a *sheaf cohomology theory*. In the first two sections, we introduce the category $\mathrm{Et}(S)$ of etale schemes $T\to S$ over a noetherian base scheme S. Here the etale schemes over a base scheme S correspond to the "bounded" unramified covering spaces of a complex analytic space.

Sheaves then are certain contravariant functors on the category $\mathrm{Et}(S)$ (with values, say, in the category of abelian groups) that satisfy a "sheaf property". We presume that the readers have a good familiarity with the usual topological sheaf theory (see e.g. Godement's book *Théorie des Faisceaux* [51]), and for the most part we leave it to them to carry over from the topological to the algebraic case the proofs of basic formalities for such things as Čech cohomology. In the subsequent paragraphs we make extensive use of Grothendieck's language of schemes and of several fundamental results, e.g. from the theory of coherent sheaves. One can find this in the "Éléments de Géométrie Algébrique" of Grothendieck and Dieudonné or in several newer, more accessible books, e.g. Hartshorne, Griffiths/Harris, Mumford, Shafarevich [60–63, 73, 53, 116, 126].

The fundamental results of this chapter are the two *base change theorems* (§6, §7) and their consequences, for example the *finiteness theorem* for cohomology groups with torsion coefficients (8.10) and the behavior of these groups under *specialization* (8.13). The corresponding theorems in the topological sheaf theory are mostly trivial. For instance, here is how the *proper base change theorem* goes in the topological case:

Let $f\colon X\to Y$ be a proper continuous mapping of locally compact spaces, and let $\mathcal{G}$ be a sheaf on X. Then the stalks $R^if_*(\mathcal{G})_y$ of the higher direct image sheaves of $\mathcal{G}$ at a point y in Y are canonically isomorphic to the cohomology groups $H^i(f^{-1}(y),\mathcal{G}|f^{-1}(y))$ of the restriction of $\mathcal{G}$ to the fiber $f^{-1}(y)$.

In the etale case these theorems do not come easily. With some technical effort they can generally be reduced either to theorems of *Galois cohomology* (Tsen's Proposition, 1.15) or to deep theorems about curves or relative curves. One fundamental result about smooth irreducible projective curves X over an algebraically closed base field is the theorem describing the group J_n of n-division points on the Jacobian variety J:

$$J_n\cong(\mathbb{Z}/n\mathbb{Z})^{2\dim J}.$$

Here n is assumed to be relatively prime to the characteristic of the base field. In fact, the first etale cohomology group $H^1(X, \mathbb{Z}/n\mathbb{Z})$ is isomorphic to J_n (see § 5). The necessary theorems about abelian varieties can be found e.g. in Mumford's book "Abelian Varieties" [117].

Unfortunately, the etale cohomology theory developed up to section 11 yields only the "right" cohomology groups with *torsion coefficients*. More effort is needed to get the right cohomology groups $H^i(X, L)$ with coefficients in a field L of characteristic zero (namely, the field $L=\mathbb{Q}_l$ of l-adic numbers). In §12 these l-adic cohomology groups are introduced by means of inverse limits:

$$H^i(X, \mathbb{Z}_l) = \varprojlim_v H^i(X, \mathbb{Z}/l^v\mathbb{Z}),$$

$$H^i(X, \mathbb{Q}_l) = H^i(X, \mathbb{Z}_l) \otimes_{\mathbb{Z}_l} \mathbb{Q}_l .$$

The main result is a finiteness theorem for these l-adic cohomology groups and for the direct images of what are called l-adic sheaves.

It hardly needs to be emphasized that homological algebra also plays a role in the development of etale cohomology theory. For us it is often convenient to use it in the form given to it by Verdier, which seems to be unfamiliar to many mathematicians. We have therefore assembled in an appendix (A II) the basic ideas on derived categories and derived functors in the sense of Verdier.

The second chapter is devoted to Poincaré duality and the proof of the fixed-point formula for the Frobenius homomorphism. The main result is Grothendieck's formula for the ζ-function $\zeta_{X_0}(s)=f(q^{-s})$ of an algebraic variety X_0 over a field with q elements. From this formula it follows that $f(t)$ is a *rational function*. This was first proved by Dwork [46] using other methods.

The theme of the third chapter is Deligne's *monodromy theory* for Lefschetz pencils. This plays a central role in Deligne's proof of the Weil conjecture. Over a sufficiently large field, and after appropriate modification, every smooth projective variety can be fibered over the projective line $\mathbb{P}^1$ as a *Lefschetz pencil*,

$$f\colon X \to \mathbb{P}^1 .$$

This fibering mapping has only finitely many singularities, which lie over a finite subset S of $\mathbb{P}^1$. They are of a specially simple nature, *ordinary double points*. To use the Leray spectral sequence for computing the cohomology groups of X, one needs information about how the fundamental group of $\mathbb{P}^1 - S$ acts on the cohomology groups of a generic fiber X_a of f. Here we may mention the *Kazhdan-Margulis* theorem about the operation on a certain part (arising from the singularities) of the middle cohomology group of X. (Take X here of even dimension.) Theorems on the cohomology of Lefschetz pencils play a large role in our induction arguments.

With the machinery developed in the first three chapters, it is not hard to understand Deligne's proof of the Weil conjecture. We present this proof in the final chapter, following the original proof very closely. To this we add the most important applications of the Weil conjecture (mainly without proof) and explain (again without proof) Deligne's generalization of the Weil conjecture.

Chapter I. The Essentials of Etale Cohomology Theory

We start this chapter with an example, due to J-P. Serre, that illuminates some of the difficulties in constructing a Weil cohomology.

Let k be an algebraically closed base field of characteristic $p>0$. We consider a "Weil cohomology theory" (see the Preface) with coefficients in a field L of characteristic 0. For a smooth projective curve of genus 1, that is, an elliptic curve, the first cohomology group $H^1(X)$ should be a vector space of dimension 2. We denote by $\mathrm{End}(X)$ the ring of endomorphisms of X, acting on X on the right. The endomorphisms act in a natural way on $H^1(X)$, and this action is compatible with products. Using the Künneth formula, one shows that the action is compatible with addition. Thus we get a representation of the ring $\mathrm{End}(X)$, and hence also of the skew field $A=\mathrm{End}(X)\otimes\mathbb{Q}$, on $H^1(X)$. If X is a supersingular curve, then A is a skew field of rank 4 over its center $\mathbb{Q}$. Such an A cannot have any two-dimensional representations over the field $\mathbb{Q}$. *Hence there does not exist any Weil cohomology theory over* $L=\mathbb{Q}$. On the other hand, for every prime number $l\neq p$ there is a faithful representation of A on the Tate module

$$T_l(X)=(\varprojlim_{\nu}\mathrm{Ker}(X\xrightarrow{l^\nu}X))\otimes\mathbb{Q},$$

and this is a two-dimensional vector space over the field $\mathbb{Q}_l$ of l-adic numbers. Thus our example does not exclude $L=\mathbb{Q}_l$ as a coefficient field for a Weil cohomology. In fact, in the proof of the Weil conjecture for abelian varieties, the dual of the Tate module of an abelian variety plays the role of the first cohomology group.

Incidentally, the fields $L=\mathbb{Q}_p$ and $\mathbb{R}$ are also excluded as coefficient fields for a Weil cohomology theory, since a quaternion skew field over $\mathbb{Q}$ has at least two ramified places. And indeed the coefficient field for crystalline cohomology in general is not $\mathbb{Q}_p$ but an extension field of it, a field of Witt vectors.

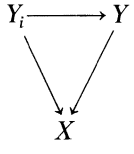
One might naturally try to construct a Weil cohomology theory by using sheaf theory for the Zariski topology. This does indeed yield reasonable cohomology groups for coherent sheaves, but these are vector spaces over the field k itself, which has characteristic p. The higher cohomology groups of constant or locally constant sheaves trivially vanish. This is connected with the fact that a connected locally trivial covering space

$$f\colon X\to Y \quad (X \text{ connected})$$

in the Zariski topology is an open embedding. We must use a somewhat different definition of (bounded) covering space in algebraic geometry: we require that f be flat and unramified. We then call X *etale* over Y.

Now on the Grothendieck's generalization, which is both simple and brilliant. To define sheaves and Čech cohomology, one does not need the whole toplogical space, just the "category of open sets" of X and the knowledge of which families of open sets form a covering of some other open set. Sheaves then are certain contravariant functors having a sheaf property with respect to these coverings. Intersections of open sets, as they appear in this definition and in the construction of Čech cohomology groups, are defined categorically as fiber products!

Here is Grothendieck's refinement of the Zariski topology. Instead of the category of Zariski-open subsets of X, we consider the category $\mathrm{Et}(X)$ of etale spaces $Y \to X$ over X. A family



in $\mathrm{Et}(Y)$ is called a covering if Y is the union of the images of the $Y_i \to Y$. Then the category of sheaves on $\mathrm{Et}(X)$ can be defined formally word-for-word as in the case of the Zariski topology. Almost immediately from the definition we get then:

The first Čech cohomology group with coefficients in the constant sheaf $\mathscr{G} = E_X$ (where E is a finite group) is isomorphic to the set of isomorphism classes of E-Galois covering spaces of X (in the sense of I.2.11). This is a first sign of the correctness of Grothendieck's approach.

In § 2 we give the precise definition of $\mathrm{Et}(X)$. The first section starts with the dual concept of an etale ring extension and the sheaf theory and sheaf cohomology for rings. In the case of a field, this gives precisely the Galois cohomology of the field.

Unfortunately, the approach to Weil cohomology that we have just sketched is still not adequate. It gives the "right" cohomology groups only for torsion coefficients (with torsion relatively prime to the characteristic of the base field). We only get the cohomology groups we want, with coefficients in the field of l-adic numbers, by passing to a projective limit and introducing denominators:

$$\varprojlim_{v} H^i(X, \mathbb{Z}/l^v\mathbb{Z}) \otimes_{\mathbb{Z}_l} \mathbb{Q}_l .$$

This passage to the limit is analogous to the construction of the algebraic fundamental group $\pi_1(X)$ as a profinite group.

§1. Etale Ring Extensions

We assemble the basic properties of *"etale ring extensions"*, giving only the briefest indications of the proofs.

References: Grothendieck, A. and Dieudonné, J. [63], Grothendieck, A. et al. [64], Raynaud, M. [125].

We introduce etale cohomology in the affine case and specify its connection with *Galois cohomology*. We state *Tsen's theorem* without proof.

Reference: Serre [141].

Finally, we give some properties of Henselian ring as well as the construction of Henselian rings by *"Henselization"* of local rings.

References: Grothendieck, A. and Dieudonné, J. [63], Raynaud, M. [125].

All rings are presumed to be *commutative*, *unital*, and *noetherian*, if nothing else is indicated by the context.

A homomorphism of rings $A \to B$ is called local if A and B are local rings and their maximal ideals $\mathfrak{m}(A)$, $\mathfrak{m}(B)$ satisfy

$$\mathfrak{m}(A) \cdot B \subset \mathfrak{m}(B).$$

A local homomorphism is called *unramified* if the following conditions are satisfied:

a) $\mathfrak{m}(A) \cdot B = \mathfrak{m}(B)$, and

b) the residue field extension $A/\mathfrak{m}(A) \to B/\mathfrak{m}(B)$ is finite and separable.

An unramified homomorphism is in particular *quasi-finite*; that is, $B/\mathfrak{m}(A) \cdot B$ is a finite-dimensional $A/\mathfrak{m}(A)$-algebra (or, equivalently, the completion $\hat{B}$ is a finitely generated $\hat{A}$-module). In general we will be interested only in *"finitely generated" local homomorphisms* $A \to B$, that is, those where B arises through localization of a finitely generated A-algebra. For such local homomorphisms it follows from quasi-finiteness (by Zariski's Main Theorem, see e.g. [125]) that B is actually the localization of an A-algebra B_0 that is finitely generated as an A-module.

1.1 Definition. A ring homomorphism $A \to B$ is called etale if the following conditions are satisfied:

a) B is a finitely generated A-algebra.

b) B is a flat A-module.

c) $A \to B$ is unramified; that is, all the localizations $A_{\mathfrak{q}} \to B_{\mathfrak{p}}$ are unramified ($\mathfrak{p} \in \operatorname{Spec} B$, $\mathfrak{q} =$ "$\mathfrak{p} \cap A$").

It follows from Zariski's Main Theorem that an etale algebra $A \to B$ is always the localization (at some multiplicative set) of an A-algebra finitely generated as an A-module.

An algebra over a field k is etale if and only if it is the Cartesian product of finitely many finite separable field extensions. Such an algebra is also called *finite separable*. More generally, we have:

1.2 Remark. A finitely generated flat algebra $A \to B$ is etale if and only if the following condition is satisfied:

For every prime ideal $\mathfrak{p} \subset A$, the algebra $B \otimes_A k(\mathfrak{p})$ is finite separable over $k(\mathfrak{p})$. (Here $k(\mathfrak{p})$ is the quotient field of $A/\mathfrak{p}$.)

Let $A \to B$ be a finitely generated local homomorphism. Ramification can be tested using a simple differential criterion: *B is unramified over A precisely when the module of Kähler differentials of B over A vanishes* [125]. This yields the following criterion for unramified ring extensions, one suitable for computation:

1.3 Proposition. *Let*

$$P_1, \ldots, P_m \in A[X_1, \ldots, X_n]$$

be polynomials over a ring A, and let $\mathfrak{p} \subset A[X_1, \ldots, X_n]$ be a prime ideal containing these polynomials. Let

$$\mathfrak{p}_0 = A \cap \mathfrak{p}.$$

The local $A_{\mathfrak{p}_0}$-algebra

$$B = \{A[X_1, \ldots, X_n]/(P_1, \ldots, P_m)\}_{\mathfrak{p}}$$

is unramified if and only if the n by n subdeterminants of the matrix

$$(\partial P_\mu / \partial X_\nu) \qquad 1 \leqq \mu \leqq m, \ 1 \leqq \nu \leqq n$$

generate the unit ideal in $A[X_1, \ldots, X_n]_{\mathfrak{p}}$.

Note. In the case where $m = n$, the above hypothesis implies that B is a relative complete intersection, and hence flat (and hence local-etale in the sense of 1.4*) over $A_{\mathfrak{p}_0}$.*

The localizations $A_{\mathfrak{q}} \to B_{\mathfrak{p}}$ of a finitely generated homomorphism $A \to B$ are in general not finitely generated. They are only finitely generated *local* homomorphisms. It is advisable therefore to modify the definition of an etale extension for local rings.

1.4 Definition. A homomorphism $A \to B$ of local rings is called local-etale if the following conditions are satisfied:

a) $A \to B$ is a finitely generated local homomorphism.

b) $A \to B$ is flat and unramified.

It follows then from the differential criterion that all localizations $A_{\mathfrak{q}} \to B_{\mathfrak{p}}$ are also local-etale. Let $A \to B$ be a finitely generated algebra. From the differential criterion it follows that the set of prime ideals $\mathfrak{p} \subset B$ for which $A_{\mathfrak{q}} \to B_{\mathfrak{p}}$ is unramified is a Zariski-open set in $\operatorname{Spec} B$. Since the flat locus is also open, we can replace "unramified" by "etale" in this statement. Hence:

Every local-etale homomorphism $A \to B$ arises by localization of a finitely generated etale A-algebra.

Let $A \to B$ be an unramified finitely generated local homomorphism. We do not assume that B is a flat A-module. We can write B as the surjective

image of an algebra homomorphism

$$A[X_1, \ldots, X_n]_{\mathfrak{m}} \to B \quad (\mathfrak{m} \text{ a maximal ideal}).$$

From the differential criterion for ramification we can deduce the existence of n polynomials in the kernel, $P_1, \ldots, P_n \in A[X_1, \ldots, X_n]$, such that the determinant of the matrix

$$(\partial P_\mu / \partial X_\nu) \qquad 1 \leqq \mu,\ \nu \leqq n$$

is a unit in $A[X_1, \ldots, X_n]_{\mathfrak{m}}$. The algebra

$$\tilde{B} = A[X_1, \ldots, X_n]_{\mathfrak{m}}/(P_1, \ldots, P_n)$$

is then flat and unramified over A, and thus local-etale (1.3).

The unramified algebra B is thus the quotient of a local-etale extension $\tilde{B}$. Hence:

In the classical case of extensions of normal integral domains $A \subset B$, the requirement of flatness is unnecessary. For then it is actually a consequence of being unramified:

1.5 Lemma. *Let $A \to B$ be a finitely generated local homomorphism. We assume that it is injective and that A is a normal ring. If B is unramified over A, then B is a local-etale A-algebra.*

Proof. With the previous notation, B is a factor ring of a local-etale A-algebra $\tilde{B}$, say $B = \tilde{B}/\mathfrak{a}$. As A is normal, so is $\tilde{B}$ (as we shall see later, 1.19), and thus without zero divisors. From the injective of $A \to B$ we conclude $\mathfrak{a} = 0$. □

Nonetheless, the concept of flatness plays a large role in the study of etale ring extensions. For one thing, rings that are not normal arise naturally; and flatness also plays a technical role even for normal rings.

What Flatness Is Good For [63]

1) Every flat local homomorphism is faithfully flat, and hence injective.

(A flat ring homomorphism $A \to B$ is called faithfully flat when the mapping $\operatorname{Spec} B \to \operatorname{Spec} A$ is surjective.)

2) Every finitely generated flat module over a local ring is free.

3) An A-module M is flat if and only if every localization $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$-module. Here we can restrict to maximal ideals.

4) Let $A \to B$ be a finitely generated flat algebra. Then the mapping $\operatorname{Spec} B \to \operatorname{Spec} A$ is open.

5) Let A be a reduced ring (for instance, an integral domain). Let M be a finitely generated module over a finitely generated A-algebra B. There exists a non zero divisor $f \in A$ for which M_f is a free A_f-module. From this fact one can deduce: The set of prime ideals $\mathfrak{p} \subset B$ for which $M_{\mathfrak{p}}$ is a flat A-module is a Zariski-open dense subset of $\operatorname{Spec} B$.

1.6 Remark. Let A be a local ring with its residue field separably algebraically closed. For a finitely generated local homomorphism $A \to B$, the following asser-

tions are equivalent:

a) It is local-etale.

b) It induces an isomorphism $\hat{A} \to \hat{B}$ on the completions.

From Remark 1.6 one can deduce:

1.7 Proposition. *Let $X \to Y$ be a mapping of affine algebraic varieties $X \subset \mathbb{C}^n$, $Y \subset \mathbb{C}^m$. Then the following two assertions are equivalent:*

a) *The mapping is locally biholomorphic in the sense of complex analysis.*

b) *The corresponding homomorphism $A(Y) \to A(X)$ of affine algebras is etale.*

1.8 Remark. (Permanence properties of etale ring extensions.)

1) A composite of etale homomorphisms is etale.

2) An etale homomorphism $A \to B$ remains etale after an arbitrary base extension $A \to C$; that is,

$$A \otimes_A C = C \to B \otimes_A C$$

is etale.

3) Let $A \to B \to C$ be ring extensions. If C/A and B/A are etale, then C/B is also etale. In particular, the tensor product of two etale A-algebras is etale.

The only nontrivial point in 1.8 is the flatness of C/B in 3). (This follows easily from 2.2.)

We denote by $\mathrm{Et}(A)$ the category of all etale extensions of A (understood as a full subcategory of the category of all A-algebras). Because of 3), all morphisms in $\mathrm{Et}(A)$ are etale.

1.9 Definition. A presheaf (of sets, of abelian groups, ...) is a covariant functor

$$\mathscr{F}: \mathrm{Et}(A) \to \mathrm{Sets}\ (= \text{category of sets}),$$
$$\mathrm{Ab}\ (= \text{category of abelian groups}),$$
$$\ldots$$

1.10 Definition. A presheaf $\mathscr{F}$ is called a sheaf if the following two conditions are satisfied:

a) $\mathscr{F}(B_1 \times B_2) \cong \mathscr{F}(B_1) \times \mathscr{F}(B_2)$ (canonically).

b) Let $B \to C$ be a faithfully flat homomorphism of etale algebras. Then the sequence

$$\mathscr{F}(B) \to \mathscr{F}(C) \rightrightarrows \mathscr{F}(C \otimes_B C)$$

is exact.

The two arrows on top of each other in b) are induced by the ring homomorphisms

$$C \to C \otimes_B C; \qquad c \mapsto c \otimes 1, \qquad c \mapsto 1 \otimes c.$$

The exactness means that $\mathscr{F}(B)$ is mapped bijectively onto the kernel of the pair of arrows:

$$\mathscr{F}(B) \cong \{s \in \mathscr{F}(C), \text{ the two images of } s \text{ agreeing in } \mathscr{F}(C \otimes_B C)\}.$$

The category $\mathrm{Et}(A)$ is small; that is, the isomorphism classes of its objects form a set. Hence the collection of all presheaves, with the natural transformations as morphisms, forms a category. We consider the category of sheaves as a full subcategory. As in the topological case, sheaves can also be defined using the idea of covering:

A finite family $\mathfrak{B}=(B\to B_i,\ i\in I)$ of etale homomorphisms is called an (etale) *covering* when the images of the $\operatorname{Spec} B_i$ cover all of $\operatorname{Spec} B$. These then form a Zariski-open covering of $\operatorname{Spec} B$. The homomorphism

$$B\to\prod_{i\in I} B_i$$

is then faithfully flat.

Using the concept of an etale covering, we can formulate both sheaf properties a) and b) as follows.

For every covering $\mathfrak{B}$ of an etale A-algebra B we have:

The elements $s\in\mathfrak{F}(B)$ *correspond one-to-one to families of elements*

$$s_i\in\mathfrak{F}(B_i),\qquad i\in I,$$

having the property

$$\mathrm{Image}(s_i)=\mathrm{Image}(s_j)\quad \textit{in}\ (B_i\otimes_B B_j).$$

(This condition must also hold for $i=j$*!)*

As in the usual (topological) sheaf theory, we can construct the sheaf $\tilde{\mathfrak{F}}$ generated by a presheaf $\mathfrak{F}$. For this we consider the category of all etale coverings of a fixed object $B\in\mathrm{Et}(A)$. A mapping between two coverings $\mathfrak{B}\to\mathfrak{B}'$,

$$\mathfrak{B}=(B\to B_i,\ i\in I);\qquad \mathfrak{B}'=(B\to B'_j,\ j\in J),$$

is given by a mapping $\sigma\colon J\to I$ of the index sets and a family of homomorphisms $B_{\sigma(j)}\to B'_j$. For each covering

$$\mathfrak{B}=(B\to B_i)$$

we denote by $\mathfrak{F}(\mathfrak{B})$ the set of all families $s_i\in\mathfrak{F}(B_i)$ with the above compatibility property. The assignment $\mathfrak{B}\mapsto\mathfrak{F}(\mathfrak{B})$ becomes in an obvious way a functor on the category of coverings. It is easy to see that:

1) For any two coverings $\mathfrak{B}$ and $\mathfrak{B}'$ there is a third covering $\mathfrak{B}''$ together with morphisms

$$\mathfrak{B}\to\mathfrak{B}'',\qquad \mathfrak{B}'\to\mathfrak{B}''.$$

2) Two morphisms $\mathfrak{B}\rightrightarrows\mathfrak{B}'$ between the same coverings induce the same mapping $\mathfrak{F}(\mathfrak{B})\to\mathfrak{F}(\mathfrak{B}')$.

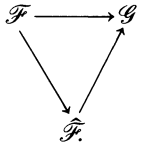
We define

$$\tilde{\mathfrak{F}}(B)=\varinjlim_{\mathfrak{B}}\mathfrak{F}(\mathfrak{B}).$$

The assignment $B\mapsto\tilde{\mathfrak{F}}(B)$ together with the obvious connecting mappings defines a presheaf $\tilde{\mathfrak{F}}$. Every morphism $\mathfrak{F}_1\to\mathfrak{F}_2$ induces a morphism $\tilde{\mathfrak{F}}_1\to\tilde{\mathfrak{F}}_2$ compatible with the natural mapping $\mathfrak{F}\to\tilde{\mathfrak{F}}$. Obviously $\mathfrak{F}$ is a sheaf precisely when the natural mapping $\mathfrak{F}\to\tilde{\mathfrak{F}}$ is an isomorphism. In general, $\tilde{\mathfrak{F}}$ is not a sheaf. We have only a weaker property:

For all $B \in \mathrm{Et}(A)$ and all etale coverings $\mathscr{B}$ of B, $\tilde{\mathscr{F}}(B) \to \tilde{\mathscr{F}}(\mathscr{B})$ is injective. If the mappings $\mathscr{F}(B) \to \mathscr{F}(\mathscr{B})$ for $\mathscr{F}$ are already injective, then $\tilde{\mathscr{F}}$ is a sheaf. From this follows:

1.11 Remark. Twofold repetition of "$\tilde{}$" applied to a presheaf $\mathscr{F}$ yields a sheaf $\hat{\mathscr{F}} = \tilde{\tilde{\mathscr{F}}}$, called the sheaf generated by $\mathscr{F}$. For every morphism of a presheaf $\mathscr{F}$ into a sheaf $\mathscr{G}$ there is a uniquely determined factorization



Thus $\mathrm{Hom}(\mathscr{F}, \mathscr{G}) \cong \mathrm{Hom}(\hat{\mathscr{F}}, \mathscr{G})$; that is, the functor "$\mathscr{F} \mapsto \hat{\mathscr{F}}$" is left adjoint to the functor embedding the category of sheaves in the category of presheaves.

As usual, one can embed the category $\mathrm{Et}(A)$ contravariantly in its functor category, i.e. in the category of all presheaves of sets, by assigning to an object $B \in \mathrm{Et}(A)$ the functor

$$\underline{B}(C) = \mathrm{Hom}_A(B, C).$$

From the descent lemma (Appendix III) we have:

1.12 Remark. If B is an object in $\mathrm{Et}(A)$, then the functor

$$\underline{B}(C) = \mathrm{Hom}_A(B, C) \quad \text{(algebra homomorphisms)}$$

is a sheaf of sets.

The category of presheaves of abelian groups is abelian:
A sequence $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact precisely when

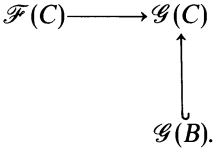
$$\mathscr{F}(B) \to \mathscr{G}(B) \to \mathscr{H}(B), \qquad B \in \mathrm{Et}(A),$$

is exact.

The construction of the "sheaf generated" functor easily gives:

The category of sheaves of abelian groups is abelian:

A sequence $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact if and only if for every $B \in \mathrm{Et}(A)$ we have: An element $s \in \mathscr{G}(B)$ lies in the kernel of $\mathscr{G}(B) \to \mathscr{H}(B)$ if and only if there is a faithfully flat etale extension $B \to C$ such that s lies in the image of $\mathscr{F}(C)$,



Warning: A sequence $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H}$ of sheaves is indeed exact in the category of sheaves if and only if it is exact in the category of presheaves. But a mapping that is surjective in the category of sheaves does not have to be surjective in the category of presheaves. If $\mathscr{F} \subset \mathscr{G}$ are sheaves of abelian groups, the quotient

sheaf is obtained from the quotient presheaf

$$B \mapsto \mathscr{G}(B)/\mathscr{F}(B)$$

by applying the functor "$\widehat{\ }$" (in this case "$\sim$" suffices). We always use $G/\mathscr{F}$ to denote the *quotient sheaf*. The functor $\mathscr{F} \mapsto \hat{\mathscr{F}}$ from the category of presheaves to the category of sheaves is exact.

The functor $\mathscr{F} \mapsto \mathscr{F}(A)$, going from the category of sheaves of abelian groups to the category of abelian groups, is left exact. Its derived functors are the etale cohomology groups,

$$H^i(A, \mathscr{F}) = R^i(\mathscr{F} \mapsto \mathscr{F}(A)).$$

Thus if $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ is an exact sequence of sheaves, we get a long exact cohomology sequence

$$\begin{array}{ccccccccc} 0 \to & \mathscr{F}(A) & \to & \mathscr{G}(A) & \to & \mathscr{H}(A) & \to & H^1(A, \mathscr{F}) & \to \ldots \\ & \| & & & & & & & \\ & H^0(A, \mathscr{F}). & & & & & & & \end{array}$$

Remark. In the category of sheaves of abelian groups, there are enough injectives; that is, every sheaf is a subsheaf of an injective sheaf.

By Grothendieck's results [54], this follows from

1) the existence and exactness of filtered direct limits,

2) the existence of a set $\mathfrak{M}$ of objects (called "generators") with the following property:

Every sheaf is a quotient sheaf of a (possibly infinite) direct sum of sheaves from $\mathfrak{M}$.

For 1): In the category of presheaves arbitrary limits exist trivially:

$$(\varinjlim \mathscr{F}_i)(U) = \varinjlim (\mathscr{F}_i(U)), \qquad (\varprojlim \mathscr{F}_i)(U) = \varprojlim \mathscr{F}_i(U).$$

The inverse limit of a system of sheaves, computed in the category of presheaves, satisfies the sheaf properties and hence is the limit in the category of sheaves. The direct limit of a system of sheaves is found by applying the "sheaf generated" functor to the direct limit constructed in the category of presheaves. To prove this, use the fact that this functor is left adjoint to the embedding functor (1.11). Since the "sheaf generated" functor is exact, the exactness of filtered direct limits follows from the corresponding exactness statement in the category of presheaves.

For 2): Let $\mathscr{F}$ be a sheaf of sets. If we assign to each object $B \in \mathrm{Et}(A)$ the free abelian group generated by $\mathscr{F}(B)$, we obtain a presheaf of abelian groups. We denote by $\mathscr{F}^{\mathrm{ab}}$ the sheaf of abelian groups generated by this presheaf. To every object $B \in \mathrm{Et}(A)$ we have assigned the sheaf $\underline{B}$ represented by B. We maintain now: If B runs through a system of representatives of the isomorphism classes of $\mathrm{Et}(A)$ (these form a set!), then the sheaves $\underline{B}^{\mathrm{ab}}$ form a system of generators. The proof follows easily from the following fact:

Let $\mathcal{F}$ be a sheaf of sets over A, and B an etale A-algebra. For every element $s \in \mathcal{F}(B)$ there is exactly one morphism $\underline{B} \to \mathcal{F}$ that takes the element

$$\mathrm{id}_B \in \underline{B}(B) = \mathrm{Hom}_A(B, B)$$

into s. □

1.13 Remark. Let M be an A-module.

1) The functor

$$B \mapsto \tilde{M}(B) = M \otimes_A B \quad \text{for } B \in \mathrm{Et}(A)$$

is a sheaf of abelian groups.

2) $H^i(A, \tilde{M}) = 0$ for $i > 0$.

In what follows we only need part 1) of this statement, which like (1.12) follows from the descent lemma. The second part comes again from the descent lemma (Appendix III) combined with Cartan's lemma (2.9). □

Galois Cohomology as a Special Case of Etale Cohomology

Let K be a field, $G = G(\bar{K}/K)$ the Galois group of the separable closure $\bar{K}$ of K. The etale category over K is especially simple:

Every etale K-algebra is the direct product of finitely many separable field extensions.

Let $\mathcal{F}$ be an etale sheaf of abelian groups over K. We set

$$\mathcal{F}(\bar{K}) = \varinjlim_{\substack{K \subset L \subset \bar{K} \\ [L:K] < \infty}} \mathcal{F}(L).$$

Here we can of course restrict ourselves to Galois extensions L.

The Galois group G acts continuously on the abelian group $\mathcal{F}(\bar{K})$; that is, the stabilizer of each element in $\mathcal{F}(\bar{K})$ has finite index in G. Galois cohomology theory studies the groups $H^*(G, \mathcal{F}(\bar{K}))$.

1.14 Remark. Assigning $\mathcal{F} \mapsto \mathcal{F}(\bar{K})$ defines an equivalence between the category of etale sheaves over K and the category of continuous G-modules. In particular we have:

$$H^q(K, \mathcal{F}) = H^q(G, \mathcal{F}(\bar{K})).$$

Example. Consider the sheaf $\mathcal{G}_m$:

$$\mathcal{G}_m(B) = B^* \quad \text{(multiplicative group)}.$$

This corresponds to the Galois module $(\bar{K})^*$. It is well known that

$$H^1(K, \mathcal{G}_m) = 0 \quad \text{(“Hilbert Theorem 90”)},$$

$$H^2(K, \mathcal{G}_m) = \mathrm{Br}(K) \quad \text{(Brauer group)}.$$

If K is a function field in one variable over an algebraically closed constant field, the Brauer group of K vanishes by *Tsen's theorem* [141]. Using this fact, one can show [141]:

1.15 Proposition. *Let K be a function field in one variable over an algebraically closed constant field. Then*

$$H^q(K, \mathcal{G}_m) = 0 \quad \text{for } q > 0.$$

The Brauer group $\mathrm{Br}(K)$ of a function field in one variable over a base field k that is separably algebraically closed – not necessarily algebraically closed – does not vanish in general. But it is always a p-torsion group (p = characteristic of k).

1.16 Supplement to Proposition 1.15. *Let K be a function field in one variable over a separably algebraically closed constant field of characteristic p. Then $H^q(K, \mathcal{G}_m)$ for $q > 0$ is a p-torsion group.*

Henselian Rings

1.17 Definition. A (noetherian) local ring A is called Henselian if every local quasi-finite homomorphism from A to a localization of a finitely generated A-algebra is finite.

We recall our terminology:

1) A ring homomorphism $A \to B$ is called *finite* if B is a finitely generated A-module.

2) A local ring homomorphism $A \to B$ is called *quasifinite* if

$$A/\mathfrak{m}(A) \to B/\mathfrak{m}(A) \cdot B$$

is finite, or equivalently when the completion $\hat{B}$ is a finite $\hat{A}$-module.

For example, complete rings are Henselian.

A local ring A is Henselian if and only if Hensel's lemma holds [125].

Hensel's Lemma. *Let $P \in A[X]$ be a monic polynomial whose residue class in $(A/\mathfrak{m})[X]$ splits as the product of two relatively prime monic polynomials,*

$$\bar{P} = \bar{Q}\bar{R}, \quad \bar{Q} \text{ and } \bar{R} \in A/\mathfrak{m}[X],$$

Then P itself is the product of two monic polynomials

$$P = QR, \quad Q \text{ and } R \in A[X],$$

such that $\bar{Q}$ and $\bar{R}$ are the residue classes of Q and R.

A Henselian ring is called *strictly Henselian* if the residue class field is separably algebraically closed.

1.18 Remark. Let A be a strictly Henselian ring. Every local-etale homomorphism $A \to B$ is an isomorphism.

It follows from this that for every faithfully flat etale homomorphism ϕ of a strictly Henselian ring into a finitely generated A-algebra there exists an algebra homomorphism $\psi: B \to A$ such that $\psi \circ \phi = \mathrm{id}_A$.

Consequently, the functor

$$\mathrm{Et}(B)\to Ab, \qquad B\mapsto \mathscr{F}(B)$$

is exact for any sheaf of abelian groups.

1.19 Corollary. *Let A be a strictly Henselian ring. Then*

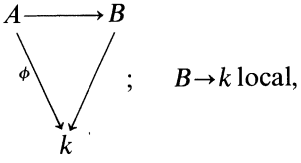
$$H^i(A, \mathscr{F})=0 \quad \text{for } i>0 \text{ and every sheaf } \mathscr{F}.$$

A generalization of 1.18 says:

The category of local-etale extensions of a Henselian ring is equivalent to the category of finite separable field extensions of $A/\mathfrak{m}(A)$.

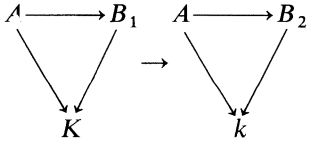
Construction of Henselian Rings

Let A be a local ring and ϕ: $A\to k$ a local homomorphism to a field. The field k is thus an extension field of the residue field of A, $A/\mathfrak{m}(A)\to k$. We consider commutative diagrams



; $B\to k$ local,

where $A\to B$ is a local-etale homomorphism. In an obvious way, these diagrams form a category: a morphism



is a local homomorphism $B_1\to B_2$ that is compatible with the arrows $A\to B_i\to k$ $(i=1, 2)$.

One can show the following:

1) There is at most one morphism between two such diagrams.

2) For every two diagrams one can find two morphisms of them into a third diagram, $B_1\to B_3$, $B_2\to B_3$. One constructs B_3 as a localization of $B_1\otimes_A B_2$ at a prime ideal.

In particular, the category of diagrams is filtered.

If we form the direct limit $\tilde{A}=\varinjlim B$ over this category of diagrams, we get a local algebra $A\to\tilde{A}$ together with a local homomorphism of $\tilde{A}$ to k, that is, an embedding $\tilde{A}/\mathfrak{m}(\tilde{A})\to k$.

1.20 Remark [125]. The ring $\tilde{A}$ is Henselian (in particular, noetherian), and is even strictly Henselian if k is separably algebraically closed.

Supplement. The extension $A\to\tilde{A}$ is faithfully flat.

We call $\tilde{A}$ the Henselization of A with respect to the specified field extension of $A/\mathfrak{m}(A)$. Two cases should be emphasized:

1) $k=A/\mathfrak{m}$. Then $\tilde{A}$ is the "smallest" Henselian extension of A, and is called simply *the Henselization* of A.

2) k is separably algebraically closed. Then $\tilde{A}$ is strictly Henselian and is called a strict Henselization of A. Up to isomorphism it does not depend on the choice of k.

Example. Let A be a ring without zero divisors that is excellent (see below), for instance an integral domain locally finitely generated over a base field. Then one can show that the Henselization of A is the subring of those elements in the completion of A that are algebraic (not necessarily integral) over A.

In conclusion, we formulate some stability properties of local-etale extensions.

1.21 Proposition [63]. *Let $A\to B$ be a local-etale homomorphism. The ring B is regular, reduced, Cohen-Macauley, or normal if and only if the same holds for A.*

Warning. This statement does not hold for absence of zero divisors!

It follows immediately that the properties of A in 1.21 carry over to the Henselization $\tilde{A}$ (with respect to any residue field extension). Another important property of a local ring A is that of being *excellent* [63], which means:

a) A is universally catenary; that is, for every locally finitely generated A-algebra B and every prime ideal $\mathfrak{p}$ of B we have

$$\dim B=\dim B_{\mathfrak{p}}+\dim B/\mathfrak{p}.$$

b) For every locally finitely generated A-algebra $A\to B$, the regular locus of the completion $\hat{B}$ of B is the inverse image of the regular locus of B. This property carries over immediately to every local-etale ring extension of A and to the Henselization $\tilde{A}$ of A (since $\hat{\tilde{A}}=\hat{A}$).

Not so immediate is:

Remark. The strict Henselization of an excellent local ring A is excellent. For lack of a reference we give a brief sketch of the proof. It is enough to prove the following special case of b):

Let $\mathfrak{q}$ be a prime ideal of $\hat{\tilde{A}}$ with $\mathfrak{q}\cap A=0$. Then $\hat{\tilde{A}}_{\mathfrak{q}}$ is regular. Here one can even assume that A is a complete integral domain of characteristic $p>0$. Then A is the factor ring of a formal power series ring $\kappa[[t_1,\ldots,t_n]]$ (over some field κ) divided by some prime ideal $\mathfrak{a}$. If k is the separable algebraic closure of κ, then

$$\hat{\tilde{A}}=k[[t_1,\ldots,t_n]]/\mathfrak{a}\,k[[t_1,\ldots,t_n]].$$

Since the singular locus of $\mathrm{Spec}(\hat{\tilde{A}})$ is closed, there is a radical ideal $\mathfrak{s}$ of $k[[t_1,\ldots,t_n]]$ that contains $\mathfrak{a}$ and is contained precisely in those prime ideals $\mathfrak{p}$ of $k[[t_1,\ldots,t_n]]$ for which $\hat{\tilde{A}}_{\mathfrak{p}}$ is *not regular*. The "singularity ideal" $\mathfrak{s}$ is

obviously invariant under the natural action of the Galois group Γ of k over κ on $k[[t_1, \ldots, t_n]]$. We conclude from this that $\mathfrak{s}$ is generated by an ideal $\mathfrak{s}_0 = \mathfrak{s} \cap \kappa[[t_1, \ldots, t_n]]$ of the ring $\kappa[[t_1, \ldots, t_n]]$. For we have: Let V be a finite-dimensional vector space over κ, for instance $V = k[[t_1, \ldots, t_n]]/(t_1, \ldots, t_n)^r$, and W a k-subvector space of $k \otimes_\kappa V$ that is invariant under the operation of Γ on $k \otimes_\kappa V$. Then W arises by coefficient extension from a κ-subspace of V.

Let $\mathfrak{q}$ be a prime ideal of $k[[t_1, \ldots, t_n]]$ that is contained in the singular locus of $\operatorname{Spec}(\hat{A}) \subset \operatorname{Spec}(k[[t_1, \ldots, t_n]])$, with $\mathfrak{q} \cap A = \mathfrak{a}$. Then $\mathfrak{s}$ would have to be $\mathfrak{a} k[[t_1, \ldots, t_n]]$, and $\hat{A}$ would be a reduced ring with empty regular locus! □

§ 2. Etale Cohomology of Schemes

We first generalize the concepts of § 1 to schemes. All schemes considered are assumed to be noetherian and separated, so long as nothing to the contrary is evident from the context.

A morphism $f: X \to Y$ of schemes is called *finitely generated*, if every point in Y has an open affine neighborhood $V \subset Y$ with the following property: the inverse image of V is a union of open affine schemes whose rings of sections are finitely generated algebras over the ring of sections of V.

If $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$ are affine schemes, this means simply that A is a finitely generated *B-algebra.*

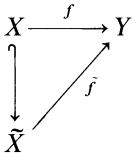
A morphism $f: X \to Y$ of schemes is called *finite* if the following conditions are satisfied for every open affine subscheme $V \subset Y$:

1) $U = f^{-1}(V)$ is affine,
2) $\Gamma(U, O_X)$ is a finitely generated *$\Gamma(V, O_Y)$-module.*

If $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$ are affine schemes, this means simply that A is a finitely generated B-module.

A morphism $f: X \to Y$ of schemes is called *quasi-finite* if it is finitely generated and all fibres $f^{-1}(y)$ are finite sets.

By "Zariski's Main Theorem" [62], *a* morphism f is quasi-finite precisely when there is a finite morphism $\tilde{f}: \tilde{X} \to Y$ and an open embedding $X \hookrightarrow \tilde{X}$ such that the diagram



is commutative.

2.1 Definition. A morphism $f: X \to Y$ of schemes is called etale if the following conditions are satisfied:

a) X is finitely generated over Y,

b) for every point $x \in X$ the morphism

$$\mathcal{O}_{Y, f(x)} \to \mathcal{O}_{X, x}$$

is local-etale.

If $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$ are affine schemes, this means that A is an etale B-algebra.

Etale morphisms are quasi-finite.

As in the affine case, a finitely generated flat morphism $f: X \to Y$ is etale precisely when the sheaf $\Omega_{X/Y}$ of relative differential forms vanishes. The diagonal mapping $\Delta: X \to X \times_Y X$ embeds the scheme X as a closed subscheme of $X \times_Y X$. Let J be the associated sheaf of ideals on $X \times_Y X$. It is well known [125] that $\Omega_{X/Y}$ is isomorphic to the restriction of J/J^2 to X. From this follows:

2.2 Remark. A flat, finitely generated morphism $X \to Y$ of schemes is etale precisely when the diagonal mapping $X \to X \times_Y X$ is an open embedding.

The permanence properties carry over from the affine case to the general one:

1) *A composite of etale morphisms is etale.*

2) *An etale morphism $X \to Y$ remains etale after an arbitrary base extension $Z \to Y$; that is,*

$$X \times_Y Z \to Z$$

is etale.

3) *Let two morphisms*

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

be given. If $g \circ f$ and g are etale, so is f.

As was already true in the affine case, only the flatness of f in 3) is nontrivial. But one gets this easily by means of Remark 2.2 (for $Y \to Z$). □

Thus in the category of all etale extensions of a scheme X, considered as a full subcategory of all X-schemes, all morphisms are etale. We denote this category by $\operatorname{Et}(X)$.

2.3 Definition. A presheaf $\mathscr{F}$ (on the category $\operatorname{Et}(X)$) of sets, abelian groups, ..., is a contravariant functor

$$\mathscr{F}: \operatorname{Et}(X) \to \begin{matrix} \text{Sets (category of sets)} \\ \text{Ab (category of abelian groups)} \\ \dots . \end{matrix}$$

2.4 Definition. *A* presheaf $\mathscr{F}$ is called a (etale) sheaf when the following conditions are satisfied:

a) $\mathscr{F}(U \dot{\cup} V) = \mathscr{F}(U) \times \mathscr{F}(V)$

b) When $U \to V$ is a surjective morphism of etale X-schemes, the sequence

$$\mathscr{F}(V) \to \mathscr{F}(U) \rightrightarrows \mathscr{F}(U \times_V U)$$

is exact.

Here by $U \dot{\cup} V$ we mean the disjoint union of the schemes U and V. We have

$$\operatorname{Spec} A \dot{\cup} \operatorname{Spec} B = \operatorname{Spec}(A \times B).$$

By an (etale) *covering* of a scheme U we mean a finite family of etale mappings

$$U_i \xrightarrow{\phi_i} U; \qquad U = \bigcup \phi_i(U_i).$$

As in the affine case, we can express the sheaf property in terms of coverings:

The elements $s \in \mathscr{F}(U)$ correspond bijectively to families of elements $s_i \in \mathscr{F}(U_i)$ whose images agree in $\mathscr{F}(U_i \times_U U_j)$.

The collection of all sheaves should again be considered as a full subcategory of the category of all presheaves.

One can again assign to every presheaf $\mathscr{F}$ a sheaf $\hat{\mathscr{F}}$ generated by it, together with a functorial mapping $\mathscr{F} \to \hat{\mathscr{F}}$ such that the mapping

$$\operatorname{Hom}(\hat{\mathscr{F}}, \mathscr{G}) \xrightarrow{\sim} \operatorname{Hom}(\mathscr{F}, \mathscr{G})$$

is bijective for all sheaves $\mathscr{G}$.

As in the case of rings, we have:

The category of sheaves of abelian groups is abelian. Arbitrary direct and inverse limits exist. Every sheaf is a subsheaf of an injective sheaf. The functor

$$\mathscr{F} \mapsto \mathscr{F}(X) = \Gamma(X, \mathscr{F})$$

from the category of sheaves of abelian groups on X to the category of abelian groups, is left exact.

We denote its derived functors by $H^i(X, \mathscr{F})$. $H^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F})$.

In particular, let $X = \operatorname{Spec} A$ be an affine scheme. One can assign to each sheaf $\mathscr{F}$ on X *a* sheaf $\mathscr{F}_0$ on A in the sense of 1.10:

$$\mathscr{F}_0(B) = \mathscr{F}(\operatorname{Spec} B).$$

This assignment is obviously an equivalence of categories. In particular, we have

$$H^i(\operatorname{Spec} A, \mathscr{F}) = H^i(A, \mathscr{F}_0).$$

Example of a sheaf. Let M be a quasicoherent module on the scheme X. One can assign to M the etale presheaf

$$M_{\text{et}}(U) = \Gamma(U, f^* M), \qquad f: U \to X \text{ etale.}$$

2.5 Proposition. *Let M be a quasicoherent module on the scheme X.*

a) *The presheaf M_{et} is a sheaf.*

b) $H^i(X, M_{\text{et}}) = H^i(X, M)$.

(On the right side here is the usual sheaf cohomology with respect to the Zariski topology.)

This is a generalization of 1.13. Here again we will only need part a).

Corollary. *Let X be a scheme. The assignment*

$$U \mapsto \Gamma(U, \mathcal{O}_U)^* \quad \textit{(group of units)}$$

for $U \in \mathrm{Et}(X)$ defines a sheaf on X.

We denote this sheaf by $\mathcal{O}^*_{X,\mathrm{et}}$, or simply by $\mathcal{O}^*_X$ if there is no risk of ambiguity.

2.6 Remark. Suppose the natural number n is invertible in $\Gamma(X, \mathcal{O}_X)$. Then the mapping

$$n\colon \mathcal{O}^*_{X,\mathrm{et}} \to \mathcal{O}^*_{X,\mathrm{et}}$$
$$f \mapsto f^n$$

is surjective in the category of sheaves.

If the natural number n is invertible in a ring A, then for every element $a \in A$ one easily constructs an etale extension $A \to B$ in which the equation $x^n = a$ has a solution, namely, $B = "A[\sqrt[n]{a}]"$; $\operatorname{Spec} B \to \operatorname{Spec} A$ is surjective. □

We denote the kernel of the mapping defined in 2.6 by

$$\mu_n = (\mu_n)_X.$$

The short exact sequence

$$0 \to (\mu_n)_X \to \mathcal{O}^*_X \to \mathcal{O}^*_X \to 0$$

is called the *Kummer sequence.*

$\mu_n(U)$ for an etale extension $U \to X$ is the group of all n-th roots of unity in $\Gamma(U, \mathcal{O}_U)$.

Let F be a set (or abelian group) and X a scheme. We can associate with F the "constant sheaf" F_X on X:

$$F_X(U) = F \times \ldots \times F = F^{n_U} \quad (U \in \mathrm{Et}\, X),$$

where n_U denotes the number of connected components of U. This is the sheaf associated with the constant presheaf given by $U \mapsto F$.

2.7 Remark. Let X be a scheme and n a natural number invertible on X. Suppose the ring $\Gamma(X, \mathcal{O}_X)$ contains a primitive n-th root of unity, that is, an element $\xi \in \Gamma(X, \mathcal{O}_X)$ whose image in $\mathcal{O}^*_{X,x}$ has order precisely n for every point x. Then ξ defines an isomorphism

$$(\mathbb{Z}/n\mathbb{Z})_X \xrightarrow{\sim} (\mu_n)_X$$
$$1 \mapsto \xi.$$

Geometric Interpretation of Some Cohomology Groups

1) From the definition of the constant sheaf F_X associated with an abelian group F it follows that

$$H^0(X, F_X) = F_X(X) = F^{n_X} \quad (X \neq \emptyset),$$

$$n_X = \text{number of connected components of } X.$$

2) Let $\mathscr{X}$ be an etale covering of the scheme X, ands let $\mathscr{F}$ be a sheaf of abelian groups on X. In the usual way then*) one can define the *Čech complex* $\check{C}^{\cdot}(\mathscr{X}, \mathscr{F})$ and the *Čech cohomology* with respect to $\mathscr{X}$.

$$H^i(\mathscr{X}, \mathscr{F}) = H^i(\check{C}^{\cdot}(\mathscr{X}, \mathscr{F})).$$

The *Čech complex* for the etale covering

$$\mathscr{X}\colon\ U_\nu \to X, \qquad \nu = 1, \ldots, n,$$

is defined as follows:

$$C^p(\mathscr{X}, \mathscr{F}) = \prod \mathscr{F}(U_{\nu_0} \times_X \ldots \times_X U_{\nu_p})$$
$$(ds)_{\nu_0, \ldots, \nu_{p+1}} = \Sigma_j (-1)^j \bar{s}_{\nu_0, \ldots, \hat{\nu}_j, \ldots, \nu_{p+1}}.$$

In this formula $\bar{s}_{\nu_0, \ldots, \hat{\nu}_j, \ldots, \nu_{p+1}}$ denotes the image of $s_{\nu_0, \ldots, \hat{\nu}_j, \ldots, \nu_{p+1}}$ in $\mathscr{F}(U_{\nu_0} \times_X \ldots \times_X U_{\nu_{p+1}})$, and the superscript "$\hat{}$" means "omit". If we take the direct limit over the category of all coverings, we get the Čech cohomology groups

$$\check{H}^i(X, \mathscr{F}) = \varinjlim_{\mathscr{X}} H^i(\mathscr{X}, \mathscr{F}).$$

Here, as in the analogous construction of the "generated sheaf" following 1.10, we must show that two mappings

$$\phi, \psi\colon\ \mathscr{X} \to \mathscr{Y}$$
$$\phi_\nu\colon\ X_\nu \to Y_{\sigma(\nu)}$$
$$\psi_\nu\colon\ X_\nu \to Y_{\tau(\nu)}$$

between coverings induce the same mapping

$$H^i(\mathscr{Y}, \mathscr{F}) \to H^i(\mathscr{X}, \mathscr{F})$$

on cohomology (cf. [51], Chap. II, Proposition 5.7.1). And in fact there is a homotopy

$$h^n\colon\ \check{C}^n(\mathscr{Y}, \mathscr{F}) \to \check{C}^{n-1}(\mathscr{X}, \mathscr{F})$$

between the two mappings

$$\check{C}^n(\mathscr{Y}, \mathscr{F}) \to \check{C}^n(\mathscr{X}, \mathscr{F})$$

induced by ϕ and ψ, namely

$$h^n(s)_{\nu_0, \ldots, \nu_{n-1}} = \Sigma_k (-1)^k \bar{s}_{\sigma\nu_0, \ldots, \sigma\nu_k, \tau\nu_k, \ldots, \tau\nu_{n-1}}.$$

Here $\bar{s}$ ... is the image of s ... under the map induced by

$$\phi_{\nu_0} \times \ldots \times \phi_{\nu_{k-1}} \times (\phi_{\nu_k}, \psi_{\nu_k}) \times \psi_{\nu_{k+1}} \times \ldots \times \psi_{\nu_{n-1}}. \quad \square$$

*) One can get an idea of the basic definitions and facts about Čech cohomology from Godement's book [57].

There exist canonical mappings

$$H^i(\mathscr{X}, \mathscr{F}) \to H^i(X, \mathscr{F}),$$
$$\check{H}^i(X, \mathscr{F}) \to H^i(X, \mathscr{F}).$$

M. Artin has shown that the Čech cohomology groups $\check{H}^i(X, \mathscr{F})$ and the sheaf cohomology groups $H^i(X, \mathscr{F})$ are canonically isomorphic, under the hypothesis that each finite subset of points of the scheme X is contained in an open affine subscheme. This hypothesis is for instance true if X is quasiprojective over a noetherian ring [12].*)

We do not intend to make any use of this result.

The following two comparison propositions carry over immediately from the topological case to the case at hand.

2.8 Proposition. *In the case $i=1$, the canonical mapping*

$$\check{H}^i(X, \mathscr{F}) \to H^i(X, \mathscr{F})$$

is an isomorphism.

2.9 Proposition ("Cartan's lemma"). *Let $\mathscr{F}$ be a sheaf of abelian groups on the scheme X, and let $\mathscr{K}$ be a class of etale morphisms $U \to X$ with the following properties:*

a) *When U and V are in $\mathscr{K}$, so is $U \times_X V$.*

b) *Every etale extension $Y \to X$ has a covering (and hence has arbitrarily fine coverings) by morphisms $U \to Y$ for which the composite $U \to X$ is in $\mathscr{K}$.*

c) *$\check{H}^i(U, \mathscr{F}) = 0$ for all $U \in \mathscr{K}$ and all $i > 0$.*

Then the canonical mapping

$$\check{H}^i(X, \mathscr{F}) \xrightarrow{\sim} H^i(X, \mathscr{F})$$

is an isomorphism for all i.

Remark on Proposition 2.9. In analogy to the topological case, we call a sheaf $\mathscr{F}$ of abelian groups on a scheme X flabby if for every etale scheme $U \to X$ over X and every finite etale covering $\mathscr{X}$ of U the cohomology groups $\check{H}^v(\mathscr{X}, \mathscr{F})$ vanish for $v \geqq 1$.

Obviously then for every etale scheme $U \to X$ we have

$$H^v(U, \mathscr{F}) = 0, \qquad v \geqq 1.$$

Every sheaf $\mathscr{F}$ can be embedded in a flabby sheaf, for instance in the canonical flabby sheaf $C^0(\mathscr{F})$ introduced by Godement (see also the proof of 12.15). Since an injective subsheaf of any sheaf always splits off as a direct summand, we get:

Every injective sheaf is flabby.

We now obtain a geometric interpretation of $H^1(X, \mathscr{O}_X^*)$. First we explain the concept of an *etale line bundle.*

*) Following Verdier, one can more generally define the Čech cohomology of "hypercoverings" and then always get agreement with the "right" cohomology.

Let $\mathscr{L}$ be an etale sheaf on X. If $\mathscr{L}(U)$ for every etale mapping $U \to X$ carries a structure of $\Gamma(U, \mathcal{O}_U)$-module, and if these structures are compatible with the restriction mappings, we call $\mathscr{L}$ an $\mathcal{O}_{X,\mathrm{et}}$-module. If $U \to X$ is an etale mapping, we define the restriction $\mathscr{L}|U$ in the obvious way. An $\mathcal{O}_{X,\mathrm{et}}$-module $\mathscr{L}$ is called an etale line bundle if there is an etale covering $U_i \to X$ such that $\mathscr{L}|U_i \cong \mathcal{O}_{U_i,\mathrm{et}}$.

We denote the set of isomorphism classes of etale line bundles by $\mathrm{Pic}_{\mathrm{et}} X$. We define a group structure on it using the tensor product.

A Cech 1-cocycle is obviously nothing but the patching data for an etale line bundle:

$$\check{H}^1(X, \mathcal{O}^*_{X,\mathrm{et}}) \cong \mathrm{Pic}_{\mathrm{et}}(X).$$

If L is an ordinary line bundle with respect to the Zariski topology, L_{et} is an etale line bundle. We thus obtain a natural mapping

$$\mathrm{Pic}(X) \to \mathrm{Pic}_{\mathrm{et}}(X),$$

where $\mathrm{Pic}(X)$ denotes the group of isomorphism classes of line bundles in the Zariski topology.

An important result of descent theory (A III) says:
The mapping $\mathrm{Pic}(X) \to \mathrm{Pic}_{\mathrm{et}}(X)$ *is an isomorphism.*

2.10 Proposition. *We have*

$$H^1(X, \mathcal{O}^*_{X,\mathrm{et}}) \cong \mathrm{Pic}(X).$$

Corollary. The first cohomology groups of $\mathcal{O}^*_X$ *(with respect to the Zariski topology) and* $\mathcal{O}^*_{X,\mathrm{et}}$ *agree.*

This comparison proposition between etale and Zariski cohomology of course does not follow from the analogous comparison theorem for coherent sheaves!

3) In a similar way we obtain from descent theory" a geometric interpretation of the first cohomology group with values in a constant sheaf.

Let F be a finite group. We denote by $\tilde{F}_X$ the group scheme defined by F over X. As a space, $\tilde{F}_X$ is the disjoint union of $|F|$ copies of X. The scheme $\tilde{F}_X$ represents the constant sheaf (of groups) F_X (see 1.12). Let X' be an etale scheme over X. Let the group F act on the X-scheme X'. We call X'/X an F-Galois covering space if the morphism

$$\tilde{F}_X \times_X X' \to X' \times_X X'$$

$$(\sigma, x') \mapsto (\sigma x', x')$$

is an isomorphism.

Two F-Galois covering spaces X'/X, X''/X are called isomorphic if there is an X-isomorphism $X' \to X''$ compatible with the action of F.

2.11 Proposition. *Let* F *be a finite abelian group. The elements of* $\check{H}^1(X, F_X)$ *correspond bijective to the isomorphism classes of* F*-Galois covering spaces* $X' \to X$.

Supplement to 2.11. *One can define* $\check{H}^1(X, F)$ *as a pointed set even for non-abelian groups* F. *Proposition* 2.11 *then remains true also for non-abelian groups* F.

Proof. Let $\mathcal{G}$ be a sheaf of sets on which the group F acts transitively and faithfully. This means that the sheaf mapping

$$F_X \times \mathcal{G} \to \mathcal{G} \times \mathcal{G}$$
$$(\sigma, t) \mapsto (t^\sigma, t)$$

is an isomorphism.

Assumption. There is an etale covering $\mathcal{V}$ with the property

$$\mathcal{G}(U) \neq \emptyset \quad \text{for } U/X \in \mathcal{V}.$$

Then for every $U \in \mathcal{V}$ there is an element $t_U \in \mathcal{G}(U)$ for which the mapping

$$F_X(U) \to \mathcal{G}(U) \quad \text{for } U/X \in \mathcal{V}$$
$$\sigma \mapsto (t_U)^\sigma$$

is bijective.

If U/X, V/X are two elements of the covering $\mathcal{V}$, then there is a uniquely determined element $\sigma(U, V) \in F_X(U \times_X V)$ with the property that

$$(t_U | U \times_X V)^{\sigma(U,V)} = t_V | U \times_X V.$$

Obviously $\{\sigma(U,V)\}$ is a 1-cocycle in the Čech complex of F_X with respect to the covering $\mathcal{V}$. This gives an element in $\check{H}^1(X, F_X)$.

We denote now by $\mathfrak{G}$ the set of all F-isomorphism classes of sheaves $\mathcal{G}$ on which F acts transitively and faithfully and which satisfy the above assumption. It is easy to see that the construction above defines a bijection

$$\mathfrak{G} \cong \check{H}^1(X, F_X).$$

Now comes the nontrivial part of the proof of 2.11.

Assertion. Every sheaf $\mathcal{G}$ in $\mathfrak{G}$ is representable by a scheme X'/X; that is,

$$\mathcal{G}(Y) \cong \mathrm{Hom}_X(Y, X') \quad \text{(canonical isomorphism)}.$$

Proof. From the assumption we have made, it follows that the sheaves

$$\mathcal{G} | U; \quad U/X \in \mathcal{V}$$

are constant and hence representable. Thus $\mathcal{G}$ is locally representable! Descent theory (A III) implies that

Every locally representable sheaf $\mathcal{G}$ is actually representable.

It is clear that the group F acts on the representing object X' and that X'/X is etale. □

Geometric Points and Stalks of Sheaves

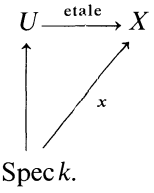
By a *geometric point* x of a scheme X with values in a separably closed field k we mean a morphism

$$x\colon \mathrm{Spec}\, k \to X;$$

that is, a point $x_0 \in X$ together with an embedding

$$\mathcal{O}_{X,x_0}/\mathfrak{m}(\mathcal{O}_{X,x_0}) \to k.$$

An *etale neighborhood* of x is a commutative diagram



The etale neighborhoods of a geometric point form a filtered category. As in the affine case (construction of the Henselization in §1), there is at most one morphism between two connected etale neighborhoods. The category is unchanged if we replace k by the separable closure of $\mathcal{O}_{X,x_0}/\mathfrak{m}(\mathcal{O}_{X,x_0})$ in k. Since the separable closure of a field is uniquely determined up to noncanonical isomorphism, the category of etale neighborhoods of a geometric point x depends only on the point $x_0 \in X$ up to (noncanonical) isomorphism.

The *strict local ring* of X at the geometric point x is defined by

$$\tilde{\mathcal{O}}_{X,x} = \varinjlim \mathcal{O}_X(U).$$

Here U runs through the category of all etale neighborhoods of x.

It is easy to see that $\tilde{\mathcal{O}}_{X,x}$ is nothing but the Henselization defined in §1 for the ring $\mathcal{O}_{X,x_0}$ with respect to the prescribed residue field extension. *In particular, $\tilde{\mathcal{O}}_{X,x}$ is a strictly Henselian ring.*

More generally, if $\mathcal{F}$ is a presheaf (of sets, abelian groups, ...), we define its *stalk* at the geometric point x to be the limit

$$\mathcal{F}_x = \varinjlim \mathcal{F}(U),$$

where again U runs through all etale neighborhoods of x.

If $\mathcal{F} \to \mathcal{G}$ is a mapping of presheaves, then passage to the limit gives maps $\mathcal{F}_x \to \mathcal{G}_x$, and $\mathcal{F} \mapsto \mathcal{F}_x$ is a functor.

Up to noncanonical isomorphism, the stalk $\mathcal{F}_x$ depends only on the point $x_0 \in X$ (and not on the special choice of embedding). If the isomorphism is not in question, we sometimes write just $\mathcal{F}_{x_0}$ instead of $\mathcal{F}_x$. In particular, it makes sense to talk about the support of a sheaf of abelian groups,

$$\text{Support}(\mathcal{F}) = \{x_0 \in X \mid \mathcal{F}_{x_0} \neq 0\}.$$

This is a subset of X, not necessarily closed.

From the construction of the generated sheaf $\hat{\mathcal{F}}$ it follows immediately that the canonical mapping $\mathcal{F} \to \hat{\mathcal{F}}$ induces isomorphisms on the stalks.

From the definition of a sheaf, the exactness of filtered direct limits, and the quasicompactness of the schemes we are considering (noetherian ones), we immediately get*):

*) The existence of enough points and stalks is a special property of the etale topology and not automatic in every "Grothendieck topology".

2.12 Remark. A mapping $\mathscr{F} \to \mathscr{G}$ of sheaves is an isomorphism if and only if it induces isomorphisms $\mathscr{F}_x \to \mathscr{G}_x$ on all stalks. A sequence $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ of sheaves of abelian groups is exact (in the category of sheaves of abelian groups) if and only if it is stalkwise exact.

In any scheme finitely generated over a field, the closed points are dense, and so in this case we have the following

Sharpening of 2.12. *Let X be a finitely generated scheme over a field k. In* 2.12 *one can restrict to the geometric points that belong to closed points of X.*

More generally: Let $X \to S$ be a finitely generated scheme over the base scheme S. In 2.12 *one can restrict x to those geometric points for which the underlying point $x_0 \in X$ is closed in its fiber.*

Direct Images and Inverse Images

Let $f\colon X \to Y$ be a morphism of schemes, $\mathscr{F}$ an (etale) sheaf on X. The *direct image*

$$(f_* \mathscr{F})(U) = \mathscr{F}(X \times_Y U), \qquad U \to Y \text{ etale},$$

is obviously an etale sheaf over Y. The functor

$$\mathscr{F} \mapsto f_* \mathscr{F}$$

is left exact in the abelian case. Its higher derived images are denoted by

$$R^i f_* \mathscr{F}.$$

In particular, if $Y = \operatorname{Spec} k$ is the spectrum of a separably closed field, one can identify sheaves on Y with abstract abelian groups (1.14). Then the sheaf $R^i f_* \mathscr{F}$ corresponds to the abelian group $H^i(X, \mathscr{F})$.

More generally, we have:

2.13 Remark. The i-th direct image $R^i f_* \mathscr{F}$ agrees with the sheaf generated by the presheaf

$$U \mapsto H^i(X \times_Y U, \mathscr{F}_U), \qquad U \to Y \text{ etale}.$$

Here we denote by $\mathscr{F}_U$ the restriction of $\mathscr{F}$ with respect to the etale mapping $X \times_Y U \to X$.

Corollary. *For a geometric point a of Y, we have*

$$(R^i f_* \mathscr{F})_a = \varinjlim H^i(X \times_Y U, \mathscr{F}_U),$$

where U runs through all etale neighborhoods of a.

Inverse Images of Sheaves

2.14 Proposition. *Let $f\colon X \to Y$ be a morphism of schemes. In a functorial way one can assign to every sheaf $\mathscr{G}$ (of sets, groups, ...) on Y a sheaf $f^*\mathscr{G}$ on X and a canonical mapping*

$$\mathscr{G} \to f_* f^* \mathscr{G}$$

so that the natural mapping

$$\mathrm{Hom}(f^*\mathscr{G}, \mathscr{F}) \to \mathrm{Hom}(\mathscr{G}, f_*\mathscr{F})$$

is bijective; that is, f^* is left adjoint to f_*.

Supplement. Let x: $\mathrm{Spec}k \to X$ be a geometric point and $y = f \circ x$ its image point in Y. There is a canonical isomorphism

$$\mathscr{G}_{f(x)} \xrightarrow{\sim} (f^*\mathscr{G})_x.$$

Corollary 1. *The functor*

$$\mathscr{G} \mapsto f^*\mathscr{G}$$

is exact.

Corollary 2. *The direct image $f_*\mathscr{J}$ of an injective sheaf is injective.*

As usual, the second corollary yields the spectral sequence for the derived functors of the composite functor $g_* \circ f_*$,

Leray Spectral Sequence:

$$R^p g_*(R^q f_* \mathscr{F}) \Rightarrow R^{p+q}((g \circ f)_* \mathscr{F}).$$

It is often useful to consider the derived functors in the sense of Verdier (Appendix II). In place of the Leray spectral sequence one then has:

Let $\mathscr{F}^\cdot$ be a complex of sheaves bounded below. Then

$$\mathrm{R}\, g_*(\mathrm{R} f_* \mathscr{F}^\cdot) = \mathrm{R}(g \circ f)_* \mathscr{F}^\cdot.$$

Here $\mathrm{R} f_* \mathscr{F}^\cdot$ and the like denote the derived direct image complexes in the sense of Verdier.

For the Construction of Inverse Images (of a Sheaf of Sets)

As in the affine case (§1), we can embed the category $\mathrm{Et}(X)$ fully faithfully in the category of sheaves of sets on X, associating with each object Z the sheaf

$$U \mapsto \mathrm{Hom}(U, Z), \qquad U/X \text{ etale.}$$

On these special sheaves the functor f^* for a morphism f: $X \to Y$ is defined by the formula

$$f^*(Z) = Z \times_Y X \quad (Z/Y \text{ etale}).$$

The representable sheaves generate the category of sheaves of sets: an arbitrary sheaf of sets $\mathscr{H}$ is the cokernel of a pair of mappings

$$\coprod_{\alpha \in A} \mathscr{F}_\alpha \rightrightarrows \coprod_{\beta \in B} \mathscr{G}_\beta,$$

where $\mathscr{F}_\alpha$ and $\mathscr{G}_\beta$ are representable sheaves*). Then $f^*\mathscr{H}$ is the cokernel of

$$\coprod_{\alpha\in A} f^*(\mathscr{F}_\alpha) \rightrightarrows \coprod_{\beta\in B} f^*(\mathscr{G}_\beta).$$

§ 3. Finite Morphisms

3.1 Lemma. *Let A be a Henselian ring* (1.17) *and $A\to B$ a finite homomorphism into an arbitrary ring B**). Then B is a cartesian product of finitely many Henselian local rings,*

$$B=\prod B_i\,.$$

The induced homomorphisms $A\to B_i$ are local.

Proof. We consider the maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ of B. The homomorphisms

$$A\to B_i = B_{\mathfrak{m}_i}$$

are local, since $A\to B$ is finite (Cohen-Seidenberg). The homomorphisms $A\to B_i$ are finite, as A is Henselian. The rings B_i are themselves Henselian. The homomorphism $B\to\prod B_i$ is injective, and by Nakayama's Lemma it is also surjective. □

3.2 Corollary. *Let A be an arbitrary local noetherian ring and $A\to B$ a finite homomorphism. Consider a separably closed extension field k of the residue field $A/\mathfrak{m}$ of A. Choose this extension large enough that for every maximal ideal $\mathfrak{n}$ of B there is an $A/\mathfrak{m}$-homomorphism of $B/\mathfrak{n}$ into k. Let $\tilde{A}$ be the strict Henselization of A with respect to $A\to k$, and let $\tilde{B}_1, \ldots, \tilde{B}_n$ be the strict Henselizations of B for all the A-homomorphisms $B\to k$. Then*

$$\tilde{A}\otimes_A B=\tilde{B}_1\times\ldots\times\tilde{B}_n\,.$$

Now we consider a finite homomorphism $A\to B$ from a strictly Henselian local ring to an arbitrary ring B, with

$$f\colon\ X=\operatorname{Spec} B\to Y=\operatorname{Spec} A$$

the corresponding morphism of spectra. The natural homomorphism $A\to k$ from A to a separably closed extension k of the residue field $A/\mathfrak{m}$ corresponds to a geometric point $y\colon \operatorname{Spec} k\to Y$. We choose k now to be "so large" that all residue fields for maximal ideals of B can be embedded in k (over $A/\mathfrak{m}$). The geometric points of X over Y, i.e. the points

$$x\colon\ \operatorname{Spec} k\to X$$

with $f\circ x=y$, correspond then uniquely to the local factors B_i in $B=\prod B_i$ (3.1).

*) In principle this was proved during the construction of generators to obtain enough injective sheaves in § 1.

**) That is, B is a finitely generated A-module.

Let $\mathscr{F}$ be a sheaf of sets on X. The rings A and B_i are strictly Henselian, and we have

$$Y=\dot{\bigcup} \operatorname{Spec}(B_i).$$

From this we get

$$(f_*\mathscr{F})_y=(f_*\mathscr{F})(Y)=\mathscr{F}(X)=\prod \mathscr{F}(\operatorname{Spec} B_i)=\prod_{x \text{ over } y} \mathscr{F}_x.$$

We shall generalize this result to an arbitrary finite morphism $f\colon X\to Y$ of schemes. Let k be a separably closed field,

$$y\colon \operatorname{Spec} k\to Y$$

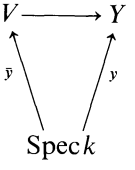
a geometric point of Y, and

$$x\colon \operatorname{Spec} k\to X, \quad f\circ x=y$$

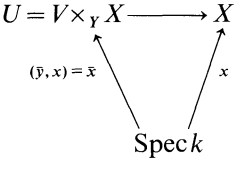
a geometric point of X over y. Let $\mathscr{G}$ be a sheaf of sets on X. Then there is a natural morphism of stalks

$$(f_*\mathscr{G})_y\to\mathscr{G}_x$$

constructed as follows. Let



be an etale neighborhood of y. Then



is an etale neighborhood of x. We have a natural mapping

$$f_*(\mathscr{G})(V)=\mathscr{G}(U)\to\mathscr{G}_x.$$

Passing to the limit, we get from this the mapping we want,

$$f_*(\mathscr{G})_y=\varinjlim f_*(\mathscr{G})(V)\to\mathscr{G}_x.$$

Here V runs through the etale neighborhoods of y. We choose k to be a "large enough" separably closed field extension of the residue field at y_0 (as in 3.2), where $y_0\in Y$ is the ordinary point underlying y. We consider all the geometric points of X over y,

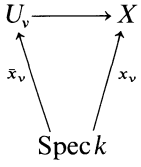
$$x_1, \ldots, x_n\colon \operatorname{Spec} k\to X, \quad f\circ x_n=y.$$

We then have a natural mapping

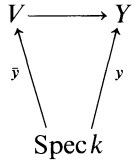
$$(f_* \mathcal{G})_y \to \prod_{v=1}^{n} \mathcal{G}_{x_v}.$$

Applying Corollary 3.2 to the strict Henselization $A = \tilde{\mathcal{O}}_{Y,y}$, we can show:

For every n-tuple of etale neighborhoods



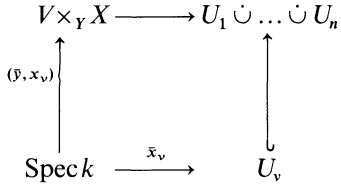
of $x_1, \ldots, x_n$ there is an etale neighborhood



and an X-morphism

$$V \times_Y X \to U_1 \dot{\cup} \ldots \dot{\cup} U_n$$

such that the diagrams

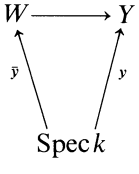


commute. It follows that the mapping

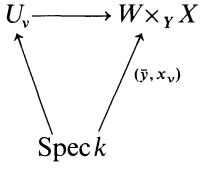
$$(f_* \mathcal{G})_y \to \prod \mathcal{G}_{x_v}$$

is surjective. We want to show that it is also injective.

Let



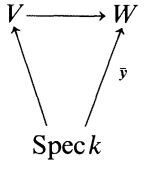
be an etale neighborhood of y, and let α, β be elements of $(f_* \mathcal{G})(W) = \mathcal{G}(W \times_Y X)$ whose images become equal in $\mathcal{G}_{x_1}, \ldots, \mathcal{G}_{x_n}$. Then there are etale neighborhoods



of $(y, x_1), \ldots, (y, x_n)$ such that the images of α and β already become equal in $\mathcal{G}(U_v)$, $v=1, \ldots, n$. Applying the previous result to the mapping

$$W \times_Y X \to W$$

and the neighborhoods $U_v \to W \times_Y X$ of $(\bar{y}, x_v)$, we obtain an etale neighborhood



of $\bar{y}$ and a $(W \times_Y X)$-morphism

$$V \times_W (W \times_Y X) = V \times_Y X \longrightarrow U_1 \dot{\cup} \ldots \dot{\cup} U_n$$
$$\searrow \quad \swarrow$$
$$W \times_Y X$$

making the appropriate diagrams commute. The images of $\alpha, \beta \in (f_* \mathcal{G})(W) = \mathcal{G}(W \times_Y X)$ are equal in $\mathcal{G}(U_1 \dot{\cup} \ldots \dot{\cup} U_n) = \prod_v \mathcal{G}(U_v)$; hence the images are equal in $\mathcal{G}(V \times_Y X) = (f_* \mathcal{G})(V)$, and hence the germs of α, β are equal. □

3.3 Proposition. *Let $X \to Y$ be a finite morphism of schemes, y: $\operatorname{Spec} k \to Y$ a geometric point of Y in a separably closed field k, and $\mathcal{G}$ a sheaf of sets on X. Let k be chosen "large enough" that the residue fields at all points of the geometric fiber*

$$\operatorname{Spec}(k) \times_Y X \to \operatorname{Spec}(k)$$

coincide with k. Then the natural mapping

$$(f_*(\mathcal{G}))_y \to \prod_{\substack{x: \operatorname{Spec}(k) \to X \\ \text{over } y}} \mathcal{G}_x$$

is an isomorphism.

In particular, for such a mapping f the functor $f_*(\mathcal{G})$ is an exact functor on the category of sheaves of abelian groups on X.

3.4 Corollary. *Let f: $X \to Y$ be a finite morphism, $\mathcal{G}$ a sheaf of abelian groups on X. Then*

$$R^v f_*(\mathcal{G}) = 0 \quad \text{for } v > 0.$$

3.5 Corollary*). *Let f: $X \to Y$ be a finite morphism, g: $Z \to Y$ an arbitrary morphism. $\mathcal{G}$ a sheaf of abelian groups on X. Consider the cartesian diagram*

*) This is a special case of the "proper base change theorem" (Theorem 6.1).

$$\begin{array}{ccc} X \times_Y Z & \xrightarrow{f'} & Z \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & Y. \end{array}$$

Then for all ν the sheaves $g^*(R^\nu f_* \mathscr{G})$ and $R^\nu(f'_*(g'^*(\mathscr{G}))$ are canonically isomorphic.

We come now to a special class of finite morphisms, the *purely inseparable* morphisms.

3.6 Lemma. *Let $f: X \to Y$ be a finite morphism of schemes. The following properties are equivalent:*

(1) *Over each point y (of the underlying topological space) of Y there lies exactly one point x of X, and the residue field extension is purely inseparable.*

(2) *Over each geometric point $\operatorname{Spec} k \to Y$ with values in an algebraically closed field k there lies exactly one geometric point $\operatorname{Spec} k \to X$.*

3.7 Definition. A finite morphism $f: X \to Y$ that has the equivalent properties in the lemma is called purely inseparable.

3.8 Example. $X_{\mathrm{red}} \to X$ is purely inseparable.

3.9 Permanence Properties. (1) Let $X \to Y$ be purely inseparable, $Z \to Y$ an arbitrary scheme over Y. Then $X \times_Y Z \to Z$ is also purely inseparable.

(2) Let $f: X \to Y$ and $g: Y \to Z$ be two finite morphisms. If f and g are purely inseparable, so is $h = g \circ f$.

If h is purely inseparable and f surjective, then the mappings f and g are also purely inseparable.

3.10 Note. It follows from Zariski's Main Theorem that in the definition of a purely inseparable mapping "finite" can be replaced by "proper."

Since a finite mapping is also a closed mapping (of the underlying topological spaces), one easily gets:

3.11 Lemma. *A purely inseparable mapping $f: X \to Y$ is a homeomorphism; in particular, it is open. This property is preserved under base extension.*

Let $f: X \to Y$ be a purely inseparable morphism, k an algebraically closed field,

$$y: \operatorname{Spec}(k) \to Y$$

a geometric point of Y, and

$$x: \operatorname{Spec}(k) \to X$$

the *unique* geometric point of X with values in k lying over y. If $\mathscr{F}$ is any sheaf on X, Proposition 3.3 shows that the stalk $f_*(\mathscr{F})_y$ of the direct image of $\mathscr{F}$ at the geometric point y is canonically isomorphic to $\mathscr{F}_x$. Moreover, if $\mathscr{G}$ is any sheaf on Y, the stalk $f^*(\mathscr{G})_x$ is canonically isomorphic to $\mathscr{G}_y$. Both isomorphisms are compatible with the adjunction mappings for the adjoint functors f_* and f^*. We get

3.12 Proposition. *Let $f: X \to Y$ be a purely inseparable mapping. The functors f_* and f^* establish an equivalence between the category of etale sheaves of sets on X and the category of etale sheaves of sets on Y. More precisely: For a sheaf $\mathscr{F}$ on X and a sheaf $\mathscr{G}$ on Y, the adjunction mappings*

$$\mathscr{G} \to f_* f^*(\mathscr{G})$$

$$\mathscr{F} \to f^* f_*(\mathscr{F})$$

are isomorphisms.

From this we see at once that the corresponding categories of sheaves of abelian groups (or of groups, ...) are equivalent.

3.13 Corollary. *Let $f: X \to Y$ be purely inseparable. $\mathscr{G}$ a sheaf of abelian groups on Y. Then*

$$H^\nu(Y, \mathscr{G}) = H^\nu(X, f^*(\mathscr{G})).$$

If we combine this corollary with the fact that cohomology is compatible with projective limits of schemes (4.18), we get:

3.14 Corollary. *Let X be a finitely generated*) scheme over a field k, and let L/k be a (possibly infinite) purely inseparable field extension. Let $\mathscr{G}$ be a sheaf of abelian groups on X and $\tilde{\mathscr{G}}$ its inverse image on $X \otimes_k L$. Then*

$$H^\nu(X, \mathscr{G}) = H^\nu(X \otimes_k L, \tilde{\mathscr{G}}).$$

The category $\mathrm{Et}(X)$ of etale schemes over a scheme X is canonically isomorphic to a full subcategory of the category of etale sheaves of sets on X, namely to the category of *representable sheaves*. In the Appendix to this section we prove the following theorem:

3.15 Representability Lemma. *An etale sheaf $\mathscr{F}$ of sets on a scheme X is representable if and only if the following conditions are satisfied:*

(1) *The stalks of $\mathscr{F}$ are finite sets.*

(2) *For each etale scheme $U \to X$ over X and every two sections $\alpha, \beta \in \mathscr{F}(U)$, the set of points $x_0 \in U$ for which the germs α_{x_0}, β_{x_0} are different is a open set. Here α_{x_0}, β_{x_0} are the images in the stalk $\mathscr{F}_{x_0}$. This second condition corresponds to the separatedness of the desired representing scheme over X!*

Let $f: X \to Y$ be a purely inseparable morphism, and let $\mathscr{F}$ be a representable sheaf on X. For a geometric point s on X and its image point t in Y, we

*) It is sufficient to require that $X \otimes_k L$ also be noetherian.

have

$$\mathscr{F}_s \cong f_*(\mathscr{F})_t .$$

As $\mathscr{F}_s$ is finite, so is the stalk $f_*(\mathscr{F})_t$. Now let

$$\alpha, \beta \in f_*(\mathscr{F})(Y) = \mathscr{F}(X)$$

be two sections; obviously then

$$\{y \in Y \mid \alpha_y \neq \beta_y\} = f(\{x \in X \mid \alpha_x \neq \beta_x\}).$$

As f is a homeomorphism and $\{x \in X \mid \alpha_x \neq \beta_x\}$ is open, $\{y \in Y \mid \alpha_y \neq \beta_y\}$ is also open. The corresponding reasoning is valid for every etale scheme $U \to Y$ and the purely inseparable extension

$$U \times_Y X \to U.$$

Thus the representability lemma implies that $f_*(\mathscr{F})$ is again representable. □

3.16 Proposition. *For a purely inseparable mapping $f: X \to Y$, the inverse image functor*

$$\mathrm{Et}(Y) \to \mathrm{Et}(X)$$
$$(U \to Y) \mapsto (U \times_Y X \to X)$$

is an equivalence of categories.

3.17 Corollary. *Let $f: X \to Y$ be purely inseparable, s a geometric point of X, and t its image in Y. Then the fundamental group (A I) satisfies*

$$\pi_1(X, s) = \pi_1(Y, t).$$

Because: The equivalence

$$\mathrm{Et}(Y) \to \mathrm{Et}(X)$$

establishes an equivalence of the categories of etale covering spaces.

Appendix. The Representability Lemma (3.15)

An etale sheaf of sets $\mathscr{G}$ on a scheme X is called *representable* if there is an etale scheme $Y \to X$ over X such that $\mathscr{G}$ is isomorphic to the sheaf $\tilde{Y}$,

$$\tilde{Y}(-) = \mathrm{Hom}_X(-, Y).$$

An etale mapping $Y \to X$ is quasi-affine. Therefore, as Grothendieck showed, descent data for Y with respect to a faithfully flat mapping $X \to Z$ are always effective. This holds in particular for any surjective etale mapping $X \to Z$ (A III). In the language of sheaves, this can also be expressed as follows:

3.18 Lemma. *Let $\mathscr{G}$ be an etale sheaf on a scheme X. Then $\mathscr{G}$ is representable if and only if it is locally representable with respect to the etale topology, i.e., if and only if every geometric point of X has an etale neighborhood $U \to X$ such that $\mathscr{G} \mid U$ is representable in* $\mathrm{Et}(U)$.

We want to prove Lemma 3.15: An etale sheaf of sets on a scheme X is representable if and only if conditions (1) and (2) in 3.15 are satisfied. Obviously these conditions are necessary for $\mathcal{G}$ to be representable. Every subsheaf of $\mathcal{G}$ satisfies conditions (1) and (2). The inverse image of such a sheaf $\mathcal{G}$ under any morphism $Y \to X$ satisfies (1) and (2), so in particular this is true for the restriction to an etale scheme $U \to X$ over X.

The points of a scheme at which the germs of two sections of a sheaf coincide form an open set. Condition 3.15(2) thus carries with it the following stronger statement:

3.15(2′). Let $\mathcal{G}$ be a sheaf of sets on a scheme X that satisfies condition (2) of 3.15. Take $s, t \in \mathcal{G}(X)$. The following subsets of X are *open and closed:*

$$\{x \in X \mid s_x = t_x\}, \qquad \{x \in X \mid s_x \neq t_x\}.$$

In particular, if X is connected, each of these sets must either be empty or coincide with all of X. The corresponding statement holds for sections over an etale scheme $U \to X$.

From 3.15(1) and 3.15(2′), we get:

3.15(1′). For every etale scheme $U \to X$, $\mathcal{G}(U)$ is *finite.*

When $\mathcal{G}$ is a sheaf on X and $Y \to X$ is an etale scheme representing a sheaf $\tilde{Y}$, then there is a natural isomorphism

$$\mathcal{G}(Y) \cong \mathrm{Hom}(\tilde{Y}, \mathcal{G}).$$

Thus in a natural way an m-tuple $s_1, \ldots, s_m \in \mathcal{G}(Y)$ defines a morphism

$$\widetilde{\underbrace{Y \dot{\cup} \ldots \dot{\cup} Y}_{m \text{ copies}}} = \tilde{Y} \coprod \ldots \coprod \tilde{Y} \to \mathcal{G}.$$

3.19 Lemma. *Suppose the sheaf $\mathcal{G}$ on the scheme X satisfies condition* (2) *in* 3.15. *Let*

$$x\colon \mathrm{Spec}\, k \to X$$

be a geometric point of X, and let $s_1, \ldots, s_m \in \mathcal{G}(X)$ be an m-tuple of sections of $\mathcal{G}$ whose germs $s_{1x}, \ldots, s_{mx}$ are pairwise distinct. We consider the mapping

$$\tilde{Y} =_{\mathrm{def}} \widetilde{X \dot{\cup} \ldots \dot{\cup} X} = \tilde{X} \coprod \ldots \coprod \tilde{X} \to \mathcal{G}$$

defined by this m-tuple. Then there is a Zariski-open neighborhood U of x such that the restriction $\tilde{Y} \mid U \to \mathcal{G} \mid U$ of this mapping to U is injective. If X is connected, we can take $U = X$.

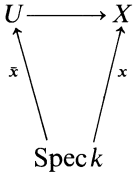
If X is irreducible, x is a generic geometric point, and

$$\{s_{1x}, \ldots, s_{mx}\} = \mathcal{G}_x,$$

then the mapping in question is also surjective.

Proof. Part 1: It follows immediately from 3.15(2′) that for U we can choose the connected component of X that contains x.

Part 2: Let $U \to X$ be a nonempty connected etale scheme over X, and $t \in \mathcal{G}(U)$. As X is irreducible and x is a generic point of X, there is a geometric point $\bar{x}$ of U lying over x. The diagram



is an etale neighborhood of x. By hypothesis then there is some v for which the image t_x of t in the stalk $\mathcal{G}_x$ coincides with s_{vx}. But then t is equal to the inverse image of s_v in $\mathcal{G}(U)$ by 3.15(2′). Thus the mapping in question is surjective. □

3.20 Corollary. *Under the hypotheses of Lemma* 3.15, *the sheaf* $\mathcal{G}$ *is representable in an etale neighborhood* $Y \to X$ *of a generic point of an irreducible component of* X. *More precisely,*

$$\widetilde{Y \dot\cup \ldots \dot\cup Y} = \mathcal{G} \mid Y.$$

3.21 Lemma. *Let* X *be a scheme,* A *a closed subset of* X, *and* $\mathcal{G}$ *a sheaf on* X. *Suppose the restriction* $\mathcal{G}|(X-A)$ *of* $\mathcal{G}$ *to* $X-A$ *is representable by an etale scheme* $Y \to X-A$ *on* $X-A$. *Suppose the stalks* $\mathcal{G}_x$ *are empty for all geometric points* x *that lie in* A. *Then* $\mathcal{G}$ *is representable on* X *by* Y *(interpreted as an etale scheme over* X*).*

Proof. Since

$$\mathrm{Hom}(\tilde{Y}, \mathcal{G}|(X-A)) = (\mathcal{G}|(X-A))(Y) = \mathcal{G}(Y) = \mathrm{Hom}(\tilde{Y}, \mathcal{G}),$$

the given isomorphism

$$\tilde{Y} \xrightarrow[X-A]{\sim} \mathcal{G} \mid X-A$$

defines a morphism $\tilde{Y} \to \mathcal{G}$ of sheaves on X. By hypothesis this mapping is an isomorphism on all stalks at geometric points in $X-A$. As the stalks Y_x and $\mathcal{G}_x$ are empty for all geometric points that lie in A, the morphism $\tilde{Y} \to \mathcal{G}$ is an isomorphism. □

Now let $\mathcal{G}$ be a sheaf of sets on the scheme X that satisfies both conditions (1) and (2) of Lemma 3.15. Let $\{s_1, \ldots, s_m\} = E \subset \mathcal{G}(X)$ be a given set of pairwise distinct sections of $\mathcal{G}$. Assume X is connected. By 3.15(2′), the germs of these sections are pairwise distinct at all geometric points of X. Hence the mapping

$$\tilde{Y} = X \dot\cup \ldots \dot\cup X \to \mathcal{G}$$

defined by these sections is injective. Let $U \to X$ be etale. If U is connected and nonempty, we set

$$\tilde{\mathcal{G}}(U) = \mathcal{G}(U) - E \mid U \subset \mathcal{G}(U).$$

If U is not connected, we split U into its connected components $U_v \to X$ and set

$$\tilde{\mathcal{G}}(U) = \prod_v \tilde{\mathcal{G}}(U_v) \subset \prod \mathcal{G}(U_v) = \mathcal{G}(\dot{\bigcup_v} U_v).$$

3.22 Lemma. *$\tilde{\mathcal{G}}$ is a subsheaf of $\mathcal{G}$, and $\mathcal{G} = \tilde{Y} \coprod \tilde{\mathcal{G}}$.*

Proof. We must show that $\tilde{\mathcal{G}}$ is compatible with the restriction mappings. Let

$$V \to U \to X$$

be etale morphisms, and assume U and V are connected and nonempty.

Assertion. The restriction mapping

$$\mathcal{G}(U) \to \mathcal{G}(V)$$

sends $\tilde{\mathcal{G}}(U)$ into $\tilde{\mathcal{G}}(V)$.

Indeed, take $s \in \mathcal{G}(U)$, and suppose the restriction $s|V$ of s to V is not contained in $\tilde{\mathcal{G}}(V)$. Then there is a section $t \in E \subset \mathcal{G}(X)$ with $t|V = s|V$. By 3.15(2'), the set of points in U at which the germs of s and $t|U$ coincide is either empty or equal to U. The inverse image of this set in V is precisely the set of all points in V at which $s|V$ and $t|V$ coincide, and thus it is equal to V. It follows that $s = t|U \in E|U$. Thus the section s was not in $\tilde{\mathcal{G}}(U)$. □

3.23 Lemma. *Let the hypotheses and notation be as in Lemma 3.22. Assume in addition that $\mathcal{G}$ is representable by an etale scheme $Z \to X$. Then $\tilde{\mathcal{G}}$ is also representable.*

Proof. The injection $\tilde{Y} \hookrightarrow \mathcal{G} = \tilde{Z}$ is induced by an X-morphism

$$Y = X \dot{\cup} \ldots \dot{\cup} X \to Z$$

and thus by m morphisms over X,

$$X \to Z.$$

The image of X in Z under each of these morphisms is open and closed, and hence so is the image of Y in Z. The complement Z' of this image in Z is open and closed in Z and represents $\tilde{\mathcal{G}}$! □

We now prove Lemma 3.15 by noetherian induction. Let $\mathcal{G}$ be a sheaf of sets on a nonempty scheme X satisfying conditions (1) and (2) of 3.15. By 3.20 there is a nonempty etale scheme $Y \to X$ over X such that the restriction $\mathcal{G}|Y$ is representable. The image W of Y in X is open and nonempty. By 3.18 we see that $\mathcal{G}|W$ is representable. Thus there is a closed subscheme A of X with the following properties:

The complement $W = X - A$ of A in X is nonempty.

The restriction $\mathcal{G}|W$ is representable.

We choose a minimal A with these properties; we must show that A is empty. Suppose not. Let

$$a\colon \operatorname{Spec} k \to X$$

be a geometric point of X that is a generic geometric point of an irreducible component of $A \subset X$.

From Corollary 3.20, we see that there is a connected etale neighborhood

$$P\colon \begin{array}{ccc} U & \longrightarrow & X \\ & \nwarrow \quad \nearrow_{a} & \\ & \operatorname{Spec} k & \end{array}$$

of a in X and sections $s_1, \ldots, s_m$ in $\mathscr{G}(U)$ such that the mapping

$$\tilde{Y} = \widetilde{U \cup \ldots \cup U} \to \mathscr{G}|U$$

defined by $s_1, \ldots, s_m$ becomes an isomorphism when restricted to $p^{-1}(A)$. We apply Lemma 3.22 to the sheaf $\mathscr{G}|U$ and the sections $s_1, \ldots, s_m$, getting

$$\mathscr{G}|U = \tilde{Y} \coprod \tilde{\mathscr{G}}.$$

Here $\tilde{\mathscr{G}}$ arises from $\mathscr{G}$ by "removing the sections $s_1, \ldots, s_m$". By assumption $\mathscr{G}|(X-A)$ is representable; then so is $\mathscr{G}|(U-p^{-1}(A))$, and by Lemma 3.23 so is $\tilde{\mathscr{G}}|(U-p^{-1}(A))$. By construction the stalks of $\tilde{\mathscr{G}}$ at geometric points of $p^{-1}(A)$ are empty. By Lemma 3.21 then $\tilde{\mathscr{G}}$ itself is representable, and hence so is $\mathscr{G}|U = \tilde{Y} \coprod \tilde{\mathscr{G}}$. Let

$$V = (X-A) \cup p(U).$$

Since $X - A \hookrightarrow V$ and $U \to V$ form an etale covering of V, it follows from Lemma 3.18 that $\mathscr{G}|V$ is also representable. But V is strictly larger than $X-A$, which contradicts the minimality of A. □

§ 4. Finiteness Conditions on Sheaves and Compatibility of Cohomology with Limits

A central concept in etale cohomology theory is the concept of a *constructible sheaf*, which will now be introduced.

As we have already seen in §1, for every sheaf of sets $\mathscr{F}$ on the scheme X there is a family X_α of etale X-schemes and a surjective sheaf mapping

$$\coprod \tilde{X}_\alpha \to \mathscr{F}$$

(where $\tilde{X}_\alpha$ is the sheaf represented by X_α). The sheaf $\mathscr{F}$ is called constructible if one can get by with a finite family. In this case $\mathscr{F}$ actually turns out to be the quotient of a representable sheaf by a representable equivalence relation (not separated, in general). Thus one can interpret $\mathscr{F}$ as an etale algebraic

space over X (not necessarily separated) in the sense of M. Artin, Knutson [91].

We recall that one can assign to every set (abelian group, ...) M a sheaf M_X on a scheme X. A sheaf $\mathscr{F}$ on X is called *constant* if it is isomorphic to a sheaf of this form. If X is connected, this means $\mathscr{F} \cong \mathscr{F}(X)_X$. Then a sheaf $\mathscr{F}$ is constant precisely when the restriction mappings

$$\mathscr{F}(U) \to \mathscr{F}(V), \qquad V \to U \text{ in } \mathrm{Et}(X),$$

are isomorphisms for nonempty connected U and V.

Let X be an irreducible scheme, a: $\mathrm{Spec}\, K \to X$ a generic geometric point of X. A sheaf $\mathscr{F}$ on X is constant precisely when the following two conditions are satisfied:

a) $\mathscr{F}(X) \to \mathscr{F}_a$ is an isomorphism.

b) If U is a connected etale neighborhood of a, then $\mathscr{F}(U) \to \mathscr{F}_a$ is injective.

Condition b) is satisfied if $\mathscr{F}$ is a subsheaf of a constant sheaf.

From the equivalence (3.16) of the categories $\mathrm{Et}(X)$ and $\mathrm{Et}(X_{\mathrm{red}})$ we get:

4.1 Remark. A sheaf $\mathscr{F}$ on a scheme X is constant if and only if its restriction to X_{red} is constant.

4.2 Definition. A sheaf $\mathscr{F}$ on a scheme X is called locally constant when there is an etale covering $U_i \to X$ such that each $\mathscr{F} \mid U_i$ is constant. If the $\mathscr{F}(U_i)$ appearing here are finite, we say we have a finite locally constant sheaf.

4.3 Definition. A sheaf $\mathscr{F}$ on a scheme X is called constructible, if X can be written as a union of finitely many locally closed subschemes $Y \subset X$ for which $\mathscr{F} \mid Y$ is finite lcally constant.

Obviously each of the following statements is equivalent to this definition (use 4.1 and noetherian induction).

4.3′. For every closed subscheme $Y \subset X$ there is an open nonempty subset $U \subset Y$ for which the inverse image $\mathscr{F} \mid U$ is finite locally constant.

4.3″. For every irreducible closed subscheme $Y \subset X$ there is an etale neighborhood $U \to Y$ of the generic point of Y for which the inverse image of $\mathscr{F}$ on U is finite locally constant.

From this we get:

4.3‴. A sheaf $\mathscr{F}$ is constructible precisely when, for every closed irreducible subscheme Y (reduced is enough), there is a nonempty etale scheme V/Y for which the inverse image of $\mathscr{F}$ on V is constructible.

Constructible Sheaves of Sets

4.4 Lemma. *Every sheaf representable by an etale scheme U/X,*

$$\mathscr{F}(V)=\mathrm{Hom}_X(V, U),$$

is a constructible sheaf of sets.

Proof. In view of the criterion 4.3″, it is enough to show that there is a nonempty etale scheme $V\to X$ for which the V-scheme

$$U\times_X V\to V$$

splits, i.e.

$$U\times_X V=V\dot\cup\ldots\dot\cup V.$$

For the proof we can assume that X is integral (because $\mathrm{Et}(X)=\mathrm{Et}(X_{\mathrm{red}})$). If in particular X is the spectrum of a field k, this statement means that for every finite separable field extension $K\supset k$ there is a finite separable extension $L\supset k$ such that

$$L\otimes_k K\cong_L L\times\ldots\times L \quad \text{(e.g., } L=\text{Galois closure of } K).$$

The general case is reduced to this by a simple limit argument, taking k specifically to be the function field of X. □

Constructible sheaves have an important compactness property.

4.5 Lemma. *Let*

$$\mathscr{F}_i\to\mathscr{F}, \qquad i\in I,$$

be a family of mappings of arbitrary sheaves to the constructible sheaf $\mathscr{F}$. We assume that it is surjective, i.e. that for every geometric point x the stalk $\mathscr{F}_x$ is the union of the images $(\mathscr{F}_i)_x$. Then there is a finite surjective subfamily

$$\mathscr{F}_{i_1},\ldots,\mathscr{F}_{i_r}\to\mathscr{F},$$

that is, one for which

$$\coprod_\nu \mathscr{F}_{i_\nu}\to\mathscr{F}$$

is surjective.

Proof. We apply noetherian induction. It is enough to show that there is a nonempty etale scheme $\phi\colon U\to X$ such that the assertion is true for $\mathscr{F}_i|U\to\mathscr{F}|U$ (and hence for $\mathscr{F}_i|\phi(U)\to\mathscr{F}|\phi(U)$) instead of for $\mathscr{F}_i\to\mathscr{F}$. We can choose $U\to X$ so that U is irreducible, $\mathscr{F}|U$ is constant, and the (finitely many) sections of $\mathscr{F}(U)$ are images of elements from $\mathscr{F}_{i_\nu}(U)$, $\nu=1,\ldots,r$. But then the assertion is clear.

For the characterization of constructible sheaves mentioned at the start, we need some exactness properties of the category of sheaves of sets (cf. §1).

1) *A* mapping $\mathscr{F}\to\mathscr{G}$ of sheaves of sets on X is surjective (in the categorical sense) if and only if, for every $s\in\mathscr{G}(U)$, U/X etale, there is a surjective etale

morphism $V \to U$ such that s is contained in the image of

$$\begin{array}{ccc} \mathscr{F}(V) & \longrightarrow & \mathscr{G}(V) \\ & & \cup \\ & & \mathscr{G}(U). \end{array}$$

This condition is equivalent to stalkwise surjectivity.

2) There exist fiber products in the category of sheaves of sets:

$$(\mathscr{F} \times_{\mathscr{H}} \mathscr{G})(U) = \mathscr{F}(U) \times_{\mathscr{H}(U)} \mathscr{G}(U).$$

3) In the category of sheaves of sets there exist cokernels for pairs of arrows

$$\mathscr{F} \rightrightarrows \mathscr{G}.$$

One gets the cokernel by taking the sheaf generated by the presheaf

$$U \mapsto \mathrm{Coker}(\mathscr{F}(U) \rightrightarrows \mathscr{G}(U)).$$

4) Let the pair of arrows $\mathscr{F} \rightrightarrows \mathscr{G}$ represent an equivalence relation, i.e.

a) the induced mapping $\mathscr{F} \to \mathscr{G} \times \mathscr{G}$ is injective, and

b) for every etale U/X, $\mathscr{F}(U) \subset \mathscr{G}(U) \times \mathscr{G}(U)$ is an equivalence relation.

Let $\mathscr{H}$ be the cokernel of the pair of arrows.

Then: Two elements $s, t \in \mathscr{G}(U)$ have the same image in $\mathscr{H}$ if and only if $(s, t) \in \mathscr{F}(U)$.

Conversely, every surjective mapping $\mathscr{G} \to \mathscr{H}$ of sheaves is the cokernel of an equivalence relation, namely

$$\mathscr{F} = \mathscr{G} \times_{\mathscr{H}} \mathscr{G} \rightrightarrows \mathscr{G}. \quad \square$$

We now come to the promised characterization of constructible sheaves.

4.6 Proposition. *For a sheaf $\mathscr{F}$ of sets on X, the following three conditions are equivalent.*

1) *$\mathscr{F}$ is constructible.*

2) *There is a sheaf $\mathscr{G} = \tilde{Y}$ representable by some etale scheme Y/X and a surjective mapping $\mathscr{G} \to \mathscr{F}$.*

3) *There is an equivalence relation $R \rightrightarrows Y$ in $\mathrm{Et}(X)$ such that $\mathscr{F}$ is the cokernel of the equivalence relation*

$$\tilde{R} \rightrightarrows \tilde{Y}$$

of representable sheaves. (The injective etale mapping $R \to Y \times Y$ is automatically an open embedding.)

Corollary 1. *Every subsheaf of a constructible sheaf is constructible (for, by the representability lemma* (3.15)*, every subsheaf of a representable sheaf is representable).*

2. *Every sheaf of sets is the filtered direct limit of its constructible subsheaves.*

Proof of 4.6 1) ⇒ 2). As was shown earlier (in the construction of generators in §1), there is *a* surjective family

$$\tilde{Y}_\alpha \to \mathscr{F}, \qquad Y_\alpha \in \mathrm{Et}(X).$$

It follows from the compactness criterion 4.5 that there are finitely many indices $\alpha_1, \ldots, \alpha_r$ such that

$$\widetilde{[Y_{\alpha_1} \dot\cup \ldots \dot\cup Y_{\alpha_r}]} = \coprod \tilde{Y}_{\alpha_\nu} \to \mathscr{F}$$

is surjective.

2) ⇒ 3). We consider a surjective mapping

$$\mathscr{G} = \tilde{Y} \to \mathscr{F}.$$

Then $\mathscr{F}$ is the cokernel of the equivalence relation

$$\mathscr{H} = \mathscr{G} \times_{\mathscr{F}} \mathscr{G} \rightrightarrows \mathscr{G}.$$

By the representability lemma 3.15, $\mathscr{H}$ is itself representable, being a subsheaf of the representable sheaf $\widetilde{Y \times Y} = \mathscr{G} \times \mathscr{G}$.

3) ⇒ 1). Suppose we have an equivalence relation

$$R \rightrightarrows Y$$

in $\mathrm{Et}(X)$. It is easy to see that if the open embedding $R \to Y \times Y$ is also closed, then the "separation property" in the representability lemma 3.15 is satisfied; thus $\mathscr{F}$ is representable and hence constructible. In the general case, there is a nonempty open subset $U \subset X$ such that the inverse images of U in R and in Y are finite over U. Hence the restriction of $R \to Y \times Y$ to U is also finite and hence closed. Thus $\mathscr{F} \mid U$ is representable and thereby constructible. The assertion follows now from 4.3″. □

Constructible Sheaves of Abelian Groups

Constructible sheaves of abelian groups are particular examples of torsion sheaves.

4.7 Definition. *A* sheaf $\mathscr{F}$ of abelian groups on X is called a torsion sheaf, provided that one of the following equivalent conditions is satisfied:

1) All stalks of $\mathscr{F}$ are torsion groups (that is, all elements have finite order).
2) $\mathscr{F}(U)$ is a torsion group for all etale schemes U/X.

The stalks (and, for that matter, the modules of sections) of a constructible sheaf are finite groups and hence torsion groups. The main result about the category of constructible sheaves of abelian groups is this:

4.8 Proposition. 1) *The constructible sheaves of abelian groups are precisely the noetherian objects in the category of all torsion sheaves.*

2) *Every torsion sheaf is the filtered direct limit of its constructible subsheaves.*

(A sheaf $\mathscr{F}$ is called noetherian when every ascending chain of subsheaves breaks off.)

Corollary. *The category of constructible sheaves is an abelian subcategory of the category of all torsion sheaves. The category of all torsion sheaves is locally noetherian* [54]; *that is,*

a) *the noetherian objects form a system of generators,*

b) *filtered direct limits are exact.*

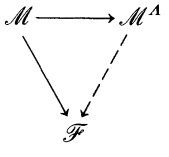
For b): Direct limits are compatible with forming stalks.

Proof of 4.8. Since a subsheaf of a constructible sheaf is constructible (Corollary to 4.6), we can deduce 4.8 from the compactness criterion 4.5 and the following considerations:

Every torsion sheaf $\mathscr{G}$ is the filtered direct limit of the subsheaves

$$\mathscr{G}_n = \mathrm{Ker}(\mathscr{G} \xrightarrow{n} \mathscr{G}), \qquad n \in \mathbb{N},$$

of n-torsion elements. For the proof of 4.8 it is sufficient to select a fixed natural number n and consider only sheaves of modules for $\Lambda = \mathbb{Z}/n\mathbb{Z}$. For every sheaf $\mathscr{M}$ of sets, we can construct (in analogy to $\mathscr{M}^{\mathrm{ab}}$ in §1) a sheaf $\mathscr{M}^{\Lambda}$ of Λ-modules together with a sheaf mapping $\mathscr{M} \to \mathscr{M}^{\Lambda}$ such that for every mapping of $\mathscr{M}$ into a sheaf $\mathscr{F}$ of Λ-modules there is a unique homomorphism $\mathscr{M}^{\Lambda} \to \mathscr{F}$ of sheaves of Λ-modules for which



commutes. This means that the forgetful functor

$$(\text{Sheaves of } \Lambda\text{-modules}) \to (\text{Sheaves of sets})$$

has a left adjoint functor.

Construction of $\mathscr{M}^{\Lambda}$. Assign to every etale U/X the free Λ-module generated by $\mathscr{M}(U)$, and consider the sheaf generated by this presheaf. Proposition 4.8 now follows easily from the properties proved for constructible sheaves and from

4.9 Lemma. *Let $\mathscr{M}$ be a sheaf of sets on the scheme Y.*

1) *When $\mathscr{M}$ is constructible, so is $\mathscr{M}^{\Lambda}$.*

2) *If $f: X \to Y$ is a morphism of schemes, then*

$$f^*(\mathscr{M}^{\Lambda}) = (f^*\mathscr{M})^{\Lambda}.$$

Proof. If M_0 is a finite set and $\mathscr{M} = (M_0)_Y$ is the associated constant sheaf, then $\mathscr{M}^{\Lambda}$ is the constant sheaf associated with the free Λ-module generated by M_0. Thus 1) follows from 2) by 4.3″.

For 2): For any sheaf $\mathscr{G}$ of Λ-modules on X, we have

$$\begin{aligned}\mathrm{Hom}_{\mathrm{Sets}}(f^*(\mathscr{M}), \mathscr{G}) &= \mathrm{Hom}_{\mathrm{Sets}}(\mathscr{M}, f_*(\mathscr{G})) \\ &= \mathrm{Hom}(\mathscr{M}^\Lambda, f_*\mathscr{G}) = \mathrm{Hom}(f^*(\mathscr{M}^\Lambda), \mathscr{G}). \quad \square\end{aligned}$$

From Proposition 4.6, 2) we get:

4.10 Proposition. *Let $\mathscr{F}$ be a constructible sheaf of Λ-modules ($\Lambda = \mathbb{Z}/n\mathbb{Z}$) on a scheme X. There is an etale scheme U over X and a surjective sheaf homomorphism*

$$(\tilde{U})^\Lambda \to \mathscr{F}.$$

Remark. The construction of $(\tilde{U})^\Lambda$ will be generalized in § 8 (8.1). For an etale morphism $f: U \to X$ we have

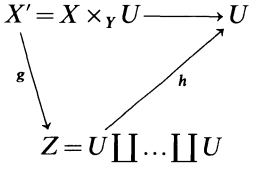
$$(\tilde{U})^\Lambda = f_!(\Lambda_U).$$

We aim now for a result (4.12) dual to (4.10), and we first remark

4.11 Lemma (Special case of the Finiteness Theorem 8.10). *Let $f: X \to Y$ be a finite morphism of schemes and $\mathscr{F}$ a constructible sheaf on X. Then $f_*\mathscr{F}$ is also constructible.*

Proof. By the base change Proposition 3.5 for finite morphisms, and criterion 4.3‴, it is enough to prove the following:

There is a nonempty etale scheme $U \to Y$ such that the statement of 4.11 is true for the inverse image of $\mathscr{F}$ on $X \times_Y U$ and the projection $X \times_Y U \to U$. We can assume that X and Y are irreducible reduced, and that f is surjective. But then there is an etale scheme $U \to Y$ and a factorization



such that g is purely inseparable. This follows immediately from the corresponding assertion for the spectra of the function fields. Our assertion is trivial for h. If $\mathscr{H}$ is a constructible sheaf on X', and thus the image of a representable sheaf, then it follows from Propositions 3.12 and 3.16 that $g_*(\mathscr{H})$ is the image of a representable sheaf, hence constructible. $\square$

4.12 Proposition. *Let $\mathscr{F}$ be a constructible sheaf of abelian groups on a scheme X. There is an injective homomorphism*

$$\mathscr{F} \to \bigoplus_{\nu=1}^{m} \mathscr{G}_\nu .$$

Here $\mathscr{G}_\nu$ is a sheaf of the following type:

There exists a finite (not necessarily surjective) morphism

$$f_\nu\colon X_\nu \to X$$

and a constant sheaf $(M)_X$ *of abelian groups on* X_ν *with the property*

$$\mathcal{G}_\nu = (f_\nu)_*((M_\nu)_{X_\nu}).$$

One can assume X *to be reduced and irreducible.*

Supplement. Let $\mathcal{F}$ *be actually a* $\mathbb{Z}/n\mathbb{Z}$*-module. Then one can always choose* $(M_\nu)_{X_\nu}$ *to be the constant sheaf* $(\mathbb{Z}/n\mathbb{Z})_X$.

(Use: Every finite $\mathbb{Z}/n\mathbb{Z}$-module is *a* submodule of a finite free $\mathbb{Z}/n\mathbb{Z}$-module.)

Proof. For simplicity we restrict to the case of a scheme that is excellent in the sense of Grothendieck [63], which is sufficient for our purposes. We show now, for every irreducible excellent scheme X and every constructible sheaf $\mathcal{G}$ on X:

There exists an irreducible scheme X', finite over X under some $f\colon X' \to X$, a constant sheaf $\mathcal{G}$ on X', and a mapping $\mathcal{F} \to f_*\mathcal{G}$ whose restriction to a non-empty open subset of X is injective.

Proposition 4.12 follows easily from this by noetherian induction.

Let Y be irreducible, and let $g\colon Y \to X$ be a finite surjective mapping. It is enough then to prove the assertion above for $g^*\mathcal{F}$ on Y instead of for $\mathcal{F}$ on X, because the adjunction mapping $\mathcal{F} \to g_* g^*\mathcal{F}$ is injective (as one sees easily on stalks) (3.3).

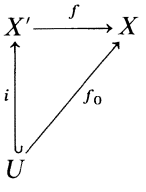
Since by assumption X is excellent, the normalization of X_{red} is finite over X. Thus for the proof of the assertion above we can assume that X is normal and irreducible.

The constructible sheaf $\mathcal{F}$ is locally constant in a Zariski-neighborhood of the generic point. Thus there is an etale mapping

$$F_0\colon U \to X, \qquad U \neq \emptyset, \quad U \text{ connected},$$

such that the inverse image $f_0^*\mathcal{F}$ is constant. U is again normal and irreducible.

By Zariski's Main Theorem, U is an open dense subscheme of a scheme X' that is finite over X.



We can assume that X' is normal and irreducible. Then we have:

Let $p\colon V \to X'$ be etale, V connected and nonempty. Then V is normal, and hence irreducible! The inverse image $p^{-1}(U)$ is then likewise connected and nonempty. From this we see that the direct image $i_* M_U$ of the constant sheaf M_U is isomorphic to the constant sheaf $M_{X'}$. The direct image sheaf $\mathcal{G} = i_*(f_0^*\mathcal{F})$ is thus constant on X'.

The adjunction mapping

$$\mathscr{F} \to (f_0)_*(f_0^* \mathscr{F}), \quad f_0 \colon U \to X,$$

becomes injective after restriction to $X - f(X' - U)$. We have

$$(f_0)_*(f_0^* \mathscr{F}) = f_*(\mathscr{G}),$$

and our assertion is proved. □

Compatibility of Cohomology with Limits

4.13 Remark. Let $\mathscr{F}$ be an injective object in the category of all

a) sheaves of abelian groups,

b) torsion sheaves,

c) sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules (or, more generally, of R-modules). In each of these cases $\mathscr{F}$ is flabby, i.e.

$$H^i(\mathscr{X}, \mathscr{F}) = 0 \quad \text{for } i > 0$$

and for every etale covering $\mathscr{X}$ of an etale scheme U over the base scheme. (Remark on 2.9).

Supplement. Each of these three categories has enough injective objects.

In case a), the category of all sheaves, we have already proved the existence of injective objects; it follows in part b) from 4.8, and likewise for part c). From Cartan's lemma 2.9 we get:

4.14 Corollary. *Let $\mathscr{F}$ be a torsion sheaf (resp. a sheaf of R-modules). One gets the same cohomology groups of $\mathscr{F}$ if one takes the derived functors of the section functor in the category of all sheaves or only in that of torsion sheaves (resp. sheaves of R-modules).*

The same is true for the functors

1) direct image f_*

2) functor of sections Γ_A with support on *a* closed set $A \subset X$.

(The functor Γ_A,

$$\Gamma_A(\mathscr{F}) = \{s \in \mathscr{F}(X), s|(X - A) = 0\},$$

is left exact. Its derived functors are denoted $H_A(X, \mathscr{F})$.)

Every torsion sheaf is a filtered direct limit of constructible sheaves. Therefore it is important to have available the following compatibility result for cohomology, which holds for compactness reasons.

4.15 Proposition. *Let $\mathscr{F} = \varinjlim \mathscr{F}_\nu$ be a filtered direct limit of arbitrary sheaves $\mathscr{F}_\nu$ on a scheme X. Then we have*

$$H^\cdot(X, \mathscr{F}) = \varinjlim H^\cdot(X, \mathscr{F}_\nu).$$

From this follows also a corresponding compatibility condition for the higher direct images of a filtered limit of sheaves.

Proof. Let $\mathscr{X}=(X_\alpha)$ be a finite etale covering of a scheme X' etale over X. The exactness of

$$\mathscr{F}_\nu(X')\to\prod_\alpha \mathscr{F}_\nu(X_\alpha)\rightrightarrows\prod_{\alpha,\beta}\mathscr{F}_\nu(X_\alpha\times_{X'}X_\beta)$$

implies the exactness of

$$\varinjlim_\nu(\mathscr{F}_\nu(X'))\to\prod_\alpha\varinjlim\mathscr{F}_\nu(X_\alpha)\rightrightarrows\prod_{\alpha,\beta}\varinjlim(\mathscr{F}_\nu(X_\alpha\times_{X'}X_\beta)).$$

Hence

$$X'\mapsto\varinjlim\mathscr{F}_\nu(X')$$

is already a sheaf, and thus the section functor is compatible with filtered direct limits.

More generally, the Čech cohomology

$$H^*(\mathscr{X},\mathscr{F})$$

for a finite etale covering is compatible with filtered direct limits. Consequently, the Čech cohomology vanishes for filtered direct limits

$$\mathscr{J}=\varinjlim_\nu\mathscr{J}_\nu$$

of injective sheaves $\mathscr{J}_\nu$. From Cartan's lemma 2.9 it follows that

$$H^i(X,\mathscr{J})=0\quad\text{for } i>0.$$

Every inductive system of sheaves has a resolution by inductive systems of injective sheaves! □

In what follows we will need a further compatibility of etale cohomology with passage to limits. Suppose we have

1) a filtered projective system (X_α) of schemes. Assume the connecting maps

$$X_\alpha\to X_\beta,\quad \alpha\geqq\beta,$$

are all affine.

2) a scheme X together with a family of affine morphisms

$$X\to X_\alpha$$

compatible with all the morphisms of the system. (As before, X and X_α are assumed to be noetherian, but the morphisms do not have to be finitely generated!)

For every index α_0 and every open affine subscheme $U_{\alpha_0}\subset X_{\alpha_0}$, assume

$$\Gamma(U,\mathcal{O}_X)=\varinjlim_{\alpha\geqq\alpha_0}\Gamma(U_\alpha,\mathcal{O}_{X_\alpha}).$$

Here we let

$$U_\alpha = U_{\alpha_0} \times_{X_{\alpha_0}} X_\alpha, \qquad U = U_{\alpha_0} \times_{X_{\alpha_0}} X.$$

We then call (X_α) *a projective system*, and X the *projective limit* of the system,

$$X = \varprojlim X_\alpha .$$

For a fixed index α_0 suppose we are given a finitely generated*) scheme

$$Y_{\alpha_0} \to X_{\alpha_0} .$$

We obtain a projection system

$$Y_\alpha = Y_{\alpha_0} \times_{X_{\alpha_0}} X_\alpha, \qquad \alpha \geqq \alpha_0,$$

and we get

$$Y = \varprojlim Y_\alpha = Y_{\alpha_0} \times_{X_{\alpha_0}} X.$$

If

$$Z_{\alpha_0} \to X_{\alpha_0}$$

is another finitely generated scheme, and

$$f_{\alpha_0} \colon\ Y_{\alpha_0} \to Z_{\alpha_0}$$

is a X_{α_0}-morphism, then f_{α_0} induces mappings

$$f_\alpha \colon\ Y_\alpha \to Z_\alpha, \qquad \alpha \geqq \alpha_0,$$

$$f \colon\ Y \to Z.$$

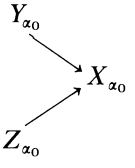
Let $\mathscr{F}_{\alpha_0}$ be an etale sheaf on X_{α_0}. We denote by $\mathscr{F}_\alpha$ resp. $\mathscr{F}$ the inverse images of $\mathscr{F}_{\alpha_0}$ on X_α $(\alpha \geqq \alpha_0)$ resp. on X. We use corresponding notations for homomorphisms of sheaves $\mathscr{F}_{\alpha_0} \to \mathscr{G}_{\alpha_0}$. By Grothendieck [63], the "etale category" satisfies the following condition of compatibility with such projective limits.

4.16 Proposition. *Let $X = \varprojlim X_\alpha$ be the projective limit of a projective system. Then:*

1) *Let $Y \to X$ be etale. There is an index α_0 and an etale scheme $Y_{\alpha_0} \to X_{\alpha_0}$ such that*

$$Y = Y_{\alpha_0} \times_{X_{\alpha_0}} X.$$

2) *Let α_0 be an index,*



two etale morphisms. Then

$$\mathrm{Hom}_X(Y, Z) = \varinjlim \mathrm{Hom}_{X_\alpha}(Y_\alpha, Z_\alpha), \qquad \alpha \geqq \alpha_0.$$

*) This is to make all the fiber products occurring noetherian, to fit our general hypothesis.

3) *Let* α_0 *be an index, and let*

$$U_{\alpha_0}^{(i)} \to X_{\alpha_0}; \quad i=1, \ldots, n$$

be a finite system of etale morphisms.

Then the following two statements are equivalent:

a) $$U^{(i)} \to X; \quad i=1, \ldots, n$$

is an etale covering.

b) *There is an index* $\alpha \geqq \alpha_0$ *such that*

$$U_{\alpha}^{(i)} \to X_{\alpha}; \quad i=1, \ldots, n$$

is an etale covering.

A similar compatibility with projective limits is valid even for arbitrary finitely generated flat or smooth morphisms. Proofs of these and similar theorems can be found in [63]. The only problem in theorems of this kind is the "depressability of flatness".

The goal of this section is the proof of the following two propositions for etale sheaves of sets resp. abelian groups on a projective limit

$$X = \varprojlim X_{\alpha}.$$

4.17 Proposition. 1) *Let* $\mathscr{F}$ *be a constructible sheaf of sets on* X. *Then there exist an index* α_0 *and a constructible sheaf* $\mathscr{F}_{\alpha_0}$ *on* X_{α_0} *whose inverse image on* X *coincides with* $\mathscr{F}$.

2) *Let* $\mathscr{F}_{\alpha_0}$ *and* $\mathscr{G}_{\alpha_0}$ *be sheaves of sets on some* X_{α_0}. *If* $\mathscr{F}_{\alpha_0}$ *is constructible, then*

$$\mathrm{Hom}(\mathscr{F}, \mathscr{G}) = \varinjlim \mathrm{Hom}(\mathscr{F}_{\alpha}, \mathscr{G}_{\alpha}), \quad \alpha \geqq \alpha_0.$$

If in particular $\mathscr{F}_{\alpha_0}$ *is representable by an etale* X_{α_0}*-scheme* U_{α_0}, *this means*

$$\mathscr{G}(U) = \varinjlim \mathscr{G}_{\alpha}(U_{\alpha}), \quad \alpha \geqq \alpha_0.$$

It follows immediately from this theorem that the corresponding statement is true in the category of sheaves of groups (abelian groups, ...). For this is the category of group objects (...) in the category of sheaves of sets.

We shall prove 4.17.2 only for constructible sheaves $\mathscr{G}$; the general case follows by Proposition 4.6, Corollary 2.

Compatibility of Cohomology with Limits

4.18 Proposition*). 1) *Let* $\mathscr{F}_{\alpha_0}$ *be a torsion sheaf on an* X_{α_0}. *Then*

$$H^*(X, \mathscr{F}) = \varinjlim H^*(X_{\alpha}, \mathscr{F}_{\alpha}), \quad \alpha \geqq \alpha_0.$$

In particular

$$\mathscr{F}(X) = \varinjlim \mathscr{F}_{\alpha}(X_{\alpha}).$$

*) Part 1) holds even for arbitrary sheaves of abelian groups (by 4.15). In part 2), it is sufficient to require that $\mathscr{F}_{\alpha_0}$ be constructible; $\mathscr{G}_{\alpha_0}$ can be an arbitrary $\mathbb{Z}/n\mathbb{Z}$-module.

2) *Let* $\mathscr{F}_{\alpha_0}$ *and* $\mathscr{G}_{\alpha_0}$ *be constructible sheaves of* $\mathbb{Z}/n\mathbb{Z}$*-modules* ($n>0$). *Then*

$$\mathrm{Ext}^{\bullet}_{\mathbb{Z}/n\mathbb{Z}}(X;\mathscr{F},\mathscr{G})=\varinjlim \mathrm{Ext}^{\bullet}_{\mathbb{Z}/n\mathbb{Z}}(X_\alpha;\mathscr{F}_\alpha,\mathscr{G}_\alpha).$$

Note. In the category of sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules, the functor

$$\mathscr{G}\to\mathrm{Hom}(\mathscr{F},\mathscr{G})$$

associated with a fixed sheaf $\mathscr{F}$ of $\mathbb{Z}/n\mathbb{Z}$-modules is left exact. Its derived functors are denoted by

$$\mathrm{Ext}^{q}_{\mathbb{Z}/n\mathbb{Z}}(X;\mathscr{F},\mathscr{G}).$$

Proof of Proposition 4.17. For representable sheaves, both statements follow immediately from Grothendieck's proposition (4.16).

For 1). We represent $\mathscr{F}$ as the cokernel of a pair of arrows between representable sheaves (4.6.3),

$$\mathscr{R}\rightrightarrows\mathscr{H}\to\mathscr{F}.$$

The pair of arrows $\mathscr{R}\rightrightarrows\mathscr{H}$ arises by extension from a pair of arrows

$$\mathscr{R}_{\alpha_0}\rightrightarrows\mathscr{H}_{\alpha_0}$$

of representable sheaves over X_{α_0} for a suitable index α_0. Since taking inverse images is exact, $\mathscr{F}$ is the inverse image of the cokernel $\mathscr{F}_{\alpha_0}$:

$$\mathscr{R}_{\alpha_0}\rightrightarrows\mathscr{H}_{\alpha_0}\to\mathscr{F}_{\alpha_0}.$$

For 2). Let $\mathscr{G}_{\alpha_0}$ be constructible.

We treat first the special case of a representable sheaf $\mathscr{F}_{\alpha_0}$: If $U_{\alpha_0}\to X_{\alpha_0}$ is an etale scheme over X_{α_0}, then

$$\mathscr{G}(U)=\varinjlim\mathscr{G}_\alpha(U_\alpha).$$

For the proof we represent $\mathscr{G}_{\alpha_0}$ as the cokernel of a representable equivalence relation

$$\mathscr{R}_{\alpha_0}\rightrightarrows\mathscr{H}_{\alpha_0}.$$

Let $\mathscr{T}_\alpha\subset\mathscr{G}_\alpha$ and $\mathscr{T}\subset\mathscr{G}$ be the images of $\mathscr{H}_\alpha$ in $\mathscr{G}_\alpha$ and $\mathscr{H}$ in $\mathscr{G}$, computed in the category of presheaves (i.e., "elementwise"). In the category of presheaves, these are isomorphic to the quotients of

$$\mathscr{R}_\alpha\rightrightarrows\mathscr{H}_\alpha\quad\text{and of}\quad\mathscr{R}\rightrightarrows\mathscr{H}$$

in the category of presheaves.

For an etale scheme $U_{\alpha_0}\to X_{\alpha_0}$ we have:

a) $\mathscr{T}(U)=\varinjlim\mathscr{T}_\alpha(U_\alpha)$.

b) For sections s, s' in $\mathscr{G}_{\alpha_1}(U_{\alpha_1})$ there is a surjective etale morphism

$$V_{\alpha_1}\to U_{\alpha_1}$$

such that the images of s, s' in $\mathscr{G}_{\alpha_1}(V_{\alpha_1})$ already are contained in $\mathscr{T}_{\alpha_1}(V_{\alpha_1})$.

From a) and b) we get first the injectivity of

$$\varinjlim_{\alpha} \mathscr{G}_\alpha(U_\alpha) \hookrightarrow \mathscr{G}(U).$$

The surjectivity comes now from the injectivity just proved, from a), and from this fact:

For every section s in $\mathscr{G}(U)$ there is a surjective etale mapping $V_{\alpha_1} \to U_{\alpha_1}$ such that the image of s in $\mathscr{G}(V)$ is already contained in $\mathscr{T}(V)$.

The *general* case: $\mathscr{F}_{\alpha_0}$ is the cokernel of a pair of mappings of representable sheaves,

$$\mathscr{R}_{\alpha_0} \rightrightarrows \mathscr{H}_{\alpha_0} \to \mathscr{F}_{\alpha_0}.$$

Since the functor $\mathrm{Hom}(-,-)$ is left exact in its first variable, assertion 4.17.2 follows from the results already proved for the representable sheaves $\mathscr{R}_{\alpha_0}$ and $\mathscr{H}_{\alpha_0}$. □

We come now to the proof of Proposition 4.18.

We restrict ourselves to the essential case of constructible sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules. If $\mathscr{F}$ is a constructible sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules, then

$$H^*(X, \mathscr{F}) = \mathrm{Ext}^*_{\mathbb{Z}/n\mathbb{Z}}(X; (\mathbb{Z}/n\mathbb{Z})_X, \mathscr{F}).$$

Hence it suffices to prove the Ext-formula 2).

Here we use the interpretation of the Ext groups as "Yoneda Ext". For that, we consider exact sequences

$$0 \to \mathscr{F} \to \mathscr{F}_1 \to \ldots \to \mathscr{F}_p \to \mathscr{G} \to 0$$

in the category of sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules. Another sequence

$$0 \to \mathscr{F} \to \mathscr{F}_1' \to \ldots \to \mathscr{F}_p' \to \mathscr{G} \to 0$$

dominates the previous one if there is a commutative diagram

$$\begin{array}{ccccccccc}
0 \to & \mathscr{F} & \to & \mathscr{F}_1' & \to \ldots \to & \mathscr{F}_p' & \to & \mathscr{G} & \to 0 \\
 & \| & & \downarrow & & \downarrow & & \| & \\
0 \to & \mathscr{F} & \to & \mathscr{F}_1 & \to \ldots \to & \mathscr{F}_p & \to & \mathscr{G} & \to 0.
\end{array}$$

The set $\mathrm{Ext}^p_{\mathbb{Z}/n\mathbb{Z}}(X; \mathscr{F}, \mathscr{G})$ is canonically isomorphic to the set of equivalence classes of such exact sequences modulo the equivalence relation generated by "dominance".

Let $\mathscr{F}$ and $\mathscr{G}$ be constructible sheaves, and thus noetherian objects in the category of sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules. For the construction of the Yoneda Ext, it is enough to consider only sequences of constructible sheaves (and dominances between them). The desired limit formula for Ext now follows from the limit formula proved for the Hom functor, from the depressability of constructible

sheaves, and from the following statement: Let

$$\mathscr{F}_{\alpha_0} \to \mathscr{G}_{\alpha_0} \to \mathscr{H}_{\alpha_0}$$

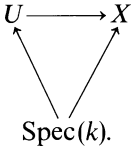
be a sequence of constructible sheaves on X_{α_0} which becomes exact when lifted to X. Then it already becomes exact after lifting to a suitable X_α, $\alpha \geqq \alpha_0$. □

Applications

Let x: Spec$k \to X$ be a geometric point of X. We denote by $\tilde{R}$ the strict local ring of X at x (§ 2). We recall that

$$\tilde{R} = \varinjlim \mathscr{O}_U(U).$$

Here the limit is taken over the category of etale neighborhoods of x,



In this, one can restrict to connected affine U.

4.19 Proposition. *Let $\tilde{\mathscr{F}}$ be the inverse image of $\mathscr{F}$ on $\tilde{Y} = Y \times_X \mathrm{Spec}(\tilde{R})$. Then*

$$R^\nu f_*(\mathscr{F})_x = H^\nu(\tilde{Y}, \tilde{\mathscr{F}}) = \varinjlim_{\substack{U \to X \\ \mathrm{Spec}(k)}} H^\nu(Y \times_X U, \mathscr{F}).$$

§ 5. Calculation of Cohomology for Curves

5.1 Proposition. *Let X be a complete connected smooth curve over an algebraically closed field k whose characteristic is relatively prime to the natural number n. Then:*

a) $H^0(X, \mu_n) \cong \mu_n(k)$.
b) $H^1(X, \mu_n) \cong {}_n\mathrm{Pic}\, X$.

Here we let ${}_n\mathrm{Pic}\, X$ denote the group of n-division points in Pic X, *i.e.*

$${}_n\mathrm{Pic}\, X = \{\mathscr{L} \in \mathrm{Pic}\, X,\ \mathscr{L}^n = \mathscr{O}_X\},$$

c) $H^2(X, \mu_n) \cong \mathbb{Z}/n\mathbb{Z}$.
d) $H^q(X, \mu_n) = 0$ *for* $q > 2$.

These isomorphisms are canonical. Only after selecting a specific primitive n-th root of unity in k do we get isomorphisms

$$(\mu_n)_X \cong (\mathbb{Z}/n\mathbb{Z})_X,$$
$$\mu_n(k) \cong \mathbb{Z}/n\mathbb{Z}.$$

Furthermore, we have [95]

$${}_n\mathrm{Pic}\, X \cong \mathbb{Z}/n\mathbb{Z} \times \ldots \times \mathbb{Z}/n\mathbb{Z} = (\mathbb{Z}/n\mathbb{Z})^{2g} \quad \text{(noncanonically)},$$

where g denotes the genus of X.

The proof uses the Kummer sequence

$$0 \to (\mu_n)_X \to \mathcal{O}_X^* \to \mathcal{O}_X^* \to 0.$$

The proof of Proposition 5.1 rests on

5.2 Lemma. *Let X be a smooth curve over an algebraically closed field k. Then*

$$H^i(X, \mathcal{O}_X^*) = 0 \quad \textit{for } i \geqq 2.$$

Note. Lemma 5.2 extends to an arbitrary finitely generated scheme X of dimension $\leqq 1$ over an algebraically closed base field. If the field k has positive characteristic p and is only separably algebraically closed, then we only have:

$$H^i(X, \mathcal{O}_X^*) \text{ is a } p\text{-torsion group for } i \geqq 2.$$

The central point of the proof (see below) is Tsen's theorem. Under the weaker hypothesis on the base field, the Brauer group of a function field in one variable does not in general vanish. But nonetheless it is a p-torsion group (1.15).

To prove Lemma 5.2, we use the sheaf $\mathcal{R}_X$ of rational functions on X. If $U \to X$ is irreducible and etale, $\mathcal{R}_X(U)$ is the field of rational functions on U, and $\mathcal{D}_X(U)$ is the group of divisors on U, i.e., the free abelian group generated by the closed points of U. There is a natural exact sequence of sheaves

$$0 \to \mathcal{O}_X^* \to \mathcal{R}_X^* \to \mathcal{D}_X \to 0.$$

Obviously Lemma 5.2 will follow from

1) $H^i(X, \mathcal{R}_X^*) = 0$ for $i > 0$,
2) $H^i(X, \mathcal{D}_X) = 0$ for $i > 0$.

Proof. We can assume that X is irreducible.

1) Let K be the function field of X, and let

$$j\colon \mathrm{Spec}\, K \to X$$

be the natural embedding. Obviously

$$\mathcal{R}_X^* = j_* \mathcal{G}_m, \qquad \mathcal{G}_m = \mathcal{O}^*_{\mathrm{Spec}\, K}.$$

We now show that

a) $$R^i j_* \mathcal{G}_m = 0 \quad \text{for } i > 0.$$

The i-th direct image is the sheaf associated with the functor

$$U \mapsto H^i(\operatorname{Spec} K \times_X U, \mathcal{G}_m).$$

Now $\operatorname{Spec} K \times_X U$ is an etale extension of $\operatorname{Spec} K$, and thus it is made up of finitely many finite separable extensions of K. Their cohomology vanishes by Tsen's theorem (1.15). By the Leray spectral sequence, this gives us

b) $$H^i(X, j_* \mathcal{G}_m) \cong H^i(\operatorname{Spec} K, \mathcal{G}_m) = H^i(K, \mathcal{G}_m).$$

This group likewise vanishes by Tsen's theorem.

2) Every closed point $x \in X$ can be uniquely interpreted as a geometric point

$$x\colon \operatorname{Spec} k \to X \quad (\text{over } k).$$

Obviously

$$\mathcal{D}_X = \bigoplus_{x \text{ a closed point}} x_*(\mathbb{Z}_{\operatorname{Spec} k}).$$

All sheaves on $\operatorname{Spec} k$ (and on etale extensions of $\operatorname{Spec} k$) are acyclic, since k is algebrically closed. It follows from this that the sheaves $x_*(\mathbb{Z}_{\operatorname{Spec} k})$ are acyclic. Therefore $\mathcal{D}_X$ is also acyclic, because cohomology is compatible with arbitrary direct sums (4.18).

With this, 5.2 is proved. □

We can now calculate the cohomology $H^*(X, \mathbb{Z}/n\mathbb{Z})$ by means of the Kummer sequence

$$0 \to (\mu_n)_X \to \mathcal{O}_X^* \to \mathcal{O}_X^* \to 0.$$

From the corollary to 5.2 we get assertion d) of 5.1, as well as

$$H^2(X, \mu_n) = \operatorname{Coker}(\operatorname{Pic} X \xrightarrow{n} \operatorname{Pic} X)$$

$$H^1(X, \mu_n) = \operatorname{Ker}(\operatorname{Pic} X \xrightarrow{n} \operatorname{Pic} X).$$

Every line bundle $\mathcal{L}$ has an associated degree $d(\mathcal{L})$. We denote by $\operatorname{Pic}^0 X$ the group of line bundles of degree 0,

$$0 \to \operatorname{Pic}^0 X \to \operatorname{Pic} X \xrightarrow{d} \mathbb{Z} \to 0.$$

$\operatorname{Pic}^0 X$ carries the structure of an abelian variety of dimension g (genus of X) over k. This abelian variety is the Jacobian variety of X [95]. The mapping

$$\operatorname{Pic}^0 X \xrightarrow{n} \operatorname{Pic}^0 X$$

is therefore surjective, and we get

$$H^2(X, \mu_n) = \operatorname{Coker}(\mathbb{Z} \xrightarrow{n} \mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$$

and

$$H^1(X, \mu_n) = \operatorname{Ker}(\operatorname{Pic}^0 X \xrightarrow{n} \operatorname{Pic}^0 X).$$

The group of n-division points of a g-dimensional abelian variety over an algebraically closed field of characteristic relatively prime to n is isomorphic to $(\mathbb{Z}/n\mathbb{Z})^{2g}$ [95].

With this, Proposition 5.1 is completely proved. □

Now let X be an *affine* smooth connected curve over the algebraically closed field k, and let n be a number relatively prime to the characteristic of k. Lemma 5.2 also holds in this case, and by using the Kummer sequence we get

1) $H^q(X, \mu_n)=0$ for $q>2$.

2) The sequence

$$\mathcal{O}_X^* \xrightarrow{n} \mathcal{O}_X^* \to H^1(X, \mu_n) \to \operatorname{Pic} X \xrightarrow{n} \operatorname{Pic} X \to H^2(X, \mu_n) \to 0$$

is exact.

We now embed X in a complete smooth connected curve $\bar{X}$. We can identify the elements of $\mathcal{O}_X(X)^*$ with the rational functions on $\bar{X}$ whose divisors are contained in the finite set $S=\bar{X}-X$. Since the group of these divisors is finitely generated, and since $k^* \xrightarrow{n} k^*$ is surjective, it follows that

$$\operatorname{Coker}(\mathcal{O}_X(X)^* \to \mathcal{O}_X(X)^*) \cong \operatorname{Ker}(H^1(X, \mu_n) \to \operatorname{Pic} X)$$

is a finite group. Furthermore, the kernel $\operatorname{Pic}(X) \xrightarrow{n} \operatorname{Pic}(X)$ is also finite:

For this, consider the short exact sequence

$$0 \to E \to \operatorname{Pic} \bar{X} \to \operatorname{Pic} X \to 0.$$

The assertion follows from the finite generation of the kernel E and the finiteness of

$$\operatorname{Ker}(\operatorname{Pic} \bar{X} \xrightarrow{n} \operatorname{Pic} \bar{X}) = \operatorname{Ker}(\operatorname{Pic}^0 \bar{X} \xrightarrow{n} \operatorname{Pic}^0 \bar{X}).$$

We conclude that $H^1(X, \mu_n)$ is also finite.

We observe now that the mapping

$$\operatorname{Pic} X \xrightarrow{n} \operatorname{Pic} X$$

in the *affine* case is surjective, since every divisor on X can be extended to a divisor of degree 0 on $\bar{X}$ (as $S \neq \emptyset$), and since the mapping

$$\operatorname{Pic}^0 \bar{X} \xrightarrow{n} \operatorname{Pic}^0 \bar{X}$$

is surjective. We thus get

$$H^2(X, \mu_n)=0.$$

In summary:

Let X be a smooth (not necessarily connected) curve over an algebraically closed field k, and let n be a number relatively prime to the characteristic of k. Then the groups $H^q(X, (\mathbb{Z}/n\mathbb{Z})_X)$ are finite for all q, and vanish for $q>2$; if X is affine, they actually vanish even for $q>1$.

Since every finite abelian group A is a direct product of cyclic groups, the corresponding statement also holds for arbitrary finite constant sheaves of order relatively prime to the characteristic of k.

We assert now more generally:

5.3 Proposition. *Let k be a separably closed field and X a finitely generated k-scheme, $\dim X \leqq 1$. Let $\mathscr{F}$ be a constructible sheaf whose sections have order relatively prime to the characteristic of k. Then:*

a) *The cohomology groups $H^q(X, \mathscr{F})$ are finite.*

b) *$H^q(X, \mathscr{F}) = 0$ for $q \geqq 3$.*

c) *If X is affine, then actually*

$$H^q(X, \mathscr{F}) = 0 \quad \text{for } q \geqq 2.$$

Statements b) and c) can be extended from constructible sheaves to torsion sheaves by passage to limits.

Two *Preliminary remarks* before the proof.

1) Let $\mathscr{F}$ be a sheaf of abelian groups on the scheme X. Suppose there exists a *finite* closed subscheme $S \xhookrightarrow{j} X$ such that

$$\mathscr{F} | U = 0, \quad U = X - S.$$

Then $\mathscr{F} = j_*(\mathscr{F} | S)$, and

$$H^q(X, \mathscr{F}) = H^q(S, \mathscr{F} | S) = 0 \quad \text{for } q > 0.$$

From this follows: Let $\mathscr{F} \to \mathscr{G}$ be a homomorphism of constructible sheaves which is an isomorphism "outside finitely many points". Then the statements in the theorem are equivalent for $\mathscr{F}$ and for $\mathscr{G}$.

2) Let $f: X' \to X$ be a finite morphism, $\mathscr{F}'$ a constructible sheaf on X'. Since the higher drived functors $R^q f_*(\mathscr{F}')$ vanish (3.4), the statements in the theorem are equivalent for $(X', \mathscr{F}')$ and for $(X, f_* \mathscr{F}')$.

Proof of Proposition 5.3. Reduction to smooth curves. We can assume at the start that k is algebraically closed and that X is reduced (because of 3.13 and 3.14). Let $f: X' \to X$ be the normalization of X. It is an isomorphism outside finitely many points. Hence the adjunction mapping

$$\mathscr{F} \to f_* f^* \mathscr{F}$$

is also an isomorphism outside finitely many points. On the basis of the preliminary remarks, it is enough to prove the Proposition for $f^* \mathscr{F}$ on X' instead of for $\mathscr{F}$ on X.

Proof of the finiteness Proposition 5.3 a). Let

$$0 \to \mathscr{F} \to \mathscr{F}_1 \to \mathscr{F}_2 \to \dots$$

be an exact sequence of sheaves. Suppose the cohomology groups of the sheaves $\mathscr{F}_\nu$ are finite. Then obviously the cohomology groups of $\mathscr{F}$ are finite.

It is enough therefore to show:

Every constructible sheaf $\mathscr{F}$ on X is embeddable in a constructible sheaf with finite cohomology groups.

Let $\mathscr{F}$ be a constructible sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules on X ($n \neq 0$ in k). Then $\mathscr{F}$ is a subsheaf of a finite direct sum of sheaves

$$f_{\nu *}((\mathbb{Z}/n\mathbb{Z})_{Y_\nu})$$

for finite morphisms

$$f_\nu\colon\ Y_\nu \to X \quad (4.12).$$

We can assume that the schemes Y_ν are normal. Then

$$\dim Y_\nu \leqq \dim X_\nu \leqq 1.$$

We have already seen that the cohomology groups

$$H^j(Y_\nu, (\mathbb{Z}/n\mathbb{Z})_{Y_\nu}) = H^j(X, f_{\nu *}((\mathbb{Z}/n\mathbb{Z})_{Y_\nu}))$$

are finite groups.

Proof of the vanishing Proposition 5.3 b). Let $\mathscr{F}$ be a torsion sheaf on an irreducible reduced curve X, and suppose its stalk $\mathscr{F}_{\bar{\eta}}$ vanishes at the generic geometric point

$$\bar{\eta}\colon\ \operatorname{Spec} \bar{K} \to X$$

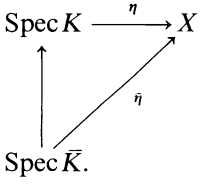
$$\bar{K} = \text{separable closure of the function field } K = \mathscr{R}(X).$$

If $\mathscr{F}$ is constructible, then the restriction of $\mathscr{F}$ to an open dense subscheme is 0, and thus the sheaf $\mathscr{F}$ is concentrated on finitely many points and

$$H^q(X, \mathscr{F}) = 0 \quad \text{for } q > 0.$$

Since an arbitrary torsion sheaf is a direct limit of constructible subsheaves, this formula extends to arbitrary torsion sheaves $\mathscr{F}$ with $\mathscr{F}_{\bar{\eta}} = 0$.

Let $\mathscr{F}$ now be an arbitrary torsion sheaf on an irreducible reduced curve X. We consider the diagram



The projective limit (in the sense of § 4) of the system $(U \times_X \operatorname{Spec} K)$, where U runs through the affine etale neighborhoods of $\bar{\eta}$, is

$$\operatorname{Spec} \bar{K} \times_X \operatorname{Spec} K = \operatorname{Spec} \bar{K}.$$

From the corollary to 2.13 combined with the limit Theorem 4.21, we get

$$R^q \eta_* (\eta^* \mathscr{F})_{\bar{\eta}} = \begin{cases} \mathscr{F}_{\bar{\eta}} & \text{for } q = 0, \\ 0 & \text{for } q > 0. \end{cases}$$

In particular, the adjunction mapping

$$\mathscr{F} \to \eta_* \eta^* \mathscr{F}$$

is an isomorphism at the generic point $\bar{\eta}$, and it is sufficient to prove

$$H^q(X, \eta_* \eta^* \mathscr{F}) = 0 \quad \text{for } q \geqq 3$$

instead of 5.3 b).

It follows from Tsen's theorem that

1) $H^q(\operatorname{Spec} K, \eta^* \mathscr{F}) = 0$, $q \geqq 2$,

2) $R^q \eta_*(\eta^* \mathscr{F}) = 0$, $q \geqq 2$.

We know besides that

3) $H^p(X, R^q \eta_*(\eta^* \mathscr{F})) = 0$ for $p > 0$, $q > 0$.

Using 1), 2), 3) in the analysis of the Leray spectral sequence

$$H^p(X, R^q \eta_* \eta^* \mathscr{F}) \Rightarrow H^{p+q}(\operatorname{Spec} K, \eta^* \mathscr{F}),$$

we get

$$H^p(X, \eta_* \eta^* \mathscr{F}) = 0 \qquad \text{for } p \geqq 3.$$

Proof of the affine vanishing Proposition 5.3c). Let X be an affine irreducible smooth nonempty curve over k. We can assume that $\mathscr{F}$ is a constructible sheaf of Λ-modules on the scheme X, with $\Lambda = \mathbb{Z}/n\mathbb{Z}$ and $n \neq 0$ in k. We have already shown that the functor $H^p(X, -)$ vanishes for $p \geqq 3$. In particular, $H^2(X, -)$ is right exact. By Proposition 4.10, the sheaf $\mathscr{F}$ is a factor sheaf of a finite direct sum of sheaves of the type $(\tilde{Y})^\Lambda$. Here $\tilde{Y}$ is the sheaf of sets represented by an etale scheme

$$g\colon\ Y \to X,$$

and $(\tilde{Y})^\Lambda$ is the sheaf of free Λ-modules generated by Y. The scheme Y is a smooth affine curve over k. We can assume that Y is nonempty. It is enough to show:

$$H^2(X, (Y)^\Lambda) = 0.$$

First a

Preliminary remark. Let

$$f\colon\ U \to V$$

be an etale morphism of schemes. There is a natural homomorphism

$$\phi\colon\ (\tilde{U})^\Lambda \to f_*(\Lambda_U).$$

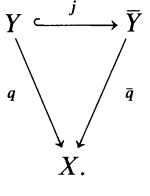
Construction of ϕ. By the definition of $(\tilde{U})^\Lambda$ we have

$$\operatorname{Hom}_\Lambda((\tilde{U})^\Lambda, f_*(\Lambda_U)) = \operatorname{Hom}(\tilde{U}, f_*(\Lambda_U)) = \Gamma(U, f_*(\Lambda_U)) = \Gamma(U \times_V U, \Lambda_U).$$

The diagonal $\Delta \subset U \times_V U$ is open and closed in $U \times_V U$. Let s be the section in $\Gamma(U \times_V U, \Lambda_U)$ that is 1 on the diagonal and 0 outside it. Define ϕ to be the homomorphism corresponding to this section.

It is easy to see that ϕ is an isomorphism when V is a point. *It follows from this that ϕ is an isomorphism when f is finite.* For in this case the direct image, $(\tilde{U})^\Lambda$ and ϕ are compatible with base change.

Back to our original situation $q\colon\ Y \to X$. We represent Y as an open dense subscheme of a scheme finite and normal over X:



Then $\bar{Y}$ is also a smooth affine curve over k. We consider the homomorphisms

$$(\tilde{Y})^{\Lambda} \to q_*(\Lambda_Y) \leftarrow (\bar{q})_*(\Lambda_{\bar{Y}}).$$

Both homomorphisms are isomorphisms outside a finite set of points. We have already shown that

$$H^2(X, (\bar{q})_*(\Lambda_{\bar{Y}})) = H^2(\bar{Y}, \Lambda_{\bar{Y}})$$

vanishes. Therefore we also have the vanishing of

$$H^2(X, (\tilde{Y})^{\Lambda}). \quad \square$$

In § 9 we shall generalize the affine vanishing theorem to affine algebraic varieties of arbitrary dimension.

§ 6. The Base Change Theorem for Proper Morphisms

Let

$$\begin{array}{ccc} X & \xrightarrow{f'} & T \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ Y & \xrightarrow{f} & S \end{array}$$

be a commutative diagram of schemes. For every sheaf $\mathscr{F}$ on Y we can define a natural mapping

$$g^* f_* \mathscr{F} \to f'_* g'^* \mathscr{F}$$

as follows: we apply the functor f_* to the adjunction mapping

$$\mathscr{F} \to g'_*(g')^* \mathscr{F},$$

getting

$$f_* \mathscr{F} \to f_* g'_*(g')^* \mathscr{F} = g_* f'_* g'^* \mathscr{F},$$

and then use the adjointness of the functors g_* and g^*.

More generally, for every i we construct a natural mapping

$$g^*(R^i f_* \mathscr{F}) \to R^i f'_*(g'^* \mathscr{F}),$$

the *base change homomorphism.*

We choose an injective resolution $\mathscr{I}^\cdot$ of $\mathscr{F}$ and an injective resolution $\mathscr{J}^\cdot$ of $g'^*(\mathscr{I}^\cdot)$, that is, a quasi-isomorphism from $g'^*(\mathscr{I}^\cdot)$ into a complex $\mathscr{J}^\cdot$ bounded below of injective sheaves. Because g'^* is exact, $g'^*(\mathscr{I}^\cdot)$ is a resolution of $g'^*(\mathscr{F})$, and hence so is $\mathscr{J}^\cdot$. We get a natural mapping

$$\begin{array}{ccc} g^* f_*(\mathscr{I}^\cdot) & \longrightarrow & f'_*(\mathscr{J}^\cdot) \\ & \searrow \quad \nearrow & \\ & f'_* g'^*(\mathscr{I}^\cdot). & \end{array}$$

We have

$$H^i(f'_*(\mathscr{I}^\cdot)) = R^i f'_*(g'^*(\mathscr{F})).$$

Since g^* is exact and

$$H^i(f'_* \mathscr{I}) = R^i f'_* g'^* \mathscr{F},$$

we have

$$H^i(g^* f_* \mathscr{I}^\cdot) = g^*(H^i f_* \mathscr{I}^\cdot) = g^*(R^i f_* \mathscr{F}). \quad \square$$

Remark. More generally, for a complex $\mathscr{F}^\cdot$ of sheaves on Y bounded below, one can use the same procedure to construct a base change homomorphism of the derived complexes in the sense of Verdier (A.II),

$$g^*(\mathrm{R} f_* \mathscr{F}^\cdot) \to \mathrm{R} f'_*(g'_* \mathscr{F}).$$

In what follows, only the case of a cartesian diagram $X = Y \times_S T$ is of interest. We call f then the *structure mapping* and g the *base change mapping.*

For the base change homomorphism, we have in an obvious way a series of compatibility conditions on the level of complexes (that is, in the derived category), for instance:

a) Compatibility under composition of the structure mappings,

b) Compatibility under composition of the base change mappings.

6.1 Theorem (Base change theorem for proper mappings). *Let $f: X \to S$ be a proper morphism of schemes, and let*

$$\begin{array}{ccc} X_T & \xrightarrow{f'} & T \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g}, \\ X & \xrightarrow{f} & S \end{array} \qquad X_T = X \times_S T$$

be a cartesian diagram. For every torsion sheaf $\mathscr{F}$ on X, the base change homomorphism

$$g^*(R^i f_* \mathscr{F}) \xrightarrow{\sim} R^i f'_*(g'^* \mathscr{F})$$

is an isomorphism.

We prove this theorem only under the following supplementary hypotheses, which are satisfied in our applications:

1) $\mathscr{F}$ is a sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules, n invertible on S.

2) The morphism $f: X \to S$ is projective. (The non-projective case can be reduced to the projective case using Chow's lemma.)

Some preliminary remarks before the proof:

1) Let $\mathscr{F}^\cdot$ be a complex of torsion sheaves on Y, bounded below. Suppose the base change theorem holds for all components $\mathscr{F}^\nu$. Then it also holds for

$\mathscr{F}^{\cdot}$:

$$g^* \mathrm{R}f_* \mathscr{F}^{\cdot} \xrightarrow{\sim} \mathrm{R}f'_*(g'^* \mathscr{F}^{\cdot}) \quad (\text{in } D(T)).$$

2) Let $f = f_1 \circ f_2$,

$$f_2 \colon X \to X'; \quad f_1 \colon X' \to S,$$

be the product of two proper morphisms.

a) If the base change theorem holds for f_1 and f_2, it also holds for f.

b) If the base change theorem holds for f and f_1, it holds with respect to f_2 for all complexes of the form $\mathrm{R}f_{1*}\mathscr{F}^{\cdot}$.

3) If T is the spectrum of a separably closed field k,

$$g = s \colon \operatorname{Spec} k \to S,$$

then the base change theorem means

6.1′:

$$\boxed{\begin{array}{l} (R^i f_* \mathscr{F})_s = H^i(X_s, \mathscr{F}_s), \\ X_s = X \times \operatorname{Spec} k \text{ (geometric fiber)} \\ \mathscr{F}_s = \mathscr{F} \mid X_s \text{ (inverse image).} \end{array}}$$

Proof of Theorem 6.1. We can assume that T is a finitely generated S-scheme (one can assume that T and S are affine and then use the limit theorem (4.18)).

By 2.12, a homomorphism of sheaves on T is an isomorphism if and only if it induces isomorphisms on all stalks. Since T/S is finitely generated, we can here restrict to the points $t_0 \in T$ that are closed in their fibers (2.12).

In addition, we know: Let

$$t \colon \operatorname{Spec} k \to T \text{ be a geometric point,}$$

$$T(t) = \operatorname{Spec} \tilde{\mathscr{O}}_{T,t}$$

the spectrum of the strict local ring at t. If $\mathscr{G}$ is a sheaf on X_T and $\mathscr{G}(t)$ is its inverse image on $X_T \times_T T(t)$, then by (4.19) we have

$$(R^i f'_* \mathscr{G})_t \cong H^i(X_T \times_T T(t), \mathscr{G}(t)).$$

Similarly one can compute the stalks of the higher direct images with respect to $f \colon X \to S$.

It follows from this that the base change theorem only needs to be proved in the following two cases.

1) The base change mapping is of the form

$$T = \operatorname{Spec} K \to \operatorname{Spec} k = S,$$

where K is a finite algebraic extension field of a separably closed field k.

2) S is the spectrum of a strictly Henselian ring, $T = \operatorname{Spec} k$ is the spectrum of the residue field, and $T \to S$ is the natural mapping.

In case 1), K is a purely inseparable extension of k. In this special case, the base change theorem was proved in 3.14.

Proof of the proper base change theorem. By the preceding remarks, we can restrict ourselves to proving the base change theorem in the following form:

$$S = \operatorname{Spec} A, \quad A \text{ a strictly Henselian ring,}$$

$$g = s\colon \operatorname{Spec} k \to S, \quad k = A/\mathfrak{m}.$$

In this case the theorem says:

$$H^i(X, \mathscr{F}) = H^i(X_s, \mathscr{F}_s),$$

$$X_s = X \times \operatorname{Spec} k, \quad \mathscr{F}_s = \mathscr{F} \mid X_s.$$

We deal first with the case where the special fiber X_s is at most one-dimensional, and to begin with we show:

Assertion. *Let* $\dim X_s \leqq 1$. *The natural mapping*

$$H^i(X, \mathbb{Z}/n) \to H^i(X_s, \mathbb{Z}/n) \quad (n \text{ invertible in } \mathcal{O}_X)$$

is

a) *bijective when* $i = 0$,
b) *surjective when* $i > 0$.

When $i > 2$, there is nothing to prove (5.3). The cases $i = 0, 1, 2$ will be handled separately.

1) $i = 0$. The assertion follows from the Zariski connectedness theorem for a proper scheme over a Henselian ring: The number of connected components of X and X_s is the same [62, Chap. III 4.33].

2) $i = 1$. In view of 2.11, it is enough to prove: Every Galois etale covering space of the special fiber can be extended to a Galois etale covering space of X.

Proof. The functor

$$\operatorname{Et}(X_n) \to \operatorname{Et}(X_{n-1}), \quad X_n = X \otimes A/\mathfrak{m}^n,$$

is an equivalence of categories (3.16). A Galois etale covering space of the special fiber $X_s = X_1$ can thus be extended inductively to the X_n. In the language of formal schemes, we can express this as follows:

Every Galois etale covering space of the special fiber can be extended to a Galois covering space of the formal completion of X along X_s.

From the fundamental existence theorem of Grothendieck [62, Chap. III 5.1.4] for proper mappings, it follows now that:

Every Galois etale covering space of the special fiber can be extended to a Galois covering space of $X \otimes \hat{A}$, where $\hat{A} = \varprojlim A/\mathfrak{m}^\nu$.

We now want to apply the Artin approximation theorem. Here a small technical difficulty occurs: in the Artin theory there are restrictions on the ring A. Admissible rings in this theory are

strict Henselizations of finitely generated algebras over $\mathbb{Z}$*).

*) A strict Henselization of a finite generated algebra over $\mathbb{Z}$ is isomorphic to the Henselization of a finite generated algebra over an excellent discret valuation ring.

But every strictly Henselian ring is a direct limit of such rings. Hence, in view of 4.18, it is enough to restrict ourselves to such rings.

We now consider the following functor F on the category of (noetherian) A-algebras B: it assigns to an algebra B the set $F(B)$ of isomorphism classes of etale Galois $\mathbb{Z}/n\mathbb{Z}$-covering spaces of $X \otimes B$. It is locally of finite presentation [10, 1.5]:

$$F(\varinjlim B_\nu) = \varinjlim F(B_\nu).$$

From the *Artin approximation theorem* [10, 1.12] it follows that:

For the given $\nu \geqq 1$ there is a Galois covering space of X that agrees over $X \otimes A/\mathfrak{m}^\nu$ with the covering space constructed for $X \otimes \hat{A}$.

3) $i=2$: The diagram (Kummer sequence)

$$\begin{array}{ccc} \operatorname{Pic} X & \longrightarrow & H^2(X, \mu_n) \\ \downarrow & & \downarrow \\ \operatorname{Pic} X_s & \longrightarrow & H^2(X_s, \mu_n) \end{array}$$

is commutative. The second line is surjective: for if k has characteristic 0, the group $H^2(X_s, \mathcal{O}_{X_s}^*)$ vanishes. In the case of positive characteristic p, it is a p-torsion group (see the note to 5.2), and p is relatively prime to n.

Since k is separably algebraically closed, the Henselian ring A contains a primitive n-th root of unity, whose image in k is also a primitive n-th root of unity. After selecting one such root, we can identify the constant sheaf $\mathbb{Z}/n\mathbb{Z}$ on X and X_s with the sheaf μ_n of n-th roots of unity. Thus it is enough to show:

Every line bundle on the special fiber X_s can be extended to all of X. As in the case $i=1$, it is enough to show:

Every line bundle on $X_n = X \otimes A/\mathfrak{m}^n$ can be extended to X_{n+1}. We show this in the following way: The same topological space underlies all the schemes X_n. We can thus consider the structure sheaf $\mathcal{O}_{X_n}$ as a sheaf (with respect to the Zariski topology) on X_s. Let

$$\mathcal{I} = \mathfrak{m}^n(\mathcal{O}_X)/\mathfrak{m}^{n+1}(\mathcal{O}_X) = \operatorname{Ker}(\mathcal{O}_{X_{n+1}} \to \mathcal{O}_{X_n}).$$

The sequence

$$\begin{array}{c} 0 \to \mathcal{I} \to \mathcal{O}_{X_{n+1}}^* \to \mathcal{O}_{X_n}^* \to 1 \\ \alpha \longmapsto 1+\alpha \end{array}$$

is exact. The sheaf $\mathcal{I}$ is coherent, and X_s is at most one-dimensional, and hence $H^2(X_s, \mathcal{I})$ vanishes. The long exact cohomology sequence then yields the surjectivity of

$$H^1(X_s, \mathcal{O}_{X_n}^*) \to H^1(X_s, \mathcal{O}_{X_{n+1}}^*).$$

Thus the assertion above is proved.

We now apply this result to schemes $X' \xrightarrow{p} X$ finite over X, rather than to X. Since the base change theorem is already known for finite morphisms

(3.5), we have: The mapping

$$H^\nu(X, \mathscr{F}) \to H^\nu(X_s, \mathscr{F}_s)$$

is

a) bijective for $\nu = 0$,

b) surjective for $\nu > 0$,

for all sheaves $\mathscr{F}$ that are isomorphic to a finite direct sum of sheaves of the form $p_*((\mathbb{Z}/n\mathbb{Z})_{X'})$. By 4.12, every constructible sheaf on X is a subsheaf of such a sheaf.

We summarize:

Every constructible sheaf $\mathscr{F}$ (of $\mathbb{Z}/n\mathbb{Z}$-modules) on X is a subsheaf of a constructible sheaf $\mathscr{G}$ with the following property: The mapping

$$H^i(X, \mathscr{G}) \to H^i(X_s, \mathscr{G}_s)$$

is

a) bijective in the case $i = 0$,

b) surjective in the case $i > 0$.

From the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(X, \mathscr{F}) & \longrightarrow & H^0(X, \mathscr{G}) & \longrightarrow & H^0(X, \mathscr{G}/\mathscr{F}) \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & H^0(X_s, \mathscr{F}_s) & \longrightarrow & H^0(X_s, \mathscr{G}_s) & \longrightarrow & H^0(X_s, (\mathscr{G}/\mathscr{F})_s) \end{array}$$

one can read off that the mapping

$$H^0(X, \mathscr{F}) \to H^0(X_s, \mathscr{F}_s) \quad \text{is injective.}$$

This holds for all constructible sheaves (of $\mathbb{Z}/n\mathbb{Z}$-modules) an hence also for all sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules.

Using the injectivity of

$$H^0(X, \mathscr{G}/\mathscr{F}) \to H^0(X_s, (\mathscr{G}/\mathscr{F})_s),$$

one can now conclude that the mapping

$$H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}_s)$$

is bijective for all sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules.

We now prove by induction on i that the mapping

$$H^i(X, \mathscr{F}) \to H^i(X_s, \mathscr{F}_s)$$

is an isomorphism for all sheaves $\mathscr{F}$ of $\mathbb{Z}/n\mathbb{Z}$-modules. Thus we assume that this has been proved for all $i < p$ and then prove the isomorphism for $i = p$. In this we can restrict ourselves to constructible sheaves $\mathscr{F}$. We here also use the fact, already proved, that the sheaf $\mathscr{F}$ can be embedded in a torsion sheaf $\mathscr{G}$ for which the mapping

$$H^i(X, \mathscr{G}) \to H^i(X_s, \mathscr{G}_s)$$

is surjective for all i.

As a first step, we show that the mapping

$$H^p(X, \mathscr{G}) \to H^p(X_s, \mathscr{G}_s)$$

is injective (and hence an isomorphism). We consider a short exact sequence

$$0 \to \mathcal{G} \to \mathcal{J} \to \mathcal{K} \to 0,$$

where we let $\mathcal{J}$ be injective in the category of $\mathbb{Z}/n\mathbb{Z}$-modules. In the diagram

$$\begin{array}{ccccccc}
H^{p-1}(X, \mathcal{J}) = H^{p-1}(\mathcal{J}) & \longrightarrow & H^{p-1}(\mathcal{K}) & \longrightarrow & H^p(\mathcal{G}) & \longrightarrow & 0 \\
\downarrow s & & \downarrow s & & \downarrow & & \downarrow \\
H^{p-1}(X_s, \mathcal{J}_s) = H^{p-1}(\mathcal{J}_s) & \longrightarrow & H^{p-1}(\mathcal{K}_s) & \longrightarrow & H^p(\mathcal{G}_s) & \longrightarrow & H^p(\mathcal{J}_s),
\end{array}$$

the first two vertical arrows are bijective (induction hypothesis). One can now show easily that the mapping $H^p(\mathcal{G}) \to H^p(\mathcal{G}_s)$ is injective (and hence an isomorphism).

We now consider the diagram of long exact cohomology sequences,

$$\begin{array}{ccccccccc}
H^{p-1}(\mathcal{G}) & \longrightarrow & H^{p-1}(\mathcal{G}/\mathcal{F}) & \longrightarrow & H^p(\mathcal{F}) & \longrightarrow & H^p(\mathcal{G}) & \longrightarrow & H^p(\mathcal{G}/\mathcal{F}) \longrightarrow \ldots \\
\downarrow s & & \downarrow s & & \downarrow & & \downarrow s & & \downarrow \\
\ldots \longrightarrow H^{p-1}(\mathcal{G}_s) & \longrightarrow & H^{p-1}((\mathcal{G}/\mathcal{F})_s) & \longrightarrow & H^p(\mathcal{F}_s) & \longrightarrow & H^p(\mathcal{G}_s) & \longrightarrow & H^p(\mathcal{G}/\mathcal{F})_s) \longrightarrow .
\end{array}$$

It follows first that $H^p(\mathcal{F}) \to H^p(\mathcal{F}_s)$ is injective. Since $\mathcal{F}$ was arbitrary, this injectivity also holds for $\mathcal{G}/\mathcal{F}$. Then surjectivity also follows from the diagram, giving us the desired bijectivity

$$H^p(\mathcal{F}) \xrightarrow{\sim} H^p(\mathcal{F}_s).$$

Using the reduction sketched at the beginning, we now know the truth of the base change theorem for those structure mappings $f: X \to S$ whose geometric fibers are at most one-dimensional. The base change theorem follows from this for those mappings f that can be written as products of proper mappings with at most one-dimensional fibers. In particular, the base change theorem is true for a cartesian product of projective lines over S,

$$X = \mathbb{P}^1_S \times \ldots \times \mathbb{P}^1_S.$$

Next we prove the base change theorem for projective space,

$$f: \mathbb{P}^n_S \to S.$$

It is well known that the n-dimensional projective space can be covered by a product of n projective lines,

$$p: \mathbb{P}^1_S \times \ldots \times \mathbb{P}^1_S \to \mathbb{P}^n_S \quad (= \mathbb{P}^1_S \times \ldots \times \mathbb{P}^1_S/\text{symmetric group}).$$

The validity of the base change theorem for sheaves of the form $p_* \mathcal{F}$ follows from (3.5). The mapping

$$\mathcal{G} \to p_* p^* \mathcal{G}$$

is injective for every sheaf $\mathscr{G}$ on $\mathbb{P}^n_S$. Hence $\mathscr{G}$ is quasi-isomorphic to a bounded-below complex of sheaves $\mathscr{F}^\bullet$ where the base change theorem is true for each $\mathscr{F}^n$. The base change theorem follows for $\mathscr{F}^\bullet$ and then for $\mathscr{G}$.

Since every projective morphism $f: X \to S$ can be embedded in a projective space $\mathbb{P}^n_S$ (if S is affine),

$$X \hookrightarrow \mathbb{P}^n_S \to S,$$

we can deduce the base change theorem for arbitrary projective morphisms. □

§ 7. The Smooth Base Change Theorem

Homotopy invariance is an important property of the singular cohomology groups. For the Čech cohomology groups, one can describe it as follows:

Let X be a paracompact topological space, J a closed finite interval in $\mathbb{R}$, J^n the n-fold cartesian product,

$$q: X \times J^n \to X$$

the projection onto X, and $\mathscr{F}$ an arbitrary sheaf of abelian groups on X. Then

$$H^\nu(X \times J^n, q^*(\mathscr{F})) = H^\nu(X, \mathscr{F}), \qquad \nu \geqq 0.$$

Equivalently,

$$R^\nu q_* \, q^*(\mathscr{F}) = \begin{cases} 0 & \nu > 0 \\ \mathscr{F} & \nu = 0. \end{cases}$$

Another way of saying this is:

The projection q *is acyclic.*

This property yields a base change theorem:

Let $f: X \to S$ be a continous mapping of paracompact spaces, $\mathscr{F}$ a sheaf of abelian groups on X and $g: T \to S$ a *"smooth"* mapping of topological spaces, that means:

For every point $t \in T$ there is a neighbourhood V of $g(t)$, a neighbourhood $U \subseteq g^{-1}(V)$ of t, a suitable n and a homeomorphism

$$\begin{array}{ccc} U & \xrightarrow{\cong} & V \times J^n \\ & \searrow \quad \swarrow & \\ & S. & \end{array}$$

Consider the cartesian diagram

$$\begin{array}{ccc} X \times_S T & \xrightarrow{f'} & T \\ \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & S. \end{array}$$

Then

$$g^*(R^\nu f_*(\mathscr{F})) = R^\nu f'_*(g'^*(\mathscr{F})).$$

In etale cohomology theory, the substitute for homotopy invariance is the *smooth base change theorem* 7.3 or the equivalent *acyclicity theorem* 7.4.

Recall the concept of a smooth mapping (of noetherian schemes). Let $A \to B$ be a finitely generated homomorphism of rings. We call B a *smooth A-algebra* if B is flat over A and, for every extension field K of a residue field of A, the ring $B \otimes_A K$ is regular. A *local homomorphism* $A \to B$ is called *smooth* if B is the localization of a smooth finitely generated A-algebra.

For strictly Henselian rings, we modify this concept: If $A \to B$ is a local homomorphism of strictly Henselian rings, then B is called a *smooth A-algebra* if B is the strict Henselization of a smooth local A-algebra. In this case B is a direct limit of smooth finitely generated A-algebras.

7.1 Lemma. *Let $A \to B$ be a local homomorphism of strictly Henselian rings. Let B be the strict Henselization of a localization of a finitely generated A-algebra. Then it is smooth if and only if B is a strict Henselization of*

$$A[T_1, \ldots, T_n]_{\mathfrak{p}}.$$

Here $\mathfrak{p}$ should be a prime ideal in the polynomial ring lying over $\mathfrak{m} = \mathfrak{m}(A)$.

If the residue fields of B and A coincide. we can take

$$\mathfrak{p} = (\mathfrak{m}, T_1, \ldots, T_n).$$

Notation. Let $A\{T_1, \ldots, T_n\}$ be the Henselization of

$$A[T_1, \ldots, T_n]_{\mathfrak{p}}, \qquad \mathfrak{p} = (\mathfrak{m}, T_1, \ldots, T_n).$$

If the residue field extension is just algebraic, and hence purely inseparable and of finite degree, we get from 7.1:

Corollary to 7.1. *If $A \to B$ is a smooth homomorphism of strictly Henselian rings and the residue field extension is algebraic, then there exists a local A-algebra A_1 with the following properties:*

a) *A_1 is a finitely generated free A-module.*
b) *The residue field extension for $A \to A_1$ is purely inseparable.*
c) $B \otimes_A A_1 = A_1\{T_1, \ldots, T_n\}$.

Proof. We first consider the case of a smooth *locally finitely generated* homomorphism $A \to B$ of local rings, not necessarily Henselian. For the modules of differentials $M(B/A)$ of B over A and $M(\bar{B}/k)$ of $\bar{B} = B/\mathfrak{m}(A)B$ over $k = A/\mathfrak{m}(A)$ we have

$$M(B/A)/\mathfrak{m}(A)\, M(B/A) = M(\bar{B}/k).$$

Since $\bar{B}$ is smooth over the field k, $M(\bar{B}/k)$ is a finitely generated free $\bar{B}$-module of rank r (Jacobian criterion). Here r is the sum of the Krull dimension of $\bar{B}$ and of the transcendence degree of $B/\mathfrak{m}(B)$ over $A/\mathfrak{m}(A)$. We choose elements $x_1, \ldots, x_r \in B$ for which the differentials $dx_1, \ldots, dx_r$ modulo $\mathfrak{m} = \mathfrak{m}(A)$ form a

basis of $M(\bar{B}/k)$, and we consider the evaluation homomorphism

$$A[T_1, \ldots, T_r] \to B$$
$$T_\nu \mapsto x_\nu .$$

Let $\mathfrak{p} \subset A[T_1, \ldots, T_r]$ be the inverse image of the maximal ideal of B. Then the module of differentials of $\bar{B}$ over $k[T_1, \ldots, T_r]_{\mathfrak{p}}$ vanishes. This ring extension is thus unramified. One shows that the Krull dimensions of the two rings are the same, and then by criterion 1.5 it follows that $\bar{B}$ is etale over $k[T_1, \ldots, T_r]_{\mathfrak{p}}$. It even follows then from this that B is etale over $A[T_1, \ldots, T_r]_{\mathfrak{p}}$, since both rings are flat over A.

If the two residue fields coincide, we can perform an appropriate modification $x_\nu \to x_\nu - a_\nu$ $(a_\nu \in A)$ to guarantee that

$$x_\nu \in \mathfrak{m}(B).$$

The assertions of the lemma follow by "strict Henselization" from what we have proved. □

Let $f: X \to Y$ be a finitely generated morphism of schemes. One shows: *The set of points $x \in X$ for which $\mathcal{O}_{Y, f(x)} \to \mathcal{O}_{X,x}$ is smooth – the set of smooth points of X over Y – is open.*

7.2 Definition. A morphism $f: X \to Y$ of schemes is called smooth if it is finitely generated and all points of X are smooth over Y.

This condition is equivalent to saying that f is a finitely generated flat morphism and that for *all* geometric points x: $\operatorname{Spec} k \to Y$ the geometric fiber $X \times_Y \operatorname{Spec} k$ is nonsingular; algebraically closed fields k suffice.

Etale mappings are smooth; conversely, a smooth mapping is etale if and only if its geometric fibers are 0-dimensional.

As in 7.1, we can show:

Remark on 7.2. Let $f: X \to Y$ be a smooth morphism and x a point of X. There is a Zariski open neighborhood U of x such that $U \to Y$ is the product of smooth morphisms

$$f_\nu: X_\nu \to X_{\nu+1}, \qquad 1 \leqq \nu < d,$$
$$f \mid U = f_d \circ \ldots \circ f_1,$$

with $\dim X_\nu / X_{\nu+1} = \max_{y \in X_{\nu+1}} \dim f_\nu^{-1}(y) \leqq 1$.

Permanence Properties of Smooth Morphisms

1) A composite of smooth morphisms is smooth.

2) A smooth morphism $X \to Y$ remains smooth after an arbitrary base extension $Z \to Y$; that is, $X \times_Y Z \to Z$ is smooth.

3) Let $X \to Y$ be an arbitrary morphism and $Z \to Y$ a faithfully flat morphism. If $X \times_Y Z \to Z$ is smooth, then $X \to Y$ is also smooth.

The goal of this section is to prove the following two theorems.

7.3 Theorem (Base change theorem for smooth base change). *Let*

$$\begin{array}{ccc} X_T & \xrightarrow{f'} & T \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & S \end{array}$$

be a cartesian diagram of (noetherian) schemes. The base change homomorphism

$$g^* R^i f_* \mathscr{F} \to R^i f'_* (g'^* \mathscr{F})$$

is an isomorphism when the following two assumptions are satisfied:

1) *g is smooth.*

2) *$\mathscr{F}$ is a torsion sheaf, and the orders of the sections of $\mathscr{F}$ are relatively prime to the residue characteristics of the scheme S.*

It follows immediately from this that if $\mathscr{F}^\cdot$ is a complex of sheaves, (cohomologically) bounded below, with cohomology sheaves satisfying assumption 2), then: The base change mapping

$$g^* \mathrm{R} f_* \mathscr{F}^\cdot \to \mathrm{R} f'_* (g'^* \mathscr{F}^\cdot)$$

is a quasi-isomorphism.

Using the compatibility of cohomology and projective limits of schemes, we get the validity of the base change theorem under the following assumptions.

In place of 1). T is the projective limit of a filtered system (T_α) of affines schemes smooth over S. In addition we must then require: X/S is finitely generated (in particular then X_T is noetherian).

7.3′ Remark. For our purposes it is sufficient to prove the smooth base change theorem only for a *quasi-projective* structure map $f: X \to S$. By means of the spectral sequence for an affine covering of X, the general case can be reduced to the case of an affine structure map, and by the limit theorems it can be reduced to the case of a finitely generated affine structure map.

7.4 Theorem (Acyclicity theorem). *Let $A \to B$ be a smooth homomorphism of strictly Henselian rings. Then the morphism g: Spec $B \to$ Spec A is acyclic; that is, for all torsion sheaves $\mathscr{F}$ on* Spec A *with torsion relatively prime to the residue characteristic of A,*

a) *$\mathscr{F} \to g_* g^* \mathscr{F}$ is an isomorphism, and*

b) *$R^i g_* (g^* \mathscr{F}) = 0$ for $i > 0$.*

Remark. One can then easily deduce: If $\phi: U \to$ Spec A is etale, and $V \to$ Spec B is its inverse image, then the acyclicity condition is also satisfied for the induced mapping $V \to U$ and every torsion-sheaf $\mathscr{G}$ on U that satisfy the stated primality

condition. This is clear for sheaves of the form $\mathscr{G} = \mathscr{F} \mid U$, and an arbitrary sheaf $\mathscr{G}$ on U is in a natural way a direct summand of

$$\phi^*(\phi_*(\mathscr{G})) = \phi_*(\mathscr{G}) \mid U \qquad (\phi \text{ etale!}).$$

In the derived category, a) and b) can be formulated as follows:

For every complex $\mathscr{F}^\cdot$ of sheaves that is cohomologically bounded below, and whose cohomology sheaves satisfy the stated relative primality condition, $\mathscr{F}^\cdot \to \mathrm{R}g_* g^* \mathscr{F}^\cdot$ is a quasi-isomorphism.

All sheaves appearing in what follows are supposed to satisfy the restriction on the torsion that is required in the smooth base change theorem and the acyclicity theorem, even when this is not explicitly mentioned.

We first show the equivalence of the acyclicity theorem and the smooth base change theorem (at least for a quasiprojective structure map f).

7.5 Lemma. *Suppose the smooth base change theorem has already been proved for the case where the structure map $f: X \to S$ is an open embedding. Then it also holds for arbitrary quasiprojective morphisms $f: X \to S$, and hence also when X/S is the projective limit of a projective system of quasiprojective spaces over S (in the sense of § 4; see the explanation preceding 4.16).*

For: A quasi-projective mapping f is a product of an open embedding j and a proper mapping $\bar{f}$,

$$f = \bar{f} \circ j.$$

But for a proper structure map $\bar{f}$ the base change theorem was already proved in the preceding section. Since

$$\mathrm{R}f_* = \mathrm{R}\bar{f}_* \circ \mathrm{R}j_*,$$

the base change property follows for f if it holds for $\bar{f}$ and for j. □

7.6 Lemma. 1) *Suppose the smooth base change theorem has been proved for all etale structure maps f. Then the acyclicity theorem follows.*

2) *Suppose the acyclicity theorem has been proved for all smooth homomorphisms*

$$\phi: A \to B$$

of strictly Henselian local rings with algebraic residue field extensions. Then the smooth base change theorem follows for all etale structure maps

$$f: X \to S.$$

Proof. (1) Let $\phi: A \to B$ be a smooth homomorphism of strictly Henselian local rings,

$$f: X \to \mathrm{Spec}\,(A) = S$$

an etale mapping, and $\mathcal{F}$ a sheaf on S. We consider the cartesian diagram

$$\begin{array}{ccc} X_T & \xrightarrow{f'} & \operatorname{Spec}(B)=T \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g=\operatorname{Spec}(\phi)} \\ X & \xrightarrow{f} & \operatorname{Spec}(A)=S. \end{array}$$

Since B is a direct limit of smooth (finitely generated) A-algebras, the sheaf $\mathcal{G}=f^*(\mathcal{F})=\mathcal{F}\mid X$ satisfies

$$\mathrm{R}f'_*\, g'^*(\mathcal{G})=g^*\,\mathrm{R}f_*(\mathcal{G}) \qquad (g'^*(\mathcal{G})=g^*(\mathcal{F})\mid X_T)$$

by the smooth base change theorem. Since A and B are strictly Henselian rings, we have

$$H^\nu(T, g^*(\mathcal{K}^\cdot))=H^\nu(S, \mathcal{K}^\cdot)$$

for every complex $\mathcal{K}^\cdot$ of sheaves on S cohomologically bounded below. We get

$$\begin{aligned} H^\nu(X_T, g^*(\mathcal{F}))&=H^\nu(X_T, g'^*(\mathcal{G}))=H^\nu(T, \mathrm{R}f'_*\, g'^*(\mathcal{G})) \\ &=H^\nu(T, g^*\,\mathrm{R}f_*(\mathcal{G}))=H^\nu(S, \mathrm{R}f_*(\mathcal{G}))=H^\nu(X, \mathcal{G})=H^\nu(X, \mathcal{F}). \end{aligned}$$

The sheaf generated by the presheaf $X\mapsto H^\nu(X_T, g^*(\mathcal{F}))$ is $\mathrm{R}^\nu g_*\, g^*(\mathcal{F})$; the sheaf generated by the presheaf $X\mapsto H^\nu(X, \mathcal{F})$ is $\mathcal{F}$ for $\nu=0$ and vanishes for $\nu>0$.

(2) We consider first the same cartesian diagram as in (1),

$$\begin{array}{ccc} X_T & \xrightarrow{f'} & \operatorname{Spec}(B)=T \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g=\operatorname{Spec}(\phi)} \\ X & \xrightarrow{f} & \operatorname{Spec}(A)=S, \end{array}$$

together with a sheaf $\mathcal{G}$ on X. We assume first that $\mathcal{G}$ is the inverse image of a sheaf $\mathcal{F}$ on S, $\mathcal{G}=f^*(\mathcal{F})$. We further consider the natural geometric point in the residue field k of B,

$$\begin{array}{ccc} t\colon \operatorname{Spec}(k) & \longrightarrow & T \\ & {\scriptstyle s}\searrow & \downarrow \\ & & S. \end{array}$$

The acyclicity theorem applied to the sheaf $\mathcal{F}$ yields

$$H^\nu(X, \mathcal{G})=H^\nu(X_T, g'^*(\mathcal{G})).$$

On the other hand, we have

$$\begin{aligned} g^*(R^\nu f_*\,\mathcal{G})_t&=(R^\nu f_*\,\mathcal{G})_s=H^\nu(X, \mathcal{G}), \\ R^\nu f'_*\, g'^*(\mathcal{G})_t&=H^\nu(X_T, g'^*(\mathcal{G})). \end{aligned}$$

The stalks $g^*(R^\nu f_*\,\mathcal{G})_t$ and $R^\nu f'_*\, g'^*(\mathcal{G})_t$ are thus canonically isomorphic. If $\mathcal{G}$ is an arbitrary sheaf on X, then it is easy to see that $\mathcal{G}$ is a direct summand

of $f^*(f_*(\mathscr{G}))$. The isomorphism thus also extends to this case. We now consider the base change situation

$$\begin{array}{ccc} X_T & \xrightarrow{f'} & T \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & S \end{array}$$

with an etale mapping f and a smooth mapping g. For a sheaf $\mathscr{G}$ on X we have the base change mapping

$$g^* R^\nu f_*(\mathscr{G}) \to R^\nu f'_* g'^*(\mathscr{G}).$$

From the preceding we get:

This base change mapping is an isomorphism on the stalks at all geometric points of T whose underlying ordinary points are closed in their fibers over S. But then the base change mapping is an isomorphism! □

We need an additional theorem essentially equivalent to the smooth base change theorem and the acyclicity theorem.

7.7 Key Lemma. *Let A be an excellent strictly Henselian local ring, $\mathfrak{p}$ a prime ideal, K the separable closure of the residue class field $\kappa(\mathfrak{p}) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$, M a finite abelian group of order relatively prime to the characteristic of the residue field of A, and*

$$A \to B$$

a local smooth homomorphism of Henselian rings. Suppose the residue field extension is algebraic and

$$\dim B/\mathfrak{m}(A)\,B \leqq 1.$$

For an intermediate field $\kappa(\mathfrak{p}) \subset L \subset K$ let

$$X_L = \operatorname{Spec}(B \otimes_A L).$$

Then

$$\varinjlim_{\substack{L \\ \kappa(\mathfrak{p}) \subset L \subset K \\ L/\kappa(\mathfrak{p}) \text{ finite}}} H^\nu(X_L, M_{X_L}) = \begin{cases} 0 & \nu > 0 \\ M & \nu = 0. \end{cases}$$

Remark. (1) $\varinjlim H^\nu(X_L, M_{X_L})$ occurs in place of the cohomology group $H^\nu(X_K, M_{X_K})$ on the possibly non-noetherian scheme X_K.

(2) The statement of the key lemma can also be interpreted in a different way. For an arbitrary sheaf $\mathscr{F}$ on $\operatorname{Spec}(\kappa(\mathfrak{p}))$ (satisfying the restriction on torsion), the induced mapping

$$h_\mathfrak{p} : X_{\kappa(\mathfrak{p})} \to \operatorname{Spec}(\kappa(\mathfrak{p}))$$

satisfies

$$R^\nu h_\mathfrak{p}^*(h_\mathfrak{p}^*(\mathscr{F})) = \begin{cases} 0 & \nu > 0 \\ \mathscr{F} & \nu = 0. \end{cases}$$

This is clear for a constant constructible sheaf. For an arbitrary constructible sheaf $\mathscr{F}$ there is a finite separable field extension L of $\kappa(\mathfrak{p})$ such that the inverse image of $\mathscr{F}$ becomes constant on $\mathrm{Spec}(L)$. The general case follows using the limit theorems of § 4.

(3) The acyclicity theorem implies the corresponding result for an arbitrary smooth strictly Henselian algebra B over an arbitrary strictly Henselian local ring A!

We postpone the proof of this key lemma to the end of this section.

One more remark on the proof of the base change theorem: It is enough to prove the smooth base change theorem under some special restrictions, as follows. We consider again the base change situation

$$\begin{array}{ccc} X_T & \xrightarrow{f'} & T \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & S. \end{array}$$

Let $\mathscr{F}$ be a sheaf on X.

(1) *By 7.5, it is sufficient to assume that f is etale.*

(2) *It is sufficient to assume that S and T are affine and that* $\dim T/S \leqq 1$.

For the base change property is of local nature with respect to S and T. Locally, g is a product of smooth mappings with fiber dimension less than or equal to 1.

(3) *We can assume that the base scheme S is excellent and has finite dimension.*

For from the compatibility of cohomology with direct limits of sheaves (4.16) and projective limits of schemes (4.19) it follows that the base change theorem only has to be proved for base schemes S that are finitely generated over $\mathbb{Z}$.

We now intend to prove the smooth base change theorem, under the special restrictions (1)–(3), using the key lemma together with induction on the dimension of the base scheme S. For this we apply the equivalence stated in Lemma 7.6 between the smooth base change theorem and the acyclicity theorem. It follows immediately from the proof of this result that the stated equivalence is also valid if we consider only base change situations satisfying (1)–(3) and the acyclicity theorem only for local smooth homomorphisms

$$A \to B$$

of excellent strictly Henselian rings with algebraic residue field extension and $\dim B/\mathfrak{m}(A)B \leqq 1$. In addition, we can assume that

$$\dim S \leqq r, \qquad \dim B \leqq r$$

for some fixed integer r. In view of these remarks, it is sufficient to prove the following induction step.

Let r be a fixed integer. Assume the smooth basis change theorem has been proved under the restrictions (1)–(3) for all base schemes S with $\dim S < r$. Sup-

pose

$$A \to B$$

is a smooth local homomorphism of strictly Henselian excellent rings with algebraic residue field extension. Suppose

$$\dim A \leqq r, \quad \dim B/\mathfrak{m}(A)\,B \leqq 1.$$

Then the acyclicity theorem holds for every bounded-below complex $\mathscr{F}^\cdot$ of sheaves on $U = \operatorname{Spec}(A)$ and the induced mapping

$$\bar{g}\colon V = \operatorname{Spec}(B) \to U = \operatorname{Spec}(A).$$

That is, the natural homomorphism

$$\mathscr{F}^\cdot \to \mathrm{R}\bar{g}_*\,\bar{g}^*(\mathscr{F}^\cdot)$$

is a quasi-isomorphism.

It is sufficient to assume that $\mathscr{F}^\cdot = \mathscr{F}$ is a single sheaf, that A is a *reduced ring*, and that

$$r \geqq 1, \quad \dim B/\mathfrak{m}(A)\,B = 1.$$

Let s be the closed point of U, S its complement in U, T the inverse image of S in V, and let $K_1, \ldots, K_m$ be the residue fields of the generic points of the irreducible components of U. We consider the diagram

$$\begin{array}{ccccc} X \times_S T = X_T & \xrightarrow{f'} & T & \xrightarrow{j'} & V = \operatorname{Spec}(B) \\ \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle \bar{g}} \\ \coprod_i \operatorname{Spec}(K_i) = X & \xrightarrow{f} & S & \xrightarrow{j} & U = \operatorname{Spec}(A) \end{array}$$

$$\bar{f} = j \circ f, \quad \bar{f}' = j' \circ f'.$$

a) *Let $\mathscr{R}^\cdot$ be a bounded-below complex of sheaves on U with cohomology sheaves $\mathscr{H}^\nu(\mathscr{R}^\cdot)$, whose stalks $\mathscr{H}^\nu(\mathscr{R}^\cdot)_\eta$ vanish at all "generic" geometric points of irreducible components of U.* The support of a constructible subsheaf of such a cohomology sheaf $\mathscr{H}^\nu(\mathscr{R}^\cdot)$ is contained in a closed subscheme $\bar{U}$ of U with $\dim \bar{U} < r$. By the induction hypothesis (and the equivalent theorem 7.6), the acyclicity theorem is true for this subsheaf, and hence also for the sheaves $\mathscr{H}^n(\mathscr{R}^\cdot)$ (limit theorems from § 4), and hence also for the complex $\mathscr{R}^\cdot$.

b) *Let $\mathscr{G}$ be a sheaf on X and $\mathscr{K}^\cdot = \mathrm{R}\bar{f}_*(\mathscr{G})$.* The scheme T is a projective limit (in the sense of § 4) of smooth affine schemes T_α over S with $\dim T_\alpha/S \leqq 1$. The scheme X/S is a projective limit of open affine subschemes of S, since S by assumption is reduced. Since $\dim S < r$, the smooth base change theorem holds by induction for the diagram

$$\begin{array}{ccc} X_T & \longrightarrow & T \\ \downarrow & & \downarrow \\ X & \longrightarrow & S. \end{array}$$

Here $T \to S$ is the base change mapping.

Hence we have

$$(\alpha) \qquad g^*(\mathsf{R}f_*\mathscr{G}) \cong \mathsf{R}f'_* g'^*(\mathscr{G}) \quad \text{(quasi-isomorphic)}.$$

The key lemma asserts that

$$(\beta) \qquad \mathscr{G} \cong \mathsf{R}g'_* g'^*(\mathscr{G}).$$

It follows from (α) and (β) that

$$\mathsf{R}f_*(\mathscr{G}) \cong \mathsf{R}g_* g^*(\mathsf{R}f_*\mathscr{G}),$$

or equivalently:

The homomorphism

$$\mathscr{K}^\cdot | S \to \mathsf{R}\bar{g}_* \bar{g}^*(\mathscr{K}^\cdot) | S$$

is a quasi-isomorphism.

But since A and B are strictly Henselian, we have

$$\mathscr{K}^\cdot_s \cong \mathsf{R}\bar{g}_* \bar{g}^*(\mathscr{K}^\cdot)_s$$

for the closed point s of U. Thus we have proved the acyclicity theorem for the complex $\mathscr{K}^\cdot$; that is, the homomorphism

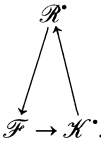
$$\mathscr{K}^\cdot \to \mathsf{R}\bar{g}_* \bar{g}^*(\mathscr{K}^\cdot)$$

is a quasi-isomorphism.

c) *The general case.* Let $\mathscr{F}$ be a sheaf on U, and $\mathscr{G}$ its inverse image on X. The mapping

$$\mathscr{F} \to \mathsf{R}\bar{f}_* \bar{f}^*(\mathscr{F}) = \mathscr{K}^\cdot$$

lies in an admissible triangle (A II)



The homomorphism $\mathscr{F} \to \mathscr{K}^\cdot$ induces a quasi-isomorphism

$$\mathscr{F}_\eta \to \mathscr{K}^\cdot_\eta$$

for all "generic" geometric points η of irreducible components of U. Hence $\mathscr{R}^\cdot$ satisfies the assumptions in a). The acyclicity theorem holds for the complex $\mathscr{R}^\cdot$ by a), and for the complex $\mathscr{K}^\cdot = \mathsf{R}\bar{f}_*(\mathscr{G})$ by b), and hence it holds also for the third "complex" $\mathscr{F}$ of the admissible triangle.

We come now to the

Proof of the key lemma 7.7. Reduction to the case $B = A\{t\}$.

We may assume

$$\dim B/\mathfrak{m}(A)B = 1.$$

Then

$$B \otimes_A A_1 = A_1\{t\},$$

where A_1 is a suitable local A-algebra that is a finitely generated free A-module. (See the corollary to 7.1.) Over the prime ideal $\mathfrak{p}$ of A there lies a prime ideal $\mathfrak{p}_1$ of A_1 with residue field $\kappa_1 \supseteq \kappa$, because A_1/A is faithfully flat.

The separable closure K_1 of κ_1 contains the separable closure K of κ. Obviously

$$\varinjlim_{\substack{L \\ \kappa \subset L \subset K_1 \\ L/\kappa \text{ finite}}} H^\nu(X_L, M_{X_L}) = \varinjlim_{\substack{L \\ \kappa \subset L \subset K_1 \\ L/\kappa \text{ finite} \\ \text{(not necessarily separable)}}} H^\nu(X_L, M_{X_L})$$

$$= \varinjlim_{\substack{\kappa_1 \subset L \subset K_1 \\ L/\kappa_1 \text{ finite}}} H^\nu(X_L, M_{X_L}) = \varinjlim_{\substack{\kappa_1 \subset L \subset K_1 \\ L/\kappa_1 \text{ finite}}} H^\nu(X_1, M_{X_{1_L}}),$$

$$X_{1_L} = \operatorname{Spec}(A_1\{t\}) \times_{\operatorname{Spec}(A_1)} \operatorname{Spec} L.$$

Thus we can assume $B = A\{t\}$. We can replace A by the integral closure of $A/\mathfrak{p}$ in its quotient field $\kappa = \kappa(\mathfrak{p})$. Without loss of generality we can thus assume that A is a normal integral domain and K is the separable closure of the quotient field κ of A. Let $X = \operatorname{Spec}(A\{t\})$. For an intermediate field $\kappa \subset L \subset K$, finite over κ, let $X_L = \operatorname{Spec}(A\{t\} \otimes_A L)$. We must show that

$$\varinjlim H^\nu(X_L, M_{X_L}) = \begin{cases} 0, & \nu > 0 \\ M, & \nu = 0. \end{cases}$$

Some Preliminary Remarks

Let A' be a local A-algebra that is finitely generated as an A-module. Then

$$A\{t\} \otimes_A A' = A\{t\}.$$

The t-adic completion of $A'\{t\}$ is the power series ring $A'[[t]]$. Hence,

$$A' \text{ integral domain} \Rightarrow A'\{t\} \text{ integral domain}$$
$$A' \text{ normal} \Rightarrow A'\{t\} \text{ normal}.$$

In particular, if $\mathfrak{p}$ is a prime ideal of A, then $\mathfrak{p}A\{t\}$ is a prime ideal of $A\{t\}$; for

$$A\{t\}/\mathfrak{p}A\{t\} = (A/\mathfrak{p})\{t\}.$$

More generally, if $\mathfrak{q}$ is a primary ideal for the prime ideal $\mathfrak{p}$ of A, then (for flatness reasons) $\mathfrak{q}A\{t\}$ is a primary ideal for the prime ideal $\mathfrak{p}A\{t\}$. In the special case where A (and hence also $A\{t\}$) is normal, and $\mathfrak{p}$ is a prime ideal of height 1 in A (so that the localization $A_\mathfrak{p}$ is then a discrete valuation ring), we actually have:

Every primary ideal for $\mathfrak{p}A\{t\}$ *is of the form* $\mathfrak{q}A\{t\}$, *where* $\mathfrak{q} \subset A$ *is a primary ideal for* $\mathfrak{p}$.

An obvious generalization of this (A still normal!) is:

Let $\mathfrak{a}$ *be a divisorial ideal of* A *(that is, all associated prime ideals have height 1), and let* $\mathfrak{b}$ *be a divisorial ideal of* $A\{t\}$ *that divides* $\mathfrak{a}A\{t\}$ (*i.e.*, $\mathfrak{b} \supset \mathfrak{a}A\{t\}$). *Then* $\mathfrak{b}$ *is generated by a divisorial ideal of* A *that divides* $\mathfrak{a}$.

Furthermore, $A_{\mathfrak{p}}$ regular $\Rightarrow A\{t\}_{\mathfrak{p}} = A\{t\} \otimes_A A_{\mathfrak{p}}$ regular.

We denote by A_L the integral closure of A in the finite extension field L of the quotient field κ of A. By assumption (A excellent), A_L is a finitely generated A-module.

The case $\nu = 0$. $A\{t\} \otimes_A L = A_L\{t\} \otimes_{A_L} L$ is an integral domain, and hence the scheme $X_L = \mathrm{Spec}\,(A\{t\} \otimes_A L)$ is connected. It follows from this that

$$H^0(X_L, M_{X_L}) = M.$$

The case $\nu \geqq 2$. $A\{t\}$ is a (filtered) direct limit of finitely generated smooth A-algebras B_α of relative dimension 1 over A,

$$A\{t\} = \varinjlim B_\alpha.$$

Therefore $\mathrm{Spec}\,(B_\alpha \otimes_A K)$ is a smooth affine curve over the separably closed field K. By 5.3, then,

$$H^\nu(\mathrm{Spec}\,(B_\alpha \otimes_A K), M_{\mathrm{Spec}(B_\alpha \otimes_A K)}) = 0, \qquad \nu \geqq 2$$

$$\varinjlim_L H^\nu(X_L, M_{X_L}) = \varinjlim_{L,\alpha} H^\nu(\mathrm{Spec}(B_\alpha \otimes_A L, M_{\mathrm{Spec}(B_\alpha \otimes_A L)})$$

$$= \varinjlim_\alpha H^\nu(\mathrm{Spec}(B_a \otimes_A K), M_{\mathrm{Spec}(B_\alpha \otimes_A K)}) = 0, \qquad \nu \geqq 2.$$

The case $\nu = 1$. Without loss of generality let $M = \mathbb{Z}/n\mathbb{Z}$ ($n \notin \mathfrak{m}(A)$). Since A contains the n-th roots of unity, $M_X \cong \mu_{nX}$. Because of the long exact sequence coming from the Kummer sequence, it is enough to show

a) $\varinjlim_L (A_L\{t\} \cdot L)^*/(A_L\{t\} \cdot L)^{*n} = 0$,

b) $\varinjlim_L {}_n\mathrm{Pic}(X_L) = 0$, ${}_n\mathrm{Pic}\, X_L = \mathrm{Ker}(\mathrm{Pic}(X_L) \xrightarrow{n} \mathrm{Pic}(X_L))$.

Here A_L is the integral closure of A in the finite separable field extension L of κ.

For a). We show: Every unit of $A\{t\} \cdot \kappa$ is the product of a unit of $A\{t\}$ and an element of the quotient field κ of A. Similarly, we will have

$$(A_L\{t\} \cdot L)^* = A_L\{t\}^* \cdot L^*,$$

and therefore

$$(A_L\{t\} \cdot L)^*/(A_L\{t\} \cdot L)^{*n} = L^*/L^{*n}.$$

Let $f(t) \in A\{t\}$ be a unit in $A\{t\} \cdot \kappa$. Then there is a nonzero element a in A with

$$f(t) \mid a \quad \text{in } A\{t\}.$$

Consequently there is a divisorial ideal $\mathfrak{a}$ in A, dividing $A \cdot a$, with

$$A\{t\} \cdot f(t) = \mathfrak{a} \cdot A\{t\}.$$

This must then be a principal ideal in A,

$$\mathfrak{a} = A \cdot \alpha.$$

It follows that

$$f(t)=\varepsilon\cdot\alpha,$$

and here ε is a unit in $A\{t\}$.

For b). We show that every line bundle $\mathscr{G}_0$ on $X_\kappa=\mathrm{Spec}(A\{t\}\cdot\kappa)$ having

$$\mathscr{G}_0^n=\mathscr{O}_{X_\kappa}$$

is itself trivial. The same then holds also for X_L in place of X_K. We denote by V the regular locus of $Y=\mathrm{Spec}(A)$. It is open, since A is an excellent local ring. The complement of V in Y has codimension at least 2, since A is normal. The inverse image $U=q^{-1}(V)$ of V in X under the natural mapping

$$q\colon \mathrm{Spec}\,A\{t\}=X\to Y=\mathrm{Spec}\,A$$

is the regular locus of X.

We set

$$X_\nu=\mathrm{Spec}(A\{t\}/t^{\nu+1}A\{t\})$$
$$U_\nu=U\cap X_\nu.$$

We have $X_0=Y$, $V=U_0$.

Since A and $A\{t\}$ are normal rings, and since $A\{t\}/t^{\nu+1}A\{t\}$ is a free A-module, we have

$$H^0(U,\mathscr{O}_U)=A\{t\},$$
$$H^0(V,\mathscr{O}_V)=A,$$
$$H^0(U_\nu,\mathscr{O}_{U_\nu})=A\{t\}/t^{\nu+1}A\{t\}.$$

The bundle $\mathscr{G}_0$ can be extended to a line bundle $\mathscr{G}$ on U (U is regular). After modification by the inverse image of a line bundle on V, one can assume that $\mathscr{G}\mid U_0$ is trivial. We assert that $\mathscr{G}^n$ is trivial on U. Since $\mathscr{G}_0^n$ is trivial on X_κ, there is a section s is $\mathscr{G}^n(U)$ and a nonzero element a in A with

$$a\mathscr{G}^n\subset\mathscr{O}_U\,s\subset\mathscr{G}^n.$$

The annihilator $\mathfrak{a}$ of $\mathscr{G}^n/\mathscr{O}_U\,s$ in $A\{t\}$ is then a divisorial ideal, and

$$\mathfrak{a}\cdot\mathscr{G}^n=\mathscr{O}_U\cdot s\qquad (U\text{ is quasi-affine!}).$$

The ideal $\mathfrak{a}$ is a divisor of the divisorial ideal $A\{t\}\cdot a$. Hence there is a divisorial ideal $\mathfrak{b}$ in A, dividing $A\cdot a$, with

$$\mathfrak{b}\cdot A\{t\}=\mathfrak{a}.$$

The ideal $\mathfrak{b}$ defines a line bundle $\mathscr{L}=\mathfrak{b}\mathscr{O}_V$ on V,

$$\mathscr{G}^n\cdot q^*(\mathscr{L})\cong\mathscr{O}_U.$$

As $\mathscr{G}^n\mid U_0$ is trivial, so is $\mathscr{L}=q^*(\mathscr{L})\mid U_0$. We thus know that the bundles

$$\mathscr{G}\mid U_0\quad\text{and}\quad\mathscr{G}^n\mid U_\nu$$

are trivial. Because of $H^1(U_\nu,\mu_{nU_\nu})=H^1(U_0,\mu_{nU_0})={}_n\mathrm{Pic}(U_0)$ (3.13) it follows that

$$\mathscr{G}\mid U_\nu=\mathscr{O}_{U_\nu}$$

for all ν.

We choose a nonzero section s of $\mathscr{G}$ over the quasi-affine scheme U and a section e of $\mathscr{G}^n$ with

$$\mathscr{O}_U \cdot e = \mathscr{G}^n.$$

Then

$$s^n = f \cdot e, \quad f \in A\{t\}.$$

Because $\mathscr{G} \mid U_v = \mathscr{O}_{U_v}$ and $H^0(U_v, \mathscr{O}_{U_v}) = A\{t\}/t^{v+1}A\{t\}$, there is for every v a unit ε_v in $A\{t\}$ such that $f \cdot \varepsilon_v$ is an n-th power modulo t^{v+1}. As this is true for all v, there is a power series $g(t)$ in $A[[t]] \supseteq A\{t\}$ with

$$f = g^n$$

and hence

$$s^n = g^n e.$$

Let $\bar{U}$ be the inverse image of U in $\mathrm{Spec}(A[[t]]) = \bar{X}$, and let $\bar{\mathscr{G}}$ be the inverse image of $\mathscr{G}$ on $\bar{U}$. Then s is divisible by g in $H^0(\bar{U}, \bar{\mathscr{G}})$, and

$$\bar{\mathscr{G}} = \mathscr{O}_{\bar{U}}(s/g)$$
$$H^0(\bar{U}, \bar{\mathscr{G}}) = A[[t]] \cdot (s/g).$$

For reasons of flatness, we have

$$H^0(\bar{U}, \bar{\mathscr{G}}) = H^0(U, \mathscr{G}) \otimes_{A\{t\}} A[[t]].$$

It follows that $\mathscr{G}$ itself is also a trivial bundle. □

§ 8. Cohomology with Compact Support; Applications of the Base Change Theorems

For what is to come, we need the cohomology groups with compact support, $H_c^v(X, \mathscr{F})$, for a torsion sheaf $\mathscr{F}$ on a finitely generated scheme X over an algebraically closed field k; more generally, we need direct images $R^v f_!(\mathscr{F})$ with compact support for a finitely generated morphism f of schemes.

The natural definition to start with for $\Gamma_c(X, \mathscr{F}) = H_c^0(X, \mathscr{F})$ is

$$\Gamma_c(X, \mathscr{F}) = \{s \in \mathscr{F}(X), \text{ support of } s \text{ proper over } k\}.$$

This definition proves to be "correct". For when $\mathbb{C}$ is the base field, this $\Gamma_c(X, \mathscr{F})$ coincides with $H_c^0(X_{\mathrm{an}}, \mathscr{F}_{\mathrm{an}})$, the group of sections with compact support of the associated topological sheaf on the associated complex analytic space (see §11).

Unfortunately, the derived functors $R^v \Gamma_c(X, \mathscr{F})$ of this functor are in general not the cohomology groups we want. For instance, let X be a smooth affine curve over an algebraically closed field. Then the derived functors of Γ_c satisfy

$$R^v \Gamma_c(X, \mathscr{F}) = \bigoplus_{a \in X} H^v_{\{a\}}(X, \mathscr{F}).$$

Here a runs through all closed points of X, and $H^v_{\{a\}}(X, \mathscr{F})$ is the v-th cohomology group with support at the point a. Let M be a finite abelian group of order prime to the characteristic of k, and $\mathscr{F} = M_X$ the associated constant

sheaf. Then

$$H^1_{\{a\}}(X, \mathscr{F}) \cong M.$$

Thus the group $R^1 \Gamma_c(X, \mathscr{F})$ is highly infinite!

On the other hand, in topology there is also another familiar description of the cohomology groups with compact support, and it leads to the correct definition in etale cohomology. Let X be an open subspace of a compact topological space $\bar{X}$, let $\mathscr{G}$ be a sheaf on X, and let $\bar{\mathscr{G}}$ be the sheaf extended by 0 to $\bar{X}$. Then

$$H^\nu_c(X, \mathscr{G}) \cong H^\nu(\bar{X}, \bar{\mathscr{G}}).$$

In etale cohomology theory one uses this formula to define cohomology with compact support, at least in the case where X can be embedded as an open subscheme of a proper scheme (over the base field). In the case $\nu = 0$ we get again the module of sections with compact support that we defined at the beginning. The cohomology groups that we get for $\nu > 0$ are no longer the derived functors of H^0_c in the sense of homological algebra. This occurs because, in contrast to the topological case, the sheaf $\bar{\mathscr{G}}$ extended by 0 from an injective sheaf $\mathscr{G}$ is in general not acyclic for the functor of sections.

In the present section we use this approach to introduce our "cohomology groups with compact support", and more generally "higher direct images with proper support", where for our purposes it would be sufficient to restrict to the quasi-projective case. The essential steps are the "extension of a sheaf by 0" and the fact that the definition of the cohomology with compact support does not depend on the choice of a compactification. The latter is a consequence of the proper base change theorem*). One main result of this section is the proof of the finiteness theorem 8.10 for higher direct images with proper support. This theorem is only seemingly more general than the finiteness theorem for the usual direct images under proper morphisms. For technical reasons in the proof (the localization principle), it is nonetheless useful to prove it in the seemingly more general form. For localization reasons, likewise, the cohomology groups with compact support play a central role in the Poincaré duality theorem.

Extension of a Sheaf by Zero

Suppose we are given an etale morphism

$$f\colon X \to Y.$$

We want to construct a functor $f_!$ from the category of sheaves of abelian groups on X to the category of sheaves of abelian groups on Y.

8.1 Definition. Let $\mathscr{F}$ be a sheaf of abelian groups on X. We denote by

$$f_!(\mathscr{F})$$

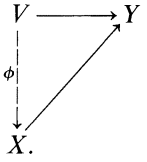
*) Actually we proved the proper base-change theorem only for projective mappings and under some restrictions about the torsion of the considered sheafs. This is sufficient for our applications.

the sheaf associated with the following presheaf on Y:

$$(V \to Y) \mapsto \bigoplus_{\phi \in \mathrm{Hom}_Y(V,X)} \mathscr{F}(\phi)$$

(with the natural restriction maps).

The direct sum is thus taken over all factorizations



These are etale, since by assumption V and X are etale over Y.

Remark. The construction given here generalizes the construction of the sheaf $(\tilde{X})^{\Lambda}$ in § 4 (4.9). For a ring Λ we have

$$f_!(\Lambda_X) = (\tilde{X})^{\Lambda}.$$

Here $\tilde{X}$ is the sheaf of sets represented by X, and $(\tilde{X})^{\Lambda}$ is the sheaf of Λ-modules that it generates.

8.2 Lemma. *Let $f: X \to Y$ be an etale mapping.*

(1) *The functor $f_!$ is left adjoint to the functor f^*; that is, for every sheaf $\mathscr{F}$ (of abelian groups, of course) on X and every sheaf $\mathscr{G}$ on Y there is a natural isomorphism of functors in $\mathscr{F}$ and $\mathscr{G}$,*

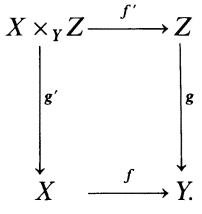
$$\mathrm{Hom}(\mathscr{F}, f^*(\mathscr{G})) \cong \mathrm{Hom}(f_!(\mathscr{F}), \mathscr{G}).$$

Caution. The functor f_* is right adjoint to f^*. The two functors f_* and $f_!$ do not in general coincide.

(2) *If $g: Y \to Z$ is another etale mapping, then*

$$(g \circ f)_! = g_! \circ f_!.$$

(3) *The functor $f_!$ is compatible with base change: Suppose we are given an arbitrary base change mapping $g: Z \to Y$. We consider the Cartesian diagram*



For every sheaf $\mathscr{F}$ of abelian groups on X there is a natural isomorphism

$$g^*(f_!\mathscr{F}) \cong f'_!(g'^*\mathscr{F}).$$

(4) *Let α: Spec$(\Omega) \to Y$ be a geometric point with values in a separably closed field Ω, and let*

$$\alpha_\nu\colon \operatorname{Spec}(\Omega)\to X \qquad \nu=1,\ldots,r$$

be the geometric points of X *over* α. *Then the stalk* $f_!(\mathscr{F})_\alpha$ *is canonically isomorphic to*

$$\prod_{\nu=1}^{r}\mathscr{F}_{\alpha_\nu}.$$

In particular, the functor $f_!$ *is exact and compatible with direct limits.*

(5) *Let* $\mathscr{F}$ *be constructible. Then* $f_!(\mathscr{F})$ *is also constructible.*

Remark. Since the functor $f_!$ is exact, its derived functor in the sense of Verdier can be defined in a particularly simple way on the whole derived category:

$$f_!(=\mathrm{R}f_!)\colon D(X)\to D(Y)$$
$$(f_!(\mathscr{K}^\cdot))^n=f_!(\mathscr{K}^n).$$

Proof of 8.2. The adjointness property (1): Since f is etale, the inverse image functor f^* is simply the natural restriction mapping. For an etale scheme $U\xrightarrow{\phi}X$ over X,

$$f^*(\mathscr{G})(\phi)=\mathscr{G}(f\circ\phi).$$

An element h in $\operatorname{Hom}(\mathscr{F},f^*(\mathscr{G}))$ is thus just a family

$$h_\phi\colon \mathscr{F}(\phi)\to\mathscr{G}(f\circ\phi), \qquad U\xrightarrow{\phi}X \text{ in } \operatorname{Et}(X)$$

with certain compatibility properties. Suppose such a system (h_ϕ) is given. For every etale mapping

$$q\colon V\to Y,$$

we get a homomorphism

$$\bigoplus_{\phi\in\operatorname{Hom}_Y(V,X)}\mathscr{F}(\phi)\to\mathscr{G}(q)$$
$$\Sigma s_\phi\mapsto\Sigma h_\phi(s_\phi).$$

The compatibility properties for the system (h_ϕ) are equivalent to saying that this system is a homomorphism of presheaves. It then induces a sheaf homomorphism

$$f_!(\mathscr{F})\to\mathscr{G}.$$

The adjointness property is an easy consequence of this consideration.

The second assertion follows immediately from the adjointness.

The Compatibility with Base Change (3)

Since the inverse image functor f^* for an etale morphism coincides with the restriction, the trivial base change theorem

$$f^*\,g_*(\mathscr{H})\cong g'_*f'^*(\mathscr{H})$$

holds in the situation of 6.2 (3). To prove 6.2 (3) now, we use the characterization of $f_!'$ by the adjointness property: we show for the (fixed) sheaf

$$\mathscr{H} = g^*(f_!(\mathscr{F}))$$

that the functors

$$\mathscr{G} \mapsto \mathrm{Hom}(\mathscr{H}, \mathscr{G})$$

and

$$\mathscr{G} \mapsto \mathrm{Hom}(g'^*(\mathscr{F}), f'^*(\mathscr{G}))$$

are isomorphic. For sheaves $\mathscr{G}$ on Z we have

$$\begin{aligned}\mathrm{Hom}(\mathscr{H}, \mathscr{G}) &= \mathrm{Hom}(g^*(f_!\mathscr{F}), \mathscr{G}) \\ &= \mathrm{Hom}(f_!\mathscr{F}, g_*\mathscr{G}) \\ &= \mathrm{Hom}(\mathscr{F}, f^* g_*\mathscr{G}) \\ &= \mathrm{Hom}(\mathscr{F}, g'_* f'^*\mathscr{G}) \\ &= \mathrm{Hom}(g'^*\mathscr{F}, f'^*\mathscr{G}).\end{aligned}$$

The fourth assertion is trivial when Y is the spectrum of a separably closed field. The general case follows by using the base change property of $f_!$.

The proof of 8.3 (5) follows the same line as the proof of the constructibility of the "free sheaf" $\mathscr{G}^{\Lambda}$ of Λ-modules generated by a constructible sheaf $\mathscr{G}$ of sets (§ 4). Because of the base change property of $f_!$ it is enough to deal with the trivial case $X = Y \dot\cup Y \dot\cup \ldots \dot\cup Y$. □

We now want to investigate the case of an open embedding $f\colon X \to Y$. Obviously the base change property of the functor $f_!$ implies that, for any sheaf $\mathscr{F}$ on X,

the restriction of $f_!(\mathscr{F})$ to X is isomorphic to $\mathscr{F}$, and
the restriction of $f_!(\mathscr{F})$ to the complement of X in Y vanishes.

These two properties characterize $f_!$.

8.3 Lemma. *Let $j\colon X \to Y$ be an open embedding, $\mathscr{F}$ a sheaf (of abelian groups) on X, $\mathscr{G}$ a sheaf on Y, and*

$$t\colon \mathscr{F} \to j^*(\mathscr{G})$$

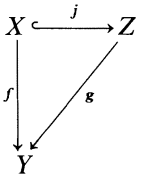
an isomorphism. Suppose the restriction of $\mathscr{G}$ to the complement of X in Y vanishes. Then $\mathscr{G}$ is canonically isomorphic to $j_!(\mathscr{F})$. More precisely, for each sheaf $\mathscr{H}$ on Y the mapping t induces an isomorphism

$$\begin{aligned}\mathrm{Hom}(\mathscr{G}, \mathscr{H}) &\xrightarrow{\cong} \mathrm{Hom}(\mathscr{F}, j^*(\mathscr{H})) \\ h &\mapsto j^*(h) \circ t.\end{aligned}$$

In the case of an open embedding j, then, $j_!(\mathscr{F})$ is just the "extension by zero".

We now want to show that $f_!(\mathscr{F})$ for etale f is in a natural way a subfunctor of the direct image functor $f_*(\mathscr{F})$, namely, the "subsheaf of sections with proper support".

Suppose we are given a commutative diagram



and a sheaf of abelian groups on X. Suppose that f here is an etale mapping and j is an open embedding. We want to construct a natural homomorphism

$$f_!(\mathscr{F}) \to g_*(j_!(\mathscr{F})).$$

The adjointness property of $f_!$ shows that we must therefore construct a homomorphism

$$\mathscr{F} \to f^*(g_*(j_!(\mathscr{F}))) = g_*(j_!(\mathscr{F}))|X.$$

Let $\phi\colon U \to X$ be an etale scheme over X. The mapping

$$(\mathrm{id}, j\circ\phi)\colon\ U \to U \times_Y Z$$

is a closed embedding. It is etale because U and $U \times_Y Z$ are etale over Z. The mapping

$$U \hookrightarrow U \times_Y Z$$

is thus an isomorphism onto an open and closed subscheme. To a section s in $\mathscr{F}(U)$ we assign the section s' in

$$g_*(j_!(\mathscr{F}))(U) = j_!(\mathscr{F})(U \times_Y Z) = \mathscr{F}(U) \times j_!(\mathscr{F})(U \times_Y Z - U)$$

that agrees with s in $\mathscr{F}(U)$ and is zero in $j_!\,\mathscr{F}(U \times_Y Z - U)$.

The collection of all homomorphisms

$$\mathscr{F}(U) \to g_*(j_!(\mathscr{F}))(U)$$

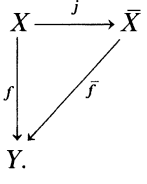
defined in this way gives the mapping we want.

If $Z = X$ and j is the identity mapping, then one can show that the mapping

$$f_!(\mathscr{F}) \to f_*(\mathscr{F})$$

is injective, and that, for every etale scheme U over Y, the set of sections of $f_!(\mathscr{F})$ over U is isomorphic to the set of sections of $\mathscr{F}(U \times_Y X) = f_*(\mathscr{F})(U)$ whose supports are proper over U (see 6.4). This remark justifies calling the functor $f_!$ the *"direct image with proper support"*.

It follows from Zariski's Main Theorem that every etale scheme $f\colon X \to Y$ can be embedded as an open subscheme of a scheme *finite* over Y:



8.4 Lemma. *Let $\mathscr{F}$ be a sheaf of abelian groups. Under the stated assumptions, the homomorphism*

$$f_!(\mathscr{F}) \to \bar{f}_* j_!(\mathscr{F})$$

is an isomorphism.

Proof. The functors $f_!$ and $\bar{f}_*$ are both compatible with base change. One shows that the construction of the homomorphism

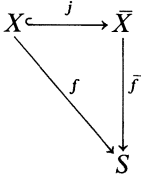
$$f_!(\mathscr{F}) \to \bar{f}_* j_!(\mathscr{F})$$

is also compatible with base change. Thus it suffices to prove the assertion in the trivial case where Y is the spectrum of a separably closed field. □

The Construction of Proper Direct Images for Compactifiable Morphisms

When nothing is said to the contrary, all sheaves occurring are to be etale torsion sheaves. Let $D(X, \text{tor})$ denote the full subcategory of the derived sheaf category on the scheme X containing the complexes whose cohomology sheaves are torsion, and let $D_+(X, \text{tor})$ denote the subcategory of $D(X, \text{tor})$ containing the complexes cohomologically bounded below.

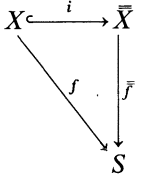
Let a morphism $f: X \to S$ be given. A *compactification* of f is a commutative diagram



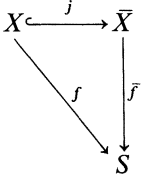
in which j is an open embedding and $\bar{f}$ is a proper morphism.

A morphism f that has such a compactification is called *compactifiable.* For example all quasiprojective morphisms, and in particular all quasifinite morphisms, are compactifiable.

One says that a compactification



dominates the compactification



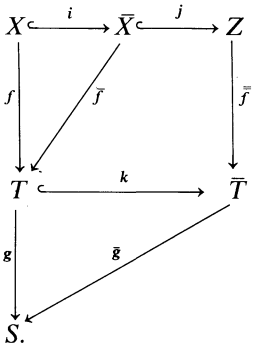
when there is a morphism $\phi: \bar{\bar{X}} \to \bar{X}$ with $\phi \circ i = j$ and $\bar{\bar{f}} = \bar{F} \circ \phi$. We record some simple properties:

For any two compactifications of a mapping f, there is always a third one that dominates them both.

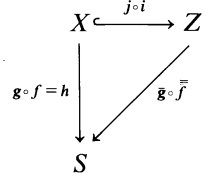
Let two mappings be given,

$$X \xrightarrow{f} T \quad \text{and} \quad T \xrightarrow{g} S.$$

Suppose g and $h=g\circ f$ are compactifiable. Then f is also compactifiable, and there is a commutative diagram

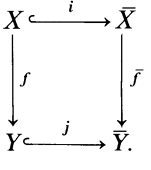


The mappings i, j, k are open embeddings, and the mappings $\bar{f}, \bar{\bar{f}}, \bar{g}$ are proper. In particular,



is a compactification of h.

8.5 Lemma. *Consider the commutative diagram*



Let $\mathscr{F}$ be a sheaf on X. Let the mappings i and j be open embeddings, f a proper mapping. Then there is a natural transformation

$$j_! f_*(\mathscr{F}) \to \bar{f}_* i_!(\mathscr{F}).$$

This induces a natural transformation of the derived functors

$$j_! \mathrm{R} f_*, \mathrm{R}\bar{f}_* i_!\colon D_+(X, \mathrm{tor}) \to D_+(Y, \mathrm{tor})$$

$$(*)\colon j_! \mathrm{R} f_*(\mathscr{F}^\cdot) \to \mathrm{R}\bar{f}_* i_!(\mathscr{F}).$$

This mapping $(*)$ is a quasi-isomorphism (isomorphism in $D(Y)$) if $\bar{f}$ is also proper.

Construction, first case. Suppose the initial diagram is cartesian. By Lemma 4.9 (compactibility of $j_!$ with inverse image functors) we have a natural isomorphism

$$\bar{f}^* j_!(f_*(\mathscr{F})) \xrightarrow{\cong} i_! f^* f_*(\mathscr{F}).$$

Composition with the adjunction mapping $f^* f_*(\mathscr{F}) \to \mathscr{F}$ yields the mapping

$$\bar{f}^* j_! f_*(\mathscr{F}) \to i_!(\mathscr{F}).$$

Since $\bar{f}_*$ is right adjoint to $\bar{f}^*$, we get from this the mapping we want,

$$j_! f_*(\mathscr{F}) \to \bar{f}_* i_!(\mathscr{F}).$$

Its restriction to Y yields an isomorphism.

The transformation thus constructed extends immediately to complexes $\mathscr{F}^\cdot$. Now let $\mathscr{F}^\cdot$ be a complex in $D_+(X, \text{tor})$, and let $i_! \mathscr{F}^\cdot \to \mathscr{J}^\cdot$ be an injective resolution of $i_! \mathscr{F}^\cdot$. Then the adjoint mapping supplies an injective resolution

$$\mathscr{F}^\cdot \to i^*(\mathscr{J}^\cdot) = \mathscr{I}^\cdot.$$

We apply the functor $\bar{f}_*$ to the natural mapping $i_!(\mathscr{I}^\cdot) \to \mathscr{J}^\cdot$ and get a morphism

$$\bar{f}_* i_!(\mathscr{I}^\cdot) \to \bar{f}_*(\mathscr{J}^\cdot),$$

which is an isomorphism on Y.

Composing with the transformation $j_! f_* \to \bar{f}_* i_!$ that was already constructed, we get a mapping

$$j_! f_*(\mathscr{I}^\cdot) \to \bar{f}_*(\mathscr{J}^\cdot),$$

which becomes an isomorphism after restriction to Y! On the other hand, $f_*(\mathscr{I}^\cdot) = \mathsf{R} f_*(\mathscr{F}^\cdot)$ and $\bar{f}_*(\mathscr{J}^\cdot) = \mathsf{R}\bar{f}_*(i_!(\mathscr{F}^\cdot))$. Passage to the cohomology sheaves yields homomorphisms

$$j_! R^\nu f_*(\mathscr{F}^\cdot) \to R^\nu \bar{f}_* i_!(\mathscr{F}^\cdot)$$

that become isomorphisms after restriction to Y.

Now let $\bar{f}$ be proper. From the proper base change theorem we get the vanishing of the restriction of $R^\nu \bar{f}_* i_!(\mathscr{F}^\cdot)$ to the complement of Y in $\bar{Y}$. In this case therefore the mappings

$$j_! R^\nu f_*(\mathscr{F}^\cdot) \to R^\nu \bar{f}_* i_!(\mathscr{F}^\cdot)$$

are isomorphisms.

The general case. We consider the natural diagram

$$\begin{array}{ccccc} X & \xrightarrow{i_1} & X_1 = \bar{X} \times_{\bar{Y}} Y & \xrightarrow{i_2} & \bar{X}, \qquad i = i_2 \circ i_1 \\ & \searrow^{f} & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle \bar{f}} \\ & & Y & \xrightarrow{j} & \bar{Y}. \end{array}$$

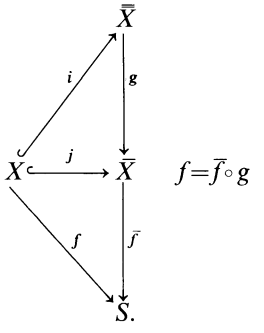
The two mappings i_1 and i_2 are open embeddings. The mapping i_1 is proper, since f is proper, and hence it represents an isomorphism onto an open and closed subscheme. Trivially

$$i_{1!}(\mathscr{F}^\cdot) = i_{1*}(\mathscr{F}^\cdot).$$

The desired results now follow from this together with the cartesian case handled at the beginning! □

We come now to the introduction of the higher direct images with proper support, as mentioned earlier.

Suppose we have a compactifiable mapping $f: X \to S$ and two compactifications. We shall assume to begin with that one of them dominates the other one. Thus we have a commutative diagram



Here i and j are open embeddings, g and $\overline{f}$ proper. Let $\mathscr{F}^\bullet$ be a complex in $D_+(X, \text{tor})$. It follows from Lemma 8.1 that there is a natural quasi-isomorphism

$$j_!(\mathscr{F}^\bullet) \xrightarrow{\cong} \mathsf{R}\, g_*(i_!(\mathscr{F}^\bullet)).$$

For a single sheaf $\mathscr{F}$, this means

$$g_*\, i_!(\mathscr{F}) = j_!(\mathscr{F}),$$
$$R^\nu g_*\, i_!(\mathscr{F}) = 0, \qquad \nu > 0.$$

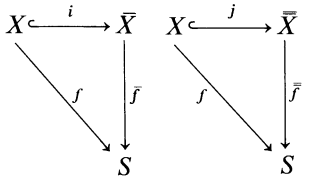
On the other hand, the functors

$$\mathsf{R}(\overline{f}\circ g)_* \quad \text{and} \quad (\mathsf{R}\overline{f}_*)\circ(\mathsf{R}\, g)_*\colon D_+(X) \to D_+(S)$$

are isomorphic in a natural way. Hence

$$\mathsf{R}\overline{f}_*\, j_!(\mathscr{F}^\bullet) \quad \text{and} \quad \mathsf{R}\overline{f}_*\, i_!(\mathscr{F}^\bullet)$$

are isomorphic in a natural way in $D_+(S)$. In the general case, for any two compactifications



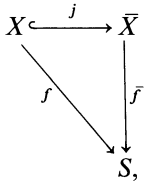
there is a third one that dominates them both. Using the preceding result, one thus obtains an isomorphism of functors

$$\mathsf{R}\overline{f}_*\, i_!(\mathscr{F}^\bullet) \cong \mathsf{R}\overline{\overline{f}}_*\, j_!(\mathscr{F}^\bullet).$$

To begin with, this isomorphism depends on the choice of the third compactification and the "domination mappings". But for any two compactifications that dominate the original pair, there is a further one that dominates all the preceding

ones, and in this way one can easily see that the isomorphism is independent of the construction chosen.

8.6 Definition (and Proposition)*). *Let $f: X \to S$ be a compactifiable mapping. Using a compactification*



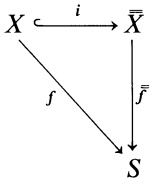
we define the ∂-functor $\mathsf{R}f_!: D_+(X, \mathrm{tor}) \to D_+(X, \mathrm{tor})$

$$\mathsf{R}f_! =_{\mathrm{def}} (\mathsf{R}\bar{f}_*) \circ j_!.$$

Concrete construction: Let $\mathscr{F}^\cdot$ be a complex in $D_+(X, \mathrm{tor})$ and let $j_!(\mathscr{F}^\cdot) \to \mathscr{I}^\cdot(j_!(\mathscr{F}^\cdot))$ be an injective resolution. Then

$$\mathsf{R}f_!(\mathscr{F}^\cdot) = \bar{f}_* \mathscr{I}^\cdot(j_!(\mathscr{F}^\cdot)) \quad \text{in } D(S).$$

The functor $\mathsf{R}f_!$ is in the following sense independent of the compactification: If



is another compactification, there is a natural isomorphism between

$$\mathsf{R}\bar{\bar{f}}_* i_!(\mathscr{F}^\cdot) \quad \text{and} \quad \mathsf{R}\bar{f}_* j_!(\mathscr{F}^\cdot)$$

(independent of all auxiliary constructions).

We call the cohomology sheaves

$$R^\nu f_!(\mathscr{F}^\cdot) =_{\mathrm{def}} \mathscr{H}^\nu(\mathsf{R}f_!(\mathscr{F}^\cdot)) = R^\nu \bar{f}_*(j_!(\mathscr{F}^\cdot))$$

the (higher) direct images with proper support. For a sheaf $\mathscr{F}$ we have in particular

$$R^0 f_!(\mathscr{F}) = \bar{f}_* j_!(\mathscr{F}).$$

For etale f, 8.4 *gives*

$$R^0 f_!(\mathscr{F}) \cong f_!(\mathscr{F}) \quad (\text{canonically}).$$

For a compactifiable morphism we define in general

$$f_!(\mathscr{F}) \cong \bar{f}_* j_!(\mathscr{F}) = R^0 f_!(\mathscr{F}).$$

*) This definition makes essential use of the proper base change theorem, which we have proved only under a restrictive hypothesis on the torsion (and for projective f). But for our purposes it suffices to construct $\mathsf{R}f_!$ under these restrictions.

If $S=\mathrm{Spec}(\Omega)$ is the spectrum of a separably closed field Ω, then we write

$$\mathsf{R}\Gamma_c(X,\mathscr{F}^\cdot)=_{\mathrm{def}}\Gamma(S,\mathsf{R}f_!(\mathscr{F}^\cdot))$$

$$H_c^\nu(X,\mathscr{F}^\cdot)=_{\mathrm{def}}\Gamma(S,R^\nu f_!(\mathscr{F}^\cdot))$$

(Cohomology groups with compact support). For a single sheaf $\mathscr{F}$ we have

$$\Gamma_c(X,\mathscr{F})=H_c^0(X,\mathscr{F}).$$

Remark and Warning. Let $\mathscr{F}$ be a torsion sheaf. Then $f_!(\mathscr{F})$ is a subsheaf of $f_*(\mathscr{F})$, as follows:

Let $U\to S$ be etale

$$f_!(\mathscr{F})(U)=\{s\in\mathscr{F}(U\times_S X)\,|\,\text{the support of } s \text{ in } U\times_S X \text{ is proper over } U\}.$$

But though the functors $R^\nu f_!(\mathscr{F})$ do indeed form an exact ∂-functor, they are not in general the derived functors of $f_!(\mathscr{F})$ in the sense of homological algebra! The reason for this is that the functor $j_!$ (unlike j_*) does not in general carry injective sheaves to acyclic ones (with respect to $\bar{f}_*$).

Remark. Let $f\colon X\to S$ be an etale morphism. By Zariski's Main Theorem, there exists a compactification

$$\begin{array}{ccc} X & \xrightarrow{\ j\ } & \bar{X} \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle \bar{f}} & \\ & S & \end{array}$$

with $\bar{f}$ a finite morphism. Since the higher derived functors vanish for a finite morphism, we have

$$R^\nu\bar{f}_*(j_!(\mathscr{F}))=0 \quad \text{for } \nu>0,$$

$$R^0\bar{f}_*(j_!(\mathscr{F}))=f_!(\mathscr{F})$$

for the $f_!$ defined at the beginning (8.1).

We need some properties of these higher direct images with proper support, and we assemble them in the following theorem.

8.7 Theorem. (1) *Base change theorem for the direct images with proper support (see also 6.1):*

We consider a cartesian diagram

$$\begin{array}{ccc} T\times_S X=Z & \xrightarrow{\ f'\ } & T \\ \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{\ f\ } & S. \end{array}$$

Let f and hence f' be compactifiable.

There is a natural base change isomorphism (in $D(T)$)

$$g^* \mathrm{R}f_!(\mathscr{F}^\cdot) \xrightarrow{\cong} \mathrm{R}f'_! g'^*(\mathscr{F}^\cdot)$$

for $\mathscr{F}^\cdot \in D_+(X, \mathrm{tor})$. In particular

$$g^* R^\nu f_!(\mathscr{F}^\cdot) \xrightarrow{\cong} R^\nu f'_! g'^*(\mathscr{F}^\cdot).$$

(2) *Direct images with proper support for composite mappings:*
Suppose we have mappings

$$X \xrightarrow{f} Y \xrightarrow{g} S$$
$$X \xrightarrow{h=g\circ f} S$$

where g, h, and hence f are compactifiable. In the derived category, there is a natural isomorphism of functors

$$\mathrm{R}g_! \circ \mathrm{R}f_! \cong \mathrm{R}h_!.$$

This yields a "Leray" spectral sequence for computing $R^\nu h_!(\mathscr{F}^\cdot)$.

(3) *The long exact sequence for an open subscheme: Let $f: X \to S$ be a compactifiable mapping, $j: U \hookrightarrow X$ an open embedding, and $i: A \hookrightarrow X$ a closed embedding. Suppose $i(A) \cup j(U) = X$ (for the underlying sets). In short, let A be a complement of U in X. Then for each torsion sheaf $\mathscr{F}$ on X, there is a functorial long exact sequence*

$$0 \to (f\circ j)_!(j^*(\mathscr{F})) \to f_!(\mathscr{F}) \to (f\circ i)_!(i^*(\mathscr{F})) \to \dots$$
$$\to R^\nu(f\circ j)_!(j^*(\mathscr{F})) \to R^\nu f_!(\mathscr{F}) \to \dots.$$

More abstractly: This sequence is the long exact sequence of cohomology sheaves for an admissible triangle in $D(S)$, that can be assigned in a functorial way to each complex $\mathscr{F}^\cdot$ in $D_+(X, \mathrm{tor})$:

$$\begin{array}{ccc} & \mathrm{R}(f\circ i)_!(i^*(\mathscr{F}^\cdot)) & \\ \swarrow & & \nwarrow \\ \mathrm{R}(f\circ j)_!(j^*(\mathscr{F}^\cdot)) & \longrightarrow & \mathrm{R}f_!(\mathscr{F}^\cdot). \end{array}$$

For the proof. (1) comes from the proper base change theorem and the "base change property" for the functor $i_!$ when i is an etale mapping (4.7).

(2) There is a commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{i_1} & \bar{X} & \xrightarrow{i_2} & Z \\ {\scriptstyle f}\downarrow & \swarrow{\scriptstyle \bar{f}} & & \swarrow{\scriptstyle \phi} & \\ Y & \xrightarrow{j} & \bar{Y} & & \\ {\scriptstyle g}\downarrow & \swarrow{\scriptstyle \bar{g}} & & & \\ S & & & & \end{array}$$

i_1, i_2, and j are open embeddings,
$\bar{f}$, $\bar{g}$, and ϕ are proper mappings.

$$g \circ f = h, \qquad \bar{g} \circ \phi = \bar{h}, \qquad i_2 \circ i_1 = i.$$

For a complex $\mathscr{F}^\cdot$ in $D_+(X, \text{tor})$, Lemma 8.1 gives a natural quasi-isomorphism

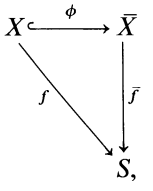
$$j_! \mathsf{R}\bar{f}_*(i_{1!}(\mathscr{F}^\cdot)) \xrightarrow{\cong} \mathsf{R}\phi_* i_{2!}(i_{1!}(\mathscr{F}^\cdot)) = \mathsf{R}\phi_*(i_!(\mathscr{F}^\cdot)).$$

On the other hand, there is a natural isomorphism (in the derived category)

$$\mathsf{R}(\bar{g} \circ \phi)_* \cong (\mathsf{R}\bar{g}_*) \circ (\mathsf{R}\phi_*).$$

Composition gives the desired isomorphism. One can show that the isomorphism is independent of the construction (again using "dominating diagrams").

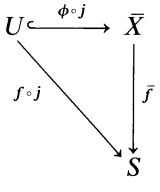
(3) We consider a compactification of f,



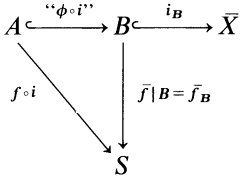
and a closed subscheme i_B: $B \hookrightarrow \bar{X}$ of $\bar{X}$ with

$$\phi(X) \cup B = \bar{X}, \qquad \phi^{-1}(B) = A.$$

Then



is a compactification of $U \to S$, and



is a compactification of $A \to S$.

For a complex $\mathscr{F}^\cdot$ on X, there is a natural exact sequence

$$(*) \qquad 0 \to (\phi \circ j)_!(j^*\mathscr{F}^\cdot) \to \phi_!(\mathscr{F}^\cdot) \to (i_B)_*(\text{“}\phi \circ i\text{”})_!(i^*(\mathscr{F}^\cdot)) \to 0.$$

The sequence we want is the long exact sequence of direct image sheaves $R^\nu \bar{f}_*$ for this short exact sequence. Here we use

$$(R^\nu \bar{f}_{B*})(\text{“}\phi \circ i\text{”})_!(\mathscr{G}) = (R^\nu \bar{f}_*)\, i_{B*}(\text{“}\phi \circ i\text{”})_!(\mathscr{G})$$

for a sheaf $\mathscr{G}$ on A. The sequence (*) can be embedded in an admissible triangle (AII, p. 295). The image triangle under $\mathsf{R}\bar{f}_*$ is the one we want. □

Here is a typical application of these properties (method of reduction to the case of curves).

8.8 Theorem. *Let $f\colon X\to S$ be a compactifiable mapping, $\mathscr{F}$ a torsion sheaf on X. Let $\dim X/S$ be the maximum of the dimensions of all geometric fibers. Then*

$$(*)\qquad R^v f_!(\mathscr{F})=0,\qquad v>2\dim X/S.$$

This easily implies the more general statement that, for any complex $\mathscr{F}^\cdot$ in $D_+(X, \mathrm{tor})$ with $\mathscr{H}^v(\mathscr{F}^\cdot)=0$ for $v\geqq r$,

$$(**)\qquad R^v f_!(\mathscr{F}^\cdot)=0\quad \textit{for}\quad v\geqq r+2\dim X/S.$$

It is enough for our purposes to make the restrictive hypothesis that there is a number m, invertible on S, such that every section of $\mathscr{F}$ on every etale scheme $U\to X$ is annihilated by an appropriate power of m.

Proof. Using noetherian induction and the long exact sequence 8.3 (3), one concludes that it is sufficient to assume $X\to S$ affine. By the base change Theorem 8.3 (1), it is enough to show that when $X\to S=\mathrm{Spec}(\Omega)$ is an affine scheme over a separably algebraically closed field Ω and $\mathscr{F}$ is a sheaf on X, then

$$H_c^v(X,\mathscr{F})=0,\qquad v>2\dim X.$$

But then there is a factorization of f into affine mappings

$$f_v\colon X_v\to X_{v+1},\qquad f=f_1\circ\ldots\circ f_n$$

with

$$\dim X_v/X_{v+1}\leqq 1.$$

By 8.3 (2), it suffices to prove the assertion (**) for the individual mappings f_v. By the base change theorem, it is thus enough to prove

$$H_c^v(X,\mathscr{F})=0\quad \text{for}\quad v\geqq 3$$

for an affine scheme $X\to\mathrm{Spec}(\Omega)$ of dimension $\dim(X)\leqq 1$. But X can be embedded as an open subscheme of a projective scheme $\bar{X}\to\mathrm{Spec}(\Omega)$ of dimension $\dim\bar{X}\leqq 1$. Our assertion thus follows from Proposition 5.3 for the scheme $\bar{X}$ (at least when the torsion of $\mathscr{F}$ is relatively prime to the characteristic of Ω. That suffices to prove the theorem under our restrictive hypothesis on the torsion). □

Our goal is to prove the following two main theorems:

8.9 Theorem. *Let*

$$f\colon X\to S$$

be a smooth proper morphism, and let $\mathscr{F}$ be a (constructible) locally constant torsion sheaf on X with torsion orders invertible on S: thus there is a number n, invertible on S, with $n\mathscr{F}=0$. Then all the sheaves $R^{\nu}f_{*}(\mathscr{F})$ are locally constant (and constructible).

8.10 Theorem. *Let*

$$f\colon X\to S$$

be a compactifiable mapping, $\mathscr{F}$ a constructible sheaf on X. Then all the sheaves $R^{\nu}f_{!}(\mathscr{F})$ are constructible. More generally, it follows immediately from this that the sheaves

$$R^{\nu}f_{!}(\mathscr{F}^{\cdot})=\mathscr{H}^{\nu}(\mathrm{R}f_{!}(\mathscr{F}^{\cdot}))$$

are constructible for any complex $\mathscr{F}^{\cdot}\in D_{+}(X,\mathrm{tor})$ with constructible cohomology sheaves.

The hypothesis on the torsion is essential in 8.9, since the proof requires the *smooth* base change theorem. We shall prove Theorem 8.10 under the same restrictive hypothesis, which is sufficient for us.

We need a new characterization of the (constructible) locally constant sheaves on a scheme X. Consider two geometric points with values in two separably algebraically closed fields k and K:

$$\mathrm{Spec}(K)\xrightarrow{a}X\xleftarrow{s}\mathrm{Spec}(k).$$

Let R be the strict Henselization with respect to s of the local ring of the structure sheaf at the ordinary point underlying s, and let $X_{(s)}=\mathrm{Spec}(R)$. Suppose

$$\begin{array}{ccc}\phi\colon\ \mathrm{Spec}(K) & \longrightarrow & X_{(s)}\\ & \searrow\quad\swarrow & \\ & X & \end{array}$$

is a morphism over X. The geometric point s is then a "specialization" of the geometric point a. Let

$$\begin{array}{ccc}U & \longleftarrow & \mathrm{Spec}(k)\\ \downarrow & \swarrow_{s} & \\ X & & \end{array}$$

be an etale neighborhood of s. If we compose ϕ with the natural mapping $X_{(s)}\to U$, we get an etale neighborhood

$$\begin{array}{ccc}\mathrm{Spec}(K) & \longrightarrow & U\\ & \searrow_{a} & \downarrow\\ & & X\end{array}$$

of a. For a sheaf $\mathcal{G}$ on X, we get a homomorphism

$$\mathcal{G}(U) \to \mathcal{G}_a = \varinjlim_{\substack{\mathrm{Spec}(K) \to V \\ a \searrow \downarrow \\ X}} \mathcal{G}(V).$$

Passing to the direct limit over all etale neighborhoods of s, we get finally a homomorphism

$$\mathcal{G}_s \to \mathcal{G}_a.$$

8.11 Definition. The homomorphism of stalks

$$\mathcal{G}_s \to \mathcal{G}_a$$

constructed in this way is called the specialization homomorphism with respect to the specialization

$$\phi\colon \begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X_{(s)} \\ & a \searrow & \downarrow \\ & & X \end{array}$$

of the point a to s.

We shall not bother to write down the simple behavior of the specialization homomorphism under formation fo direct images and inverse images of a sheaf.

8.12 Lemma. *A sheaf $\mathcal{G}$ on a scheme S is a (constructible) locally constant sheaf if and only if*

a) *all specialization homomorphisms are isomorphisms, and*
b) *all stalks of $\mathcal{G}$ are finite.*

We leave the simple proof to the reader.

8.13 Lemma. *Let*

$$f\colon X \to S$$

be a smooth proper morphism, and let $\mathcal{F}$ be a locally constant sheaf on X satisfying the condition on torsion assumed in Theorem 8.10. Then the specialization homomorphisms are isomorphisms for all direct images $R^v f_*(\mathcal{F})$.

Proof. Without loss of generality we can assume that $S = \mathrm{Spec}(A)$ is the spectrum of a strictly Henselian local ring with residue field k,

$$s\colon \mathrm{Spec}(k) \to S$$

is the "special point" of S, and

$$a\colon \operatorname{Spec}(K)\to S$$

is some geometric point.

We consider the diagram

$$\begin{array}{ccccc}
\operatorname{Spec}(K)\times_S X = X_a & \xrightarrow{j} & X & \xleftarrow{i} & X_s = \operatorname{Spec}(k)\times_S X \\
\downarrow{\scriptstyle f_a} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_s} \\
\operatorname{Spec}(K) & \xrightarrow{a} & S & \xleftarrow{s} & \operatorname{Spec}(k).
\end{array}$$

The restriction mapping

$$R^\nu f_*(\mathscr{F})_a = H^\nu(X_a, j^*(\mathscr{F})) \leftarrow H^\nu(X, \mathscr{F}) = R^\nu f_*(\mathscr{F})_s$$

is the specialization homomorphism. We must show that it is an isomorphism. We first show, using the smooth base change theorem, that

$$R^\nu j_* j^*(\mathscr{F}) = 0, \qquad \nu > 0,$$

and that the natural mapping

$$i^* j_* j^*(\mathscr{F}) \leftarrow i^*(\mathscr{F})$$

is an isomorphism. In other words,

$$i^*(\mathscr{F}) \to i^*(\mathsf{R} j_*(j^*(\mathscr{F})))$$

is an isomorphism in $D(X_s)$. For this we do not assume that f is proper. Since the assertion is of local nature with respect to the etale topology on X, we can assume that $\mathscr{F}$ is constant,

$$\mathscr{F} = M_X, \qquad M \text{ a finite abelian group.}$$

By the smooth base change theorem, it is enough to show that

$$R^\nu a_*(M_{\operatorname{Spec}(K)}) = 0, \qquad \nu > 0,$$

$$(a_* M_{\operatorname{Spec}(K)})(S) = M.$$

But this is clear, since K is separably closed and A is strictly Henselian. Now let f be smooth and proper. By the proper base change theorem, the formula $i^*(\mathscr{F}) = i^*(\mathsf{R} j_*(j^*(\mathscr{F})))$ that we have just proved gives us

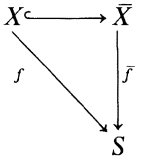
$$\begin{aligned} H^\nu(X, \mathscr{F}) &= H^\nu(X_s, i^*(\mathscr{F})) = H^\nu(X_s, i^*(\mathsf{R} j_*(j^*(\mathscr{F})))) \\ &= H^\nu(X, \mathsf{R} j_*(j^*(\mathscr{F}))) = H^\nu(X_a, j^*(\mathscr{F})). \quad \square \end{aligned}$$

We come now to the proof of the finiteness Theorem 8.6; Theorem 8.9 then also follows from this theorem and Lemmas 8.12 and 8.13.

Let $X \xrightarrow{f} S$ be a compactifiable morphism, $\mathscr{F}$ a constructible sheaf on X, or more generally $\mathscr{F}^\cdot$ a complex in $D_+(X, \text{tor})$. For simplicity we make

the restrictive hypothesis on the torsion of $\mathscr{F}$ (and correspondingly for $\mathscr{F}^{\cdot}$) as in 8.9.

1) Let f be quasi-finite. Then f has a compactification



with $\bar{f}$ a finite mapping. The assertion then follows by 3.3 and 3.4.

2) *Let X be a smooth curve over S:* then f is smooth and $\dim X/S=1$. Every constructible sheaf $\mathscr{F}$ has a resolution

$$\ldots\to\mathscr{F}_{\nu}\to\ldots\to\mathscr{F}_{0}\to\mathscr{F}\to 0$$

where (see § 4) the sheaves $\mathscr{F}_{\nu}$ have the form

$$p_!(M_{\tilde{X}})$$

$$p\colon\ \tilde{X}\to X \text{ etale, } \tilde{X} \text{ affine}$$

$$M=(\mathbb{Z}/m\mathbb{Z}).$$

Because of the restriction on the torsion of $\mathscr{F}$, we can assume here that m is invertible on S. Since $R^{\nu}f_!$ vanishes for large ν ($\nu\geqq 3$), it is enough to prove the assertion for sheaves of this type. Since $R^{\nu}f_!(p_!(M_X))=R^{\nu}(f\circ p)_!(M_X)$, we can assume without loss of generality that

$$X \text{ is affine,}$$

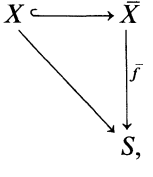
$$\mathscr{F}=M_X \text{ is a constant sheaf.}$$

We use noetherian induction on S. Because of the base change theorem for $\mathsf{R}f_!$, it is enough to show that if $S=\mathrm{Spec}(A)$ is integral, then there is a nonempty open affine subscheme $U\subset S$ such that the sheaves

$$R^{\nu}f_!(M_X)\,|\,U$$

are constructible. For the proof we can shrink S and (3.12) replace it by a scheme T purely inseparable over S.

After an appropriate such change of S we can assume that there is an open embedding over S,



where $\bar{X}$ is a smooth projective curve over S and the complement of X in $\bar{X}$ is quasi-finite over S.

Because of the exact sequence in 8.3 (3), it is enough to show that $R^{\nu}\bar{f}_*M_{\bar{X}}=R^{\nu}\bar{f}_!M_{\bar{X}}$ is locally constant (and constructible).

We have already shown that the specialization mappings on $R^\nu \bar{f}_* M_{\bar{X}}$ are isomorphisms (8.13). By Lemma 8.12 and the proper base change theorem, it remains to show only that for a geometric point

$$\alpha\colon \operatorname{Spec}(K) \to S$$

the groups

$$H^\nu(\operatorname{Spec}(K) \times_S \bar{X}, M_{\operatorname{Spec}(\Omega) \times_S \bar{X}}) = R^\nu \bar{f}_*(M_{\bar{X}})_\alpha$$

are finite. But this is a result from § 5 (5.3).

3) As in the proof of Theorem 8.8, we use 8.7 to reduce the general case to the case of a curve.

Using the exact sequence in 8.7 and noetherian induction, one sees that X can be assumed to be affine. But then f can be decomposed into a product

$$f = f_n \circ \ldots \circ f_0$$

where f_0 is a closed embedding (embedding in "the" affine space of dimension n over S) and the $f_\nu\colon X_\nu \to X_{\nu+1}$ are smooth mappings with $\dim X_\nu / X_{\nu+1} = 1$. By 8.7 (2),

$$\mathrm{R}f_!(\mathscr{F}^\cdot) = \mathrm{R}f_{n!} \circ \ldots \circ \mathrm{R}f_{0!}(\mathscr{F}^\cdot). \quad \square$$

To conclude this section, we use the proper base change theorem to deduce an identity of the Künneth formula type.

We first make a preliminary remark. Let $f\colon X \to S$ be a proper morphism. Then by Theorem 8.8 the cohomological dimension of the functor f_* on the category of torsion sheaves is finite:

$$R^\nu f_*(\mathscr{F}) = 0 \qquad \nu > 2 \dim X/S.$$

One can conclude from this that every complex $\mathscr{F}^\cdot$ of torsion sheaves has an acyclic resolution with respect to f_*, that is, a quasi-isomorphism

$$\mathscr{F}^\cdot \to \mathscr{J}^\cdot(\mathscr{F}^\cdot)$$

into a complex $\mathscr{J}^\cdot(\mathscr{F}^\cdot)$ of torsion sheaves with

$$R^\nu f_*(\mathscr{J}^j) = 0 \qquad \nu > 0.$$

Using such resolutions, one can extend the derived functor in the sense of Verdier (Appendix II),

$$\mathrm{R}f_*\colon D_+(X, \mathrm{tor}) \to D_+(S, \mathrm{tor}),$$

in a natural way to the full derived category of the torsion sheaves,

$$\mathrm{R}f_*\colon D(X, \mathrm{tor}) \to D(S, \mathrm{tor})$$
$$\mathscr{F}^\cdot \mapsto f_*(\mathscr{J}^\cdot(\mathscr{F}^\cdot))$$

(see Hartshorne [72], Chapter I).

If the mapping $f\colon X \to S$ is only compactifiable, then using a compactification one again defines the functor

$$\mathrm{R}f_!\colon D(X, \mathrm{tor}) \to D(S, \mathrm{tor}).$$

In the following, we consider only sheaves of modules over the fixed ring $\Lambda = \mathbb{Z}/n\mathbb{Z}$. For simplicity we shall also assume that n is invertible on all the schemes occurring.

In Appendix II we have introduced the derived tensor product of two complexes $\mathscr{F}^\cdot \in D_-(X, \Lambda)$ and $\mathscr{G}^\cdot \in D(X, \Lambda)$ on a scheme X using a resolution bounded above,

$$\mathscr{P}^\cdot(\mathscr{F}^\cdot) \to \mathscr{F}^\cdot,$$

where the complex $\mathscr{P}^\cdot(\mathscr{F}^\cdot)$ consists of stalkwise Λ-flat sheaves $\mathscr{P}^\nu(\mathscr{F}^\cdot)$:

$$\mathscr{F}^\cdot \overset{\mathbb{L}}{\otimes} \mathscr{G}^\cdot = \mathscr{P}^\cdot(\mathscr{F}^\cdot) \otimes \mathscr{G}^\cdot.$$

Here $(\mathscr{P}^\cdot(\mathscr{F}^\cdot) \otimes \mathscr{G}^\cdot)^n = \bigoplus_{p+q=n} \mathscr{P}^p(\mathscr{F}^\cdot) \otimes \mathscr{G}^q$.

8.14 Proposition. *Let*

$$f\colon X \to S$$

be a compactifiable mapping, $\mathscr{F}^\cdot$ a complex $\in D_-(S, \Lambda)$, and $\mathscr{G}^\cdot$ a complex $\in D(X, \Lambda)$. Then in the derived category there is a natural isomorphism

$$\mathscr{F}^\cdot \overset{\mathbb{L}}{\otimes} \mathrm{R}f_!(\mathscr{G}^\cdot) \xrightarrow{\cong} \mathrm{R}f_!(f^*(\mathscr{F}^\cdot) \overset{\mathbb{L}}{\otimes} \mathscr{G}^\cdot).$$

Remark (1). The essential part of the proper base change theorem is a special case of this formula. Just apply the result to the following situation: S is the spectrum of a strictly Henselian ring, and the complex $\mathscr{F}^\cdot$ is the sheaf with stalk Λ at the closed point and zero outside this point.

Remark (2). If $f\colon X \to S$ is an arbitrary morphism with finite homological dimension with respect to f_* on the category of sheaves of Λ-modules, then the same method yields another isomorphism

$$\mathscr{F}^\cdot \overset{\mathbb{L}}{\otimes} \mathrm{R}f_*(\mathscr{G}^\cdot) \xrightarrow{\cong} \mathrm{R}f_*(f^*(\mathscr{F}^\cdot) \overset{\mathbb{L}}{\otimes} \mathscr{G}^\cdot)$$

for a complex $\mathscr{G}^\cdot$ in $D(X, \Lambda)$ and a complex $\mathscr{F}^\cdot$ in $D_-(S, \Lambda)$, if we make the additional assumption that the cohomology sheaves of $\mathscr{F}^\cdot$ are *locally constant*.

Proof. It is enough to assume that f is *proper*. For a sheaf $\mathscr{F}$ on S and a sheaf $\mathscr{G}$ on X, we have

$$f^*(\mathscr{F} \otimes f_*(\mathscr{G})) = f^*(\mathscr{F}) \otimes f^* f_*(\mathscr{G}).$$

By composing with the adjunction homomorphism

$$f^*(f_*(\mathscr{G})) \to \mathscr{G},$$

we get a homomorphism

$$f^*(\mathscr{F} \otimes f_*(\mathscr{G})) \to f^*(\mathscr{F}) \otimes \mathscr{G},$$

and this induces a natural homomorphism

$$\mathscr{F} \otimes f_*(\mathscr{G}) \to f_*(f^*(\mathscr{F}) \otimes \mathscr{G}).$$

This homomorphism extends to the category of complexes of sheaves. We assert:

Let $\mathscr{F}$ be a Λ-flat sheaf on S and $\mathscr{G}$ an f_*-acyclic sheaf on X. Then:

a) $R^\nu f_*(f^*(\mathscr{F})\otimes\mathscr{G})=0 \quad \nu>0$,

b) $f_*(f^*(\mathscr{F})\otimes\mathscr{G})=\mathscr{F}\otimes f_*(\mathscr{G})$.

By the proper base change theorem, it is enough to prove both assertions only when $S=\mathrm{Spec}(\Omega)$ is the spectrum of a separably closed field. But then $\mathscr{F}=M_S$ is a constant sheaf for some flat Λ-module M. Since $\Lambda=\mathbb{Z}/n\mathbb{Z}$ is an artinian ring, M is actually projective, and hence it is a direct summand of a free module. Both assertions then come from the compatibility of $H^\nu(X,-)$ with arbitrary direct sums.

Proof of the proposition. We can assume that $\mathscr{F}^\cdot$ is a complex of Λ-flat sheaves and $\mathscr{G}^\cdot$ is a complex of f_*-acyclic sheaves (A II). Then $f^*(\mathscr{F}^\cdot)$ is also a complex of Λ-flat sheaves.

From a) and the compatibility of $R^\nu f_*$ with direct sums, it follows that

$$f^*(\mathscr{F}^\cdot)\otimes\mathscr{G}^\cdot$$

is a complex of f_*-acyclic sheaves.

Consequently,

$$\mathrm{R}f_*(f^*(\mathscr{F}^\cdot)\underset{=}{\otimes}\mathscr{G}^\cdot)=f_*(f^*(\mathscr{F})\otimes\mathscr{G}^\cdot).$$

From b) then, because of the compatibility of f_* with direct sums, it follows that

$$f_*(f^*(\mathscr{F}^\cdot)\otimes\mathscr{G}^\cdot)=\mathscr{F}^\cdot\otimes f_*(\mathscr{G}^\cdot)=\mathscr{F}^\cdot\underset{=}{\otimes}\mathrm{R}^\cdot f_*(\mathscr{G}). \quad \square$$

8.15 Definition. A complex $\mathscr{G}^\cdot$ on a scheme X is called perfect, if there is a *bounded* complex $\mathscr{K}^\cdot$ of flat constructible sheaves and a quasi-isomorphism of complexes

$$\mathscr{K}^\cdot\to\mathscr{G}^\cdot.$$

Remark on Definition 8.15. Suppose the complex $\mathscr{G}^\cdot$ is bounded above, and all cohomology sheaves of $\mathscr{G}^\cdot$ are constructible. Then there is a complex $\mathscr{K}^\cdot$ of flat constructible sheaves, bounded above, and a quasi-isomorphism

$$\mathscr{K}^\cdot\to\mathscr{G}^\cdot.$$

Under these conditions, $\mathscr{G}^\cdot$ is perfect if and only if the following condition is satisfied:

There is a natural number m such that for all finite Λ-modules M we have

$$\mathrm{Tor}_\nu^\Lambda(M_X,\mathscr{G}^\cdot)=\mathscr{H}^{-\nu}(M_X\underset{=}{\otimes}\mathscr{G}^\cdot)=0 \qquad \text{for } \nu>m.$$

For it follows from this condition that the cohomology sheaves

$$\mathscr{H}^\nu(\mathscr{K}^\cdot)$$

vanish for $\nu<-m$ and that the image sheaf $\mathscr{B}^{-m}$ of $\mathscr{K}^\cdot$ in $\mathscr{K}^{-m}$ is flat. Hence

$$0\to\mathscr{B}^{-m}\to\mathscr{K}^{-m}\to\mathscr{K}^{-m+1}\to\ldots$$

is a finite flat resolution of $\mathscr{G}^\cdot$.

8.16 Corollary. *Let*

$$f\colon X \to S$$

be a compactifiable morphism, and let $\mathscr{G}^\cdot$ be a perfect complex on X. Then $\mathsf{R}^\cdot f_!(\mathscr{G}^\cdot)$ is also a perfect complex.

Proof. It follows from the finiteness Theorem 8.10 that the cohomology sheaves of $\mathsf{R}^\cdot f_!(\mathscr{G}^\cdot)$ are constructible. It follows from Theorem 8.14 that $\mathsf{R}^\cdot f_!(\mathscr{G}^\cdot)$ is contained in $D_-(X, \Lambda)$. We can assume that $\mathscr{G}^\cdot$ is a bounded complex of flat sheaves $\mathscr{G}^\nu$,

$$\mathscr{G}^\nu = 0 \quad \text{for } \nu < -m.$$

From the Künneth formula in Proposition 8.14 it follows then that for every Λ-module M,

$$\begin{aligned}\operatorname{Tor}_\nu^\Lambda(M_S, \mathsf{R}^\cdot f_!(\mathscr{G}^\cdot)) &= \mathscr{H}^{-\nu}(M_S \overset{\mathsf{L}}{\otimes} \mathsf{R}^\cdot f_!(\mathscr{G}^\cdot)) \\ &= R^\nu f_!(M_X \otimes \mathscr{G}^\cdot) = 0 \quad \text{for } \nu > +m. \quad \square\end{aligned}$$

8.17 Corollary. *Let X and Y be finitely generated schemes over Ω, a separably closed ground field. Let $\mathscr{F}$ be a sheaf of Λ-modules on X and $\mathscr{G}$ a sheaf of Λ-modules on Y. Suppose one of the two sheaves is flat. Then in the derived category there is a natural isomorphism*

$$\mathsf{R}\Gamma_c(X, \mathscr{F}) \overset{\mathsf{L}}{\otimes} \mathsf{R}\Gamma_c(Y, \mathscr{G}) \cong \mathsf{R}\Gamma_c(X \times Y, p^*(\mathscr{F}) \otimes q^*(\mathscr{G})).$$

Here p and q are the projections of $X \times Y$ onto X and Y.

Remark. If X and Y are complete schemes, the induced natural mapping

$$H_c^i(X, \mathscr{F}) \otimes H_c^j(Y, \mathscr{G}) \to H_c^{i+j}(X \times Y, p^*(\mathscr{F}) \otimes q^*(\mathscr{G}))$$

can be described as

$$\alpha \otimes \beta \mapsto p^*(\alpha) \smile q^*(\beta)$$

in terms of the cup product (see A II)

$$H^i(X \times Y, p^*(\mathscr{F})) \otimes H^j(X \times Y, q^*(\mathscr{G})) \to H^{i+j}(X \times Y, p^*(\mathscr{F}) \otimes q^*(\mathscr{G})).$$

If all the cohomology groups with compact support of $\mathscr{F}$ or of $\mathscr{G}$ are flat Λ-modules, the isomorphism 8.17 in the derived category means we have an isomorphism of Λ-modules:

$$\boxed{\bigoplus_{i+j=n} H_c^i(X, \mathscr{F}) \otimes H_c^j(Y, \mathscr{G}) = H^n(X \times Y, p^*(\mathscr{F}) \otimes q^*(\mathscr{G})).}$$

On the proof of Corollary 8.17. Use the proper base change theorem for the base change

$$X \to \operatorname{Spec}(\Omega)$$

and the Künneth formula 8.14 for the mappings

$$X \times Y \to X \quad \text{and} \quad X \to \operatorname{Spec}(\Omega). \quad \square$$

§ 9. The Cohomological Dimension of Affine Algebraic Schemes

Let X be a complex analytic Stein variety of *complex* dimension n. Using Morse theory, one can show that X has the homotopy type of an n-dimensional CW-complex. It follows from this that, for every locally constant sheaf $\mathscr{F}$ on X, the cohomology groups $H^\nu(X, \mathscr{F})$ vanish for $\nu > n$.

The goal of this section is to prove the following analogue of this theorem in etale cohomology theory.

9.1 Theorem. *Let X be a finitely generated affine scheme over an separably algebraically closed base field k. Let $\mathscr{F}$ be a torsion sheaf on X with torsion relatively prime to the characteristic of k. Let n be a given natural number, and suppose the stalk $\mathscr{F}_a = 0$ for every geometric point a of X with*

$$\dim \bar{a} > n.$$

(Here $\bar{a}$ is the closed irreducible subvariety of X with generic point a.) Then

$$H^p(X, \mathscr{F}) = 0 \quad \text{for } p > n.$$

We prove this by induction on n.

First, some preliminary remarks: We can assume without loss of generality that k is algebraically closed (base field extension). Since every torsion sheaf is a direct limit of constructible subsheaves, and every subsheaf of $\mathscr{F}$ satisfies the same conditions as $\mathscr{F}$, we can assume that $\mathscr{F}$ is a constructible sheaf. If in this case we then replace X by the closure of the set of points a with $F_a \neq 0$, we can even assume that

$$\dim X \leqq n.$$

All the sheaves we consider here will be sheaves of Λ-modules, $\Lambda = \mathbb{Z}/m\mathbb{Z}$, with m relatively prime to the characteristic of k. The case $n = 1$ was already treated in § 5 (5.3). Assume $n \geqq 2$. Using the induction hypothesis, we first prove:

9.2 Lemma. *Let A be a regular affine k-algebra, $\mathfrak{m}$ a maximal ideal, R the strict Henselization of the local ring $A_{\mathfrak{m}}$, t a regualar parameter of R, and $\mathscr{G}$ an etale Λ-sheaf on $S = \operatorname{Spec}(R_t)$. Suppose*

$$\dim R \leqq n.$$

Then

$$H^r(S, \mathscr{G}) = 0 \quad \text{for } r > n.$$

Proof. We can assume that t is contained in $\mathfrak{m}$, that $\mathscr{G}$ is constructible, and (after replacing $\operatorname{Spec}(A)$ by an etale neighborhood of $\mathfrak{m}$) that $\mathscr{G}$ is the inverse image of an etale Λ-sheaf $\mathscr{G}_0$ on $\operatorname{Spec}(A)$. We can further assume that

$$\dim A = \dim A_{\mathfrak{m}} = \dim R \leqq n.$$

We consider the system of connected etale neighborhoods $\operatorname{Spec}(B)$ of $\mathfrak{m}$ (more precisely, of an associated geometric point), setting

$$R = \varinjlim B.$$

Let

$$B(t) = B \otimes_{k[t]} k(t).$$

Then $B(t)$ is an affine $k(t)$-algebra with dimension

$$\dim B(t) \leqq \dim B - 1 = \dim A - 1 \leqq n-1$$
$$\uparrow$$
(regularity hypothesis).

We have

$$R_t = R \otimes_{k[t]} k(t) = \varinjlim B(t).$$

Let $\mathscr{G}_B$ be the inverse image of $\mathscr{G}_0$ on

$$Y_B = \operatorname{Spec}(B(t)).$$

Then

$$H^r(\operatorname{Spec}(R_t), \mathscr{G}) = \varinjlim H^r(Y_B, \mathscr{G}_B).$$

It is enough to show that for $r > n$

$$H^r(Y_B, \mathscr{G}_B)$$

vanishes. We consider the Leray spectral sequence

$$E_2^{pq} = H^p(\operatorname{Spec}(k(t)), R^q h_*(\mathscr{G}_B)) \Rightarrow H^{p+q}(Y_B, \mathscr{G}_B)$$

associated with the mapping

$$h\colon Y_B \to \operatorname{Spec}(k(t)).$$

By Tsen's theorem (1.15),

$$H^p(\operatorname{Spec}(k(t)), -)$$

vanishes for $p > 1$. Let $\overline{k(t)}$ be the separable algebraic closure of $k(t)$. We consider the geometric point

$$a\colon \operatorname{Spec}\overline{(k(t))} \to \operatorname{Spec}(k(t)).$$

It follows from the induction hypothesis that

$$R^q f_*(\mathscr{G}_B)_a = H^q(\operatorname{Spec}(Y_B \otimes_{k(t)} \overline{k(t)}, -) = 0 \quad \text{for } q > n-1.$$

This yields

$$E^{pq} = 0 \quad \text{for } p+q > n,$$
$$H^r(Y_B, \mathscr{G}_B) = 0 \quad \text{for } r > n,$$

and thus Lemma 9.2 is proved. □

For our induction proof now we assume without loss of generality that $\dim X = n \geqq 2$. Let $A = k[X_1, \ldots, X_n]$ be the polynomial ring over k in the variables $X_1, \ldots, X_n$, and let $B = k[X_1, \ldots, X_{n-1}]$. The Noether normalization theorem says that there is a *finite* mapping

$$X \to \operatorname{Spec}(A).$$

It is enough to prove the statement of the theorem for the direct image of $\mathscr{F}$ on $\operatorname{Spec}(A)$. Thus we may assume that $X = \operatorname{Spec}(k[X_1, \ldots, X_n])$. But then,

using the induction hypothesis and the Leray spectral sequence, we can deduce the assertion in the theorem from:

9.3 Lemma. *Consider the natural mapping*

$$f\colon X=\operatorname{Spec} B[X_n]\to \operatorname{Spec}(B)=Y.$$

Let a be a geometric point of $\operatorname{Spec}(B)$ *with* $\dim \bar{a}>n-q$. *Then*

$$(R^q f_*\mathscr{F})_a=0.$$

Proof. We distinguish two cases:

a) $\dim \bar{a}>0$. Let $\mathfrak{p}$ be the prime ideal of B associated with the geometric point a. By hypothesis we have $\dim \bar{a}=s=$ transcendence degree$_k$ $(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})$. We select elements $t_1,\ldots,t_s\in B$ whose images in $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ form a separating transcendence basis of $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ (we have assumed k algebraically closed).

Let K be the separable algebraic closure of $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$. Then $\tilde{B}=B\otimes_{k(t_1,\ldots,t_s)}K$ is an affine K-algebra. Since $t_1,\ldots,t_s$ are algebraically independent over k, we have

$$\dim \tilde{B}=\dim B-s=n-1-s\leqq n-2.$$

There is a geometric point

$$\tilde{a}\colon \operatorname{Spec}(K)\to\operatorname{Spec}(\tilde{B})$$

of $\operatorname{Spec}(\tilde{B})$ lying over $a\colon \operatorname{Spec}(K)\to\operatorname{Spec}(B)$. Since K is separable over $k(t_1,\ldots,t_s)$, $\tilde{B}$ is a direct limit of etale algebras over B. Therefore the strict Henselization of $\tilde{B}$ with respect to $\tilde{a}$ coincides with that of B with respect to a. Set $S=\operatorname{Spec}(\tilde{B})$, $T=\operatorname{Spec}(\tilde{B}[X_n])$, and $\tilde{\mathscr{F}}=$ the inverse image of $\mathscr{F}$ on T. Consider

$$\tilde{f}=f\times_Y S\colon T\to S.$$

Then

$$(R^q f_*\mathscr{F})_{\tilde{a}}=(R^q\tilde{f}_*\tilde{\mathscr{F}})_{\tilde{a}}=\varinjlim_{U\to S} H^q(U\times_S T,\tilde{\mathscr{F}}_U).$$

Here $\tilde{\mathscr{F}}_U$ is the inverse image of $\tilde{\mathscr{F}}$ on $U\times_S T$, and U runs through the affine etale neighborhoods of a.

$$\dim U\times_S T=\dim S+1=n-s<n,\qquad n-s<n-(n-q)=q.$$

By the induction hypothesis, $H^q(U\times_S T,\tilde{\mathscr{F}}_U)$ vanishes for all U, and hence so does $(R^q f_*\mathscr{F})_a$.

b) $\dim \bar{a}=0$, $q>n\geqq 2$. We consider the natural open embedding

$$j\colon X\hookrightarrow \mathbb{P}^1\times_{\operatorname{Spec}(k)} Y$$

of $X=\operatorname{Spec}(B[X_n])$ in the product of $Y=\operatorname{Spec}(B)$ with the projective line. Let

$$\bar{f}\colon \mathbb{P}^1\times_{\operatorname{Spec}(k)} Y\to Y$$

be the projection onto Y, $\bar{f}\circ j=f$.

Because of the Leray spectral sequence, it is enough to show

$$R^r\bar{f}_*(R^s j_*\mathscr{F})=0\quad\text{for } r+s=q>n\geqq 2.$$

For $s=0$: $R^r\bar{f}_*(-)=0$ for $r>2$ by (8.8).

For $s>0$: $R^s j_*(-)$ is concentrated on the complement "$\infty \times Y$" $\cong Y$ of X in $\mathbb{P}^1 \times_{\mathrm{Spec}(k)} Y$. It is enough then to assume that

$$r=0, \qquad s=q>n.$$

We must show that

$$R^q j_*(\mathscr{F})_{(\infty, a)}=0$$

for every "closed" geometric point a of Y. We have

$$R^q j_*(\mathscr{F})_{(\infty, a)}=H^q(\mathrm{Spec}(R_t), \mathscr{G}),$$

where $t=1/X_n$, R is the strict Henselization of the local ring of $\mathbb{P}^1 \times_{\mathrm{Spec}(k)} Y$ at the point (∞, a), and $\mathscr{G}$ is the inverse image of $\mathscr{F}$ on $\mathrm{Spec}(R_t)$. We can apply Lemma 9.2. □

Using the Poincaré duality theorem (Chap. II, § 1), one obtains a dual result for the cohomology groups with compact support. Let X be a smooth affine algebraic variety of dimension n over a separably closed field k, and let $\mathscr{F}$ be a locally constant torsion sheaf on X with torsion relatively prime to the characteristic of k. Then

$$H_c^v(X, \mathscr{F})=0 \qquad \text{for } v \leqq n-1.$$

9.4 Corollary ("The Weak Lefschetz Theorem"). *Let $\bar{X} \subset \mathbb{P}^N$ be a projective scheme of dimension n over a separably closed field k. Let $\mathscr{F}$ be a torsion sheaf on $\bar{X}$ with torsion relatively prime to the characteristic of k. Let $h \subset \mathbb{P}^N$ be a hyperplane, $A=\bar{X} \cap h$ a hyperplane section. Suppose the complement X of A in $\bar{X}$ is smooth, and the restriction of $\mathscr{F}$ to X is locally constant. Then the mappings*

$$H^v(\bar{X}, \mathscr{F}) \to H^v(A, \mathscr{F} \mid A)$$

are bijective for $v<n-1$ and injective for $v=n-1$.

For the proof. Use the long exact sequence

$$H_c^v(X, \mathscr{F}) \to H^v(\bar{X}, \mathscr{F}) \to H^v(A, \mathscr{F} \mid A) \to \ldots . \quad \square$$

§ 10. Purity Theorems

In what follows, $\Lambda=\mathbb{Z}/m\mathbb{Z}$ will be fixed. We consider only sheaves of Λ-modules and only schemes on which m is invertible.

Let X be a scheme and Y a closed subscheme of X,

$$j: X-Y=U \to X.$$

Further, let $\mathscr{F}$ be an etale sheaf on X. We then have the functors Γ_Y (sections with support in Y) and $\boldsymbol{\Gamma}_Y$ (sheaf of sections with support in Y):

$$\Gamma_Y \mathscr{F}=\mathrm{Ker}(\mathscr{F}(X) \to \mathscr{F}(U)),$$
$$\boldsymbol{\Gamma}_Y \mathscr{F}=\mathrm{Ker}(\mathscr{F} \to j_* j^* \mathscr{F}).$$

For the derived functors of Γ_Y we also write

$$H^v_Y(X, \mathscr{F}) = R^v \Gamma_Y(\mathscr{F}).$$

We have the usual long exact sequences

$$0 \to \Gamma_Y(\mathscr{F}) \to \mathscr{F}(X) \to \mathscr{F}(U) \to H^1_Y(X, \mathscr{F}) \to H^1(X, \mathscr{F}) \to H^1(U, \mathscr{F}) \to \dots,$$

$$0 \to \Gamma_Y(\mathscr{F}) \to \mathscr{F} \to j_* j^* \mathscr{F} \to R^1 \Gamma_Y(\mathscr{F}) \to 0$$

$$R^v j_* j^* \mathscr{F} \cong R^{v+1} \Gamma_Y(\mathscr{F}) \quad \text{for } v \geqq 1.$$

10.1 Theorem. *Let $X \to S$ and $Y \to S$ be smooth schemes over a base scheme S, with*

$$\begin{array}{ccc} Y & \xrightarrow{j} & X \\ & \searrow & \downarrow \\ & & S \end{array}$$

a closed embedding over S. Suppose Y has constant codimension d in all geometric fibers of $X \to S$. Let $\mathscr{F}$ be a locally constant constructible sheaf on X. Then

$$R^v \Gamma_Y \mathscr{F} = \begin{cases} 0 & \text{for } v \neq 2d \\ (R^{2d} \Gamma_Y \Lambda_X) \otimes \mathscr{F} & \text{for } v = 2d. \end{cases}$$

The sheaf $R^{2d} \Gamma_Y \Lambda_X$ is concentrated on Y, locally constant there, and (etale) locally isomorphic to Λ_Y. This $R^{2d} \Gamma_Y \Lambda_X$ is invariant under base change of S.

Proof. For simplicity, we shall assume in addition that the stalks of $\mathscr{F}$ are free Λ-modules. Let a be a point of Y. Let A and B denote the strict Henselizations of the local rings of X and Y at a, and let C be the strict Henselization of the local ring of S at the image point of a,

$$\begin{array}{ccc} B & \longleftarrow & A \\ & \nwarrow & \uparrow \\ & & C. \end{array}$$

Since B is smooth over C, it follows from Lemma 7.1 that there is a section

$$\begin{array}{ccc} B & \longleftarrow & A \\ & \nwarrow & \uparrow \\ & & C. \end{array}$$

It is easy to see that A is also smooth over B. Since A and B have the same residue fields, it follows from 7.1 that

$$B = A\{T_1, \dots, T_d\}.$$

Theorem 10.1 is of local nature with respect to X. The sheaves $R^v \Gamma_Y \mathscr{F}$ have support in Y.

For the proof of 10.1, we can (after these preliminary remarks) assume without loss of generality that we have the following special situation:

$$X = \mathbb{A}_S^d \quad \text{(affine space of dimension } d \text{ over } S\text{)},$$
$$Y = S \hookrightarrow X = \mathbb{A}_S^d$$
$$s \mapsto \text{“}(s, 0)\text{”}.$$

Moreover, we can assume that

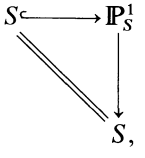
$$\mathscr{F} = \Lambda_X.$$

For a closed imbedding $i: Z \hookrightarrow X$, a closed subscheme $Y \subseteq Z$, and a sheaf $\mathscr{F}$ on Z we have

$$\Gamma_Y = \Gamma_Y \circ \Gamma_Z,$$
$$\mathrm{R}\Gamma_Y = \mathrm{R}\Gamma_Y \circ \mathrm{R}\Gamma_Z$$
$$i_* \mathrm{R}\Gamma_Y(\mathscr{F}) = \mathrm{R}\Gamma_Y\, i_*(\mathscr{F}).$$

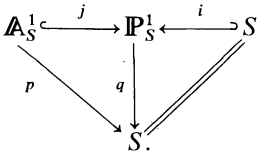
Thus it is enough to treat the case $d = 1$.

Instead of $S \hookrightarrow \mathbb{A}_S^1$, we can treat the case



(“mapping to the point at infinity”)

with $\mathbb{P}_S^1 - S = \mathbb{A}_S^1$:



10.2 Lemma. *For every complex $\mathscr{R}^\bullet$ of sheaves on S bounded below, the natural mapping*

$$\mathscr{R}^\bullet \to \mathrm{R}p_*\, p^* \mathscr{R}^\bullet$$

is a quasi-isomorphism.

The theorem follows from the lemma:

One shows first, by a connectedness argument, that

$$j_*(\Lambda_{\mathbb{A}_S^1}) = \Lambda_{\mathbb{P}_S^1}.$$

$R^\nu j_* \Lambda_{\mathbb{A}_S^1}$ is concentrated on $S \subset \mathbb{P}_S^1$ for $\nu > 0$, and hence

$$R^\mu q_* R^\nu j_* \Lambda_{\mathbb{A}_S^1} = \begin{cases} 0 & \text{for } \mu > 0,\ \nu > 0, \\ R^\nu j_* \Lambda_{\mathbb{A}_S^1} | S & \text{for } \mu = 0,\ \nu > 0. \end{cases}$$

Finally

$$R^v q_* \Lambda_{\mathbb{P}^1_S} = \begin{cases} 0 & \text{for } v \neq 0, 2 \\ \Lambda_S & \text{for } v = 0, \\ \mu_{mS}^{-1} & \text{for } v = 2 \text{ (Kummer sequence).} \end{cases}$$

Applying the Leray spectral sequence for $q \circ j = p$, we get

$$R^v j_* \Lambda_{\mathbb{A}^1_S}|_S = \begin{cases} 0 & \text{for } v > 1, \\ \mu_{mS}^{-1} & \text{for } v = 1. \end{cases}$$

From this we get

$$R^v \Gamma_S \Lambda_{\mathbb{P}^1_S} = \begin{cases} 0 & \text{for } v \neq 2, \\ i_* \mu_{mS}^{-1} & \text{for } v = 2. \end{cases}$$

This isomorphism is compatible with base extension.

Proof of the lemma. It is sufficient to suppose that $\mathscr{R}^\cdot = \mathscr{F}$ is a constructible sheaf. The problem is of local nature with respect to S. Hence the assertion will follow for a base scheme S once it is proved for all strict Henselizations at all points of S. In particular, we can assume that S has finite Krull dimension. We make an induction on the dimension $d = d(S)$ of S. We can assume that S is the spectrum of a reduced strictly Henselian ring R. Let s be the unique closed point, $U = S - \{s\}$.

If $d = 0$, then R is a separably algebraically closed field. Thus $\mathrm{Pic}(\mathbb{A}^1_S) = 0$, $H^0(\mathbb{A}^1_S, \mathcal{O}_{\mathbb{A}^1_S}) = R^*$, and hence $H^1(\mathbb{A}^1_S, \mathbb{Z}/r\mathbb{Z}) = 0$ for r relatively prime to the characteristic. From this follows the vanishing of $H^1(\mathbb{A}^1_S, F)$ for all Λ-modules F. In addition, $H^v(\mathbb{A}^1_S, F) = 0$ for $v \geqq 2$.

Say $d > 0$. Then $\dim U < d$. We consider the natural diagram

$$\begin{array}{ccc} \mathbb{A}^1_U & \xrightarrow{\bar{j}} & \mathbb{A}^1_S \\ \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle p} \\ U & \xrightarrow{j} & S. \end{array}$$

We consider the mapping cone $\mathscr{K}^\cdot$ for the natural mapping $\mathscr{F} \to \mathrm{R}j_* j^* \mathscr{F}$ and the admissible triangle

$$\begin{array}{ccc} & \mathscr{K}^\cdot & \\ \swarrow & & \nwarrow \\ \mathscr{F} & \longrightarrow & \mathrm{R}j_* j^* \mathscr{F}. \end{array}$$

Since the mapping $\mathscr{F} \to \mathrm{R}j_* j^* \mathscr{F}$ is a quasi-isomorphism outside s, the cohomology sheaves of $\mathscr{K}^\cdot$ are concentrated on s. Therefore

$$\mathscr{K}^\cdot \to \mathrm{R}p_* p^* \mathscr{K}^\cdot$$

is a quasi-isomorphism. It remains to show that

$$\mathrm{R}j_* j^* \mathscr{F} \to \mathrm{R}p_* p^* \mathrm{R}j_* j^* \mathscr{F}$$

is also a quasi-isomorphism.

It follows from the smooth base change theorem that $p^* \mathrm{R} j_* j^* \mathscr{F}$ is quasi-isomorphic to $\mathrm{R}\bar{j}_* q^* j^* \mathscr{F}$.

By the induction hypothesis ($\dim U < \dim S$!), the natural mapping

$$j^* \mathscr{F} \to \mathrm{R} q_* \, q^* j^* \mathscr{F}$$

is a quasi-isomorphism, and hence so is

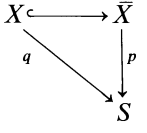
$$\mathrm{R} j_* j^* \mathscr{F} \to \mathrm{R} j_* \mathrm{R} q_* \, q^* j^* \mathscr{F}.$$

But we have

$$\begin{aligned} \mathrm{R} j_* \mathrm{R} q_* \, q^* j^* \mathscr{F} &\cong \mathrm{R} p_* \mathrm{R}\bar{j}_* q^* j^* \mathscr{F} \\ &\cong \mathrm{R} p_* \, p^* \mathrm{R} j_* j^* \mathscr{F} \end{aligned}$$

in the derived category. □

10.3 Corollary. *Suppose we have a diagram*



with X open in $\bar{X}$, $Y = \bar{X} - X$ smooth over S, and $\bar{X} \to S$ smooth and proper. Let $\bar{\mathscr{G}}$ be locally constant and constructible on $\bar{X}$, $\mathscr{G} = \bar{\mathscr{G}} \mid X$. Then $R^\nu q_* \mathscr{G}$ is locally constant and constructible.

§ 11. Comparison Theorems Between Etale Cohomology and Singular Cohomology

In this section we consider finitely generated schemes over the field $\mathbb{C}$ of complex numbers. It is known then that there is a naturally constructed functor from the category of these schemes to the category of complex spaces; it assigns a complex space X_{an} to each scheme X, and to each morphism $f: X \to Y$ of schemes it assigns a holomorphic mapping $f_{\mathrm{an}}: X_{\mathrm{an}} \to Y_{\mathrm{an}}$. There is a natural morphism of local ringed spaces

$$X_{\mathrm{an}} \to X,$$

compatible with the functor, that induces an isomorphism

$$\mathrm{Hom}(X, Z) \to \mathrm{Hom}(X_{\mathrm{an}}, Z)$$

for every complex space Z (the morphisms here are in the category of local ringed spaces). In particular, we identify the set $X_{\mathbb{C}}(\mathbb{C}) = \mathrm{Hom}_{\mathrm{Spec}(\mathbb{C})}(\mathrm{Spec}(\mathbb{C}), X)$ of geometric points of X having values in $\mathbb{C}$ ($\cong$ the set of closed points of X) with the points of X_{an}.

Let $p: U \to X$ be an etale scheme over the scheme X. Then $p_{\mathrm{an}}: U_{\mathrm{an}} \to X_{\mathrm{an}}$ is a local isomorphism. Here $U \to X$ is an isomorphism if and only if $U_{\mathrm{an}} \to X_{\mathrm{an}}$ is an isomorphism. Now a scheme is connected in the Zariski topology if and

only if the associated analytic space is connected. Hence the connected components of U (and thus also the open and closed subsets of U) that are mapped isomorphically onto X by p correspond bijectively to the open and closed subsets of U_{an} that are mapped isomorphically onto X_{an}. Thus

$$\mathrm{Hom}_X(X, U) \cong \mathrm{Hom}_{X_{\text{an}}}(X_{\text{an}}, U_{\text{an}}).$$

Since $X \to X_{\text{an}}$ is compatible with cartesian products, we can thus conclude:

11.1 Lemma. *Let $U \to X$ and $V \to X$ be etale morphisms. Then the mapping*

$$\mathrm{Hom}_X(U, V) \to \mathrm{Hom}_{X_{\text{an}}}(U_{\text{an}}, V_{\text{an}})$$

is an isomorphism.

For a complex space Z, let us denote by $\mathrm{Et}(Z)$ the category of complex spaces

$$q\colon Y \to Z$$

over Z for which q is a local isomorphism. Then the functor

$$\mathrm{Et}(X) \to \mathrm{Et}(X_{\text{an}})$$
$$(U \to X) \mapsto (U_{\text{an}} \to X_{\text{an}})$$

is a *fully faithful embedding* of categories.

Let Z be an analytic space. We shall embed the category of sheaves on Z in the category of contravariant functors

$$\mathrm{Et}(Z) \to \mathrm{Sets}.$$

Let $p\colon U \to Z$ be a local isomorphism. We set

$$\mathscr{G}(U \to Z) \text{“}=\text{”} \mathscr{G}(U) =_{\text{def}} p^*(\mathscr{G})(U).$$

For a mapping

$$\phi\colon \begin{array}{ccc} U & \longrightarrow & V \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & Z & \end{array} \quad \text{in } \mathrm{Et}(Z),$$

the map $\mathscr{G}(\phi)\colon \mathscr{G}(V) \to \mathscr{G}(U)$ is the map induced by ϕ,

$$\Gamma(V, q^*(\mathscr{G})) \to \Gamma(U, p^*(\mathscr{G})).$$

Remark. a) This functor

$$\mathrm{Et}(Z) \to \mathrm{Sets}$$

can also be defined in a different way:

To a space $p\colon Y \to Z$ in $\mathrm{Et}(Z)$ we can assign a sheaf $\tilde{Y}$ on Z,

$$\tilde{Y}(U) = \mathrm{Hom}_X(U, Y) \qquad U \text{ open in } Z.$$

There is then a natural isomorphism

$$\mathscr{G}(Y) = p^*(\mathscr{G})(Y) \xrightarrow{\cong} \mathrm{Hom}(\tilde{Y}, \mathscr{G})$$

for every sheaf $\mathscr{G}$ on Z.

b) Every mapping between two sheaves on Z extends in a natural way to a natural transformation of contravariant functors on $\mathrm{Et}(Z)$. We obtain a fully faithful embedding of the category of sheaves on Z into the category of contravariant functors

$$\mathrm{Et}(Z) \to \mathrm{Sets}.$$

c) If $\mathcal{G}$ is a sheaf of (abelian) groups, then the functor extended to $\mathrm{Et}(Z)$ is in a natural way a functor with values in the category of (abelian) groups.

11.2 Lemma. *Let $\mathcal{G}$ be a sheaf on the analytic space Z. Then the following "sheaf properties" hold for the functor extension of $\mathcal{G}$ to* $\mathrm{Et}(Z)$*:*

1) *Let $Y_1 \to Z$ and $Y_2 \to Z$ be in* $\mathrm{Et}(Z)$*. Then $\mathcal{G}(Y_1 \cup Y_2) = \mathcal{G}(Y_1) \times \mathcal{G}(Y_2)$. (Similarly for an infinite disjoint union.)*

2) *Let $U \to V$ be a surjective mapping in* $\mathrm{Et}(Z)$*. Then the induced sequence*

$$\mathcal{G}(V) \to \mathcal{G}(U) \rightrightarrows \mathcal{G}(U \times_V U)$$

is exact.

Proof. Part (1) is clear.

Part (2) follows from $\tilde{U} \to \tilde{V}$ is a surjective mapping of sheaves, and we have $\widetilde{U \times_V U} = \tilde{U} \times_{\tilde{V}} \tilde{U}$. Hence

$$\mathrm{Hom}(\tilde{V}, G) \to \mathrm{Hom}(\tilde{U}, G) \rightrightarrows \mathrm{Hom}(\tilde{U} \times_{\tilde{V}} \tilde{U}, G)$$

is exact. □

Now let X be a scheme over $\mathbb{C}$. There is then a "direct image functor" from the category of sheaves on X_{an} to the category of etale sheaves,

$$\mathcal{G} \mapsto \mathcal{G}_{\mathrm{al}}:$$

for $U \to X$ etale,

$$\mathcal{G}_{\mathrm{al}}(U) = \mathcal{G}(U_{\mathrm{an}}),$$

and for a mapping $\phi\colon V \to U$ in $\mathrm{Et}(X)$,

$$\mathcal{G}(\phi) = \mathcal{G}(\phi_{\mathrm{an}})\colon\ \mathcal{G}_{\mathrm{al}}(U) = \mathcal{G}(U_{\mathrm{an}}) \to \mathcal{G}_{\mathrm{al}}(V) = \mathcal{G}(V_{\mathrm{an}}).$$

The functor $\mathcal{G}_{\mathrm{al}}$ is an etale sheaf by Lemma 11.2. Similarly, every mapping of sheaves on X_{an} defines an natural mapping of the associated etale sheaves on $\mathrm{Et}(X)$.

11.3 Proposition and Definition. *The functor just constructed has a left adjoint functor, the "inverse image functor"*

$$\mathcal{G} \mapsto \mathcal{G}_{\mathrm{an}}$$

from the category of etale sheaves on X to the category of sheaves on X_{an}. Adjointness property:

$$\mathrm{Hom}(\mathcal{G}_{\mathrm{an}}, \mathcal{F}) = \mathrm{Hom}(\mathcal{G}, \mathcal{F}_{\mathrm{al}})$$

for sheaves $\mathcal{F}$ on X_{an}.

This functor is defined on the category of representable sheaves as follows:

Let $Y \to X$ be etale, $\tilde{Y}$ the associated etale sheaf

$$\tilde{Y}(-) = \mathrm{Hom}(-, Y).$$

Then $(\tilde{Y})_{an}$ *is analytically isomorphic to* $(\widetilde{Y_{an}})$, *that is, to the sheaf defined by the analytic space* $Y_{an} \to X_{an}$.

Remark. The functor $\mathcal{G} \to \mathcal{G}_{an}$ induces a functor from the category of sheaves of (abelian) groups to the category of sheaves of (abelian) groups. For a sheaf of groups is a group object in the category of all sheaves of sets, and the functor $\mathcal{G} \mapsto \mathcal{G}_{an}$ is compatible with finite cartesian products (11.4). For a finite group M, we have $(M_X)_{an} = M_{X_{an}}$.

Proof. Let $\mathcal{G}$ be an etale sheaf on X. It is enough to prove that the covariant functor

$$\mathcal{F} \mapsto \mathrm{Hom}(\mathcal{G}, \mathcal{F}_{al})$$

from the category of sheaves on X_{an} to the category of sets is representable by a sheaf $\mathcal{G}_{an}$ on X_{an}.

a) Let $\mathcal{G}$ be representable. There is thus an etale scheme $Y \to X$ with $\tilde{Y} = \mathcal{G}$. Then

$$\mathrm{Hom}(\tilde{Y}, \mathcal{F}_{al}) = \mathcal{F}_{al}(Y) = \mathcal{F}(Y_{an}) = \mathrm{Hom}(\widetilde{Y_{an}}, \mathcal{F}).$$

Thus $\widetilde{Y_{an}}$ is the desired sheaf.

b) An arbitrary etale sheaf $\mathcal{G}$ is a direct (unfiltered) limit of representable sheaves $\mathcal{G}_\alpha = \tilde{Y}_\alpha$,

$$\mathcal{G} = \varinjlim_\alpha \tilde{Y}_\alpha .$$

Then by a) the sheaf

$$\mathcal{G}_{an} = \varinjlim_\alpha (\widetilde{Y_{\alpha an}})$$

has the desired property.

11.4 Proposition. *The functor* $\mathcal{G} \mapsto \mathcal{G}_{an}$ *has the following properties:*

(1) *For an etale scheme* $U \to X$ *over the base scheme* X, *there is a natural mapping*

$$\begin{array}{ccc} \mathcal{G}(U) & \to & \mathcal{G}_{an}(U_{an}) \\ \| & & \| \\ \mathrm{Hom}(\tilde{U}, \mathcal{G}) & & \mathrm{Hom}(\widetilde{U_{an}}, \mathcal{G}_{an}) \end{array}$$

$$s \mapsto s_{an}$$

(2) *The functor is compatible with arbitrary direct limits.*

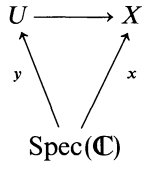
(3) *It is compatible with finite projective limits.*

(4) *It is compatible with formation of inverse images, i.e.: For a morphism* $f: X \to Y$ *of schemes over* $\mathbb{C}$ *and an etale sheaf on* Y,

$$(f_{an})^*(\mathcal{G}_{an}) = (f^*(\mathcal{G}))_{an} .$$

(5) *If* $x \in \mathrm{Hom}_{\mathrm{Spec}(\mathbb{C})}(\mathrm{Spec}(\mathbb{C}), X)$ *is a geometric point of the scheme* X, *then the stalks* $\mathcal{G}_x$ *and* $(\mathcal{G}_{an})_x$ *are canonically isomorphic.*

Remark. Realization of this last isomorphism: Let



be an etale neighborhood of x and $s \in \mathcal{G}(U)$. For a sufficiently small connected open neighborhood V of x in X_{an}, there is exactly one section $q\colon V \to U_{\mathrm{an}}$ that sends x to y. We assign to s the germ of $\mathcal{G}_{\mathrm{an}}(q)(s_{\mathrm{an}})$ in $\mathcal{G}_{\mathrm{an}x}$. Passing to the limit over all etale neighborhoods, we get the desired mapping $\mathcal{G}_x \to \mathcal{G}_{\mathrm{an}x}$.

Proof. (2) follows from the adjointness property.

(4) follows from the adjointness property and the compatibility of the functor $\mathcal{F} \mapsto \mathcal{F}_{\mathrm{al}}$ with direct images.

(5) only needs to be proved for the case where the space X consists of a single point, by (4), and it is clear then because $\mathrm{Et}(X) \cong \mathrm{Et}(X_{\mathrm{an}})$.

(3) follows from (5). □

11.5 Proposition. *Let $\mathcal{F}$ and $\mathcal{G}$ be two sheaves on a scheme X. Then the mapping*

$$\mathrm{Hom}(\mathcal{F}, \mathcal{G}) \to \mathrm{Hom}(\mathcal{F}_{\mathrm{an}}, \mathcal{G}_{\mathrm{an}})$$

is injective. It is actually bijective if $\mathcal{G}$ is constructible. In particular, for an etale scheme $U \to X$ over X and a constructible sheaf $\mathcal{G}$,

$$\mathcal{G}(U) \cong \mathcal{G}_{\mathrm{an}}(U_{\mathrm{an}}).$$

Proof. The injectivity follows from 11.4 (5). The bijectivity for a constructible sheaf $\mathcal{G}$:

It suffices to assume that $\mathcal{F}$ is representable, and thus we have to show

$$\mathcal{G}(U) \cong \mathcal{G}_{\mathrm{an}}(U_{\mathrm{an}}).$$

Say for instance $U = X$. The bijectivity is clear in any case if $\mathcal{G}$ is also representable. In the general case, $\mathcal{G}$ is at least representable on a Zariski-open dense set. It follows by noetherian induction that for two sections s, $t \in \mathcal{G}_{\mathrm{an}}(X_{\mathrm{an}})$ the set of points $x \in X_{\mathrm{an}}$ at which the germs s_x and t_x agree is constructible in the Zariski topology and hence actually Zariski-open. By 11.4 (5), then, for every $s \in \mathcal{G}_{\mathrm{an}}(X_{\mathrm{an}})$ and every geometric point $x \in \mathrm{Hom}_{\mathrm{Spec}(\mathbb{C})}(\mathrm{Spec}(\mathbb{C}), X)$, there is a Zariski-open neighborhood U and an element $t \in G(U)$ with $t_{\mathrm{an}} = s | U_{\mathrm{an}}$. From this the assertion follows easily. □

We now consider the derived category $D(X)$ for the category of all etale sheaves of abelian groups on the scheme X and the derived category $D(X_{\mathrm{an}})$ for the category of all sheaves of abelian groups on X_{an}. Since the functor $\mathcal{G} \mapsto \mathcal{G}_{\mathrm{an}}$ is exact (11.4), it automatically extends to a cohomological functor

$$D(X) \to D(X_{\mathrm{an}})$$

$$\mathcal{K}^{\cdot} \mapsto \mathcal{K}^{\cdot}_{\mathrm{an}}.$$

We now consider a morphism

$$f\colon X \to S$$

of schemes (over $\mathbb{C}$, ...) and a sheaf $\mathscr{F}$ on X. There is a natural mapping

$$f_*(\mathscr{F}) \to (f_{\mathrm{an}*}(\mathscr{F}_{\mathrm{an}}))_{\mathrm{al}}$$

given as follows:

Let $U \to S$ be etale. We have a natural mapping

$$f_*(\mathscr{F})(U) = \mathscr{F}(U \times_S X) \to \mathscr{F}_{\mathrm{an}}(\underbrace{U_{\mathrm{an}} \times_{S_{\mathrm{an}}} X_{\mathrm{an}}}_{(U \times_S X)_{\mathrm{an}}}) = f_{\mathrm{an}*}(\mathscr{F}_{\mathrm{an}})(U_{\mathrm{an}}) = (f_{\mathrm{an}*}(\mathscr{F}_{\mathrm{an}}))_{\mathrm{al}}(U).$$

The adjoint mapping is a mapping

$$f_*(\mathscr{F})_{\mathrm{an}} \to f_{\mathrm{an}*}(\mathscr{F}_{\mathrm{an}}).$$

If the sheaf $\mathscr{F}$ is a sheaf of abelian groups, then this mapping is a homomorphism of sheaves of abelian groups. It extends immediately to complexes, and by injective resolutions it yields a natural transformation of functors $D_+(X) \to D(Y_{\mathrm{an}})$.

Let $\mathscr{K}^\cdot$ be a complex in $D_+(X)$, $\mathscr{K}^\cdot \to \mathscr{I}(\mathscr{K}^\cdot)$ an injective resolution of $\mathscr{K}^\cdot$, and $\mathscr{K}^\cdot_{\mathrm{an}} \to \mathscr{I}(\mathscr{K}^\cdot_{\mathrm{an}})$ an injective resolution of $\mathscr{K}^\cdot_{\mathrm{an}}$. Since $\mathscr{K}^\cdot_{\mathrm{an}} \to (\mathscr{I}(\mathscr{K}^\cdot))_{\mathrm{an}}$ is also a quasi-isomorphism, there is a homotopy-commutative diagram

$$\begin{array}{ccc} \mathscr{K}^\cdot_{\mathrm{an}} & \longrightarrow & \mathscr{I}(\mathscr{K}^\cdot_{\mathrm{an}}) \\ & \searrow & \uparrow \\ & & (\mathscr{I}(\mathscr{K}^\cdot))_{\mathrm{an}}. \end{array}$$

Using the mapping constructed before, we get a homomorphism

$$(\mathrm{R}f_*(\mathscr{K}^\cdot))_{\mathrm{an}} = (f_*(\mathscr{I}^\cdot(\mathscr{K}^\cdot)))_{\mathrm{an}} \to f_{\mathrm{an}*}(\mathscr{I}(\mathscr{K}^\cdot_{\mathrm{an}})) = \mathrm{R}f_{\mathrm{an}*}(\mathscr{K}^\cdot_{\mathrm{an}}).$$

Passage to the cohomology sheaves yields

$$(R^\nu f_*(\mathscr{F}))_{\mathrm{an}} \to R^\nu f_{\mathrm{an}*}(\mathscr{F}_{\mathrm{an}}).$$

In the case when $S = \mathrm{Spec}(\mathbb{C})$, the mappings here constructed are equivalent to mappings

$$\mathrm{R}\Gamma(X, \mathscr{K}^\cdot) \to \mathrm{R}\Gamma(X_{\mathrm{an}}, \mathscr{K}^\cdot_{\mathrm{an}}),$$

$$H^\nu(X, \mathscr{F}) \to H^\nu(X_{\mathrm{an}}, \mathscr{F}_{\mathrm{an}}).$$

Let

$$\begin{array}{ccc} X & \overset{j}{\hookrightarrow} & \bar{X} \\ & {\scriptstyle f}\searrow & \downarrow{\scriptstyle \bar{f}} \\ & & S \end{array}$$

be a compactification of f and $\mathscr{F}$ a sheaf on X (§ 8). It is easy to see (e.g., 11.4 (5)) that $(j_!(\mathscr{F}))_{\mathrm{an}}$ is the "extension by zero" of $\mathscr{F}_{\mathrm{an}}$ to all of $\bar{X}_{\mathrm{an}}$,

$$(j_!(\mathscr{F}))_{\mathrm{an}} = j_{\mathrm{an}!}(\mathscr{F}_{\mathrm{an}}).$$

It is known that the higher direct images $R^\nu \bar{f}_{\mathrm{an}*}(j_{\mathrm{an}!}(\mathscr{F}_{\mathrm{an}}))$ are canonically isomorphic to the derived functors $R^\nu f_{\mathrm{an}!}(\mathscr{F}_{\mathrm{an}})$ of the direct image functor $f_!(\mathscr{F}_{\mathrm{an}})$ with compact support. The corresponding fact is true for complexes. Combining this with the transformations previously constructed, we thus get a natural transformation of functors

$$D_+(X, \mathrm{tor}) \to D(S_{\mathrm{an}}),$$

namely

$$(\mathsf{R}f_!(\mathscr{K}^\cdot))_{\mathrm{an}} \to \mathsf{R}f_{\mathrm{an}!}(\mathscr{K}^\cdot_{\mathrm{an}}),$$

from this again homomorphisms

$$R^\nu f_!(\mathscr{F})_{\mathrm{an}} \to R^\nu f_{\mathrm{an}!}(\mathscr{F})$$

and, in the case $S = \mathrm{Spec}(\mathbb{C})$,

$$\mathsf{R}\Gamma_c(X, \mathscr{K}^\cdot) \to \mathsf{R}\Gamma_c(X_{\mathrm{an}}, \mathscr{K}^\cdot_{\mathrm{an}}),$$

$$H^\nu_c(X, \mathscr{F}) \to H^\nu_c(X_{\mathrm{an}}, \mathscr{F}_{\mathrm{an}}).$$

We do not bother to state and derive the compatibility properties for composition of mappings and base change, nor the long exact cohomology sequence for the direct images $R^n f_!$ with compact support for an open subscheme.

11.6 Theorem (Comparison Theorem). *Let $f\colon X \to S$ be a morphism of schemes and $\mathscr{K}^\cdot$ a complex in $D_+(X, \mathrm{tor})$.*

(1) *The mapping*

$$(\mathsf{R}f_!(\mathscr{K}^\cdot))_{\mathrm{an}} \to \mathsf{R}f_{\mathrm{an}!}(\mathscr{K}^\cdot)$$

is an isomorphism in $D(X_{\mathrm{an}})$. In particular,

$$R^\nu f_!(\mathscr{F})_{\mathrm{an}} \cong R^\nu f_{\mathrm{an}!}(\mathscr{F})$$

$$H^\nu_c(X, \mathscr{F}) \cong H^\nu_c(X_{\mathrm{an}}, \mathscr{F}_{\mathrm{an}}).$$

(2) *Suppose in addition that the complex $\mathscr{K}^\cdot$ has constructible cohomology sheaves. Then the mapping*

$$(\mathsf{R}f_*(\mathscr{K}^\cdot))_{\mathrm{an}} \to \mathsf{R}f_{\mathrm{an}*}(\mathscr{K}^\cdot)$$

is an isomorphism.

We shall content ourselves with proving part (1).

For the proof of (2): Using part (1) and Hironaka's theory of the resolution of singularities of algebraic varieties, one can show that it is enough to treat the following special case of (2):

$$X \to S$$

is an open embedding in a smooth scheme S over $\mathbb{C}$, the complement of X in S is a smooth divisor, and $\mathscr{K}^\cdot = \mathscr{F} = (\mathbb{Z}/n\mathbb{Z})_X$ is a constant sheaf. One then uses the purity theorem (§ 10) and its analytic analogue.

Proof of (1). As in the proof of Theorem 8.8, we can use Theorem 8.7 and its (trivial) analytic analogue to reduce the proof of (1) to the following special case:

Let X be a smooth irreducible projective curve over $\mathbb{C}$, $\mathscr{F}=(\mathbb{Z}/n\mathbb{Z})_X$ the constant sheaf. Then the mapping

$$H^\nu(X,\mathscr{F})\to H^\nu(X_{\mathrm{an}},\mathscr{F}_{\mathrm{an}})$$

is an isomorphism for every ν. Since $\mathbb{C}$ contains the n-th roots of unity, $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to the sheaf μ_{nX} of n-th roots of unity, and thus to the kernel of

$$\begin{aligned}\mathscr{O}_X^*&\to\mathscr{O}_X^*\\ a&\mapsto a^n.\end{aligned}$$

Proof of the special case. For a locally isomorphic mapping $q\colon Z\to X_{\mathrm{an}}$, we have $q^*(\mathscr{O}_{X_{\mathrm{an}}})=\mathscr{O}_Z$. Thus

$$\mathscr{O}_{X_{\mathrm{an}}}(Z)=\mathscr{O}_Z(Z).$$

Let $U\to X$ be etale. Then there is a natural mapping

$$\mathscr{O}_X^*(U)=\mathscr{O}_U^*(U)\to\mathscr{O}_{U_{\mathrm{an}}}^*(U_{\mathrm{an}})=\mathscr{O}_{X_{\mathrm{an}}}^*(U_{\mathrm{an}})=(\mathscr{O}_{X_{\mathrm{an}}}^*)_{\mathrm{al}}(U).$$

Thus we have constructed a homomorphism

$$\mathscr{O}_X^*\to(\mathscr{O}_{X_{\mathrm{an}}}^*)_{\mathrm{al}}.$$

The adjoint mapping is a homomorphism

$$(\mathscr{O}_X^*)_{\mathrm{an}}\to(\mathscr{O}_{X_{\mathrm{an}}}^*).$$

This homomorphism induces homomorphisms

$$H^\nu(X,\mathscr{O}_X^*)\to H^\nu(X_{\mathrm{an}},\mathscr{O}_{X_{\mathrm{an}}}^*).$$

Assertion. These homomorphisms are isomorphisms.

$$\begin{aligned}&\nu\geqq 2\colon\quad H^\nu(X_{\mathrm{an}},\mathscr{O}_{X_{\mathrm{an}}}^*)=H^\nu(X,\mathscr{O}_X^*)=0\\ &\nu=0\colon\quad H^0(X_{\mathrm{an}},\mathscr{O}_{X_{\mathrm{an}}}^*)=H^0(X,\mathscr{O}_X^*)=\mathbb{C}^*\end{aligned}\tag{5.2}$$

$n=1$: Using Čech cohomology, one shows that the homomorphism

$$H^1(X,\mathscr{O}_X^*)\to H^1(X_{\mathrm{an}},\mathscr{O}_{X_{\mathrm{an}}}^*)$$

is equivalent to the natural homomorphism

$$\mathrm{Pic}(X)\to\mathrm{Pic}(X_{\mathrm{an}}).$$

But it follows from the GAGA theorems of Serre that this homomorphism is an isomorphism.

Using the Kummer sequence, one then gets the desired isomorphism statement for the sheaf $\mathscr{F}=\mu_{nX}$. □

§12. l-adic Sheaves

Up to this section, we have been concerned essentially only with torsion sheaves, usually even with a restrictive hypothesis on the torsion; all our main results have been proved only for such sheaves. But naturally the cohomology groups of such sheaves are torsion groups. Thus we have not reached our goal of obtaining Weil cohomology groups with coefficients in a field of characteristic zero. This failing cannot be remedied by using more general sheaves.

Indeed, one can show the following:

Let X be a finitely generated scheme of dimension n over an algebraically closed base field, and let $\mathscr{F}$ be an arbitrary etale sheaf on X. Then the cohomology groups $H^v(X, \mathscr{F})$ for $v > n$ are torsion groups. If X is normal and $\mathscr{F}$ is a locally constant sheaf, then in fact all the cohomology groups

$$H^v(X, \mathscr{F}) \quad \text{for } v > 0$$

are torsion groups. The reason for this is that all higher Galois cohomology groups for an arbitrary Galois module are torsion groups.

An illuminating example is given by a smooth projective connected curve X of genus g over an algebraically closed base field k. Let l be a prime number different from the characteristic of k, let $\mathbb{Z}_l$ be the ring of l-adic integers, and let $\mathbb{Q}_l$ be the field of fractions of $\mathbb{Z}_l$. By the methods of Section 5, one can show

$$H^v(X, \mathbb{Q}_{lX}) = \begin{cases} \mathbb{Q}_l & \text{for } v = 0 \\ 0 & \text{for } v > 0, \end{cases}$$

$$H^v(X, \mathbb{Z}_{lX}) = \begin{cases} \mathbb{Z}_l & \text{for } v = 0 \\ 0 & \text{for } v = 1 \\ (\mathbb{Q}_l/\mathbb{Z}_l)^{2g} & \text{for } v = 2 \\ \mathbb{Q}_l/\mathbb{Z}_l & \text{for } v = 3 \\ 0 & \text{for } v > 3. \end{cases}$$

However, a different approach leads to the "right" cohomology groups:

$$\varprojlim_n H^v(X, (\mathbb{Z}/l^n\mathbb{Z})_X) = \begin{cases} \mathbb{Z}_l & \text{for } v = 0 \\ (\mathbb{Z}_l)^{2g} & \text{for } v = 1 \\ \mathbb{Z}_l & \text{for } v = 2 \\ 0 & \text{for } v > 2 \end{cases}$$

$$\varprojlim_n H^v(X, (\mathbb{Z}/l^n\mathbb{Z})_X) \otimes \mathbb{Q}_l = \begin{cases} \mathbb{Q}_l \\ \mathbb{Q}_l^{2g} \\ \mathbb{Q}_l \\ 0. \end{cases}$$

The first projective limit is thus not the same as

$$H^v(X, \varprojlim (\mathbb{Z}/l^n\mathbb{Z})_X),$$

for we have

$$\varprojlim (\mathbb{Z}/l^n\mathbb{Z})_X = \mathbb{Z}_{lX}\,.$$

More generally, in what follows we shall consider certain projective systems of torsion sheaves, and we shall define the cohomology groups of these systems as projective limits of the cohomology groups of the individual members of the system. As the previous example shows, we are "forbidden" to replace these by the cohomology groups of the projective limits of the sheaves. We must remain in the category of projective systems.

The crucial theorem of this section is the finiteness theorem for the proper direct images of A−R l-adic projective systems. All the other theorems are concerned with deriving identities between sheaves and can be deduced fairly easily by applying the corresponding theorems for torsion sheaves to the members of the projective system. We mention only the most important ones.

At the end of the section, cohomology groups with coefficients in $\mathbb{Q}_l$ will be constructed from the groups with coefficients in $\mathbb{Z}_l$ by introducing denominators.

In what follows, l will be a fixed prime number. We shall only consider schemes on which this prime is invertible. Let

$$\Lambda_n = \mathbb{Z}/l^{n+1}\mathbb{Z},$$

and let

$$\Lambda = \varprojlim_n \Lambda_n$$

be the ring of l-adic integers.

We first want to introduce a category of projective systems of torsion Λ-modules that will be equivalent to the category of finite Λ-modules. We first consider the abelian category of projective systems

$$F = (F_n, u_n)_{n\in\mathbb{Z}} \text{“}=\text{”} (F_n), \qquad u_n\colon\ F_n \to F_{n-1}$$

where the F_n are torsion Λ-modules with $F_n = 0$ for $n \leqq n_0$. For each integer r we define the *shifted* system

$$F[r] = (F_{n+r}, u_{n+r})_{n\in\mathbb{Z}}.$$

12.1 Definition. The projective system $F = (F_n)$ satisfies

a) the Mittag-Leffler condition (abbreviated ML), if for every n there is some $t \geqq n$ with

$$\text{Image}(F_m \to F_n) = \text{Image}(F_t \to F_n) \qquad \text{for all } m \geqq t;$$

b) the Mittag-Leffler-Artin-Rees condition (abbreviated MLAR), if there is some $t \geqq 0$ with

$$\text{Image}(F[r] \to F) = \text{Image}(F[t] \to F) \qquad \text{for } r \geqq t.$$

We further call such a system $F = (F_n)$ a null system if there is a $t \geqq 0$ for which

$$F[t] \to F$$

is the zero mapping.

If we have a short exact sequence of projective systems,

$$0 \to (F_n) \to (G_n) \to (H_n) \to 0,$$

and the system (F_n) of kernels satisfies the condition ML, then the sequence

$$0 \to \varprojlim_n F_n \to \varprojlim_n G_n \to \varprojlim_n H_n \to 0$$

is exact.

Example. If all the modules F_n are of finite length, the condition ML is automatically satisfied.

We now form the quotient category of this category of projective systems by the subcategory of null systems.

12.2 Definition of the A – R category of projective systems. The objects of this category are again the systems considered above. The morphisms in this category are

$$\mathrm{Hom}_{\mathrm{AR}}(F, G) = \varinjlim_{r \geqq 0} \mathrm{Hom}(F[r], G).$$

Composition is carried out in the obvious way.

An A – R morphism $F \to G$ is thus realized by a morphism of projective systems

$$f\colon\ F[r] \to G.$$

The kernel and cokernel of this homomorphism, formed in the category of projective systems, are also the kernel and cokernel in the A – R category. Thus the A – R category is abelian.

We also remark that f represents an isomorphism in the A – R category, an A – R *isomorphism*, if and only if its kernel and cokernel are null systems.

12.3 Definition. A projective system $F = (F_n, u_n)$ is called an l-adic system when the following hold:

a) $F_n = 0 \quad n < 0$,

b) $l^{n+1} F_n = 0$ for all n,

c) For all $n > 0$, u_n induces an isomorphism

$$F_{n+1} \otimes_{\Lambda_{n+1}} \Lambda_n = F_{n+1}/l^{n+1} F_{n+1} \cong F_n,$$

d) For all n, F_n is a module of finite length.

The projective system $(F_n) = F$ is called A – R l-adic, if it is A – R-isomorphic to an l-adic system.

It is easy to see: Let F be an l-adic system, and let $G = (G_n)$ be a system with $l^{n+1} G_n = 0$ for all n. Then

$$\mathrm{Hom}(F, G) = \mathrm{Hom}_{\mathrm{AR}}(F, G).$$

Let $F = (F_n)$ be a projective system. Obviously then $\varprojlim_n F_n = \varprojlim_n F_{n+r}$. The projective limit

$$F = (F_n) \mapsto \varprojlim F = \varprojlim_n F_n$$

is hence in a natural way a functor from the A−R category of projective systems to the category of Λ-modules. It is exact on the subcategory of systems (F_n) with finite modules F_n, since for such modules the Mittag-Leffler condition is automatically satisfied.

Now let $F=(F_n)$ be an l-adic system. It is easy to see that

$$M=\varprojlim_n F_n$$

is a finite Λ-module, and we have

$$F_n=M/l^{n+1}M.$$

The functor $\varprojlim$ thus establishes an equivalence between the A−R category of all A−R l-adic system and the category of finitely generated Λ-modules. Let $\phi\colon M\to N$ be a homomorphism of finitely generated Λ-modules, K the kernel of this homomorphism, and $\bar{K}$ the cokernel. By the Artin-Rees lemma, then, the homomorphism

$$K/l^{n+1}K\to \mathrm{Ker}\,(M/l^{n+1}M\to N/l^{n+1}N)$$

of projective systems is an A−R isomorphism. Moreover, the sequence

$$M/l^{n+1}M\to N/l^{n+1}N\to \bar{K}/l^{n+1}\bar{K}\to 0$$

is exact for all n. The kernel and cokernel of a homomorphism of l-adic systems are thus A−R l-adic.

12.4 Proposition. *The projective limit* $\varprojlim F=\varprojlim_n F_n$ *establishes an equivalence between the* A−R *category of all* A−R l*-adic projective systems* F *and the category of finitely generated* Λ*-modules. The category of* A−R l*-adic projective systems is an exact subcategory of the* A−R *category: the kernel and cokernel of an* A−R *homomorphism of* A−R l*-adic systems are again* A−R l*-adic. Moreover, let* $0\to F\to G\to H\to 0$ *be a short exact sequence in the* A−R *category. Suppose that* $l^{n+t}G_n=0$ *for some appropriate* t *and all* n*. If then* F *and* H *are* A−R l*-adic, so is* G*.*

12.5 Lemma. *Consider a projective system of complexes of* Λ*-modules,*

$$(K_n^\bullet),\qquad K_n^\bullet=0 \text{ for } n<0.$$

Suppose $K_n^\bullet$ *is annihilated by* l^{n+1} *and is a complex of flat* Λ_n*-modules. Suppose the complexes are uniformly bounded:*

$$K_n^\nu=0 \quad \text{when } \nu>r \text{ or } \nu<-r.$$

Suppose the cohomology modules $H^\nu(K_n^\bullet)$ *are finite. Suppose that*

$$K_{n+1}^\bullet\otimes_{\Lambda_{n+1}}\Lambda_n\to K_n^\bullet$$

is a quasi-isomorphism for each $n=0,\ldots$*. Then there is a bounded complex* $K_n^\bullet$ *of finitely generated free* Λ*-modules together with quasi-isomorphisms*

$$K^\bullet/l^{n+1}K^\bullet\to K^\bullet$$

for which the diagram

$$K^\cdot/l^{n+2}K^\cdot \to K^\cdot_{n+1}$$
$$K^\cdot/l^{n+1}K^\cdot \to K^\cdot_n$$

is homotopy-commutative. In particular, the projective systems $(H^\nu(K^\cdot_n))_{n\in\mathbb{Z}}$ *of cohomology modules of this complex are* A – R *l-adic.*

For the proof. Without loss of generality we can assume: All the modules K^ν_n are themselves finite, and the connecting mappings $K_{n+1} \to K_n$ are surjective. Let $H^\nu(K^\cdot_n)=0$ for $\nu>m$. We can then actually assume: $K^\nu_n=0$ for $\nu>m$. By hypothesis then $H^m(K^\cdot_{n+1})\otimes \Lambda_n = H^m(K^\cdot_n)$. This says that $(H^m(K^\cdot_n))_{n\in\mathbb{Z}}$ is an l-adic system. We choose a surjective mapping $F \to M = \varprojlim_n H^m(K^\cdot_n)$ from a finitely generated free Λ-module onto $\varprojlim_n H^m(K^\cdot_n)$. The mapping

$$\varprojlim_n K^m_n \to \varprojlim_n H^m(K^\cdot_n) = M$$

is surjective. The homomorphism $F \to M$ can thus be lifted to a homomorphism $F \to \varprojlim_n K^m_n$. We consider the new projective system $(\bar{K}^\cdot_n)$ of complexes,

$$\bar{K}^\cdot_n\colon \ \ldots \to K^{m-2}_n \to K^{m-1}_n \oplus F/l^{n+1}F \to K^m_n \to 0.$$

It satisfies the same hypotheses as $(K^\cdot_n)$, and we have $H(\bar{K}^\cdot_n)=0$ for $\nu>m-1$. The proof of 12.5 now follows easily by induction on m.

Now let X be a scheme on which l is invertible. We want to transfer to the category of etale torsion sheaves with l-torsion the constructions carried out in the category of abelian l-torsion groups. In contrast to the case treated previously, we can no longer pass to the projective limits. As in Definition 12.2, we introduce the A – R category of projective systems $\mathscr{F}=(\mathscr{F}_n)$ of l-torsion sheaves. The sheaves $\mathscr{F}_n$ are l-torsion sheaves, i.e.

$$\mathscr{F}_n = \bigcup_\nu \operatorname{Ker}(F_n \xrightarrow{l^\nu} F_n).$$

We suppose $\mathscr{F}_n=0$ for $n<n_0$. For two systems $\mathscr{F}=(\mathscr{F}_n)$, $\mathscr{G}=(\mathscr{G}_n)$ of l-torsion sheaves, we define

$$\operatorname{Hom}_{AR}(\mathscr{F},\mathscr{G}) = \varprojlim_r \operatorname{Hom}(\mathscr{F}[r],\mathscr{G}).$$

Here again $\mathscr{F}[r]=(\mathscr{F}_{n+r})$. Definition 12.1 can be extended to the case of sheaves. We extend Definition 12.3 to sheaves:

12.6 Definition. A projective system $(\mathscr{F}_n)_{n\in\mathbb{Z}}$ of etale sheaves on a scheme X is called an l-adic sheaf if all the sheaves $\mathscr{F}_n$ are constructible and the following conditions are satisfied:

$$\mathscr{F}_n = 0 \quad \text{for } n<0,$$
$$l^{n+1}\mathscr{F}_n = 0 \quad \text{for } n\geqq 0,$$
$$\mathscr{F}_{n+1}\otimes_{\Lambda_{n+1}}\Lambda_n = \mathscr{F}_{n+1}/l^{n+1} \xrightarrow{\cong} \mathscr{F}_n.$$

A system $\mathscr{F}=(\mathscr{F}_n)$ is called an A−R l-adic sheaf if it is isomorphic in the A−R category to an l-adic sheaf. Thus there is then an l-adic sheaf $\mathscr{G}=(\mathscr{G}_n)$ and a homomorphism of projective systems

$$\mathscr{G}[r]\to\mathscr{F}$$

for which the kernel and cokernel are null systems.

Again for an l-adic sheaf $\mathscr{G}=(\mathscr{G}_n)$ and a projective system $\mathscr{F}=(\mathscr{F}_n)$ of sheaves with $l^{n+1}\mathscr{F}_n=0$ we clearly have

$$\mathrm{Hom}(\mathscr{G},\mathscr{F})=\mathrm{Hom}_{\mathrm{AR}}(\mathscr{G},\mathscr{F}).$$

Examples of l-adic sheaves:

12.7 Definition. (1) Let M be a finitely generated Λ-module. The projective system $M_X=((M/l^{n+1}M)_X)$ is an l-adic sheaf.

(2) *Let* $\mu_{l^n,X}=\mathrm{Ker}\,(\mathcal{O}_X^*\xrightarrow{l^n}\mathcal{O}_X^*)$ be the etale sheaf of l^n-th roots of unity.

$$Z_l(1)=(\mu_{l^{n+1},X};\ \mu_{l^{n+2}X}\to\mu_{l^{n+1}X}),\ Z_l(m)=(\mu_{l^{n+1},X}^{\otimes m}).$$
$$\xi\mapsto\xi^l$$

are l-adic sheafs.

(3) Let $\mathscr{F}$ be a constructible l-torsion sheaf, $l^m\mathscr{F}=0$ for a natural number m. The system of sheaves

$$\mathscr{F}_n=\begin{cases}\mathscr{F}/l^{n+1}\mathscr{F} & \text{for } n\geqq 0\\ 0 & \text{for } n<0\end{cases}$$

is stationary and again represents an l-adic sheaf. The functor

$$\mathscr{F}\mapsto(\mathscr{F}_n)$$

identifies the category of constructible l-torsion sheaves with a full subcategory of the category of all l-adic sheaves:

$$\mathscr{F}\text{ “}=\text{”}(\mathscr{F}_n).$$

Clearly:

12.8 Remark. Let $\mathscr{F}=(\mathscr{F}_n)$ be a projective system of constructible sheaves with $l^{n+1}\mathscr{F}_n=0$ and $\mathscr{F}_n=0$ for $n<0$. Then $\mathscr{F}$ is an A−R l-adic sheaf if and only if:

a) The condition *ARML* is satisfied: there is an $r>0$ with

$$\mathrm{Image}\,(\mathscr{F}[t]\to\mathscr{F})=\mathrm{Image}\,(\mathscr{F}[r]\to\mathscr{F})\qquad\text{for } t\geqq r,$$

and

b) We consider the projective system of universal images

$$(\bar{\mathscr{F}}_n):\ \bar{\mathscr{F}}_n=\mathrm{Image}\,(\mathscr{F}_{n+t}\to\mathscr{F}_n),\qquad t\geqq r,$$

which exists because of a). Then there is a natural number $s\geqq 0$ with

$$\bar{\mathscr{F}}_{n+t}/l^{n+1}\bar{\mathscr{F}}_{n+t}\cong\bar{\mathscr{F}}_{n+s}/l^{n+1}\bar{\mathscr{F}}_{n+s},\qquad n\geqq 0,\ t\geqq s.$$

The system $(\bar{\mathscr{F}}_{n+s}/l^{n+1}\bar{\mathscr{F}}_{n+s})$ is then an l-adic sheaf isomorphic to $\mathscr{F}$.

In particular, for a system $\mathscr{F}=(\mathscr{F}_n)$ of sheaves with $l^{n+1}\mathscr{F}_n=0$ for $n\geqq 0$ and $\mathscr{F}_n=0$ for $n<0$, we have

(1) The system $(\mathscr{F}_n)$ is an A−R l-adic sheaf if and only if it is an l-adic sheaf locally with respect to the etale topology.

(2) Let A be a closed subscheme of X and U its complement. Then $\mathscr{F}$ is an A−R l-adic sheaf precisely when the restricted systems

$$(\mathscr{F}_n|U) \quad \text{and} \quad (\mathscr{F}_n|A)$$

are so.

(3) Suppose the scheme X is connected and the sheaves $\mathscr{F}_n$ are constructible and locally constant. Let s: Spec $(\Omega)\to X$ be a geometric point. Then $\mathscr{F}$ is an A−R l-adic sheaf if and only if the projective system of stalks $(\mathscr{F}_{ns})$ is an A−R l-adic systems of Λ-modules.

Remark. For a geometric point s,

$$(\mathscr{F}_n)\mapsto(\mathscr{F}_{ns})$$

defines an exact functor from the A−R category of sheaves on X to the A−R category of Λ-modules.

If $(\mathscr{F}_n)$ is l-adic, resp. A−R l-adic, then so is $(\mathscr{F}_{ns})$. Then for an A−R l-adic sheaf we define the stalk

$$\mathscr{F}_s=\varprojlim \mathscr{F}_{ns}.$$

We have:

An A−R l-adic sheaf $\mathscr{F}$ vanishes in the A−R category iff all stalks vanish.

12.9 Definition. An l-adic sheaf $\mathscr{F}=(\mathscr{F}_n)$ is locally constant if all the sheaves $\mathscr{F}_n$ are locally constant.

Example. The sheaves M_X and $\mathbb{Z}_l(n)$ from 12.7.

12.10 Proposition. *Let $\mathscr{F}=(\mathscr{F}_n)$ be an l-adic sheaf on a scheme X. Then there is a dense open subscheme U for which the restricted system $\mathscr{F}|U=(\mathscr{F}_n|U)$ is locally constant.*

Proof. We may assume that X is irreducible; then we must simply find a nonempty open subscheme for which $F|U$ is locally constant. It suffices to find a nonempty open subscheme U such that $\mathscr{F}_0$ and the kernels $l^{n+1}\mathscr{F}_{n+1}$ of the surjective mappings $\mathscr{F}_{n+1}\to\mathscr{F}_n$ are locally constant on U. But the mapping

$$\mathscr{F}_{n+1}\xrightarrow{l^{n+1}} l^{n+1}\mathscr{F}_{n+1}$$

factors through $\mathscr{F}_{n+1}\to\mathscr{F}_0(\cong\mathscr{F}_{n+1}/l\mathscr{F}_{n+1})$. The ascending chain of kernels $\mathscr{C}_n$ of

$$\mathscr{F}_0\to l^{n+1}\mathscr{F}_{n+1}$$

is stationary, since $\mathscr{F}_0$ is noetherian (§4). We can therefore find a nonempty open subscheme U on which $\mathscr{F}_0$ and all the sheaves $\mathscr{C}_n$ are locally constant, and hence all the sheaves $l^{n+1}\mathscr{F}_{n+1}$ will be so too.

We shall now show that the A−R *category of* A−R *l-adic sheaves is an exact subcategory of the* A−R *category of sheaves, and hence it is an abelian category.*

12.11 Proposition. a) *Let* $\mathscr{F} \to \mathscr{G}$ *be an* A−R *homomorphism of two* A−R *l-adic sheaves on the scheme* X. *Then the kernel and cokernel are also* A−R *l-adic sheaves.*

b) *Suppose we have a short exact sequence in the* A−R *category,*

$$0 \to \mathscr{F} = (\mathscr{F}_n) \to \mathscr{G} = (\mathscr{G}_n) \to \mathscr{H} = (\mathscr{H}_n) \to 0.$$

Suppose $l^{n+t}\mathscr{G}_n = 0$ *for a suitable* t *and all* n. *Suppose* $\mathscr{F}$ *and* $\mathscr{H}$ *are* A−R *l-adic; then* $\mathscr{G}$ *is* A−R *l-adic.*

Proof. a) We can assume that

$$\phi\colon \mathscr{F} = (\mathscr{F}_n) \to \mathscr{G} = (\mathscr{G}_n)$$

is a homomorphism of l-adic sheaves. By 12.8, we can carry out the proof by noetherian induction. Let X be irreducible. We must show that there is a nonempty open subscheme U such that the kernel and cokernel of ϕ on U are A−R l-adic sheaves. By Proposition 12.10, we can assume by shrinking U that $\mathscr{F}$ and $\mathscr{G}$ are locally constant. But then the kernels $\mathscr{K}_n$ and cokernels $\bar{\mathscr{K}}_n$ of

$$\mathscr{F}_n \to \mathscr{G}_n$$

are locally constant. For a geometric point s of X, the systems of stalks $(\mathscr{K}_{ns})$ and $(\bar{\mathscr{K}}_{ns})$ are A−R l-adic Λ-modules (12.4). By 12.8(3) then both systems $(\mathscr{K}_n)$ and $(\bar{\mathscr{K}}_n)$ are A−R l-adic sheaves.

For b): We can assume that $l^{n+1}\mathscr{G}_n = 0$ and that the sequence

$$0 \to \mathscr{F}_n \to \mathscr{G}_n \to \mathscr{H}_n \to 0$$

are exact. Without loss of generality we can further assume that the structural homomorphism in all three systems are surjective, that $\mathscr{H}$ is actually an l-adic sheaf, and that the sheaves $\mathscr{F}_n$ and $\mathscr{G}_n$ are constructible. We then show (using Proposition 12.13) that for the l-adic sheaf $\mathscr{H}$ there is a constant $c \geqq 1$ such that the natural mappings

$$\operatorname{Tor}_1^{\Lambda_{n+t+1}}(\mathscr{H}_{n+t+1}, \Lambda_n) \to \operatorname{Tor}_1^{\Lambda_{n+t}}(\mathscr{H}_{n+t}, \Lambda_n)$$

are isomorphisms for n, $t \geqq c$. Then we verify that the conditions in Remark 12.8 hold for the projective system $\mathscr{G}$.

We make a further remark. Let $\mathscr{F} \hookrightarrow \mathscr{G}$ be an A−R injective homomorphism of l-adic sheaves on an irreducible scheme X. Thus in the A−R category $\mathscr{F}$ is a "subsheaf" of $\mathscr{G}$. Let $\mathscr{G}$ be locally constant, and suppose the stalk $\mathscr{F}_a$ of $\mathscr{F}$ vanishes at a generic geometric point. For another geometric point s then each specialization mapping (8.11)

$$(\mathscr{G}_{ns}) \to (\mathscr{G}_{na})$$

is bijective, and hence

$$(\mathscr{F}_{ns}) \to (\mathscr{F}_{na})$$

is injective. It follows that $\mathscr{F}$ vanishes.

Suppose now we have an ascending chain of A−R l-adic subsheaves $\mathscr{F}^{(\nu)}$ of the A−R l-adic sheaf $\mathscr{G}$ on an irreducible scheme X:

$$\mathscr{F}^{(1)} \hookrightarrow \mathscr{F}^{(2)} \hookrightarrow \ldots \hookrightarrow \mathscr{G}.$$

The ascending chain of their stalks at a generic geometric point a is then stationary, since the Λ-module $\mathscr{G}_a$ is noetherian:

$$\mathscr{F}_a^{(m)} = \mathscr{F}_a^{(m+1)} = \ldots \hookrightarrow \mathscr{G}_a.$$

We choose a nonempty open subscheme U such that $\mathscr{G}/\mathscr{F}^{(m)}$ is locally constant on U. From the foregoing it follows that

$$\mathscr{F}^{(m)} | U = \mathscr{F}^{(m+1)} | U = \ldots \quad \text{(in the A−R category)}.$$

By noetherian induction we have:

12.12 Proposition. *The category of* A−R *l-adic sheaves is noetherian.*

We apply this theorem to the ascending chain of kernels

$$\mathscr{K}^{(n)} = \operatorname{Ker}(\mathscr{G} \xrightarrow{l^n} \mathscr{G})$$

in an l-adic sheaf $\mathscr{G}$, and we get:

For every l-adic sheaf $\mathscr{G}$ there is an A−R exact sequence

$$0 \to \mathscr{H} \to \mathscr{G} \to \mathscr{F} \to 0$$

with l-adic sheaves $\mathscr{H}$ and $\mathscr{F}$ for which

a) $\mathscr{F} = (\mathscr{F}_n)$ is l-torsion free, i.e. $\mathscr{F} \xrightarrow{l} \mathscr{F}$ is A−R injective;

b) $\mathscr{H} = (\mathscr{H}_n)$ is a torsion sheaf, i.e. there is a natural number n with $l^{n+1}\mathscr{H} = 0$.

One then shows easily:

The sheaves $\mathscr{F}_n$ (i.e., their stalks) are Λ_n-flat. We have $\mathscr{H}_{n+1} = \mathscr{H}_{n+2} = \ldots$ for $n \geqq n_0$, and $\mathscr{H}$ can be identified with the constructible torsion sheaf $\mathscr{H}_{n+1}$ (see 12.7(3)).

12.13 Proposition. *For every l-adic sheaf $\mathscr{G}$ there is an ordinary etale l-torsion sheaf $\mathscr{H}$, a torsion-free l-adic sheaf $\mathscr{F} = (\mathscr{F}_n)$ and an* A−R *exact sequence*

$$0 \to \mathscr{H} \to \mathscr{G} \to \mathscr{F} \to 0.$$

Here $\mathscr{F}_n$ is Λ_n-flat.

As in Lemma 12.5, we have:

12.14 Lemma. *Let*

$$(\mathscr{K}_n^{\cdot})$$

be a projective system of complexes of sheaves. Assume the following hypotheses hold:

$$l^{n+1}\mathscr{K}_n^{\cdot} = 0 \quad \text{for all } n.$$

The sheaves $\mathscr{K}_n$ are Λ_n-flat. All cohomology sheaves $\mathscr{H}^{\nu}(\mathscr{K}_n^{\cdot})$ are constructible. The complexes are uniformly bounded.

$$\mathscr{K}_n^{\nu} = 0 \quad \text{when } \nu < -r \text{ or } \nu > +r.$$

For all n the induced mapping

$$\mathscr{K}^{\cdot}_{n+1} \otimes_{\Lambda_{n+1}} \Lambda_n = \mathscr{K}^{\cdot}_{n+1}/l^{n+1}\mathscr{K}^{\cdot}_{n+1} \to \mathscr{K}^{\cdot}_n$$

is a quasi-isomorphism.

Then the projective systems

$$(\mathscr{H}^{\nu}(\mathscr{K}^{\cdot}_n))_{n\in\mathbb{Z}}$$

of cohomology sheaves are A – R *l-adic sheaves.*

Proof. By 12.8(3) and 12.5, it is enough to show the following: Let X be irreducible. Then there is a nonempty open subscheme U on which all the sheaves $\mathscr{H}^{\nu}(\mathscr{K}^{\cdot}_n)$ are locally constant. We consider the short exact sequence

$$0 \to l^{n+1}\mathscr{K}_{n+1} = \mathscr{C}^{\cdot}_{n+1} \to \mathscr{K}^{\cdot}_{n+1} \to \mathscr{K}_{n+1}/l^{n+1}\mathscr{K}^{\cdot}_{n+1} = \bar{\mathscr{K}}_n \to 0$$

and the quasi-isomorphism $\bar{\mathscr{K}}^{\cdot}_n \to \mathscr{K}^{\cdot}_n$, and we get the long exact sequence

$$\ldots \to \mathscr{H}^{m-1}(\mathscr{K}^{\cdot}_n) \to \mathscr{H}^m(\mathscr{C}^{\cdot}_{n+1}) \to \mathscr{H}^m(\mathscr{K}^{\cdot}_{n+1}) \to \mathscr{H}^m(\mathscr{K}^{\cdot}_n) \to \ldots.$$

We further consider the commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathscr{C}^{\cdot}_{n+1} & \longrightarrow & \mathscr{K}^{\cdot}_{n+1} & \longrightarrow & \bar{\mathscr{K}}^{\cdot}_n & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow{\scriptstyle l} & & \uparrow & & \\
0 & \longrightarrow & l^n\bar{\mathscr{K}}_n & \longrightarrow & \bar{\mathscr{K}}^{\cdot}_n & \longrightarrow & \mathscr{K}^{\cdot}_n/l^n\mathscr{K}^{\cdot}_n & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathscr{C}^{\cdot}_n & \longrightarrow & \mathscr{K}^{\cdot}_n & \longrightarrow & \bar{\mathscr{K}}^{\cdot}_{n+1} & \longrightarrow & 0.
\end{array}$$

By hypothesis the lower vertical arrows are quasi-isomorphisms. The induced mapping

$$l^n\bar{\mathscr{K}}^{\cdot}_n \xrightarrow{l} l^{n+1}\mathscr{K}^{\cdot}_{n+1} = \mathscr{C}^{\cdot}_{n+1}$$

is an isomorphism ($\mathscr{K}^{\cdot}_{n+1}$ is flat!). We obtain a commutative diagram

$$\begin{array}{ccc}
\mathscr{H}^{m-1}(\mathscr{K}^{\cdot}_n) & \longrightarrow & \mathscr{H}^m(\mathscr{C}^{\cdot}_{n+1}) \\
\uparrow & & \uparrow{\scriptstyle \mathbb{R}} \\
\mathscr{H}^{m-1}(\mathscr{K}^{\cdot}_{n-1}) & \longrightarrow & \mathscr{H}^m(\mathscr{C}^{\cdot}_n).
\end{array}$$

The constructible sheaf

$$\mathscr{H}^m(\mathscr{C}^{\cdot}_{n+1}) \cong \mathscr{H}^m(\mathscr{C}^{\cdot}_n) \cong \mathscr{H}^m(\mathscr{C}^{\cdot}_{n-1}) \cong \ldots \cong \mathscr{H}^m(\mathscr{K}^{\cdot}_0)$$

is noetherian. Hence for $n \geqq n_0$ the induced mapping

$$\text{Image}\,[\mathscr{H}^{m-1}(\mathscr{K}^{\cdot}_{n+1}) \to \mathscr{H}^m(\mathscr{C}^{\cdot}_n)] \to \text{Image}\,[\mathscr{H}^{m-1}(\mathscr{K}^{\cdot}_n) \to \mathscr{H}^m(\mathscr{C}^{\cdot}_{n+1})]$$

is an isomorphism. By shrinking X we can thus assume that for all m and for all $n \geqq 0$ the sheaves $\text{Image}\,[\mathscr{H}^{m-1}(\mathscr{K}^{\cdot}_n) \to \mathscr{H}^m(\mathscr{C}^{\cdot}_{n+1})]$ and the sheaves $\mathscr{H}^m(\mathscr{C}^{\cdot}_n) \cong \mathscr{H}^m(\mathscr{K}^{\cdot}_0)$ are locally constant. But then all the sheaves $\mathscr{H}^m(\mathscr{K}^{\cdot}_n)$ are locally constant! □

Now let $f: X \to Y$ be a morphism of noetherian schemes. We then have in a natural way an inverse image functor

$$\mathscr{F} = (\mathscr{F}_n) \mapsto f^*(\mathscr{F}) = (f^*(\mathscr{F}_n))$$

from the A – R category of projective systems of sheaves on Y to the A – R category of projective sheaves on X. This functor is exact and takes l-adic sheaves to l-adic sheaves, resp. A – R l-adic sheaves to A – R l-adic sheaves.

Similarly we define the direct image functors $R^v f_*$ "componentwise" on the A – R category,

$$\mathscr{F} = (\mathscr{F}_n) \mapsto (R^v f_*(\mathscr{F}_n)) = R^v f_*(\mathscr{F}),$$

and for compactifiable mappings f also the direct images with compact support,

$$\mathscr{F} = (\mathscr{F}_n) \mapsto R^v f_!(\mathscr{F}).$$

Componentwise we associate with every short exact sequence in the A – R category of sheaves on X a long exact sequence for the direct images resp. direct images with compact support on Y.

The functors $(R^n f_*)$ and the functors $(R^n f_!)$ thus each form a system of ∂-functors on the A – R category. For a compactifiable mapping $f: X \to Y$ and a closed subscheme A of X with complement U, we similarly define componentwise the long exact sequence relating the proper direct images of A – R systems on A, U, and X.

For the composite $f \circ g$ of two mappings

$$X \xrightarrow{g} Y \xrightarrow{f} Z,$$

one gets the Leray spectral sequences

$$R^p f_* R^q g_* \Rightarrow R^n (f \circ g)_*$$
$$R^p f_! R^q g_! \Rightarrow R^n (f \circ g).$$

The two base change theorems immediately yield corresponding theorems for A – R systems.

The only nontrivial theorem is:

12.15 Theorem. *Let $f: X \to S$ be a compactifiable mapping, and $\mathscr{F} = (\mathscr{F}_n)$ an* A – R *l-adic sheaf on X. Then the system $R^v f_!(\mathscr{F}) = (R^v f_!(\mathscr{F}_n))$ are* A – R *l-adic sheaves.*

12.16 Remark. Let $S = \mathrm{Spec}(\Omega)$, Ω a separably closed field. Then we define

$$H_c^v(X, \mathscr{F}) = \varprojlim_n H_c^v(X, \mathscr{F}_n).$$

By 12.15 and 12.4, these Λ-modules are finitely generated.

Proof. We can assume that $\mathscr{F} = (\mathscr{F}_n)$ is an l-adic sheaf. By Proposition 12.13 we can assume that $\mathscr{F}$ is torsion-free, i.e. the sheaves $\mathscr{F}_n$ are Λ_n-flat. We can further assume that f is proper.

We shall show that the projective system $(\mathrm{R}^\cdot f_*(\mathscr{F}_n))$ in $D(S, \mathrm{tor})$ is represented

by a projective system $(\mathscr{K}_n^{\cdot})$ of complexes and maps of complexes which satisfy the hypotheses of Lemma 12.14. The assertion of the theorem then follows from that lemma.

To construct this complex, it is convenient to use the canonical *Godement resolution* of a sheaf, familiar from topology. We choose a set M of geometric points

$$\alpha\colon \operatorname{Spec}(\Omega_\alpha) \to X \qquad (\Omega_\alpha \text{ algebraically closed})$$

that for every point $x \in X$ contains at least one geometric point associated with x. For every sheaf $\mathscr{F}$ then we define a sheaf $\mathscr{C}^0(\mathscr{F})$ and an injection

$$\mathscr{F} \to \mathscr{C}^{\cdot}(\mathscr{F})$$

by

$$\mathscr{C}^0(\mathscr{F}) = \prod_{\alpha \in \mathfrak{M}} \alpha^*(\mathscr{F}_{\alpha, \operatorname{Spec}(\Omega)}).$$

For an etale scheme $\phi\colon U \to X$ over X then we have

$$\mathscr{C}^0(\mathscr{F})(U) = \prod_{\beta \circ \phi \in \mathfrak{M}} \mathscr{F}_\beta.$$

The mapping

$$\mathscr{F}(U) \to \mathscr{C}^0(\mathscr{F})(U)$$

is induced then by the natural mappings of the group of sections into the stalks. This $\mathscr{C}^0(\mathscr{F})$ is a functor on the category of all sheaves on X with the following properties:

(1) The functor $\mathscr{C}^0(\mathscr{G})$ is exact.

(2) $\mathscr{G}_\alpha$ is a direct summand of $\mathscr{C}^0(\mathscr{G})_\alpha$ for every geometric point α of X.

(3) $\mathscr{C}^0(\mathscr{G})$ for every $\mathscr{G}$ is flabby in the sense of § 2 (Remark on 2.9), and in particular it is acyclic for every morphism $f\colon X \to S$. If $\mathscr{G}$ is a sheaf of A-modules over a ring A and the stalks $\mathscr{G}_\alpha$ are A-injective, then $\mathscr{C}^0(\mathscr{G})$ is actually an injective sheaf of A-modules.

(4) If A is a noetherian ring and $\mathscr{G}$ a flat sheaf of A-modules, then $\mathscr{C}^0(\mathscr{G})$ is A-flat.

The construction given can be iterated,

$$\mathscr{C}^1(\mathscr{G}) = \mathscr{C}^0(\operatorname{Coker}(\mathscr{G} \to \mathscr{C}^0(\mathscr{G})), \ldots.$$

For every sheaf $\mathscr{G}$ we get a flabby resolution

$$\mathscr{C}^{\cdot}(\mathscr{G}),$$

the *Godement resolution*, depending functorially on $\mathscr{G}$. For every natural number m we define the truncated resolution

$$\mathscr{C}^{\cdot}_{(m)}(\mathscr{G})\colon \mathscr{C}^0(\mathscr{G}) \to \mathscr{C}^1(\mathscr{G}) \to \ldots \to \operatorname{Image}(\mathscr{C}^{m-1}(\mathscr{G}) \to \mathscr{C}^m(\mathscr{G})) \to 0.$$

Then $\mathscr{C}^{\cdot}_{(m)}(\mathscr{G})$ is an *exact* functor of $\mathscr{G}$. Under the hypothesis (4) above, $\mathscr{C}^{\cdot}_{(m)}(\mathscr{G})$ is actually A-flat.

Now back to our original situation.

Let $\mathscr{F} = (\mathscr{F}_n)$ be an l-adic sheaf where the sheaves $\mathscr{F}_n$ are constructible flat sheaves of Λ_n-modules. Say $\dim X/S \leqq r$.

For every torsion sheaf $\mathscr{G}$ on X then

$$R^v f_*(\mathscr{G})=0, \qquad v>2\cdot r=m.$$

Therefore $\mathscr{C}^\cdot_{(m)}(\mathscr{F}_n)$ is an f_*-acyclic resolution of $\mathscr{F}_n$. One sees easily that the complex

$$\mathscr{C}^\cdot_{(m)}(\mathscr{F}_{n+1})\otimes_{\Lambda_{n+1}}\Lambda_n=\mathscr{C}^\cdot_{(m)}(\mathscr{F}_{n+1})/l^{n+1}\mathscr{C}^\cdot_{(m)}(\mathscr{F}_{n+1})$$

is canonically isomorphic to $\mathscr{C}^\cdot_{(m)}(\mathscr{F}_{n+1}/l^{n+1}\mathscr{F}_{n+1})=\mathscr{C}^\cdot_{(m)}(\mathscr{F}_n)$.

Now let $\mathscr{G}^\cdot$ be the complex on X that consists of the single sheaf $\mathscr{C}^r_{(m)}(\mathscr{F}_{n+1})$, and $\mathscr{F}'$ the Λ_{n+1}-flat resolution

$$\ldots\to\Lambda_{n+1\,S}\xrightarrow{l}\Lambda_{n+1\,S}\xrightarrow{l^{n+1}}\Lambda_{n+1\,S}$$

of the constant sheaf Λ_{nS} on S. Applying Proposition 8.14 to this complex, we find that $f_*\mathscr{C}^\cdot_{(m)}(\mathscr{F}_{n+1})$ is a complex of Λ_{n+1}-flat sheaves and that $f_*\mathscr{C}^\cdot_{(m)}(\mathscr{F}_{n+1})\otimes_{\Lambda_{n+1}}\Lambda_n$ is canonically isomorphic to $f_*\mathscr{C}^\cdot_{(m)}(\mathscr{F}_n)$. Thus the projective system of complexes

$$\mathscr{K}^\cdot_n=f_*\mathscr{C}^\cdot_{(m)}(\mathscr{F}_n)=\mathsf{R}f_*(\mathscr{F}_n)$$

satisfies the hypotheses of Lemma 12.14! □

Now a few more remarks on the passage from etale cohomology to singular cohomology (see § 11):

Here we consider schemes X over the field $\mathbb{C}$ complex numbers. In § 11 we constructed a functor from the category of etale sheaves on X to the category of sheaves on the topological space X_{an}. The every projective system $\mathscr{F}=(\mathscr{F}_n)$ of torsion sheaves on X we can associate the system $(\mathscr{F}_{n\,\mathrm{an}})$ on X_{an}. This gives us an exact functor from the A−R category of sheaves on X to the corresponding A−R category on X_{an}. If $\mathscr{F}$ is an A−R l-adic sheaf, we set

$$\mathscr{F}_{\mathrm{an}}=\varprojlim_n\mathscr{F}_{n\,\mathrm{an}},$$

and this gives us a functor from the A−R category of A−R sheaves on X to the category of sheaves on X_{an}. For an A−R l-adic sheaf, we can show

$$(\varprojlim_n\mathscr{F}_{n\,\mathrm{an}})_x=\varprojlim_n\mathscr{F}_{n\,\mathrm{an}\,x}=\varprojlim_n\mathscr{F}_{nx}, \qquad x\in X_{\mathrm{an}}.$$

For a mapping $f\colon X\to S$, we have

$$R^v f_{\mathrm{an}!}(\varprojlim_n\mathscr{F}_{n\,\mathrm{an}})=\varprojlim_n R^v f_!(\mathscr{F}_{n\,\mathrm{an}}).$$

A corresponding statement holds for $R^v f_*$. From this we see that for A−R l-adic sheaves again $\mathscr{F}\to\mathscr{F}_{\mathrm{an}}$ is exact and compatible with inverse images. Moreover, we get a comparison theorem between etale and singular cohomology corresponding to Proposition 11.7. But we shall make no use of it.

A Simple Special Case

Let M be a finitely generated Λ-module. Then we have

$$H^v_c(X, M_X)=H^v_c(X_{\mathrm{an}}, M).$$

To finish this section, we shall describe the quotient category of the category of all A – R l-adic sheaves by the subcategory of l-torsion sheaves.

Let $\mathbb{Q}_l$ be the quotient field of the ring $\Lambda = \varprojlim_n \mathbb{Z}/l^n\mathbb{Z}$ of all l-adic integers.

We introduce the category of *sheaves* of $\mathbb{Q}_l$-vector spaces on a scheme X as follows:

The objects in the category of sheaves of $\mathbb{Q}_l$-vector spaces are the same as in the A – R category of projective systems $\mathscr{F} = (\mathscr{F}_n)$ of l-torsion sheaves. If $\mathscr{F}$ is such a system, we denote by $\mathscr{F} \otimes \mathbb{Q}_l$ this system considered as a sheaf of $\mathbb{Q}_l$-vector spaces. The morphisms in the category of sheaves of $\mathbb{Q}_l$-vector spaces are defined by

$$\mathrm{Hom}(\mathscr{F} \otimes \mathbb{Q}_l, \mathscr{G} \otimes \mathbb{Q}_l) = \mathrm{Hom}_{\mathrm{A-R}}(\mathscr{F}, \mathscr{G}) \otimes_\Lambda \mathbb{Q}_l.$$

The composition of morphisms is carried out in the obvious way. This category is again abelian.

The natural functor from the A – R category into the category of sheaves of $\mathbb{Q}_l$-vector spaces is exact.

A sheaf of $\mathbb{Q}_l$-vector spaces is called *constructible* resp. *locally constant* if it is isomorphic to a sheaf $\mathscr{F} \otimes \mathbb{Q}_l$ for an A – R l-adic sheaf $\mathscr{F}$ resp. a locally constant l-adic sheaf $\mathscr{F}$.*)

It follows from Proposition 12.13 that every constructible sheaf of $\mathbb{Q}_l$-vector spaces is isomorphic to a sheaf $\mathscr{F} \otimes \mathbb{Q}_l$ with $\mathscr{F}$ a torsion-free l-adic sheaf.

Most sheaf-theoretic concepts and theorems extend in an obvious way to sheaves of $\mathbb{Q}_l$-vector spaces.

We define the stalk at a geometric point α: $\mathrm{Spec}(\Omega) \to X$ as

$$(\mathscr{F} \otimes \mathbb{Q}_l)_\alpha = \mathscr{F}_\alpha \otimes \mathbb{Q}_l.$$

Further:

$$\begin{aligned}
\Gamma(X, \mathscr{F} \otimes \mathbb{Q}_l) &= \Gamma(X, \mathscr{F}) \otimes \mathbb{Q}_l \\
\Gamma_c(X, \mathscr{F} \otimes \mathbb{Q}_l) &= \Gamma_c(X, \mathscr{F}) \otimes \mathbb{Q}_l \\
H^\nu(X, \mathscr{F} \otimes \mathbb{Q}_l) &= H^\nu(X, \mathscr{F}) \otimes \mathbb{Q}_l \\
H^\nu_c(X, \mathscr{F} \otimes \mathbb{Q}_l) &= H^\nu_c(X, \mathscr{F}) \otimes \mathbb{Q}_l \\
R^\nu f_*(\mathscr{F} \otimes \mathbb{Q}_l) &= R^\nu f_*(\mathscr{F}) \otimes \mathbb{Q}_l \\
R^\nu f_!(\mathscr{F} \otimes \mathbb{Q}_l) &= R^\nu f_!(\mathscr{F}) \otimes \mathbb{Q}_l, \qquad \text{and so on.}
\end{aligned}$$

Examples are the locally constant sheaves

$$\mathbb{Q}_l(n)_X = \Lambda(n)_X \otimes \mathbb{Q}_l.$$

*) Warning: Our notion of a locally constant sheaf of $\mathbb{Q}_l$-vector spaces is in general not local with respect to the etale topology. In French terminology one uses now the notion "lisse"; previously "constant tordu". Maybe a better notion would have been "smooth".

Chapter II. Rationality of Weil ζ-Functions

The goal of this chapter is to prove the rationality of the Weil ζ-function of an algebraic variety over a finite field, or more generally of the L-series for constructible sheaves (Theorem 4.4). Following Grothendieck, we will derive the rationality from a fixed point formula of Lefschetz type for the Frobenius morphism (Proposition 4.2).

The Poincaré duality theorem is the basis for the proof of the fixed point theorem. In the first section we will prove the duality theorem for etale cohomology groups on smooth algebraic varieties over algebraically closed base fields (Theorem 1.13).

In the second section we use the duality theorem to construct the etale cohomology class of an algebraic cycle,

$$\mathrm{cl}_X(Z) \in H^{2d}(X, \mathbb{Z}/n\mathbb{Z}).$$

Here d is the codimension of the cycle Z in the variety X.

If we consider a morphism $f\colon Y \to X$ of two smooth manifolds, then the inverse image $f^*(Z)$ in the sense of intersection theory satisfies a compatibility formula:

$$\mathrm{cl}_Y(f^*(Z)) = H^{2d}(f)(\mathrm{cl}_X(Z)).$$

(Proposition 2.3, for the case of divisors, which is all that we need.) This formula is used as follows:

Let $f\colon X \to X$ be a morphism of a smooth projective curve into itself, and let

$$\Gamma_f\colon X \to X \times X$$

be the graph mapping. We use the Künneth formula to compute the cohomology class of the diagonal $\Delta \subset X \times X$ and then apply the compatibility formula to determine the cohomology class of the fixed point cycle $\Gamma_f^*(\Delta)$. In this way we get the Lefschetz fixed point formula 2.9.

The goal of Section 3 is to extend this fixed point formula to the following situation:

Let $f\colon X \to X$ be a morphism of a smooth algebraic variety into itself with only finitely many fixed points, all of which have multiplicity 1. Consider also a constructible sheaf of flat $\mathbb{Z}/n\mathbb{Z}$-modules and a homomorphism

$$\phi\colon f^*(\mathscr{G}) \to \mathscr{G}.$$

The pair (f, ϕ) induces homomorphisms

$$H_c^i(f, \phi)\colon H_c^i(X, \mathscr{G}) \to H_c^i(X, \mathscr{G}).$$

What we want is a "trace formula"

$$\sum_{a\in \mathrm{Fix}(f)} \mathrm{Tr}(\phi_a)=\sum(-1)^i \mathrm{Tr}(H_c^i(f, \phi)).$$

Here for simplicity we assume that the cohomology groups are projective $\mathbb{Z}/n\mathbb{Z}$-modules. In other cases one needs the concept of the trace of a mapping of complexes. Using a method of Grothendieck, we can complete the proof for the case of a curve when f is the Frobenius morphism of X and ϕ is the Frobenius homomorphism of $\mathscr{G}$ (Theorem 3.10).

Here is the idea of the proof: We first assume that X has an unramified Galois covering space $Y\to X$ with Galois group Γ such that the inverse image of $\mathscr{G}$ on Y is constant:

$$p^*(\mathscr{G})=F_Y; \qquad F \text{ a free } (\mathbb{Z}/n\mathbb{Z})\text{-module.}$$

Then Γ acts on F. If this action is trivial, the formula we want follows from the fixed point theorem of Section 2 for all the morphisms

$$\sigma\circ \mathrm{Fr}_Y, \qquad \sigma\in\Gamma.$$

(Fr_Y is the Frobenius morphism of Y.) The general case can be reduced to this special one.

Passage to limits yields a corresponding trace formula for l-adic sheaves.

§1. Poincaré Duality

In what follows we consider only noetherian schemes on which a fixed given natural number n is invertible. We set

$$\Lambda=\mathbb{Z}/n\mathbb{Z}.$$

Λ is an *injective* Λ-module!

All sheaves considered will be etale sheaves of Λ-modules if nothing is said to the contrary.

The goal of this section is to prove the Poincaré duality theorem for sheaves on a smooth projective scheme X of constant dimension d over a separably closed base field k.

As usual, we denote by μ_{nX} the sheaf of n-th roots of unity on X and by μ_{nX}^d its d-fold tensor product, $\mu_{nX}\otimes\ldots\otimes\mu_{nX}$. Since k is separably closed, these sheaves are *noncanonically* isomorphic to the constant sheaf $(\mathbb{Z}/n\mathbb{Z})_X$. In what follows we shall construct a canonical homomorphism – a *trace mapping* –

$$S=S_{X/k}\colon\ H^{2d}(X, \mu_n^d)\to\mathbb{Z}/n\mathbb{Z}$$

and show that it is an isomorphism when X is connected. We consider then the cup product (A II)

$$\mathrm{Ext}^p(\mathscr{G}, \mu_{nX}^d)\times H^q(X, \mathscr{G})\to H^{p+q}(X, \mu_{nX}^d)$$

$$(\alpha, \beta)\mapsto \alpha\smile\beta.$$

Composing this with the trace mapping, we get a pairing:

$$\mathrm{Ext}^p_\Lambda(\mathscr{G}, \mu_{nX}^d) \times H^{2d-p}(X, \mathscr{G}) \to \Lambda$$
$$(\alpha, \beta) \mapsto S(\alpha \smile \beta).$$

Here is our main result (Theorem 1.13):

This pairing is nondegenerate; i.e. it sets up an isomorphism

$$\mathrm{Ext}^p_\Lambda(\mathscr{G}, \mu_{nX}^d) \xrightarrow{\cong} \mathrm{Hom}_\Lambda(H^{2d-p}(X, \mathscr{G}), \Lambda).$$

More generally, we shall prove a corresponding result also for schemes X over k that are not complete. In place of $H^{2d-p}(X, \mathscr{G})$ then we have the cohomology group "with compact support", $H_c^{2d-p}(X, \mathscr{G})$.

As we want to use the *"Technique of fibering by curves"* to construct the trace mapping, it is essential for us to aim for a relative version of the trace mapping:

For every smooth compactifiable morphism $f\colon X \to Y$ with constant fiber dimension $d(X/Y) = d$, we will construct a homomorphism

$$R^{2d} f_!(\mu_{nX}^d) \to \Lambda_X,$$

the relative trace mapping (1.6). This will be done in several steps.

We first consider the case of an *etale* mapping

$$f\colon X \to Y.$$

In the first chapter (I 8.1), we constructed a left adjoint functor $f_!$ (the direct image functor with compact support) for the inverse image functor f^*. Thus for every sheaf $\mathscr{F}$ on Y there is an adjunction mapping

$$S_{X/Y}\colon f_! f^*(\mathscr{F}) \to \mathscr{F},$$

which for every sheaf $\mathscr{G}$ on X induces an isomorphism

$$\begin{array}{ccc} \mathrm{Hom}(\mathscr{G}, f^*(\mathscr{F})) & \xrightarrow{\cong} & \mathrm{Hom}(f_!(\mathscr{G}), \mathscr{F}) \\ & \searrow \quad \nearrow & \\ & \mathrm{Hom}(f_!(\mathscr{G}), f_! f^*(\mathscr{F})). & \end{array}$$

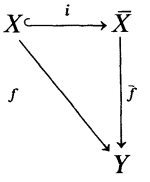
This *trace mapping* is a natural transformation of functors. It induces a natural mapping

$$H_c^\nu(X, f^*(\mathscr{F})) = H_c^\nu(Y, f_! f^*(\mathscr{F})) \to H_c^\nu(Y, \mathscr{F}).$$

In Chapter I, § 8 we defined the functor $f_!$ more generally for compactifiable morphisms, in particular for *quasi-finite* ones

$$f\colon X \to Y.$$

For then there is a compactification, i.e. a commutative diagram



where i is an open embedding and $\bar{f}$ actually a *finite* mapping, and we let

$$f_!(\mathscr{G}) = f_* i_!(\mathscr{G}).$$

Let

$$\alpha\colon \operatorname{Spec}(\Omega) \to Y$$

be a geometric point of Y in an algebraically closed field Ω, and let $f^{-1}(\alpha)$ be the finite set of geometric points

$$\beta\colon \operatorname{Spec}(\Omega) \to X$$

over α. Then there is a canonical isomorphism on stalks

$$f_!(\mathscr{G})_\alpha = \bigoplus_{\beta \in f^{-1}(\alpha)} \mathscr{G}_\beta$$

(Chap. I, § 3). Let f be actually *quasi-finite and flat*. Let $\tilde{\mathcal{O}}_{Y,\alpha}$ be the strict local ring of Y at α, i.e. the strict Henselization of the corresponding ring of the structure sheaf $\mathcal{O}_Y$ with respect to α, and let $\tilde{\mathcal{O}}_{X,\beta}$ be the strict local ring of X at $\beta \in f^{-1}(\alpha)$. Then we have: $\tilde{\mathcal{O}}_{X,\beta}$ is a *finitely generated free* $\tilde{\mathcal{O}}_{Y,\alpha}$-module. We denote by $n(\beta)$ the rank of $\tilde{\mathcal{O}}_{X,\beta}$ over $\tilde{\mathcal{O}}_{Y,\alpha}$.

1.1 Lemma. *Generalization of the trace mapping: Let $f\colon X \to Y$ be a quasi-finite flat morphism, $\mathscr{F}$ a sheaf on Y. There is a uniquely determined trace mapping*

$$S = S_{X/Y}\colon f_! f^*(\mathscr{F}) \to \mathscr{F}$$

defined as follows on the stalk of a geometric point α of Y:

$$(f_! f^* \mathscr{F})_\alpha = \bigoplus_{\beta \in f^{-1}(\alpha)} \mathscr{F}_\alpha \to \mathscr{F}_\alpha$$
$$\oplus\, s_\beta \mapsto \sum n(\beta)\, s_\beta .$$

If f is etale, then this mapping agrees with the adjunction mapping.

We now consider the Kummer sequence on a scheme X (I, 2.6):

$$0 \to \mu_{nX} \to \mathcal{O}_X^* \xrightarrow{\ n\ } \mathcal{O}_X^* \to 0.$$

Here $\mathcal{O}_X^*$ is the sheaf of units of the structure sheaf $\mathcal{O}_X$ on $\operatorname{Et}(X)$, and so exceptionally is not a sheaf of Λ-modules. The sequence induces a connecting homomorphism

$$\operatorname{Pic}(X) = H^1(X, \mathcal{O}_X^*) \to H^2(X, \mu_{nX}).$$

1.2 Definition. Let $\mathscr{L}$ be a line bundle on X. We denote the image of $\mathscr{L}$ under the mapping $\operatorname{Pic}(X) \to H^2(X, \mu_{nX})$ by

$$\operatorname{cl}(\mathscr{L}).$$

If $f: X \to S$ is a morphism, then we write

$$\mathrm{cl}_{X/S}(\mathscr{L}) \in \Gamma(S, R^2 f_*(\mu_{nX}))$$

for the natural image of $\mathrm{cl}(\mathscr{L})$ in $\Gamma(S, R^2 f_*(\mu_{nX}))$.

Let $f: \tilde{X} \to X$ be a finite flat morphism. We denote the norm mapping by

$$N: f_*(\mathcal{O}_{\tilde{X}}^*) \to \mathcal{O}_X^*.$$

Then the following diagram is exact and commutative:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & f_*(\mu_{n\tilde{X}}) = f_!(f^*(\mu_{nX})) & \longrightarrow & f_*(\mathcal{O}_{\tilde{X}}^*) & \xrightarrow{n} & f_*(\mathcal{O}_{\tilde{X}}^*) & \longrightarrow & 0 \\
 & & \downarrow S & & \downarrow N & & \downarrow N & & \\
0 & \longrightarrow & \mu_{nX} & \longrightarrow & \mathcal{O}_X^* & \xrightarrow{n} & \mathcal{O}_X^* & \longrightarrow & 0.
\end{array}$$

From this we get a commutative diagram of connecting homomorphisms

$$\begin{array}{ccc}
\mathrm{Pic}(\tilde{X}) & \longrightarrow & H^2(\tilde{X}, \mu_{n\tilde{X}}) = H^2(X, f_!(\mu_{n\tilde{X}})) \\
\downarrow N & & \downarrow S \\
\mathrm{Pic}(X) & \longrightarrow & H^2(X, \mu_{nX}).
\end{array}$$

1.3 Lemma. *Let $f: \tilde{X} \to X$ be a finite flat morphism and $\mathscr{L}$ a line bundle on X. If again*

$$S_{\tilde{X}/X} = S: H^2(\tilde{X}, \mu_{n\tilde{X}}) \to H^2(X, \mu_{nX})$$

denotes the mapping induced by the trace mapping, then the norm) $N(\mathscr{L})$ of the line bundle $\mathscr{L}$ satisfies:*

$$\boxed{\mathrm{cl}(N(\mathscr{L})) = S_{\tilde{X}/X}(\mathrm{cl}(\mathscr{L}))}.$$

We first construct a trace mapping for smooth curves $f: X \to \mathrm{Spec}(k)$ over an algebraically closed base field. Using the Kummer sequence for a smooth, irreducible, projective curve, we get an isomorphism

$$\mathrm{Pic}(X)/n\,\mathrm{Pic}(X) \xrightarrow{\cong} H^2(X, \mu_{nX}).$$

But $\mathrm{Pic}(X)/n\,\mathrm{Pic}(X)$ is canonically isomorphic to $\Lambda = \mathbb{Z}/n\mathbb{Z}$. Thus we have obtained a natural isomorphism

$$S_{X/k}: H^2(X, \mu_{nX}) \xrightarrow{\cong} \Lambda$$

which is characterized by the condition

$$S_{X/k}(\mathrm{cl}(\mathscr{L})) = 1$$

for every line bundle $\mathscr{L}$ of degree 1 (see Chap. I, 5.1).

*) Highest exterior power of the direct image.

If $X \neq \emptyset$ is irreducible but not complete, then X can be embedded in a unique smooth irreducible projective curve

$$j\colon X \hookrightarrow \bar{X}.$$

Since the complement of X in $\bar{X}$ is finite, j induces an isomorphism

$$H_c^2(X, \mu_{nX}) \xrightarrow{\cong} H^2(\bar{X}, \mu_{n\bar{X}}).$$

Composing with $S_{\bar{X}/k}$, we get a trace mapping

$$S_{X/k}\colon H_c^2(X, \mu_{nX}) \xrightarrow{\cong} \Lambda.$$

If X is not connected, then we get

$$S_{X/k}\colon H_c^2(X, \mu_{nX}) \to \Lambda$$

by adding the trace mappings on the connected components.

In this case of curves, the mapping $H_c^2(X, \mu_{nX}) \to \Lambda$ that we have constructed is equivalent to a mapping

$$\boxed{S_{X/k}\colon R^2 f_!(\mu_{nX}) \to \Lambda_{\mathrm{Spec}(k)}.}$$

We now want to construct the relative trace mapping step-by-step for arbitrary smooth compactifiable morphisms of constant fiber dimension.

1.4 Definition. A finitely generated flat morphism $f\colon X \to S$ of schemes will be called of type S if there is a natural number d such that all the nonempty geometric fibers are nonsingular and pure d-dimensional.

In particular, morphisms of type S are smooth. The fibers do not have to be connected. Etale morphisms are of type $S(d=0)$.

The number $d = d(X/S)$ is the *relative dimension of X over S.*

Notation. If $f\colon X \to S$ is a morphism of schemes and d the largest dimension of a geometric fiber, we set

$$\mathscr{T}_{X/S} = (\mu_n)_X^{\otimes d}.$$

The following lemmas are tools for constructing a universal trace mapping.

1. Composition of Trace Mappings

Let

$$g\colon X \to T, \quad h\colon T \to S, \quad f = h \circ g\colon X \to S$$

be morphisms of type S. The vanishing theorem (Chap. I, 8.8) says that

$$R^\nu g_!(\mathscr{F}) = 0 \quad \text{if } \nu > 2d(X/T)$$

$$R^\nu h_!(\mathscr{G}) = 0 \quad \text{if } \nu > 2d(T/S).$$

The Leray spectral sequence therefore induces natural isomorphisms

$$R^{2[d(X/T)+d(T/S)]} f_!(\mathscr{F}) \xrightarrow{\cong} R^{2d(T/S)} h_!\left[R^{2d(X/T)} g_!(\mathscr{F})\right]$$

for every torsion sheaf $\mathscr{F}$ on X.

We note also that for a sheaf $\mathscr{F}$ on X and a locally constant locally free sheaf $\mathscr{G}$ on T, there is a natural isomorphism (I 8, 14)

$$R^v g_!(\mathscr{F} \otimes g^*(\mathscr{G})) \cong R^v g_!(\mathscr{F}) \otimes \mathscr{G}.$$

Now suppose we have homomorphisms

$$\alpha\colon\ R^{2d(X/T)} g_!(\mathscr{T}_{X/T}) \to \Lambda_T$$

and

$$\beta\colon\ R^{2d(T/S)} h_!(\mathscr{T}_{T/S}) \to \Lambda_S .$$

We have

$$\mathscr{T}_{X/S} = \mu_n^{d(X/T)} \otimes \mu_n^{d(T/S)} = \mathscr{T}_{X/T} \otimes g^*(\mathscr{T}_{T/S}).$$

Then we have mappings

$$R^{2d(X/S)} f_!(\mathscr{T}_{X/S}) \xrightarrow{\cong} R^{2d(T/S)} h_![R^{2d(X/T)} g_!(\mathscr{T}_{X/T} \otimes g^*(\mathscr{T}_{T/S}))]$$

$$\xrightarrow{\cong} R^{2d(T/S)} h_![R^{2d(X/T)} g_!(\mathscr{T}_{X/T}) \otimes \mathscr{T}_{T/S}]$$

$$\longrightarrow R^{2d(T/S)} h_!(\mathscr{T}_{T/S}) \to \Lambda_S .$$

We denote the composite mapping by $\beta \square \alpha$.

1.5 Remark. Let $g\colon X \to T$, $h\colon T \to S$ be morphisms of type S. Two trace mappings

$$\alpha\colon\ R^{2d(X/T)} g_!(\mathscr{T}_{X/T}) \to \Lambda_T ,$$
$$\beta\colon\ R^{2d(T/S)} h_!(\mathscr{T}_{T/S}) \to \Lambda_S$$

can be composed to give a trace mapping

$$\beta \square \alpha\colon\ R^{2d(X/S)}(h \circ g)_!(\mathscr{T}_{X/S}) \to \Lambda_S .$$

If α and β are isomorphisms, so is $\beta \square \alpha$.

2. Base Change for Trace Mappings

Let

$$\begin{array}{ccc} X_T & \xrightarrow{f'} & T \\ {\scriptstyle g'}\downarrow & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & S \end{array}$$

be a cartesian diagram of schemes. Let the morphism f, and hence also f', be of type S. Set $d = d(X/S) = d(X_T/T)$.

$$g'^*(\mathscr{T}_{X/S}) = \mathscr{T}_{X_T/T} .$$

We say that two trace mappings α for X/S and β for X_T/T, are compatible with base change if the following diagram is commutative

$$\begin{array}{ccc} g^* R^{2d} f_!(\mathscr{T}_{X/S}) & \cong & R^{2d} f_!(\mathscr{T}_{X/T}) \\ \downarrow{\scriptstyle g^*(\alpha)} & & \downarrow{\scriptstyle \beta} \\ g^*(\Lambda_S) & = & \Lambda_T . \end{array}$$

The isomorphism in the first line comes from the proper base change theorem.

We are now in a position to characterize the trace mapping axiomatically.

1.6 Theorem. *One can assign to every (compactifiable) morphism $f\colon X \to S$ of type S a trace mapping*

$$S_{X/S}\colon R^{2d} f_!(\mathscr{T}_{X/S}) \to \Lambda_S$$

which has the following properties and is uniquely determined by them:

1) *$S_{X/S}$ is compatible with base change.*

2) *$S_{X/S}$ is transitive, i.e. compatible with the composition of trace mappings in the sense of* 1.5.

3) *If f is etale ($d=0$), then*

$$S_{X/S}\colon f_! f^*(\Lambda_S) \to \Lambda_S$$

is the canonical mapping.

4) *If $f\colon X \to S$ is a smooth curve over the spectrum S of an algebraically closed field k, then*

$$S_{X/S}\colon R^2 f_!(\mathscr{T}_{X/S}) \to \Lambda_S (\sim H^2(X, \mu_{nX}) \to \Lambda)$$

is the mapping $S_{X/k}$ that we considered before, the one coming from the Kummer sequence.

In addition, we prove one part of a relative duality theorem:

1.7 Proposition. If all fibres are connected and nonempty, then $S_{X/S}$ is an isomorphism.

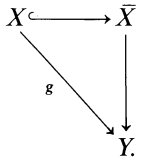
We shall first show that the composition property 2) of 1.6 is valid for etale morphisms and the trace mappings that we have already constructed for curves over an algebraically closed field.

1.8 Lemma. *Let Y be a smooth curve over an algebraically closed field k, and $g\colon X \to Y$ an etale mapping. Then*

$$S_{X/k} = S_{Y/k} \square S_{X/Y} .$$

Proof. We can assume that Y is projective and connected, and that X is a nonempty open subspace of a smooth connected curve $\bar{X}$ that is a ramified

covering space of Y,



$\bar{X}$ is finite and flat over Y. For a point $\alpha \in X$, and the line bundle $\mathscr{L}(\alpha)$ of degree 1 associated with the divisor (α), we have by Lemma 1.3 that

$$\mathrm{cl}(N(\mathscr{L}(\alpha))) = S_{\bar{X}/Y}\,\mathrm{cl}(\mathscr{L}(\alpha)).$$

$N(\mathscr{L}(\alpha)) = \mathscr{L}(g(\alpha))$ also has degree 1. The composition property follows now from the characterization of the mappings $S_{\bar{X}/k}$ and $S_{Y/k}$. □

Next we define the trace mapping for the affine line $\mathbb{A}^1_S$ over an arbitrary scheme S. Let $X = \mathbb{P}^1_S$ be the projective line over S, and $\mathscr{L}$ the line bundle corresponding to the zero section. We get a sheaf mapping

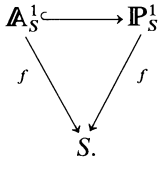
$$\begin{aligned} (\mathbb{Z}/n\mathbb{Z})_S &\to R^2 f_*(\mu_{nX}) \\ 1 &\mapsto \mathrm{cl}_{X/S}(\mathscr{L}). \end{aligned} \tag{1.2}$$

1.9 Lemma. *Let $f\colon \mathbb{P}^1_S \to S$ be the projective line over a scheme S, and let D be the zero section. The assignment $1 \mapsto \mathrm{cl}_{\mathbb{P}^1_S/S}(\mathscr{L}(D))$ defines a sheaf isomorphism*

$$(\mathbb{Z}/n\mathbb{Z})_S \xrightarrow{\sim} R^2 f_*(\mu_{nX}).$$

If S is the spectrum of an algebraically closed field, then this is clear. The general case can be reduced to this by base change. □

We can consider the affine line $\mathbb{A}^1_S$ as an open subscheme of the projective line,



If we compose the inverse of the mapping in 1.9 with the natural mapping

$$R^2 f_!(\mu_{n\mathbb{A}^1_S}) \to R^2 f_*(\mu_{n\mathbb{P}^1_S}),$$

we get a mapping

$$S_{X/S}\colon\ R^2 f_!(\mu_{nX}) \to (\mathbb{Z}/n\mathbb{Z})_S\,.$$

Since $\mathbb{P}^1_S - \mathbb{A}^1_S$ is finite over S, this is likewise an isomorphism. If S is the spectrum of an algebraically closed field k, then $S_{X/S}$ agrees with the trace mapping $S_{X/k}$ already introduced.

As the next case, we consider the affine space of dimension m over S:

$$\mathbb{A}^m_S = \mathbb{A}^1_S \times \ldots \times \mathbb{A}^1_S \to S.$$

This mapping is the product of m projections,

$$\mathbb{A}_S^m = \mathbb{A}^1_{\mathbb{A}_S^{m-1}} \to \mathbb{A}_S^{m-1} \to \ldots \to \mathbb{A}^1_S \to S.$$

We define

$$S_{\mathbb{A}^m_S/S} = S_{\mathbb{A}^1_S/S} \square S_{\mathbb{A}^2_S/\mathbb{A}^1_S} \square \ldots \square S_{\mathbb{A}^m_S/\mathbb{A}^{m-1}_S}.$$

The trace thus defined is independent of the order in which the projections are taken:

Indeed, for an algebraically closed base field k, the group $\mathrm{Gl}_m(k)$ of linear transformations of $X = \mathbb{A}^m_{\mathrm{Spec}(k)}$ operates trivially on $H_c^{2m}(X, \mu_{nX}^m) \cong \Lambda$, since it has no proper subgroups of finite index. In particular, the interchanges of factors in $\mathbb{A}^1_{\mathrm{Spec}(k)} \times \ldots \times \mathbb{A}^1_{\mathrm{Spec}(k)}$ induce the identity in $H_c^{2m}(X, \mu_{nX}^m)$.

We now want to consider morphisms

$$f\colon X \to S$$

of type S that have a factorization

$$X \xrightarrow{g} \mathbb{A}^d_S \longrightarrow S$$

with g an etale mapping. Here g will be given by a sequence $x_1, \ldots, x_d$ of d elements in $\Gamma(X, \mathcal{O}_X)$. If we change the order in which these elements are taken, we obtain a mapping

$$g' = \phi \circ g, \qquad \phi \in \mathrm{Gl}_d.$$

We define a trace, depending on this system of elements:

$$S_{X/S}^{(x_1,\ldots,x_d)} = S_{\mathbb{A}^d_S/S} \square S_{X/\mathbb{A}^d_S}\colon\ R^{2d} f_!(T_{X/S}) \to \Lambda_S.$$

This mapping is *independent* of the *ordering* of the elements $x_1, \ldots, x_d$, since for an algebraically closed field k the group $\mathrm{Gl}_d(k)$ operates trivially on $H_c^{\nu}(\mathbb{A}^d_{\mathrm{Spec}(k)}, \mu^d_{n\mathbb{A}^d_{\mathrm{Spec}(k)}})$.

1.10 Lemma. *Let $f\colon X \to S$ be a morphism of schemes. Suppose there exist elements $x_1, \ldots, x_d \in \Gamma(X, \mathcal{O}_X)$ such that the mapping*

$$X \to \mathbb{A}^d_S$$

that they define is etale. Then:

a) *The trace mapping $S_{X/S}^{(x_1,\ldots,x_d)}$ does not depend on the choice of the elements $x_1, \ldots, x_d$.*

b) *If S is the spectrum of an algebraically closed field k and X is a curve over S, then $S_{X/S}^{(x_1)}$ agrees with the trace $S_{X/K}$ already defined for curves.*

Proof. b) follows from Lemma 1.8.

a) In the case where $d(X/S) = 1$, a) follows from b) by the base change theorem. In general case, we can assume that S is the spectrum of an algebraically closed field k and that X is irreducible. Let U be an nonempty open subset of X. Since $H_c^{\nu}(X - U, F)$ vanishes for $\nu > 2d(X/S) - 2$, the mapping

$$H_c^{2d}(U, \mathcal{T}_{X/S}) \to H_c^{2d}(X, \mathcal{T}_{X/S})$$

is an isomorphism. Thus our assertion is of local nature with respect to X. Let a be a closed point of X, $\mathfrak{m}$ the maximal ideal of the regular local ring

$\mathcal{O}_{X,a}$. Let $(x_1, \ldots, x_d)$ and $(y_1, \ldots, y_d)$ be two systems for which the corresponding mappings $X \to \mathbb{A}_S^d$ are etale. We assert: If we replace X by a sufficiently small Zariski-open neighborhood of a, then there is a finite chain of systems of elements

$$(x_1, \ldots, x_d) = (x_1^0, \ldots, x_d^0), (x_1^{(1)}, \ldots, x_d^{(1)}), \ldots, (x_1^{(N)}, \ldots, x_d^{(N)}) = (y_1, \ldots, y_d)$$

such that all the systems induce etale mappings

$$X \to \mathbb{A}_S^d$$

and such that each system $(x_1^{(\nu+1)}, \ldots, x_d^{(\nu+1)})$ arises from $(x_1^{(\nu)}, \ldots, x_d^{(\nu)})$ by the replacement of just one element.

We can assume that the elements $x_1, \ldots, x_d, y_1, \ldots, y_d$ are contained in $\mathfrak{m}$ (replace x_ν by $x_\nu - a_\nu$, with $a_\nu \in k$, $x_\nu \equiv a_\nu$ modulo $\mathfrak{m}$, ...). But then $(x_1, \ldots, x_d)$ and $(y_1, \ldots, y_d)$ are regular systems of parameters for $\mathcal{O}_{X,a}$. But in a regular local ring of dimension d, one can start from any regular system of parameters and obtain any other one in d steps, changing only one element at each step (Steinitz exchange theorem applied to the k-vector space $\mathfrak{m}/\mathfrak{m}^2$). Any regular system of parameters for $\mathcal{O}_{X,a}$ defines an etale mapping into $\mathbb{A}_S^d$ in a sufficiently small Zariski-open neighborhood of a.

Thus it is enough to show that

$$S_{X/S}^{(x_1, \ldots, x_{d-1}, y)} = S_{X/S}^{(x_1, \ldots, x_{d-1}, z)}.$$

But this equality follows from the identity

$$S_{X/\mathbb{A}_S^{d-1}}(y) = S_{X/\mathbb{A}_S^{d-1}}(z)$$

that has already been proved. □

Thus for an morphism $X \to S$ that has a factorization through an etale mapping $X \to \mathbb{A}_S^d$, given by $x_1, \ldots, x_d$, we can define

$$S_{X/S} = S_{X/S}^{(x_1, \ldots, x_d)}.$$

Not every morphism $f\colon X \to S$ of type S can be factored in the specified way. But we do have:

There is a finite Zariski-open covering

$$\phi_i\colon U_i \hookrightarrow X$$

such that the restricted mappings $U_i \to S$ have such factorizations through etale mappings.

Indeed, let a be a point that is closed in its fiber, and $s = f(a)$ its image point in S. For a sufficiently small neighborhood U of a there are elements $x_1, \ldots, x_d \in \Gamma(U, \mathcal{O}_X)$ whose differentials in the module of differentials of $\mathcal{O}_{X,a}$ over $\mathcal{O}_{S,s}$ form a basis over the local ring $\mathcal{O}_{X,a}$. (See also Chap. I, 1.3.) After we shrink U, the induced mapping

$$U \to \mathbb{A}_S^d$$

will be etale.

We get a trace mapping for arbitrary morphisms of type S now from the following localization principle.

1.11 Lemma. *Let $f: X \to S$ be a morphism of schemes with maximal fiber dimension d. Let $\mathscr{F}$ be a sheaf on X, and let*

$$\phi_i: U_i \hookrightarrow X$$

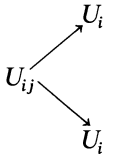
be an etale covering. We set

$$\phi_{ij}: U_{ij} = U_i \times_X U_j \to X$$
$$f_i = f \circ \phi_i, \quad f_{ij} = f \circ \phi_{ij}.$$

The sequence

$$\bigoplus_{i,j} R^{2d} f_{ij!}(\mathscr{F} \mid U_{ij}) \to \bigoplus_{i} R^{2d} f_{i!}(\mathscr{F} \mid U_i) \to R^{2d} f_!(\mathscr{F}) \to 0$$

induced by the projection mappings



is exact.

This follows from the right exactness of the functor $R^{2d} f_!$ and the exactness of the sequence

$$\bigoplus_{i,j} \phi_{ij!}(\mathscr{F} \mid U_{ij}) \rightrightarrows \bigoplus_{k} \phi_{k!}(\mathscr{F} \mid U_k) \to \mathscr{F} \to 0.$$

In this way we get a trace mapping $S_{X/S}$ for all (compactifiable) mappings $X \to S$ of type S. The compatibility properties required in Theorem 1.6 and the isomorphism assertion of Proposition 1.7 can be derived directly from the construction using the localization principle or can be reduced to the case of curves by base change and the properties of the composite trace (1.5). □

Now we want to define the trace mapping on the level of complexes (Appendix II). For simplicity we restrict ourselves to the case of a smooth algebraic variety of constant dimension over an algebraically closed field k,

$$f: X \to \mathrm{Spec}(k) = S, \quad d(X/S) = d.$$

$\mathrm{R}\Gamma_c(X, \mathscr{T}_{X/S})$ is quasi-isomorphic to a complex $\mathscr{K}^\cdot$ with $\mathscr{K}^\nu = 0$, $\nu > 2d$. In the derived category of complexes of Λ-modules, therefore, we have a homomorphism

$$\mathrm{R}\Gamma_c(X, \mathscr{T}_{X/S}) \to H^{2d}(\mathscr{K}^\cdot)[-2\mathrm{d}] = \mathrm{H}_c^{2\mathrm{d}}(\mathrm{X}, \mathscr{T}_{X/S})[-2d].$$

Here we write $\mathscr{K}^\cdot[m]$ for the shifted complex $T^m(\mathscr{K}^\cdot)$ (Appendix II). After shifting and composing with the trace mapping, we obtain a morphism in the derived category,

$$S_{X/S}: \mathrm{R}\Gamma_c(X, \mathscr{T}_{X/S}[2d]) \to \Lambda.$$

Moreover, there is a natural transformation for every sheaf $\mathscr{F}$ on X,

$$\mathrm{R\,Hom}(\mathscr{F}, \mathscr{T}_{X/S}[2d]) \to \mathrm{R\,Hom}(\mathrm{R}\Gamma_c(X, \mathscr{F}), \mathrm{R}\Gamma_c(X, \mathscr{T}_{X/S}[2d]).$$

By composing with $\mathrm{R}\,\mathrm{Hom}(\mathrm{Id}, S_{X/S})$ and using the fact $\mathrm{R}\,\mathrm{Hom}(-, \Lambda) = \mathrm{Hom}(-, \Lambda)$ (Λ is injective), we obtain the *duality mapping*,

1.12 $$\Delta_{X/S}\colon \mathrm{R}\,\mathrm{Hom}(\mathscr{F}, \mathscr{T}_{X/S}[2d]) \to \mathrm{Hom}_\Lambda(\mathrm{R}\Gamma_c(X, \mathscr{F}), \Lambda).$$

1.13 Poincaré Duality Theorem.

$$\Delta_{X/S}\colon \mathrm{R}\,\mathrm{Hom}(\mathscr{F}, \mathscr{T}_{X/S}[2d]) \to \mathrm{Hom}_\Lambda(\mathrm{R}\Gamma_c(X, \mathscr{F}), \Lambda)$$

is a quasi-isomorphism for all sheaves $\mathscr{F}$ on X.

The induced mapping

$$\Delta'_{X/S}\colon \mathrm{Ext}^p(\mathscr{F}, \mathscr{T}_{X/S}) \to \mathrm{Hom}_\Lambda(H_c^{2d-p}(X, \mathscr{F}), \Lambda)$$

is an isomorphism of ∂-functors.

Remark. The duality isomorphism $\mathrm{Ext}^p(\mathscr{L}, \mathscr{T}_{X/S}) \xrightarrow{\cong} \mathrm{Hom}_\Lambda(H_c^{2d-p}(\mathscr{L}, \Lambda))$ can also be described in terms of a pairing given by cup product and the trace mapping $S_{X/S}$ (A II):

The pairing induced by cup product and the trace mapping,

$$\mathrm{Ext}^p(\mathscr{F}, \mu_X^d) \times H_c^{2d-p}(X, \mathscr{F}) \to H_c^{2d}(X, \mu_X^d) \to \Lambda$$
$$(u, v) \mapsto u \smile v \qquad \mapsto S(u \smile v),$$

is nondegenerate.

We carry out the proof only for constructible sheaves $\mathscr{F}$. The general case follows by passage to limits.

Let $j\colon U \to X$ be an etale mapping and $\mathscr{F} = j_!(\Lambda_U)$. Then

$$\mathrm{Ext}^p(\mathscr{F}, \mathscr{T}_{X/S}) = \mathrm{Ext}^p(\Lambda_U, j^*(\mathscr{T}_{X/S})) = \mathrm{Ext}^p(\Lambda_U, \mathscr{T}_{U/S}) = H^p(U, \mathscr{T}_{U/S})$$

(adjointness property of $j_!$),

$$H_c^q(X, \mathscr{F}) \cong H_c^q(U, \Lambda_U).$$

It follows from the properties of the trace that under this isomorphism $\Delta'_{X/S}$ goes to $\Delta'_{U/S}$. For $\mathscr{F} = j_!(\Lambda_U)$ then in place of $\Delta'_{X/S}$ we have

$$\Delta'_{U/S}\colon H^p(U, \mathscr{T}_{U/S}) \to \mathrm{Hom}_\Lambda(H_c^{2d-p}(U, \Lambda_U), \Lambda).$$

Suppose now we have two etale mappings,

$$V \xrightarrow{i} U \xrightarrow{j} X.$$

Then on the leftz side the homomorphism

$$(j \circ i)_!(\Lambda_V) \to j_!(\Lambda_U)$$

induces the restriction mapping

$$H^p(U, \Lambda_U) \to H^p(V, \Lambda_V),$$

and on the right it yields the mapping induced by the natural mapping

$$H_c^q(V, \Lambda_V) \to H_c^q(U, \Lambda_U).$$

From Theorem 1.7 then we see that for $\mathscr{F} = j_! \Lambda_U$ the mapping

$$\Delta'_{X/S}\colon \operatorname{Ext}^0(\mathscr{F}, \mathscr{T}_{X/S}) = \Gamma(U, \mathscr{T}_{U/S}) \to \operatorname{Hom}_\Lambda(H_c^{2d}(U, \Lambda_U), \Lambda)$$

is an isomorphism.

If $\mathscr{F}$ is a constructible sheaf, then (Chap. I, 4.10) there always exists a finite family of etale mappings

$$j_\alpha\colon U_\alpha \to X$$

and a surjective mapping

$$\bigoplus_\alpha j_{\alpha!}(\Lambda_{U_\alpha}) \to \mathscr{F}.$$

Let a be a geometric point of X. Then

$$\varinjlim_U H^p(U, \mathscr{T}_{U/S}) = 0 \quad \text{for } p \neq 0,$$

where $U \to X$ runs through the etale neighborhoods of a.

We shall later prove a corresponding "dual" result for $H_c^q(U, \Lambda_U)$:

1.14 Fundamental Lemma. *Let U be a smooth variety of constant dimension d over an algebraically closed field k, and a geometric point of U. Then there is an etale neighborhood $j\colon V \to U$ of a such that the natural mapping*

$$H^q(U, j_! \Lambda_V) = H_c^q(V, \Lambda_V) \to H_c^q(U, \Lambda_U)$$

(induced by the trace $S_{V/U}$) is zero for $q \neq 2d$.

The duality theorem follows then from

1.15 Lemma. *Suppose*

$$F^\nu, G^\nu\colon \mathfrak{A} \to Ab$$

are two contravariant (exact) ∂-functors from an abelian category $\mathfrak{A}$ to the category Ab of abelian groups, and $F^\nu \to G^\nu$ is a given natural transformation of ∂-functors. Let $\mathfrak{M}$ be a class of objects in $\mathfrak{A}$ with the following properties:

a) *For every A in $\mathfrak{A}$ there is a surjective mapping $M \to A$ with M in $\mathfrak{M}$. If α is an element in $F^\nu(A)$ or $G^\nu(A)$, $\nu \neq 0$, then $M \to A$ can actually be chosen so that the image of α in $F^\nu(M)$ resp. $G^\nu(M)$ is zero.*

b) *For every M in $\mathfrak{M}$,*

$$F^0(M) \to G^0(M) \quad \text{is an isomorphism.}$$

Furthermore, $F^\nu(A) = G^\nu(A) = 0$ for $\nu < 0$ and all A in $\mathfrak{A}$.

Assertion. $F^\nu(A) \to G^\nu(A)$ is an isomorphism for all A in $\mathfrak{A}$ and all ν.

For the Proof. The case $\nu = 0$: Both functors F^0 and G^0 are left exact. An arbitrary object A is the cokernel of a homomorphism $M_1 \to M_2$ of objects in M. It follows by the five-lemma that $F^0(A) \to G^0(A)$ is an isomorphism. The rest follows by induction on ν. □

We come now to the proof of Lemma 1.14. We write X again instead of U. Since k is algebraically closed, we have $\Lambda_X \cong \mu_{nX}$ for all X. We need only consider points a: $\mathrm{Spec}(k) \to X$ over k. We carry out the proof first for curves, thus for $d(X/S)=1$. This implies the duality theorem for curves. Using this duality theorem we then prove 1.14 for $d(X/S)>1$ by induction on $d(X/S)$.

Let $X \to \mathrm{Spec}(k)=S$ be a smooth irreducible curve and a: $\mathrm{Spec}(k) \to X$ a geometric point. Let $Y \to X$ be an irreducible etale neighborhood of a. We consider smooth irreducible compactifications $\bar{X}$, $\bar{Y}$ of X and Y and obtain a commutative diagram

$$\begin{array}{ccc} Y & \longrightarrow & \bar{Y} \\ \downarrow & & \downarrow \\ X & \longrightarrow & \bar{X}. \end{array}$$

Here $\bar{Y} \to \bar{X}$ is a finite flat mapping, but possibly ramified outside Y.

Let $E=\bar{X}-X$, $F=\bar{Y}-Y$. We consider the commutative diagram

$$\begin{array}{ccccccccc} \ldots & \longrightarrow & H^0(F, \mu_n) & \longrightarrow & H^1_c(Y, \mu_n) & \longrightarrow & H^1(\bar{Y}, \mu_n) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ \ldots & \longrightarrow & H^0(E, \mu_n) & \longrightarrow & H^1_c(X, \mu_n) & \longrightarrow & H^1(\bar{X}, \mu_n) & \longrightarrow & 0. \end{array}$$

If F is nonempty, then $H^0_c(Y, \mu_n)$ vanishes. We describe the vertical arrows induced by the trace mapping given in Lemma 1.1:

$$H^1(\bar{Y}, \mu_n) = {}_n\mathrm{Jac}(\bar{Y}) = \mathrm{Ker}(\mathrm{Jac}(\bar{Y}) \xrightarrow{n} \mathrm{Jac}(\bar{Y})) \text{ *)}$$

$$H^1(\bar{X}, \mu_n) = {}_n\mathrm{Jac}(\bar{X}).$$

The mapping $H^1(\bar{Y}, \mu_n) \to H^1(\bar{X}, \mu_n)$ is induced by the natural mapping of the Jacobian varieties $\mathrm{Jac}(X) \to \mathrm{Jac}(Y)$ (the Albanese mapping; see also Lemma 1.3).

Let $Z_n=\mu_{nS}(S)$ be the group of n-th roots of unity. We consider

$$H^0(F, \mu_n) = \bigoplus_{b \in F} H^0(b, \mu_n) \to H^0(E, \mu_n) = \bigoplus_{\bar{b} \in E} H^0(\bar{b}, \mu_n)$$

$$\cup\mathrm{I} \qquad\qquad \cup\mathrm{I}$$

$$H^0(b, \mu_n) = Z_n \xrightarrow{e_b} Z_n = H^0(\bar{b}, \mu_n); \quad j(b)=\bar{b} \text{ in } E$$

$$\searrow 0; \quad j(b) \text{ not in } E.$$

Here e_b is the ramification degree of b.

We get what we want by repeat application of the following

*) We use the fact that the Jacobian variety $\mathrm{Jac}(X)$ of a curve, i.e. its Albanese variety, is in a natural way isomorphic to the Picard variety $\mathrm{Pic}^0(X)$.

1.16 Lemma. *Let $\bar{X}$ be a smooth irreducible projective curve.*

a) *There is a finite unramified covering space $\bar{Y}\to\bar{X}$ such that* $\mathrm{Ker}(\mathrm{Jac}(\bar{Y})\to\mathrm{Jac}(\bar{X}))$ *contains all n-division points.*

b) *There is a finite mapping $\bar{Y}\to\bar{X}$ of a smooth irreducible curve $\bar{Y}$ into $\bar{X}$ that is unramified over a prescribed point a and is such that n divides the ramification degree of every point of $\bar{Y}$ over a prescribed finite set E of points of $\bar{X}$ (not containing a).*

For a). Take for $\bar{Y}$ a component of the inverse image of $\bar{X}$ under the mapping $\mathrm{Jac}(\bar{X})\xrightarrow{n}\mathrm{Jac}(\bar{X})$.

For b). Let R be the function field of $\bar{X}$ and ϕ a rational function of order 0 at a and of order 1 at all points of E. Take for $\bar{Y}$ the normalization of $\bar{X}$ in $R(\sqrt[n]{\phi})$. □

This completes the proof of the duality theorem in the case of curves. An analysis of the proof shows that it used the isomorphism between the Picard and Albanese varieties for curves. Thus in this case the duality theorem ultimately depends on the *self-duality of the Jacobi variety of a curve.*

Before we come to the general case, we want to reformulate the local acyclicity theorem for smooth mappings without using Henselizations. Let $f\colon X\to S$ be of type S, $d(X/S)=1$, with s a geometric point of S and a a geometric point of X over s. To begin with, we still assume that S is the spectrum of a strictly Henselian ring and s is the closed point of S. For every geometric point t of S then there is an etale neighborhood U of a such that the geometric fibers F of $X\to S$ over t and $\tilde{F}$ of $U\to S$ over t satisfy

$$\mathrm{Image}[H^\nu(F,\Lambda_F)\to H^\nu(\tilde{F},\Lambda_{\tilde{F}})]=\begin{cases}0, & \nu\neq 0\\ \Lambda, & \nu=0.\end{cases}$$

Here the finiteness of these cohomology groups was used (Chap. I, 5.3). Dualy,

$$H_c^\nu(\tilde{F},\mu_{n\tilde{F}})\to H_c^\nu(F,\mu_{nF})$$

is the zero mapping for $\nu\neq 2$.

The image of $H_c^2(\tilde{F},\mu_{n\tilde{F}})$ in $H_c^2(F,\mu_{nF})$ is mapped isomorphically onto Λ by the trace mapping. But by the base change theorem, this means that stalkwise we have, for $\nu\geq 2$,

$$\prod_{j:U\to X}\mathrm{Image}[R^\nu(f\circ j)_!(\mu_n)\to R^\nu f_!(\mu_n)]=0.$$

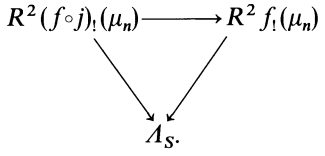
Here the intersection runs over all etale neighborhoods of a. Since $R^\nu f_!(\mu_n)$ is constructible, there is thus an etale neighborhood $U\to X$ of a for which the natural mapping

$$R^\nu(f\circ j)_!(\mu_n)\to R^\nu f_!(\mu_n)$$

is zero for $\nu\geq 2$. For $\nu=2$ we can similarly conclude:

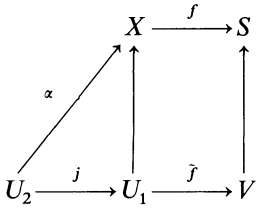
The image of $R^2(f\circ j)_!(\mu_n)$ in $R^2 f_!(\mu_n)$ is mapped isomorphically onto Λ_S by the trace mapping, and thus the natural mapping $R^2(f\circ j)_!(\mu_n)\to R^2 f_!(\mu_n)$ has

a factorization over Λ_S,



If S is not the spectrum of a strictly Henselian ring, then through Henselization of S at s we get:

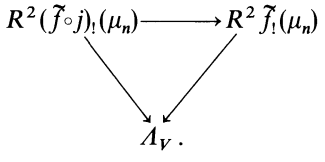
There are etale neighborhoods $U_i \to X$ of a $(i=1, 2)$, an etale neighborhood $V \to S$ of s, and a commutative diagram



of neighborhoods such that

$\nu \neq 2$: $R^\nu(\tilde{f}\circ j)_!(\mu_n) \to R^\nu \tilde{f}_!(\mu_n)$ is the zero mapping,

$\nu = 2$: There is a factorization



Let $S \xrightarrow{g} T = \mathrm{Spec}(k)$ be another mapping of type S, k an algebraically closed field,

$$d(S/T) < d(X/T) = d.$$

Suppose Lemma 1.14 has already been proved for dimension $d-1$. Then by hypothesis there is an etale neighborhood

$$W \to V$$

of s in V such that the mapping

$$H_c^\nu(W, \Lambda_W) \to H_c^\nu(V, \Lambda_V)$$

is the zero mapping for $\nu \neq 2(d-1)$. From this we get that the mapping in the Leray spectral sequence

$$H_c^p(S, R^q(f\circ\beta)_!(\mu_n)) \to H_c^p(S, R^q f_!(\mu_n))$$

is the zero mapping for $p+q \neq 2d$ (where β: $U_3 = U_2 \times_V W \to X$ is the induced map). Since the length of the Leray filtration is bounded, we eventually obtain Lemma 1.14 for $X \to \mathrm{Spec}(k)$ by repeated application of this result.

Locally, every mapping $X \to \mathrm{Spec}(k)$ of type S with $d(X/\mathrm{Spec}(k)) > 1$ has a factorization

$$X \to S \to \mathrm{Spec}(k)$$

where

$$X \to S \quad \text{and} \quad S \to \mathrm{Spec}(k)$$

are both of type S, $d(X/S) = 1$, and $d(S/\mathrm{Spec}(k)) < d(X/\mathrm{Spec}(k))$. Lemma 1.14 follows from this by induction. With this, the duality theorem is completely proved. □

In conclusion, we formulate the duality theorem for the category of sheaves of $\mathbb{Q}_l$-vector spaces (Chap. I, §12).

Let l be a prime number invertible on X. Passing to limits and forming quotients, we first obtain trace mappings

$$H_c^{2d}(X, \mathbb{Z}_l(d)) \to \mathbb{Z}_l,$$

$$H_c^{2d}(X, \mathbb{Q}_l)) \to \mathbb{Q}_l,$$

Now we consider a locally constant locally free l-adic sheaf $\mathscr{F} = (\mathscr{F}_n)$; thus $\mathscr{F}_n$ here is a locally constant locally free $\mathbb{Z}/l^{n+1}\mathbb{Z}$-module. Then

$$\mathsf{Hom}(\mathscr{F}, \mathbb{Z}_l(d)) = (\mathsf{Hom}(\mathscr{F}_n, \mu_{l^{n+1}}^d))_n$$

is a locally constant locally free l-adic sheaf, and

$$\mathsf{Hom}(\mathscr{F} \otimes \mathbb{Q}_l, \mathbb{Q}_l(d)) = \mathsf{Hom}(\mathscr{F}, \mathbb{Z}_l(d)) \otimes \mathbb{Q}_l$$

is a locally constant constructible sheaf of $\mathbb{Q}_l$-vector spaces. Passing to limits, we get a cup product and hence a pairing

$$H^p(X, \mathsf{Hom}(\mathscr{F}, \mathbb{Z}_l(d)) \times H_c^{2d-p}(X, \mathscr{F}) \to \mathbb{Z}_l.$$

If follows from the previous results that the induced mapping

$$H^p(X, \mathsf{Hom}(\mathscr{F}, \mathbb{Z}_l(d)) \to \mathrm{Hom}(H_c^{2d-p}(X, \mathscr{F}), \mathbb{Z}_l)$$

has only torsion modules for its kernel and cokernel. In general it is not an isomorphism (the reason for this is that $H_c^v(X, \mathscr{F}_n)$ in general is not a free $\mathbb{Z}/l^{n+1}\mathbb{Z}$-module).

Passing to quotients, we get:

1.17 Proposition. *For every locally constant locally free constructible sheaf $\mathscr{F} \otimes \mathbb{Q}_l$ of $\mathbb{Q}_l$-vector spaces, the natural pairing*

$$H^p(X, \mathsf{Hom}(\mathscr{F} \otimes \mathbb{Q}_l, \mathbb{Q}_l(d)) \times H_c^{2d-p}(X, \mathscr{F} \otimes \mathbb{Q}_l) \to \mathbb{Q}_l$$

is nondegenerate.

Note. 1) One can also deduce a duality theorem for l-adic sheaves on the level of complexes. Unlike $\mathbb{Z}/l^n\mathbb{Z}$, the ring $\mathbb{Z}_l$ is not injective over itself. Hence in this theory one must replace $\mathbb{Z}_l$ by an injective resolution.

2) Using the relative trace map we have constructed, one can also state a relative version of the duality theorem in the obvious way. The proof of this relative duality theorem can be found in [157].

§ 2. Cohomology Classes of Algebraic Cycles

In this section we consider algebraic varieties over the spectrum $S=\mathrm{Spec}(k)$ of an algebraically closed field k. As in § 1, let $\Lambda=\mathbb{Z}/n\mathbb{Z}$ with n relatively prime to the chatracteristic of k.

Let Y_0 be a reduced algebraic variety of constant dimension t over S, Y_0 the dense open subspace of regular points on Y. We have $\dim(Y-Y_0)<t$. The natural mapping

$$H_c^{2t}(Y_0, \mu_n^t)\to H_c^{2t}(Y, \mu_n^t) \qquad (\mu_n^t=\mu_n^{\otimes t})$$

is an isomorphism. Composing its inverse with the trace mapping

$$H_c^{2t}(Y_0, \mu_n^t)\to\Lambda,$$

we get a trace mapping

$$S_Y\colon\ H_c^{2t}(Y, \mu_n^t)\to\Lambda.$$

More generally, one can define a trace mapping for a non-reduced algebraic variety of constant dimension t,

$$S_Y=\sum_\nu e_\nu S_Y(\nu)\colon\ H_c^{2t}(Y, \mu_n^t)=H_c^{2t}(Y_{\mathrm{red}}, \mu_n^t)=\bigoplus_\nu (H_c^{2t}(Y^{(\nu)}, \mu_n^t)\to\Lambda.$$

Here the $Y^{(\nu)}$ are the irreducible components of Y_{red} and the numbers e_ν are the multiplicities of Y at the generic points of the components $Y^{(\nu)}$.

Let X be a smooth algebraic variety of constant dimensioon d over S,

$$j\colon\ Y\hookrightarrow X$$

an irreducible reduced closed subvariety of codimension s.

We compose the restriction mapping

$$j^*\colon\ H_c^{2(d-s)}(X, \mu_n^{d-s})\to H_c^{2(d-s)}(Y, \mu_n^{d-s})$$

with the trace mapping

$$S_Y\colon\ H_c^{2(d-s)}(Y, \mu_n^{d-s})\to\Lambda$$

and get an element α in

$$\mathrm{Hom}_\Lambda(\to H_c^{2(d-s)}(X, \mu_n^s), \Lambda).$$

By the duality theorem, this group is canonically isomorphic to $H^{2s}(X, \mu_n^s)$.

We obtain a cohomology class in $H^{2s}(X, \mu_n^s)$ as the image of α under the duality isomorphism.

2.1 Definition. The cohomology class thus constructed is called the cohomology class associated with $Y\subset X$.

Notation. $\mathrm{cl}_X(Y) \in H^{2s}(X, \mu_n^s)$.

Extending linearly, we get a homomorphism

$$\mathrm{cl}_X\colon\ Z^s(X) \to H^{2s}(X, \mu_n^s)$$

from the group $Z^s(X)$ of algebraic cycles of codimension s on X to the cohomology group $H^{2s}(X, \mu_n^s)$.

Remark. Using the characterization of the duality mapping by the trace and the cup product $\smile$, one finds that cl_X is characterized by the following properties for positive cycles $y \in Z^s(x)$: If y is the cycle associated with a (not necessarily reduced) subvariety $j\colon Y \hookrightarrow X$, then

$$S_x(\mathrm{cl}_X(y) \smile v) = S_Y(j^*(v))$$

for all $\mathrm{H}_c^{2(d-s)}(\mathrm{X}, \mu_n^{d-s})$.

If X is already defined over a subfield k_0 of k, and y is a cycle rational over k_0, then

$$\mathrm{cl}_X(y) \in H^{2s}(X, \mu_n^s)^{\mathrm{Gal}(k/k_0)}.$$

The sheaf of ideals $\mathscr{J}$ of an irreducible reduced subvariety Y of X of codimension 1 is a line bundle. We set $\mathscr{L}(Y) = \mathscr{J}^{-1}$. Extending linearly, we get a homomorphism

$$Z^1(X) \to \mathrm{Pic}(X),$$
$$D \mapsto \mathscr{L}(D).$$

2.2 Proposition. *Let $D \in Z^1(X)$ be a divisor on X, and $\mathscr{L}(D)$ the associated line bundle. Let $\mathrm{cl}(\mathscr{L}(D))$ be again the first Chern class of $\mathscr{L}(D)$, defined using the Kummer sequence (see Definition 1.2). Then*

$$\mathrm{cl}_X(D) = \mathrm{cl}(\mathscr{L}(D)).$$

2.3 Corollary. *The cohomology class $\mathrm{cl}_X(D)$ of a divisor $D \in Z^1(X)$ depends only on the rational equivalence class. The mapping $\mathrm{cl}_X(D)$ is compatible with construction of inverse images in the sense of intersection theory:*

For $f\colon \tilde{X} \to X$ we have

$$f^*(\mathrm{cl}_X(D)) = \mathrm{cl}_{\tilde{X}}(f^*(D)).$$

Remark. The assertion in the corollary can also be proved for cycles of arbitrary codimension.

For the proof we can assume that $D = Y$ is an irreducible reduced subvariety of codimension one and that X is irreducible. Let

$$j\colon\ Y \hookrightarrow X$$

be the embedding of Y in X.

We refine the statement of the proposition in order to be able to localize it better. We shall take the two cohomology classes in $H^2(X, \mu_n)$ that we have

associated with the divisor Y and represent them in a natural way as images of cohomology classes in $H^2_Y(X, \mu_n)$ (cohomology with support in Y); then we will show that these latter classes are the same.

Let $\mathscr{R}_X$ be the sheaf of rational functions on the "etale topos" of X (Chap. I, § 5), $\mathscr{R}_X^*$ the sheaf of invertible rational functions, and $\mathscr{O}_X^*$ the sheaf of invertible holomorphic functions (exceptionally, these are not sheaves of Λ-modules).

The line bundle $\mathscr{L}(Y)$ associated with Y is in a natural way contained in $\mathscr{R}_X$,

$$\mathscr{O}_X \subset \mathscr{L}(Y) \subset \mathscr{R}_X .$$

There is a covering $\{U_i\}$ by Zariski-open subsets U_i of X and rational functions $f_i \in \mathscr{R}_X^*(U_i)$ with

$$\mathscr{O}_{U_i} \cdot f_i = \mathscr{L}(D) | U_i .$$

The family $\{f_i\}$ defines a section $\alpha \in \Gamma(X, \mathscr{R}_X^*/\mathscr{O}_X^*)$, whose support is contained in Y because $\mathscr{L}(D)|(X-Y) = \mathscr{O}_X(X-Y)$. The section α is independent of the choice of the covering and the functions f_i.

For the short exact sequence

$$0 \to \mathscr{O}_X^* \to \mathscr{R}_X^* \to \mathscr{R}_X^*/\mathscr{O}_X^* \to 0$$

and the Kummer sequence

$$0 \to \mu_n \to \mathscr{O}_X^* \xrightarrow{\ n\ } \mathscr{O}_X^* \to 0,$$

we consider the corresponding long exact cohomology sequences with support in Y:

$$\ldots \to \Gamma_Y(X, \mathscr{R}_X^*/\mathscr{O}_X^*) \to H^1_Y(X, \mathscr{O}_X^*) \to \ldots$$
$$\ldots \to H^1_Y(X, \mathscr{O}_X^*) \to H^2_Y(X, \mu_n) \to \ldots .$$

2.4 Definition. We denote the image of α under the composite mapping

$$\Gamma_Y(X, \mathscr{R}_X^*/\mathscr{O}_X^*) \to H^2_Y(X, \mu_n)$$

by $\mathrm{cl}^0(Y)$ $(\in H^2_Y(X, \mu_n))$.

Now for the sheaf

$$\mathscr{F} = j_* j^*(\mu_{nX}^{d-1}) = j_*(\mu_{nY}^{d-1})$$

we consider the duality isomorphism

$$\Delta\colon \mathrm{Ext}^1(\mathscr{F}, \mu_n^d) \xrightarrow{\ \cong\ } \mathrm{Hom}_\Lambda(H_c^{2d-1}(X, \mathscr{F}), \Lambda).$$

We have canonical isomorphisms

$$H_c^m(X, \mathscr{F}) \cong H_c^m(Y, \mu_{nY}^{d-1})$$
$$\mathrm{Ext}^i(\mathscr{F}, \mu_{nX}^d) = H^i_Y(X, \mathsf{Hom}(\mu_{nX}^{d-1}, \mu_{nX}^d)) \cong H^i_Y(X, \mu_{nX}).$$

The duality isomorphism goes over into an isomorphism

2.5. $$\Delta\colon H^i_Y(X, \mu_{nX}) \xrightarrow{\ \cong\ } \mathrm{Hom}_\Lambda(H_c^{2d-1}(Y, \mu_{nY}^{d-1}), \Lambda).$$

We consider Δ for $i = 2$.

2.6 Definition and Lemma. *We denote by* $\mathrm{cl}'^0(Y)$ *the image of*

$$S_Y \in \mathrm{Hom}_\Lambda(H_c^{2d-1}(Y, \mu_{nY}^{d-1}), \Lambda)$$

under

$$\Delta^{-1}\colon \mathrm{Hom}_\Lambda(H_c^{2d-1}(Y, \mu_{nY}^{d-1}), \Lambda) \to H_Y^2(X, \mu_n).$$

The natural image of $\mathrm{cl}'^0(Y) \in H_Y^2(X, \mu_n)$ *in* $H^2(X, \mu_n)$ *is* $\mathrm{cl}_X(Y)$.

Theorem 2.2 follows now from

2.7 Proposition. *Under the previous hypotheses,*

$$\mathrm{cl}^0(Y) = \mathrm{cl}'^0(Y).$$

This last assertion can be "localized".

Let $f\colon \tilde{X} \to X$ be an etale mapping with $\tilde{Y} = f^{-1}(Y)$ irreducible and nonempty. Then there is a commutative diagram

$$\begin{array}{ccc} H_c^{2(d-1)}(\tilde{Y}, \mu_n^{d-1}) & \xrightarrow{S_{\tilde{Y}}}_{\cong} & \Lambda \\ \downarrow & \nearrow_{S_Y} & \\ H_c^{2(d-1)}(Y, \mu_n^{d-1}). & & \end{array}$$

Thus the mapping

$$H_c^{2(d-1)}(\tilde{Y}, \mu_n^{d-1}) \to H_c^{2(d-1)}(Y, \mu_n^{d-1})$$

is also an isomorphism.

Using the duality theorem (see 1.13), we derive from this that

$$H_{\tilde{Y}}^2(\tilde{X}, \mu_n) \leftarrow H_Y^2(X, \mu_n)$$

is an isomorphism.

Since cl^0 and cl'^0 are functorial under such mappings, the assertion of the theorem for (X, Y) is equivalent to that for $(\tilde{X}, \tilde{Y})$.

Thus we have obtained a localization principle that allows us to restrict to the case

$$\begin{aligned} X &= \mathbb{A}^{d-1} \times \mathbb{P}^1 && \mathbb{A} \text{ affine line} \\ Y &= \mathbb{A}^{d-1} \times 0 && \mathbb{P}^1 \text{ projective line.} \end{aligned}$$

For "locally" (in an etale neighborhood of the generic point of Y) this situation is isomorphic to the original situation (see also I.7.1).

We consider the mapping

$$\begin{aligned} j\colon \mathbb{A}^{d-1} &\to \mathbb{A}^{d-1} - \mathbb{P}^1 \\ a &\mapsto (a, 0) \end{aligned}$$

and the projections

$$\mathbb{A}^{d-1} - \mathbb{P}^1 \xrightarrow{p} \mathbb{A}^{d-1}$$
$$\xrightarrow{q} \mathbb{P}^1.$$

We have $H^\nu(\mathbb{A}^{d-1}, \Lambda) = 0$ for $\nu \neq 0$ (I.10.2), and so

$$H_c^\nu(\mathbb{A}_{d-1}, \mu_n^{d-1}) = 0 \quad \text{for } \nu \neq 2(d-1).$$

Using the Künneth formula (I.8.17), we get that the mapping

$$p^*\colon H_c^{2(d-1)}(\mathbb{A}^{d-1}, \mu_n^{d-1}) \to H_c^{2(d-1)}(\mathbb{A}^{d-1} \times \mathbb{P}^1, \mu_n^{d-1})$$

is an isomorphism and that

$$u \in H_c^{2(d-1)}(\mathbb{A}^{d-1}, \mu_n^{d-1}), \qquad v \in H^2(\mathbb{P}^1, \mu_n)$$

satisfy

$$S_{\mathbb{A}^{d-1} \times \mathbb{P}^1}(p^*(u) \smile q^*(v)) = S_{\mathbb{A}^{d-1}}(u)\, S_{\mathbb{P}^1}(v)$$

(see 1.6). Let

$$\alpha = \mathrm{cl}(\mathscr{L}(0)) \in H^2(\mathbb{P}^1, \mu_n).$$

It follows from the characterization of the trace for curves that

$$S_{\mathbb{P}^1}(\alpha) = 1$$

$$\mathrm{cl}(\mathscr{L}(Y)) = \mathrm{cl}(q^*(\mathscr{L}(0))) = q^*(\alpha).$$

Assertion. $\mathrm{cl}(\mathscr{L}(Y)) = q^*(\alpha)$ *equals* $\mathrm{cl}_{\mathbb{A}^{d-1} \times \mathbb{P}^1}(Y)$!!

Indeed, for $u = p^*(u_0) \in H_c^{2(d-1)}(X, \mu_n^{d-1})$ we have

$$\begin{aligned} S_{\mathbb{A}^{d-1} \times \mathbb{P}^1}(q^*(\alpha) \smile u) &= S_{\mathbb{A}^{d-1} \times \mathbb{P}^1}(p^*(u_0) \smile q^*(\alpha)) \\ &= S_{\mathbb{A}^{d-1}}(u_0)\, S_{\mathbb{P}^1}(\alpha) = S_{\mathbb{A}^{d-1}}(u_0) \\ &= S_{\mathbb{A}^{d-1}}(j^*(u)). \end{aligned}$$

On the other hand, since $(\mathbb{A}^{d-1} \times \mathbb{P}^1) - Y \cong \mathbb{A}^d$,

$$H_Y^2(\mathbb{A}^{d-1} \times \mathbb{P}^1, \mu_n) \cong H^2(\mathbb{A}^{d-1} \times \mathbb{P}^1, \mu_n).$$

Hence also

$$\mathrm{cl}^0(Y) = \mathrm{cl}'^0(Y). \quad \square$$

From the compatibility of $\mathrm{cl}_X(-)$ with formation of inverse images of divisors in the sense of intersection theory, we can deduce a fixed point theorem of the Lefschetz type. To this end we first compute the cohomology class of the diagonal

$$D \subset X \times X.$$

We choose a fixed primitive n-th root of unity, $\rho \in k$. For every variety X, ρ defines natural isomorphisams

$$\mu_{nX} \to \Lambda_X$$
$$\rho \mapsto 1$$

and

$$\mu_{nX}^{\otimes s} \to \Lambda_X^{\otimes s} \cong \Lambda_X$$

that are compatible with formation of inverse images. Using these isomorphisms, we make the identifications

$$H^v(X, \mu_{nX}^s) = H^v(X, \Lambda_X),$$
$$H_c^v(X, \mu_{nx}^s) = H_c^v(X, \Lambda_X).$$

The trace mapping then is a mapping

$$S_X\colon\ H_c^{2\dim X}(X, \Lambda_X) \to \Lambda.$$

As before, let X be smooth of constant dimension d. In addition, let X be complete.

For simplicity, we shall assume that all the cohomology groups $H^v(X, \Lambda)$ are free Λ-modules. For example, this is true for curves.

Let

$$p\colon\ X \times X \to X, \qquad q\colon\ X \times X \to X$$

be the projections onto the first and second factors respectively,

$$j\colon\ X \hookrightarrow X \times X$$

the diagonal mapping, $j(X) = D$ the diagonal.

One can compute $H^\cdot(X \times X, \Lambda)$ using the Künneth formula (I.8.17):

$$H^\cdot(X) \otimes H^\cdot(X) \to H^\cdot(X \times X)$$
$$\alpha \otimes \beta \mapsto p^*(\alpha) \smile q^*(\beta)$$

is an isomorphism.

The trace mapping $S_{X \times X}$ is the product of the trace mappings for the factors X (1.6):

$$S_{X \times X}(p^*(\alpha) \smile q^*(\beta)) = S_X(\alpha)\, S_X(\beta), \qquad \alpha, \beta \in H^{2d}(X, \Lambda).$$

For each $0 \leqq v \leqq 2d$ we choose a basis $e_1^v, e_2^v, \ldots$ for $H^v(X, \Lambda)$ and a dual basis $f_1^v, f_2^v, \ldots$ for $H^{2d-v}(X, \Lambda)$,

$$S_X(e_i^v \smile f_j^v) = \delta_{ij}.$$

This is possible because the pairing $S_X(u \smile v)$ establishes the duality isomorphism $H^v(X, \Lambda_X) \to \mathrm{Hom}(H^2d - v(X, \Lambda_X), \Lambda)$.

2.8 Proposition. *Under the stated hypotheses, the cohomology class of the diagonal D in $X \times X$ can be calculated:*

$$\mathrm{cl}_{X \times X}(D) = \sum_{i,v} p^*(f_i^v) \smile q^*(e_i^v).$$

Proof. For the proof, we use the characterization of the cohomology class of a positive cycle in terms of cup products and trace (Remark on Definition 2.1).

Thus we must show that element

$$\alpha = \sum_{i,\nu} p^*(f_i^\nu) \smile q^*(e_i^\nu) \in H^{2d}(X \times X, \Lambda)$$

satisfies

$$S_{X\times X}(\alpha \smile u) = S_X(j^*(\alpha)) \quad \text{for all } u \in H^{2d}(X \times X, \Lambda).$$

It is sufficient to prove this formula for the basis elements

$$u = p^*(e_r^\mu) \smile q^*(f_s^\mu) \in H^{2d}(X \times X, \Lambda).$$

We have

$$p^*(f_i^\nu) \smile q^*(e_i^\nu) \smile p^*(e_r^\mu) \smile q^*(f_s^\mu)$$
$$= p^*(\underbrace{e_r^\mu \smile f_i^\nu}_{\in H^{2d-\nu-\mu}(X,\Lambda)}) \smile q^*(\underbrace{e_i^\nu \smile f_s^\mu = 0}_{\in H^{2d-(\mu-\nu)}(X,\Lambda)}) \quad \text{for } \nu \neq \mu,$$

$$S_{X\times X}(p^*(f_i^\nu) \smile q^*(e_i^\nu) \smile p^*(e_r^\nu) \smile q^*(f_s^\nu))$$
$$= S_X(e_r^\nu \smile f_i^\nu) \cdot S_X(e_i^\nu \smile f_s^\nu) = \delta_{ir} \cdot \delta_{is}$$

$$S_{X\times X}(\alpha \smile p^*(e_r^\mu) \smile q^*(f_s^\mu)) = \sum_{i,\nu} \delta_{\nu\mu}\, \delta_{ri}\, \delta_{is} = \delta_{rs}.$$

On the other hand, we have

$$S_X(j^*(p^*(e_r^\mu) \smile q^*(f_s^\mu))) = S_X(p \circ j)^*(e_r^\mu) \smile (q \circ j)^*(f_k^\mu))$$
$$= S_X(e_r^\mu \smile f_s^\mu) = \delta_{rs}. \quad \square$$

Now let

$$h\colon X \to X$$

be a morphism,

$$\Gamma_h\colon X \to X \times X$$
$$x \mapsto (x, h(x))$$

the associated graph mapping.

Let the endomorphism $h^* | H^\nu(X, \Lambda)$ induced by h on $H^\nu(X, \Lambda)$ be given in the basis $\{e_i^\nu\}$ by the matrix $M^{(\nu)} = (c_{rs}^{(\nu)})$,

$$h^*(e_i^\nu) = \sum_j c_{ji}^{(\nu)}\, e_j^\nu.$$

We have

$$S_X(\Gamma_h^*(\mathrm{cl}_{X\times X}(D))) = \sum_\nu (-1)^\nu \sum S_X(h^*(e_i^\nu) \smile f_i^\nu)$$
$$= \sum (-1)^\nu \sum_{i,j} c_{ji}^{(\nu)}\, \delta_{ij} = \sum_\nu (-1)^\nu \,\mathrm{Trace}(M^{(\nu)}).$$

If X is a curve, then D is a divisor. We have proved:

$$\Gamma_h^*(\mathrm{cl}_{X\times X}(D)) = \mathrm{cl}_X(\Gamma_h^*(D)).$$

But it follows from the definition of the trace that for curves,

$$S_X(\mathrm{cl}_X(\mathrm{Divisor})) = \deg(\mathrm{Divisor}) \quad \text{in } \Lambda = \mathbb{Z}/n\mathbb{Z}.$$

2.9 Theorem (Lefschetz Fixed Point Formula). *Let $h\colon X\to X$ be an endomorphism of a smooth complete curve over an algebraically closed field, $\Gamma_h\colon X\to X\times X$ the associated graph mapping. Then the inverse image $\Gamma_h^*(D)$ of the diagonal $D\subset X\times X$, formed in the sense of intersection theory (modulo rational equivalence) ([49]) satisfies*

$$\deg(\Gamma_h^*(D))=\sum_\nu(-1)^\nu\,\mathrm{Trace}(h^*\mid H^\nu(X,\Lambda))\qquad \text{in } \mathbb{Z}/n\mathbb{Z}=\Lambda.$$

Remark. If h has only isolated fixed points, then $\deg(\Gamma_h^*(D))$ is the number of fixed points of h, counted with multiplicities. If h is the identity mapping, then for an irreducible curve of genus g we get

$$2-2g=\sum_\nu(-1)^\nu\dim H^\nu(X,\Lambda).$$

If we use the compatibility of the cohomology class of an algebraic cycle of arbitrary dimension with formation of inverse images in the sense of intersection theory, we get the fixed point formula also for smooth complete manifolds of arbitrary dimension.

§ 3. The Fixed Point Formula of Grothendieck and Nielsen-Wecken for the Frobenius Homomorphism of a Curve

We need a generalization of the trace and the characteristic polynomial of an endomorphism of a finitely generated free module.

Let P be a finitely generated projective module over a commutative ring A, and

$$f\colon P\to P$$

an endomorphism. We choose any projective module Q for which $P\times Q=F$ is a free module. We consider the mapping

$$F_P\colon P\times Q\to P\times Q$$
$$(a,b)\mapsto(f(a),0).$$

We then define the trace and characteristic polynomial of f by

$$\mathrm{Tr}_A(f)=\mathrm{Trace}(f_F)\in A,$$
$$\det(1-tf)=\det(\mathrm{Id}_F-tf_F)\in A[t].$$

The constructions are independent of the choice of Q.

Now let $P^\cdot$ be a bounded complex of finitely generated projective A-modules, and

$$f^\cdot\colon P^\cdot\to P^\cdot$$

a homomorphism of complexes. We set

$$\mathrm{Tr}_A(f^\cdot)=\sum_\nu(-1)^\nu \mathrm{Tr}_A(f^\nu)\in A,$$

$$\det(1-tf^\cdot)=\prod_\nu \det(1-tf^\nu)^{(-1)^\nu}\in A[[t]].$$

One can show:

a) If all the cohomology groups of $P^\cdot$ vanish, then

$$\mathrm{Tr}_A(f^\cdot)=0, \quad \det(1-tf^\cdot)=1.$$

b) Let $Q^\cdot$ be another bounded complex of finitely generated projective A-modules. Suppose the following diagram of mappings of complexes is commutative up to homotopy:

$$\begin{array}{ccc} P^\cdot & \xrightarrow{f^\cdot} & P^\cdot \\ {\scriptstyle\phi^\cdot}\downarrow & & \downarrow{\scriptstyle\phi^\cdot} \\ Q^\cdot & \xrightarrow{g^\cdot} & Q^\cdot. \end{array}$$

Let $C^\cdot(\phi^\cdot)$ be the mapping cone of $\phi^\cdot$. Together with the homotopy occurring, $f^\cdot$ and $g^\cdot$ define in a natural way a homomorphism of complexes

$$h^\cdot\colon\ C^\cdot(\phi^\cdot)\to C^\cdot(\phi^\cdot).$$

Then we have

$$\mathrm{Tr}_A(f^\cdot)+\mathrm{Tr}_A(h^\cdot)=\mathrm{Tr}_A(g^\cdot),$$

$$\det(1-tf^\cdot)\cdot\det(1-th^\cdot)=\det(1-tg^\cdot).$$

It follows from this in particular that the trace and characteristic polynomial depend only on the homotopy class of the mapping, and that these constructions can even be defined for a certain class of mappings in the derived category $D(A)$ of the category of A-modules.

3.1 Definition. Let B be a ring, not necessarily commutative. A complex (of left modules) $K^\cdot\in D(B)$ is called perfect if there is a bounded complex $P^\cdot$ of finitely generated projective B-modules and a quasi-isomorphism

$$P^\cdot\to K^\cdot.$$

Back to the commutative case.

3.2 Definition. Let $K^\cdot$ be a perfect complex in $D(A)$, and

$$f^\cdot\colon\ K^\cdot\to K^\cdot$$

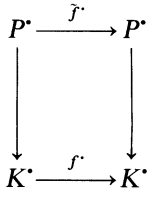
a morphism in $D(A)$. We choose a bounded complex $P^\cdot$ of finitely generated projective modules and a quasi-isomorphism

$$P^\cdot\to K^\cdot.$$

Then there is a homomorphism of complexes

$$\tilde{f}^{\cdot}: P^{\cdot} \to P^{\cdot}$$

such that the diagram

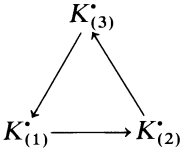


is commutative in $D(A)$. We set

$$\mathrm{Tr}_A(f^{\cdot}) = \mathrm{Tr}_A(\tilde{f}^{\cdot}),$$
$$\det(1 - tf^{\cdot}) = \det(1 - t\tilde{f}^{\cdot}).$$

Properties a) and b) of the trace and characteristic polynomial of an endomorphism of a bounded complex of finitely generated projective A-modules imply that this definition is independent of the choice of $P^{\cdot} \to K^{\cdot}$ and of $\tilde{f}^{\cdot}$. Moreover, we get:

3.3 Lemma. *The trace and characteristic polynomial are additive in the following sense Let*

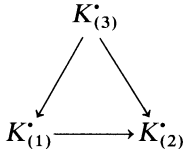


be an admissible triangle in $D(A)$, and let $f^{\cdot}_{(v)}: K^{\cdot}_{(v)} \to K^{\cdot}_{(v)}$ be mappings in $D(A)$. Suppose two of the three complexes are perfect. Then the third is also perfect.

1) *The following additivity formulae*

$$\mathrm{Tr}_A(f^{\cdot}_{(1)}) + \mathrm{Tr}_A(f^{\cdot}_{(3)}) = \mathrm{Tr}_A(f^{\cdot}_{(2)}),$$
$$\det(1 - tf^{\cdot}_{(1)}) \cdot \det(1 - tf^{\cdot}_{(3)} = \det(1 - tf^{\cdot}_{(2)}).$$

2) *are valid, if the mappings $f^{\cdot}_{(v)}$ define a true endomorphism of the triangle in the following sense Let $E(A)$ be the (abelian) category of endomorphisms $M \to M$ of A modules and $D(E(A))$ its derived Category. Each admissible triangle of $D(E(A))$ defines in an obvious way an admissible triangle*



in $D(A)$ and a system $f^{\cdot}_{(\nu)}\colon K^{\cdot}_{(\nu)}\to K^{\cdot}_{(\nu)}$ of mappings in $D(A)$. A system of mapping which arises in this way is called a true endomorphism of this triangle in $D(A)$.)*

The concept of trace can also be defined for certain noncommutative rings. Let G be a finite group, $A[G]$ the group ring of G over A. The center $\mathfrak{Z}(A[G])\subset A[G]$ has a basis over A the "conjugacy classes" $[\sigma]$:

$$\sigma\in G\colon\ [\sigma]=\sum_{\substack{\tau \text{ in } G\\ \text{conjugate}\\ \text{to}\ \sigma}}\tau.$$

We consider the "trace mapping"

$$S_3\colon\ \mathfrak{Z}(A[G])\to A$$
$$[\sigma]\mapsto 1.$$

Let F be a finitely generated free (left) $A[G]$-module,

$$f\colon\ F\to F$$

an $A[G]$-homomorphism. After selection of an $A[G]$-basis, f is given by a matrix M with coefficients in $A[G]$. We can represent M uniquely as

$$M=\sum_{\sigma\in G}\sigma M_\sigma$$

where the M_σ are matrices with entries in A. We define

$$\mathrm{Tr}_G(f)=\sum_{\sigma}\mathrm{Trace}(M_\sigma)[\sigma]\in\mathfrak{Z}(A[G]).$$

This construction is independent of the choice of basis.

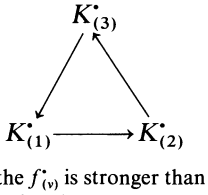
3.4 Definition and Lemma. *The same construction procedure as in the commutative case extends the concept of trace to endomorphisms of perfect complexes in $D(A[G])$. For every homomorphism*

$$f^{\cdot}\colon\ K^{\cdot}\to K^{\cdot}\quad \text{in } D(A[G])$$

of a perfect $A[G]$-complex, one gets in a natural way an element

$$\mathrm{Tr}_G(f^{\cdot})\in\mathfrak{Z}(A[G]).$$

Tr_G *is additive If the mappings $f^{\cdot}_{(\nu)}\colon K^{\cdot}_{(\nu)}\to K^{\cdot}_{(\nu)}$ in $D(A[G])$ define a true endomorphism in the same sense described in $\rangle.\langle$ of a triangle*



*) This compatibility condition for the $f^{\cdot}_{(\nu)}$ is stronger than the obvious condition of commutativity in $D(A)$. The latter is note sufficient for the additivity. The notation of a true endomorphism of a triangle is defined in an obvious way for each abelian category instead of the category of modules. The result which we need in the text is the following. Let F be *a* for example left exact covariant functor of an abelian category A with enough injectives into another abelian category. If the morphisms $f^{\cdot}_1$, $f^{\cdot}_2$, $f^{\cdot}_3$ define a true endomorphism of a triangle in $D(A)$ then the derived morphisms $\mathfrak{R}f^{\cdot}_1$, $\mathfrak{R}f^{\cdot}_2$, $\mathfrak{R}f^{\cdot}_3$ define also a true endomorphism.

in $D(A[G])$ and two of these three complexes are perfect, then the third one is also perfect, and

$$\mathrm{Tr}_G(f^{\cdot}_{(1)})+\mathrm{Tr}_G(f^{\cdot}_{(3)})=\mathrm{Tr}_G(f^{\cdot}_{(2)}).$$

We need some further properties of Tr_G:

3.5 Lemma. *Let $K^{\cdot}$ be a perfect complex in $D(G[A])$ and*

$$f\colon\ K^{\cdot}\to K^{\cdot}k$$

a morphism in $D(A[G])$.

(1) *Let σ be an element in G, $e([\sigma])$ the number of elements in the centralizer of σ in G. Multiplication by σ defines an A-homomorphism*

$$\sigma\colon\ K^{\cdot}\to K^{\cdot}.$$

Then

$$\sigma\circ f^{\cdot}=f^{\cdot}\circ\sigma\colon\ K^{\cdot}\to K^{\cdot}\quad (\text{in } D(A)).$$

Let $\mathrm{Tr}_G(f^{\cdot})=\sum a_{[\sigma]}[\sigma]$.

Then $K^{\cdot}$ is also perfect as a complex of A-modules, and

$$\mathrm{Tr}_A(\sigma\circ f^{\cdot})=e([\sigma])\,a_{[\sigma^{-1}]}.$$

(2) *Let Γ^G denote the functor of forming G-invariants. $\mathsf{R}\Gamma^G$ its derived functor in the sense of Verdier. Then $f^{\cdot}$ induces a homomorphism*

$$\mathsf{R}\Gamma^G(f^{\cdot})\colon\ \mathsf{R}\Gamma^G(K^{\cdot})\to\mathsf{R}\Gamma^G(K^{\cdot})$$

in $D(A)$.

The complex $\mathsf{R}\Gamma^G(K^{\cdot})$ is perfect in $D(A)$, and

$$S_3(\mathrm{Tr}_G(f^{\cdot}))=\mathrm{Tr}_A(\mathsf{R}\Gamma^G(f^{\cdot})).$$

(3) *Let M be a finitely generated $A[G]$-module that is projective as an A-module. Let*

$$\psi\colon\ M\to M$$

be an $A[G]$-homomorphism. Choose a fixed representative $\sigma(\alpha)$ in each conjugacy class α of G, and define an action (depending on this choice) of $\mathfrak{Z}(A[G])$ on the A-module M by

$$\alpha(m)=\sigma(\alpha)\,m.$$

In this way each element of $\mathfrak{Z}(A[G])$ defines an A-homomorphism of M to itself. Let

$$\Big(\sum_{\sigma\in G}a_\sigma\cdot\sigma\Big)^*=\sum_{\sigma\in G}a_\sigma\cdot\sigma^{-1}.$$

We consider the G-homomorphism

$$f^{\cdot}\otimes\psi\colon\ K^{\cdot}\otimes_A M\to K^{\cdot}\otimes_A M.$$

Then $K^{\cdot}\otimes_A M$ is a perfect $A[G]$-module, and

$$\mathrm{Tr}_A(\mathrm{Tr}_G(f^{\cdot})^*\circ\psi)=S_3(\mathrm{Tr}_G(f^{\cdot}\otimes\psi)).$$

By (2) *therefore*

$$\mathrm{Tr}_A(R\Gamma^G(f^\cdot \otimes \psi)) = \mathrm{Tr}_A(\mathrm{Tr}_G(f^\cdot)^* \circ \psi).$$

Here $\mathrm{Tr}_G(f^\cdot) \in \mathfrak{Z}(A[G])$ *is interpreted in an* A*-endomorphism of* M.

For the proof of (3). Without loss of generality we can assume that $K^\cdot = F$ is a finite free $A[G]$-module and M is a free A-module.

Typical case: $F = A[G]$.

Let M_{tr} denote M considered as a trivial G-module. We have a G-isomorphism

$$H\colon\ A[G] \otimes_A M_{\mathrm{tr}} \to A[G] \otimes_A M$$

$$\sigma \otimes m \mapsto \sigma \otimes \sigma m.$$

Let $f(1) = \sum a_\tau \cdot \tau$. Then

$$h^{-1} \circ (f \otimes \psi) \circ h(\sigma \otimes m) = \sum_\tau \sigma\tau \otimes a_\tau\, \tau^{-1}\, \psi(m),$$

$$S_3(\mathrm{Tr}_G(f \otimes \psi)) = \mathrm{Tr}_A(\sum_\alpha (\sum_{\tau \in \alpha} a_\tau)\, \sigma(\alpha^{-1}) \circ \psi),$$

α running through the conjugacy classes of G.

Now let A and hence $A[G]$ be noetherian on both sides. Then every finitely generated flat left $A[G]$-module is projective. From this, as in I 8.15 (see the associated Remark), we get the following criterion for perfectness:

3.6 Lemma. *Let* $K^\cdot$ *be a complex of finitely generated* $A[G]$*-modules, bounded above. Suppose there is some* v_0 *such that for every finitely generated right* $A[G]$*-module* M *one has*

$$\mathrm{Tor}_v^{A[G]}(M, K^\cdot) = 0 \quad \text{for } v \geq v_0.$$

Then $K^\cdot$ *is a perfect* $A[G]$*-complex.*

The Frobenius Homomorphism ([67], SGA 5, Chap. XV)

In the following, κ is a finite field of characteristic p with $q = p^s$ elements, and k is its algebraic closure. For every algebraic variety X over κ the Frobenius morphism with respect to κ,

$$\mathrm{Fr}_X\colon\ X \to X,$$

is defined over κ. In the affine case $X = \mathrm{Spec}(A)$, Fr_X is given by the κ-homomorphism

$$A \leftarrow A$$

$$a^q \longleftarrow a.$$

Fr_X commutes with all κ-morphisms of algebraic varieties over κ.

Let $U \to X$ be a morphism. Since the diagram

$$\begin{array}{ccc} \mathrm{Fr}_U\colon\ U & \longrightarrow & U \\ \downarrow & & \downarrow \\ \mathrm{Fr}_X\colon\ X & \longrightarrow & X \end{array}$$

is commutative, there is an induced mapping

$$\mathrm{Fr}_{U|X}\colon\ U\to X_{\mathrm{Fr}_X}\times_X U=\mathrm{Fr}_{X^{-1}}(U)$$

(X considered as a scheme over X via Fr_X).

3.7 Lemma. *If $U\to X$ is etale, then $\mathrm{Fr}_{U|X}\colon U\to X_{\mathrm{Fr}_X}\times_X U$ is an isomorphism, i.e. the diagram above is cartesian.*

Let $\mathcal{G}$ be a sheaf of sets on X. For every etale mapping $U\to X$, there is a mapping

$$\mathcal{G}(U)\xrightarrow{\mathrm{Fr}_{\bar{U}}|_X}\mathcal{G}(\mathrm{Fr}_{X^{-1}}(U))=(\mathrm{Fr}_X)_*(\mathcal{G})(U).$$

The collection of these mappings defines an *isomorphism*

$$\mathcal{G}\to(\mathrm{Fr}_X)_*(\mathcal{G}).$$

Because

$$\mathrm{Hom}(\mathcal{G},(\mathrm{Fr}_X)_*(\mathcal{G}))=\mathrm{Hom}(\mathrm{Fr}_X^*(\mathcal{G}),\mathcal{G}),$$

this isomorphism induces a mapping

$$\mathrm{Fr}_X^*(\mathcal{G})\to\mathcal{G}.$$

If $\mathcal{G}$ is representable by a scheme $V\to X$ etale over X, this mapping corresponds precisely to the isomorphism

$$(\mathrm{Fr}_{V|X})^{-1}\colon\ \mathrm{Fr}_{X^{-1}}(V)\to V.$$

3.8 Definition and Lemma. *The map thus constructed is called the (geometric) Frobenius homomorphism of $\mathcal{G}$ (with respect to κ)*

$$\mathrm{Fr}_{\mathcal{G}}\colon\ \mathrm{Fr}_X^*(\mathcal{G})\to\mathcal{G}.$$

$\mathrm{Fr}_{\mathcal{G}}$ is an isomorphism and compatible with all morphisms of sheaves.

3.9 Remark. $\mathrm{Fr}_{\mathcal{G}}$ has a series of compatibility properties: For a κ-morphism $f\colon Y\to X$. $\mathrm{Fr}_{\mathcal{G}}$ is compatible with formation of inverse images, direct images, higher direct images, and with the "higher direct images with compact support". By d-fold repeated application, one gets a homomorphism

$$\mathrm{Fr}_{\mathcal{G}}^d\colon\ (\mathrm{Fr}_X^d)^*(\mathcal{G})\to\mathcal{G}.$$

If κ' is a field extension of degree d of κ, then

$$\mathrm{Fr}_X^d\otimes\kappa'\colon\ X\otimes\kappa'\to X\otimes\kappa'$$

is precisely the Frobenius morphism $\mathrm{Fr}_{X\otimes\kappa'}$ of $X\otimes\kappa'$ with respect to κ'. The corresponding fact holds for the inverse image $\mathcal{G}_{\kappa'}$ of $\mathcal{G}$ on $X\otimes\kappa'$ and the induced mapping

$$(\mathrm{Fr}_{\mathcal{G}}^d)_{\kappa'}\colon\ (\mathrm{Fr}_{X\otimes\kappa'})^*(\mathcal{G}_{\kappa'})\to\mathcal{G}_{\kappa'}$$

$$(\mathrm{Fr}_{\mathcal{G}}^d)_{\kappa'}=\mathrm{Fr}_{\mathcal{G}_{\kappa'}};$$

Let U_0 be an algebraic scheme over κ and $\mathscr{G}_0$ a sheaf on U_0. We denote by $U = U_0 \otimes k$ the extension of U_0 to the algebraic closure k of κ, and by $\mathscr{G}$ the inverse image of $\mathscr{G}_0$ on U. After field extension, Fr_{U_0} and $\mathrm{Fr}_{\mathscr{G}_0}$ induce mappings

$$\mathrm{Fr}_U = \mathrm{Fr}_{U_0} \otimes k\colon\ U \to U$$

and

$$\mathrm{Fr}_{\mathscr{G}}\colon\ Fr_U^*(\mathscr{G}) \to \mathscr{G}.$$

We consider fixed powers of Fr_U and $\mathrm{Fr}_{\mathscr{G}}$:

$$F_U = \mathrm{Fr}_U^d\colon\ U \to U.$$

$$F_{\mathscr{G}} = \mathrm{Fr}_{\mathscr{G}}^d\colon\ (\mathrm{Fr}_U^d)^*(\mathscr{G}) \to \mathscr{G}.$$

These arise from $\mathrm{Fr}_{U_0}^d$ and $\mathrm{Fr}_{\mathscr{G}_0}^d$ by field extension. The pair of mappings $(F_U, F_{\mathscr{G}})$ induces a mapping

$$\mathsf{R}\Gamma_c(F_U, F_{\mathscr{G}})\colon\ \mathsf{R}\Gamma_c(U, \mathscr{G}) \to \mathsf{R}\Gamma_c(U, \mathscr{G}).$$

Then

$$\mathsf{R}\Gamma_c(F_U, F_{\mathscr{G}}) = \mathsf{R}\Gamma_c(\mathrm{Fr}_U^d, \mathrm{Fr}_{\mathscr{G}}^d) = (\mathsf{R}\Gamma_c(F_U, F_{\mathscr{G}}))^d.$$

3.10 Theorem ([66, SGA 4 1/2, p. 100, Theorem 5.3). *Let U_0 be a smooth curve over κ and $\mathscr{G}_0$ a constructible sheaf (of $\Lambda = \mathbb{Z}/n\mathbb{Z}$-modules) whose stalks are projective Λ-modules. Then for the inverse image $\mathscr{G}$ of $\mathscr{G}_0$ on U,*

$$\mathsf{R}^\cdot\Gamma_c(U, \mathscr{G}) \text{ is a perfect complex of } \Lambda\text{-modules.}$$

For a power $(F_U, F_{\mathscr{G}}) = (\mathrm{Fr}_U^d, \mathrm{Fr}_{\mathscr{G}}^d)$ of $(\mathrm{Fr}_U, \mathrm{Fr}_{\mathscr{G}})$,

$$\mathrm{Tr}_\Lambda(\mathsf{R}\Gamma_c(F_U, F_{\mathscr{G}})) = \sum_{a \in \mathrm{Fix}(F_U)} \mathrm{Tr}_\Lambda(F_{\mathscr{G}_a}).$$

Here $\mathrm{Fix}(F_U)$ is the set of geometric fixed points $\rightleftharpoons$ with values in $k \leftrightharpoons$ of $F_U = \mathrm{Fr}_U^d$. (For each point $a \in \mathrm{Fix}(F_U)$, F induces a stalk mapping

$$F_{\mathscr{G}_a}\colon\ \mathscr{G}_a = F_U^*(\mathscr{G})_a \to \mathscr{G}_a.)$$

Proof. The complex $K^\cdot = \mathsf{R}^\cdot\Gamma_c(U, \mathscr{G})$ is perfect (I.8.16).

The fixed point formula is trivial for sheaves that are concentrated on finitely many points. Moreover, if $0 \to \mathscr{G}_0 \to \mathscr{H}_0 \to \mathscr{M}_0 \to 0$ is a short exact sequence of sheaves on U_0 whose stalks are projective, then the inverse image sequence on U,

$$0 \to \mathscr{G} \to \mathscr{H} \to \mathscr{M} \to 0,$$

is exact and induces an admissible triangle in $D(\Lambda)$,

$$\begin{array}{ccc} & \mathsf{R}\Gamma_c(U, \mathscr{M}) & \\ \swarrow & & \nwarrow \\ \mathsf{R}\Gamma_c(U, \mathscr{G}) & \longrightarrow & \mathsf{R}\Gamma_c(U, \mathscr{H}). \end{array}$$

Since the trace is additive (3.3), it is enough to prove the theorem for two of the three sheaves.

If $j_0\colon U_0' \hookrightarrow U_0$ is an open dense subscheme of the curve U_0, then $\mathscr{G}_0/j_{0!}j_0^*(\mathscr{G}_0)$ is concentrated on finitely many points; hence it is enough to prove the theorem for $j_{0!}j_0^*(\mathscr{G}_0)$, and thus it is enough to prove it for $j_0^*(\mathscr{G}_0)$ instead of $\mathscr{G}_0$. Shrinking U_0, then, we can assume that

$$\mathscr{G}_0 \text{ is locally constant,}$$

$$\mathrm{Fr}_U^d \text{ has no fixed points.}$$

In addition, we can assume that U_0 is irreducible.

We then have to prove:

$$\mathrm{Tr}_\Lambda(\mathrm{R}\Gamma_c(F_U, F_{\mathscr{G}})) = 0.$$

We embed U_0 as an open dense subspace of a smooth irreducible projective curve X_0. Since $\mathscr{G}_0$ is locally constant and U_0 is irreducible, there is a smooth irreducible projective curve Y_0, a finite mapping

$$Y_0 \xrightarrow{p_0} X_0,$$

and a finite group G of automorphisms of Y_0 over X_0 that acts on Y_0 on the right with the following properties:

$X_0 = Y_0/G$,
G acts without fixed points on $V_0 = p_0^{-1}(U_0)$,
the inverse image of $\mathscr{G}_0$ on V_0 is constant, i.e. of the form M_{V_0}.

Here M is a finitely generated projective Λ-module on which G acts. Thus we have the following situation:

$$\begin{array}{ccc} V_0 & \xrightarrow{i_0} & Y_0 \\ {\scriptstyle q_0}\downarrow & & \downarrow{\scriptstyle p_0} \\ U_0 & \xrightarrow{j_0} & X_0 = Y_0/G. \end{array}$$

Now G acts on $q_{0*}(M_{V_0})$, and we have $\mathscr{G}_0 = q_{0*}(M_{V_0})^G$. After extensions to k, we have a corresponding situation

$$\begin{array}{ccc} V & \xrightarrow{i} & Y \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ V/G = U & \xrightarrow{j} & X = Y/G. \end{array}$$

Here G acts without fixed points on $V = p^{-1}(U)$, and $\mathscr{G} = q_*(M_V)^G$.

We consider the category of G-sheaves on Y (and similarly on V).

3.11 Definition. For an etale scheme s: $W \to Y$ and an element $\sigma \in G$, we denote by $W_\sigma \to Y$, or W_σ for short, the etale scheme $\sigma^{-1} \circ (W \to Y)$. This W_σ is canonically isomorphic to the inverse image $\sigma^{-1}(W)$ of W under σ: $Y \to Y$.

A G-sheaf on Y is a sheaf $\mathscr{H}$ (of sets or modules) on Y together with the following additional structure: For each etale object $W \to Y$ and each element σ in the group G, a mapping

$$S_{W,\sigma}\colon\ \mathscr{H}(W) \to \mathscr{H}(W_\sigma)$$

is prescribed in such a way that for every morphism $W' \to W$ of etale spaces over Y the natural diagram

$$\begin{array}{ccc} \mathscr{H}(W') & \longrightarrow & \mathscr{H}(W'_\sigma) \\ \downarrow & & \downarrow \\ \mathscr{H}(W) & \longrightarrow & \mathscr{H}(W_\sigma) \end{array}$$

is commutative and for every two elements $\sigma, \tau \in G$ we have

$$S_{(W_\sigma),\tau} \circ S_{W,\sigma} = S_{W,\tau\sigma}$$

($(W_\sigma)_\tau = W_{\tau\sigma}$, as G acts on Y on the right).

The G-sheaves form a category in the obvious way.

Remark. The mappings $S_{W,\sigma}$ define a morphism

$$\mathscr{H} \to \sigma_*(\mathscr{H}).$$

We want to assemble some properties of G-sheaves of Λ-modules. If $\mathscr{H}$ is a G-sheaf of abelian groups on Y, then $\mathsf{R}^\cdot \Gamma(Y, \mathscr{H})$ is in a natural way a complex of $\Lambda[G]$-modules.*) If $\mathscr{H}$ is a G-sheaf on V, then $i_!(\mathscr{H})$ is a G-sheaf on Y, and so

$$\mathsf{R}^\cdot \Gamma_c(V, \mathscr{H}) = \mathsf{R}(\Gamma(Y, i_!(\mathscr{H}))$$

is a complex in $D(\Lambda[G])$. The group G acts on $q_*(\mathscr{H})$. For a $\Lambda[G]$-module N, the constant sheaf N_V is a G-sheaf on V.

3.12 Lemma. *Let N and P be two finitely generated $\Lambda[G]$-modules, both Λ-projective. Then*

a) $\mathsf{R}\Gamma^G \mathsf{R}\Gamma_c(V, N_V) = \mathsf{R}\Gamma^G \mathsf{R}\Gamma_c(Y, i_!(N_V)) = \mathsf{R}\Gamma_c(U, q_*(N_V)^G)$,

b) $\mathsf{R}\Gamma_c(V, (N \otimes_\Lambda P)_V) \cong {}_G N \otimes_\Lambda \mathsf{R}\Gamma_c(V, P_V)$.

Back to our sheaf $\mathscr{G} = q_*(M_V)^G$.

As in the case of a commutative base ring (I.8.16), we can use the Künneth formula and Lemma 3.6 to show that $\mathsf{R}\Gamma_c(V, M_V)$ is a perfect complex of $\Lambda[G]$-

*) One shows: The category of (abelian) G-sheaves has enough injective objects, and these are injective in the category of all abelian sheaves (without G-structure). Indeed, the forgetful functor has a left adjoint functor.

modules. The complex

$$K^{\cdot} = \mathrm{R}\Gamma_c(V, \Lambda_V)$$

of $\Lambda[G]$-modules is likewise perfect.

From 3.12 b) we get

$$K^{\cdot} \otimes_{\Lambda} M \cong_G \mathrm{R}\Gamma_c(V, M_V),$$

and from 3.12 a) we get

$$\mathrm{R}\Gamma_c(U, \mathscr{G}) = \mathrm{R}\Gamma^G(K^{\cdot} \otimes_{\Lambda} M).$$

Thus by Lemma 3.5 (3) it is enough to show that

$$\mathrm{Tr}_G(\mathrm{R}\Gamma_c(F_V, F_{\Lambda_V})) = 0.$$

Here

$$F_{\Lambda_V}\colon\ F_V^*(\Lambda_V) = \Lambda_V \to \Lambda_V$$

is the identity mapping; G acts trivially on Λ.

We consider now another number n' relatively prime to the characteristic of κ and divisible by n. For $\Lambda' = \mathbb{Z}/n'\mathbb{Z}$ then we have:

The image of the trace

$$\mathrm{Tr}_G(\mathrm{R}\Gamma_c(F_V, F_{\Lambda_V'})) \in \mathfrak{Z}(\Lambda'[G]) \subseteq \Lambda'[G]$$

under the natural mapping

$$\Lambda'[G] \to \Lambda[G] = \Lambda'[G]/n\Lambda'[G]$$

is precisely the trace

$$\mathrm{Tr}_G(\mathrm{R}\Gamma_c(F_V, F_{\Lambda_V})) \in \mathfrak{Z}(\Lambda[G])$$

(cf. proof of 4.1).

Hence it is sufficient to show that, for all rings $\Lambda = \mathbb{Z}/n\mathbb{Z}$ with n relatively prime to the characteristic,

$$\#(G)\ \mathrm{Tr}_G(\mathrm{R}\Gamma_c(F_V, F_{\Lambda_V})) = 0.$$

Here by hypothesis

$$F_U\colon\ U = V/G \to U$$

has no fixed points.

3.13 Lemma. *For every element $\sigma \in G$ and the prescribed power $F_Y = \mathrm{Fr}_Y^d$, the Frobenius mapping*

$$\sigma \circ F_V = F_V \circ \sigma\colon\ Y \to Y$$

has only fixed points of multiplicity one. These all lie in the comlement of V.

The mapping

$$\sigma \circ F_{\Lambda}\colon\ (\sigma \circ F_Y)^*(\Lambda_Y) = \Lambda_Y \to \Lambda_Y$$

is the identity. We consider the exact sequence

$$0 \to i_!(\Lambda_V) \to \Lambda_Y \to \mathscr{M} \to 0,$$

$$\mathrm{Tr}_{\Lambda}\mathrm{R}\Gamma_c(\sigma \circ F_V, \sigma \circ F_{\Lambda_V}) = \mathrm{Tr}_{\Lambda}\mathrm{R}\Gamma(\sigma \circ F_Y, \sigma \circ F_{\Lambda_Y}) - \mathrm{Tr}_{\Lambda}\mathrm{R}\Gamma(\sigma \circ F_Y, \sigma \circ F_{\mathfrak{M}}).$$

The sheaf $\mathfrak{M}$ is concentrated on finitely many points, and

$$\mathrm{Tr}_\Lambda \mathrm{R}\Gamma(\sigma \circ F_Y, \sigma \circ F_{\mathfrak{M}}) = \# \mathrm{Fix}(\sigma \circ F_Y).$$

From the Lefschetz fixed point formula for $\sigma \circ F_Y$ (2.9) we get

$$\mathrm{Tr}_\Lambda \mathrm{R}\Gamma(\sigma \circ F_Y, \sigma \circ F_{\Lambda_Y}) = \# \mathrm{Fix}(\sigma \circ F_Y).$$

Thus $\mathrm{Tr}_\Lambda \mathrm{R}\Gamma_c(\sigma \circ F_V, \sigma \circ F_{\Lambda_V})$ vanishes.

Let

$$\mathrm{Tr}_G(\mathrm{R}\Gamma_c(F_V, F_{\Lambda_V})) = \Sigma a_{[\sigma]}[\sigma].$$

By Theorem 3.5 (1),

$$e([\sigma])\, a_{[\sigma^{-1}]} = 0,$$

and hence

$$\#(G) \cdot a_{[\sigma^{-1}]} = 0.$$

As this holds for all $\sigma \in G$, we have proved

$$\#(G) \cdot \mathrm{Tr}_G(\mathrm{R}\Gamma_c(F_V, F_{\Lambda_V})) = 0. \quad \square$$

We now want to generalize Theorem 3.10. Since the trace is additive, we get the corresponding result when we replace the sheaf $\mathscr{G}_0$ by a bounded complex $\mathscr{H}_0^\cdot$ of constructible sheaves with flat stalks. It is in fact enough to assume that $\mathscr{H}_0^\cdot$ is a perfect complex in the sense of Chap. I, Def. 8.15. For then $\mathscr{H}_0^\cdot$ is quasi-isomorphic to a complex with the sharper properties. We generalize the result to varieties of arbitrary dimension.

3.14 Theorem. *Lefschetz Fixed Point Formula for the Frobenius Morphism.*

Let $f_0\colon U_0 \to V_0$ be a morphism of algebraic schemes over the finite field κ, and let $\mathscr{H}_0^\cdot$ be a perfect complex on U_0. All stalks of this complex are then perfect complexes of Λ-modules. By extension to the algebraic closure k of κ, one gets a morphism

$$f\colon U \to V$$

and a complex $\mathscr{H}^\cdot$ as inverse image on U. By base extension of a power of the Frobenius mapping, one gets

$$\mathrm{Fr}_U^d\colon U \to U,$$

$$\mathrm{Fr}_{\mathscr{H}^\cdot}^d\colon (\mathrm{Fr}_U^d)^*(\mathscr{H}^\cdot) \to \mathscr{H}^\cdot.$$

The same holds true for V and the image complex $\mathscr{K}^\cdot = \mathrm{R}f_!(\mathscr{H}^\cdot)$. Then $\mathscr{K}^\cdot$ also has perfect stalk complexes, and

$$\sum_{a \in \mathrm{Fix}(\mathrm{Fr}_U^d)} \mathrm{Tr}_\Lambda(\mathrm{Fr}_{\mathscr{H}^\cdot, a}^d) = \sum_{b \in \mathrm{Fix}(\mathrm{Fr}_V^d)} \mathrm{Tr}_\Lambda(\mathrm{Fr}_{\mathscr{K}^\cdot, b}^d)$$

For $V_0 = \mathrm{Spec}(\kappa)$, this yields:

3.15 Corollary.

$$\boxed{\sum_{a \in \mathrm{Fix}(\mathrm{Fr}_U^d)} \mathrm{Tr}_\Lambda(\mathrm{Fr}_{\mathscr{H}^\cdot, a}^d) = \mathrm{Tr}_\Lambda \mathrm{R}\Gamma_c(\mathrm{Fr}_U^d, \mathrm{Fr}_{\mathscr{H}^\cdot}^d).}$$

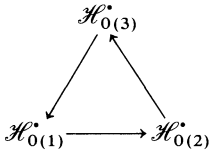
The proof comes from repeated application of the following facts. First, one can replace U_0, V_0 by $(U_0)_{\mathrm{red}}$, $(V_0)_{\mathrm{red}}$. Further:

(1) The assertion of the theorem is true for quasi-finite mappings.

(2) The assertion is true for smooth mappings whose fibers have dimension one.

(3) If $f_0 = h_0 \circ g_0$ and the theorem is true for h_0 and g_0, then it is true also for f_0.

(4) Let



be a triangle in the derived category of the category of sheaves on U_0. If the theorem is true for two of the complexes, it also holds for the third.

In particular we have (see also I 8.7 (3)).

(5) Let U_0' be an open subspace of U_0, and A_0 the complement of U_0' in U_0. If the theorem is true for $U_0' \to V_0$ and for $A_0 \to V_0$, it is true for $U_0 \to V_0$.

For point (2): Let Z_0' be the subscheme of fixed points of $\mathrm{Fr}^d_{V_0}$: $V_0 \to V_0$ and $Z_0 = (Z_0')_{\mathrm{red}}$. Here Z_0' is the inverse image of the diagonal of $V_0 \times V_0$ under the graph mapping of $\mathrm{Fr}^d_{V_0}$, Z_0 is a smooth variety of dimension zero, and $U_0 \times_{V_0} Z_0$ is a smooth curve over κ. The assertion for $U_0 \times_{V_0} Z_0 \to Z_0$ follows from Theorem 3.10 and its generalization to complexes. The desired formula for $U_0 \to V_0$ then follows by the base change theorem.

All other points, and the proof of the fixed point formula, are now clear. □

§4. Grothendieck's Formula for L-Series of l-adic Sheaves

As in the previous section, let κ be a finite field and k its algebraic closure. In what follows, we consider not sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules but rather l-adic sheaves (Chap. I, §12). Many of the definitions and results in the previous section carry over immediately to l-adic sheaves by passage to the limit. For example, there is an obvious way to define the Frobenius homomorphism $\mathrm{Fr}\,\mathscr{G}_0$ for an l-adic sheaf $\mathscr{G}_0$ on a scheme over κ.

We want to formulate the trace formula for the Frobenius homomorphism from the previous section (Corollary 3.15) in a somewhat different way for l-adic sheaves. Let $\Lambda = \varprojlim \mathbb{Z}/l^{\nu}\mathbb{Z}$ be the ring of l-adic numbers, $\mathbb{Q}_l$ its quotient field, V a finitely generated Λ-module, and $\phi\colon V \to V$ an endomorphism. Define the trace and characteristic polynomial of ϕ by

$$\mathrm{Tr}_{\Lambda}(\phi) = \mathrm{Tr}_{\mathbb{Q}_l}(\phi \otimes \mathbb{Q}_l) \in \Lambda \subset \mathbb{Q}_l,$$

$$\det(1 - t\phi) = \det(1 - t\phi \otimes \mathbb{Q}_l) \in \Lambda[t].$$

In other words, the trace and characteristic polynomial of ϕ are the trace and characteristic polynomial of the endomorphism induced on V modulo l-torsion.

For a perfect complex $\mathscr{K}^\cdot$ of Λ-modules and an endomorphism

$$\phi^\cdot\colon \mathscr{K}^\cdot \to \mathscr{K}^\cdot \quad \text{in } D(\Lambda),$$

we have

$$\mathrm{Tr}_\Lambda(\phi^\cdot)=\sum_\nu (-1)^\nu H^\nu(\phi^\cdot),$$

$$\det(1-t\phi^\cdot)=\prod_\nu \det(1-tH^n(\phi^\cdot))^{(-1)^\nu}.$$

Here $H^\nu(\phi^\cdot)$ is the endomorphism induced by $\phi^\cdot$ on the cohomology group $H^\nu(\mathscr{K}^\cdot)$.

Let X be an algebraic scheme over the separably closed base field k, and $\mathscr{G}=(\mathscr{G}_n)$ a torsion-free l-adic sheaf on X (Chap. I, §12). The sheaves $\mathscr{G}_n$ of the projective system then are constructible sheaves of $\Lambda_n=\mathbb{Z}/l^{n+1}\mathbb{Z}$-modules with Λ_n-flat stalks. Let

$$\phi\colon X\to X$$

be a k-morphism, and let

$$f\colon \phi^*(\mathscr{G})\to\mathscr{G}$$

be a homomorphism of l-adic sheaves, i.e. a compatible system of homomorphisms

$$f_\nu\colon \phi^*(\mathscr{G}_\nu)\to\mathscr{G}_\nu.$$

4.1 Lemma. *Consider the mappings*

$$\mathrm{R}\Gamma_c(\phi,f_\nu)\colon \mathrm{R}\Gamma_c(X,\mathscr{G}_\nu)\to\mathrm{R}\Gamma_c(X,\mathscr{G}_\nu)$$

and

$$H_c^m(\phi,f)=\varprojlim H_c^m(\phi,f_\nu)\colon\ H_c^m(X,\mathscr{G})=\varprojlim H_c^m(X,\mathscr{G}_\nu)\to H_c^m(X,\mathscr{G}_\nu)$$

induced by the pair (ϕ,f). Then the image of

$$\sum_m (-1)^m \mathrm{Tr}\, H_c^m(\phi,f)$$

in Λ_n is precisely the trace

$$\mathrm{Tr}\,\mathrm{R}\Gamma_c(\phi,f_n).$$

Remark. The derived complex $\mathrm{R}\Gamma_c(X,\mathscr{G}_\nu)$ is formed for the sheaf $\mathscr{G}_\nu$ in the category of sheaves of Λ_ν-modules. It is then a perfect complex of Λ_ν-modules (I.8.16).

Proof. In Chapter I, §12, it was shown (proof of Theorem 12.15 and Lemma 12.5) that there is a bounded complex $\mathscr{K}^\cdot$ of finitely generated free Λ-modules such that the system of derived complexes and derived mappings

$$\ldots\to\mathrm{R}\Gamma_c(X,\mathscr{G}_{n+1})\to\mathrm{R}\Gamma_c(X,\mathscr{G}_n)\to\ldots$$

is represented by the projective system

$$\ldots\to\mathscr{K}^\cdot/l^{n+2}\mathscr{K}^\cdot\to\mathscr{K}^\cdot/l^{n+1}\mathscr{K}^\cdot\to\ldots.$$

The mappings $R\Gamma_c(\phi, f_n)$ then are represented by mappings of complexes

$$h_n^\cdot\colon \mathscr{K}^\cdot/l^{n+1}\mathscr{K}^\cdot \to \mathscr{K}^\cdot/l^{n+1}\mathscr{K}^\cdot.$$

The diagrams

$$\begin{array}{ccc} h_{n+1}^\cdot\colon \mathscr{K}^\cdot/l^{n+2}\mathscr{K}^\cdot & \longrightarrow & \mathscr{K}^\cdot/l^{n+2}\mathscr{K}^\cdot \\ \downarrow & & \downarrow \\ h_n^\cdot\colon \mathscr{K}^\cdot/l^{n+1}\mathscr{K}^\cdot & \longrightarrow & \mathscr{K}^\cdot/l^{n+1}\mathscr{K}^\cdot \end{array}$$

are commutative up to homotopy. Since the complexes $\mathscr{K}^\cdot/l^{n+1}\mathscr{K}^\cdot$ are complexes of free Λ_n-modules for every n, we can modify the mappings on the complexes homotopically step by step until the diagrams actually become *commutative.* After this modification, we can form the projective limit

$$h^\cdot\colon \varprojlim_n h_n^\cdot\colon \mathscr{K}^\cdot \to \mathscr{K}^\cdot.$$

We have:

$$H_c^m(X, \mathscr{G}_n) = H^m(\mathscr{K}^\cdot/l^{n+1}\mathscr{K}^\cdot),$$

$$H_c^m(X, \mathscr{G}) = \varprojlim_n H^m(\mathscr{K}^\cdot/l^{n+1}\mathscr{K}^\cdot) = H^m(\mathscr{K}^\cdot),$$

$$H_c^m(\phi, f) = H^m(h^\cdot),$$

$$\operatorname{Tr} h^\cdot = \Sigma(-1)^m \operatorname{Tr} H_c^m(\phi, f).$$

This lemma makes it possible to carry over the fixed point formula proved in the previous section to l-adic sheaves.

Let X_0 be an algebraic scheme over the finite field κ and $(\mathscr{G}_0) = (\mathscr{G}_{0\nu})$ a constructible l-adic sheaf on X_0. Let

$$\mathrm{Fr}_{X_0}\colon X_0 \to X_0$$

again be the Frobenius morphism of X_0 over κ. The system

$$\mathrm{Fr}_{\mathscr{G}_{0\nu}}\colon \mathrm{Fr}_{X_0}^*(\mathscr{G}_{0\nu}) \to \mathscr{G}_{0\nu}$$

defines the *Frobenius homomorphism*

$$\mathrm{Fr}_{\mathscr{G}_0}\colon \mathrm{Fr}_{X_0}(\mathscr{G}_0) \to \mathscr{G}_0.$$

By base field extension from k to the algebraic closure we get an algebraic scheme X over k, an l-adic constructible sheaf $\mathscr{G}$ on X, and the "Frobenius mappings"

$$\mathrm{Fr}_X\colon X \to X,$$

$$\mathrm{Fr}_{\mathscr{G}}\colon \mathrm{Fr}_X^*(\mathscr{G}) \to \mathscr{G}.$$

We consider a fixed power of Fr_{X_0}, $\mathrm{Fr}_{\mathscr{G}_0}$ resp. Fr_X, $\mathrm{Fr}_{\mathscr{G}}$:

$$\mathrm{Fr}_X^d\colon X \to X, \qquad \mathrm{Fr}_{\mathscr{G}}^d\colon (\mathrm{Fr}_X^d)^*(\mathscr{G}) \to \mathscr{G}.$$

Let a be a geometric fixed point of Fr_X^d:

$$a \in \mathrm{Fix}(\mathrm{Fr}_X^d) = \mathrm{Hom}_{\mathrm{Spec}(\kappa)}(X_0, \mathrm{Spec}\,\kappa_d)$$
$$\kappa \subset \kappa_d \subset \kappa, \quad [\kappa_d : \kappa] = d.$$

Then $\mathrm{Fr}_{\mathscr{G}}$ induces a homomorphism

$$\mathrm{Fr}_{\mathscr{G},a}^d : \mathscr{G}_a \to \mathscr{G}_a,$$

and Fr_X^d and Fr^d *induce a* Λ-endomorphism on the cohomology groups of $\mathscr{G}$ with compact support:

$$H_c(\mathrm{Fr}_X^d, \mathrm{Fr}_{\mathscr{G}}^d) = (H_c^\nu(\mathrm{Fr}_X, \mathrm{Fr}_{\mathscr{G}}))^d : H_c(X, \mathscr{G}) \to H_c^\nu(X, \mathscr{G}).$$

4.2 Proposition. *Under the stead hypotheses,*

$$\sum_{a \in \mathrm{Fix}(\mathrm{Fr}_X^d)} \mathrm{Tr}_\Lambda(\mathrm{Fr}_{\mathscr{G},a}^d) = \sum_\nu (-1)^\nu \,\mathrm{Tr}((H_c^\nu(\mathrm{Fr}_X, \mathrm{Fr}_{\mathscr{G}}))^d).$$

Remark. $\mathrm{Fix}(\mathrm{Fr}_X^d)$ is canonically isomorphic to the set of geometric points of X_0 over κ with coordinates in $\kappa_d \subset k$, the field extension of degree d over κ in k.

Remark. It is not necessary to assume that $\mathscr{G}$ is torsion-free (and thus has free stalks)! For without changing the formula, we can replace $\mathscr{G}_0$ by $\mathscr{G}_0$ modulo l-torsion (I.12.13).

Let $\mathrm{Ab}(X_0)$ be the set of closed points of the scheme X_0. We also consider the set of geometric points of X_0 resp. of X with coordinates in k,

$$X_0(k) = \mathrm{Hom}_{\mathrm{Spec}(\kappa)}(\mathrm{Spec}(k), X_0),$$
$$X(k) = \mathrm{Hom}_{\mathrm{Spec}(k)}(\mathrm{Spec}(k), X).$$

These two sets of points are canonically isomorphic,

$$X_0(k) = X(k).$$

For a geometric point $a \in X_0(k) = X(k)$, the stalks of $\mathscr{G}_0$ and $\mathscr{G}$ at a are canonically isomorphic,

$$\mathscr{G}_{0a} = \mathscr{G}_a.$$

For every geometric point $a \in X_0(k)$ there is an underlying closed point $|a|$ in X_0. The mapping

$$X(k) = X_0(k) \to \mathrm{Ab}(X_0)$$
$$a \mapsto |a|$$

is surjective. When two geometric points a, a' belong to the same closed point $\alpha = |a| = |a'|$, the fields $\kappa(a)$, $\kappa(a') \subset k$ associated with a and a' are the same,

$$\kappa(a) = \kappa(a') = \kappa(\alpha),$$
$$d(a) = d(|a|) = [\kappa(\alpha) : \kappa].$$

Over every point $\alpha \in \mathrm{Ab}(X_0)$ there lie exactly $d(\alpha)$ geometric points. Let G_α be the Galois group of k over $\kappa(\alpha)$. Then G_α contains the Frobenius automorphism

$$f_\alpha = f_a \colon k \to k, \qquad \alpha = |a|,$$
$$x \mapsto x^{\#(\kappa(\alpha))}.$$

The stalk $\mathcal{G}_{0a} = \mathcal{G}_a$ is a $G_{|a|}$-module. The G_α-modules $\mathcal{G}_a$ for all geometric points a over α are isomorphic. If $f_{a,\mathcal{G}_a} \colon \mathcal{G}_a \to \mathcal{G}_a$ is the endomorphism induced by f_a on $\mathcal{G}_a$, then it follows that

$$\det(1 - t f_{a,\mathcal{G}_a}^{-1})$$

depends only on the underlying point $\alpha = |a|$. For each α in $\mathrm{Ab}(X_0)$ we choose a geometric point a over α and write

$$f_{\alpha,\mathcal{G}_\alpha} = f_{a,\mathcal{G}_a}.$$

We have:

$a \in X(k)$ is a fixed point of Fr_X^d precisely when $d(a) | d$, i.e. when $\kappa(a)$ is contained in the field κ_d of degree d over κ. We have the formula

$$f_{a,\mathcal{G}_a}^{-1} = \mathrm{Fr}_{\mathcal{G},a}^{d(a)}.$$

4.3 Definition. The L-series for an l-adic constructible sheaf $\mathcal{G}_0$ on an algebraic variety X_0 over κ is

$$Z_{\mathcal{G}_0}(t) = \prod_{\alpha \in \mathrm{Ab}(X_0)} \frac{1}{\det(1 - f_{\alpha,\mathcal{G}_\alpha}^{-1} t^{d(\alpha)})}.$$

We want to determine the logarithmic derivative of $Z_{\mathcal{G}_0}$. For an endomorphism ϕ of a finite Λ-module, the logarithmic derivative of $\det(1 - t\phi)^{-1}$ is

$$\sum_{\nu=1}^{\infty} \mathrm{Tr}_\Lambda(\phi^\nu)\, t^{-1}.$$

Hence we get

$$\begin{aligned}
\frac{Z'_{\mathcal{G}_0}}{Z_{\mathcal{G}_0}}(t) &= \sum_{\alpha \in \mathrm{Ab}(X_0)} \sum_{\nu=1}^{\infty} d(a)\, \mathrm{Tr}_\Lambda(F_{\alpha,\mathcal{G}_\alpha}^{-\nu})\, t^{\nu d(\alpha) - 1} \\
&= \sum_{a \in X(k)} \sum_{\nu=1}^{\infty} \mathrm{Tr}_\Lambda(f_{a,\mathcal{G}_a}^{-\nu})\, t^{\nu d(a) - 1} \\
&= \sum_{a \in X(k)} \sum_{\nu=1}^{\infty} \mathrm{Tr}_\Lambda(\mathrm{Fr}_{\mathcal{G},a}^{\nu d(a)})\, t^{\nu d(a) - 1} \\
&= \sum_{n=1}^{\infty} \sum_{\substack{a \in X(k) \\ d(a) | n}} \mathrm{Tr}_\Lambda(\mathrm{Fr}_{\mathcal{G},a}^{n})\, t^{n-1} \\
&= \sum_{n=1}^{\infty} \sum_{a \in \mathrm{Fix}(\mathrm{Fr}_X^n)} \mathrm{Tr}_\Lambda(\mathrm{Fr}_{\mathcal{G},a}^{n})\, t^{n-1}.
\end{aligned}$$

It follows immediately then from Proposition 4.2 that $Z'_{\mathscr{G}_0}/Z_{\mathscr{G}_0}$ is the logarithmic derivative of

$$\prod_{\nu} \det(1 - H_c^{\nu}(\mathrm{Fr}_X, \mathrm{Fr}_{\mathscr{G}}))^{(-1)^{\nu-1}}.$$

4.4 Theorem. *Let $\mathscr{G}_0$ be a constructible l-adic sheaf on a finitely generated scheme X_0 over the finite field κ. Let $\mathscr{G}$ be the inverse image of $\mathscr{G}_0$ on $X = X_0 \times_{\mathrm{Spec}(\kappa)} \mathrm{Spec}(k)$. Then*

$$Z_{\mathscr{G}_0}(t) = \prod_{\nu} \det(1 - H_c^{\nu}(\mathrm{Fr}_X, \mathrm{Fr}_{\mathscr{G}}))^{(-1)^{\nu-1}},$$

$$(Z'_{\mathscr{G}_0}/Z_{\mathscr{G}_0})(t) = \sum_{n=1}^{\infty} \sum_{a \in \mathrm{Fix}(\mathrm{Fr}_X^n)} \mathrm{Tr}_{\Lambda}(\mathrm{Fr}_{G,a}^n)\, t^{n-1}.$$

In particular, $Z_{\mathscr{G}_0}$ is a rational function!

4.5 Corollary. *Let $f_0 \colon X_0 \to Y_0$ be a morphism of finitely generated schemes over κ, and let $\mathscr{G}_0$ be a constructible l-adic sheaf on X_0. Then*

$$Z_{\mathscr{G}_0}(t) = \prod_{\nu} Z_{R^{\nu} f_{0!}(\mathscr{G}_0)}(t)^{(-1)^{\nu+1}}.$$

4.6 Corollary. *Let $Z_{X_0}(t)$ be the L-series for the constant sheaf Λ on X_0 and $N_n(X_0)$ the number of geometric points of X_0 with coordinates in the field κ_n with $[\kappa_n : \kappa] = n$. Then*

$$Z'_{X_0}(t)/Z_{X_0}(t) = \sum_{n=1}^{\infty} N_n\, t^{n-1}.$$

Chapter III. The Monodromy Theory of Lefschetz Pencils

In this chapter, we shall present the essential results of SGA 7 II [69], as far as they are necessary for the proof of the Weil conjecture.

§1. Lefschetz Pencils

In what follows we shall prove the existence of Lefschetz pencils. We shall show: For every smooth projective variety X over an infinite field k, there is a birationally equivalent variety $\tilde{X}$ and a fibering $\tilde{X} \to \mathbb{P}^1$ over the projective line that is singular at only finitely many points. The singularities are of particularly simple nature (ordinary double points). The fibers are hyperplane sections of X for an appropriate projective embedding. We first describe the singularities that occur.

Let V be a finite dimensional vector space over a field k. A quadratic form Q on V is a mapping

$$Q\colon\ V \to k$$

which after choice of a basis $e_0, \ldots, e_n$ can be written in the form

$$Q(x) = \sum_{0 \leqq i,j \leqq n} a_{ij}\, x_i\, x_j; \qquad x = x_0\, e_0 + \ldots + x_n\, e_n$$

with coefficients $a_{ij} \in k$. Only when the characteristic of k is different from 2 can one get the matrix of coefficients to be symmetric. Associated to Q is the symmetric bilinear form

$$B(x, y) = Q(x+y) - Q(x) - Q(y).$$

We have

$$B(x, x) = 2\,Q(x).$$

As usual, we call B *nondegenerate* if for every $x \in V$, $x \neq 0$, there is some $y \in V$ with $B(x, y) \neq 0$. The set X of zeros of Q in the projective space $\mathbb{P}(V) \cong \mathbb{P}_k^n$ is the *quadric* belonging to Q. One can show:

1.1 Lemma. *Let Q be a quadratic form on the finite dimensional vector space V. The following two assertions are equivalent:*

a) *The associated bilinear form is nondegenerate.*

b) *The associated quadric X is smooth, and if the characteristic is 2, the dimension of V is even.*

1.2 Definition. The quadratic form Q is called nonsingular, when the associated quadric is smooth.

1.3 Definition. Let X be an algebraic scheme over the field k, and let a be a closed point of X with local ring

$$A = \mathcal{O}_{X,a}.$$

Case 1. Let k be algebraically closed. Then a is called an ordinary double point when the completion $\hat{A}$ of A has the following form:

$$\hat{A} \cong k[[X_1, \ldots, X_n]]/(f)$$

where

$f \in \mathfrak{m}^2$, $\mathfrak{m}$ = maximal ideal of $k[[X]]$,
$f = Q + R$, $R \in \mathfrak{m}^3$,
Q is a nonsingular quadratic form.

Case 2. For a general k with algebraic closure $\bar{k}$, the point a is called an ordinary double point when all the points of $X \otimes_k \bar{k}$ lying over a are ordinary double points.

The condition given in Case 1 of this definition can also be formulated without using the completion.

1.3$_1$ *Variant of 1.3. Case* 1. Let the ring A be represented as a factor ring of a locally finitely generated regular k-algebra R with maximal ideal $\mathfrak{m}(R)$:

$$A \cong R/\mathfrak{a}.$$

We assume that this representation is a minimal one (such representations always exist):

$$\mathfrak{a} \subset \mathfrak{m}(R)^2.$$

Let $x_1, \ldots, x_n$ be a regular system of parameters for $\mathfrak{m}(R)$. Then a is an ordinary double point if and only if $\mathfrak{a} = (f)$ is a principal ideal and there is a nonsingular quadratic form Q on k^n with

$$f \equiv Q(x_1, \ldots, x_n) \mod \mathfrak{m}^3.$$

1.3$_2$ *Remark.* Ordinary double points are isolated singularities.

Now we consider a smooth projective algebraic variety over a field k and an embedding

$$j\colon X \hookrightarrow \mathbb{P}^r_k$$

into r-dimensional projective space. We denote by $\check{\mathbb{P}}^r$ the projective space dual to $\mathbb{P}^r$. The geometric points of $\check{\mathbb{P}}^r$ are the hyperplanes of $\mathbb{P}^r$.

1.4 Definition. The embedding

$$j\colon X \hookrightarrow \mathbb{P}^r$$

is called a *Lefschetz embedding* if there is a closed subset $A \subset \check{\mathbb{P}}^r$ of codimension $\geqq 2$ such that the following conditions are satisfied by every hyperplace H with coordinates in the algebraic closure $\bar{k}$ of k that does not lie in A:

a) H does not contain any connected components of X.

b) The (scheme-theoretic) intersection $H \cap X$ contains at most one singular point, and when such a point exists it is an ordinary double point.

For the construction of Lefschetz embeddings, one needs the *Segre embedding* of degree d ($\geqq 1$),

$$\mathbb{P}^r \to \mathbb{P}^N, \qquad N = \binom{d+r}{r} - 1,$$

defined by the monomials of degree d:

$$(x_0, \ldots, x_r) \mapsto (\ldots, x_0^{d_0} \ldots x_r^{d_r}, \ldots), \sum_{j=0}^{n} d_j = d.$$

Existence of Lefschetz Embeddings (SGA 7 II, exp. XVII)

1.5 Proposition. *Let X be an irreducible smooth projective variety, $\dim X \geqq 1$, over an infinite base field k. Let $j\colon X \hookrightarrow \mathbb{P}^r$ be a projective embedding. Denote by*

$$j(d)\colon X \hookrightarrow \mathbb{P}^N$$

the composition of j with the Segre embedding $\mathbb{P}^r \to \mathbb{P}^N$ of degree d. Then:

a) *In the case of characteristic 0, $j = j(1)$ is a Lefschetz embedding.*

b) *$j(d)$ is always a Lefschetz embedding for $d \geqq 2$.*

As we are concerned only with the existence of some Lefschetz embedding, we shall only prove that $j(d)$ for $d \geqq 3$ is a Lefschetz embedding.

The proof comes from the following Lemmas 1.6–1.9. We may assume that k is algebraically closed.

1.6 Lemma. *The set*

$$Y = Y(j) = \{(x, H) \in X \times \check{\mathbb{P}}^r,\ H \text{ touches } X \text{ at } x\}$$

is an irreducible smooth closed subvariety of dimension $r-1$ of $X \times \check{\mathbb{P}}^r$ (resp. $= \emptyset$, when $X = \mathbb{P}^r$). More precisely:

Let $\mathcal{J}$ be the sheaf of ideals in $\mathcal{O}_{\mathbb{P}^r}$ associated with (X, j). Then Y is isomorphic to the projective bundle space of the conormal bundle $(\mathcal{J}/\mathcal{J}^2)|X$.

The proof is elementary algebraic geometry.

Remark. A hyperplane H touches X at x if and only if it contains the projective tangent space, i.e., when either it contains the manifold X or the scheme-theoretic intersection $X \cap H$ is singular.

1.7 Lemma. *With notation as in* 1.6, *the set*

$$\tilde{Y} = \tilde{Y}(j) = \{(x, H) \in Y(j),\ x \text{ not an ordinary double point of } X \cap H\}$$

is Zariski-closed in Y.

We forgo the elementary proof.

We now consider (for an algebraically closed ground field) the Segre embedding

$$j(d)\colon\ X \to \mathbb{P}^N, \qquad N = \binom{r+d}{d} - 1.$$

There is a one-to-one correspondence between the *hyperplanes* in $\mathbb{P}^N$ and the *hypersurfaces of degree d* in $\mathbb{P}^r$: If h is a hyperplane in $\mathbb{P}^N$ and $\check{h}$ the corresponding hypersurface of degree d in $\mathbb{P}^r$, then

$$j^{-1}(\check{h}) = j(d)^{-1}(h).$$

1.8 Lemma. *Assume* $d \geqq 2$. *For every closed point* $a \in X$, *there is a hypersurface F of degree d which touches X at a and is such that a is an ordinary double point in* $X \cap F$. *In particular,*

$$\dim \tilde{Y}(j(d)) \leqq N - 2 \quad (= \dim Y(j(d)) - 1).$$

Proof. We can choose a homogeneous coordinate system $y_0, \ldots, y_r$ in such a way that $(1, 0, \ldots, 0)$ are the homogeneous coordinates of $a \in X$ and the functions $(y_1/y_0), \ldots, (y_n/y_0)$ $(n = \dim X)$ form a regular system of parameters for X at a. There exists a nonsingular quadratic form Q in the variables $(y_1/y_0), \ldots, (y_n/y_0)$. We can take as our hypersurface the set of zeros of the homogeneous polynomial $y_0^{d-2} Q(y_1, \ldots, y_n)$. □

1.9 Lemma (Notations as in 1.5). *The set*

$$Z(d) = \{(x, y, H) \in X \times X \times \check{\mathbb{P}}^N;\ x \neq y,\ H \text{ touches } X \text{ at } x \text{ and } y\}$$

is Zariski closed in $(X \times X\text{-Diagonal}) \times \check{\mathbb{P}}^N$, *and*

$$\dim Z(d) \leqq N - 2 \quad \textit{for } d \geqq 3.$$

Proof. The Zariski closure follows from Lemma 1.6. For the bound on the dimension, it suffices to show that every fiber of the projection mapping

$$Z(d) \to X \times X\text{-Diagonal}$$

has dimension at most $N - 2 - 2 \cdot \dim X$.

The fiber of this projection over a pair of points (a, b) is the collection of all hypersurfaces of degree d that touch the projective tangent space of X at a and of X at b. The required bound then follows from the

Assertion. Suppose we have two linear subspaces V, $W \subset \mathbb{P}^r$ and points $a \in V$, $b \in W$, $a \neq b$. We consider the vector space $\bar{U}$ of all homogeneous equations f

of degree d (including $f=0$). This space has dimension $N+1$. The subspace of all equations f whose sets of zeros $F_f \subset \mathbb{P}^r$ touch the space V at a and the space W at b is a linear subspace U of $\bar{U}$, and

$$\operatorname{codim} U \geqq \dim V + \dim W + 2.$$

Proof. Let $a_0 = a$, $a_1, \ldots, a_{\dim V}$ be independent points in V, and $b_0 = b$, $b_1, \ldots, b_{\dim W}$ independent points in W. The condition on F_f is

a) $a_0, b_0 \in F_f$.

b) $a_v \in T_a(F_f)$ (tangent space) for $v \geqq 1$, $b_v \in T_b(F_f)$ for $v \geqq 1$.

These conditions (there are $\dim V + \dim W + 2$ of them) are linear conditions. We must show that they are linearly independent. Obviously it suffices to handle the case where $V = W = \mathbb{P}^r$. Then they say that

$$a, b \in F_f, \quad F_f \text{ is singular at } a \text{ and } b,$$

$$T_a(F_f) = \mathbb{P}^r.$$

After suitable change of coordinates, we can assume that

$$a = (1, 0, \ldots, 0), \quad b = (0, 1, 0, \ldots, 0)$$

(homogeneous coordinates),

$$F = \sum a_\nu X^\nu; \quad X^\nu = X_0^{\nu_0} \ldots X_r^{\nu_r}, \quad \sum \nu_j = d.$$

Here is what the conditions now say:

$$a, b \in F_f \text{ means } a_{(d,0,\ldots,0)} = a_{(0,d,0,\ldots,0)} = 0.$$

F_f singular at a, b means

$$(\partial f/\partial X_j)(1, 0, \ldots, 0) = (\partial f/\partial X_j)(0, 1, 0, \ldots, 0) = 0, \quad 0 \leqq j \leqq r.$$

The coefficients of $X_0^{d-1} X_j$ and $X_1^{d-1} X_j$ must therefore vanish. The vector space of equations f that satisfy these conditions has codimension $2r+2$ (recall that $d \geqq 3$). Thus the $2r+2$ linear conditions are linearly independent. □

With this, Lemmas 1.6–1.9 are proved, and hence also Theorem 1.6 under the stricter hypothesis $d \geqq 3$: for we just consider the union A of the Zariski closures of the projections of $\tilde{Y}(j(d))$ and $Z(d)$ in $\check{\mathbb{P}}^N$. Lemmas 1.8 and 1.9 show that $\dim A \leqq N-2$. By the definition of $\tilde{Y}(j(d))$ and $Z(d)$, every hyperplane not contained in A meets X in just one ordinary double point.

Lefschetz Pencils

We start with a Lefschetz embedding

$$j\colon X \hookrightarrow \mathbb{P}^r$$

of an irreducible smooth projective variety X of positive dimension over an infinite field k. We consider lines (one-dimensional linear subspaces) $\check{D}$ in $\check{\mathbb{P}}^r$. These lines correspond one-to-one to linear subspaces $D \subset \mathbb{P}^r$ of codimension 2. (One can get D as the intersection of two hyperplanes in $\check{D}$.) We now choose

a line $\check{D}$ for which the following two conditions are satisfied:

a) $\check{D} \cap A = \emptyset$ (notation as in 1.4)
b) D intersects X transversally.

Both conditions are generically satisfied. The *"general"* line has empty intersection with A, since the codimension of A is greater than 1. Condition b) is generically satisfied because of the classical theorem of algebraic geometry that a general hyperplane intersects X transversally (and consequently the intersection is again smooth). Thus conditions a) and b) are satisfied by "almost all" lines.

We now consider the following (reduced) projective subvariety of $X \times \check{D}$:

$$\tilde{X} = \{(x, H), x \in H\}.$$

We denote the two projections by

$$f\colon \tilde{X} \to \check{D},$$
$$\pi\colon \tilde{X} \to X.$$

The fibers of f are precisely the intersections of X with the hyperplanes in $\check{D}$. One can show that $\tilde{X}$ arises from X by blowing up the subvariety $D \cap X$ (which by assumption is smooth). In particular, $\tilde{X}$ is nonsingular.

1.10 Proposition. *Let X be a smooth projective algebraic variety of positive dimension over a field k. If k is infinite, there is an (everywhere defined) birational morphism*

$$\pi\colon \tilde{X} \to X$$

from a smooth projective variety $\tilde{X}$ and a mapping

$$f\colon \tilde{X} \to \mathbb{P}^1_k$$

that is singular at no more than finitely many points. Each fiber $f^{-1}(x)$ contains at most one singular point, and any such point occurring is an ordinary double point.

Supplement to 1.9. *Let the base field k be finite. The statement of Proposition* 1.9 *remains true after a finite extension of the base field.*

Remark. Let

$$\pi\colon \tilde{X} \to X$$

be a birational morphism of irreducible smooth projective varieties over an algebraically closed field k. The canonical mapping

$$H^p(X, \mathbb{Z}/l\mathbb{Z}) \to H^p(\tilde{X}, \mathbb{Z}/l\mathbb{Z}) \quad (l, \operatorname{char} k) = 1$$

is injective.

Proof of 1.10. 1) $p = 2 \dim X$. The assertion follows from the existence of the trace mapping.

2) $p<2\dim X$. This case is reduced to the first case by using the cup-product pairing

$$H^p(X, \mathbb{Z}/l\mathbb{Z}) \times H_c^q(X, \mathbb{Z}/l\mathbb{Z}) \to H_c^{2\dim X}(X, \mathbb{Z}/l\mathbb{Z}).$$

We need the fact that the cup product is nondegenerate and compatible with inverse images. □

For the proof of the Weil conjecture, we can by 1.10 replace the variety X by the fibered variety $\tilde{X}$,

$$f\colon \tilde{X} \to \mathbb{P}_k^1.$$

The computation of the cohomology of $\tilde{X} \otimes_k \bar{k}$ can be reduced (by the Leray spectral sequence) to the computation of the cohomology of certain sheaves on $\mathbb{P}_k^1$ (namely, the higher direct images of $(\mathbb{Z}/l\mathbb{Z})_{\tilde{X}}$). The goal of our monodromy investigations is to describe these sheaves.

§ 2. Classification of Nondegenerate Double Points

In this section we want to classify the singularities of a Lefschetz pencil, i.e. the ordinary double points. For simplicity we restrict ourselves to the case of "nondegenerate" double points. This is sufficient for our application to Lefschetz pencils with odd fibre dimension in § 4.

2.1 Definition. Let A be a local ring, $V = A e_1 \oplus \ldots \oplus A e_n$ a finitely generated free A-module. A mapping $Q\colon V \to A$ is called a quadratic form if it can be expressed in coordinates as

$$Q(x_1 e_1 + \ldots + x_n e_n) = \sum a_{ij} x_i x_j.$$

Q is called *nondegenerate* if the associated bilinear form

$$B(v, w) = Q(v+w) - Q(v) - Q(w)$$

is nondegenerate, that is, when B induces an isomorphism

$$V \xrightarrow{\cong} \mathrm{Hom}_A(V, A)$$

$$v \mapsto e_v, \qquad e_v(w) = B(v, w).$$

A homogeneous polynomial $P(x) = P(x_1, \ldots, x_n) \in A[x_1, \ldots, x_n]$ of degree 2 defines a quadratic form on A^n in a natural way. We identify P with this quadratic form.

We then interpret the associated bilinear form B as a polynomial

$$B(x, y) = B(x_1, \ldots, x_n; y_1, \ldots, y_n) \in A[x_1, \ldots, x_n, y_1, \ldots, y_n].$$

Such a nondegenerate quadratic form $P(x_1, \ldots, x_n)$ defines a *smooth hypersurface* F of dimension $n-2$ in the projective space of dimension $n-1$ over A. If A is strictly Henselian, then for $n \geqq 2$ this hypersurface always has a rational point

with coordinates in A. Thus there are elements $a_1, \ldots, a_n \in A$ that generate the unit ideal in A and satisfy

$$P(a_1, \ldots, a_n) = 0.$$

As in the case of a base field, one can deduce the classification theorem from this:

2.2 Proposition. *Let $Q: V \to A$ be a nondegenerate quadratic form over a strictly Henselian local ring A. Let $n = \dim V$. Then*

Case 1). If $n = 2m$, V has a basis $e_1, \ldots, e_m, f_1, \ldots, f_m$ with

$$Q(x_1 e_1 + \ldots + x_m e_m + y_1 f_1 + \ldots + y_m f_m) = \sum_{i=1}^{m} x_i y_i .$$

Case 2). If $n = 2m+1$, V has a basis $e_1, \ldots, e_m, f_1, \ldots, f_m, g$ with

$$Q(x_1 e_1 + \ldots + x_m e_m + y_1 f_1 + \ldots + y_m f_m + z g) = \sum_{i=1}^{m} x_i y_i + z^2.$$

The second case can only occur when the residue field characteristic of A is different from 2.

The isomorphism class of the "quadratic module" (V, Q) thus depends only on the dimension of V.

We first want to classify the completions of the local rings at (nondegenerate) ordinary double points.

2.3 Proposition. *Let A be a local ring, $F(x) = F(x_1, \ldots, x_n)$ a power series in the power series ring $A[[x_1, \ldots, x_n]]$, and $\mathfrak{a}$ the ideal generated by the variables $x_1, \ldots, x_n$ in $A[[x_1, \ldots, x_n]]$. Assume that there is a nondegenerate quadratic form $Q(x)$ over A and an element a in A with*

$$F(x) \equiv Q(x) + a \quad \text{modulo } \mathfrak{a}^{m+1}$$

for some integer $m \geqq 2$.

Then there are elements $h_1, \ldots, h_n$ in $\mathfrak{a}^m$ with

$$F(x_1 + h_1, \ldots, x_n + h_n) = Q(x) + a.$$

In other words, the ring $A[[x]]/(F(x))$ is A-isomorphic to $A[[x]]/(Q(x) + a)$.

Proof. Since $A[[x]]$ is complete in the $\mathfrak{a}$-adic topology, it is enough to show that for every m there are elements $h_1, \ldots, h_n$ in $\mathfrak{a}^m$ with

$$F(x+h) = F(x_1 + h_1, \ldots, x_n + h_n) \equiv Q(x) + a \quad \text{modulo } \mathfrak{a}^{m+2}.$$

Let $B(x, y) = \sum_{i=1}^{n} b_i(x)\, y_i$ be the bilinear form associated with $Q(x)$. Since it is nondegenerate, the linear polynomials $b_\nu(x)$ generate the ideal $\mathfrak{a}$. By hypothesis

we have

$$F(x)=Q(x)+a-q(x), \qquad q(x)\in\mathfrak{a}^{m+1}.$$

For an arbitrary n-tuple $h=(h_1, \ldots, h_n)$ in $\mathfrak{a}^m$,

$$Q(h)\in\mathfrak{a}^{m+2}$$

$$q(x+h)\equiv q(x) \quad \text{modulo } \mathfrak{a}^{m+2}$$

$$\begin{aligned} F(x+h)&=Q(x+h)+a-q(x+h)\\ &=Q(x)+a+B(x,h)-q(x+h)+Q(h)\\ &\equiv Q(x)+a+\sum_{i=1}^{n} b_i(x)\,h_i-q(x) \quad \text{modulo } \mathfrak{a}^{m+2}. \end{aligned}$$

Since $\mathfrak{a}^{m+1}=\mathfrak{a}^m\, b_1(x)+\ldots+\mathfrak{a}^m\, b_n(x)$, there are elements $h_1, \ldots, h_n$ in $\mathfrak{a}^m$ with

$$b_1(x)\,h_1+\ldots+b_n(x)\,h_n=q(x). \quad \square$$

2.4 Proposition. *Let A be a complete local ring, $\mathfrak{m}$ its maximal ideal, $Q(x_1, \ldots, x_n)$ a nondegenerate quadratic form, a, $a_1, \ldots, a_n$ elements in $\mathfrak{m}$, and $F(x_1, \ldots, x_n)=F(x)$ a power series of the form*

$$F(x)\equiv a+\sum_{\nu=1}^{n} a_\nu x_\nu+Q(x) \quad \text{modulo } (x_1, \ldots, x_n)^3.$$

Then there is an element $a'\in\mathfrak{m}$ and a nondegenerate quadratic form Q' with

$$A[[x_1, \ldots, x_n]]/(F(x))\cong_A A[[x_1, \ldots, x_n]]/(Q'(x)+a').$$

Proof. By Proposition 2.3, it is enough to show that there is an n-tuple $\alpha=(\alpha_1, \ldots, \alpha_n)$ of elements in $\mathfrak{m}$ with

$$F(x+\alpha)\equiv a'+Q'(x) \quad \text{modulo } (x_1, \ldots, x_n)^3.$$

This congruence is equivalent to the system of equations

$$(\partial F/\partial x_1)(\alpha)=0, \ldots, (\partial F/\partial x_n)(\alpha)=0.$$

The existence of a solution α follows from the implicit function theorem, since the matrix $(\partial^2 F/\partial x_i\,\partial x_j)(0)$ is nondegenerate (Q is nondegenerate!). $\square$

We come now to the geometric case.

We introduced the concept of an ordinary double point in Definition 1.3. We resume from that point.

2.5 Definition. Let

$$\phi\colon A\to B$$

be a locally finitely generated local homomorphism.

Case 1. Let $A=k$ be algebraically closed. Then ϕ resp. B/k is called nondegenerately quadratically singular when the completion $\hat{B}$ of B over k is isomorphic

to

$$k[[x_1, \ldots, x_n]]/(f(x_1, \ldots, x_n)),$$

where f is a power series that is congruent modulo $(x_1, \ldots, x_n)^3$ to a nondegenerate quadratic form Q.

Case 2. Let k be the algebraic closure of the residue field of A. Then B over A resp. ϕ is called nondegenerately quadratically singular if ϕ is flat and there is a localization $(B \otimes_A k)_\mathfrak{p}$ for which $B \to (B \otimes_A k)_\mathfrak{p}$ is local and $k \to (B \otimes_A k)_\mathfrak{p}$ is nondegenerately quadratically singular.

2.6 Definition. Let $f: X \to S$ be a finitely generated morphism of schemes. A point a of X is called a *nondegenerate double point* of f if

$$\mathcal{O}_{S, f(a)} \to \mathcal{O}_{X, a}$$

is nondegenerately quadratically singular, (2.5).

Here is our main result.

2.7 Proposition. *Let $A \to B$ be a nondegenerately quadratically singular homomorphism* (2.5). *Assume that A is strictly Henselian with maximal ideal $\mathfrak{m} = \mathfrak{m}(A)$. Let $Q(x_0, \ldots, x_n)$, $n = \dim B - \dim A$, be a nondegenerate quadratic form over A.*

Assertion. There is an element $a \in \mathfrak{m}$ with the following property: Let $\mathfrak{M}$ be the maximal ideal generated by $x_0, \ldots, x_n$ and $\mathfrak{m}$, and let

$$R = (A[x_0, \ldots, x_n]/(Q(x) + a))_\mathfrak{M}.$$

The Henselizations of B and R are isomorphic over A,

$$\tilde{B} \cong_A \tilde{R}.$$

Remark. By 2.2, it is enough to prove the isomorphism for *one* quadratic form.

The proof also works in the case of a (not strictly) Henselian ring A, provided that the residue fields of A and B are the same.

Here is the geometric form of this theorem:

2.8 Proposition. *Let $S = \operatorname{Spec}(A)$ be the spectrum of a strictly Henselian ring with maximal ideal $\mathfrak{m}$, and*

$$f: X \to S$$

a finitely generated morphism. Let the point a in X be a nondegenerate double point over $\mathfrak{m}$ of f. Let $n = \dim_a f^{-1}(f(a)) + 1$. Let Ω be a separably closed extension field of the residue field of A, and let

$$\alpha: \operatorname{Spec}(\Omega) \to X \qquad (\textit{over } \operatorname{Spec} A)$$

be a geometric point of X centered at a. Consider a nondegenerate quadratic form $Q(X_1, \ldots, x_n)$, an element u in $\mathfrak{m}$, and the S-scheme

$$Y = \operatorname{Spec}(A[x_1, \ldots, x_n]/(Q + u)).$$

Let

$$\beta\colon \operatorname{Spec}(\Omega)\to Y \quad (\textit{over } \operatorname{Spec} A)$$

be the geometric point of Y under which the elements $x_1, \ldots, x_n$ are sent to zero.

For a suitable choice of the element u in $\mathfrak{m}$, there are etale neighborhoods U of α and V of β that are isomorphic over S!

For the proof of Proposition 2.7, it is sufficient to assume that A is the strict Henselization of a locally finitely generated ring over $\mathbb{Z}$. The approximation theorem of M. Artin [10] holds for such a ring. To prove 2.8, we need the concept of the Jacobi ideal of a local homomorphism $\phi\colon A\to B$:

Let $A\to B$ be locally finitely generated and r the minimal number of generators for the universal module of differentials $M(B/A)$ of B over A. The Jacobi ideal $j(B/A)\subset B$ is the annihilator of the r-th exterior power of $M(B/A)$. More generally, one can define the Jacobi ideal in a similar way when $A\to B$ is not locally finitely generated, so long as there is a finitely generated universal module of differentials. There is one, for instance, when B is complete and the residue field extension is finitely generated, for then it agrees with the "continuous" differentials. The properties of the module of differentials imply the following properties of the Jacobi ideal: Let $A\to B$ be locally finitely generated, $A\to C$ a local homomorphism, and D a localization of $B\otimes_A C$ lies over C. Then

$$j(B/A)\cdot D = j(D/C).$$

The completions $\hat{A}$ and $\hat{B}$ of A and B satisfy

$$j(\hat{B}/\hat{A}) = j(B/A)\cdot \hat{B}.$$

If $B=A[[x_1, \ldots, x_n]]/(F(x))$, then the Jacobi ideal $j(B/A)$ is generated by the images in B of the partial derivatives $\partial F/\partial x_i$.

If $Q(x)$ is a nondegenerate quadratic form with associated bilinear form $\sum_{i=1}^{n} b_i(x)\, y_i$, and $F=Q+a$ for an element a in A, then

$$\partial F/\partial x_i = b_i(x).$$

If F is merely congruent modulo $(x_1, \ldots, x_n)^3$ to $Q+a$, then at least

$$\partial F/\partial x_i \equiv b_i(x) \quad \text{modulo } (x_1, \ldots, x_n)^2.$$

The elements b_i generate the ideal $(x_1, \ldots, x_n)$, and therefore the partial derivatives $\partial F/\partial x_i$ also generate the ideal $(x_1, \ldots, x_n)$. Hence:

2.9. $\boxed{B/j(B/A) = A/aA.}$

We come now to the proof of Proposition 2.7.

First, let $A=k$ be a separably closed field, L the algebraic closure of k. Since L/k is purely inseparable, $B\otimes_k L$ is already local. Since B/k is nondegenerately singular, the completion $\widehat{B\otimes_k L}$ of $B\otimes_k L$ satisfies

$$\widehat{B\otimes_k L} \cong_L L[[x_0, \ldots, x_n]]/(F(x)), \qquad n=\dim B.$$

Here $F(x)$ is congruent modulo $(x_0, \ldots, x_n)^3$ to a nondegenerate quadratic form. By 2.9, we have

$$L = \widehat{B \otimes L}/j(\widehat{B \otimes L}/L) = B/j(B/k) \otimes_k L.$$

Hence we have

$$B/j(B/k) = k.$$

Thus the residue field of B is k, and $j(B/k)$ coincides with the maximal ideal of B. Moreover B is a complete intersection and the maximal ideal is generated precisely by $n+1 = \dim B + 1$ elements. Therefore,

$$\hat{B} \cong_k k[[x_0, \ldots, x_n]]/(G(x)).$$

Comparing with $\widehat{B \otimes L}$, we see that $G(x)$ is also congruent modulo $(x_0, \ldots, x_n)^3$ to a nondegenerate quadratic form.

Now we consider the general case: let $\phi: A \to B$ be nondegenerately quadratically singular, A strictly Henselian with maximal ideal $\mathfrak{m}$. Let $\alpha_0, \ldots, \alpha_n$ be elements of B that (modulo $\mathfrak{m}B$) form a minimal generating system for the maximal ideal of $B/\mathfrak{m}B$. We consider the $\hat{A}$-homomorphism

$$\hat{A}[[x_0, \ldots, x_n]] \to \hat{B}$$
$$x_i \mapsto \alpha_i .$$

Since B is flat over A, it follows from the corresponding property of $B/\mathfrak{m}B$ over $A/\mathfrak{m}$ that this homomorphism is surjective, that the kernel is generated by one element $F(x)$, and that F satisfies

$$F(x) \equiv Q(x) + a + \sum_{i=0}^{n} a_i x_i \quad \text{modulo } (x_0, \ldots, x_n)^3,$$

$$a, a_i \in \mathfrak{m},$$

$Q(x)$ a nondegenerate quadratic form.

From Proposition 2.4, then,

$$\hat{B} \cong_{\hat{A}} \hat{A}[[x_0, \ldots, x_n]]/(Q'(x) + a'),$$

$$a' \in \mathfrak{m},$$

Q' a nondegenerate quadratic form.

By 2.9,

$$\hat{A}/\hat{A}\, a' = \hat{B}/j(\hat{B}/\hat{A}) = \widehat{B/j(B/A)}.$$

Since A is Henselian, then, we have

$$B/j(B/A) = A/A\, a''.$$

The elements a' and a'' differ by a unit ε in $\hat{A}$: $a'' = \varepsilon a'$. Multiplying $Q'(x) + a'$ by ε, we can assume that $a' \in A$. By Proposition 2.2, we can assume that $Q'(x)$ is already contained in $A[x_0, \ldots, x_n]$. Thus $Q'(x) + a' \in A[x_0, \ldots, x_n]$. We then get Proposition 2.7 from Artin's approximation theorem. □

Note. In analogy to Proposition 2.8, one can also prove a corresponding classification theorem for ordinary double points a of the mapping $f: X \to S = \mathrm{Spec}(A)$, or in other words for the case of even fibre dimension and residue characteristic 2. In that situation, A must be replaced by a suitable quadratic extension. In place of $Q(x)+a$ occurs $Q(x)+a+l(x)$ for a suitable linear form $l(x)$! $Q(x)$ is merely a nonsingular quadratic form (1.1, 1.2). One can reduce this case to the one already treated (cf. SGA 7, II).

§ 3. The Monodromy Formalism

The goal of the monodromy theory, as we already mentioned, is to describe the higher direct images of the sheaf Λ_X, $\Lambda = \mathbb{Z}/r\mathbb{Z}$, r invertible on $\mathbb{P}^1$, under a Lefschetz fibering (§1)

$$f: \tilde{X} \to \mathbb{P}^1.$$

The local theory studies the specialization homomorphism (I.8.11)

$$(R^p f_* \Lambda_{\tilde{X}})_s \to (R^p f_* \Lambda_{\tilde{X}})_\eta .$$

Here η is to be a generic point of $\mathbb{P}^1$ and s a special point. This specialization homomorphism is an isomorphism when the fiber $\tilde{X}_s$ is smooth. We must therefore investigate the case where $\tilde{X}_s$ contains a singularity, an ordinary double point. This problem is of local nature with respect to the base space $\mathbb{P}^1$. Thus it suffices to study a corresponding "local situation"

$$f: X \to \mathrm{Spec}\, R.$$

Here R is the spectrum of a strictly Henselian discrete valuation ring; f is proper, flat, and smooth except for one ordinary double point on the special fiber. The specialization homomorphism has the form

$$H^p(X_s, \Lambda) \to H^p(X_\eta, \Lambda).$$

Here X_s and X_η denote the special and generic fibers of f. We shall restrict ourselves to the case of an odd fiber dimension $n = 2m+1$. It turns out that the specialization homomorphism is an isomorphism for $p \neq n, n+1$. In dimension $p = n$ it is injective, and the cokernel is generated by one element, called the *vanishing cycle*. The cohomology of the generic fiber is acted on by the *monodromy group*, that is, the Galois group $G(\bar{K}/K)$ of the separable closure $\bar{K}$ of the quotient field K of R. It is of fundamental importance that one can describe the monodromy operation on the vanishing cycle. The result of this theory are the *Picard-Lefschetz formulas* (4.3). They are the etale analogue of the classical Picard-Lefschetz formulas, which deal with proper complex families

$$f: X \to \Delta, \qquad \Delta = \{q \in \mathbb{C}, |q| < 1\}$$

over the unit disc that are smooth apart from a single double point over the origin. In that case the monodromy operation is the action of the fundamental

group

$$\pi_1(\Delta - \{0\}, a) \cong \mathbb{Z}, \qquad a \neq 0$$

on the cohomology of the fiber over a.

In the following formalism, local monodromy theory is carried through somewhat more generally, at the level of complexes and for f not proper. The reason is of technical nature. The formalism shows that it is possible to read off the essential information about the monodromy in an "arbitrarily small" etale neighborhood of the singular point in X_s. Using the classification of ordinary double points, one can show that every pencil $X \to S$ is etale locally isomorphic to a certain pencil of quadrics. It thus suffices to prove the refined Picard-Lefschetz formulas (on the level of complexes (4.8–4.10)) for these pencils of quadrics. The computation for these special pencils of quadrics is carried out in § 4. This gives us the Picard-Lefschetz formulas up to a constant. This constant is determined in § 5, § 6 by comparison with the analytic theory.

Let R be a strictly Henselian discrete valuation ring with

$$\text{Residue field} \quad k = R/\mathfrak{m},$$

$$\text{Quotient field} \quad K = Q(R),$$

and let

$$s\colon \operatorname{Spec} k \to \operatorname{Spec} R,$$

$$\eta\colon \operatorname{Spec} \bar{K} \to \operatorname{Spec} R$$

$$\bar{K} = \text{separable closure of } K$$

be "the two" geometric points of $S = \operatorname{Spec} R$.

Let

$$f\colon X \to \operatorname{Spec} R$$

be a finitely generated, say quasiprojective, flat scheme over R with fibers

$$X_s = X \times_S \operatorname{Spec}(k) \quad \text{(special fiber)}$$

$$X_\eta = X \times_S \operatorname{Spec}(\bar{K}) \quad \text{(generic fiber)}$$

$$\begin{array}{ccccc} X_\eta & \xrightarrow{j} & X & \xleftarrow{i} & X_s \\ \downarrow{\scriptstyle f_\eta} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_s} \\ \operatorname{Spec}(\bar{K}) & \xrightarrow{\eta} & S & \xleftarrow{s} & \operatorname{Spec}(k) \end{array}$$

We always assume that all fibers of X at all points have dimension

$$n = n(X/S).$$

Let r be a natural number relatively prime to the characteristic of k, and set

$$\Lambda = \mathbb{Z}/r\mathbb{Z}.$$

In what follows, we will consider only sheaves of Λ-modules. In particular, an "injective sheaf" will mean an injective object in the category of all sheaves of Λ-modules.

Let $\mathscr{F}$ be a sheaf (of Λ-modules) on X. We are interested above all in the following case:

a) f is proper and smooth apart from one singularity in the special fiber X_s. This singularity is an ordinary double point.

b) $\mathscr{F}=\Lambda_X$.

Because of the proper base change theorem (I.8.7), we have canonical isomorphisms

$$(\mathrm{R}^\bullet f_!\mathscr{F})_s \xrightarrow{\sim} H_c^\bullet(X_s, \mathscr{F}_s),$$

$$(\mathrm{R}^\bullet f_!\mathscr{F})_\eta \xrightarrow{\sim} H_c^\bullet(X_\eta, \mathscr{F}_\eta).$$

The specialization mapping (I.8.11)

$$(\mathrm{R}^\bullet f_!\mathscr{F})_s \to (\mathrm{R}^\bullet f_!\mathscr{F})_\eta$$

induces a mapping

$$H_c^\bullet(X_s, \mathscr{F}_s) \to H_c^\bullet(X_\eta, \mathscr{F}_\eta)$$

which we shall also refer to as the specialization mapping.

The cohomology of the generic fiber can be expressed as the cohomology of a complex on the total space X. Indeed, for an arbitrary sheaf $\mathscr{G}$ on X_η we have

$$\mathrm{R}\Gamma(X_\eta, \mathscr{G}) = \mathrm{R}\Gamma(X, \mathrm{R}j_*\mathscr{G}).$$

Together with the restriction mapping to the fiber, this gives us

$$\mathrm{R}\Gamma(X_\eta, \mathscr{G}) \to \mathrm{R}\Gamma(X_s, i^*\mathrm{R}j_*\mathscr{G})$$

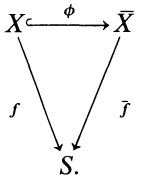
resp.

$$H^\nu(X_\eta, \mathscr{G}) \to H^\nu(X_s, i^*\mathrm{R}j_*\mathscr{G}).$$

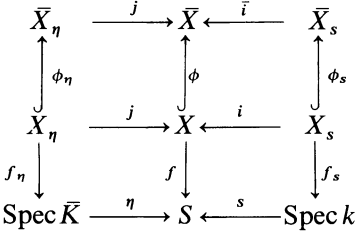
This mapping is an isomorphism for proper f. We want to construct in a similar way a mapping in the reverse direction for cohomology with compact support,

$$H_c^i(X_s, i^*\mathrm{R}j_*\mathscr{G}) \to H_c^i(X_\eta, \mathscr{G}).$$

To compute the cohomology with compact support, we consider a compactification



We get a diagram



$(\bar{X}_\eta = \bar{X} \times_S \mathrm{Spec}\,\bar{K},\ \bar{X}_s = \bar{X} \times_S \mathrm{Spec}\,k)$.

Let $\mathscr{G}$ be a sheaf on X_η. In Chap. I (I.8.1) we defined the sheaf "extended by zero" $(\phi_\eta)_!(\mathscr{G})$ on $\bar{X}_\eta$. The cohomology with compact support for $\mathscr{G}$ is by definition (I, § 8)

$$H_c^\cdot(X_\eta, \mathscr{G}) = H^\cdot(\bar{X}_\eta, (\phi_\eta)_!(\mathscr{G})).$$

The restriction of $(\phi_\eta)_!(\mathscr{G})$ to the open subscheme X_η is canonically isomorphic to $\mathscr{G}$. Because moreover X_η is the full inverse image of the open subscheme $X \subset \bar{X}$ under $\bar{j}$, there is a canonical isomorphism

$$\mathsf{R}\, j_*(\mathscr{G}) \xrightarrow{\sim} \phi^*(\mathsf{R}\, \bar{j}_*((\phi_\eta)_!(\mathscr{G})))$$

in the derived category. Adjoint to this is a mapping

$$\phi_!(\mathsf{R}\, j_* \mathscr{G}) \to \mathsf{R}\, \bar{j}_*((\phi_\eta)_!(\mathscr{G})).$$

We restrict this mapping to the special fiber $\bar{X}_s$. Since extension by zero is compatible with formation of inverse images, we get

$$(\phi_s)_!(i^* \mathsf{R}\, j_* G) \to \bar{i}^* \mathsf{R}\, \bar{j}_*((\phi_\eta)_!(\mathscr{G})),$$

and from this

$$\begin{array}{ccc} H^\nu(\bar{X}_s, (\phi_s)_!(i^* \mathsf{R}\, j_* \mathscr{G})) & \to & H^\nu(\bar{X}_s, \bar{i}^* \mathsf{R}\, \bar{j}_*((\phi_\eta)_!(\mathscr{G}))) \\ \| & & \\ H_c^\nu(X_s, i^* R\, j_* G). & & \end{array}$$

The morphism

$$H^\nu(\bar{X}_\eta, (\phi_\eta)_!(\mathscr{G})) \to H^\nu(\bar{X}_s, \bar{i}^* \mathsf{R}\, \bar{j}_*((\phi_\eta)_!(\mathscr{G})))$$

constructed at the start (with $\bar{X}/S$ in place of X/S and $(\phi_\eta)_!(\mathscr{G})$ in place of $\mathscr{G}$) is an isomorphism, since $\bar{X}/S$ is proper. We compose its inverse with the map just constructed and finally get

$$H_c^\nu(X_s, i^* \mathsf{R}\, j_* \mathscr{G}) \to H_c^\nu(X_\eta, \mathscr{G}).$$

The special case $\mathscr{G} = \mathscr{F}_\eta = j^* \mathscr{F}$ ($\mathscr{F}$ a sheaf on X) is of interest for us. The previous constructions yield natural mappings

$$\boxed{\begin{array}{l} H^\nu(X_\eta, \mathscr{F}_\eta) \to H^\nu(X_s, i^* \mathsf{R}\, j_* \mathscr{F}_\eta) \\ H_c^\nu(X_\eta, \mathscr{F}_\eta) \leftarrow H_c^\nu(X_s, i^* \mathsf{R}\, j_* \mathscr{F}_\eta) \end{array}}$$

that are isomorphisms inverse to each other for proper X/S.

For what is to come, it is useful to refine the construction of $i^* \mathsf{R}\, j_* \mathscr{F}_\eta$ so that we can follow the *action of the Galois group* $G(\bar{K}/K)$ on the cohomology of the generic fiber. For this we consider a *concrete injective resolution* of the sheaf $\mathscr{F}$,

$$\mathscr{F} \to \mathscr{J}^\cdot, \qquad \mathscr{J}^\nu = 0 \quad \text{for } \nu < 0.$$

(Thus $\mathscr{J}^\cdot$ is a positive complex in the category of sheaves of Λ-modules on X.) From this we get a resolution

$$\mathscr{F}_\eta \to \mathscr{J}_\eta^\cdot.$$

This is an acyclic resolution of $\mathscr{F}_\eta$, for we have:

3.1 Lemma. *Let $\mathscr{J}$ be an injective sheaf (of Λ-modules) on X. Every etale mapping*

$$U \to X_\eta,$$

satisfies

$$H^i(U, \mathscr{J}_\eta | U) = 0 \quad \text{for } i > 0.$$

(A sheaf $\mathscr{J}_\eta$ with this property is also called flabby (Chapter I, Remark after Proposition 2.9). In fact, more is true: $\mathscr{J}_\eta$ is actually injective.)

Proof. X_η and therefore U is a projective limit of schemes etale over X. The restriction of an injective sheaf to an etale scheme remains injective. The assertion follows then from the limit theorems (I, § 4). □

3.2 Notation.

$$\mathscr{A}^\cdot = \mathscr{A}^\cdot(\mathscr{F}, f) = i^* j_*(\mathscr{J}_\eta^\cdot), \qquad \mathscr{A}^\cdot(f) = \mathscr{A}^\cdot(\Lambda_X, f).$$

From 3.1 follows

3.3 Remark. The complex $\mathscr{A}^\cdot$ represents $i^* \mathrm{R} j_* \mathscr{F}_\eta$ in the derived category.

Functorial properties of $\mathscr{A}^\cdot$: The adjunction mapping

$$\mathscr{J}^\cdot \to j_* j^* \mathscr{J}^\cdot = j_* \mathscr{J}_\eta^\cdot$$

induces a mapping

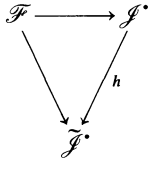
$$\phi\colon \mathscr{F}_s \to \mathscr{A}^\cdot.$$

The pair $(\mathscr{A}^\cdot, \phi)$ is unique in the following sense.

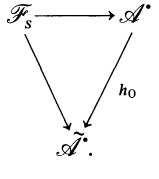
Let

$$\mathscr{F} \to \tilde{\mathscr{J}}^\cdot \quad (\text{with } \tilde{\mathscr{J}}^\nu = 0 \text{ for } \nu < 0)$$

be another injective resolution, and $(\tilde{\mathscr{A}}, \tilde{\phi})$ the associated pair. It is well known that there exists a mapping of complexes $h\colon \mathscr{J}^\cdot \to \tilde{\mathscr{J}}^\cdot$ such that the diagram



is commutative in the category of complexes. The mapping h is uniquely determined up to homotopy. It induces a commutative diagram



The mapping h_0 is a quasi-isomorphism and *uniquely determined by this construction up to homotopy.*

3.4 Remark. The assignment

$$\mathscr{F} \mapsto (\mathscr{F}_s, \phi, \mathscr{A}^\cdot)$$

is a functor from the category of sheaves (of Λ-modules) on X into the following category of triples: The objects are triples $(\mathscr{H}, \phi, \mathscr{K}^\cdot)$, where $\mathscr{H}$ is a sheaf on X_s, $\mathscr{K}^\cdot$ is a positive complex of sheaves, and $\phi\colon \mathscr{H} \to \mathscr{K}^\cdot$ is a mapping of complexes. The morphisms are commutative diagrams

$$\begin{array}{ccc} \mathscr{H} & \xrightarrow{\phi} & \mathscr{K}^\cdot \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta^\cdot} \\ \tilde{\mathscr{H}} & \xrightarrow{\tilde{\phi}} & \tilde{\mathscr{K}}^\cdot \end{array}$$

where α is a sheaf homomorphism and $\beta^\cdot$ is a homotopy class of mappings of complexes.

We have constructed natural mappings

$$H^i(X_\eta, \mathscr{F}_\eta) \to H^i(X_s, \mathscr{A}^\cdot)$$
$$H^i_c(X_\eta, \mathscr{F}_\eta) \leftarrow H^i_c(X_s, \mathscr{A}^\cdot)$$

that are isomorphisms (inverse to each other) in the proper case. If we compose the second of these mappings with the mapping

$$H^i_c(X_s, \mathscr{F}_s) \to H^i_c(X_s, \mathscr{A}^\cdot)$$

induced by ϕ, we get the specialization mapping

$$H^i_c(X_s, \mathscr{F}_s) \to H^i_c(X_\eta, \mathscr{F}_\eta)$$

considered at the outset.

The Action of the Galois Group $G = G(\bar{K}/K)$

Each element σ of the Galois group of $\bar{K}$ over K defines an automorphism $\operatorname{Spec}\bar{K} \to \operatorname{Spec}\bar{K}$, again denoted by σ, and hence defines an automorphism

$$\mathrm{id}_X \times \sigma\colon\ X \times_K \operatorname{Spec}\bar{K} \to X \times_K \operatorname{Spec}\bar{K}.$$

For simplicity we once again denote this by

$$\sigma\colon\ X_\eta \to X_\eta.$$

The Galois group acts on the scheme X_η on the right (if, as usual, it acts on $\bar{K}$ on the left). We describe the action of G on the cohomology $H^\cdot(X_\eta, \mathscr{F}_\eta)$. First, a

Preliminary Remark. Let $Y \xrightarrow{\phi} Z$ be a morphism of schemes, $\mathcal{G}$ a sheaf on Y, and $\mathcal{H}$ a sheaf on Z. If

$$\mathcal{H} \to \phi_* \mathcal{G} \quad (\text{or } \phi^* \mathcal{H} \to \mathcal{G})$$

is a homomorphism of sheaves, it induces a homomorphism of the cohomology groups

$$H^*(Z, \mathcal{H}) \to H^*(Y, \mathcal{G}).$$

We apply this to the morphism

$$\sigma\colon\ X_\eta \to X_\eta .$$

We have $j \circ \sigma = j$. Hence we have a canonical isomorphism

$$\sigma^* \mathcal{F}_\eta \to \mathcal{F}_\eta \qquad (\mathcal{F}_\eta = j^* \mathcal{F}),$$

and this induces a homomorphism

$$H^*(X_\eta, \mathcal{F}_\eta) \to H^*(X_\eta, \mathcal{F}_\eta).$$

The Galois group acts on the cohomology groups on the left. Our goal is to describe this representation of G precisely.

First we must describe the action of G on the level of the complex $\mathcal{A}^*$. For this we need the concept of a G-sheaf, where G is a profinite group and (as is enough for our purpose) G acts *trivially* on a scheme Z. In this case a G-sheaf is nothing but a sheaf of $\Lambda[G]$-modules on Z. We can formulate this as follows:

A G-sheaf is a sheaf $\mathcal{F}$ of Λ-modules where G acts on every module of sections $F(U)$, $U \to Z$ etale, in such a way that this action is compatible with restriction. We require the usual continuity property:

The stabilizer of an element $s \in \mathcal{F}(U)$ is open in G (and hence has finite index).

Back to our situation $Z = X$, $G = G(\bar{K}/K)$. The Galois group acts trivially on X. We show that $j_* j^* \mathcal{F} = j_* \mathcal{F}_\eta$ ($\mathcal{F}$ a sheaf on X) has a natural G-sheaf structure. We have already assigned a mapping

$$H^\nu(X_\eta, \mathcal{F}_\eta) \to H^\nu(X_\eta, \mathcal{F}_\eta)$$

to each $\sigma \in G$. This mapping is defined more generally for any etale scheme U/X in place of X and $\mathcal{F}/U$ in place of $\mathcal{F}$. In the special case $\nu = 0$ we get a family of mappings

$$\begin{array}{ccc} \mathcal{F}_\eta(U_\eta) & \to & \mathcal{F}_\eta(U_\eta) \qquad (U_\eta = U \times \operatorname{Spec} \bar{K}) \\ \| & & \\ (j_* \mathcal{F}_\eta)(U). & & \end{array}$$

This family represents a sheaf mapping

$$j_* \mathcal{F}_\eta \to j_* \mathcal{F}_\eta .$$

The assignment

$$\mathcal{F} \mapsto j_* j^* \mathcal{F} = j_* \mathcal{F}_\eta$$

is a functor from the category of sheaves on X to the category of G-sheaves on X. The adjunction mapping

$$\mathscr{F} \to j_* j^* \mathscr{F}$$

is G-equivariant, if one lets G act trivially on $\mathscr{F}$.

We recall the construction of the complex $\mathscr{A}^\cdot$. First we consider an injective resolution (on X),

$$\mathscr{F} \to \mathscr{J}^\cdot, \qquad \mathscr{J}^\nu = 0 \quad \text{for } \nu < 0.$$

Applying the functor $j_* j^*$ to this, we get a morphism

$$j_* \mathscr{F}_\eta \to j_* \mathscr{J}_\eta^\cdot, \qquad \mathscr{J}_\eta^\nu = j^* \mathscr{J}^\nu.$$

This is an *equivariant* morphism from the G-sheaf $j_* \mathscr{F}_\eta$ to a complex of G-sheaves. In particular, the composite morphism

$$\mathscr{F} \to j_* j^* \mathscr{F} = j_* \mathscr{F}_\eta \to j_* \mathscr{J}_\eta^\cdot$$

is equivariant. We restrict this mapping to the special fiber X_s.

We let the Galois group G act trivially on X_s (as on X). The restriction $i^* \mathscr{H}$ of a G-sheaf $\mathscr{H}$ on X is a G-sheaf on X_s. The morphism

$$i^* \mathscr{F} = \mathscr{F}_s \to i^* j_* \mathscr{J}_\eta^\cdot \quad (= \mathscr{A}^\cdot(\mathscr{F}, f))$$

is an equivariant morphism of the trivial G-sheaf $\mathscr{F}_s$ into a complex of G-sheaves.

3.5 Remark. The assignment

$$\mathscr{F} \mapsto (\mathscr{F}_s, \phi, \mathscr{A}^\cdot)$$

from Remark 2.4 is a functor from the category of sheaves (of Λ-modules) to an "equivariant triple category": The objects are the equivariant homomorphisms

$$\phi\colon \mathscr{H} \to \mathscr{K}^\cdot$$

of a G-trivial sheaf $\mathscr{H}$ on X_s to a positive complex $\mathscr{K}^\cdot$ of G-sheaves. The morphisms are the commutative diagrams

$$\begin{array}{ccc} \mathscr{H} & \longrightarrow & \mathscr{K}^\cdot \\ \downarrow{\scriptstyle a} & & \downarrow{\scriptstyle \beta^\cdot} \\ \tilde{\mathscr{H}} & \longrightarrow & \tilde{\mathscr{K}}^\cdot \end{array}$$

where $\beta^\cdot$ is an (equivariant) homomorphism of complexes uniquely determined modulo G-equivariant homotopy.

The natural mapping

$$H^\cdot(X_\eta, \mathscr{F}_\eta) \to H^\cdot(X_s, \mathscr{A}^\cdot)$$

is G-equivariant. If one applies this construction to a compactification of X/S, one sees that

$$H_c^\cdot(X_\eta, \mathscr{F}_\eta) \leftarrow H_c^\cdot(X_s, \mathscr{A}^\cdot)$$

is also in a natural way G-equivariant.

We now consider the mapping cone of the mapping of complexes

$$\phi\colon \mathscr{F}_s \to \mathscr{A}^\cdot.$$

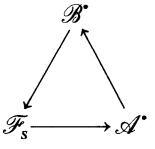
3.6 Definition. Let the complex $\mathscr{B}^\cdot(\mathscr{F}, f)$ be the mapping cone of ϕ, concretely given as the complex

$$\ldots \to 0 \to \mathscr{F}_s \xrightarrow{\phi} \mathscr{A}^0 \to \mathscr{A}^1 \to \ldots$$

(0-th place)

We set $\mathscr{B}^\cdot(f) = \mathscr{B}^\cdot(\Lambda_X, f)$. In accordance with its later significance, we call $\mathscr{B}^\cdot$ the complex of *vanishing cycles* (cf. footnote to Theorem 4.3).

The admissible triangle



gives a long exact cohomology sequence. If X/S is proper, this runs

$$\ldots \to H^i(X_s, \mathscr{F}_s) \to H^i(X_\eta, \mathscr{F}_\eta) \to H^i(X_s, \mathscr{B}^\cdot)$$
$$\to H^{i+1}(X_s, \mathscr{F}_s) \to H^{i+1}(X_\eta, \mathscr{F}_\eta) \to \ldots$$

The hypercohomology of the complex $\mathscr{B}^\cdot$ thus describes the deviation between the cohomology of the generic fiber and the special fiber. We investigate the complex $\mathscr{B}^\cdot = \mathscr{B}^\cdot(f)$ in the case of the constant sheaf $\mathscr{F} = \Lambda_X$.

For the cartesian diagram

$$\begin{array}{ccc} X_\eta & \xrightarrow{j} & X \\ \downarrow & & \downarrow f \\ \operatorname{Spec} \bar{K} & \xrightarrow{\eta} & S \end{array}$$

and the constant sheaf $\Lambda_{\operatorname{Spec}\bar{K}}$ we have the base change mapping

$$\begin{array}{ccc} f^*(\eta_* \Lambda_{\operatorname{Spec}\bar{K}}) & \longrightarrow & j_* \mathscr{J}_\eta \\ \| & & \| \\ f^*(R\eta_* \Lambda_{\operatorname{Spec}\bar{K}}) & \longrightarrow & R j_* \Lambda_X. \end{array}$$

By the smooth base change theorem (I § 7), we have

3.7 Remark. The base change mapping

$$f^*(\eta_* \Lambda_{\operatorname{Spec}\bar{K}}) \to j_* \mathscr{J}_\eta^\cdot$$

is a quasi-isomorphism at all smooth points of X/S. In particular,

$$\Lambda_{X_s} \to \mathscr{A}^\cdot(f) \quad (= i^* j_* \mathscr{J}_\eta^\cdot)$$

is a quasi-isomorphism on the smooth locus of X_s. It follows from the definition of the complex $\mathscr{B}^\cdot = \mathscr{B}^\cdot(f)$ that the restriction of $\mathscr{B}^\cdot$ to the smooth locus is exact (even at the 0-th place).

We assume that the mapping $f: X \to S$ is smooth apart from a single point $a \in X_s$.

3.8 Lemma. *Let the mapping $f: X \to S$ be smooth apart from a single point $a \in X_s$. Then the complex $\mathscr{B}^\cdot_x$ is exact for all geometric points x apart from a. In particular, there are canonical isomorphisms*

$$H^i_c(X, \mathscr{B}^\cdot) = H^i(X, \mathscr{B}^\cdot) = H^i_{\{a\}}(X, \mathscr{B}^\cdot) = H^i(\mathscr{B}^\cdot_a).$$

(We identify a with the corresponding geometric point with values in the algebraic closure of the residue field k.)

The Variation

The Galois group $G = G(\bar{K}/K)$ acts on the complex $\mathscr{A}^\cdot$. Since G acts trivially on $\mathscr{F}_s$, for every $\sigma \in G$ the mapping

$$\begin{array}{ccccccccccc}
\ldots \to 0 & \longrightarrow & 0 & \longrightarrow & \mathscr{A}^0 & \longrightarrow & \mathscr{A}^1 & \longrightarrow & \mathscr{A}^2 & \to \ldots \\
\uparrow & & \uparrow{\scriptstyle 0} & & \uparrow{\scriptstyle \sigma-1} & & \uparrow{\scriptstyle \sigma-1} & & \uparrow{\scriptstyle \sigma-1} & \\
\ldots \to 0 & \longrightarrow & \mathscr{F}_s & \longrightarrow & \mathscr{A}^0 & \longrightarrow & \mathscr{A}^1 & \longrightarrow & \mathscr{A}^2 & \to \ldots
\end{array}$$

"$(\sigma - 1)\, x = \sigma x - x$" is a mapping of complexes. We will denote it by

$$\sigma - 1: \mathscr{B}^\cdot \to \mathscr{A}^\cdot.$$

We apply to the mapping $\mathscr{B}^\cdot \xrightarrow{\text{"}\sigma-1\text{"}} \mathscr{A}^\cdot$ the functor $H^d_{\{a\}}(X_s, -)$ (hypercohomology with support at a). If a is the only singularity of the mapping f, and thus $\mathscr{B}^\cdot$ is exact apart from a (3.8), we have a canonical isomorphism

$$H^\nu_{\{a\}}(X_s, \mathscr{B}^\cdot) \cong H^\nu(\mathscr{B}^\cdot_a).$$

In addition,

$$H^\nu(\mathscr{B}^\cdot_a) = H^\nu(\mathscr{A}^\cdot_a) \quad \text{for } \nu \neq 0.$$

Moreover, there are canonical mappings

$$H^\nu_{\{a\}}(X_s, \mathscr{A}^\cdot) \to H^\nu_c(X_s, \mathscr{A}^\cdot) \to H^\nu_c(X_\eta, \mathscr{F}_\eta)$$

(see 3.4).

3.9 Definition. Let $n > 0$. We denote by $\operatorname{var} \sigma$ the mapping

$$H^n(\mathscr{B}^\cdot_a) = H^n(\mathscr{A}^\cdot_a) \to H^n_{\{a\}}(X_s, \mathscr{A}^\cdot), \qquad n = \dim X/S,$$

induced by $\sigma - 1: \mathscr{B}^\cdot \to \mathscr{A}^\cdot$.

3.10 Remark. Let σ_a be the mapping induced on $H^d_{\{a\}}(X_s, \mathscr{A}^\cdot)$ by $\sigma: \mathscr{A}^\cdot \to \mathscr{A}^\cdot$. Then

$$\operatorname{var}(\sigma\tau) = \operatorname{var}\sigma + \operatorname{var}\tau + (\sigma_a - 1)(\operatorname{var}\tau).$$

This formula follows trivially from the following formula for mappings of the complex $\mathscr{A}^\bullet$ to itself:

$$\sigma\tau - 1 = (\sigma - 1) + (\tau - 1) + (\sigma - 1)(\tau - 1).$$

3.11 Remark. There is a canonical isomorphism

$$H^\nu(\mathscr{A}_a) = \varinjlim H^\nu(U_\eta, \mathscr{F}_\eta).$$

Here U runs through the etale neighborhoods of a in X, and U_η denotes the inverse image of U in X_η. In particular, there is a canonical mapping

$$H^\nu(X_\eta, \mathscr{F}_\eta) \to H^\nu(\mathscr{A}_a^\bullet).$$

Proof. Use

$$H^\nu(\mathscr{A}_a^\bullet) = H^\nu((j_* \mathscr{J}^\bullet)_a) = \varinjlim_U H^\nu(U_\eta, \mathscr{F}_\eta).$$

3.12 Remark. The diagram

$$\begin{array}{ccc} H_c^n(X_\eta, \mathscr{F}_\eta) & \xrightarrow{\sigma - 1} & H_c^n(X_\eta, \mathscr{F}_\eta) \\ \downarrow & & \uparrow \\ H^n(\mathscr{A}_a) & \xrightarrow{\mathrm{var}\sigma} & H_{\{a\}}^n(X_s, \mathscr{A}) \end{array}$$

is commutative.

3.13 Remark. There is a natural pairing

$$H^n(\mathscr{A}_a^\bullet) \times H_{\{a\}}^n(X_s, \mathscr{A}^\bullet) \to \Lambda(-n) = \mu_n^{\otimes -n}, \qquad n = \dim(X/S).$$

For the construction we apply the usual duality pairing

$$H^n(U_\eta, \Lambda) \times H_c^n(U_\eta, \Lambda) \to \Lambda(-n)$$

for arbitrary etale neighborhoods $U \to X$ of a. Using the restriction mapping

$$H_c^\nu(X_\eta, \Lambda) \to H_c^\nu(U_\eta, \Lambda)$$

and remark 3.11, we get the desired pairing by passage to the limit.

3.14 Remark. Let $S = \operatorname{Spec} R$ be the spectrum of a strictly Henselian valuation ring, and let $f\colon X \to S$ be a finitely generated flat morphism that is smooth outside a single point $a \in X$. Let the special fiber have the same dimension at all points. Let ϕ be an etale mapping over S,

$$\begin{array}{ccc} Y & \xrightarrow{\phi} & X \\ & {\scriptstyle f'} \searrow \quad \swarrow {\scriptstyle f} & \\ & S & \end{array} \qquad \phi^{-1}(a) = \{b\}.$$

Assertion. There exist canonical isomorphisms

$$H^{\cdot}_{\{b\}}(Y_s, \mathscr{A}(f')) \cong H^{\cdot}_{\{a\}}(X_s, \mathscr{A}^{\cdot}(f))$$
$$H^{\cdot}(\mathscr{A}^{\cdot}_b(f')) \cong H^{\cdot}(\mathscr{A}^{\cdot}_a(f)).$$

These are compatible with

a) the variation,
b) the canonical pairing,
c) the long exact sequence.

For: The restriction of $\mathscr{A}^{\cdot}(f)$ to Y is a representative of $\mathscr{A}^{\cdot}(f')$ in the triple category (3.5).

The goal of the following sections is to determine the groups $H^{\nu}_{\{a\}}(X_s, \mathscr{A}^{\cdot}(f))$, $H^{\nu}(\mathscr{A}^{\cdot}_a(f))$ and the variation mapping in the special case of an ordinary double point a. For our application we need only the (harder) case of odd fiber dimension. The main result are the Picard-Lefschetz formulas (Theorem 4.3).

§ 4. The Picard-Lefschetz Formulas

We put the main result of monodromy theory, the Picard-Lefschetz formulas, right at the beginning of this section. As in the previous section, let R be a strictly Henselian discrete valuation ring, K the quotient field of R, and $\bar{K}$ its separable closure. We first define the fundamental character of the monodromy group $G = G(\bar{K}/K)$. By hypothesis, r is a natural number invertible in R. Therefore R contains the group $\Lambda(1) = \mu_r(R)$ of r-th roots of unity. This is canonically isomorphic to the group $\mu_r(k)$ of r-th roots of unity in the residue field $k = R/\mathfrak{m}$.

4.1 Definition. We define the fundamental character

$$\chi\colon\ G \to \Lambda(1)$$

by the formula

$$\chi(\sigma) = \sigma(\sqrt[r]{\pi})/\sqrt[r]{\pi}.$$

Here π is a generator of the maximal ideal $\mathfrak{m}$ of R. More generally, if $\lambda \in R$ is a nonzero element of order ν, we set

$$\chi_\lambda = \chi^\nu.$$

Obviously

$$\chi_\lambda(\sigma) = \sigma(\sqrt[r]{\lambda})/\sqrt[r]{\lambda}.$$

4.2 Properties of the fundamental character. 1) *Since the order of Λ is relatively prime to the residue field characteristic, every homomorphism ϕ of G into an Λ-module M factors uniquely through χ:*

Thus there is an element $u \in M(-1) = M \otimes \Lambda(-1)$ *with*

$$\phi(\sigma) = \chi(\sigma) \cdot u.$$

2) *Let* R *be the strict Henselization of a Henselian valuation ring* R_0 *with quotient field* K_0 *and residue field* k_0,

$$\tilde{G} = \mathrm{Gal}(\bar{K}/K_0), \qquad \Gamma \equiv \tilde{G}/G = \mathrm{Gal}(K/K_0) \cong \mathrm{Gal}(k/k_0).$$

Then Γ *acts on the group* $\Lambda(1)$ *of r-th roots of unity, and an element* $\tilde{\tau}$ *in* $\tilde{G}$ *with image* τ *in* Γ *satisfies*

$$\chi(\tilde{\tau}\sigma\tilde{\tau}^{-1}) = \tau(\chi(\sigma)) \quad \text{for } \sigma \in G.$$

4.3 Theorem. *The Picard-Lefschetz Formulas. Let* $S = \mathrm{Spec}(R)$ *be the spectrum of a strictly Henselian valuation ring* R *(with residue field* k, *quotient field* K, ...*). Let*

$$f\colon X \to S$$

be a proper flat morphism of odd fiber dimension $n = 2m+1$ *with special fiber* X_s *and generic geometric fiber* X_η. *Let* f *be singular at just one point,* a, *and let this point be an ordinary double point. By* § 2, *the Henselization of* $\mathcal{O}_{X,a}$ *is* R*-isomorphic to*

$$R\{X_0, \ldots, X_n\} \Big/ \left(\sum_{\nu=0}^{m} X_\nu X_{\nu+1+m} + \lambda \right), \qquad 0 \neq \lambda \in \mathfrak{m}(R).$$

Then

$$(0) \qquad H^\nu(X_s, \Lambda) \xrightarrow{\cong} H^\nu(X_\eta, \Lambda), \qquad \nu \neq n,\ n+1.$$

There is a "vanishing cycle" *)

$$\delta \in H^n(X_\eta, \Lambda)(m) = H^n(X_\eta, \Lambda) \otimes \Lambda(m)$$

and a "covanishing cycle" $\delta^* \in H^{n+1}(X_s, \Lambda(n-m))$ *such that the sequence*

$$x \mapsto \langle x, \delta \rangle$$

$$(1) \quad 0 \to H^n(X_s, \Lambda) \to H^n(X_\eta, \Lambda) \to \Lambda(m-n) \to H^{n+1}(X_s, \Lambda) \to H^{n+1}(X_\eta, \Lambda) \to 0$$

$$x \mapsto x \cdot \delta^*$$

is exact.

The Galois group G *of the algebraic closure of* $\bar{K}$ *over* K *acts trivially on* $H^{n+1}(X_\eta, \Lambda)$ *and acts on* $H^n(X_\eta, \Lambda)$ *by*

$$(2) \qquad \boxed{\sigma(x) = x + (-1)^{m+1} \chi_\lambda(\sigma) \langle x, \delta \rangle \delta}$$

---

*) This name comes from the dual description of monodromy theory using *homology groups* for a topological local Lefschetz pencil $f\colon X \to \Delta$ over the unit circle. The vanishing cycle δ is then a cycle in the homology group $H_n(f^{-1}(t), \mathbb{Z})$ of a general fiber $f^{-1}(t)\,(t \neq 0)$ that is sent to zero, and thus vanishes, under the "specialization mapping"

$$H_n(f^{-1}(t), \mathbb{Z}) \to H_n(f^{-1}(0), \mathbb{Z})$$

into the homology group of the degenerate fiber $f^{-1}(0)$.

with

$$\sigma(\sqrt[r]{\lambda}) = \chi_\lambda(\sigma)\sqrt[r]{\lambda}.$$

Remark. Let l be a prime number invertible in R. All constructions in this section are compatible with the homomorphisms induced on cohomology groups by $\mathbb{Z}/l^{r+1}\mathbb{Z} = \Lambda' \to \Lambda = \mathbb{Z}/l^r\mathbb{Z}$. By passage to the projective limit, one gets a vanishing cycle δ in

$$H^n(X_\eta, \hat{\mathbb{Z}}_l(m)) = \varinjlim_r H^n(X_\eta, \mu_{l^r}^{\otimes m}),$$

and similarly a covanishing cycle, for which the results corresponding to (0), (1), and (2) are valid. Formula (2) with $\chi_l\colon G \to \varprojlim_\nu \mu_{l^\nu} = \hat{\mathbb{Z}}_l(1)$ uniquely determines the element $\delta \in H^n(X_\eta, \hat{\mathbb{Z}}_l(m))$ modulo torsion up to sign!

Supplement. If the fiber dimension is even and a simply an ordinary double point, then one can likewise deduce (0) and (1), even without transcendental methods. The monodromy formula (2) then says

$$\sigma(x) = x + (-1)^m \tfrac{1}{2}[\varepsilon(\sigma) - 1]\langle x, \delta\rangle\, \delta.$$

Here ε is a character of order $\leqq 2$,

$$\varepsilon\colon G \to \{\pm 1\} \quad \text{(see SGA 7 II, Exp. XV, Th. 3.4).}$$

Notes. Let R be the strict Henselization of a merely Henselian ring R_0 with quotient field K_0 and residue field k_0. Let $f_0\colon X_0 \to S_0 = \mathrm{Spec}(R_0)$ be a scheme over S_0 with $f = f_0 \times_{S_0} S\colon X_0 \times_{S_0} S = X \to S$. The point a then has coordinates in k_0. Let $\tilde{G} = \mathrm{Gal}(\bar{K}/K_0)$, $\Gamma = \tilde{G}/G = \mathrm{Gal}(K/K_0) \cong \mathrm{Gal}(k/k_0)$. Then Γ and hence $\tilde{G}$ act on $\Lambda(1)$ and on all the cohomology groups $H^\cdot(X_s, \Lambda(-))$, and $\tilde{G}$ acts on $H^\cdot(X_\eta, \Lambda(-))$. Since $\langle \delta, \delta\rangle = 0$, δ is contained in the group of invariants $H^n(X_\eta, \Lambda(m))^G$. From the construction of the cohomology classes δ, δ^* carried out in this section using algebraic cycles, one can conclude that after a suitable etale extension the class δ for all rings $\Lambda = \mathbb{Z}/l^\nu\mathbb{Z}$ will lie in $H^n(X_\eta, \Lambda(m))^{\tilde{G}}$, and δ^* in $H^{n+1}(X_s, \Lambda(n-m))^{\tilde{G}} = H^{n+1}(X_s, \Lambda(n-m))^{\tilde{G}}$. Under these conditions, the exact sequence (1) is actually an exact *sequence of $\tilde{G}$-modules.*

At the end of this section we give an invariant construction of the vanishing cycle. But this needs the "local" formalism of the previous section.

To prove the Picard-Lefschetz formulas, we shall first investigate special pencils, namely pencils of quadrics. For these pencils $X \to S$ (of odd fiber dimension) we determine the complexes $\mathscr{A}^\cdot$ and $\mathscr{B}^\cdot$ introduced in the previous paragraph and the variation mapping. However, for this we use a fundamental lemma (4.10) that will be proved in §§ 5, 6 by transcendental methods. We get a refined (local) version of the Picard-Lefschetz formulas for the pencils of quadrics (4.8, 4.9). Thanks to the localization principle in § 3, we can deduce from this the refined version for arbitrary Lefschetz pencils and thereby prove Theorem 4.3.

The Cohomology of Quadrics

For what follows, we need some results on the cohomology of quadrics over a separably closed base field k. One can for instance deduce them easily from

the corresponding results on singular cohomology of quadrics over the field of complex numbers by using the comparison theorem (Chap. I §12) and the method of specialization (I, § 8). Since the trace mapping is defined relatively (Chap. II 1.6), the formation of the cycle class of an algebraic cycle (II 2.1) is compatible with specialization. Similarly, one can easily show the compatibility of the comparison mapping with the algebraic and topological trace mappings. Therefore results about cohomology classes of cycles can also be derived by this method.

We summarize the necessary results on quadrics without proof.

Without loss of generality we can assume that the base field k is algebraically closed. For a Λ-module V, let $V(m)$ denote $V \otimes_\Lambda \Lambda(1)^{\otimes m}$. After selecting a primitive m-th root of unity, one has an isomorphism

$$\Lambda(1) \cong \Lambda, \qquad V(m) \cong V.$$

Let Q be a quadratic form in $n+2$ variables. Its set of zeros is an n-dimensional subvariety X in the projective space $\mathbb{P}^{n+1}$ over k, a quadric. If X is smooth, then for a suitable choice of coordinates Q has the following normal form:

a) $Q = \sum_{\nu=0}^{m} x_\nu x_{\nu+m+1}$, $n = 2m$

b) $Q = \sum_{\nu=0}^{m} x_\nu x_{\nu+m+1} + x_{n+1}^2$, $n = 2m+1$.

We consider a hyperplane $H \subset \mathbb{P}^{n+1}$ and the two linear subspaces

$$L_1 \colon\ x_0 = x_1 = \ldots = x_m = 0$$
$$L_2 \colon\ x_0 = x_1 = \ldots = x_{m-1} = x_{2m+1} = 0.$$

In case a), we have $L_1, L_2 \subset X$. Let

$$\xi = \mathrm{cl}_{\mathbb{P}^{n+1}}(H) \in H^2(\mathbb{P}^{n+1}, \Lambda(1)) = H^2(\mathbb{P}^{n+1}, \Lambda)(1).$$

The inverse image of this class in $H^2(X, \Lambda(1))$ is (II 2.3)

$$\overline{\xi} = \mathrm{cl}_X(H \cap X) \in H^2(X, \Lambda(1)) = H^2(X, \Lambda)(1).$$

Here $H \cap X$ is computed with multiplicities!

In case a), i.e. for $n = 2m$, we have

$$\mathrm{cl}_X(L_1) = \theta_1 \in H^n(X, \Lambda(m)) = H^n(X, \Lambda)(m),$$
$$\mathrm{cl}_X(L_2) = \theta_2 \in H^n(X, \Lambda(m)) = H^n(X, \Lambda)(m).$$

The cohomology groups of the projective space satisfy

$$H^\nu(\mathbb{P}^{n+1}, \Lambda) = 0 \qquad \text{for } 2 \nmid \nu,$$
$$H^2(\mathbb{P}^{n+1}, \Lambda(\nu)) = \Lambda \cdot \xi^\nu \cong \Lambda \qquad \text{for } 0 \leqq \nu \leqq n+1.$$

Here ξ^ν is the ν-th power of ξ under the cup product.

4.4 Proposition. *Let $X \subset \mathbb{P}^{n+1}$ be a smooth quadric of dimension n over k, taken in the normal form* a) *or* b). *Then*

$$\begin{aligned} H^\nu(X, \Lambda) &= 0 \quad \text{for } 2 \nmid \nu, \\ H^{2\nu}(X, \Lambda) &\cong \Lambda \quad \text{for } n \neq 2\nu,\ 0 \leqq \nu \leqq n \\ &\Lambda \oplus \Lambda \quad \text{for } n = 2\nu. \end{aligned}$$

More precisely,

$$\begin{aligned} H^{2\nu}(X, \Lambda(n)) &= \Lambda \cdot \bar{\xi}^\nu && (0 \leqq 2\nu < n), \\ H^{2\nu}(X, \Lambda(n)) &= \Lambda \cdot \bar{\xi}_\nu \quad \text{with} \quad 2\bar{\xi}_\nu = \bar{\xi}^\nu && (n < 2\nu \leqq 2n), \end{aligned}$$

$$\text{“}\bar{\xi}_\nu = \tfrac{1}{2}\bar{\xi}^\nu\text{”}.$$

For $n = 2m$,

$$H^n(X, \Lambda(m)) = \Lambda \cdot \theta_1 \oplus \Lambda \cdot \theta_2 \quad \text{with} \quad \theta_1 + \theta_2 = \bar{\xi}^m.$$

Remark (for the proof). The weak Lefschetz theorem (I 9.2) gives us the isomorphism

$$H^\mu(\mathbb{P}^{n+1}, \Lambda) \xrightarrow{\cong} H^\mu(X, \Lambda) \quad \text{in case } \mu < n.$$

For $\mu > n$, we determine $H^\mu(X, \Lambda)$ from $H^{2n-\mu}(X, \Lambda)$ using Poincaré duality.

Since X has degree two, one can conclude

$$\begin{aligned} S_{X/k}(\bar{\xi}^\nu \cdot \bar{\xi}^{n-\nu}) &= S_{X/k}(\bar{\xi}^n) = 2 \\ 2\bar{\xi}_\nu &= \bar{\xi}^\nu. \end{aligned}$$

It remains only to determine $H^n(X, \Lambda)$. The rank comes from computing the Euler-Poincaré characteristic. For this, one decomposes X into cells that are isomorphic to cartesian products of affine lines and pointed affine lines.

We consider once again the projective quadric $X \subset \mathbb{P}^{n+1}$ of odd dimension $n = 2m+1$. Contained in it as the hyperplane section with the hyperplane $x_{n+1} = 0$ is the quadric $Y \subset \mathbb{P}^n$ of even dimension $2m$ given by the equation

$$\sum_{\nu=0}^{m} x_\nu x_{\nu+m+1} = 0.$$

The complement is the smooth affine quadric X_e given by the equation

$$\sum_{\nu=0}^{m} x_\nu x_{\nu+m+1} + 1 = 0$$

in the $(n+1)$-dimensional affine space $\mathbb{A}^{n+1}$. We study the long exact cohomology sequence

$$(*) \quad \ldots \to H_c^\nu(X_e, \Lambda(\ )) \to H^\nu(X, \Lambda(\ )) \to H^\nu(Y, \Lambda(\ )) \xrightarrow{\partial} H_c^{\nu+1}(X_e, \Lambda(\ )) \to \ldots$$

The mappings

$$H^{2\nu}(X, \Lambda(\nu)) \to H^{2\nu}(Y, \Lambda(\nu))$$
$$\bar{\xi}^{\nu} \mapsto \bar{\xi}^{\nu}$$
$$\text{resp. } 2\bar{\xi}_{\nu} \mapsto 2\bar{\xi}_{\nu} \quad \text{(even for rings } \mathbb{Z}/2r!)$$

are isomorphisms for $\nu \neq m$, $\nu < n$.

For odd numbers μ, both $H^{\mu}(X, \Lambda)$ and $H^{\mu}(Y, \Lambda)$ vanish. It follows that

$$H_c^{\nu}(X_e, \Lambda) = 0 \quad \text{for } \nu \neq n,\ 2n$$
$$H_c^{n}(X_e, \Lambda(m)) \cong H^{n-1}(Y, \Lambda(m))/H^{n-1}(X, \Lambda(m)).$$

Finally, the trace mapping $S_{X_e/k}$ yields a canonical isomorphism

$$S_{X_e/k}\colon\ H_c^{2n}(X_e, \Lambda(n)) \xrightarrow{\cong} \Lambda$$

or equivalently:

$$H_c^{2n}(X_e, \Lambda) \xrightarrow{\cong} \Lambda(-n).$$

4.5 Proposition. *The cohomology groups of the affine smooth quadric X_e with equation*

$$\sum_{\nu=0}^{m} x_{\nu} x_{\nu+m+1} + 1 = 0$$

satisfy

$$H_c^{\nu}(X_e, \Lambda) = 0 \quad \text{for } \nu \neq n = 2m+1,\ 2n$$
$$H_c^{2n}(X_e, \Lambda) = \Lambda(-n)$$

(canonically by means of the trace mapping)

$$H_c^{n}(X_e, \Lambda(m)) = \Lambda \cdot \theta \cong \Lambda \cdot \theta_1 + \Lambda \cdot \theta_2 / \Lambda \underbrace{(\theta_1 + \theta_2)}_{= \bar{\xi}^{m}}.$$

Here θ is the image of one of the two cycles θ_1, θ_2 in $H^{2m}(Y, \Lambda(m))$ under the connecting homomorphism

$$\partial\colon\ H^{2m}(Y, \Lambda(m)) \to H_c^{n}(X_e, \Lambda(m))$$

in the long cohomology sequence (*). The "quasicycle" θ is uniquely determined by this up to sign.

4.6 Proposition. *Let Z be the singular affine quadric of dimension $n = 2m+1$ given by the equation*

$$\sum_{\nu=0}^{m} x_{\nu} x_{\nu+m+1} = 0$$

in $(n+1)$-dimensional affine space $\mathbb{A}^{n+1}$.

Let $\mathscr{A}^{\cdot}$ be a complex of sheaves on Z with only finitely many nonzero cohomology sheaves, and assume they are constant sheaves of Λ-modules when restricted

to $Z-\{0\}$. *Then*

(1) $H^v(Z, \mathscr{A}^\cdot) \cong H^v(\{0\}, \mathscr{A}^\cdot) = H^v(\mathscr{A}_0^\cdot)$,

(2) $H^v_{\{0\}}(Z, \mathscr{A}^\cdot) \xrightarrow{\equiv} H_c(Z, \mathscr{A}^\cdot)$.

Proof. It suffices to undertake the proof for the cohomology sheaves of $\mathscr{A}^\cdot$ in place of $\mathscr{A}^\cdot$. We can even restrict ourselves to the case $\mathscr{A}^\cdot = F_Z$, F a Λ-module. We conclude by using a homotopy argument: We consider the mappings

$$\begin{array}{ccccc} & & & & (x, \lambda) \mapsto \lambda x \\ X & \underset{f_1}{\overset{f_0}{\rightrightarrows}} & X \times \mathbb{A} & \xrightarrow{m} & X \\ & & \downarrow p & & \\ & & X & & \end{array}$$

$f_0(x) = (x, 0)$, $f_1(x) = (x, 1)$,

p the projection on the first factor.

It follows from the smooth base change theorem (Chap. I, 10.2) that the projection p induces an isomorphism

$$H^\cdot(X, F) \xrightarrow{\cong} H^\cdot(X \times \mathbb{A}, F).$$

It follows that the mappings f_0 and f_1, and hence also the mappings $m \circ f_0$ and $m \circ f_1$, induce the same mappings on cohomology groups. But $m \circ f_0$ is the mapping to the point zero, $m \circ f_1$ the identity mapping of X. Hence (1) follows.

For the proof of (2). In $(n+1)$-dimensional projective space $\mathbb{P}^{n+1}$, with homogeneous coordinates $x_0, \ldots, x_{2m+1}, z$, we consider the projective quadric X given by

$$\sum_{\nu=0}^{m} x_\nu x_{\nu+m+1} = 0.$$

Let Y be the intersection with the hyperplane $z=0$. Then $Z = X - Y$. It follows again by a homotopy argument that

$$H^\cdot(X - \{(0, 0, \ldots, 0, 1)\}, F) \xrightarrow{\cong} H^\cdot(Y, F).$$

The desired isomorphisms come from the commutative diagram of long exact cohomology sequences

$$\begin{array}{ccccccc} \ldots \to H^v_{\{(0,\ldots,1)\}}(X, F) & \longrightarrow & H^v(X, F) & \longrightarrow & H^v(X - \{(0, \ldots, 1)\}, F) \to \ldots \\ \downarrow & & \| & & \downarrow \\ \ldots \to H^v_c(Z, F) & \longrightarrow & H^v(X, F) & \longrightarrow & H^v(Y, F) \to \ldots . \end{array}$$ □

We return now to the situation of Section 3.

Let R be a strictly Henselian discrete valuation ring with residue field $k=R/\mathfrak{m}$ and quotient field K. Let $\bar{K}$ be the separable closure.

$$S=\operatorname{Spec} R,$$

$$s\colon \operatorname{Spec}(k)\to S \qquad \text{special point,}$$

$$\eta\colon \operatorname{Spec}(\bar{K})\to S \qquad \text{generic point.}$$

We consider the projective quadric $X\subset \mathbb{P}_S^{n+1}$ of relative dimension $n=2m+1$ over S given by the homogeneous equation

$$\sum_{\nu=0}^{m} x_\nu x_{\nu+m+1}+\lambda x_{n+1}^2=0, \qquad 0\neq\lambda\in\mathfrak{m}.$$

$$\begin{array}{ccccc} X\times_S\operatorname{Spec}(\bar{K})=X_\eta & \xrightarrow{j} & X & \xleftarrow{i} & X_s=X\times_S\operatorname{Spec}(k) \\ \downarrow{\scriptstyle f_\eta} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_s} \\ \operatorname{Spec}(\bar{K}) & \longrightarrow & S & \longleftarrow & \operatorname{Spec}(k). \end{array}$$

In X we consider the hyperplane section "at infinity",

$$Y\subset \mathbb{P}_S^n\colon \sum_{\nu=0}^{m} x_\nu x_{\nu+m+1}=0,$$

and the affine complement

$$X_e=X-Y\subset \mathbb{A}_S^{n+1}\colon \sum_{\nu=0}^{m} x_\nu x_{\nu+m+1}+\lambda=0,$$

$$Y_s=Y\times_S\operatorname{Spec}(k), \qquad X_{es}=X_e\times_S\operatorname{Spec}(k)\subset X_s,$$

$$Y_\eta=Y\times_S\operatorname{Spec}(\bar{K}), \qquad X_{e\eta}=X_e\times_S\operatorname{Spec}(\bar{K})\subset X_\eta.$$

Y is smooth over S. X is flat over S and smooth apart from the point $a=(0,\ldots,1)$ in the special fiber. This point is a nondegenerate double point of X over S.

Let G be the Galois group of $\bar{K}$ over K. In § 3 we introduced an admissible triangle of complexes of G-equivariant sheaves over X_s coming from the sheaf Λ_X, (3.2, 3.6)

$$\begin{array}{ccc} & \mathscr{B}^\cdot & \\ \swarrow & & \nwarrow \\ \Lambda_X & \longrightarrow & \mathscr{A}^\cdot. \end{array}$$

Here $\mathscr{A}^\cdot$ is quasi-isomorphic to $i^*\,\mathrm{R}j_*(\Lambda_X)$, and $\mathscr{B}^\cdot$ is the mapping cone of the natural mapping

$$\Lambda_X\to\mathscr{A}^\cdot.$$

Since a is the only singularity of the mapping f, the cohomology sheaves of the complex $\mathscr{B}^\cdot$ of vanishing cycles are concentrated on the point a (3.8).

4.7 Lemma. *The following composite mappings (cf. 3.11) are isomorphisms:*
(1) $H^{\cdot}_{\{a\}}(X_s, \mathscr{A}^{\cdot}) = H^{\cdot}_{\{a\}}(X_{es}, \mathscr{A}^{\cdot}) \to H^{\cdot}_c(X_{e\eta}, \Lambda)$,
(2) $H^{\cdot}(X_{e\eta}, \Lambda) \to H^{\cdot}(X_{es}, \mathscr{A}^{\cdot}) \to H^{\cdot}(\mathscr{A}^{\cdot}_a)$.

Proof. We have already shown in 4.6 that the mappings

$$H^{\nu}_{\{a\}}(X_{es}, \mathscr{A}^{\cdot}) \to H^{\nu}_c(X_{es}, \mathscr{A}^{\cdot})$$

and

$$H^{\nu}(X_{es}, \mathscr{A}^{\cdot}) \to H^{\nu}(\mathscr{A}^{\cdot}_a)$$

are isomorphisms. For indeed the complex $\mathscr{A}^{\cdot}$ is quasi-isomorphic to Λ_X away from the singularity a.

By the base change theorem, the base change homomorphism for the diagram

$$\begin{array}{ccc} X_\eta & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{h}\colon \operatorname{Spec} \bar{K} & \longrightarrow & S \end{array} \qquad \leftarrow \text{base change mapping}$$

$$\uparrow$$

structure mapping

$$f^*(\mathscr{F}) \to \mathscr{K}^{\cdot} = \mathrm{R}\, j_*(\Lambda X_\eta),$$

$$\mathscr{F} = h_* \Lambda_{\operatorname{Spec} \bar{K}}$$

is a quasi-isomorphism at all points apart from the unique singularity of f. Hence there exists an admissible triangle

$$\begin{array}{ccc} & \mathscr{E}^{\cdot} & \\ \swarrow & & \nwarrow \\ f^*(\mathscr{F}) & \longrightarrow & \mathscr{K}^{\cdot} \end{array}$$

such that the cohomology sheaves of the complex $\mathscr{E}^{\cdot}$ are concentrated on a. The inverse image $i^* \mathscr{K}^{\cdot}$ coincides in the derived category with the complex $\mathscr{A}^{\cdot}$ (3.3). We consider the mappings

$$f_e = f | X_e \colon\ X_e \to S$$

$$f_Y = f | Y \colon\ Y \to S.$$

For the proof of (2). We must show that the natural homomorphism

$$H^{\cdot}(X_{e\eta}, \Lambda) \cong H^{\cdot}(X_e, \mathscr{K}^{\cdot}) \xrightarrow{b} H^{\cdot}(X_{es}, \mathscr{A}^{\cdot})$$

is an isomorphism. This would follow from the proper base change theorem if f_e were proper. In any case this base change property holds for the complex $\mathscr{E}^{\cdot}$ in place of $\mathscr{K}^{\cdot}$, since its cohomology is concentrated on a. It therefore suffices to show that

$$H^{\cdot}(X_e, f^*(\mathscr{F})) \to H^{\cdot}(X_{es}, i^*(f^*(\mathscr{F})))$$

is an isomorphism. For this we consider the long cohomology sequence

$$\ldots \to H^\nu(Y, \mathrm{R}\Gamma_Y(f^*(\mathscr{F})) \to H^\nu(X, f^*(\mathscr{F})) \to H^\nu(X_e, f^*\mathscr{F})) \to \ldots$$

and the corresponding sequence on X_s. Here Γ_Y is the functor that assigns to a sheaf on X the subsheaf of sections with support in Y. The cohomology sheaves $\mathscr{H}^\nu(\mathrm{R}\Gamma_Y(-)) = \mathscr{H}^\nu_Y(X, (-))$ are its derived functors (see Chap. I, §10). Since Y and X are smooth over the base scheme S at all points of Y, every inverse image $f^*(\mathscr{F})$ of a sheaf of Λ-modules on the base scheme S has the base change property

$$i^*(\mathscr{H}^\bullet_Y(X, f^*(\mathscr{F}))) \xrightarrow{\cong} \mathscr{H}^\bullet_{Y_s}(X_s, i^*(f^*(\mathscr{F}))).$$

This base change theorem was proved in I.10.1 for locally constant sheaves on X instead of $f^*(\mathscr{F})$. The proof carries over to the present case, an inverse image of a sheaf on S (see I.10.2).

The desired base change property follows from the proper base change theorem for the two structure mappings

$$X \to S, \qquad Y \to S$$

together with the stated long exact sequence.

We come to the proof of (1). We consider

$$H^\bullet_c(X_{es}, \mathscr{A}^\bullet) \cong H^\bullet((S, \mathrm{R}f_{e!}(\mathscr{K}^\bullet)) \xrightarrow{b} H^\bullet_c(X_{e\eta}, \Lambda).$$

The mapping b would be an isomorphism if $X_e \to S$ were proper. As that is not the case, we argue as follows. The complexes $\mathscr{K}^\bullet | Y$ and $\mathrm{R}(j | Y_\eta)_*(\Lambda_{Y_\eta})$ are quasi-isomorphic, since they are both quasi-isomorphic to

$$f^*(\mathscr{F}) | Y.$$

This follows from the smooth base change theorem, as Y and X are smooth at all points of Y over S.

The isomorphism desired comes out now using the long exact sequence

$$\ldots \to H^\nu(S, \mathrm{R}^\bullet f_{e!}(\mathscr{K}^\bullet)) \to H^\nu(S, \mathrm{R}^\bullet f_* (\mathscr{K}^\bullet)) \to H^\nu(S, \mathrm{R}^\bullet f_{Y*}\mathscr{K}^\bullet) \to \ldots,$$

the corresponding sequence over $\mathrm{Spec}(\bar{K})$, and the isomorphisms

$$H^\bullet(X_\eta, \Lambda) \cong H^\bullet(X, \mathscr{K}^\bullet) = H^\bullet(S, \mathrm{R}f_*(\mathscr{K}^\bullet))$$

$$H^\bullet(Y_\eta, \Lambda) \cong H^\bullet(Y, \mathrm{R}(j | Y)_*(\Lambda_Y)) = H^\bullet(S, \mathrm{R}f_{Y*}(\mathscr{K}^\bullet)).$$

Thus Lemma 4.7 is proved. □

We denote by

$$\delta \in H^n_{\{a\}}(X_s, \mathscr{A}^\bullet(m)) = H^n_{\{a\}}(X_s, \mathscr{A}^\bullet)(m)$$

the inverse image of $\theta \in H^n_c(X_{e\eta}, \Lambda(m))$ (cf. 4.5) under the isomorphism induced by 4.7 (1). The element δ, like θ, is determined uniquely up to sign. It follows from the Poincaré duality theorem for $X_{e\eta}$ that the pairing introduced in § 3.13 between $H^n_{\{a\}}(X, \mathscr{A}^\bullet)$ and $H^n(\mathscr{B}^\bullet_a)$ with values in $\Lambda(-n)$ is nondegenerate. Hence

there is an element

$$\delta^* \in H^n(\mathscr{B}_a^\cdot)(n-m) \quad \text{with} \quad \langle \delta^*, \delta \rangle = 1.$$

By Propositions 4.5 and 4.7, we get:

4.8 Proposition. *The pencil of quadrics $f: X \to S$ with odd fiber dimension $n = 2m+1$ and the associated complexes $\mathscr{A}^\cdot$ (3.2) and $\mathscr{B}^\cdot$ (3.6) satisfy*

$$H^\nu_{\{a\}}(X_s, \mathscr{A}^\cdot) = \begin{cases} 0 & \nu \neq n,\ 2n \\ \Lambda(-n) & \nu = 2n \\ \Lambda(-m)\,\delta & \nu = n \end{cases}$$

$$H^\nu(\mathscr{B}_a^\cdot) = \begin{cases} 0 & \nu \neq n \\ \Lambda(m-n)\,\delta^* & \nu = n. \end{cases}$$

Moreover, $\langle \delta^*, \delta \rangle = 1$.

Remark.

$$H^\nu(\mathscr{A}_a^\cdot) = \begin{cases} \Lambda & \nu = n \\ H^\nu(\mathscr{B}_a^\cdot) & \nu \neq 0. \end{cases}$$

We consider the variation mapping

$$\begin{array}{ccc} \text{“}\sigma - 1\text{”} = \mathrm{Var}(\sigma): & H^n(\mathscr{B}_a^\cdot) & \longrightarrow H^n_{\{a\}}(X_s, \mathscr{A}_a^\cdot) \\ & \| 2 & \| 2 \\ & H^n(X_{e\eta}, \Lambda) & H^n_c(X_{e\eta}, \Lambda) \end{array}$$

introduced in 3.8 for $\sigma \in \mathrm{Gal}(\bar{K}/K)$. For two elements σ, τ in G, we have by 3.11 the formula

$$\mathrm{Var}(\sigma\tau) = \mathrm{Var}(\sigma) + \mathrm{Var}(\tau) + (\sigma - 1) \circ \mathrm{Var}(\tau)$$

$$\sigma - 1: \ H^n_{\{a\}}(X_s, \mathscr{A}_a^\cdot) \to H^n_{\{a\}}(X_s, \mathscr{A}^\cdot).$$

We claim that G acts trivially on $H^n_{\{a\}}(X_s, \mathscr{A}^\cdot)$. Indeed, there is (4.5, 4.7) a surjective G-equivariant mapping

$$H^{2m}(Y_\eta, \Lambda) \xrightarrow{\partial} H^n(X_{e\eta}, \Lambda) \cong H^n_{\{a\}}(X_s, \mathscr{A}^\cdot).$$

But G acts trivially on $H^n(Y_\eta, \Lambda)$. Indeed, the specialization mapping

$$H^n(Y_\eta, \Lambda) \to H^n(Y_s, \Lambda)$$

is an isomorphism, since Y is smooth over S, and G acts trivially on $H^n(Y_s, \Lambda)$.

For the pencil of quadrics $X \to S$ of odd fiber dimension $n > 0$, we have proved:

4.9 Proposition. *Any two elements σ, τ in $G = G(\bar{K}/K)$ satisfy*

$$\mathrm{Var}(\sigma\tau) = \mathrm{Var}(\sigma) + \mathrm{Var}(\tau).$$

In other words, the mapping

$$\mathrm{Var}: \ G \to \mathrm{Hom}(H^n(\mathscr{B}_a^\cdot), H^n_{\{a\}}(X_s, \mathscr{A}^\cdot))$$

is a homomorphism. It therefore factors through the canonical character $\chi: G \to \Lambda(1)$ (see 4.2). *Thus it follows from Proposition* 4.7 *that there is an element* c_n *in* Λ *with*

$$\mathrm{Var}(\sigma)(x) = c_n\, \chi(\sigma)\langle x, \delta\rangle\, \delta.$$

Here $\langle\ ,\ \rangle$ is the nondegenerate pairing between $H^n(\mathscr{B}_a^\cdot) = \Lambda(m-n)\,\delta^*$ and $H^n_{\{a\}}(X_s, \mathscr{A}^\cdot) = \Lambda(-m)\,\delta$.

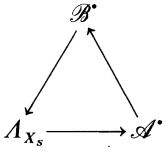
With transcendental methods, we shall determine this constant for our quadric $\sum_{v=0}^{m} x_v\, x_{v+m+1} + \lambda\, x_{n+1}^2 = 0$ (§ 5 and § 6):

4.10 Fundamental Lemma.

$$\boxed{\begin{aligned} c_n &= (-1)^{m+1} \cdot \text{order of } \lambda \\ \mathrm{Var}(\sigma)(x) &= (-1)^{m+1}\, \chi_\lambda(\sigma)\langle x, \delta\rangle\, \delta. \end{aligned}}$$

The results of 4.8 and 4.9 for pencils of quadrics are of etale local nature by § 3 (3.14); therefore, by the classification theorem for nondegenerate double points, they can in essence be carried over to an arbitrary pencil $X \to S$ of odd fiber dimension whose only singularity is an ordinary double point (and hence nondegenerate – § 2).

We come now to the proof of the Picard-Lefschetz formulas (4.3) for a proper morphism $f: X \to S$ satisfying the conditions stated in 4.3. For this we consider the triangle



and the resulting long exact cohomology sequence

$$\ldots \to H^n(X_s, \Lambda) \longrightarrow H^n(X_s, \mathscr{A}^\cdot) \longrightarrow H^n(X_s, \mathscr{B}^\cdot) \xrightarrow{\partial} H^{n+1}(X_s, \Lambda) \to \ldots$$

$$\| \qquad\qquad\qquad \|$$

$$H^n(X_\eta, \Lambda) \qquad\qquad \Lambda(m-n)\,\delta^*$$

For simplicity we use δ, δ^* again to denote the images of δ, δ^* under the mappings

$$H^n_{\{a\}}(X_s, \mathscr{A}^\cdot)(m) \to H^n(X_\eta, \Lambda)(m) \quad \text{(cf. 3.11)}$$

$$H^n(X_s, \mathscr{B}^\cdot)(n-m) \xrightarrow{\partial \otimes \Lambda(n-m)} H^{n+1}(X_s, \Lambda)(n-m).$$

These give us the vanishing cycle and covanishing cycle required in 4.3. The Picard-Lefschetz formulas 4.3 now follow immediately from 4.8, 4.9 (carried

over to arbitrary pencils) and the fundamental lemma 4.10 that is still to be proved.

Invariant Characterization of the Vanishing Cycle

The vanishing cycle δ in $H^n_{\{a\}}(\mathscr{A}^\cdot_a)(m)$ resp. in $H^n(X, \Lambda)(m)$ can also be obtained up to sign by an invariant construction, not using an etale local isomorphism with our pencil of quadrics. Let $A = \tilde{\mathcal{O}}_{s,a}$ be the Henselization of the local ring $\mathcal{O}_{X,a}$ at the nondegenerate double point a of X_s, and let $\tilde{X}_{s(a)} = \mathrm{Spec}(A)$. By blowing up the maximal ideal $\mathfrak{m}$ of A, we get a regular scheme Z over $\tilde{X}_{s(a)}$. The inverse image F of a in Z is a smooth quadric

$$F = \mathrm{Proj}(A/\mathfrak{m} \oplus \mathfrak{m}/\mathfrak{m}^2 \oplus \ldots)$$

of dimension $2m = n-1$, and $Z - F = \tilde{X}_{s(a)} - \{a\}$. We consider the composite mapping ϕ,

$$\begin{aligned} H^{2m}(F, \Lambda(m)) &\cong H^{2m}(Z, \Lambda(m)) \to H^{2m}(Z - F, \Lambda(m)) \\ &\cong H^{2m}(\tilde{X}_{s(a)} - \{a\}, \Lambda(m)) \xrightarrow{\partial} H^{2m+1}_{\{a\}}(\tilde{X}_{s(a)}, \Lambda(m)) \\ &\cong H^{2m+1}_{\{a\}}(X_s, \Lambda(m)) \to H^{2m+1}_{\{a\}}(X_s, \mathscr{A}^\cdot(m)). \end{aligned}$$

Up to sign then the vanishing cycle δ in $H^{2m+1}_{\{a\}}(X_s, \mathscr{A}^\cdot)$ is the image under ϕ of each of the two cohomology classes

$$\theta_1, \theta_2 \in H^{2m}(F, \Lambda(m))$$

given in Proposition 4.4. They do not depend on the choice of projective coordinates. Indeed, the orthogonal group of the nondegenerate quadratic form defining F acts transitively on the linear subspaces of F of dimension m. It has two connected components. Hence the cohomology class $\mathrm{cl}_F(L)$ for any such linear subspace L is either θ_1 or θ_2. Again it suffices to carry out the simple proof for our pencil of quadrics.

§ 5. Computation of the Algebraic Monodromy Using Topological Monodromy

We intend to determine the constant (4.10) not yet settled in the Picard-Lefschetz formulas over the base ring $R = \mathbb{C}\{T\}$ by comparison with the transcendental theory. For this, we must transfer the algebraic formalism of monodromy to the analytic case, at least for a pencil of quadrics.

Let

$$f\colon X \to \mathbb{C}$$

be a flat holomorphic mapping from a smooth analytic manifold X to the complex plane $\mathbb{C}$. Assume that f is smooth apart from one point a that lies in the fiber $X_0 = f^{-1}(0) \subseteq X$.

In what follows, we shall develop a transcendental formalism of monodromy, analogous to the algebraic case.

For this we consider the universal covering space

$$\mathbb{C} \to \mathbb{C}^*$$
$$\lambda \mapsto e^{2\pi i \lambda}$$

of the pointed complex plane $\mathbb{C}^* = \mathbb{C} - \{0\}$. The fundamental group $\pi_1(\mathbb{C}^*) = \pi_1$ can be identified with the cover transformation group of

$$\mathbb{C} \to \mathbb{C}^*.$$

It consists of the translations

$$\mathbb{C} \to \mathbb{C}$$
$$\lambda \mapsto \lambda + n, \quad n \in \mathbb{Z};$$

thus

$$\pi_1 \cong \mathbb{Z}.$$

We denote by σ_0 the canonical generating element

$$\lambda \mapsto \lambda + 1.$$

We extend the covering mapping to a continuous mapping

$$\bar{\mathbb{C}} \to \mathbb{C},$$

where $\bar{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ and the mapping is continuous, surjective, and "logarithmically ramified" over the origin. A neighborhood basis for ∞ in $\bar{\mathbb{C}}$ is given by the sets

$$H_r = \{z \in \mathbb{C} \mid \operatorname{Im} z > r\} \cup \{\infty\}.$$

The point ∞ is mapped to zero. We extend the action of π_1 on $\mathbb{C}$ to a continuous action on $\bar{\mathbb{C}}$ over $\mathbb{C}$ by

$$\sigma(\infty) = \infty.$$

Remark. $\bar{\mathbb{C}}$ is metrizable but not locally compact.

Using the mapping $f: X \to \mathbb{C}$, we define spaces

$$\bar{X} = X \times_{\mathbb{C}} \bar{\mathbb{C}}, \qquad \bar{X}^* = \bar{X} - \bar{f}^{-1}(0),$$

and we get the commutative diagram

$$\begin{array}{ccccc} \bar{X}^* & \xrightarrow{\bar{j}} & \bar{X} & \xleftarrow{\bar{i}} & \bar{f}^{-1}(\infty) = X_0 \\ \downarrow{\scriptstyle \bar{f}^*} & & \downarrow{\scriptstyle \bar{f}} & & \\ \mathbb{C} & \xrightarrow{j} & \bar{\mathbb{C}}. & & \end{array}$$

The group π_1 naturally acts on $\bar{X}^*$ and $\bar{X}$, acting trivially on $X_0 \subseteq \bar{X}$.

We start with a sheaf $\mathcal{G}$ on X. The inverse image $\bar{\mathcal{G}}$ on $\bar{X}$ is in a natural way a π_1-sheaf. We consider an equivariant resolution $\mathcal{I}^\cdot$ of $\bar{\mathcal{G}} | \bar{X}^*$ by flabby

π_1-sheaves, e.g. the canonical flabby resolution [51] Chap. II 4.3. The complex

$$\mathscr{A}^\cdot = \bar{i}^* \bar{j}_* (\mathscr{I}^\cdot)$$

is an equivariant complex of π_1-sheaves on X_0. In the derived category of the category of π_1-sheaves on X_0, it represents the complex

$$\bar{i}^* \mathrm{R} \bar{j}_* (\bar{\mathscr{G}} | \bar{X}^*).$$

As $\bar{X}$ is locally metrizable, $\mathscr{A}^\cdot$ is actually flabby [51], Chap. II, § 3.3, Cor. 2. There is a natural equivariant mapping

$$\mathscr{G} | X_0 = \bar{\mathscr{G}} | X_0 \to \mathscr{A}^\cdot.$$

We now assume that $\mathscr{G}$ is a constant sheaf,

$$\mathscr{G} = F_X, \quad F \text{ an abelian group},$$

$$\bar{\mathscr{G}} = F_{\bar{X}}, \quad \bar{\mathscr{G}} | \bar{X}^* = F_{\bar{X}^*}, \quad \bar{\mathscr{G}} | X_0 = F_{X_0}.$$

For a point $b \in X_0$, one can describe the cohomology groups of the complex $\mathscr{A}_b^\cdot$ formed by the stalks of $\mathscr{A}^\cdot$ at the point b:

5.1 Lemma.

$$H^\nu(\mathscr{A}_b^\cdot) = \varinjlim_U H^\nu(U \cap \bar{X}^*, F).$$

Here U runs through a fundamental system of neighborhoods of b in $\bar{X}$, and $H^\nu(U \cap \bar{X}^*, F)$ is the ν-th singular cohomology group of $U \cap \bar{X}^*$ with coefficients in F.

5.2 Lemma. *Assume the mapping $f: X \to \mathbb{C}$ is smooth in a neighborhood of the point $b \in X_0 = f^{-1}(0)$. Then there is a neighborhood V of b in X_0, a disc*

$$D = \{z \in \mathbb{C} \mid |z| < e^{-2\pi r}\},$$

and a neighborhood of b in X isomorphic to

$$V \times D.$$

This isomorphism is compatible with the projection onto $D \subset \mathbb{C}$ and with the mapping f. In particular, there is a neighborhood U of b in $\bar{X}$ and a homeomorphism

$$\begin{array}{ccc} U & \xrightarrow{\cong} & V \times (\{z \in \mathbb{C} \mid \mathrm{Im}\, z > r\} \cup \{\infty\}) \\ & \searrow \quad \swarrow & \\ & \bar{\mathbb{C}} & \end{array}$$

over $\bar{\mathbb{C}}$.

From this we can deduce the existence of a neighborhood filter for b in $\bar{X}$ consisting of neighborhoods U of the type above with V contractible.

5.3 Corollary. *Let $b \in X_0$ be a point of the special fibre different from a (the unique singular point of f). Then $H^\nu(\mathscr{A}_b^\cdot)$ vanishes for $\nu \neq 0$. The cohomology*

sheaves $\mathscr{H}^\nu(\mathscr{A}^\cdot)$ *of the complex* $\mathscr{A}^\cdot$ *are thus concentrated on* a *for* $\nu \neq 0$. *The mapping* $F_X \to \mathscr{H}^0(\mathscr{A}^\cdot)$ *is an isomorphism outside* a.

5.3a Remark. For the only case of interest later, a family of quadrics of positive dimension over $\mathbb{C}$, the last map is also an isomorphism at a.

Let $\mathscr{B}^\cdot$ be the mapping cone of the mapping $F_X \to \mathscr{A}^\cdot$. It is the extended complex

$$0 \to F_X \to \mathscr{A}^0 \to \mathscr{A}^1 \to \ldots .$$

By Corollary 4.3, all its cohomology sheaves are concentrated on a. It is likewise in a natural way an equivariant complex.

As an element $\sigma \in \pi_1(\mathbb{C}^*)$ acts trivially on F_X,

$$\sigma - 1 \colon \mathscr{A}^\cdot \to \mathscr{A}^\cdot$$

in a natural way induces a mapping

$$\mathscr{B}^\cdot \to \mathscr{A}^\cdot .$$

We apply the functor $H^\cdot_{\{a\}}(X_0, -)$ (hypercohomology with support on $\{a\}$) to this mapping, getting a mapping

$$H^\nu_{\{a\}}(X_0, \mathscr{B}^\cdot) \to H^\nu_{\{a\}}(X_0, \mathscr{A}^\cdot).$$

Since by Corollary 5.3 the cohomology sheaves of $\mathscr{B}^\cdot$ are concentrated on the point a, we have $H^\nu_{\{a\}}(X_0, \mathscr{B}^\cdot)$ canonically isomorphic to $H^\nu(\mathscr{B}^\cdot_a)$. For $\nu \neq 0$, $H^\nu(\mathscr{B}^\cdot_a) = H^\nu(\mathscr{A}^\cdot_a)$.

5.4 Definition. For an element $\sigma \in \pi_1(\mathbb{C}^*)$, the mapping

$$\text{“}\sigma - 1\text{”}\colon H^\nu(\mathscr{B}^\cdot_a) \to H^\nu_{\{a\}}(X_0, \mathscr{A}^\cdot)$$

resp. for $\nu \neq 0$

$$\text{“}\sigma - 1\text{”}\colon H^\nu(\mathscr{A}^\cdot_a) \to H^\nu_{\{a\}}(X_0, \mathscr{A}^\cdot)$$

obtained as above is denoted $\mathrm{Var}(\sigma)$ (see 3.9).

Because of 5.3a, only the case $\nu \neq 0$ will later be of interest.

5.5 Remark. The complex $\mathscr{A}^\cdot = \bar{i}^* \mathrm{R}\bar{j}_* F_{\bar{X}^*}$ is “uniquely determined” in the derived category of π_1-sheaves on X_0, as is the morphism

$$F_X \to \mathscr{H}^0(\mathscr{A}^\cdot).$$

Thus $\mathrm{Var}(\sigma)$ does not depend on the choice of the resolution $\mathscr{I}^\cdot$.

We shall not enumerate the obvious functorial properties of $\mathrm{Var}(\sigma)$. In particular, $\mathrm{Var}(\sigma)$ is of local nature and thus depends only on the type of singularity of f at a.

In what follows, we shall determine $\mathrm{Var}(\sigma)$ when a is an ordinary double point in X_0 and X_0 is of odd dimension at a. As the question is of local nature,

it suffices to handle the case of a family of quadrics

$$f\colon X \to \mathbb{C}$$

of odd dimension $2m+1$. Here X is the "relative quadric over $\mathbb{C}$"

$$\begin{array}{ccc} (z,\lambda) & X=\{(z,\lambda)=(z_1,\ldots,z_n,\lambda)\in\mathbb{C}^{n+1} \mid z_1^2+\ldots+z_n^2=z^2=\lambda\} & \\ \downarrow & \downarrow & \\ \lambda & \mathbb{C} & n=2(m+1). \end{array}$$

We consider again the projective quadric

$$Y \to \mathbb{C}$$

defined in homogeneous coordinates by

$$\sum_{i=1}^{m+1} z_i\, z_{i+m+1} = \lambda\, z_0^2.$$

We denote by $Y^{\mathrm{aff}} \subset Y$ the affine part of Y defined by $z_0 \neq 0$. The affine equation for Y^{aff} is

$$\sum_{i=1}^{m+1} z_i\, z_{i+m+1} = \lambda.$$

We denote by Y^{∞} the complement of Y^{aff} in Y,

$$Y^{\infty}\colon \sum_{i=1}^{m+1} z_{i+m+1}\, z_i = 0.$$

5.6 Lemma. *The mapping $f\colon X \to \mathbb{C}$ is smooth outside the point $0=(0,0)$. The special fiber*

$$X_0 = f^{-1}(0) = \{z\in\mathbb{C}^n \mid z^2=0\}$$

is a cone with vertex $z=0$.

There is an isomorphism (linear transformation) over $\mathbb{C}$,

$$\begin{array}{ccc} X & \xrightarrow{\sim} & Y^{\mathrm{aff}} \\ & \searrow \quad \swarrow & \\ & \mathbb{C}. & \end{array}$$

$Y \to \mathbb{C}$ is smooth at all points of Y^{∞}; Y^{∞} is smooth over $\mathbb{C}$.

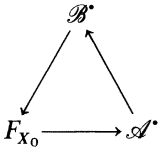
We consider again the extension of $f\colon X \to \mathbb{C}$ to $\bar{\mathbb{C}}$,

$$\begin{array}{ccccc} \bar{X}^* & \xhookrightarrow{j} & \bar{X} & \xhookleftarrow{i} & X_0 \\ \downarrow & & \downarrow & & \\ \mathbb{C} & \hookrightarrow & \bar{\mathbb{C}}. & & \end{array}$$

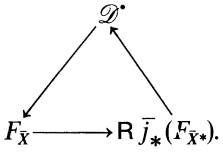
To determine $\mathrm{Var}(\sigma)$, we must study

$$\mathscr{A}^{\cdot} = \bar{i}^{*}(\mathrm{R}\,\bar{j}_{*}\,F_{\bar{X}^{*}})$$

and the complex $\mathscr{B}^{\cdot}$ defined by the admissible triangle



in the derived category of $\pi_1(\mathbb{C}^{*})$-sheaves on X_0. We define a complex $\mathscr{D}^{\cdot}$ by the admissible triangle



Then $\mathscr{D}^{\cdot}|X_0 = \mathscr{B}^{\cdot}$. Hence $\mathscr{D}^{\cdot}$ has cohomology sheaves concentrated on the point $0=(0,\infty)$.

5.7 Lemma. *The mapping*

$$(z,\lambda)\mapsto(e^{i\pi\lambda}z,\lambda)$$

establishes an isomorphism (not equivariant!) over $\mathbb{C}$,

$$X_1\times\mathbb{C}\xrightarrow{\cong}\bar{X}^{*},\qquad X_1=f^{-1}(1)$$

$$\mathbb{C}.$$

As $\bar{\mathbb{C}}$ is not locally compact, we define the functor $\bar{f}_{!}$ (direct image) with proper support as follows:

Let U be open in $\bar{\mathbb{C}}$. Let $\Phi_0(U)$ be the family of closed subsets A of $\bar{f}^{-1}(U)$ that lie properly over U. They have the following additional property if ∞ is contained in U: There is a number $r>0$ and a constant c with the property that

$$(z,\lambda)\in A,\qquad \mathrm{Im}(\lambda)>r\Rightarrow|z|\leqq c.$$

The functor $\bar{f}_{!}$ is now defined as follows:

$$\bar{f}_{!}\mathscr{G}(U)=\Gamma_{\Phi_0}(\mathscr{G},\bar{f}^{-1}(U)).$$

Here Γ_{Φ_0} is the functor of sections with support in the family Φ_0 (cf. [51], Chap. II, 3.1). We also set

$$\Phi=\{A\cap\bar{X}^{*},\ A\in\Phi_0(\bar{\mathbb{C}})\}.$$

The sets

$$\bar{H}_r=H_r\cup\{\infty\},\qquad H_r=\{z\in\mathbb{C};\ \mathrm{Im}\,z>r\}$$

form a neighborhood basis of ∞ in $\bar{\mathbb{C}}$. We list some formulas:

A) $$H^\nu(X_1, F) = H^\nu(\bar{X}^*, F) = H^\nu(\bar{f}^{-1}(H_r), F),$$
$$H_c^\nu(X_1, F) = H_\Phi^\nu(\bar{X}^*, F) = H^\nu_{\Phi\cap\bar{f}^{-1}(H_r)}(\bar{f}^{-1}(H_r), F).$$

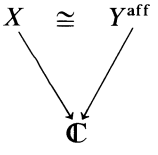
These formulas follow from Lemma 5.7 (and 5.10).

B) *Base change theorems*

$$\varinjlim_r H^\nu_{\Phi\cap\bar{f}^{-1}(H_r)}(\bar{f}^{-1}(H_r), F) = R^\nu \bar{f}_!(\mathsf{R}\,\bar{j}_*(F_{\bar{X}^*}))_\infty = H_c^\nu(X_0, \mathscr{A}^\bullet).$$

$$\varinjlim_r H^\nu(f^{-1}(H_r), F) = R^\nu \bar{f}_*(\mathsf{R}^\bullet \bar{j}_*(F_{\bar{X}^*}))_\infty = H^\nu(X_0, \mathscr{A}^\bullet).$$

The base change theorem for the complex $\mathsf{R}\,\bar{j}_*(F_{\bar{X}^*})$ and the functor $R^\nu \bar{f}_*$ comes from the corresponding theorems for $\mathscr{D}^\bullet$ (whose cohomology sheaves are concentrated on 0) and for $F_{\bar{X}}$. Use the smoothness of the projective quadric $Y \to \mathbb{C}$ at the points of Y^∞, the smoothness of $Y^\infty \to \mathbb{C}$, and the isomorphism



(Lemma 5.6 and Lemma 5.2).

C) *The natural mappings*

$$H^\nu_{\{0\}}(X_0, \mathscr{A}^\bullet) \to H_c^\nu(X_0, \mathscr{A}^\bullet)$$
$$H^\nu(X_0, \mathscr{A}^\bullet) \to H^\nu(\mathscr{A}_0^\bullet)$$

are isomorphisms.

Because: X_0 is a cone with vertex 0. The corresponding isomorphism theorems hold for F_X and $\mathscr{B}^\bullet$.

We sum up:

5.8 Proposition. *We have natural isomorphisms*

$$H_c^\nu(X_1, F) \xleftarrow{\cong} H_\Phi^\nu(\bar{X}^*, F) \cong H_c^\nu(X_0, \mathscr{A}^\bullet) \cong H^\nu_{\{0\}}(X_0, \mathscr{A}^\bullet)$$
$$H^\nu(X_1, F) \xleftarrow{\cong} H^\nu(\bar{X}^*, F) \cong H^\nu(X_0, \mathscr{A}^\bullet) \cong H^\nu(\mathscr{A}_a^\bullet).$$

In particular, $\mathscr{H}^0(\mathscr{A}^\bullet) = F_X$, and thus $\mathscr{H}^0(\mathscr{B}^\bullet) = 0$. *The variation* $\mathrm{Var}(\sigma)$ *can be interpreted in a natural way as a mapping*

$$\mathrm{Var}(\sigma)\colon\ H^\nu(X_1, F) \to H_c^\nu(X_1, F).$$

We want to determine $\mathrm{Var}(\sigma_0)$ for the generating element $\sigma_0\colon z \mapsto z+1$. The case of "universal" coefficients

$$F = \mathbb{Z}$$

will suffice. For this we need some equivariant isomorphisms.

5.9 Definition. Let M be the tangent bundle of the $(n-1)$-sphere $S \subset \mathbb{R}^n$,

$$M = \left\{(x, y) \in \mathbb{R}^{2n},\ x^2 = \sum_1^n x_\nu^2 = 1,\ x\,y = \sum_1^n x_\nu\, y_\nu = 0\right\}$$

$$M' = \{(x, y) \in M;\ y \neq 0\} \quad (M \text{ minus the zero section}).$$

For $r > 0$, we set

$$M(\geqq r) = \{(x, y) \in M;\ |y|^2 \geqq r\},$$

$$M(\leqq r) = \{(x, y) \in M;\ |y|^2 \leqq r\},$$

$$M(r) = \{(x, y) \in M:\ |y|^2 = r\},$$

$$M(1) \xrightarrow{\cong} \left\{z \in \mathbb{C}^n;\ z^2 = \sum_1^n z_\nu^2 = 0,\ |z|^2 = 2\right\}$$

$$(x, y) \mapsto z = x + i\,y.$$

5.10 Lemma.

a) $X_1 = f^{-1}(1) \cong M$

$$z = x + i\,y \mapsto (x/\sqrt{(1+|y|^2)},\ y).$$

b) $M' \cong M(1) \times \mathbb{R}^+$

$$(x, y) \mapsto ((x, y/|y|),\ |y|^2).$$

c) *The multiplicative group* $\{c \in \mathbb{C};\ |c| = 1\}$ *acts on* $M(1)$ *and on* M' *by*

$$T(c):\ M(1) \to M(1)$$

$$(x, y) \mapsto (x', y');\ x' + i\,y' = c(x + i\,y)$$

$$T(c):\ M' = M(1) \times \mathbb{R}^+ \to M'$$

$$(m, t) \mapsto (T(c)\,m,\ t).$$

We make a more careful study of $\bar{X} = \{(z, \lambda) \in \mathbb{C}^n \times \bar{\mathbb{C}};\ z^2 = e^{2\pi i \lambda}\}$ $(e^{2\pi i \infty} = 0)$ and the natural action of $\pi_1(\mathbb{C}^*)$,

$$\sigma_0:\ \bar{X} \to \bar{X}$$

$$(z, \lambda) \mapsto (z, \lambda + 1).$$

5.11 Definition. For $(z, \lambda) \in \bar{X}$, we set $\rho(z, \lambda) = \frac{1}{2}(|z|^2 - |e^{2\pi i \lambda}|)$. This function is continuous and nonnegative. If $\lambda \neq \infty$, $\phi = \mathrm{Re}(\lambda)$, and $z' = x' + i\,y' = e^{-\pi i \phi} z$, then also

$$\rho(z, \lambda) = y'^2.$$

5.12 Lemma. *We have the following equivariant homeomorphism over* $\bar{\mathbb{C}}$:

$$\{(z, \lambda) \in \bar{X};\ \rho(z, \lambda) \neq 0\} = \bar{X}' \longrightarrow M' \times \bar{\mathbb{C}}$$

$$\bar{X}' \to \bar{\mathbb{C}} \quad \text{and} \quad M' \times \bar{\mathbb{C}} \to \bar{\mathbb{C}}$$

$$(z,\lambda)\mapsto\begin{cases}\left(T(e^{i\pi\phi})\left[\dfrac{\mathrm{Re}(e^{-i\pi\phi}z)}{\sqrt{r+\rho(z,\lambda)}},\ \mathrm{Im}(e^{-i\pi\phi}z)\right],\lambda\right) & \text{for } \lambda\neq\infty\\ \left(\dfrac{\mathrm{Re}(z)}{|\mathrm{Im}(z)|},\ \mathrm{Im}\,z,\ \infty\right)\text{“}=\text{”}\left(\dfrac{\mathrm{Re}(z)}{|\mathrm{Im}\,z|},\ \mathrm{Im}\,z\right) & \text{for } \lambda=\infty.\end{cases}$$

$$\phi=\mathrm{Re}(\lambda);\qquad r=|e^{2\pi i\lambda}|.$$

Here $\pi_1(\mathbb{C}^*)$ acts trivially on M'.

We choose $0<\varepsilon<1$ and a monotonically increasing C^∞ function

$$h\colon \mathbb{R}\to\mathbb{R}$$

with

$$h(t)=\begin{cases}0 & \text{for } t\leqq\varepsilon/2\\ 1 & \text{for } t\geqq\varepsilon\end{cases}$$

$$h'(t)\neq 0\qquad \varepsilon/2<t<\varepsilon.$$

5.13 Lemma. *There is an equivariant C^∞-isomorphism over $\mathbb{C}$,*

$$\bar{X}^*\xrightarrow{\ \cong\ }M\times\mathbb{C}$$

$$(z,\lambda)\mapsto\left(T(e^{i\pi\phi h(\rho(z,\lambda))})\left[\frac{\mathrm{Re}(e^{-i\pi\phi}z)}{\sqrt{r+\rho(z,\lambda)}},\ \mathrm{Im}(e^{-\pi i\phi}z)\right],\lambda\right)$$

$$r=|e^{2\pi i\lambda}|,\qquad \phi=\mathrm{Re}(\lambda).$$

Here we set $T(e^{i\pi\phi(h(\rho(z,\lambda)))})=\mathrm{Id}$ for $\rho(z,\lambda)=0$.

The generating element σ_0 of $\pi_1(\mathbb{C}^*)$ acts on M by

$$T\colon M\to M$$

$$T(x,y)=T(e^{i\pi(1+h(|y|^2))})(x,y).$$

In particular,

$$T(x,y)=\begin{cases}(-x,-y) & |y|^2\leqq\varepsilon/2\\ (x,y) & |y|^2\geqq\varepsilon.\end{cases}$$

T rotates the vectors x, y in the (x,y)-plane through the angle $-\pi(1+h(|y|^2))$.

We now consider the following topological situation. Let A be a closed subspace of a topological space X. Let G, a group generated by a single element T, act on X. Assume that T induces the identity mapping on A. Further, let $\mathcal{F}^\cdot$ be an equivariant complex of G-sheaves on X that is bounded below, and assume that G acts trivially on $\mathcal{F}^\cdot|A$. Let Φ be the family of those subsets of $U=X-A$ that are closed in all of X. For simplicity, we assume the following restrictive hypotheses: X is paracompact, and T actually acts trivially on a neighborhood of A and on the restriction of $\mathcal{F}^\cdot$ to this neighborhood. We choose an equivariant soft resolution $\mathcal{M}^\cdot$ of $\mathcal{F}^\cdot$ such that G acts trivially on the restriction of $\mathcal{M}^\cdot$ to a suitable neighborhood of A. The canonical flabby

Godement-resolution has this property.

$$U \xhookrightarrow{j} X \xhookleftarrow{i} A.$$

There is an equivariant short exact sequence

$$0 \to \Gamma_\Phi(X, \mathscr{M}^\cdot) \to \Gamma(X, \mathscr{M}^\cdot) \to \Gamma(A, \mathscr{M}^\cdot) \to 0.$$

Since by assumption T acts trivially on $\Gamma(A, \mathscr{M}^\cdot)$, $T-1$ induces a mapping of complexes

$$\text{“}T-1\text{”}\colon\ \Gamma(X, \mathscr{M}^\cdot) \to \Gamma_\Phi(X, \mathscr{M}^\cdot).$$

Passing to cohomology, we get a homomorphism

$$H^\cdot(X, \mathscr{F}^\cdot) \to H^\cdot_\Phi(X, \mathscr{F}^\cdot).$$

Using the canonical flabby Godement resolution, one can show that the mapping thus constructed does not depend on the choice of the resolution $\mathscr{M}^\cdot$!

For $H^\cdot_\Phi(X, \mathscr{F}^\cdot)$ we also write $H^\cdot(X, A, \mathscr{F}^\cdot)$.

5.14 Definition. The mapping constructed under the stated hypotheses will be denoted

$$\mathrm{Var}_{\mathrm{top}}(T)\colon\ H^\cdot(X, \mathscr{F}^\cdot) \to H^\cdot(X, A, \mathscr{F}^\cdot) = H^\cdot_\Phi(X, \mathscr{F}^\cdot).$$

5.15 Remark. a) There is a long exact sequence of cohomology groups

$$\ldots \to H^\nu(X, A, -) \to H^\nu(X, -) \to H^\nu(A, -) \to \ldots.$$

b) If X is a C^∞-manifold, T a C^∞-mapping, $\mathscr{F}^\cdot = \mathbb{C}_X$, then for the construction of $\mathrm{Var}_{\mathrm{top}}(T)$ we can use the DeRham complex $\Omega^\cdot_X$:

If the closed differential form ω represents a cohomology class $\alpha \in H^\nu(X, \mathbb{C})$, then the differential form

$$T^*(\omega) - \omega,$$

whose support lies in $X - A$, represents the cohomology class

$$\mathrm{Var}_{\mathrm{top}}(T)\,(\alpha) \in H^\nu_\Phi(X, \mathbb{C}).$$

Back to the family of quadrics

$$f\colon\ X \to \mathbb{C}.$$

By Proposition 5.8 we have natural isomorphisms

$$H^\nu_c(X_1, F) \cong H^\nu_{\{0\}}(X_0, \mathscr{A}^\cdot)$$

$$H^\nu(X_1, F) \cong H^\nu(X_0, \mathscr{A}^\cdot) = H^\nu(\mathscr{A}^\cdot_0), \qquad \nu \neq 0.$$

By Lemma 5.10 a) there is an isomorphism

$$X_1 = f^{-1}(1) \cong M \quad \text{(tangent bundle of the } (n-1)\text{-sphere)}.$$

Further, we obviously have that the natural mapping

$$H^\nu(M, M(\geqq 1), F) \to H^\nu_c(M, F)$$

is an isomorphism.

5.16 Theorem. *Let* $\sigma_0: \lambda \mapsto \lambda + 1$ *be the generating element of* $\pi_1(\mathbb{C}^*)$, *and*

$$T = T(e^{i\pi(1+h(|y|^2))}): \ M \to M$$

the automorphism from Lemma 5.13. *Then the diagram*

$$\begin{array}{ccccc} H^\nu(M, F) & \cong & H^\nu(X_1, F) & \cong & H^\nu(\mathscr{A}_0) \\ \Big\downarrow{\scriptstyle \nu \neq 0} & & & & \Big\downarrow{\scriptstyle \mathrm{Var}(\sigma_0)} \\ H^\nu(M, M(\geqq 1), F) & \cong & H_c^\nu(M, F) & \cong & H_c^\nu(X_1, F) \cong H_{\{0\}}^\nu(X_0, \mathscr{A}^\bullet) \end{array} \qquad \mathrm{Var}_{\mathrm{top}}(T)$$

is commutative, and thus $\mathrm{Var}(\sigma_0)$ *is identified as the variation of* T *in the topological sense.*

We shall prove this theorem in what follows. We consider the homeomorphism from Lemma 5.12,

$$\begin{array}{ccc} \bar{X} \supseteq \bar{X}' = \{u \in \bar{X};\ \rho(u) \neq 0\} & \cong & M' \times \bar{\mathbb{C}} \\ \cup\mid & & \cup\mid \\ \bar{X}(\geqq r) = \{u \in \bar{X};\ \rho(u) \geqq r\} & \cong & M(\geqq r) \times \bar{\mathbb{C}} \end{array} \qquad (r > 0)$$

We set

$$\bar{X}(<r) = \{u \in \bar{X};\ \rho(u) < r\},$$

$$\bar{X}^*\begin{pmatrix} \geqq r \\ < r \end{pmatrix} = \bar{X}^* \cap \bar{X}\begin{pmatrix} \geqq r \\ < r \end{pmatrix},$$

$$X_0(\geqq r) = X_0 \cap \bar{X}(\geqq r).$$

Now we choose ε, $0 < \varepsilon < 1$, as in Lemma 5.12, so $T \mid M(\geqq \varepsilon) = \mathrm{Id}$. In addition, we choose a number r such that

$$\varepsilon < r < 1.$$

We consider the following equivalence relation $\mathscr{R}$ on $\bar{X}$: two elements $u, v \in \bar{X}$ are called equivalent if either they are equal or they both lie in $\bar{X}(\geqq r)$ and their images in $M(\geqq r) \times \bar{\mathbb{C}}$ have the same projection to $M(\geqq r)$. We set

$$\hat{\bar{X}} = \bar{X}/\mathscr{R}, \qquad \hat{\bar{X}}^* = \bar{X}^*/\mathscr{R} \subseteq \hat{\bar{X}}.$$

On $\hat{\bar{X}}$ we define a topology, one that turns out to be weaker than the quotient topology, by giving a basis: A subset of $\hat{\bar{X}}$ belongs to this basis when its inverse image V in $\bar{X}$ either lies in $\bar{X}(<r)$ and is open there or when V lies in $\bar{X}' = M' \times \bar{\mathbb{C}}$ and is of the form $U \times \bar{\mathbb{C}}$ with U open in M'. The topology correspondingly defined on $\bar{X}^*$ is the induced topology. The two spaces $\hat{\bar{X}}$ and $\hat{\bar{X}}^*$ are metrizable, and hence paracompact; for by construction $\hat{\bar{X}}$ is a Hausdorff space, has a countable basis for its topology and is completely regular, i.e. for every point $a \in X$ and every neighborhood U of a there is a continuous function $\phi: \hat{\bar{X}} \to [0, 1]$ with $\phi(a) = 0$ and $\phi(x) = 1$ for x not in U [Tichonov-Uryson theorem; P. Uryson Math. Ann., 94 (1925)]. The image $\hat{\bar{X}}(\geqq r)$ of $\bar{X}(\geqq r)$ is closed in $\hat{\bar{X}}$, and the image $\hat{\bar{X}}^*(\geqq r)$ of $\hat{\bar{X}}^*(\geqq r)$ is closed in $\hat{\bar{X}}^*$.

By Lemma 5.13, we have an equivariant continuous mapping

$$\bar{X}^* \xrightarrow{\cong} M \times \mathbb{C} \to M \cong X_1.$$

This projection is an inverse mapping for the (not equivariant) embedding

$$M \cong X_1 = \bar{f}^{-1}(0) \subset \bar{X}^*.$$

The homeomorphism

$$\sigma\colon \bar{X} \to \bar{X}, \qquad \sigma \in \pi_1(\mathbb{C}^*),$$

induces homeomorphisms

$$\hat{\bar{X}} \to \hat{\bar{X}}, \qquad \hat{\bar{X}}^* \to \hat{\bar{X}}^*.$$

The diagram

$$M \leftarrow \bar{X}^* \hookrightarrow \bar{X} \hookleftarrow X_0$$

induces a commutative diagram of continuous equivariant mappings,

5.17.

$$\begin{array}{ccccccc}
M & \longleftarrow & \bar{X}^* & \xhookrightarrow{\bar{j}} & \bar{X} & \xhookleftarrow{\bar{i}} & X_0 \\
\| & & \downarrow{\scriptstyle \tilde{q}} & & \downarrow{\scriptstyle q} & & \| \\
M & \longleftarrow & \hat{\bar{X}}^* & \xrightarrow{\bar{j}_1} & \hat{\bar{X}} & \xhookleftarrow{\bar{i}_1} & X_0 \\
\uparrow & & \uparrow & & \uparrow & & \uparrow \\
M(\geqq 1) & \cong & \hat{\bar{X}}^*(\geqq 1) & \cong & \hat{\bar{X}}(\geqq 1) & \cong & X_0(\geqq 1).
\end{array}$$

The mapping $\bar{i}_1$, like $\bar{i}$, is a closed embedding; but $\bar{j}_1$ is not an open embedding, as $\bar{j}$ is. On M, the element $\sigma_0 \in \pi_1(\mathbb{C}^*)$ acts by the mapping

$$T\colon M \to M$$

defined in Lemma 5.13, and $\sigma_0\colon \hat{\bar{X}} \to \hat{\bar{X}}$ is the identity in a neighborhood $\hat{\bar{X}}(\geqq r)$ of $\hat{\bar{X}}(\geqq 1)$.

5.18 Lemma. *The constant sheaves on $\bar{X}$ resp. $\bar{X}^*$ for an abelian group F satisfy*

$$R^\nu q_* F_{\bar{X}} = \begin{cases} 0 & \text{for } \nu > 0 \\ F_{\bar{X}} & \text{for } \nu = 0; \end{cases} \qquad R^\nu \tilde{q}_* F_{\bar{X}} = \begin{cases} 0 & \text{for } \nu > 0 \\ F_{\bar{X}^*} & \text{for } \nu = 0. \end{cases}$$

Proof. We treat the case of the sheaf $F_{\bar{X}}$ on $\bar{X}$. The proof for the case $F_{\bar{X}^*}$ is similar. It suffices to show that for every point $v \in \hat{\bar{X}}$ there exists a neighborhood basis of neighborhoods U with

$$\varinjlim_U H^\nu(q^{-1}(U), F_{\bar{X}}) = \begin{cases} 0 & \text{for } \nu > 0 \\ F & \text{for } \nu = 0. \end{cases}$$

The case

$$v = q(u), \qquad u \in \bar{X}(< r)$$

is trivial, as q maps the space $\bar{X}(<r)$ topologically onto an open subset of $\hat{\bar{X}}$.

In the other case

$$v=q(u), \quad u=(a,\lambda)\in\bar{X}(\geqq r)\text{“}=\text{”}M(\geqq r)\times\bar{\mathbb{C}},$$

there exists a basis of neighborhoods U of the following form:

$$q^{-1}(U)=V\times\bar{\mathbb{C}}\subset\bar{X}'\text{“}=\text{”}M'\times\bar{\mathbb{C}}.$$

Here V runs through a neighborhood basis of contractible neighborhoods of a in M'. One sees easily that

$$H^v(V\times\bar{\mathbb{C}}, F)=\begin{cases}0 & \text{for } v>0\\ F & \text{for } v=0.\end{cases} \quad \square$$

Using a flabby equivariant resolution of $F_{\bar{X}^*}$ we construct the equivariant complex

$$\mathscr{A}^\bullet=\bar{i}^*\,\bar{j}_*(\mathscr{J}^\bullet)=\bar{i}^*\,\mathrm{R}\,\bar{j}_*(F_{\bar{X}}) \quad \text{on } X_0.$$

We can carry out the same construction with the mappings $\bar{i}_1$, $\bar{j}_1$: we choose a flabby equivariant resolution $\mathscr{J}_1^\bullet$ of $F_{\hat{\bar{X}}^*}$ on $\hat{\bar{X}}^*$. We shall actually assume that $\pi_1(\mathbb{C}^*)$ acts *trivially* on $\mathscr{J}_1^\bullet|\hat{\bar{X}}(\geqq r')$, $1>r'>r$. This is possible, since $\pi_1(\mathbb{C}^*)$ acts trivially on a neighborhood $\hat{\bar{X}}(\geqq r)$ of $\hat{\bar{X}}(\geqq r')$. We can for instance use the canonical flabby Godement resolution.

$$\mathscr{A}_1^\bullet=\bar{i}_1^*\,\bar{j}_{1*}\,\mathscr{J}_1^\bullet=i_1^*\,\mathrm{R}\,\bar{j}_{1*}\,F_{\hat{\bar{X}}^*}.$$

The cohomology sheaves of $\mathrm{R}\,\bar{j}_*(F_{\bar{X}^*})$ are either constant or concentrated on the point $0\in X_0$ (5.3, 5.3a, 5.8).

Thus it follows from Lemma 5.18 that

$$\mathrm{R}\,\tilde{q}_*(F_{\bar{X}^*})\cong F_{\hat{\bar{X}}^*},$$

$$\mathrm{R}\,q_*(\mathrm{R}\,\bar{j}_*\,F_{\bar{X}^*})\cong\mathrm{R}\,\bar{j}_{1*}(F_{\bar{X}^*})$$

in the derived category of $\pi_1(\mathbb{C}^*)$-sheaves; the complexes $\mathscr{A}^\bullet$ and $\mathscr{A}_1^\bullet$ are isomorphic in the derived category of $\pi_1(\mathbb{C}^*)$-sheaves on X_0. Thus $\mathrm{Var}(\sigma_0)$ can be constructed using $\mathscr{A}_1^\bullet$ in place of $\mathscr{A}^\bullet$ (see Definition 5.4). By the assumption on $\mathscr{J}_1^\bullet$ we have $\pi_1(\mathbb{C}^*)$ acting trivially on $\mathscr{A}_1^\bullet|X_0(\geqq r')$ and indeed trivially on the restriction to a neighborhood $X_0(\geqq r')$ of $X_0(\geqq 1)$. As X_0 is a cone, we further have (cf. p. 216, C)

$$H^v(X_0,\mathscr{A}_1^\bullet)=H^v(\mathscr{A}_{1(0)}^\bullet),$$

$$H^v_{\{0\}}(X_0,\mathscr{A}_1^\bullet)=H^v(X_0, X_0(\geqq 1),\mathscr{A}_1^\bullet)=H_c^v(X_0,\mathscr{A}_1).$$

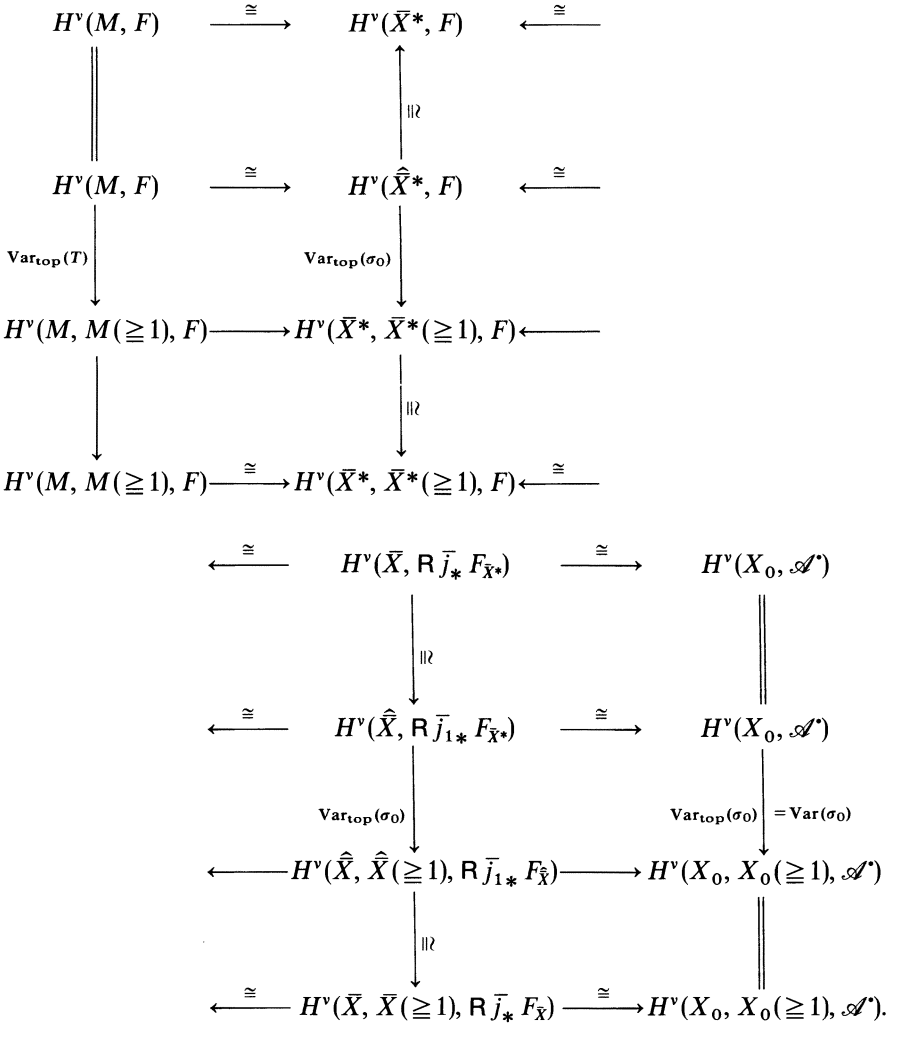
With this identification, $\mathrm{Var}(\sigma_0)$ is a mapping

$$\mathrm{Var}(\sigma_0):\ H^v(X_0,\mathscr{A}_1^\bullet)\to H^v(X_0, X_0(\geqq 1),\mathscr{A}_1^\bullet), \qquad v\neq 0.$$

It is easy to think through the fact that

5.19. $\mathrm{Var}(\sigma_0)=\mathrm{Var}_{\mathrm{top}}(\sigma_0):\ H^v(X_0,\mathscr{A}_1^\bullet)\to H^v(X_0, X_0(\geqq 1),\mathscr{A}_1^\bullet).$

Diagram (5.17) yields a commutative diagram



The isomorphisms of the top and bottom lines come from Proposition 5.8. Here we use

$$H^\nu(M, M(\geqq 1), F) = H_c^\nu(M, F) = H_c^\nu(X_1, F)$$
$$H^\nu(\bar{X}^*, \bar{X}^*(\geqq 1), F) = H_\Phi^\nu(\bar{X}^*, F).$$

and further similar isomorphisms.

The vertical isomorphisms come from Lemma 5.18 because of

$$\bar{X}(\geqq 1) = \hat{\bar{X}}(\geqq 1) \times \mathbb{C}, \ldots$$

and the long exact sequence for a pair of spaces. The diagram yields then a proof of Theorem 5.16. □

We now want to determine

$$\mathrm{Var}_{\mathrm{top}}(T)\colon\ H^{\nu}(M,\mathbb{Z})\to H^{\nu}(M,M(\geqq 1),\mathbb{Z})=H_c^{\nu}(M,\mathbb{Z}),\qquad \nu\neq 0$$

for the "universal" coefficient system $\mathbb{Z}$.

We consider the projection

$$p\colon\ M\to S$$

of the tangent bundle M onto the $(n-1)$-sphere. It induces isomorphisms

$$H^{\nu}(X,\mathbb{Z})\xrightarrow{\cong} H^{\nu}(M,\mathbb{Z}),$$

and hence

$$H^{\nu}(M,\mathbb{Z})\cong\begin{cases}0 & \text{for } \nu\neq 0,\ n-1\\ \mathbb{Z} & \text{for } \nu=n-1.\end{cases}$$

It thus suffices to study the case $\nu=n-1$. The natural mapping

$$H^{\nu}(M,\mathbb{Z})\to H^{\nu}(M,\mathbb{C})$$

is injective. Since T is the identity in a neighborhood of $M(\geqq 1)$, we can use the DeRham complex Ω_M for the computation of $\mathrm{Var}_{\mathrm{top}}(T)$ (see Definition 5.14, Remark 5.15 b).

We look for some orientation on S and M. One is defined on M by the isomorphism $M\cong f^{-1}(1)=X_1$ and the complex orientation $(i^{\dim X_1}\,dz_1\wedge d\bar{z}_1\wedge\ldots)$; it differs from the natural orientation on the sphere resp. the tangent bundle M by $(-1)^{\dim M(\dim M-1)/2}$. Let ω be a differential form of degree $n-1$ on S with

$$\int_S \omega=1.$$

It represents a generator of $H^{n-1}(S,\mathbb{Z})$. Hence $T^*(\omega)$ represents one of the two generating elements δ of $H^{n-1}(M,\mathbb{Z})$, and $\mathrm{Var}_{\mathrm{top}}(T)\,(\delta)$ is represented by $T^*(p^*(\omega))-p^*(\omega)$.

We want to compute

$$\int_M p^*(\omega)(T^*(p^*(\omega))-p^*(\omega)).$$

By Fubini's theorem,

$$\int_M T^*(p^*(\omega))\,p^*(\omega)=\int_S \varepsilon(x)\,\omega,$$

$$\varepsilon(x)=\pm\int_{M_x=p^{-1}(x)}(p\circ T)^*(\omega).$$

We consider the restriction of $p\circ T$ to M_x,

$$p\circ T\,|\,M_x=q\colon\ M_x\to S.$$

q is a C^{∞}-isomorphism outside $q^{-1}(\pm x)$. Consequently,

$$\int_M q^*(\omega)=\pm\int_S\omega\quad\left(=-\int_S\omega=-1\right).$$

Thus

$$\varepsilon(x) = \pm 1 \quad \text{(more careful computation shows } \varepsilon(x) = -1\text{)}.$$

By continuity, the sign does not depend on x. We get:

$$\int_S \varepsilon(x)\,\omega = \pm \int_S \omega = \pm 1,$$

$$\int_M T^*(p^*(\omega)) - p^*(\omega) = \pm 1.$$

5.20 Theorem. *Let $\langle\ ,\ \rangle$ be the duality pairing between $H^{n-1}(M, \mathbb{Z})$ and $H_c^{n-1}(M, \mathbb{Z})$ for some orientation. Let δ be a generator of*

$$H^{n-1}(M, \mathbb{Z}) \cong \mathbb{Z}.$$

Let δ' be the generator of $H_c^{n-1}(M, \mathbb{Z})$ with $\langle \delta, \delta' \rangle = 1$. Then

$$\mathrm{Var}_{\mathrm{top}}(T)\text{“}=\text{”}\mathrm{Var}_{\mathrm{top}}\colon\ H^{n-1}(M, \mathbb{Z}) \to H_c^{n-1}(M, \mathbb{Z})$$

satisfies

$$\mathrm{Var}_{\mathrm{top}}(x) = \pm \langle x, \delta' \rangle\, \delta'.$$

(Computing the orientation yields

$$\mathrm{Var}_{\mathrm{top}}(x) = (-1)^{m+1} \langle x, \delta' \rangle, \qquad n-1 = 2m+1.)$$

We now want to compare the algebraic variation with that defined analytically.

If X is a finitely generated scheme over $\mathbb{C}$, resp. f a morphism of such schemes, we denote by X_{an} the associated analytic space, resp. f_{an} the associated mapping of analytic spaces. There is an exact functor

$$\mathscr{F} \mapsto \mathscr{F}_{\mathrm{an}},$$

compatible with direct limits, from the category of etale torsion sheaves on X to the category of sheaves on the topological space underlying X_{an}. We identify the set $X(\mathbb{C})$ of geometric points with values in $\mathbb{C}$ (over $\mathbb{C}$) with the set of points of X_{an}. For a geometric point $s \in X(\mathbb{C})$, there is a natural isomorphism

$$\mathscr{F}_s \cong (\mathscr{F}_{\mathrm{an}})_s.$$

We do not bother to list the further compatibility properties of this functor (cf. Chap. I, §11).

Let $S = \mathrm{Spec}\,\mathbb{C}[t]$ be the affine line, $0 \hookrightarrow S$ the origin. Then $S_{\mathrm{an}} = \mathbb{C}$ is the complex plane. We consider again the logarithmically ramified covering space

$$\begin{array}{ccc} \bar{\mathbb{C}} & \longrightarrow & \mathbb{C} \\ \downarrow & & \downarrow \\ \mathbb{C} & \xrightarrow{e^{2\pi i \lambda}} & \mathbb{C}^*. \end{array}$$

The polynomial ring $\mathbb{C}[t]$ is contained in $\mathcal{O}_{\mathbb{C},0}$, the ring of germs of holomorphic functions at the origin of $S_{\text{an}}=\mathbb{C}$. We have a natural injection of $\mathcal{O}_{\mathbb{C},0}$ into the field $\mathcal{M}$ of meromorphic functions in an arbitrary small connected pointed neighborhood $H_r=\{\lambda\in\mathbb{C},\ \operatorname{Im}\lambda>r\}\subset\bar{\mathbb{C}}$ of ∞ in $\bar{\mathbb{C}}$.

$$\mathbb{C}[t]\subset\mathcal{O}_{\mathbb{C},0}\hookrightarrow\mathcal{M}$$
$$f(t)\mapsto f(e^{2\pi i\lambda}).$$

Let K be the field of those functions in $\mathcal{M}$ that are algebraic over the quotient field of $\mathbb{C}[t]$. We consider finitely generated etale $\mathbb{C}[t]$-algebras A that are contained in $\mathcal{O}_{\mathbb{C},0}$.

The residue field mapping $\mathcal{O}_{\mathbb{C},0}\to\mathbb{C}$ defines a geometric point with values in $\mathbb{C}$, and the spectrum $U=\operatorname{Spec}A$,

$$\begin{array}{ccc}\operatorname{Spec}A=U & & \\ & \searrow & \downarrow \\ 0 & \hookrightarrow & S,\end{array}$$

is an etale neighborhood of $0\to S$. The ring

$$\varinjlim_{A} A=\bigcup A=R\subset\mathcal{O}_{\mathbb{C},0}$$

is the (strict) Henselization of $\mathcal{O}_{S,0}$ at 0. For given A and a natural number n, we consider further the A-subalgebra B of K given by

$$B=A[\sqrt[n]{(t)}]=A[e^{2\pi i\lambda/n}]\subset K.$$

The mapping $\operatorname{Spec}(B)\to\operatorname{Spec}(A)$ is ramified only over 0. Over this point of U there lies exactly one point of $\operatorname{Spec}(B)$. Let

$$U_n^*=\operatorname{Spec}B-\{0\}=\operatorname{Spec}B[1/t].$$

Then

$$\varinjlim B[1/t]=\bigcup B[1/t]=K.$$

The group $\pi_1(\mathbb{C}^*)$ of cover transformations of $\mathbb{C}\xrightarrow{e^{2\pi i\lambda}}\mathbb{C}^*$ acts on $\mathcal{M}$:

$$f(\lambda)\mapsto f(\lambda+m).$$

It acts trivially on the subring $\mathcal{O}_{\mathbb{C},0}$. This action carries K to itself:

The element

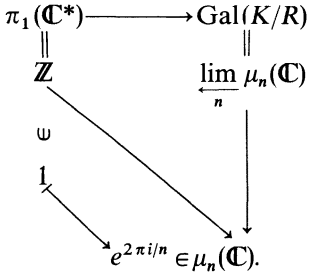
$$\sigma\colon\ \lambda\mapsto\lambda+m$$

in $\pi_1(\mathbb{C}^*)$ has the effect

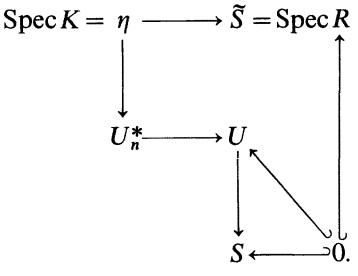
$$\sigma(f)=f\quad\text{for } f\in R\subset\mathcal{O}_{\mathbb{C},0},$$
$$\sigma(\sqrt[n]{t})=\sigma(e^{2\pi i\lambda/n})=(e^{2\pi i/n})^m\sqrt[n]{t}.$$

Thus we have a natural injection

5.21



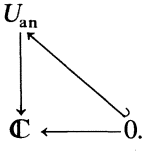
Here $\mu_n(\mathbb{C})$ is the group of n-th roots of unity in $\mathbb{C}$. We have a natural diagram



We consider the corresponding analytic situation:

$$H_r = \{\lambda \in \mathbb{C} \mid \mathrm{Im}\, \lambda > r\}$$

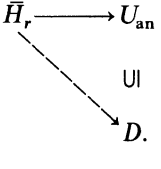
$$\bar{H}_r = H_r \cup \{\infty\}$$



Since $U_{\mathrm{an}} \to \mathbb{C}$ is unramified, there is a number $r > 0$ and an open neighborhood D of 0 in U_{an} that under $U_{\mathrm{an}} \to \mathbb{C}$ is mapped isomorphically onto the open circle of radius $e^{-2\pi r}$ around 0,

$$D \text{“} = \text{”} \{t \in \mathbb{C};\ |t| < e^{-2\pi r}\}.$$

The mapping $\bar{\mathbb{C}} \to \mathbb{C}$ induces a mapping



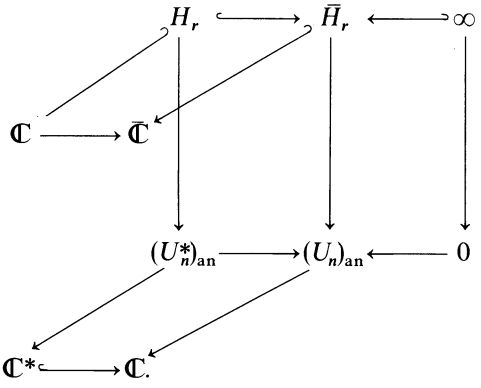
The embedding of the algebra of U_n induces then a holomorphic mapping

$$H_r \to (U_n^*)_{\mathrm{an}}.$$

The image D' is a finite unramified cover of $D-\{0\}$ of degree n,

$$\begin{array}{ccc} H_r/G_n = D' & \longrightarrow & D-\{0\} = H_r/G,\ G_n \subseteq G = \pi_1(\mathbb{C}^*) \\ \cap\!| & & \cap\!| \\ (U_n^*)_{\mathrm{an}} & \longrightarrow & (U_n)_{\mathrm{an}}. \end{array}$$

We have the diagram



Now we consider the affine algebraic quadric $g\colon Y \to S$ over S,

$$Y = \mathrm{Spec}\left(\mathbb{C}[t, z_1, \ldots, z_n] \Big/ \left(\sum_{\nu=1}^{n} z_\nu^2 - t\right)\right).$$

Then

$$f = g_{\mathrm{an}}\colon\ X = Y_{\mathrm{an}} \to \mathbb{C}$$

is the analytic quadric treated before. We set

$$\begin{aligned} \bar{X}_{\bar{H}_r} &= X \times_{\mathbb{C}} \bar{H}_r \subset \bar{X}, \\ \bar{X}^*_{\bar{H}_r} &= X \times_{\mathbb{C}} H_r \subset \bar{X}^*, \\ \bar{X}^*_{U_n} &= X \times_{\mathbb{C}} (U_n^*)_{\mathrm{an}}, \\ X_U &= X \times_{\mathbb{C}} U_{\mathrm{an}} \end{aligned}$$

and get a commutative diagram

5.22.

$$\begin{array}{ccccc} \bar{X}^* & \xhookrightarrow{\bar{j}} & X & \xhookleftarrow{\bar{i}} & X_0 \\ \cup\!| & & \cup\!| & & \| \\ \bar{X}^*_{\bar{H}_r} & \longrightarrow & \bar{X}_{\bar{H}_r} & \longleftarrow & X_0 \\ \downarrow & & \downarrow & & \| \\ \bar{X}^*_{U_n} & \xrightarrow{\bar{j}_{U,n}} & X_U & \xhookleftarrow{\bar{i}_U} & X_0. \end{array}$$

In the algebraic case, we set

$$\tilde{Y} = Y \times_S \tilde{S}, \qquad Y_\eta = Y \times_S \eta; \qquad Y_0 = g^{-1}(0),$$
$$Y^*_{U_n} = Y \times_S U^*_n, \qquad Y_U = Y \times_S U$$

and get a commutative diagram

5.23.

$$\begin{array}{ccccc}
Y_\eta & \xrightarrow{j} & \tilde{Y} & \xleftarrow{i} & Y_0 \\
\downarrow & & \downarrow & & \| \\
Y^*_{U_n} & \xrightarrow{j_{U,n}} & Y_U & \xleftarrow{i_U} & Y_0 \\
 & & \downarrow & & \downarrow \\
 & & Y & \longleftarrow & Y_0 .
\end{array}$$

Passing to the analytic version of the middle line here gives the lower line of the analytic diagram (5.22),

$$X^*_{U_n} \to X_U \leftarrow X_0 .$$

Let F be a finite abelian group, $\mathscr{I}^\cdot$ an injective resolution of the constant sheaf F_Y on Y, and $\mathscr{J}^\cdot$ an injective resolution of $\mathscr{I}^\cdot_{\mathrm{an}}$ on $X = Y_{\mathrm{an}}$,

$$\mathscr{I}^\cdot_{\mathrm{an}} \to \mathscr{J}^\cdot .$$

Then $\mathscr{J}^\cdot$ is also an injective resolution of $F_X = (F_Y)_{\mathrm{an}}$. The inverse image $\mathscr{I}^\cdot_\eta$ of $\mathscr{I}^\cdot$ on Y_η is then in a natural way an equivariant complex. It is an injective resolution of F_{Y_η}. The corresponding statement holds for the inverse image $\mathscr{I}^\cdot_{U,n}$ of $\mathscr{I}^\cdot$ on $Y^*_{U_n}$.

In the analytic case, the inverse image $\bar{\mathscr{J}}^\cdot$ of $\mathscr{J}^\cdot$ on $\bar{X}^*$ is in a natural way an equivariant injective resolution of $F_{\bar{X}^*}$ on $\bar{X}^*$. The corresponding properties hold for the inverse image $\mathscr{J}^\cdot_{U,n}$ of $\mathscr{J}^\cdot$ on $\bar{X}^*_{U_n}$.

The quasi-isomorphism

$$\mathscr{I}^\cdot_{\mathrm{an}} \to \mathscr{J}^\cdot$$

gives us a natural equivariant quasi-isomorphism

$$(\mathscr{I}^\cdot_{U,n})_{\mathrm{an}} \to \mathscr{J}^\cdot_{U,n}$$

on $\bar{X}^*_{U,n} = (Y_{U,n})_{\mathrm{an}}$.

We set

$$\mathscr{A}^\cdot = i^* j_* (\mathscr{I}^\cdot_\eta),$$
$$\mathscr{A}^\cdot_{U,n} = i^*_U j_{U,n*} (\mathscr{I}^\cdot_{U,n})$$

and get a commutative diagram

$$\begin{array}{ccc}
\mathscr{A}^\cdot_{U,n} & \longrightarrow & \mathscr{A}^\cdot \\
\uparrow & \nearrow & \\
F_{Y_0} . & &
\end{array}$$

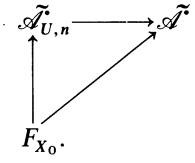
Correspondingly, in the analytic case we define

$$\tilde{\mathscr{A}}^{\cdot} = \bar{i}^{*}\, \bar{j}_{*}(\bar{\mathscr{I}}^{\cdot}),$$
$$\tilde{\mathscr{A}}^{\cdot}_{U,n} = \bar{i}^{*}\, \bar{j}_{U,n\,*}(\bar{\mathscr{I}}^{\cdot}_{U,n}).$$

Since $\tilde{\mathscr{A}}^{\cdot}$ can also be defined using the restriction $\bar{\mathscr{I}}^{\cdot} | \bar{X}^{*}_{\bar{H}_r}$ and

$$\bar{X}^{*}_{\bar{H}_r} \hookrightarrow \bar{X}_{\bar{H}_r} \hookleftarrow X_0$$

we get a natural commutative diagram



From the quasi-isomorphism

$$(\mathscr{I}^{\cdot}_{U,n})_{\mathrm{an}} \to \mathscr{J}^{\cdot}_{U,n}$$

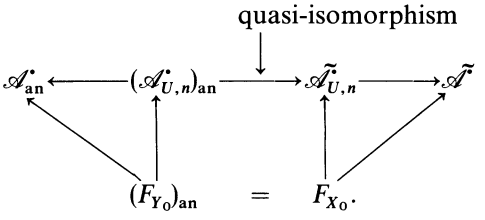
we get an equivariant comparison mapping

$$(\mathscr{A}^{\cdot}_{U,n})_{\mathrm{an}} \to \tilde{\mathscr{A}}^{\cdot}_{U,n}.$$

It is a quasi-isomorphism. Apply the "non-proper comparison theorems" for etale and analytic cohomology (I, 11.6) to $\mathrm{R}\, j_{U,n\,*}$.

Summing up, we get an equivariant diagram on X_0,

5.24.

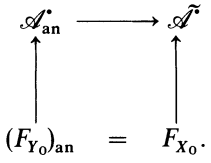


We assert:

A) $\varinjlim_{U,n} \mathscr{A}^{\cdot}_{U,n} = \mathscr{A}^{\cdot}$.

B) The mapping of complexes $\varinjlim_{U,n} \tilde{\mathscr{A}}^{\cdot}_{U,n} \to \tilde{\mathscr{A}}^{\cdot}$ is a quasi-isomorphism. These two assertions, which we shall prove immediately, yield the desired comparison theorem for the variation.

5.25 Proposition. *There is a natural equivariant quasi-isomorphism*



As the cohomology sheaves of $\mathscr{A}^{\cdot}$ are constructible, this yields a commutative diagram

$$\begin{array}{ccc} H^{\nu}(\mathscr{A}_0^{\cdot}) & \xrightarrow{\cong} & H^{\nu}(\tilde{\mathscr{A}}_0^{\cdot}) \\ \Big\downarrow \text{algebraically defined} \;\Big|\; \text{Var}(\sigma) & & \text{Var}(\sigma) \;\Big|\; \text{analytically defined} \Big\downarrow \\ H_c^{\nu}(Y_0, \mathscr{A}^{\cdot}) = H_{\{0\}}^{\nu}(Y_0, \mathscr{A}^{\cdot}) & \xrightarrow{\cong} & H_{\{0\}}^{\nu}(X_0, \tilde{\mathscr{A}}^{\cdot}) = H_c^{\nu}(X_0, \tilde{\mathscr{A}}^{\cdot}) \end{array}$$

Here we use the natural mapping

$$\pi_1(\mathbb{C}^*) \to \mathrm{Gal}(K/R)$$

constructed in 5.21.

Proof of A) and B). A) comes from the theorems on the etale cohomology of direct limits of rings (I 4.18).

B) $\varinjlim \mathscr{H}^0(\tilde{\mathscr{A}}_{U,n}^{\cdot}) = (\varinjlim \mathscr{H}^0(\mathscr{A}_{U,n}^{\cdot}))_{\mathrm{an}} = (F_{Y_0})_{\mathrm{an}}$

$$= F_{X_0} = \mathscr{H}^0(\tilde{\mathscr{A}}^{\cdot}).$$

For $\nu \neq 0$, $b \neq 0$,

$$\varinjlim H^{\nu}(\tilde{\mathscr{A}}_{U,n,b}^{\cdot}) = \varinjlim H^{\nu}(\mathscr{A}_{U,n,b}^{\cdot}) = 0 = H^{\nu}(\tilde{\mathscr{A}}_b^{\cdot}).$$

It remains to show that

$$\varinjlim H^{\nu}(\tilde{\mathscr{A}}_{U,n,0}^{\cdot}) = H^{\nu}(\tilde{\mathscr{A}}_0^{\cdot}) \quad \text{for } \nu \neq 0.$$

For this, we consider the diagram

$$\begin{array}{ccccc} \lambda & \mathbb{C} & \hookrightarrow & \bar{\mathbb{C}} & \hookleftarrow \infty \\ \downarrow & \downarrow & & \downarrow & \downarrow \\ t = e^{2\pi i \lambda/n} & \mathbb{C}^* & & & \\ \downarrow & \downarrow & & & \\ t^n & \mathbb{C}^* & \subset & \mathbb{C} & \hookleftarrow 0. \end{array}$$

By extension with the analytic quadric, we get

$$\begin{array}{ccccc} (z, \lambda) & \bar{X}^* & \xrightarrow{\bar{j}} & \bar{X} & \xleftarrow{\bar{i}} X_0 \\ \downarrow & & \searrow^{j} & \downarrow & \| \\ (z, e^{2\pi i \lambda/n}) & X_n^* & \xrightarrow{j_n} & X & \xleftarrow{i} X_0. \end{array}$$

$$(z, t) \mapsto (z, t^n)$$

We must show that

$$\varinjlim_{0 \in U, n} H^{\nu}(j_n^{-1}(V), F) = \varinjlim_{0 \in U} H^{\nu}(j^{-1}(V), F)$$

(see also Lemma 5.1).

Here V runs through a neighborhood basis of 0 in X, and hence its inverse image in $\bar{X}$ runs through a neighborhood basis of $0 \in X_0 \subset \bar{X}$ in $\bar{X}$.

Proof. For $\rho, c > 0$, we set

$$V = \{(z, t) \in X;\ |t| < e^{-2\pi\rho},\ |z|^2 - |t| < 2c^2\}.$$

Using the identification

$$\bar{X}^* \cong M \times \mathbb{C} \quad \text{(see Lemma 5.12)}$$

$$(z, \lambda) \mapsto \left(\frac{\mathrm{Re}(e^{-i\pi\phi} z)}{\sqrt{r + \rho(z, \lambda)}},\ \mathrm{Im}(e^{-i\pi\phi} z),\ \lambda \right)$$

$$\phi = \mathrm{Re}(\lambda), \qquad r = |e^{2\pi i \lambda}|, \qquad \rho(z, \lambda) = \tfrac{1}{2}(|z|^2 - r)$$

we get

$$j^{-1}(V) \cong \{(x, y) \in M;\ |y| < c\} \times H_\rho$$

$$j_n^{-1}(V) = j^{-1}(V)/G_n$$

with

$$G_n = n\mathbb{Z} \subseteq \mathbb{Z} = \pi_1(\mathbb{C}^*).$$

The cohomology groups

$$H^\nu(j^{-1}(V), F)$$

are thus finite. There is a spectral sequence

$$H^p(G_n, H^q(j^{-1}(V), F)) \Rightarrow H^{p+q}(j_n^{-1}(V), F).$$

The assertion now follows from

5.26 Lemma. *Set $G = \mathbb{Z}$, $G_n = n\mathbb{Z} \subseteq \mathbb{Z}$. Let A be a finite abelian group on which G acts. Then*

$$\varinjlim_n H^\nu(G_n, A) = \begin{cases} A & \text{for } \nu = 0 \\ 0 & \text{for } \nu \neq 0. \end{cases}$$

With this we have proved assertion *B*. □

The computation of the analytic variation for our pencil of quadrics (5.19) and the comparison theorem 5.24 give us the main result of this section:

5.27 The fundamental Lemma (4.10) *holds for the projective quadric defined by the equation*

$$\sum x_n x_{n+m+1} + t x_{n+1}^2 = 0, \qquad (n = 2m + 2)$$

over the ring $\mathbb{C}\{t\}$: The algebraic variation mapping is of the form

$$\mathrm{Var}(\sigma)(x) = c_n \chi_\lambda(\sigma) \langle x, \delta \rangle \delta$$

with

$$c_n = \pm 1 \quad (= (-1)^{m+1}).$$

§ 6. The Behaviour of the Monodromy Mapping Under Change of Base Ring

In the previous section, we used topological methods to determine the variation mapping of a quadric over the strictly Henselian base ring $\mathbb{C}\{t\}$. To prove the fundamental lemma 4.10 in the general case, it remains to study the behavior of the variation $\mathrm{Var}(\sigma)$ under base ring extension and above all under specialization.

We first want to treat the case of a base ring extension. For this, we consider an *injective* local homomorphism

$$\phi\colon\ R \hookrightarrow R_1$$

of strictly Henselian discrete valuation rings with residue fields $k=R/\mathfrak{m}$, $k_1=R_1/\mathfrak{m}_1$, quotient fields K, K_1, and an R-homomorphism

$$\phi\colon\ \bar{K} \hookrightarrow \bar{K}_1$$

from the separable closure $\bar{K}$ of K to the separable closure $\bar{K}_1$ of K_1,

$$\mathfrak{m}=R\pi, \qquad \mathfrak{m}_1=R_1\pi_1,$$

$$\phi(\pi)=\pi_1^e\cdot\text{unit}.$$

Let G be the Galois group of $\bar{K}$ over K and G_1 the Galois group of $\bar{K}_1$ over K_1. The embedding $\phi\colon \bar{K}\hookrightarrow \bar{K}_1$ induces a homomorphism

$$\psi\colon\ G_1 \to G.$$

The characters introduced in § 4,

$$\chi_\lambda\colon\ G\to\Lambda(1), \qquad 0\neq\lambda\in\mathfrak{m},$$

$$\chi_\mu\colon\ G_1\to\Lambda(1), \qquad 0\neq\mu\in\mathfrak{m}_1,$$

satisfy

$$\boxed{\chi_\lambda(\psi(\sigma))=\chi_{\phi(\lambda)}(\sigma).}$$

In particular,

$$\chi_\pi(\psi(\sigma))=e\cdot\chi_{\pi_1}(\sigma).$$

We consider the commutative diagram

$$\begin{array}{ccccc}
 & & \mathrm{Spec}(R_1) & & \\
 & & \| & & \\
\mathrm{Spec}(\bar{K}_1) & \xrightarrow{\eta_1} & T & \xleftarrow{s_1} & \mathrm{Spec}(k_1) \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{Spec}(\bar{K}) & \xrightarrow{\eta} & S & \xleftarrow{s} & \mathrm{Spec}(k) \\
 & & \| & & \\
 & & \mathrm{Spec}(R). & &
\end{array}$$

We again study our quadric

$$f\colon X \to S = \operatorname{Spec}(R)$$

with the homogeneous equation

$$\sum_{i=0}^{m} x_i x_{i+m+1} + \lambda x_{n+1}^2 = 0, \qquad 0 \neq \lambda \in \mathfrak{m}.$$

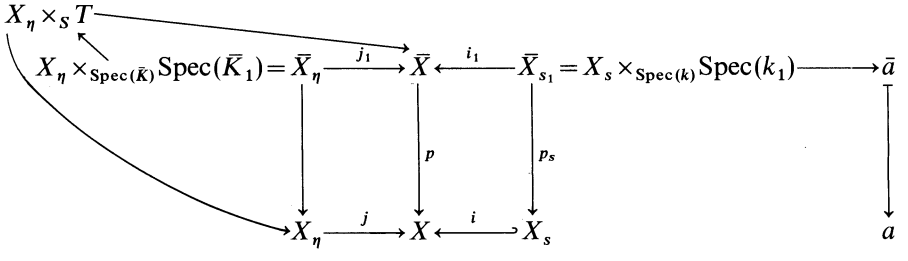
By extending the base scheme, we get a quadric

$$\bar{f}\colon \bar{X} = X \times_S T \to T = \operatorname{Spec} R_1$$

over T with equation

$$\sum_{i=0}^{m} x_i x_{i+m+1} + \phi(\lambda) x_{n+1}^2 = 0.$$

The double point $\bar{a}$ of $\bar{X}$ lies over the double point a of X. With the notations of § 3 and § 4, we get a commutative diagram



Using the base change mapping, we get a commutative diagram, equivariant with respect to the homomorphism $G_1 \to G$:

$$\begin{array}{ccccc} p_s^*(\Lambda_{X_s}) & \longrightarrow & p_s^*(\mathscr{A}^\cdot(f)) & \longrightarrow & p_s^*(\mathscr{B}^\cdot(f)) \\ \downarrow & & \downarrow & & \downarrow \\ \Lambda_{\bar{X}_{s_1}} & \longrightarrow & \mathscr{A}^\cdot(\bar{f}) & \longrightarrow & \mathscr{B}^\cdot(\bar{f}) \end{array}$$

Here $\mathscr{A}^\cdot(f)$" = "$i^* \mathrm{R}^\cdot j_*(\Lambda_{X_\eta})$, $\mathscr{A}^\cdot(\bar{f}) = i_1^* \mathrm{R}\, j_{1*}(\Lambda_{X_{\eta_1}})$, ... are the complexes constructed in § 3 (3.2, 3.6). With the isomorphisms of Lemma 4.7, we have a commutative diagram

$$\begin{array}{ccc} H^\cdot(X_{e\eta}, \Lambda) & \xrightarrow{\cong} & H^\cdot(\mathscr{A}^\cdot(f)_a) = H^\cdot(p_s^*(\mathscr{A}^\cdot(f))_a) \\ \downarrow & & \swarrow \\ H^\cdot(\bar{X}_{e\eta}, \Lambda) & \xrightarrow{\cong} & H^\cdot(\mathscr{A}^\cdot(\bar{f})_{\bar{a}}) \end{array}$$

On the other hand, the etale cohomology groups are invariant under the base field extension $\bar{K} \to \bar{K}_1$, even though X_{eh} is not proper over $\bar{K}$:

$$H^\cdot(X_{e\eta}, \Lambda) \xrightarrow{\cong} H^\cdot(X_{e\eta} \times_{\operatorname{Spec}(\bar{K})} \operatorname{Spec}(\bar{K}_1), \Lambda) = H^\cdot(\bar{X}_{e\eta_1}, \Lambda).$$

This follows from the smooth base change theorem and the invariance theorem for purely inseparable field extensions I 3.14. Indeed, $\bar{K}_1$ is a purely inseparable extension of an (infinite) separable field extension L of $\bar{K}$. The latter is a direct limit of (finitely generated) smooth k-algebras. Thus we have:

The complexes $\mathscr{A}^\cdot(\bar{f})$ *and* $p_s^*(\mathscr{A}^\cdot(f))$, *and hence also the mapping cones* $\mathscr{B}^\cdot(\bar{f})$ *and* $p_s^*(\mathscr{B}^\cdot(f))$ *are quasi-isomorphic.* The same considerations, applied to the schemes X_s and $X_s - \{a\}$ and the base field extension $k \to k_1$, show that

$$H^\cdot_{\{a\}}(X_s, \mathscr{A}^\cdot(f)) \xrightarrow{\cong} H^\cdot_{\{\bar{a}\}}(\bar{X}_{s_1}, p_s^*(\mathscr{A}^\cdot(f))).$$

With the isomorphisms of Lemma 4.7 and the element θ of Proposition 4.5 we have a commutative diagram

$$\begin{array}{ccc} H^n_{\{a\}}(X_s, \mathscr{A}^\cdot(f))(m) & \xrightarrow{\cong} & H^n_c(X_{e\eta}, \Lambda)(m) \\ \Big\| & \delta \longrightarrow \theta & \Big| \\ \Big\| & \Big\downarrow \quad \Big\downarrow & \Big| \\ \Big\| & \delta \longrightarrow \theta & \Big| \\ \Big\downarrow & & \Big\downarrow \\ H^n_{\{\bar{a}\}}(\bar{X}_s, \mathscr{A}^\cdot(\bar{f}))(m) & \xrightarrow[\cong]{} & H^n_c(\bar{X}_{e\eta}, \Lambda)(m) \end{array}$$

We obtain:

6.1 Proposition. *The behavior of the variation under the base ring extension*

$$\begin{array}{ccc} R & \longrightarrow & R_1 \\ \cap\!| & & \cap\!| \\ \phi\colon \bar{K} & \longrightarrow & \bar{K}_1 \end{array}$$

$$\psi\colon \mathrm{Gal}(\bar{K}_1/K_1) \to \mathrm{Gal}(\bar{K}/K)$$

is described by a natural commutative diagram

$$\begin{array}{ccc} H^n_{\{a\}}(X_s, \mathscr{A}^\cdot(f)) & \xrightarrow{\cong} & H^n_{\{\bar{a}\}}(\bar{X}_{s_1}, \mathscr{A}^\cdot(\bar{f})) \\ \Big\uparrow \mathrm{Var}(\psi(\sigma)) & & \Big\uparrow \mathrm{Var}(\sigma) \\ H^n(\mathscr{B}^\cdot(f)_a) & \xrightarrow{\cong} & H^n(\mathscr{B}^\cdot(\bar{f})_{\bar{a}}). \end{array}$$

The horizontal mappings are compatible with the pairing introduced in § 3. *The vanishing cycle* δ *of* X *is mapped to that of* $\bar{X}$.

6.2 Corollary. *For our quadrics*

$$X\colon \sum x_i x_{i+m+1} + \lambda x_{n+1}^2 = 0,$$
$$\bar{X}\colon \sum x_i x_{i+m+1} + \phi(\lambda) x_{n+1}^2 = 0,$$

a constant c *in* Λ *will satisfy*

$$\mathrm{Var}(\sigma)(-) = c \cdot \chi_\lambda(\sigma) \langle -, \delta \rangle \delta, \qquad \forall \sigma \in G,$$

if and only if it satisfies

$$\mathrm{Var}(\sigma)(-)=c\cdot\chi_{\phi(\lambda)}(\sigma)\langle -,\delta\rangle\,\delta,\qquad \forall\sigma\in G_1.$$

In the previous section we determined the variation constant (up to sign) for the universal quadric

$$\sum_{i=0}^{m} x_i\,x_{i+m+1}+t\,x_{n+1}^2=0$$

over $\mathbb{C}\{t\}$. Together with 6.2, this yields a proof of the fundamental lemma 4.10 for every base ring with residue field of characteristic zero!

Now we want to study the behavior of the variation under specialization. For this we need to modify the construction of the complex $\mathscr{A}^\cdot(f)$. We again consider our initial situation

$$\mathrm{Spec}(\bar{K})\longrightarrow\mathrm{Spec}(R)\xleftarrow{s}\mathrm{Spec}(k).$$

We restrict ourselves to the coefficient rings

$\Lambda=\mathbb{Z}/l^r\mathbb{Z}$, l prime and different from the characteristic of the residue field k.

We want to replace the field $\bar{K}$ by the subfield

$$K_0=\bigcup_{v}K(\sqrt[l^v]{\pi})\subseteq K.$$

Here π generates the maximal ideal of R. Let G again be the Galois group of $\bar{K}$ over K, $G_0\subset G$ the Galois group of $\bar{K}$ over K_0, and $\Gamma=G/G_0$ the Galois group of K_0 over K. Γ is canonically isomorphic to $\hat{\mathbb{Z}}_l(1)=\varprojlim_v \mu_{l^v}(R)$,

$$\Gamma\xrightarrow{\cong}\hat{\mathbb{Z}}_l(1).$$

Our character $\chi\colon G\to\Lambda(1)$ is the composite of the projection $G\to\Gamma$ and the natural mapping $\Gamma=\hat{\mathbb{Z}}_l(1)\to\Lambda(1)$. The group G_0 is a projective limit of finite groups with order prime to l!

Let $X\to\mathrm{Spec}(R)=S$ be our quadric, and $\eta_0\colon\mathrm{Spec}(K_0)\to S$ the mapping induced by $R\subset K_0$. We get a commutative diagram

$$\begin{array}{ccccc} & & X_\eta & \xrightarrow{i} & X \\ & & \downarrow{\scriptstyle q} & & \| \\ X\times_S\mathrm{Spec}(K_0)=X_{\eta_0} & \xrightarrow{j_0} & X & \xleftarrow{i} & X_s. \end{array}$$

In the construction of $\mathscr{A}^\cdot(f)$ (3.2), we replace X_η by X_{η_0}: Let $\Lambda_X\to\mathscr{I}^\cdot$ be an injective resolution of Λ_X (by injective Λ_X-moduls $\mathscr{I}^v$) with $\mathscr{I}^v=0$ for $v<0$. Then

$$\Lambda_{X_{\eta_0}}\to\mathscr{I}^\cdot_{\eta_0}=j_0^*(\mathscr{I}^\cdot)$$

is a Γ-equivariant flabby resolution of $\Lambda_{X_{\eta_0}}$ and

$$\Lambda_{X_\eta}\to j^*(\mathscr{I}^\cdot)=q^*(\mathscr{I}^\cdot_{\eta_0})$$

is a G-equivariant flabby resolution of Λ_{X_η}. In analogy with $\mathscr{A}^\cdot(f)=i^* j_*(\mathscr{I}_\eta^\cdot)$, we construct the complex of Γ-sheaves $\mathscr{A}_0^\cdot(f)=i^* j_{0*}(\mathscr{I}_{\eta_0}^\cdot)$, and we get a commutative diagram of *equivariant* mappings of complexes

$$\begin{array}{ccc} \Lambda_{X_s} & \longrightarrow & \mathscr{A}^\cdot(f) \\ \| & & \uparrow \\ \Lambda_{X_s} & \longrightarrow & \mathscr{A}_0^\cdot(f) \end{array}$$

One shows

$$(q_* q^*(\mathscr{I}_{\eta_0}^\cdot))^{G_0}=\mathscr{I}_{\eta_0}^\cdot,$$

whence

$$\mathscr{A}_0^\cdot(f)=\mathscr{A}^\cdot(f)^{G_0}.$$

As G_0 is a profinite group with "order relatively prime" to the prime number l, the cohomology sheaves of the complexes $\mathscr{A}^\cdot(f)$, $\mathscr{A}_0^\cdot(f)$ satisfy

$$\mathscr{H}^\nu(\mathscr{A}^\cdot(f))^{G_0}=\mathscr{H}^\nu(\mathscr{A}_0^\cdot(f)).$$

As the variation $\mathrm{Var}(\sigma)$ for $\sigma\in G_0$ vanishes (4.9), G_0 acts trivially on $\mathscr{H}^n(\mathscr{A}^\cdot(f))_a$ and hence on $\mathscr{H}^\nu(\mathscr{A}^\cdot(f))$ for all ν.

$$\mathscr{H}^\nu(\mathscr{A}_0^\cdot(f)) \xrightarrow{\cong} \mathscr{H}^\nu(\mathscr{A}^\cdot(f)).$$

6.3 Lemma. *The homomorphism of complexes*

$$\mathscr{A}_0^\cdot(f)\to\mathscr{A}^\cdot(f),$$

is equivariant with respect to $G\to\Gamma$ and is a quasi-isomorphism. For computing the variation, the complex $\mathscr{A}_0^\cdot(f)$ and the mapping cone $\mathscr{B}_0^\cdot(f)$ of the mapping $\Lambda_X\to\mathscr{A}_0^\cdot(f)$ can be used in place of the complex $\mathscr{A}^\cdot(f)$ and $\mathscr{B}^\cdot(f)$. Thus the separable closure $\bar{K}$ of K can be replaced by the field $K_0=\bigcup_\nu K(\sqrt[l^\nu]{\pi})\subset K$.

We shall now study families of Henselian rings and quadrics over these rings, in order to compare the variation constant over an arbitrary ring R (of characteristic $p>0$) with that over a Henselian ring with a residue field of characteristic 0.

Let o be a strictly Henselian valuation ring, and $o\{t\}=A$ the Henselization of the polynomial ring $o[t]$ localized at the ideal generated by the variable t and the maximal ideal of o. Assume the prime number l ($\Lambda=\mathbb{Z}/l^r\mathbb{Z}$) is different from the residue field characteristic. In the separable closure of the quotient field of A, we now consider the ring

$$A[\sqrt[l^\nu]{t}],$$

its localization $A_\nu(t)=A[\sqrt[l^\nu]{t}]_t$ by the powers of t, and the union

$$A(t)=\bigcup_\nu A_\nu(t)=\varinjlim_\nu A_\nu(t).$$

Obviously we have o-isomorphisms

$$\begin{array}{ccc} A_t & \xrightarrow{\cong} & A_v(t) \\ \cup\!| & & \cup\!| \\ A & \xrightarrow{\cong} & A[\sqrt[l^v]{t}] \\ t & \mapsto & \sqrt[l^v]{t}. \end{array}$$

6.4 Lemma. *The rings $A_v(t)$ and $A(t)$ are principal ideal rings – in particular, noetherian.*

Proof. $A[\sqrt[l^v]{t}]$ is a local UFD of dimension 2, and hence $A[\sqrt[l^v]{t}]_t$ is a UFD of dimension 1. We pass to the completions $\hat{o}$ of o and

$$\widehat{A[\sqrt[l^v]{t}]} = \hat{o}[[\sqrt[l^v]{t}]] \quad \text{of } A[\sqrt[l^v]{t}].$$

A power series

$$\sum_{i=0}^{\infty} a_i (\sqrt[l^v]{t})^i = \sum_{i=0}^{\infty} a_i (\sqrt[l^{v+\mu}]{t})^{il^\mu}$$

with a nonzero constant term $a_0 \in o$ of order s (with respect to the valuation of o) splits into at most s prime factors in the ring $A[\sqrt[l^{v+\mu}]{t}]$, for any natural number μ. This implies our assertion for the ring $\bigcup A_v(t) = A(t)$. □

Now we consider the homomorphisms

$$\begin{array}{ccc} A \hookrightarrow A(t), & & A \to o \\ & & t \mapsto 0 \end{array}$$

and the induced maps of affine schemes,

$$S(t) = \mathrm{Spec}(A(t)) \xrightarrow{\eta} S = \mathrm{Spec}(A) \xleftarrow{s} S_0 = \mathrm{Spec}(o).$$

6.5 Lemma.

a) $R^v \eta_* (\Lambda_{S(t)}) = 0$ for $v > 0$.

b) *The natural mapping $\Lambda_{S_0} \to s^* \eta_* \Lambda_{S(t)}$ is an isomorphism.*

Proof. $A(t)$ is a direct limit of module-finitely generated A_t-algebras. Hence $R^v \eta_* (\Lambda_{S(t)}) | \mathrm{Spec}(A_t)$ vanishes for $v > 0$. We therefore prove assertions a), b) only for the stalks of $R^v \eta_* (\Lambda_{S(t)})$ at the two essentially different geometric points of S_0. Let B be a strict Henselization of the valuation ring $A_{(At)}$, the localization of A at the prime ideal (At), let K be the quotient field of B, and

$$K_0 = \bigcup_v K[\sqrt[l^v]{t}] = A(t) \otimes_A B.$$

We must show:

(1) $H^0(\mathrm{Spec}(K_0), \Lambda) = H^0(S(t), \Lambda) = \Lambda$,
(2) $H^\nu(\mathrm{Spec}(K_0), \Lambda) = H^\nu(S(t), \Lambda) = 0$ for $\nu > 0$.

The first statement is clear. We prove the second one for $S(t)$; the proof runs similarly for $\mathrm{Spec}(K_0)$.

The purity theorem (I.10.1) implies

$$\mathscr{H}^\nu_{S_0}(S, \Lambda) = 0 \quad \text{for } \nu \neq 2.$$

Since o is strictly Henselian,

$$H^\nu_{S_0}(S, \Lambda) = \Gamma(S_0, \mathscr{H}^\nu_{S_0}(S, \Lambda)) = 0 \quad \text{for } \nu \neq 2.$$

Using the long exact sequence

$$\ldots \to H^\nu_{S_0}(S, \Lambda) \to H^\nu(S, \Lambda) \to H^\nu(S - S_0, \Lambda) \to \ldots$$

we get $H^\nu(\mathrm{Spec}(A_0(t), \Lambda) = H^\nu(S - S_0, \Lambda) = 0$ for $\nu \geqq 2$. Similarly: $H^\nu(\mathrm{Spec}(A_j(t), \Lambda) = 0$, $\nu \geqq 2$,

$$H^\nu(S(t), \Lambda) = \varinjlim_j H^\nu(\mathrm{Spec}(A_j(t), \Lambda)) = 0 \quad \text{for } \nu \geqq 2.$$

As the divisor class group of $A(t)$ is trivial (6.4), the Kummer sequence gives

$$H^1(S(t), \Lambda) = A(t)^*/A(t)^{*l^r} = 0.$$

Indeed, every unit $u \in A(t)^*$, the group of units of $A(t)$, is the product of a unit ε from a ring $A[\sqrt[l^\nu]{t}]$ and a power $(\sqrt[l^\nu]{t})^s$ for a sufficiently large number ν. In the strictly Henselian ring $A[\sqrt[l^\nu]{t}]$, ε has an l^r-th root. The element $(\sqrt[l^\nu]{t})^s$ has an l^r-th root in $A_{\nu+r}(t)$. □

We now consider the projective quadric

$$f\colon X \to S = \mathrm{Spec}\, A = \mathrm{Spec}\, o\{t\}$$

of relative dimension $n = 2m+1$ over S given by the homogeneous equation

$$\sum_{i=0}^{m} x_i x_{i+m+1} + t x_{n+1}^2 = 0$$

together with the natural commutative diagram

$$\begin{array}{ccccccc}
X \times_S S(t) = X_\eta & \xrightarrow{j} & X & \xleftarrow{i} & X_0 = X \times_S S_0 & \supseteq & \bar{S}_0 \\
\downarrow{\scriptstyle f_\eta} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_0} & \swarrow & \\
S(t) & \xrightarrow{\eta} & S & \xleftarrow{s} & S_0 & &
\end{array}$$

As in § 4, let $Y \subset X$ be the intersection of X with the hyperplane $H\colon \{x_{n+1} = 0\}$, and let $X_e = X - Y$ be the affine quadric with equation

$$\sum_{i=0}^{m} x_i x_{i+m+1} + t = 0,$$

$Y_\eta = Y \times_S S(t)$, $X_{e\eta} = X_e \times_S S(t)$, ...

$\bar{S}_0$ is the closed subscheme of $X_{e0} \subset X_0$ described by

$$x_0 = \ldots = x_n = 0.$$

The mapping f_0 induces an isomorphism $\bar{S}_0 \xrightarrow{\cong} S_0$. The quadric X is smooth over S outside the points of $\bar{S}_0$. The group $\Gamma = \varprojlim_\nu \mu_{l^\nu}(A)$ acts in a natural way on $S(t)$ over S and hence also on X_η over X. We let Γ act trivially on X and X_0.

We want to extend the construction of the complexes $\mathscr{A}^\cdot$, $\mathscr{B}^\cdot$ and the variation, which we carried through in § 3, to this quadric X over a two-dimensional base scheme. For details we refer to § 3. Let

$$\Lambda_X \to \mathscr{I}^\cdot$$

be an injective resolution of the constant sheaf Λ_X on X with $\mathscr{I}^\nu = 0$ for $\nu < 0$. Then

$$\Lambda_{X_\eta} = j^*(\Lambda_X) \to j^*(\mathscr{I}^\cdot) = \mathscr{I}_\eta^\cdot$$

is a Γ-equivariant resolution of Λ_X on X_η; $\mathscr{I}_\eta^\cdot$ is a flabby resolution, so the sheaves $\mathscr{I}_\eta^\nu$ are acyclic with respect to the functor of sections on each etale scheme over X_η. The complex $i^* j_*(\mathscr{I}_\eta^\cdot)$ is a complex of Γ-equivariant sheaves on X_0, and the natural mapping

$$\Lambda_{X_0} \to i^* j_*(\mathscr{I}_\eta^\cdot)$$

is Γ-equivariant. Here Γ acts trivially on Λ_{X_0}.

6.6 Definition. We denote by $\mathscr{A}^\cdot(f)$ the complex $i^* j_*(\mathscr{I}_\eta^\cdot)$ of Γ-sheaves, and by $\mathscr{B}^\cdot(f)$ the mapping cone of the Γ-equivariant natural mapping

$$\Lambda_{X_s} \to \mathscr{A}^\cdot(f).$$

6.7 Remark. $\mathscr{A}^\cdot(f)$ represents the complex $i^* \mathrm{R}^\cdot j_*(\Lambda_{X_\eta})$ in the derived category.

Since f is smooth outside $\bar{S}_0 \cong S_0$, it follows by the smooth base change theorem and Lemma 6.5 (see 3.6) that we have:

6.8 Lemma. *The mapping*

$$f^*(\eta_*(\Lambda_{S(t)})) \to j_*(\mathscr{I}_\eta^\cdot)$$

is a quasi-isomorphism outside $\bar{S}_0$. *In particular,*

$$\Lambda_{X_0} \to \mathscr{A}^\cdot(f)$$

is a quasi-isomorphism outside $\bar{S}_0$. *The cohomology sheaves* $\mathscr{H}^\nu(\mathscr{B}^\cdot(f))$ *of* $\mathscr{B}^\cdot(f)$ *are thus concentrated on* $\bar{S}_0$!

We come now to the construction of the variation mapping (see 3.9).

Since Γ acts trivially on Λ_X, for every element $\sigma \in \Gamma$ there is a mapping of complexes "$\sigma - 1$": $\mathscr{B}^\cdot(f) \to \mathscr{A}^\cdot(f)$ such that the diagram

$$\begin{array}{ccc} \mathscr{A}^\cdot(f) & \longrightarrow & \mathscr{B}^\cdot(f) \\ {\scriptstyle \sigma-1}\downarrow & \swarrow & \\ \mathscr{A}^\cdot(f) & & \end{array}$$

becomes commutative. We want to apply to this mapping the functor $\mathscr{H}^\cdot_{\bar{S}_0}(X_0, -)$ of derived cohomology sheaves with support in $\bar{S}_0$. Since the cohomology sheaves of $\mathscr{B}^\cdot(f)$ are concentrated on $\bar{S}_0$, we have

$$\mathscr{H}^\cdot_{\bar{S}_0}(X_0, \mathscr{B}^\cdot(f)) = \mathscr{H}^\cdot(\mathscr{B}^\cdot(f)).$$

6.9 Definition (cf. 3.9). Let $\sigma \in \Gamma$. The mapping "$\sigma - 1$": $\mathscr{B}^\cdot(f) \to \mathscr{A}^\cdot(f)$ induces a mapping on the cohomology sheaves with support in $\bar{S}_0$,

$$\mathscr{H}^n(\mathscr{B}^\cdot(f)) \to \mathscr{H}^n_{\bar{S}_0}(X_0, \mathscr{A}^\cdot(f)),$$

which we denote by $\mathrm{Var}(\sigma)$.

For a closer study of the complex $\mathscr{A}^\cdot(f)$, we carry over Lemma 4.7 to the situation here. The proof runs just as in § 4, using Lemma 6.8.

6.10 Lemma. *In the derived category of sheaves on S_0" = "$\bar{S}_0$, there are natural isomorphisms (see 4.7)*

$$(1) \qquad \mathrm{R}^\cdot \Gamma_{\bar{S}_0}(X_0, \mathscr{A}^\cdot(f))|S_0 \xrightarrow{\cong} s^* \mathrm{R}^\cdot \eta_*(\mathrm{R}^\cdot f_{e\eta!}(\Lambda_{X_{e\eta}}))$$

$$(2) \qquad s^* \mathrm{R}^\cdot \eta_*(\mathrm{R}^\cdot f_{e\eta *}(\Lambda_{X_{e\eta}})) \xrightarrow{\cong} \mathscr{A}^\cdot(f)|S_0.$$

Here $f_{e\eta} = f_\eta | X_{e\eta}$: $X_{e\eta} \to S(t)$, and $\Gamma_{\bar{S}_0}(X_0, -)$ is the functor "subsheaf of sections with support on $\bar{S}_0$".

To apply this isomorphism, we need the results corresponding to Propositions 4.4 and 4.5 for the quadrics

$$f_\eta \colon X_\eta \to S(t),$$
$$f_{Y\eta} = f_\eta | Y_\eta \colon Y_\eta \to S(t),$$
$$f_{e\eta} \colon X_e \to S(t),$$

over $S(t)$, in particular:

In $\Gamma(S(t), R^{2m} f_{Y\eta *}(\Lambda(m)_{X_\eta}))$ there are elements θ_1 and θ_2 that, under passage to the geometric fibers of $X \to S(t)$, are sent to the cohomology classes θ_1, θ_2 of algebraic cycles given in Theorem 4.1. By comparison with the geometric fibres then we get

$$R^{2m} f_{Y\eta *}(\Lambda_{X_\eta}) = \Lambda_{S(t)}(-m)\,\theta_1 \oplus \Lambda_{S(t)}(-m)\,\theta_2.$$

Then we consider again the long exact cohomology sequence relating $R^\nu f_{e\eta!}$, $R^\nu f_{\eta *}$, and $R^\nu f_{Y\eta *}$, and using the base change theorems and the results on the cohomology of the geometric fibers we get

$$R^\nu f_{e\eta!}(\Lambda_{X_e}) = \begin{cases} 0, & \nu \neq n,\ 2n \\ \Lambda_{S(t)}(-m)\,\theta, & \nu = n \\ \Lambda_{S(t)}(-n), & \nu = 2n. \end{cases}$$

Here θ is the image of (say) the class θ_1 under the connecting homomorphism. The isomorphism for $\nu = 2n$ is furnished by the relative trace mapping (II 1.7). By a suitable "cell decomposition" of $X_{e\eta}$, we get

$$R^\nu f_{e\eta *}(\Lambda_{X_{e\eta}}) \cong \begin{cases} 0, & \nu \neq 0,\ n \\ \Lambda_{S(t)}, & \nu = 0, \\ \Lambda_{S(t)}, & \nu = n. \end{cases}$$

The last isomorphism is noncanonical. We sum up:

6.11 Lemma. *There is a cohomology class θ in $\Gamma(S(t), R^n f_{e\eta!}(\Lambda(m)_{X_e}))$ which, under passage to the geometric fibres of $X_{e\eta} \to S(t)$, is sent to the class given in Proposition* 4.5 *(constructed as image of θ_1, say). Furthermore,*

$$R^\nu f_{e\eta!}(\Lambda_{X_{e\eta}}) = \begin{cases} 0, & \nu \neq n,\ 2n \\ \Lambda_{S(t)}(-m)\,\theta, & \nu = n \\ \Lambda_{S(t)}(-n), & \nu = 2n. \end{cases}$$

$$R^\nu f_{e\eta *}(\Lambda_{X_{e\eta}}) \cong \begin{cases} 0, & \nu \neq 0,\ n \\ \Lambda_{S(t)}, & \nu = 0,\ n \\ (\text{noncanonical for } \nu = n). \end{cases}$$

Combined with Lemma 6.5, this gives:

6.12 Corollary

$$s^*\, R^\nu\, \eta_*(\mathsf{R}^\cdot f_{e\eta *}(\Lambda_{X_{e\eta}})) \cong \begin{cases} 0, & \nu \neq 0,\ n \\ \Lambda_{S_0}, & \nu = 0,\ n \\ (\text{noncanonical for } \nu = n). \end{cases}$$

$$s^*\, R^\nu\, \eta_*(\mathsf{R}^\cdot f_{e\eta!}(\Lambda_{X_{e\eta}})) = \begin{cases} 0, & \nu \neq n,\ 2n \\ \Lambda_{S_0}(-n), & \nu = 2n \\ \Lambda_{S_0}(-m)\,\theta, & \nu = n \end{cases}$$

with a naturally constructed class

$$\theta \in \Gamma(S_0, s^*\, R^n\, \eta_*(\mathsf{R}^\cdot f_{e\eta!}(\Lambda(m)_{X_{e\eta}}))).$$

Using the isomorphism from Lemma 6.10, we thus get:

6.13 Proposition. *There is a "naturally constructed" element*

$$\delta \in \Gamma(S_0, \mathscr{H}^n_{S_0}(X_0, \mathscr{A}^\cdot(f))(m))$$

with

$$\mathscr{H}^n_{\bar{S}_0}(X_0, \mathscr{A}^\cdot(f))|S_0 = \Lambda_{S_0}(-m)\,\delta \quad (\cong \Lambda_{S_0}(-m)).$$

Further,

$$\mathscr{H}^n(\mathscr{B}^\cdot(f))|S_0 = \mathscr{H}^n(\mathscr{A}^\cdot(f))|S_0 \cong \Lambda_{S_0} \quad \text{(noncanonically)}.$$

Let R be a strictly Henselian discrete valuation ring with residue field k and quotient field K, and let

$$h\colon\ A \to R$$

be a not necessarily local homomorphism taking t to a generator π of the maximal ideal of R. Then

$$R \otimes_A A/tA = k$$

$$R \otimes_A A(t) = \bigcup_\nu K[\sqrt[l^\nu]{\pi}] = K_0 \quad \text{(see 6.3)}.$$

The Galois group of K_0/K is canonically isomorphic to Γ. Let $T = \mathrm{Spec}(R)$, and let

$$f(h)\colon\ X(h) \to T$$

be the projective quadric $X \times_S T$ over T given by the homogeneous equation

$$\sum_{i=0}^{m} x_i x_{i+m+1} + \pi x_{n+1}^2 = 0.$$

We consider the commutative diagram

$X(h) \times_T \mathrm{Spec}(K_0)$

$$\begin{array}{ccccccc}
 & & X_\eta & \xrightarrow{j} & X & \xleftarrow{i} & X_0 \\
 & & \uparrow{\scriptstyle q} & & \uparrow{\scriptstyle q} & & \uparrow{\scriptstyle q} \\
= X_\eta \times_S T = X(h)_\eta & \xrightarrow{j(h)} & X(h) & \xleftarrow{i(h)} & X(h)_s = X_0 \times_S T = X(h) \times_T \mathrm{Spec}(k) & & \\
\downarrow & & \downarrow{\scriptstyle f(h)} & & \downarrow & & \\
\mathrm{Spec}(K_0) & \longrightarrow & T & \longleftarrow & \mathrm{Spec}(k) & &
\end{array}$$

The inverse image of $\bar{S}_0$ in $X(h)_s$ is precisely the unique singular point $a(h)$ of $X(h)_s$. We identify it with a geometric point

$$\mathrm{Spec}(k) \to X(h)_s$$

and for simplicity write

$$a(h)\colon\ \mathrm{Spec}(k) \to S_0 \subseteq X_0$$

also for the image point in S_0" = "$S_0 \subseteq X_0$. Using the base change mapping, we get natural Γ-equivariant morphisms of complexes on $X(h)_s$:

$$
\begin{array}{ccc}
q^*(\Lambda_{X_0}) & = & \Lambda_{X(h)_s} \\
\downarrow & & \downarrow \\
q^*(\mathscr{A}(f)) & \longrightarrow & \mathscr{A}_0(f(h)) \quad \text{(quasi-isomorphic to } \mathscr{A}(f(h))) \\
\downarrow & & \downarrow \\
q^*(\mathscr{B}(f)) & \longrightarrow & \mathscr{B}_0(f(h)) \quad \text{(quasi-isomorphic to } \mathscr{B}(f(h))).
\end{array}
\tag{$*$}
$$

Here $\mathscr{A}_0(f(h))$ and $\mathscr{B}_0(f(h))$ are the two complexes introduced in Lemma 6.3 that can be used in place of $\mathscr{A}(f(h))$ and $\mathscr{B}(f(h))$ for computing the variation of $f(h)$. From this we get a commutative diagram for $\sigma \in \Gamma$:

"$\delta \longrightarrow \delta$" in the twisted groups

$$
\begin{array}{ccc}
\mathscr{H}^n_{S_0}(X_0, \mathscr{A}(f))_{a(h)} & \longrightarrow & H^n_{\{a(h)\}}(\mathscr{A}_0(f(h))) \cong H^n_{\{a(h)\}}(\mathscr{A}(f(h))) \\
\uparrow \mathrm{Var}(\sigma) & & \uparrow \mathrm{Var}(\sigma) \\
\mathscr{H}^n(\mathscr{B}(f))_{a(h)} & \longrightarrow & H^n(\mathscr{B}_0(f(h))_{a(h)}) \cong H^n(\mathscr{B}(f(h)_{a(h)}))
\end{array}
$$

The horizontal mappings in the upper diagram are quasi-isomorphisms, and hence they are isomorphisms in the lower diagram for the following "general" homomorphism

$$\omega\colon\ A = o\{t\} \to \Omega\{t\}.$$

Here Ω is the separable closure of the quotient field of o; $\Omega\{t\}$ is a direct limit of etale A-algebras!

The sheaves $\mathscr{H}^n_S(X_0, \mathscr{A}(f))|S_0$ and $\mathscr{H}^n(\mathscr{B}(f))|S_0$ are (by 6.13) *constant* on S_0. Thus all stalks of $\mathscr{H}^n_{S_0}(X_0, \mathscr{A}(f))|S_0$ are canonically isomorphic, and all stalks of $\mathscr{H}^n(\mathscr{B}(f))|S_0$ are canonically isomorphic.

6.14 Theorem. *There are natural mappings*

$$H^n_{\{a(\omega)\}}(X(\omega)_s, \mathscr{A}_0(f(\omega))) \xrightarrow{\sim} H^n_{\{a(h)\}}(X(h)_s, \mathscr{A}_0(f(h)))$$

and

$$H^n(\mathscr{B}_0(f(\omega))_{a(\omega)} \xrightarrow{\sim} H^n(\mathscr{B}_0(f(h))_{a(h)})$$

such that for every $\sigma \in \Gamma$ the diagram

"$\delta \longrightarrow \delta$" in the twisted groups

$$
\begin{array}{ccc}
H^n_{\{a(\omega)\}}(X(\omega)_s, \mathscr{A}_0(f(\omega))) & \longrightarrow & H^n_{\{a(h)\}}(X(h)_s, \mathscr{A}_0(f(h))) \\
\uparrow \mathrm{Var}(\sigma) & & \uparrow \mathrm{Var}(\sigma) \\
H^n(\mathscr{B}_0(f(\omega))_{a(\omega)} & \longrightarrow & H^n(\mathscr{B}_0(f(h))_{a(h)})
\end{array}
$$

is commutative. They are compatible with the pairing $\langle\ ,\ \rangle$ *introduced in* § 2. *The first mapping sends vanishing cycles to vanishing cycles.*

From Proposition 6.1 and Theorem 6.14 we get immediately:

To prove the fundamental Lemma 4.10, *and thereby the Picard-Lefschetz formulas (in the case of odd fiber dimension), it suffices to determine the variation* $\mathrm{Var}(\sigma)$ *for the projective quadric given by the equation*

$$\sum_{i=0}^{m} x_i x_{i+m+1} + t x_{n+1}^2 = 0$$

over the ring $\mathbb{C}\{t\}$!

This was carried out by transcendental methods in § 5.

§ 7. The Global Monodromy Theory of Lefschetz Pencils

In this section we shall use properties of the fundamental group (see Appendix I: 15 and 16) to derive from the local Picard-Lefschetz formulas (Theorem 4.3) some results on the cohomology of a Lefschetz fibering of an algebraic variety that will be essential for the proof of the Weil conjecture. We want to carry this out not in the category of torsion sheaves or of l-adic sheaves but in the category of sheaves of $\mathbb{Q}_l$-vector spaces (I, §12). Here l is again a fixed prime number invertible on all the schemes in question, and $\mathbb{Q}_l$ is the field of l-adic numbers.

As a reminder: Let X be a connected scheme and $\alpha\colon \mathrm{Spec}(\Omega) \to X$ a geometric point. Then the category of locally constant sheaves of $\mathbb{Q}_l$-vector space is equivalent to the category of continuous representations of the fundamental group $\pi_1(X, \alpha)$ on finite-dimensional $\mathbb{Q}_l$-vector spaces (A I.8). The stalk $\mathcal{G}_\alpha$ of a locally constant sheaf $\mathcal{G}$ of $\mathbb{Q}_l$-vector spaces is a $\pi_1(X, \alpha)$-module. Assigning

$$\mathcal{G} \mapsto \mathcal{G}_\alpha$$

establishes the equivalence of the two categories.

We denote the inverse functor by

$$V \mapsto \mathcal{G}(V), \qquad \mathcal{G}(V)_\alpha \underset{\pi(X,\alpha)}{\simeq} V.$$

The Picard-Lefschetz formulas obviously carry over immediately from the case of l-adic sheaves (4.3 Remark) to the case of $\mathbb{Q}_l$-sheaves.

Now we consider a smooth irreducible projective variety X over an algebraically closed field k of characteristic p and a Lefschetz embedding $X \hookrightarrow \mathbb{P}^N$ (1.4).

Let the dimension of X be even,

$$\dim X = n+1, \qquad n = 2m+1.$$

Let $F \subset \check{\mathbb{P}}^n$ be the dual variety of X, all the hyperplanes $H \subset \mathbb{P}^n$ that touch X at any point. F is the image of the manifold $Y \subset X \times \check{\mathbb{P}}^n$ in Lemma 1.6. We now consider a Lefschetz fibering of X, as described in Lemma 1.9:

$$\begin{array}{ccc} X & \longleftarrow & \tilde{X} \\ & & \downarrow f \\ & & \mathbb{P}^1 = D \end{array}$$

The parameter space D is thought of as a line $D \subset \check{\mathbb{P}}^N$ in the dual projective space. Let it meet F only transversally, at smooth points. Then f is singular only over the points of $A = F \cap D \subset D$. Over every point $s \in A$ there lies exactly one singular point in the fiber $f^{-1}(s)$. It is an ordinary double point, actually nondegenerate since n is odd. We identify the points of A with geometric points $\mathrm{Spec}(k) \xrightarrow[\mathrm{Spec}(k)]{} D$.

Let Ω be the separable closure of the function field of D and let

$$\omega\colon\ \mathrm{Spec}(\Omega) \to D - A \subset D$$

be the corresponding geometric point. For a point $s \in A$, let $R(s)$ be the (strict) Henselization $\tilde{\mathcal{O}}_{D,s}$ of the local ring $\mathcal{O}_{D,s}$ with respect to s, and $D(s) = \mathrm{Spec}(\tilde{\mathcal{O}}_{D,s})$. There is then always a geometric point in $D(s)$ over ω,

$$\begin{array}{ccc} & & D(s) \\ & \overset{\tilde{\omega}}{\nearrow} & \downarrow \\ \mathrm{Spec}(\Omega) & \xrightarrow{\omega} & D \end{array}$$

induced by a homomorphism

$$\phi\colon\ R(s) \to \Omega$$

that identifies Ω with the separable closure of the quotient field K of $R(s)$. We consider the natural mapping

$$\pi_1(D(s) - \{s\}, \tilde{\omega}) = \mathrm{Gal}(\Omega/K) \to \hat{\mathbb{Z}}^{(p)}(1) = \varprojlim_{p \nmid r} \mu_r(k)$$
$$\sigma \mapsto \rho(\sigma) \mapsto \bar{\rho}(\sigma) \in \mu_r(k)$$

with

$$\sigma(\sqrt[r]{(\phi(\pi))}) = \bar{\rho}(\sigma)(\sqrt[r]{(\phi(\pi))} \quad (4.1).$$

Let $\mathfrak{K}_s$ be the kernel. Here π generates the maximal ideal of $R(s)$.

We have the following characterization of the *tame* fundamental group $\pi_1^t(D - A, \omega)$ (A I.13):

$$\begin{array}{ccc} \pi_1(D(s) - \{s\}, \tilde{\omega}) & \longrightarrow & \pi_1(D - A, \omega) \\ \downarrow & & \downarrow \\ \gamma_{\tilde{\omega}, s}\colon\ \hat{\mathbb{Z}}^{(p)}(1) & \dashrightarrow & \pi_1^t(D - A, \omega) \end{array}$$

π_1^t is the quotient group by the smallest closed normal subgroup of $\pi_1(D-A, \omega)$ that contains the images of all the kernels $\mathfrak{K}_s$ (for all $s \in A$ and $\tilde{\omega}$). The induced homomorphism $\gamma_{\tilde{\omega}, s}$ depends only on the point $s \in A$ up to conjugation. We also write

$$\gamma_s\colon \hat{\mathbb{Z}}^{(p)}(1) \to \pi_1^t(D-A, \omega).$$

A (continuous) representation of $\pi_1(D-A, \omega)$ on a finite dimensional vector space V over $\mathbb{Q}_l$, resp. the $\pi_1(D-A, \omega)$-module V, is called *tamely ramified* if the representation factors through $\pi_1^t(D-A, \omega)$. That is equivalent to saying that the kernel $\mathfrak{K}_s$ acts trivially on V for all $s \in A$ (and $\tilde{\omega}$). A locally constant constructible sheaf $\mathscr{F}$ (of $\mathbb{Q}_l$-vector spaces) on $D-A$ is called tamely ramified when the stalk $\mathscr{F}_\omega$ is a tame $\pi_1(D-A, \omega)$-module.

Every point $s \in A$ gives rise to a local Lefschetz pencil of the type studied in the previous sections:

$$\begin{array}{ccccc} \tilde{X}_{D(s)} = \tilde{X} \times_D D(s) & & & & \\ \downarrow{\scriptstyle f_{D(s)}} & & & & \\ \mathrm{Spec}(\Omega) & \xrightarrow{\tilde{\omega}} & D(s) & \xleftarrow{\text{“}s\text{”}} & \mathrm{Spec}(k) \end{array}$$

The generic resp. special geometric fibers of $f_{D(s)}$ are canonically isomorphic to the generic fiber of f over ω, $\tilde{X}_\omega = \tilde{X} \times_D \mathrm{Spec}(\Omega)$, resp. the fiber $\tilde{X}_s = f^{-1}(s)$ of f over s.

We have here the situation of § 3:

$$\begin{array}{ccccc} & & \tilde{X} & & \\ & & \uparrow & & \\ \tilde{X}_{D(s)} \times_{D(s)} \mathrm{Spec}(\Omega) = \tilde{X}_\omega & \longrightarrow & \tilde{X}_{D(s)} & \longleftarrow & \tilde{X}_s = \tilde{X}_{D(s)} \times_{D(s)} \mathrm{Spec}(k) \\ \downarrow & & \downarrow{\scriptstyle f_{D(s)}} & & \downarrow \\ \mathrm{Spec}(\Omega) & \xrightarrow{\tilde{\omega}} & D(s) & \xleftarrow{s} & \mathrm{Spec}(k) \end{array}$$

By the proper base change theorem, $R^v f_*(\mathbb{Q}_{l\tilde{X}}(-))_\omega$ is canonically isomorphic to $H^v(\tilde{X}_\omega, \mathbb{Q}_{l\tilde{X}_\omega}(-)) = H^v(\tilde{X}_\omega, \mathbb{Q}_{l\tilde{X}_\omega})(-)$, and likewise $R^v f_*(\mathbb{Q}_{l\tilde{X}}(-))_s$ is canonically isomorphic to $H^v(\tilde{X}_s, \mathbb{Q}_{l\tilde{X}_\omega}(-))$.

Once again we formulate the Picard-Lefschetz formulas (4.3):

7.1 Theorem. *Let V be the $\pi_1(D-A, \omega)$-module $R^n f_*(\mathbb{Q}_{l\tilde{X}})_\omega = H^n(\tilde{X}_\omega, \mathbb{Q}_l)$.*

(0) *The sheaves $R^v f_*(\mathbb{Q}_{l\tilde{X}})$ are locally constant for $v \neq n, n+1$, hence constant on D, since the fundamental group of D vanishes.*

(1) *For each $s \in A$ there is a "vanishing cycle" δ_s in $V(m)$ that depends up to conjugation (and up to sign) only on s and not on the choice of $\tilde{\omega}$, a cohomology class δ_s^* in $R^{n+1} f_*(\mathbb{Q}_{l\tilde{X}})_s(n-m)$, and an exact sequence between the specialization mappings with respect to $\tilde{\omega}$* (I 8.7),

$$
\begin{aligned}
& & a &\to \langle a, \delta_s\rangle \\
0 \to R^n f_*(\mathbb{Q}_l\tilde{X})_s &\to R^n f_*(\mathbb{Q}_l\tilde{X})_\omega &&\to \mathbb{Q}_l(m-n) \quad \to \\
&\to R^{n+1} f_*(\mathbb{Q}_l\tilde{X})_s \to R^{n+1} f_*(\mathbb{Q}_l\tilde{X})_\omega &&\to 0 \\
& & \mathbb{Q}_l(m-n) \ni \lambda &\to \lambda\, \delta_s^*.
\end{aligned}
$$

(2) *The sheaves* $R^n f_*(\mathbb{Q}_l\tilde{X})$ *and* $R^{n+1} f_*(\mathbb{Q}_l\tilde{X})$ *are locally constant on* $D-A$. $\pi_1(D-A, \omega)$ *acts trivially on* $R^{n+1} f_*(\mathbb{Q}_l\tilde{X})$ *and tamely on* $R^n f_*(\mathbb{Q}_l\tilde{X})_\omega$. *For* $x \in R^n f_*(\mathbb{Q}_l\tilde{X})_\omega$ *and* $u \in \mathbb{Z}^{(p)}(1)$,

$$\boxed{\gamma_s(u)\, x = x - (-1)^m\, \bar{u}\langle x, \delta_s\rangle\, \delta_s\,.}$$

Here $\bar{u}$ *is the natural image of* u *in* $\mathbb{Z}_l(1) \subset \mathbb{Q}_l(1)$. *Obviously* δ_s *is uniquely determined up to sign by the choice of* γ_s.

For 7.1: Since f is smooth and proper over $D-A$, all the sheaves $R^\nu f_*(\mathbb{Q}_l\tilde{X})$ are locally constant on $D-A$. For $\nu \neq n$, $n+1$, the specialization mappings of these sheaves with respect to $s \in A$ and associated ω are isomorphisms. Hence they are locally constant on D (I 8.8). The groups $\pi_1(D(s)-\{s\}, \tilde{\omega})$ act trivially on $H^{n+1}(\tilde{X}_\omega, \mathbb{Q}_l)$. As they and their conjugates generate the fundamental group $\pi_1(D-A, \omega)$, this group also acts trivially. The action of $\pi_1(D(s)-\{s\}, \omega)$ on $H^n(\tilde{X}_\omega, \mathbb{Q}_l\tilde{X})$ factors by 4.3 through $\chi\colon \pi_1(D(s)-\{s\}, \omega) \to \hat{\mathbb{Z}}_l(1)$. Thus $\pi_1(D-A, \omega)$ acts tamely. Since $\tilde{X}$ is regular *at all points*, $\chi_\lambda = \chi$ in formula (2) of Theorem 4.3. □

7.2 Definition.

$$E = \sum_{s \in A,\, \sigma \in \pi_1^t(D-A, \omega)} \mathbb{Q}_l(-m)\, \sigma(\delta_s) \subset V$$

is called the space of vanishing cycles in V.

Remark. For suitable choice of the point $\tilde{\omega}$ for each point $s \in A$, the images of the homomorphisms γ_s, $s \in A$, already generate $\pi_1^t(D-A, \omega)$ topologically. We have then

$$E = \sum_{s \in A} \mathbb{Q}_l(-m)\, \delta_s\,.$$

7.3 Proposition. *All the vanishing cycles* δ_s *are conjugate up to sign. In particular,* E *vanishes if and only if one vanishing cycle* δ_s *is zero!*

Proof. We embed the Lefschetz pencil $\tilde{X} \to D$ in the family of all hyperplane sections of X:

$$
\begin{array}{ccccc}
\tilde{X} & \longrightarrow & Z \subset X \times \check{\mathbb{P}}^N & & \\
\downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \\
\mathrm{Spec}(\Omega) & \xrightarrow{\omega} & D & \hookrightarrow & \check{\mathbb{P}}^N.
\end{array}
$$

Here $Z = "\{(x, h) \in X \times \check{\mathbb{P}}^N \mid x \in h\}"$. The mapping g is smooth outside the dual variety $F \subset \check{\mathbb{P}}^N$ of hyperplanes $h \in \check{\mathbb{P}}^N$ that touch x in at least one point. $F \cap D = A$. F is irreducible. We can assume that the codimension of F is equal to one, as otherwise by our assumptions $A = F \cap D$ is empty. F meets D only at smooth points, and there transversally. By the definition of a Lefschetz embedding, there is a closed subset F' of codimension greater than or equal to two in $\check{\mathbb{P}}^N$ with the following property: For $x \in F - F'$, g has exactly one singular point in $g^{-1}(x)$. It is an ordinary double point and hence actually nondegenerate (1.4). By considering the corresponding local Lefschetz pencil over the strict Henselization $\tilde{\mathcal{O}}_{\check{\mathbb{P}}^N, a}$ of the local ring $\mathcal{O}_{\check{\mathbb{P}}^N, a}$ of $\check{\mathbb{P}}^N$ at the generic point a of F, we get: The stalk $R^n g_*(\mathbb{Q}_{lZ})_\omega$ is a tame $\pi_1(\check{\mathbb{P}}^N - F, \omega)$-module. We consider the surjective mapping (A I Theorem 16)

$$q\colon\ \pi_1^t(D - A, \omega) \to \pi_1^t(\check{\mathbb{P}}^N - F, \omega).$$

It follows from the proper base change theorem that

$$V = R^n f_*(\mathbb{Q}_{l\tilde{X}})_\omega" = "R^n g_*(\mathbb{Q}_{lX})_\omega .$$

Therefore the representation of $\pi_1^t(D - A, \omega)$ factors through q. On the other hand (A I.16), all the homomorphisms

$$q \circ \gamma_s\colon\ \hat{\mathbb{Z}}^{(p)}(1) \to \pi_1^t(\check{\mathbb{P}}^N - F, \omega)$$

are conjugate in $\pi_1^t(\check{\mathbb{P}}^N - F, \omega)$. Thus for s, $s' \in A$ there is an element $\sigma \in \pi_1^t(D - A, \omega)$ with

$$\gamma_{s'}(u)\, x = \sigma\, \gamma_s(u)\, \sigma^{-1}\, x \qquad \text{for all } u \in \hat{\mathbb{Z}}^{(p)}(1), \quad x \in V.$$

It follows that

$$\gamma_{s'}(u)\, x = x - (-1)^m\, \bar{u} \langle x, \sigma\, \delta_s \rangle\, \sigma\, \delta_s$$

for all $u \in \hat{\mathbb{Z}}^{(p)}(1)$, $x \in V$. Thus:

$$\sigma\, \delta_s = \pm \delta_{s'} . \quad \square$$

The representation of $\pi_1^t(D - A, \omega)$ on V is compatible with the pairing

$$\langle\ ,\ \rangle\colon\ V \times V \to \mathbb{Q}_l(-n)$$

given by Poincaré duality on $V = H^n(\tilde{X}_\omega, \mathbb{Q}_{l\tilde{X}_\omega})$. The form $\langle\ ,\ \rangle$ is alternating, since n is odd. We thus actually have a representation in the symplectic group $\mathrm{Sp}(V)$ with respect to $\langle\ ,\ \rangle$:

$$\rho\colon\ \pi_1^t(D - A, \omega) \to \mathrm{Sp}(V).$$

The form $\langle\ ,\ \rangle$ induces a nondegenerate alternating bilinear form again on $E/E \cap E^\perp$.

7.4 Corollary. *The induced representation of $\pi_1^t(D - A, \omega)$ on $E/E \cap E^\perp$ is absolutely irreducible.*

Proof. Let L be an extension field of $\mathbb{Q}_l$ and M a $\pi_1^t(D - A, \omega)$-stable vector subspace of $E \otimes L$ that is not contained in $(E \cap E^\perp) \otimes L$. Then there is an element

x in M and a point s in A with $\langle x, \delta_s\rangle \neq 0$,

$$\gamma_s(u)\, x - x = (-1)^{m+1}\, \bar{u}\langle x, \delta_s\rangle\, \delta_s \in M \qquad \text{for all } u \in \hat{\mathbb{Z}}^{(p)}(1).$$

It follows from this that δ_s lies in M, and so $E \otimes L \subset M$ by 7.3. □

7.5 (Kazhdan-Margulis). *The image of the group* $\pi_1^t(D-A, \omega)$ *under the induced mapping*

$$\pi_1^t(D-A, \omega) \to \mathrm{Sp}(E/E \cap E^\perp)$$

is open in the symplectic group $\mathrm{Sp}(E/E \cap E^\perp)$ *of* $E/E \cap E^\perp$ *with respect to the induced form* $\langle\ ,\ \rangle$.

Proof. For simplicity we select an isomorphism $\hat{\mathbb{Z}}_l \xrightarrow{\cong} \hat{\mathbb{Z}}_l(1)$. Let $M = E/E \cap E^\perp \neq 0$, G the image of $\pi_1^t(D-A, \omega)$ in the symplectic group $\mathrm{Sp}(M)$, and $\mathfrak{g}$ the Lie algebra of G. We must show that $\mathfrak{g}$ coincides with the Lie algebra of $\mathrm{Sp}(M)$. Let W be the set of vectors $\theta \in M$ for which the linear mapping

$$N(\theta)\colon\ M \to M$$

$$x \mapsto \langle x, \theta\rangle\, \theta$$

already lies in $\mathfrak{g}$. Then W is not empty, as it contains all the vanishing cycles. We have

$$\mathbb{Q}_l\, W \subset W,$$

$$u, v \in W \Rightarrow u \perp v \quad \text{or} \quad u + v \in W.$$

(Consider $\langle u, v\rangle\, N(u+v) = \langle u, v\rangle (N(u) + N(v)) + [N(v), N(u)] \in \mathfrak{g}$.) Let U be a maximal nonzero vector space contained entirely in W and θ an element of W that is not orthogonal to all of U. Then $(U - U \cap \theta^\perp) + \mathbb{Q}_l\, \theta$ is contained in W. Since W is Zariski-closed, the vector space $U + \mathbb{Q}_l\, \theta$ is thus also contained in W, and hence θ is an element of U. Thus we have

$$W = U \cup U^\perp \cap W.$$

But it follows from this that U is stable under the transformation

$$x \mapsto x + \lambda\langle x, \theta\rangle\, \theta$$

for all θ in W and $\lambda \in \mathbb{Q}_l$. But such transformations generate the group G topologically, since the images of the homomorphisms γ_s together with their conjugates generate $\pi_1^t(D-A, \omega)$ topologically. As M is an irreducible G-module, we thus have:

$$M = U = W.$$

On the other hand, the Lie algebra of the symplectic group is generated by the collection of all the linear mappings $N(v)$ for $v \in M$! □

We now want to use Theorem 7.1 to derive some exact sequences for the sheaves $R^\nu f_*(\mathbb{Q}_{l\tilde{X}})$ and several locally constant sheaves on $D-A$ associated with submodules and quotient modules of V.

7.6 Theorem. *We denote by j the embedding of $D-A$ in D. The direct image sheaves $R^\nu f_*(\mathbb{Q}_{l\tilde{X}})$ for $n \neq n$, $n+1$ are locally constant, and hence constant on D. For the two remaining direct images, we have:*

Case 1. $E=0$. Then $R^n f_*(\mathbb{Q}_{l\tilde{X}})$ is constant. The exact sequence in 7.1 (2) *gives a natural exact sequence*

$$0 \to \bigoplus_{s\in A} \mathbb{Q}_l(m-n)_{\{s\}} \to R^{n+1} f_*(\mathbb{Q}_{l\tilde{X}}) \to j_*(\mathscr{G}(R^{n+1} f_*(\mathbb{Q}_{l\tilde{X}})_\omega)) \to 0.$$

Here $\mathbb{Q}_l(m-n)_{\{s\}}$ is the sheaf concentrated on the point s with stalk $\mathbb{Q}_l(m-n)$.

Case 2. $E \neq 0$. Then $R^{n+1} f_*(\mathbb{Q}_{l\tilde{X}})$ is constant. We have

$$R^n f_*(\mathbb{Q}_{l\tilde{X}}) = j_* j^*(R^n f_*(\mathbb{Q}_{l\tilde{X}})) = j_*(\mathscr{G}(V)).$$

a) *Suppose $E \subset E^\perp$. Then* 7.1 (2) *yields an exact sequence*

$$0 \to j_*(\mathscr{G}(E^\perp)) \to R^n f_*(\mathbb{Q}_{l\tilde{X}}) \to j_*(\mathscr{G}(V/E^\perp)) \to \bigoplus_{s\in A} \mathbb{Q}_l(m-n)_{\{s\}} \to 0.$$

b) *Suppose E is not contained in $E^\perp$. Then* 7.1 (2) *yields two exact sequences*

$$0 \to j_*(\mathscr{G}(E)) \to R^n f_*(\mathbb{Q}_{l\tilde{X}}) \to j_*(\mathscr{G}(V/E)) \to 0,$$
$$0 \to j_*(\mathscr{G}(E \cap E^\perp)) \to j_*(\mathscr{G}(E)) \to j_*(\mathscr{G}(E/E \cap E^\perp)) \to 0.$$

The sheaves $j_*(\mathscr{G}(R^{n+1} f_*(\mathbb{Q}_{l\tilde{X}})))_\omega$ in Case 1 and $j_*(\mathscr{G}(E^\perp))$, $j_*(\mathscr{G}(V/E^\perp))$ in Case 2a and $j_*(\mathscr{G}(V/E))$, $j_*(\mathscr{G}(E \cap E^\perp))$, $j_*(\mathscr{G}(E/E \cap E^\perp))$ in Case 2b are constant on D.

Note. In actuality Case 2 is not thus complicated. The *hard Lefschetz Theorem* (Chap. IV, 5.6), proved by Deligne using the Weil conjecture, implies that $E \cap E^\perp = 0$, and thus $E \oplus E^\perp = V$.

Moreover, it follows that the Leray spectral sequence

$$E_2^{pq} = H^p(D, R^q f_*(\mathbb{Q}_{l\tilde{X}})) \Rightarrow H^{p+q}(\tilde{X}, \mathbb{Q}_l)$$

is degenerate at the E_2 term.

For the proof of Theorem 7.6. The images of the homomorphisms γ_s and their conjugate homomorphisms (if necessary) generate $\pi_1^t(D-A, \omega)$ topologically (A I.15). Therefore it follows from the Picard-Lefschetz formulas that $\pi_1^t(D-A, \omega)$ acts trivially on $E^\perp$ and V/E. The associated sheaves $\mathscr{G}(E^\perp)$ and $\mathscr{G}(V/E)$ are therefore constant on $D-A$, and their direct images under j are constant on D.

For the second case. In this case, every vanishing cycle δ_s is different from zero (7.3). The Picard-Lefschetz formulas show that for every point $s \in A$,

$$j_* j^*(R^n f_*(\mathbb{Q}_{l\tilde{X}}))_s = V^{\pi_1(D(s)-\{s\}, \bar{\omega})} = \delta_s^\perp = R^n f_*(\mathbb{Q}_{l\tilde{X}})_s.$$

Suppose E is not contained in $E^\perp$. By 7.3, then, $\mathbb{Q}_l(-m)\,\delta_s$ is not contained in $E^\perp$ for any $s\in A$, and hence E is not contained in $\delta_s^\perp = R^n f_*(\mathbb{Q}_{l\tilde{X}})_s$ for any $s\in A$. As $R^n f_*(\mathbb{Q}_{l\tilde{X}})_s$ has codimension one in $V=R^n f_*(\mathbb{Q}_{l\tilde{X}})_\omega$, the mapping

$$R^n f_*(\mathbb{Q}_{l\tilde{X}})_s \to j_*(\mathcal{G}(V/E))_s = V/E$$

is surjective. The mapping

$$j_*(\mathcal{G}(E)) \to j_*(\mathcal{G}(E/E\cap E^\perp))$$

is surjective, since for every point $s\in A$ we have

$$j_*(\mathcal{G}(E))_s = E^{\pi_1(D(s)-\{s\},\bar{\omega})} = E\cap\delta_s^\perp,$$
$$j_*(\mathcal{G}(E/E\cap E^\perp))_s = E\cap\delta_s^\perp/E\cap E^\perp. \quad \square$$

Let k be the separable closure of a subfield k_0.

We want to suppose that the Lefschetz pencil $\tilde{X}\to D$ is already defined over k_0:

$$\begin{array}{ccccc}
\tilde{X}_0 & \longleftarrow & \tilde{X}=\tilde{X}_0\times\mathrm{Spec}(k) & & \\
\downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle f=f_0\times\mathrm{Spec}(k)} & & \\
D_0=\mathbb{P}^1_{k_0} & \longleftarrow & D & \overset{\omega}{\longleftarrow} & \mathrm{Spec}(\Omega) \\
\downarrow & & \downarrow & & \\
\mathrm{Spec}(k_0) & \longleftarrow & \mathrm{Spec}(k) & &
\end{array}$$

We still denote by ω the geometric points

$$\mathrm{Spec}(\Omega)\to D_0 \quad \text{and} \quad \mathrm{Spec}(\Omega)\to\mathrm{Spec}(k_0)$$

induced by ω. Let A be the inverse image of a set A_0 of points

$$s\colon\ \mathrm{Spec}(k_0)\to D_0$$

with coordinates in k_0. We identify A_0 with A. The singular points of f_0 are then rational over k_0. We denote by $D_0(s)$ the spectrum of the (not strict) Henselization of the local ring $\mathcal{O}_{D_0,s}$ at $s\in A_0$.

$$\begin{array}{ccccc}
D & \longleftarrow & D(s)=D_0(s)\times\mathrm{Spec}(k) & \overset{\tilde{\omega}}{\longleftarrow} & \mathrm{Spec}(\Omega) \\
\downarrow & & \downarrow & \swarrow{\scriptstyle \tilde{\omega}} & \\
D_0 & \longleftarrow & D_0(s) & &
\end{array}$$

We likewise use s to denote the special points $\mathrm{Spec}(k)\to D(s)$ and $\mathrm{Spec}(k_0)\to D_0(s)$.

We consider the exact sequences

$$\begin{array}{ccccccc}
0 & \longrightarrow & \pi_1(D(s)-\{s\}, \tilde{\omega}) & \longrightarrow & \pi_1(D_0(s)-\{s\}, \tilde{\omega}) & \longrightarrow & \\
 & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \pi_1(D-A, \omega) & \longrightarrow & \pi_1(D-A, \omega) & \longrightarrow & \\
 & & & & & \longrightarrow & \pi_1(\mathrm{Spec}(k_0), \omega)=\mathrm{Gal}(k/k_0) \longrightarrow 0 \\
 & & & & & & \| \\
 & & & & & \longrightarrow & \pi_1(\mathrm{Spec}(k_0), \omega) \longrightarrow 0.
\end{array}$$

The image G_s of $\pi_1(D_0(s)-\{s\}, \tilde{\omega})$ in $\pi_1(D_0-A_0, \tilde{\omega})$ thus satisfies

$$\pi_1(D-A, \omega)\cdot G_s = G_s\, \pi_1(D-A, \omega) = \pi_1(D_0-A_0, \omega).$$

For the homomorphism

$$\gamma_s\colon \hat{\mathbb{Z}}^{(p)}(1)=\hat{\mathbb{Z}}^{(p)}(1)(k)=\varprojlim_{p\nmid r} \mu_r(k)\to\pi_1^t(D-A, \omega),$$

an element σ in G_s (or more precisely, the image in $\pi_1^t(D_0-A_0, \omega)$ with the image $\bar{\sigma}$ in $\mathrm{Gal}(k/k_0)$) satisfies

$$(*) \qquad \sigma\, \gamma_s(u)\, \sigma^{-1} = \gamma_s(\bar{\sigma}(u)).$$

$V=R^n f_*(\mathbb{Q}_{l\tilde{X}})_\omega$ is canonically isomorphic to $R^n f_{0*}(\mathbb{Q}_{l\tilde{X}_0})_\omega$ and is thus actually a $\pi_1(D_0-A_0, \omega)$-module. For the vanishing cycle δ_s, equation $(*)$ implies

$$\gamma_s(u)x = x-(-1)^m\, \bar{u}\langle x, \sigma^{-1}\delta_s\rangle\, \sigma^{-1}\delta_s$$

$$\sigma^{-1}\delta_s = \pm\delta_s,$$

and thus

$$\pi_1(D_0-A_0, \omega)\, \delta_s = \pi_1(D-A, \omega)\cdot G_s\, \delta_s \subseteq E.$$

E, $E^\perp$, and all the other vector spaces occurring in Theorem 7.6 are actually $\pi_1(D_0-A_0, \omega)$-modules, and thus the associated locally constant sheaves are already defined in a natural way on D_0-A_0.

After a finite extension of k_0, we can assume that

$$\sigma\, \delta_s = \delta_s, \qquad s\in G_s.$$

Likewise, by a finite extension of k_0 we can make sure that the mappings

$$\mathbb{Q}_l(m-n)\to R^{n+1}f_*(\mathbb{Q}_{l\tilde{X}})_s = R^{n+1}f_{0*}(\mathbb{Q}_{l\tilde{X}_0})_s \qquad \text{for all } s\in A$$

are $\pi_1(\mathrm{Spec}(k_0), \omega)$-equivariant. (See also the construction of the class δ_s^* in 4.3. One can also use the fact that these mapping factor through the equivariant mapping $H^{n+1}_{\{a\}}(\tilde{X}_s, \mathbb{Q}_l)\to H^{n+1}(\tilde{X}_s, \mathbb{Q}_l)$ and compute the local cohomology groups $H^{n+1}_{\{a\}}(-,-)$ at the double point a_s over s: for instance, $H^{n+1}_{\{a\}}(\tilde{X}_s, \mathbb{Q}_l) = \mathbb{Q}_l(m-n)$ after a quadratic extension.)

7.7 Proposition. *Let k be the separable closure of a subfield k_0. After a finite extension of k_0, the following can be guaranteed:*

(1) *The Lefschetz pencil* $\tilde{X} \to D$ *arises by base field extension from a pencil* $\tilde{X}_0 \to D_0 = \mathbb{P}^1_{k_0}$ *defined over* k_0.

(2) *The singular points of* f_0 *are rational over* k_0.

(3) *All the sheaves and exact sequences occurring in Theorem 7.6 are already defined in a natural way on* D_0 *resp.* $D_0 - A_0$.

Remark. Naturally $\mathbb{Q}_l(v)$ cannot be identified with $\mathbb{Q}_l$ over k_0!

Chapter IV. Deligne's Proof of the Weil Conjecture

§1. Formulation of the Weil Conjecture

In this section we shall formulate the Weil conjecture about the ζ-function of a smooth projective variety over a finite field and fix the notation used in what follows. We continue from where we left off in the last section of the second chapter.

Let κ be a finite field of q elements, k the algebraic closure of κ, and l a prime number that does not divide q. We consider algebraic varieties (i.e., finitely generated schemes) S_0 over κ. These will always be written with the subscript "0", and their extension to k without the subscript "0":

$$S = S_0 \times_{\operatorname{Spec}\kappa} \operatorname{Spec} k.$$

We use the corresponding notation for mappings and sheaves. Let $\mathscr{G}_0$ be a constructible l-adic sheaf on the algebraic variety S_0 over κ and $\mathscr{G}$ its inverse image on S. We recall the construction of the Weil-Grothendieck L-series for the sheaf $\mathscr{G}_0$ (Chap. II, 4.3):

$$Z_{\mathscr{G}_0}(t) = \prod_{\alpha \in \operatorname{Ab}(S_0)} \frac{1}{\det(1 - t^{d(\alpha)} f_{\alpha,(\mathscr{G}_0)_\alpha}^{-1})}.$$

Here α runs through a system of representatives of geometric points

$$\alpha\colon \operatorname{Spec} k \to S_0 \quad (\text{over } \kappa)$$

for the set of closed points $\operatorname{Ab}(S_0)$ of S_0, and $d(\alpha)$ is the degree of the field of coordinates $k(\alpha) \subset k$ of α over the base field κ. Finally,

$$\begin{aligned} f_\alpha\colon\ & k \to k \\ & x \mapsto x^{\#\kappa(\alpha)} \end{aligned}$$

is the Frobenius morphism in the Galois group

$$G_\alpha = \operatorname{Gal}(k/\kappa(\alpha)).$$

This group acts on the stalk $(\mathscr{G}_0)_\alpha$, with the homomorphism

$$f_{\alpha,(\mathscr{G}_0)_\alpha}\colon (\mathscr{G}_0)_\alpha \to (\mathscr{G}_0)_\alpha$$

corresponding to the element $f_\alpha \in G_\alpha$. The formal power series $Z_{\mathscr{G}_0}(t)$ is multiplicative under short exact sequences of l-adic sheaves and is identically 1 for torsion sheaves. It therefore depends only on the $\mathbb{Q}_l$-sheaf associated with $\mathscr{G}_0$ (Chap. I,

§12). If nothing is explicitly said to the contrary, all sheaves from now on will be (constructible) sheaves of $\mathbb{Q}_l$-vector spaces.

We want to look in more detail at the special case of *locally constant* $\mathbb{Q}_l$-sheaves.

Let ω be a ("generic", say) geometric point of S over k with values in a separably closed extension field Ω of k. ω induces in an obvious way geometric points of S_0 and of $\operatorname{Spec}\kappa$ over κ, and for simplicity we shall denote them also by ω. Let S be irreducible. Then we have the exact sequence of fundamental groups

$$\begin{array}{ccccccccc} 0 & \to & \pi_1(S,\omega) & \to & \pi_1(S_0,\omega) & \to & \pi_1(\operatorname{Spec}\kappa,\omega) & \to & 0 \\ & & & & & & \| & & \\ & & & & & & \operatorname{Gal}(k/\kappa) & & \end{array}$$

and a canonical isomorphism

$$\operatorname{Gal}(k/\kappa)\cong\hat{\mathbb{Z}}=\varprojlim_{0<m\in\mathbb{Z}}\mathbb{Z}/m\mathbb{Z}$$

$$(x\mapsto x^q)\mapsto 1.$$

Suppose we are given a continuous representation of $\pi_1(S_0,\omega)$ on a finite-dimensional $\mathbb{Q}_l$-vector space. This representation defines a locally constant sheaf $\mathscr{G}_0=\mathscr{G}_0(V)$ (A I.8). We write

$$Z_V(t)=Z_{\mathscr{G}_0}(t).$$

This L-series can be described as follows using the representation itself. Let

$$\alpha\colon\ \operatorname{Spec}k\to S_0\quad(\text{over }\kappa)$$

be a geometric point with coordinate field $\kappa(\alpha)\subset k$. The embedding $\kappa(\alpha)\subset k$ defines a *geometric point* $j\colon\operatorname{Spec}(k)\to\operatorname{Spec}(\kappa(\alpha))$. This induces a homomorphism

$$\operatorname{Gal}(k/\kappa(\alpha))=\pi_1(\operatorname{Spec}\kappa(\alpha),j)\to\pi_1(S_0,\alpha)\to\pi_1(S_0,\omega).$$

The isomorphism $\pi_1(S_0,\alpha)\to\pi_1(S_0,\omega)$ is determined up to conjugation (Appendix I).

Let the image of the Frobenius isomorphism (over $\kappa(\alpha)$)

$$\begin{aligned} &k\to k\\ &x\mapsto x^{q^{d(\alpha)}}\quad(d(\alpha)=[\kappa(\alpha)\colon\kappa]) \end{aligned}$$

in $\pi_1(S_0,w)$ be ϕ_α; ϕ_α is determined up to conjugation and acts on V. We have

$$\det(1-f_{\alpha,(\mathscr{G}_0)_\alpha}^{-1}t)=\det(1-\phi_\alpha^{-1}t)$$

and hence

$$Z_V(t)=\prod_{\alpha\in\operatorname{Ab}(S_0)}\frac{1}{\det(1-\phi_\alpha^{-1}t^{d(\alpha)})}.$$

The degree $d(\alpha)$ is precisely the image of ϕ_α in $\hat{\mathbb{Z}}$.

We recall Grothendieck's formula (II 4.5).

1.1 Proposition. *Let $\mathscr{G}_0$ be a constructible $\mathbb{Q}_l$-sheaf on S_0. Then*

$$Z_{\mathscr{G}_0}(t)=\prod_{\nu}\det(1-t\,\mathrm{Fr}^*\,|\,H^\nu(S,\mathscr{G}))^{(-1)^{\nu+1}}.$$

Here $\mathrm{Fr}^*\,|\,H^\nu(S,\mathscr{G})$ *denotes the mapping induced on the cohomology groups* (Chap. II, § 4) *by the Frobenius mapping*

$$\mathrm{Fr}_S\colon\ S\to S$$

$$\mathrm{Fr}_{\mathscr{G}}\colon\ (\mathrm{Fr}_S)^*(\mathscr{G})\to\mathscr{G}.$$

The conjecture stated by Weil and proved by Deligne for the L-series $Z_{S_0}(t)$ of the constant sheaf $\mathbb{Q}_{lS_0}$ runs as follows:

1.2 Theorem (Deligne). *Let S_0 be smooth and projective over the finite field κ with q elements.*

1) *The polynomials*

$$P_\nu(t)=\det(1-t\,\mathrm{Fr}^*\,|\,H^\nu(S,\mathbb{Q}_l))$$

in $\mathbb{Q}_l[t]$ have rational integer coefficients. These are independent of l.

2) *The eigenvalues λ of $\mathrm{Fr}^*\,|\,H^\nu(S,\mathbb{Q}_l)$, and thus the reciprocal roots of $P_\nu(t)$, all have the complex absolute value*

$$|\lambda|=q^{\nu/2}.$$

(The reciprocal roots of $P_\nu(t)$ are algebraic integers. It follows from 1) and 2) that all conjugates of λ have absolute value $q^{\nu/2}$.)

3) *There is a functional equation for*

$$Z_{S_0}(t)=\prod_{\nu}P_\nu(t)^{(-1)^{\nu+1}},$$

namely

$$Z_{S_0}(1/q^n\,t)=\varepsilon\,q^{\frac{1}{2}n\chi(S)}\,t^{\chi(S)}\,Z_{S_0}(t).$$

Here

$$\chi(S)=\sum_{\nu}(-1)^\nu\dim H^\nu(S,\mathbb{Q}_l)$$

and

$$\varepsilon=\begin{cases}1 & \text{if } 2\nmid n\\ (-1)^N & \text{if } 2\mid n\end{cases}$$

where N is the multiplicity of the eigenvalue $q^{n/2}$ of $\mathrm{Fr}^*\,|\,H^n(S,\mathbb{Q}_l)$.

Remark. a) Let S be irreducible. Then

$$P_0(t)=1-t,\qquad P_{2\dim S_0}(t)=1-q^{\dim S_0}\,t.$$

b) If S_0 arises by good reduction of a smooth projective variety Y over $\mathbb{C}$, then the degree of P_ν is equal to the ν-th Betti number of Y,

$$\deg(P_\nu)=\dim H^\nu(Y,\mathbb{C}).$$

The heart of Deligne's theorem is the second assertion about the complex absolute values of the roots of the polynomials $P_\nu(t)$ and their conjugates. Indeed,

the functional equation is a simple consequence of the Poincaré duality theorem (Chap. II, 1.13). Remark b) comes from the specialization theorem 8.5 of Chapter I and the comparison theorem with singular cohomology (Chap. I, 11.6).

We shall now show that assertion 1) on the integrality of the coefficients of the polynomials $P_n(t)$, and hence Theorem 1.2, is a consequence of the following, seemingly weaker theorem:

Theorem 1.2′. *Assume the hypotheses of Theorem 1.2. Then all the eigenvalues of the Frobenius homomorphisms*

$$\mathrm{Fr}^* | H^\nu(S, \mathbb{Q}_{lS}) \colon H^\nu(S, \mathbb{Q}_{lS}) \to H^\nu(S, \mathbb{Q}_{lS})$$

are algebraic numbers. These eigenvalues and all their conjugates λ have complex absolute value

$$|\lambda| = q^{\nu/2}.$$

Let $\bar{\mathbb{Q}}_l$ be the algebraic closure of the field $\mathbb{Q}_l$ of l-adic numbers. The Galois group $\mathrm{Gal}(\bar{\mathbb{Q}}_l/\mathbb{Q})$ of $\bar{\mathbb{Q}}_l$ over the field of rational numbers acts in a natural way on the coefficients in the ring $\bar{\mathbb{Q}}_l[[t]]$ of formal power series over $\bar{\mathbb{Q}}_l$, and hence on its fraction field. The power series

$$Z_{S_0}(t) = \prod_\nu P_\nu(t)^{(-1)^{\nu+1}}$$

has rational integer coefficients from its definition as an infinite product. Take $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}_l/\mathbb{Q})$. Then $\sigma(Z_{S_0}(t)) = Z_{S_0}(t)$, and so

$$\prod_{\nu=0}^{2\dim S_0} \sigma(P_\nu(t))^{(-1)^{\nu+1}} = \prod_{\nu=0}^{2\dim S_0} P_\nu(t)^{(-1)^{\nu+1}}.$$

But it follows from Theorem 1.2′ that the polynomials $P_\nu(t)$ and $\sigma(P_\mu(t))$ for $\nu \neq \mu$ must be relatively prime! Because of the normalization $P_\nu(0) = \sigma(P_\nu)(0) = 1$, it thus follows that

$$P_\nu(t) = \sigma(P_\nu(t)) \quad \text{for all } \sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}_l/\mathbb{Q}).$$

Therefore the coefficients of these polynomials lie in $\mathbb{Q}$. We want to show that they are actually integers. For an arbitrary prime p let $\bar{\mathbb{Q}}_p$ be the algebraic closure of the field $\mathbb{Q}_p$ of p-adic numbers and $|x|_p$ the p-adic absolute value defined on $\bar{\mathbb{Q}}_p$. Let

$$P(t) = \prod_{2 \nmid \nu} P_\nu(t), \qquad Q(t) = \prod_{2 | \nu} P_\nu(t).$$

It follows from Theorem 1.2′ that $P(t)$ and $Q(t)$ have no common roots and thus are relatively prime.

$$P(t) = Q(t) \cdot Z_{S_0}(t).$$

Let $\lambda \in \bar{\mathbb{Q}}_p$ be a root of $Q(t)$ with $|\lambda|_p < 1$. As $Z_{S_0}(t)$ has integer coefficients, $Z_{S_0}(x)$ converges for all $x \in \bar{\mathbb{Q}}_p$ with $|x|_p < 1$. Hence

$$P(\lambda) = Q(\lambda)\, Z_{S_0}(\lambda) = 0.$$

But that is a contradiction. The reciprocal roots of $Q(t)$ are thus *integral* algebraic numbers. Because $Q(0)=1$, the rational coefficients of $Q(t)$ are actually integers. The same is true for the coefficients of $P(t)=Q(t)\,Z_{S_0}(t)$. As the polynomials $P_\nu(t)$ are relatively prime to each other, it follows from $P_\nu(0)=1$ and the familiar Gauss's Lemma that all the polynomials $P_\nu(t)$ also have integer coefficients. □

It suffices to prove Theorem 1.2′ after base field extension. We can assume that S_0 is absolutely irreducible.

The following sections are devoted to presenting Deligne's proof of Theorem 1.2′.

§ 2. The Fundamental Estimate

In this section we assume that S_0 is a smooth, absolutely irreducible affine curve over the finite field κ. In addition, we again suppose that there is given a continuous representation

$$\rho\colon\ \pi_1(S_0,\omega)\to \mathrm{GL}(V)$$

on a finite-dimensional $\mathbb{Q}_l$-vector space V. Let $\mathscr{G}_0$ be the locally constant sheaf associated with V, and $\mathscr{G}$ its inverse image on $S=S_0\times \mathrm{Spec}\,k$.

We want to study the eigenvalues of the Frobenius homomorphism on $H_c^i(S,\mathscr{G})$ under certain hypotheses on (V,ρ). Here only the cases $i=1$ and 2 occur; indeed, these cohomology groups vanish for $i>2$, as S is one-dimensional, and for $i=0$ as S is affine. In particular, the L-series for $\mathscr{G}_0$ has the form

$$Z_{\mathscr{G}_0}(t)=P^1(t)/P^2(t),$$
$$P^i(t)=\det(1-t\,\mathrm{Fr}^*\,|\,H_c^i(S,\mathscr{G})).$$

It falls out immediately from the construction of the sheaf $\mathscr{G}_0$ that there is a canonical isomorphism

$$H^0(S,\mathscr{G})=V^\pi,\qquad \pi=\pi_1(S,\omega).$$

The group

$$\mathrm{Gal}(k/\kappa)=\pi_1(S_0,\omega)/\pi_1(S,\omega)$$

acts on the space V^π of π-invariants. The Frobenius element $\phi\in\mathrm{Gal}(k/\kappa)$ corresponds to the "arithmetic Frobenius homomorphism"

$$\bar\rho(\phi)\colon\ V^\pi\to V^\pi.$$

This is the inverse of the "geometric Frobenius homomorphism"

$$\mathrm{Fr}^*\colon\ H^0(S,\mathscr{G})\to H^0(S,\mathscr{G}).$$

In §1 we assigned to each geometric point

$$\alpha\colon\ \mathrm{Spec}\,k\to S\quad(\text{over }\kappa)$$

an element

$$\phi_\alpha\in\pi_1(S_0,\omega)$$

uniquely determined up to conjugacy, and thereby also a "local Frobenius homomorphism"

$$\rho(\phi_\alpha)\colon\ V \to V \quad (V \cong \mathscr{G}_\alpha)$$

likewise determined uniquely up to conjugacy. It is immediately clear that *the local Frobenius homomorphism* $\rho(\phi_\alpha)$ *leaves* V^π *invariant and acts on* V^π *as the* $d(\alpha)$-th *power of the arithmetic Frobenius homomorphism* $\bar{\rho}(\phi)^{d(\alpha)}$,

$$\bar{\rho}(\phi)^{-1} = \mathrm{Fr}^*.$$

2.1 Remark. Suppose the characteristic polynomial of a local Frobenius homomorphism

$$\rho(\phi_\alpha)\colon\ V \to V$$

has rational coefficients (that is, lying in $\mathbb{Q}$, not just in $\mathbb{Q}_l$). Then the eigenvalues of

$$\mathrm{Fr}^*\colon\ H_c^i(S, \mathscr{G}) \to H_c^i(S, \mathscr{G})$$

are algebraic numbers (that is, lying in $\bar{\mathbb{Q}}$, not just in $\bar{\mathbb{Q}}_l$).

Proof. We know from §1 that the L-series $Z_{\mathscr{G}_0}(t)$, the quotient of the two polynomials occurring, has rational coefficients. Hence it suffices to treat the case $i=2$. We apply *Poincaré duality:*

$$H_c^2(S, \mathscr{G}) \cong \mathrm{Hom}_{\mathbb{Q}_l}(H^0(S, \check{\mathscr{G}}), \mathbb{Q}_l) \otimes \mathbb{Q}_l(-1).$$

This isomorphism is compatible with the action of the Frobenius homomorphism Fr* (which acts nontrivially on $\mathbb{Q}_l(-1)$). It is easy to see that the dual sheaf $\check{\mathscr{G}}$ is the locally constant sheaf associated with the dual space

$$V^* = \mathrm{Hom}_{\mathbb{Q}_l}(V, \mathbb{Q}_l)$$

(equipped with the representation contragredient to ρ). We get

$$H_c^2(S, \mathscr{G}) = V_\pi \otimes \mathbb{Q}_l(-1).$$

Here

$$V_\pi = (V^{*\pi})^*$$

is the largest *quotient space* of V on which $\pi_1(S, \omega)$ acts trivially. The natural projection $V \to V_\pi$ is compatible with the action of $\rho(\phi_\alpha)$ on V and of $\bar{\rho}(\phi)^{d(\alpha)}$ on V_π. It follows from the hypothesis of 2.1 that the eigenvalues of $\bar{\rho}(\phi)$ on V_π are algebraic numbers, and hence that the eigenvalues of $\mathrm{Fr}^*\colon H_c^2(S, \mathscr{G}) \to H_c^2(S, \mathscr{G})$ are algebraic. □

In the rest of this section we require the following

2.2 Hypotheses. 1) *The characteristic polynomials of the local Frobenius homomorphisms*

$$\rho(\phi_\alpha)\colon\ V \to V$$

have rational coefficients.

2) *There is a nondegenerate alternating bilinear form*

$$\psi\colon V\times V\to\mathbb{Q}_l,\qquad \psi(x,y)=\langle x,y\rangle$$

and a number $a\in\mathbb{Z}$ *such that*

$$\langle\rho(\sigma)x,\rho(\sigma)y\rangle=q^{-ad(\sigma)}\langle x,y\rangle\qquad\text{for }\sigma\in\pi_1(S_0,\omega).$$

The image of $\pi_1(S,\omega)$ *in the symplectic group* $\mathrm{Sp}(\psi)$ *is Zariski-dense.*

As a reminder:

$d(\sigma)$ is the image of σ in $\hat{\mathbb{Z}}$ under the mapping

$$\pi_1(S_0,\omega)\to\mathrm{Gal}(k/\kappa)\cong\hat{\mathbb{Z}}.$$

When $\sigma=\phi_\alpha$, then $d(\sigma)=d(\alpha)$, the degree of the geometric point α.

Note. In our applications (§ 4), we will have

$$\mathscr{G}_0=R^nf_*((\mathbb{Q}_l)_{X_0}),$$

where $f_0\colon X_0\to S_0$ is the smooth part of a Lefschetz pencil of odd fiber dimension $n=a$. The pairing postulated in 2.2 is the duality pairing. The fact that the Hypotheses 2.2 are satisfied comes from the monodromy theory (see § 3, § 4).

In what follows, besides (V,ρ) we also need the higher tensor power representations $(V^{\otimes k},\rho^{\otimes k})=\rho_k$. The locally constant sheaves associated with them are precisely

$$\mathscr{G}_0^{\otimes k}=\mathscr{G}_0\otimes_{\mathbb{Q}_l}\cdots\otimes\mathscr{G}_0,\quad\mathscr{G}^{\otimes k}.$$

We denote their L-series by

$$Z_k(t)=P_k^1(t)/P_k^2(t),$$
$$P_k^i(t)=\det(1-t\,\mathrm{Fr}^*\,|\,H_c^i(S,\mathscr{G}^{\otimes k})).$$

The roots of $P_k^i(t)$ are algebraic numbers, as the Hypotheses of 2.1 carry over to tensor powers.

2.3 Lemma. *Assume the Hypotheses* (2.2). *The eigenvalues* λ *of*

$$\mathrm{Fr}^*\colon (V^{\otimes k})_\pi\to(V^{\otimes k})_\pi$$

(that is, all their complex conjugates) have complex absolute value

$$|\lambda|=q^{ka/2}.$$

Corollary. *The poles of the L-series*

$$Z_k(t)=P_k^1(t)/P_k^2(t)$$

all lie on the circle of radius

$$|t|=q^{-(1+ka/2)}.$$

(The corollary follows from $H_c^2(S,\mathscr{G}^{\otimes k})=(V^{\otimes k})_\pi\otimes\mathbb{Q}_l(-1)$.)

Proof of 2.3. We write

$\mathrm{Sp}(\psi)$ for the symplectic group of ψ,

$\mathrm{A}(\psi)$ for the group of symplectic similarities, and

$$\mathrm{A}_0(\psi) \subset \mathrm{A}(\psi)$$

for the subgroup generated by $\mathrm{Sp}(\psi)$ and the "homotheties"

$$v \mapsto a\,v, \qquad a \in \mathbb{Q}_l^*.$$

$\mathrm{A}_0(\psi)$ *has finite index in* $\mathrm{A}(\psi)$.

By hypothesis, the image of the group $\pi = \pi_1(S, \omega)$ is Zariski-dense in $\mathrm{Sp}(\psi)$. Hence the group $\mathrm{A}(\psi)/\mathrm{Sp}(\psi)$ acts on $(V^{\otimes k})_\pi$. As this group is abelian, for every eigenvalue λ_0 of a fixed element, say of $\rho_k(\phi_\alpha)$, there exists a character

$$\lambda\colon\ \mathrm{A}(\psi)/\mathrm{Sp}(\psi) \to \bar{\mathbb{Q}}_l, \qquad \lambda(\rho_k(\phi_\alpha)) = \lambda_0.$$

$\lambda(g)$ is an eigenvalue of g on $(V^{\otimes k})_\pi \otimes \bar{\mathbb{Q}}_l$ for all $g \in \mathrm{A}(\psi)/\mathrm{Sp}(\psi)$. (Consider the eigenspace of $\rho(\phi_\alpha)$ for λ_0 in $(V^{\otimes k})_\pi \otimes \bar{\mathbb{Q}}_l$. The abelian group $\mathrm{A}(\psi)/\mathrm{Sp}(\psi)$ acts on this and has an eigenvector.)

We want to compare the character λ with the determinant character

$$\chi\colon\ \mathrm{A}(\psi)/\mathrm{Sp}(\psi) \to \mathbb{Q}_l^*$$
$$u \mapsto \det u.$$

The characters $\lambda^{\dim V}$ and χ^k obviously agree on the subgroup $\mathrm{A}_0(\psi)/\mathrm{Sp}(\psi)$ generated by the homotheties $(v \mapsto a\,v)$. As this has finite index in $\mathrm{A}(\psi)/\mathrm{Sp}(\psi)$, it follows that

The character $\lambda^n \chi^{-k}$, $n = \dim V$, *has finite order.*

In particular, the n-th power of the eigenvalue λ_0 (of $\rho(\phi_\alpha)$) differs from $\det(\rho(\phi_\alpha))^k$ only by a root of unity. By hypothesis $\rho(\phi_\alpha)$ is a symplectic similarity with the distorsion factor $q^{-ad(\alpha)}$. Hence the determinant of $\rho(\phi_\alpha)$ is $q^{-nad(\alpha)/2}$. We get

$$|\lambda_0| = q^{-ad(\alpha)k/2},$$

and Assertion 2.3 follows from this ($\mathrm{Fr}^* = \rho(\phi^{-1})$, $\phi^{d(\alpha)} = \phi_\alpha$). □

We now use Lemma 2.3 to obtain a statement about the eigenvalues of

$$\rho(\phi_\alpha)\colon\ V \to V.$$

The L-series $Z_k(t)$ for $(V^{\otimes k}, \rho^{\otimes k})$ is by definition the product of the local factors

$$\det(1 - t^{d(\alpha)}\, \rho^{\otimes k}(\phi_\alpha^{-1}) \mid V^{\otimes k})^{-1}.$$

We assume now that k is *even*. Then these local factors are power series in t with nonnegative rational coefficients in their expansions and leading coefficient 1. This follows from the formulas

$$\det(1 - t\,\tau)^{-1} = \exp\left(\sum_{n=1}^{\infty} t^n \operatorname{Tr}(\tau^n)/n\right)$$

$$\operatorname{Tr}(\tau \otimes \tau) = \operatorname{Tr}(\tau)^2 \qquad \text{for } \tau \in M_n(\mathbb{Q}_l).$$

Consequently the radius of convergence for each factor is at least as large as that for the product $Z_k(t)$, thus at least $q^{-(1+ka/2)}$. The poles of the factors thus lie outside this circle. Let λ be an eigenvalue of $\rho(\phi_\alpha)$. Then λ^k is an eigenvalue of $\rho^{\otimes k}(\phi_\alpha)$. For even k it follows that

$$|\lambda^k| \geqq q^{-(1+ka/2)d(\alpha)}.$$

Taking the k-th root and passing to the limit as $k \to \infty$, we get

$$|\lambda| \geqq q^{-ad(\alpha)/2}.$$

It is easy to see that the Hypotheses 2.2 carry over from (V, ρ) to the contragredient representation, with $-a$ in place of a. Hence we get

$$|\lambda^{-1}| \geqq q^{ad(\alpha)/2},$$

and thus actually

$$|\lambda| = a^{-ad(\alpha)/2}.$$

We record this:

2.4 Lemma. *Under the Hypotheses* 2.2, *the eigenvalues of the local Frobenius homomorphisms* $\rho(\phi_\alpha)$ *are algebraic numbers whose conjugates have the complex absolute value*

$$|\lambda| = q^{-ad(\alpha)/2}.$$

We apply this lemma to estimate the radius of convergence of the infinite product

$$\prod_{\alpha \in \mathrm{Ab}(S)} \det(1 - t^{d(\alpha)} \rho(\phi_\alpha^{-1}) | V).$$

If we denote the eigenvalues of $\rho(\phi_\alpha^{-1})$ by

$$\lambda_1(\alpha), \ldots, \lambda_n(\alpha),$$

the product can be written in the form

$$\prod [(1 - \lambda_1(\alpha)\, t^{d(\alpha)}) \ldots (1 - \lambda_n(\alpha)\, t^{d(\alpha)})].$$

It converges when the series

$$\sum_{1 \leqq i \leqq h} \sum_{\alpha} |\lambda_i(\alpha)|\, t^{d(\alpha)} = n \sum_{\alpha} q^{d(\alpha)a/2}\, t^{d(\alpha)}$$

converges (absolutely).

Assertion. The radius of convergence of this series is $q^{-(1+a/2)}$.

Proof. If S_0 is the affine line, there exist exactly $q^n - q^{n-1}$ points of a given degree $n = d(\alpha)$. (This case is sufficient for us, as in our applications S_0 is an open subset of the projective line. For the general case, view S as a cover of the affine line and prove that the number of points for a given degree can be bounded by (constant)$\cdot q^n$.) The series above is essentially equal to

$$\sum_{n=0}^{\infty} [q^{1+a/2}\, t]^n,$$

and the assertion is evident. □

Thus the L-series has no zeros (and also no poles) in the circle

$$|t| < q^{-(1+a/2)}.$$

The roots of $P_1(t)$ must therefore lie outside this circle (use 2.3). This gives us the *fundamental estimate:*

2.5 Proposition. *Under the Hypotheses* 2.2, *all eigenvalues of*

$$\mathrm{Fr}^*\colon\ H_c^1(S, \mathscr{G}) \to H_c^1(S, \mathscr{G})$$

are algebraic numbers; the complex conjugates λ *satisfy the estimate*

$$|\lambda| \leqq q^{a/2+1}.$$

The idea of considering higher tensor powers of (V, ρ) and their L-series, and thereby improving the estimates available, was suggested by a classical work of Rankin [122] on the Ramanujan conjecture. In this work Rankin used the L-series

$$\sum \tau(n)^2\, n^{-s}$$

instead of $\sum \tau(n)\, n^{-s}$ (the Mellin transform of the modular form Δ) to estimate the Ramanujan function τ.

§ 3. A Rationality Proposition

Let S_0 be an *absolutely* irreducible algebraic variety of dimension one over the finite field κ. As in the first section, we consider the short exact sequence of fundamental groups,

$$0 \to \pi_1(S, \omega) \to \pi_1(S_0, \omega) \to \pi_1(\mathrm{Spec}(\kappa), \omega) = \mathrm{Gal}(k/\kappa) \to 0$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \| \wr$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \hat{\mathbb{Z}}$$
$$\sigma \mapsto d(\sigma).$$

Here ω again denotes a geometric point of S in a separably closed extension field Ω of k and also the induced points of S_0 and $\mathrm{Spec}(\kappa)$.

We consider a finite-dimensional vector space V over the field $\mathbb{Q}_l$ of l-adic numbers together with a nondegenerate alternating bilinear form ψ with values in $\mathbb{Q}_l$ and a continuous representation of $\pi_1(S_0, \omega)$ on V satisfying

$$\psi(\sigma x, \sigma y) = q^{nd(\sigma)}\, \psi(x, y), \qquad q = \#(\kappa).$$

Here n is a fixed prescribed nonzero integer. In particular, the image of $\pi_1(S, \omega)$ lies in the symplectic group with respect to ψ.

If α is a geometric point of S_0 over κ with values in k, then again as in the first section we denote by ϕ_α the associated "arithmetic" Frobenius element in $\pi_1(S_0, \omega)$, determined up to conjugacy,

$$d(\phi_\alpha) = d(\alpha).$$

3.1 Proposition. *Suppose the image of $\pi_1(S, \omega)$ is open in the symplectic group $\mathrm{Sp}_\psi(\mathbb{Q}_l)$ with respect to ψ with coordinates in $\mathbb{Q}_l$. Let $\lambda_1, \ldots, \lambda_r, \mu_1, \ldots, \mu_s$ be non zero elements in the algebraic closure $\bar{\mathbb{Q}}_l$ of $\mathbb{Q}_l$,*

$$\lambda_i \neq \mu_j \quad \text{for all } i, j.$$

Suppose the rational function

$$R_\alpha(t) = \frac{\prod(1-\lambda_i^{-d(\alpha)} t)}{\prod(1-\mu_i^{-d(\alpha)} t)} \det(1-\phi_\alpha^{-1} t)$$

for every geometric point

$$\alpha\colon \mathrm{Spec}(k) \to S_0 \text{ over } \kappa$$

is actually contained in $\mathbb{Q}(t)$, not just in $\bar{\mathbb{Q}}_l(t)$. Then the coefficients of the polynomial $\det(1-\phi_\alpha^{-1} t)$ are rational numbers for all geometric points.

Proof. Let g be an element of the Galois group $\mathrm{Gal}(\bar{\mathbb{Q}}_l/\mathbb{Q})$. Since $R_\alpha(t)$ lies in $\mathbb{Q}(t)$, we have

$$\frac{\det(1-\phi_\alpha^{-1} t)\prod(1-\lambda_i^{-d(\alpha)} t)(1-g(\mu_i)^{-d(\alpha)} t)}{\prod(1-\mu_i^{-d(\alpha)} t)}$$
$$= \prod(1-g(\lambda_i)^{-d(\alpha)} t)\; g(\det(1-\phi_\alpha^{-1} t)) \in \bar{\mathbb{Q}}_l[t].$$

In the subsequent Lemma 3.2, we shall show that the compact group $\pi_1(S_0, \omega)$ contains *a closed subset F of measure zero, closed under conjugation, with the following property:*

When $\sigma \in \pi_1(S_0, \omega)$ is not contained in F, then none of the elements $\mu_i^{d(\sigma)}$ for which μ_i is a l-adic unit, is eigenvalue of σ: $V \to V$. We consider also the finite set E of natural numbers $m > 1$ defined as follows:

m is in E if λ_i/μ_j is a root of unity of order $m(>1)$,

and

m is in E if $\mu_i \neq g(\mu_j)$ and $\mu_i/g(\mu_j)$ is a root of unity of order m.

The set $\bigcup_{m \in E} m \cdot \hat{\mathbb{Z}}$ is then a *proper* closed subset of $\hat{\mathbb{Z}}$. Let U be the inverse image of the nonempty open subset $\hat{\mathbb{Z}} - \bigcup_{m \in E} m \cdot \hat{\mathbb{Z}}$ in $\pi_1(S_0, \omega)$. Then $W = U - (F \cap U)$ is an open subset of positive Haar measure, invariant under conjugation, in $\pi_1(S_0, \omega)$. The Čebotarev density theorem for function fields in one variable over finite constant fields implies that there is a geometric point α of S_0 over κ for which ϕ_α lies in W. All eigenvalues of the transformations $\sigma\colon V \to V$, $\sigma \in \pi_1(S_0, \omega)$ are l-adic units, because $\pi_1(S_0, \omega)$ is compact. Since ϕ_α does not lie in F, the polynomial $\prod(1-\mu_i^{-d(\phi_\alpha)} t)$ is relatively prime to $\det(1-\phi_\alpha^{-1} t)$. Since $d(\alpha) = d(\phi_\alpha)$ does not lie in $\bigcup_{m \in E} m \cdot \hat{\mathbb{Z}}$, no number m in E divides the degree

$d(\alpha)$. Thus the polynomial $\prod(1-\mu_i^{-d(\alpha)}t)$ is also relatively prime to $\prod(1-\lambda_i^{-d(\alpha)}t)$. It follows that

$$\prod(1-\mu_i^{-d(\alpha)}t)=\prod(1-g(\mu_i)^{-d(\alpha)}t).$$

From this we actually get

$$\prod(1-\mu_i t)=\prod(1-g(\mu_i)t).$$

For otherwise there would be elements μ_i, $g(\mu_j)$ with

$$\mu_i \neq g(\mu_j), \qquad (\mu_i/g(\mu_j))^{d(\alpha)}=1;$$

and then $d(\alpha)$ would be contained in $\bigcup_{m\in E} m\cdot\hat{\mathbb{Z}}$, contradicting the choice of α!

We have shown that $\prod(1-\mu_i t)$ is invariant under the action of the Galois group of $\bar{\mathbb{Q}}_l$ over $\mathbb{Q}$, and hence it has rational coefficients. Thus $\prod(1-\mu_i^m t)$ is contained in $\mathbb{Q}[t]$ for all integers m.

In exactly the same way, we can show that the polynomials $\prod(1-\lambda_i^m t)$ are contained in $\mathbb{Q}[t]$ for all integers m. But by assumption, the rational functions $R_\alpha(t)$ are contained in $\mathbb{Q}(t)$ for all the geometric points α in question. This gives us the assertion of Proposition 3.1. □

We come now to the lemma already mentioned.

Let $\mathrm{Sp}(\psi)$ be the algebraic group, defined over $\mathbb{Q}_l$, of all symplectic transformations with respect to ψ; and let $\mathrm{A}(\psi)$, also defined over $\mathbb{Q}_l$, be the group of symplectic similarities. We have then the identity

$$\psi(\sigma x, \sigma y)=\chi(\sigma)\, y(x, y)$$

for an algebraic character χ of this group $\mathrm{A}(\psi)$. As usual, we denote by $\mathrm{Sp}(\psi)(L)$ resp. $\mathrm{A}(\psi)(L)$ the groups of geometric points with coordinates in the extension field L of $\mathbb{Q}_l$. These groups are in a natural way locally compact groups, if L is finite over $\mathbb{Q}_l$.

For any l-adic unit α in $\bar{\mathbb{Q}}_l$ and any element m in $\hat{\mathbb{Z}}$, the limit

$$\alpha^m = \lim_{\mathbb{Z}\ni v\to m} \alpha^v$$

exists. As q^n is an l-adic unit, we can define the set

$$G=\{(m, \sigma)\in\hat{\mathbb{Z}}\times \mathrm{A}(\psi)(\mathbb{Q}_l),\ \chi(\sigma)=q^{nm}\}.$$

Then G is a locally compact group. By hypothesis there is a homomorphism

$$\pi_1(S_0, \omega)\to G$$
$$\sigma\mapsto(d(\sigma), \text{“}\sigma\colon V\to V\text{”}).$$

The image of this mapping is open in G. For by assumption the image of $\pi_1(S, \omega)$ is open in $\mathrm{Sp}(\psi)(\mathbb{Q}_l)$, and the homomorphism

$$d\colon \pi_1(S_0, \omega)\to\hat{\mathbb{Z}}$$

is surjective.

3.2 Lemma. *Let α be an l-adic unit in $\bar{\mathbb{Q}}_l$. We consider the closed subset, invariant under conjugation, of those elements σ in $\pi_1(S_0, \omega)$ for which $\alpha^{d(\sigma)}$ is an eigenvalue*

of the induced transformation

$$\sigma\colon\ V\to V.$$

We assert: This set is a null set with respect to the Haar measure.

Proof. Because the image of $\pi_1(S_0, \omega)$ is open in G, it suffices to show that the subset of those elements (m, σ) in G for which α^m is an eigenvalue of σ has measure zero. Using Fubini's Theorem, one gets this assertion from the following considerations. Let m be a fixed chosed element in $\hat{\mathbb{Z}}$ and H the homogeneous space for the algebraic group $\mathrm{Sp}(\psi)$, defined over $\mathbb{Q}_l$, consisting of those symplectic similarities g for which

$$\chi(g)=q^{nm}.$$

This homogeneous space is a smooth absolutely irreducible algebraic subvariety of $\mathrm{A}(\psi)$. The set $H(\mathbb{Q}_l)$ of points with coordinates in $\mathbb{Q}_l$ is nonempty. Let M be the Zariski-closed subset of H, defined over $\mathbb{Q}_l$, of those transformations that have α^m or a $\mathbb{Q}_l$-conjugate of it as an eigenvalue.

One sees easily that M is a *proper* subspace of H:

$$M(\bar{\mathbb{Q}}_l)\neq H(\bar{\mathbb{Q}}_l).$$

It follows that $M(\mathbb{Q}_l)$ is a null set with respect to an invariant measure on $H(\mathbb{Q}_l)$. This assertion is "analytically" of local nature and follows from the fact that a proper analytic subset of an open subset of $\mathbb{Q}_l^r$ is a "Lebesgue" null set. On the other hand, we have:

$M(\mathbb{Q}_l)$ is the set of elements in $H(\mathbb{Q}_l)$ for which α^m is an eigenvalue! □

§ 4. Proof of Deligne's Theorem

We use the notation of the first section.

4.1 Lemma. *Let X_0 be a smooth projective algebraic variety of constant dimension over the finite field κ of q elements. Let the dimension d of X_0 be even.*

Let $\alpha\in\bar{\mathbb{Q}}_l$ be an eigenvalue of the Frobenius homomorphism

$$\mathrm{Fr}^*\colon\ H^d(X, \mathbb{Q}_l)\to H^d(X, \mathbb{Q}_l)$$

induced by the Frobenius mapping

$$\mathrm{Fr}_X\colon\ X\to X\quad (X=X_0\times_{\mathrm{Spec}(\kappa)}\mathrm{Spec}(k)).$$

Then α is an algebraic number. The complex absolute values of all complex conjugates α' of α satisfy the estimate

$$q^{d/2-1/2}\leqq|\alpha'|\leqq q^{d/2+1/2}.$$

Proof. It follows from the Poincaré duality theorem that it suffices to prove the upper estimate. Further, it is clear that the statements of the theorem are

unchanged by finite extension of the base field. In particular, it suffices (after finite extension of the base field and reduction to the connected components) to assume that X_0 is absolutely irreducible. The proof proceeds now by induction on the dimension d. The initial step $d=0$ is clear.

Take then $d>0$, and thus $d\geqq 2$. For the induction step, we use the theory of Lefschetz pencils developed in Chapter III, particularly the global results of the final Section 7. The existence theorem for Lefschetz fiberings (III, 1.10) states:

After a finite extension of the base field (which is not significant for our proof), we can find algebraic varieties and morphisms defined over κ

$$\begin{array}{ccc} X_0 & \longleftarrow & \tilde{X}_0 \\ & & \downarrow{\scriptstyle f_0} \\ & & D_0 \supseteq A_0, \end{array} \qquad U_0 = D_0 - A_0 \overset{j_0}{\hookrightarrow} D_0$$

with the following properties:

$\tilde{X}_0$ is a smooth, absolutely irreducible projective variety over κ, and the morphism $\tilde{X}_0 \to X_0$ is birational;

D_0 is the projective line over κ, and f_0 is a flat projective mapping with connected fibers of dimension

$$d-1=n=2m+1;$$

A_0 is a finite set of κ-rational geometric points of the line D_0 such that f_0 is smooth over $U_0=D_0-A_0$, while over every point of A_0 it has exactly one singular point, likewise defined over κ. All these singular points are ordinary double points, hence nondegenerate for dimension reasons.

We can assume that there is a point

$$\alpha_0\colon \operatorname{Spec}(\kappa)\to D_0-A_0 \text{ over } \kappa$$

and, for a suitable projective embedding defined over κ, a smooth hyperplane section defined over κ

$$Y_0 \hookrightarrow (\widetilde{X_0})_\alpha = f_0^{-1}(\alpha)$$

of the fiber $(X_0)_\alpha$.

Let $\omega_0\colon \operatorname{Spec}(\Omega)\to D_0-A_0$ be a "generic" geometric point over k with values in the separably closed extension field Ω of k (as in III § 7). After base extension to k, we get

$$\begin{array}{ccc} X & \longleftarrow & \tilde{X} \\ & & \downarrow{\scriptstyle f} \\ & & D \supseteq A, \end{array} \qquad U = D - A \overset{j}{\hookrightarrow} D$$

$$\alpha\colon \operatorname{Spec}(k)\to D, \qquad \omega\colon \operatorname{Spec}(\Omega)\to D.$$

By Chap. III, 1.10 the mapping

$$H^d(X, \mathbb{Q}_l) \to H^d(\tilde{X}, \mathbb{Q}_l)$$

is injective. It is compatible with the Frobenius mapping, since $\tilde{X} \to X$ arises by base extension from $\tilde{X}_0 \to X_0$. Thus it suffices to prove the statement for $\tilde{X}_0$ instead of for X_0. We compute $H^d(\tilde{X}, \mathbb{Q}_l)$ using the Leray spectral sequence

$$E_2^{pq} = H^p(D, R^q f_*(\mathbb{Q}_{l\tilde{X}})) \Rightarrow H^{p+q}(\tilde{X}, \mathbb{Q}_l).$$

It also is compatible with the Frobenius mapping. Thus it suffices to prove the estimate in the Lemma for the eigenvalues of the Frobenius mapping on the terms E_2^{pq} of the spectral sequence for $p+q=d$. Only the cases

$$E_2^{0,d},\ E_2^{1,d-1},\ \text{and}\ E_2^{2,d-2}$$

occur. Let

$$E \subseteq V = H^n(\tilde{X}_\omega, \mathbb{Q}_l)$$

be the space of vanishing cycles (III, 7.2). On V there is defined the nondegenerate alternating pairing of Poincaré duality,

$$\langle u, v \rangle = S_{X/\mathrm{Spec}(\Omega)}(u \smile v),$$

with values in $\mathbb{Q}_l(-n)$.

We denote by ψ the induced nondegenerate alternating bilinear form on $E/E \cap E^\perp$. After selecting a non-canonical isomorphism $\mathbb{Q}_l(1) \xrightarrow{\cong} \mathbb{Q}_l$, we can assume that the values of ψ lie in $\mathbb{Q}_l$. The fundamental group $\pi_1(D_0 - A_0, \omega_0)$ acts on E, and we have

$$\psi(\sigma(u), \sigma(v)) = q^{-nd(\sigma)}\, \psi(u, v);\ \sigma \in \pi_1(D_0 - A_0, \omega_0).$$

Let $\mathscr{E}_0$ be the locally constant sheaf defined on $U_0 = D_0 - A_0$ that corresponds to the $\pi_1(D_0 - A_0, \omega_0)$-module $E/E \cap E^\perp$, and $\mathscr{E}$ its inverse image on $D - A$.

We now study the action of the Frobenius mapping on the individual terms of the spectral sequence E_2^{pq}, $p+q=d=n+1$, after a finite base field extension if need be, using the exact sequences from III, 7.6 and III, 7.7.

a) The term $E_2^{2,d-2} = H^2(D, R^{n-1} f_*(\mathbb{Q}_{l\tilde{X}}))$

The sheaf $R^{n-1} f_*(\mathbb{Q}_{l\tilde{X}})$ is locally constant on D (III, 7.6). Since $\pi_1(D, \alpha)$ vanishes, this sheaf is the constant sheaf for the vector space $H^{n-1}(\tilde{X}_\alpha, \mathbb{Q}_l)$.

$$\begin{aligned} E_2^{2,d-2} &= H^2(D, \mathbb{Q}_l) \otimes H^{n-1}(\tilde{X}_\alpha, \mathbb{Q}_l) \\ &= \mathbb{Q}_l(-1) \otimes H^{n-1}(\tilde{X}_\alpha, \mathbb{Q}_l). \end{aligned}$$

These equalities are canonical and hence compatible with the action of the geometric Frobenius mapping.

On $\mathbb{Q}_l(-1)$, the geometric Frobenius homomorphism has the eigenvalue q. It follows from the weak Lefschetz theorem (Chap. I, § 9) that the restriction mapping to the hyperplane section Y

$$H^{n-1}(\tilde{X}_\alpha, \mathbb{Q}_l) \to H^{n-1}(Y, \mathbb{Q}_l)$$

is injective. From the induction hypothesis for the variety Y_0 of dimension $d-2$ over κ, we find that the eigenvalues of the geometric Frobenius mapping

$$H^{n-1}(\tilde{X}_\alpha, \mathbb{Q}_l) \to H^{n-1}(\tilde{X}_\alpha, \mathbb{Q}_l)$$

are algebraic numbers whose complex conjugates γ have complex absolute value $|\gamma|$ satisfying

$$q^{d/2-1/2-1} \leqq |\gamma| \leqq q^{d/2+1/2-1}.$$

The eigenvalues of the Frobenius mapping on $E_2^{2,d-2}$ arise from these by multiplication by q!

b) The term $E_2^{0,d} = H^0(D, R^{n+1} f_*(\mathbb{Q}_{l\tilde{X}}))$

According to Theorem 7.6 and Proposition 7.7 of Chapter III, there is an exact sequence

$$0 \to r(\bigoplus_{s \in A} \mathbb{Q}_l(m-n)_s) \to R^{n+1} f_*(\mathbb{Q}_{l\tilde{X}}) \to \mathscr{G} \to 0$$

$$r = \begin{cases} 0 & \text{for } E \neq 0 \\ 1 & \text{for } E = 0. \end{cases}$$

After base field extension, we can assume it is in a natural way "defined over κ". Here $\mathbb{Q}_l(m-n)_s$ is the sheaf concentrated on the point s with stalk $\mathbb{Q}_l(m-n)$, and $\mathscr{G}$ is a locally constant (and hence constant) sheaf on D corresponding to the vector space $H^{n+1}(\tilde{X}_\alpha, \mathbb{Q}_l) = R^{n+1} f_*(\mathbb{Q}_{l\tilde{X}})_\alpha$.

$$H^0(D, \mathscr{G}) = H^{n+1}(\tilde{X}_\alpha, \mathbb{Q}_l).$$

The long exact sequence associated with this short exact sequence yields the exact sequence

$$0 \to r \cdot \#(A) \cdot \mathbb{Q}_l(m-n) \to E_2^{0,d} \to H^{n+1}(\tilde{X}_\alpha, \mathbb{Q}_l) \to 0.$$

The geometric Frobenius homomorphism acts on $\mathbb{Q}_l(m-n)$ by multiplication by

$$q^{n-m} = q^{d/2}.$$

Using Poincaré duality on $\tilde{X}_\alpha$ and the weak Lefschetz theorem, we get a natural *surjective* mapping

$$\mathrm{Hom}(H^{n-1}(Y, \mathbb{Q}_l), \mathbb{Q}_l) \otimes \mathbb{Q}_l(-n) \to H^{n+1}(X_\alpha, \mathbb{Q}_l).$$

But the eigenvalues of the geometric Frobenius homomorphism on the first term are the elements

$$\gamma^{-1} q^n \in \bar{\mathbb{Q}}_l.$$

Here γ runs through the eigenvalues of the Frobenius homomorphism on $H^{d-2}(Y, \mathbb{Q}_l)$. The desired estimates now follow from the induction hypothesis.

We come now to the decisive estimate,

c) The term $E_2^{1,d-1} = H^1(D, R^n f_*(\mathbb{Q}_{l\tilde{X}}))$

We have several different cases to consider.

If $E = 0$, the sheaf $R^n f_*(\mathbb{Q}_{l\tilde{X}})$ is locally constant on D, hence constant (III, 7.6, Case 1). As the first cohomology group of a constant sheaf on the projective

line vanishes, we have in this case

$$E_2^{1,d-1}=0.$$

Assume now $E\neq 0$. According to III 7.6, we again have two cases to distinguish:

$$E\subset E^{\perp} \quad \text{or} \quad E\not\subseteq E^{\perp}.$$

We shall treat only the case with $E\not\subseteq E^{\perp}$; in the other case, the proof proceeds similarly using the exact sequence in III 7.6a).

Assume now $E\not\subseteq E^{\perp}$. After base field extension, then, there are two short exact sequences, defined over κ in a natural way (III 7.6, 7.7):

$$0\to\mathscr{A}\to R^n f_*(\mathbb{Q}_{l\tilde{X}})\to\mathscr{B}\to 0$$
$$0\to\mathscr{C}\to\mathscr{A}\to j_*(\mathscr{E})\to 0.$$

The sheaves $\mathscr{B}$ and $\mathscr{C}$ are locally constant, hence constant on D; $\mathscr{E}$ is the locally constant sheaf on $D-A$ that is associated with the $\pi_1(D-A,\omega)$-module $E/E\cap E^{\perp}$. Since the first cohomology groups of the sheaves $\mathscr{B}$ and $\mathscr{C}$ vanish, the long exact cohomology sequences give us the exact sequences

$$H^1(D,\mathscr{A})\to H^1(D,R^n f_*(\mathbb{Q}_{l\tilde{X}}))\to 0$$
$$0\to H^1(D,\mathscr{A})\to H^1(D,j_*(\mathscr{E})).$$

If $A=\emptyset$, then $j_*(\mathscr{E})=\mathscr{E}$ is also constant, and $H^1(d,R^n f_*(\mathbb{Q}_{l\tilde{X}}))$ vanishes. Suppose A is nonempty. Then $D-A=S$ is affine. It suffices to estimate the eigenvalues of the Frobenius mapping on $H^1(D,j_*(\mathscr{E}))$. As the natural mapping

$$H_c^1(S,\mathscr{E})\to H^1(D,j_*(\mathscr{E}))$$

is surjective, it suffices to estimate the eigenvalues of the Frobenius mapping on $H_c^1(S,\mathscr{E})$. We want to apply the fundamental estimate 2.5 to $\mathscr{G}=\mathscr{E}$, and for that we must establish the two Hypotheses in 2.2. It follows from the theorem of Kazhdan-Margulis (III, 7.5) that *the image of the fundamental group* $\pi_1(S,\omega)$ *in the symplectic group of the alternating bilinear form* ψ *on* $E/E\cap E^{\perp}$ *is open! Thus Hypothesis* 2.2, 2 *is satisfied.*

It remains still to establish the rationality hypothesis 2.2, 1), i.e.: For every geometric point

$$\beta\colon \operatorname{Spec}(k)\to S_0=D_0-A_0 \text{ over } \kappa,$$

the polynomial $\det(1-\phi_\beta^{-1}t)$ is contained in $\mathbb{Q}[t]$, not just in $\mathbb{Q}_l[t]$. Here, as in the previous section, $\phi_\beta\in\pi_1(D_0-A_0,\omega_0)$ is a representative of the arithmetic Frobenius conjugacy class for the point β in $\pi_1(D_0-A_0,\omega_0)$; $\pi_1(D_0-A_0,\omega_0)$ acts on $E/E\cap E^{\perp}$.

We now deduce the desired rationality property easily from the rationality criterion 3.1: The first hypothesis of Proposition 3.1 follows again from the Kazhdan-Margulis theorem. The L-series

$$Z_{\tilde{X}_{0\beta}}(t)=\prod_\nu \det(1-\mathrm{Fr}^*\,|\,H^\nu(\tilde{X}_\beta,\mathbb{Q}_l)\,t)^{\cdot(-1)^{\nu+1}}$$

of the variety $\tilde{X}_{0\beta}$ over $k(\beta)$ lies in $\mathbb{Q}(t)$. The second hypothesis then follows from the multiplicativity of determinants in short exact sequences, the constancy of the sheaves $\mathscr{B}$, $\mathscr{C}$, and $R^\nu f_*(\mathbb{Q}_{l\tilde{X}})$ for $\nu \neq n$, and the following lemma.

4.2 Lemma. *Let $\mathscr{G}_0$ be a locally constant sheaf on S_0 whose inverse image $\mathscr{G}$ on S becomes constant. Then there are units $\lambda_1, \ldots, \lambda_r$ in $\bar{\mathbb{Q}}_l$ such that every geometric point*

$$\beta\colon \operatorname{Spec}(k) \to S_0 \text{ over } \kappa$$

satisfies

$$\det(1 - f_{\beta,\mathscr{G}_0}^{-1} t) = \prod (1 - \lambda_i^{-d(\beta)} t).$$

Here

$$f_{\beta,\mathscr{G}_0}\colon \mathscr{G}_{0\beta} \to \mathscr{G}_{0\beta}$$

is again the arithmetic Frobenius element from §1.

Proof. Let $\mathscr{G}_0$ be the sheaf associated with the $\pi(U_0, \omega_0)$-module W, on which $\pi_1(U, \omega)$ acts trivially. Then the factor group $\operatorname{Gal}(k/\kappa)$ acts on W. We have

$$\det(1 - f_{\beta,\mathscr{G}_0}^{-1} t) = \det(1 - f^{-d(\beta)} t).$$

Here $f\colon W \to W$ is the mapping induced by the Frobenius isomorphism of k over κ, and $f^{d(\beta)}$ is thus the mapping induced by the Frobenius isomorphism of k over $\kappa(\beta)$. This finishes the proof of 4.2 and therefore also of 4.1. □

By a simple trick we shall now obtain an essential sharpening of Lemma 4.1.

4.3 Proposition. *Let X_0 be a smooth, absolutely irreducible projective variety of any dimension d over κ. Then the eigenvalues of the Frobenius homomorphism*

$$\mathrm{Fr}^*\colon H^d(X, \mathbb{Q}_l) \to H^d(X, \mathbb{Q}_l)$$

are algebraic numbers. The complex absolute values of all the complex conjugates of these eigenvalues are equal to $q^{d/2}$.

Proof. Let $r \geqq 1$ be an arbitrary even natural number. We can then apply Lemma 4.1 to the variety

$$Z_0 = X_0 \times_{\operatorname{Spec}(\kappa)} \cdots \times X_0 \quad (r \text{ factors}).$$

Let

$$(q_i)_0\colon Z_0 \to X_0$$

be the projection onto the i-th factor. It follows from the Künneth formula (I, 8.17) that the homomorphism

$$H^q(X, \mathbb{Q}_l) \otimes_{\mathbb{Q}} \ldots \otimes H^d(X, \mathbb{Q}_l) \to H^{rd}(Z, \mathbb{Q}_l)$$
$$u_1 \otimes \ldots \otimes u_r \mapsto q_1^*(u_1) \smile \ldots \smile q_r^*(u_r)$$

is injective. Let γ be an eigenvalue of the Frobenius homomorphism

$$H^d(X, \mathbb{Q}_l) \to H^d(X, \mathbb{Q}_l).$$

Then γ^r is an eigenvalue of the Frobenius homomorphism

$$H^{rd}(Z, \mathbb{Q}_l) \to H^{rd}(Z, \mathbb{Q}_l).$$

It follows from Lemma 4.1 that the element γ^r is an algebraic number, and hence γ itself is; and every complex conjugate γ' of γ satisfies the estimate

$$q^{d/2-1/2r} \leqq |\gamma'| \leqq q^{d/2+1/2r}.$$

Passage to the limit gives the equation

$$|\gamma'| = q^{d/2}.$$

We can now prove the Weil conjecture (1.2′) by induction on the dimension of the smooth projective variety S_0 over κ. Take $d > 0$. After base field extension, we can assume that S_0 is absolutely irreducible and has a smooth hyperplane section Y_0 defined over κ in a suitable projective embedding.

It follows from the weak Lefschetz theorem (I, 9.2) that for $v < d$ the homomorphism

$$H^v(X, \mathbb{Q}_l) \to H^v(Y, \mathbb{Q}_l)$$

is injective. The induction hypothesis for Y therefore gives us the desired statement about the eigenvalues of the Frobenius homomorphism on $H^v(X, \mathbb{Q}_l)$.

By applying the Poincaré duality theorem, we get from this the desired statement for the eigenvalues of the Frobenius homomorphism on $H^v(X, \mathbb{Q}_l)$ for $v > d$. The remaining case $v = d$ is the content of the Proposition 4.3 that we just proved. □

§ 5. Generalizations

By a refinement of his methods, Deligne was able to prove Theorem 1.2 in an essentially more general form. We want to sketch these more general results briefly. They are the content of the second part of Deligne's work on the Weil conjecture [35].

Let X_0 be a finitely generated scheme over the field κ (with q elements) and $\mathcal{G}_0$ a constructible sheaf of $\mathbb{Q}_l$-vector spaces on X_0. We consider a geometric point

$$\alpha\colon \operatorname{Spec}(k) \to X_0$$

of X_0 over κ with values in the closure k of κ and with field of coordinates $\kappa(\alpha)$. As in Section 1,

$$f_\alpha \in \operatorname{Gal}(k/\kappa(\alpha))$$

is the Frobenius isomorphism of k over $\kappa(\alpha)$. It acts on the stalk $\mathcal{G}_\alpha$.

5.1 Definition. Let n be an integer. We call the sheaf $\mathcal{G}$ pointwise pure of weight n if for all such geometric points α of X_0 the eigenvalues of

$$f_\alpha^{-1}\colon \mathcal{G}_\alpha \to \mathcal{G}_\alpha$$

are algebraic numbers whose complex conjugates λ have complex absolute value

$$|\lambda| = q^{nd(\alpha)/2}, \qquad d(\alpha) = [\kappa(\alpha):\kappa].$$

The sheaf $\mathscr{G}_0$ is called *mixed of weight* $\leqq n$ when $\mathscr{G}_0$ has a filtration

$$0 = \mathscr{F}_0^{(0)} \subset \mathscr{F}_0^{(1)} \subset \ldots \subset \mathscr{F}_0^{(r)} = \mathscr{G}_0$$

for which all factor sheaves $\mathscr{F}_0^{(\nu)}/\mathscr{F}_0^{(\nu-1)}$ are pointwise pure of weight $\leqq n$.

Deligne's general theorem says:

5.2 Theorem. *Let*

$$f_0\colon X_0 \to S_0$$

be a morphism of finitely generated schemes over the finite field κ, *and let* $\mathscr{G}_0$ *be a mixed sheaf of weight* $\leqq n$ *on* X_0. *Then the direct image sheaves*

$$R^i f_!(\mathscr{G}_0)$$

are mixed of weight $\leqq n+i$.

We shall denote by $\tilde{H}^i(X, \mathscr{G})$ the image of the natural mapping $H_c^i(X, \mathscr{G}) \to H^i(X, \mathscr{G})$. Here again $X = X_0 \times_{\mathrm{Spec}(\kappa)} \mathrm{Spec}(k)$ and $\mathscr{G}$ is the inverse image of $\mathscr{G}_0$ on X.

5.3 Corollary. *Suppose the scheme* X_0 *is smooth over* κ *and the sheaf* $\mathscr{G}_0$ *is locally constant and pointwise pure of weight n. The Frobenius mapping of* X_0 *over* κ *induces a homomorphism*

$$\mathrm{Fr}^*\colon \tilde{H}^i(X, \mathscr{G}) \to \tilde{H}^i(X, \mathscr{G}).$$

The eigenvalues of Fr* *are algebraic numbers, and their complex cojugates* λ *have absolute value*

$$|\lambda| = q^{(i+n)/2} \qquad q = \#(\kappa).$$

For Deligne's proof of the theorem, the monodromy theory of Lefschetz pencils and the fundamental estimate in § 2 again play a decisive role. In addition, Deligne needs a generalization of the classical theorem of Hadamard and De La Vallée Poussin that Dedekind L-series do not vanish on the boundary of the critical strip. For example, one gets:

5.4 Proposition. *Let* X_0 *be a smooth curve over* κ *and* $\mathscr{G}_0$ *a sheaf on* X_0 *pointwise pure of weight n. Then the complex conjugates* λ *of the eigenvalues of*

$$\mathrm{Fr}^*\colon H_c^1(X, \mathscr{G}) \to H_c^1(X, \mathscr{G})$$

satisfy the inequality

$$|\lambda| < q^{(n/2)+1} \quad \text{(strictly smaller!)}.$$

In this connection, Deligne succeeded in using the Weil conjecture and the generalization of the theorem of Hadamard and De La Vallée Poussin to prove the *hard Lefschetz theorem:*

5.5 Theorem. *Let* X *be a smooth irreducible projective scheme over an arbitrary algebraically closed base field* K. *Let* L *be an ample line bundle on* X *and*

$$\eta = \mathrm{cl}_X(L) \in H^2(X, \mathbb{Q}_l(1))$$

the associated cohomology class (II, 2.2). *Let* $0 < i \leqq n = \dim X$. *Then the mapping*

$$\eta^i \colon H^{n-i}(X, \mathbb{Q}_l) \to H^{n+i}(X, \mathbb{Q}_l(i))$$
$$a \mapsto \eta^i \smile a$$

is an isomorphism.

5.6 Corollary. *Let*

$$\tilde{X} \to D = \mathbb{P}^1$$

be a Lefschetz pencil over K (III, 1.9), *and let* ω *be a generic geometric point of D over K. Let* $\dim \tilde{X} = n+1$, *and let* $E \subset H^n(\tilde{X}_\omega, \mathbb{Q}_l)$ *be the space of vanishing cycles* (III, 7.2, *for* $2 \nmid n+1$). *Then the orthogonal complement* $E^\perp$ *of E under Poincaré duality satisfies*

$$E \cap E^\perp = 0.$$

§ 6. Applications

The Weil conjecture immediately yields estimates for the number of rational points on an algebraic variety over a finite field.

Let X_0 be a smooth, absolutely irreducible projective algebraic variety over the finite field κ of q elements and N_n the number of geometric points (over κ) with values in the field extension κ_n of degree n. In Chap. II, § 4, we proved the Grothendieck formula

$$N_n = \sum_{i,j} (-1)^i \lambda_{ij}^n .$$

Here

$$\lambda_{i,1}, \ldots, \lambda_{i,b_i}$$

are the eigenvalues of the Frobenius homomorphism

$$H^i(X, \mathbb{Q}_l) \to H^i(X, \mathbb{Q}_l) \quad (0 \leqq i \leqq 2d,\ d = \dim X).$$

The Weil conjecture yields the asymptotic formula

$$N_n = 1 + q^{dn} + r_n ,$$
$$|r_n| \leqq \sum_{0 < i < 2d} b_i \, q^{ni/2} = O(q^{n(d-1/2)}).$$

These formulas refine estimates that *A.* Weil derived from the Weil conjecture for curves, which he had proved. One gets still more precise estimate for hypersurfaces. Let

$$X_0 \hookrightarrow \mathbb{P}^{d+1}$$

be a smooth hypersurface of dimension $d \geqq 1$. By the weak Lefschetz theorem, the induced mapping

$$H^i(\mathbb{P}^{d+1}_\kappa, \mathbb{Q}_l) \to H^i(X, \mathbb{Q}_l)$$

is an isomorphism for $i<d$ and injective for $i=d$. As these homomorphisms are compatible with the Frobenius homomorphism, we get:

6.1 Theorem. *Let*

$$b=\dim H^d(X_0, \mathbb{Q}_l)$$

be the middle Betti number of the smooth hypersurface X_0. (It depends only on the degree on X_0, and can be computed over $\mathbb{C}$.) Set

$$b'=\begin{cases} b & d \text{ odd} \\ b-1 & d \text{ even.} \end{cases}$$

Then

$$|N_n-[q^{n(d+1)}-1]/[q^n-1]|\leqq b'\, q^{nd/2}.$$

Here $[q^{n(d+1)}-1]/[q^n-1]$ is precisely the number of points of the projective space $\mathbb{P}^d$ with values in κ_n.

There are less immediate applications of the Weil conjecture related to the estimation of exponential sums.

6.2 Theorem. *Let P be a polynomial in n variables of degree d with coefficients in κ, and let P_d be the homogeneous component of degree d. We assume that d is relatively prime to the characteristic of κ and that the projective hypersurface defined by P_d is smooth. Then for any nontrivial additive character $\psi\colon \kappa\to\mathbb{C}^*$,*

$$\Big|\sum_{x_1,\ldots,x_n}\psi(P(x_1,\ldots,x_n))\Big|\leqq(d-1)^n\, q^{n/2}.$$

6.3 Theorem (Estimate for generalized Kloostermann sums). *Let*

$$\psi\colon \kappa\to\mathbb{C}^*$$

be a nontrivial additive character of the finite field κ with q elements, and let a be a nonzero element in κ. Then for $n\geqq 1$,

$$\Big|\sum_{x_1 x\ldots x_n=a}\psi(x_1+\ldots+x_n)\Big|\leqq n\, q^{(n-1)/2}.$$

We shall skip the reduction of these two estimates to the Weil conjecture. (See SGA $4\frac{1}{2}$: *Applications de la formule des traces aux sommes trigonometriques.*)

The Ramanujan Conjecture

An application of quite different nature is the proof of the famous Ramanujan conjecture, and of its generalization by Petersson, through reduction to the Weil conjecture. This reduction is also due to Deligne, following previous work by Eichler [48], Ihara [84], and Shimura [149].

We first formulate the Ramanujan-Petersson conjecture.

Let $\Gamma\subset \mathrm{SL}(2,\mathbb{Z})$ be a subgroup of finite index in the elliptic modular group which contains a principal congruence subgroup

$$\Gamma(N)=\left\{M=\begin{pmatrix}a & b\\ c & d\end{pmatrix}\equiv\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \bmod N\right\}.$$

A cusp form of weight $k\in\mathbb{Z}$ is a holomorphic function on the upper half-plane with the transformation property

$$f\left(\frac{a z+b}{c z+d}\right)=(c z+d)^k f(z) \quad \text{for all } M\in\Gamma$$

that vanishes at all cusps of Γ, i.e.

$$\lim_{\mathrm{Im} z\to\infty}(c z+d)^{-k} f(M z)=0 \quad \text{for } M\in \mathrm{SL}(2,\mathbb{Z}).$$

The vector space $S_k(\Gamma)$ of cusp forms is finite dimensional. Every cusp form $f\in S_k(\Gamma)$ has a Fourier expansion

$$f(z)=\sum_{n=1}^{\infty} a_n e^{2\pi i n z/N}.$$

6.4 Theorem. *The Fourier coefficients of a cusp form satisfy the estimate*

$$a_n=O(n^{(k-1)/2+\varepsilon}), \quad (\varepsilon>0).$$

(The Ramanujan conjecture concerns the special case

$$e^{2\pi i z}\prod_{n=1}(1-e^{2\pi i n z})^{24} \quad (k=12).)$$

The case $k=1$ is a special case and was proved by Deligne and Serre [39] without the Weil conjecture. We assume therefore $k\geqq 2$. One can easily see that it suffices to prove the conjecture for the groups

$$\Gamma=\Gamma_1(N)=\left\{\begin{pmatrix}a & b\\ c & d\end{pmatrix}\equiv\begin{pmatrix}1 & *\\ 0 & 1\end{pmatrix}\right\}, \quad N\geqq 3.$$

The assumption $N\geqq 3$ guarantees that Γ acts freely on the upper half plane.

The connection between the Ramanujan-Petersson conjecture and the Weil conjecture rests on the theory of Hecke operators. For a natural number n relatively prime to N, one defines the Hecke operator

$$T(n)=T_{N,k}(n)\colon\ S_k(\Gamma_1(N))\to S_k(\Gamma_1(N))$$

by the formula

$$(T(n)f)(z)=n^{k/2-1}\sum_{M:\Gamma_1(N)\backslash O(N,n)}(\det M)^{k/2}(c z+d)^{-k} f\left(\frac{a z+b}{c z+d}\right).$$

Here M runs through a set of representatives of the finitely many left "cosets" of the action of $\Gamma_1(N)$ on the set

$$O(N,n)=\left\{\begin{pmatrix}a & b\\ c & d\end{pmatrix}\equiv\begin{pmatrix}1 & *\\ 0 & n\end{pmatrix} \bmod N,\ a d-b c=n\right\}.$$

The group $\Gamma_1(N)$ is a normal subgroup of finite index in

$$\Gamma_0(N)=\left\{\begin{pmatrix}a & b\\ c & d\end{pmatrix}\equiv\begin{pmatrix}* & *\\ 0 & *\end{pmatrix} \bmod N\right\}.$$

The factor group is canonically isomorphic to $(\mathbb{Z}/n\mathbb{Z})^*$,

$$\Gamma_0(N)/\Gamma_1(N)\to(\mathbb{Z}/n\mathbb{Z})^*$$

$$\begin{pmatrix}a & b\\ c & d\end{pmatrix}\mapsto a.$$

The group $(\mathbb{Z}/n\mathbb{Z})^*$ hence acts in an obvious way on $S_k(\Gamma_1(N))$. To a number $n\in\mathbb{Z}$, $(n, N)=1$, is assigned the operator

$$I(n)\colon S_k(\Gamma_1(N))\to S_k(\Gamma_1(N)).$$

The Ramanujan-Petersson conjecture is equivalent to:

6.5 Theorem. *Let $k\geqq 2$, and let p be a prime number. The reciprocal roots of the polynomial*

$$\det(1-T(p)\,t+I(p)\,p^{k-1}\,t^2)$$

are algebraic numbers, and their complex conjugates have absolute value

$$|\lambda|=p^{(k-1)/2}.$$

For completeness we should mention that one must also define the "bad" Hecke operators $T(p)$ for $p|N$. But this is handled directly, without the Weil conjecture, and so we shall not go into it here.

To set up the connection with the Weil conjecture, we must define the Hecke operators algebraically. It is well known that there exists a moduli scheme

$$\mathcal{M}\to\operatorname{Spec}\mathbb{Z}[1/N]$$

over the ring $\mathbb{Z}[1/N]$ that classifies the isomorphism classes (E, s) of elliptic curves with distinguished N-division point of order N. Let

$$f\colon\ \mathcal{E}\to\mathcal{M}$$

be the universal elliptic curve with N-division points. We denote by

$$f_{\mathrm{an}}\colon\ \mathcal{E}_{\mathrm{an}}\to\mathcal{M}_{\mathrm{an}}$$

the "analytification" of the universal elliptic curve. We denote by

$$\mathcal{G}_{\mathrm{an}}=S_m(R^1(f_{\mathrm{an}})_*(\mathbb{Z})),\qquad m=k-2,$$

the m-th symmetric power of the sheaf

$$R^1(F_{\mathrm{an}})_*(\mathbb{Z}).$$

We denote by

$$H_0^1(\mathcal{M}_{\mathrm{an}},\mathcal{G}_{\mathrm{an}})=\operatorname{Image}[H_c^1(.,.)\to H^1(.,.)]$$

the image of the cohomology with compact support in the usual cohomology under the natural mapping. It was shown by G. Shimura that there is a canonical isomorphism

$$H_0^1(\mathcal{M}_{\mathrm{an}},\mathcal{G}_{\mathrm{an}})\otimes_{\mathbb{Z}}\mathbb{C}=S_k(\Gamma_1(N))\oplus\overline{S_k(\Gamma_1(N))}.$$

The algebraically defined modular correspondences (which we shall not define here) induce endomorphisms

$$\tilde{T}(p)\colon\ H_0^1(\mathscr{M}_{\mathrm{an}}, \mathscr{G}_{\mathrm{an}}) \to H_0^1(\mathscr{M}_{\mathrm{an}}, \mathscr{G}_{\mathrm{an}}).$$

The restriction of $\tilde{T}(p) \otimes \mathbb{C}$ to $S_k(\Gamma_1(N))$ agrees with the Hecke operator $T(p)$. The analogous statement holds for the operators

$$\tilde{I}(p)\colon\ H_0^1(\mathscr{M}_{\mathrm{an}}, \mathscr{G}_{\mathrm{an}}) \to H_0^1(\mathscr{M}_{\mathrm{an}}, \mathscr{G}_{\mathrm{an}})$$

that are induced by the automorphisms

$$(E, s) \to (E, p\, s)$$

of the moduli scheme.

To set up the connection with the Weil conjecture, we reduce the universal elliptic curve modulo p and get the universal elliptic curve (with distinguished N-division point) over the field $\kappa = \mathbb{Z}/p\mathbb{Z}$,

$$f_0\colon\ \mathscr{E}_0 \to \mathscr{M}_0 \quad (= \mathscr{M} \otimes \kappa).$$

Extension to the algebraic closure $\bar{\kappa}$ of κ gives

$$\bar{f}_0\colon\ \bar{\mathscr{E}}_0 \to \bar{\mathscr{M}}_0 \quad (= \mathscr{M} \otimes \bar{\kappa}).$$

We consider the l-adic cohomology groups

$$H_0^1(\bar{\mathscr{M}}_0, \bar{\mathscr{G}}_0),\ \mathscr{G}_0 = S_m\, R^1(f_0)_*(\mathbb{Q}_l), \qquad \bar{\mathscr{G}}_0 = \mathscr{G}_0 \otimes \bar{\kappa}.$$

$$(H_0^1(.,.) = \mathrm{Image}[H_c^1(.,.) \to H^1(.,.)]).$$

It follows from the base change theorem and comparison theorem that

$$H_0^1(\bar{\mathscr{M}}_0, \bar{\mathscr{G}}_0) = H_0^1(\mathscr{M}_{\mathrm{an}}, \mathscr{G}_{\mathrm{an}}) \otimes_{\mathbb{Z}} \mathbb{Q}_l\,.$$

Besides $\tilde{T}(p)$ and $\tilde{I}(p)$, the Frobenius homomorphism F_p also acts on the $\mathbb{Q}_l$-vector spaces $H_0^1(\bar{\mathscr{M}}_0, \bar{\mathscr{G}}_0)$, and we have the *Eichler-Shimura relations:*

$$\det(1 - \tilde{T}(p)\, t + \tilde{I}(p)\, p^{k-1}\, t^2) = \det(t\, F_p - 1)\, \det(t(p^{k-1}/F_p)\, \tilde{I}(p) - 1).$$

The Ramanujan conjecture now follows immediately from Deligne's generalized version of the Weil conjecture (which we have not proved in this book): for the sheaf $\mathscr{G}_0$ is obviously pointwise pure of weight m (this follows from the Weil conjectures for the fibers of f_0). It follows then from the generalized Weil conjecture that the eigenvalues λ of F_p and then also of $p^{k-1}/F_p\, \tilde{I}(p)$ have absolute value

$$|\lambda| = p^{(m+1)/2} = p^{(k+1)/2},$$

whence the Ramanujan conjecture holds.

In Deligne's original version, instead of this the ordinary Weil conjecture was applied to a smooth projective compactification of the m-fold fibre product

$$\mathscr{E}_0 \times_{\mathscr{M}_0} \ldots \times \mathscr{E}_0.$$

The existence of such a compactification was proved by means of Néron models of elliptic curves. The later generalizations that Deligne found make this roundabout path unnecessary.

§ 7. The Weil Conjecture and the Standard Conjectures on Algebraic Cycles

Grothendieck's original idea was to obtain the Weil conjecture as a consequence of conjectures on algebraic cycles. These cycle conjectures go far beyond the Weil conjecture and are a central problem of contemporary algebraic geometry.

In what follows, let X be a smooth, irreducible n-dimensional projective algebraic variety over the algebraically closed field k.

We suppose that we have selected an "orientation", i.e. an isomorphism

$$\hat{\mathbb{Z}}_l \to \hat{\mathbb{Z}}_l(1).$$

This isomorphism yields isomorphisms

$$\mathbb{Q}_l \cong \mathbb{Q}_l(v)$$

and

$$H^i(X, \mathbb{Q}_l) \cong H^i(X, \mathbb{Q}_l(v)).$$

We consider the $\mathbb{Q}$-vector space inside

$$H^{2i}(X) = H^{2i}(X, \mathbb{Q}_l)$$

that is spanned by the cohomology classes of subvarieties of codimension i (over the field of rational numbers). We denote this by

$$C^i(X) \subset H^{2i}(X).$$

One conjectures that $C^i(X)$ is a finite-dimensional $\mathbb{Q}$-vector space. Let

$$\eta \in C^1(X)$$

be the cohomology class of a hyperplane section.

Deligne's Hard Lefschetz Theorem (which we stated without proof in § 5) says that the multiplication mapping (cup product) with the i-th power of η yields an isomorphism

$$H^{n-i}(X) \xrightarrow{\eta^i} H^{n+i}(X) \qquad (0 \leqq i \leqq n).$$

It is easy to show that this isomorphism takes algebraic cycles to algebraic cycles.

7.1 Conjecture of Lefschetz Type. *Multiplication by η^{n-2j} defines an isomorphism*

$$C^j(X) \xrightarrow{\sim} C^{n-j}(X), \qquad 0 \leqq 2j \leqq n.$$

From this conjecture one can easily deduce the following existence statement for cycles.

7.2 Consequence of Conjecture 7.1. *Consider the Künneth decomposition*

$$H^{2n}(X \times X) = \bigoplus_{i=0}^{2n} H^i(X) \otimes H^{2n-i}(X).$$

let $\Delta \in H^{2n}(X \times X)$ *be the cohomology class of the diagonal, with Künneth decomposition*

$$\Delta = \sum_{i=0}^{2n} \Delta_i .$$

Assertion: $\boxed{\Delta_i \in C^n(X \times X).}$

Remark. It follows in general from 7.1 that the Künneth components of an algebraic cycle in a cartesian product $X \times Y$ are again algebraic.

Messing and Katz [89], using the Weil conjecture, have proved that 7.2 is valid when k is the algebraic closure of a finite field.

The second central conjecture on algebraic cycles is the "Conjecture of Hodge Type".

For $0 \leqq i \leqq n$, define the primitive part

$$P^i(X) \subset H^i(X)$$

as the kernel of the multiplication mapping with η^{n-i+1}. On $H^{2i}(X)$, consider the symmetric bilinear form

$$(x, y) \mapsto (-1)^i S_{X/k}(x \cdot y \cdot \eta^{n-2i}).$$

7.3 Conjecture of Hodge Type. *The restriction of the bilinear form above to*

$$P^{2i}(X) \cap C^i(X)$$

is positive definite.

Remark. In the characteristic-zero case, this conjecture is proved. It follows from Hodge theory.

Grothendieck has pointed out that the Weil conjecture can be deduced from the two conjectures on cycles by means of certain arguments of Weil and Serre. The further significance of the cycle conjectures can be seen e.g. from the fact that they imply that two algebraic cycles are numerically equivalent if and only if their cohomology classes are the same.

Appendices

A I. The Fundamental Group

In this appendix we shall list some necessary facts about the fundamental group of a scheme. A more detailed study can be found e.g. in SGA 1 [64] or in [121]. We shall in particular use the fundamental group to describe the locally constant etale sheaves.

Let X be a scheme and

$$s\colon \operatorname{Spec}(\Omega)\to X$$

a geometric point (Ω separably closed). A scheme

$$f\colon Y\to X$$

is called a *covering space* if f is an etale and finite (equivalently, proper) mapping. The covering spaces of X form a full subcategory of the category $\operatorname{Et}(X)$ of all schemes etale over X. We consider the functor of geometric points over s:

$$Y(s)=\operatorname{Hom}_X(\operatorname{Spec}(\Omega), Y).$$

A pointed covering space (pointed over the base point s) is a pair

$$(Y, \alpha)$$

consisting of a covering space Y of X and a geometric point $\alpha\in Y(s)$. A mapping of pointed covering spaces

$$f\colon (Y_1, \alpha_1)\to(Y_2, \alpha_2)$$

is an X-morphism

$$f\colon Y_1\to Y_2$$

with

$$f\circ\alpha_1=\alpha_2.$$

Some Properties of Covering Spaces

(1) Let (Y_1, α_1) and (Y_2, α_2) be two pointed covering spaces with Y_1 connected. Then there is at most one mapping

$$(Y_1, \alpha_1)\to(Y_2, \alpha_2).$$

(2) For any two nonempty connected covering spaces $X_\nu\to X$, $\nu=1, 2$, there is a third nonempty connected covering space

$$X_3\to X$$

that dominates the two given ones; that is, there are X-morphisms

$$X_3 \to X_\nu, \qquad \nu = 1, 2.$$

(3) Let $Y \to X$ be a connected nonempty covering space of X and $G(X/Y)$ the group of automorphisms of Y over X and n the degree of the covering. Here $n = \# Y(\beta)$ for every geometric point of X for which $Y(\beta) \neq \emptyset$ (e.g. $\beta = s$, if X is connected). Because of (1), we have

$$\# G(Y/X) \leqq n.$$

Y/X is called *Galois* if

$$\# G(Y/X) = n.$$

Then because of (1), $G(Y/X)$ acts transitively on every "geometric fiber" $Y(b)$. $G(Y/X)$ always acts *on the right*. For every nonempty covering space $Z \to X$ there is a Galois covering space $Y \to X$ that dominates it, i.e. such that there is an X-morphism

$$Y \to Z.$$

(4) Let $Z \to X$ and $Y \to X$ be two Galois covering spaces of X, and

$$f\colon Z \to Y$$

an X-morphism. For every "cover transformation" $\sigma \in G(Z/X)$ there is exactly one element $\bar{\sigma} \in G(Y/X)$ with $\bar{\sigma} \circ f = f \circ \sigma$. The mapping thus defined,

$$G(Z/X) \to G(Y/X)$$

$$\sigma \mapsto \bar{\sigma},$$

is a surjective homomorphism. The kernel is precisely the "cover transformation group" $G(Z/Y)$ of the Galois covering space $Z \to Y$ of Y.

Now we consider the filtered category of all Galois pointed covering spaces (Y, α) of X (always over the prescribed base point s of X). Because of (1), there is at most one morphism between two objects of this category. The set of isomorphism classes is thus a partially ordered set.

A I.1 Definition of the fundamental group of X:

$$\pi_1(X, s) = \varprojlim_{(Y,\alpha)} G(Y/X).$$

Here (Y, α) runs over the category of pointed Galois covering spaces of X. It follows from the properties of this projective limit that the natural mapping

$$\pi_1(X, s) \to G(Y/X)$$

is surjective. The kernel is $\pi_1(Y, \alpha)$.

$\pi_1(X, s)$ furnished with the limit topology ($G(Y/X)$ discrete) is a compact topological group, a *profinite group*. The open normal subgroups are precisely the subgroups $\pi_1(Y, \alpha)$.

Remark. The connected component X_0 of X that contains s obviously satisfies

$$\pi_1(X, s) = \pi_1(X_0, s).$$

Example. Let X be a normal irreducible scheme, $F(X)$ the field of meromorphic functions on X, and Ω its separable closure. The embedding $F(X) \to \Omega$ provides a "generic" geometric point

$$a\colon \operatorname{Spec}(\Omega) \to X.$$

We consider all finite extension fields L of $F(X)$ that are contained in Ω and for which the normalization of X in L is unramified over X. The union M of all these fields is Galois over $F(X)$, and there is a natural isomorphism

$$\pi_1(X, a) = \operatorname{Gal}(M/F(X))$$

for the Galois group of M over $F(X)$.

A I.2 The fundamental group is independent of the base point in the following sense: Let X be connected. Let $\tilde{s}\colon \operatorname{Spec}(\tilde{\Omega}) \to X$ be another geometric point. The properties of the category of pointed Galois covering spaces (Y, α) (α over s), the finiteness of $Y(\tilde{s})$, and the surjectivity of $Y_1(\tilde{s}) \to Y_2(\tilde{s})$ imply that there exists an element γ in

$$\varprojlim_{(Y,\alpha)} Y(\tilde{s}).$$

This γ is given by a coherent system of geometric points

$$\gamma(Y, \alpha)\colon \operatorname{Spec}(\Omega) \to Y$$

over $\tilde{s}$. The functor

$$(Y, \alpha) \mapsto (Y, \gamma(Y, \alpha))$$

establishes an equivalence between the category of pointed Galois covering spaces (Y, α) with α over s and the category of pointed Galois covering spaces (Y, β) with β over $\tilde{s}$. We obtain an isomorphism (dependent on the choice of γ)

$$\phi(\gamma)\colon \pi_1(X, s) \xrightarrow{\cong} \pi_1(X, \tilde{s}).$$

One can show that $\pi_1(X, s)$ acts transitively on $\varprojlim_{(Y,\alpha)} Y(\tilde{s})$, thus showing that $\phi(\gamma)$ *is uniquely determined up to inner automorphisms of* $\pi_1(X, s)$.

A I.3 $\pi_1(X, s)$ is a covariant functor on the category of pointed schemes: Suppose we have a commutative diagram

$$\begin{array}{ccc} \tilde{X} & \xleftarrow{\tilde{s}} & \operatorname{Spec}(\tilde{\Omega}) \\ \downarrow & & \downarrow p \\ X & \xleftarrow{s} & \operatorname{Spec}(\Omega). \end{array}$$

Here $\tilde{\Omega}$ is any separably closed extension field of Ω. We consider the following functor from the category of pointed Galois covering spaces of (X, s) to the category of pointed Galois covering spaces of $(\tilde{X}, \tilde{s})$:

$$(Y, \alpha) \mapsto (\tilde{Y}, \tilde{\alpha}).$$

Here $\tilde{Y}$ is the connected component of $\tilde{X} \times_X Y$ that contains the geometric point $\tilde{\alpha} = (\tilde{s}, \alpha \circ p)$: $\mathrm{Spec}(\Omega) \to \tilde{X} \times_X Y$.

This functor together with the natural injection $G(\tilde{Y}/\tilde{X}) \to G(Y/X)$ furnishes a homomorphism of the projective limits,

$$\pi_1(\tilde{X}, \tilde{s}) \to \pi_1(X, s).$$

A I.4 Classification of the covering spaces of X: Let X be connected. For an arbitrary covering space $Y \to X$ then there is a pointed Galois covering space (Z, α) with X-mappings onto every component of Y (properties (2) and (3)). It follows from (1) then that: The natural mapping

$$\begin{aligned} \mathrm{Hom}_X(Z, Y) &\to Y(s) \\ f &\mapsto f \circ \alpha \end{aligned}$$

is bijective.

The group $G(Z/X) = \pi_1(X, s)/\pi_1(Z, \alpha)$ acts on $\mathrm{Hom}_X(Z, Y)$ on the left, and hence so does $\pi_1(X, s)$. This action carries over to $Y(s)$. The action of $\pi_1(X, s)$ does not depend on the choice of (Z, α). Thus $Y(s)$ is in a natural way a finite set on which $\pi_1(X, s)$ acts continuously on the left. Continuity here means that there is an open subgroup of $\pi_1(X, s)$ that acts trivially.

It is easy to show:

A I.5 Proposition. *Let X be connected. The assignment*

$$Y \mapsto Y(s)$$

establishes an equivalence between the category of covering spaces of X and the category of finite continuous $\pi_1(X, s)$-sets.

A I.6 The description of locally constant etale sheaves on a connected scheme X using $\pi_1(X, s)$: Every constant (constructible) sheaf is representable by a covering space of X. Therefore every locally constant sheaf $\mathcal{G}$ is also representable by a covering space $Y \to X$ (see the Appendix to I, § 3, Lemma 3.18). If $\mathcal{G}$ is a locally constant sheaf of abelian groups, and thus a group object in the category of etale sheaves, then $Y \to X$ is an abelian group object in the category of covering spaces, and so $Y(s)$ is an abelian group object in the category of $\pi_1(X, s)$-sets. Moreover, $Y(s)$ is canonically isomorphic to the stalk $\mathcal{G}_s$.

A I.7 Proposition. *For every locally constant (constructible) sheaf $\mathcal{G}$ of abelian groups, the stalk $\mathcal{G}_s$ is in a natural way a (continuous) $\pi_1(X, s)$-module. The assignment*

$$\mathcal{G} \mapsto \mathcal{G}_s$$

establishes an equivalence between the category of locally constant (constructible) sheaves of abelian groups and the category of finite continuous $\pi_1(X, s)$-modules.

Let l be a prime number invertible on X. By passing to projective limits and to the quotient field $\mathbb{Q}_l$ of the ring $\mathbb{Z}_l$ of l-adic integers, we get:

A I.8 Proposition. *There is a natural equivalence between the category of all locally constant l-adic sheaves* $\mathcal{G}$ (§12) *and the category of all finitely generated* $\mathbb{Z}_l$*-modules on which* $\pi_1(X, s)$ *acts continuously (with respect to the l-adic topology): The stalk* $\mathcal{G}_s$ *is a continuous* $\pi_1(X, s)$*-module. The equivalence is established by the functor*

$$\mathcal{G} \mapsto \mathcal{G}_s .$$

Likewise, $\pi_1(X, s)$ *acts continuously on the stalk* $\mathcal{H}_s = \mathcal{G}_s \otimes \mathbb{Q}_l$ *of a locally constant sheaf* $\mathcal{H} = \mathcal{G} \otimes \mathbb{Q}_l$ *of* $\mathbb{Q}_l$*-vector spaces. The functor*

$$\mathcal{H} \mapsto \mathcal{H}_s$$

establishes an equivalence between the category of locally constant sheaves of $\mathbb{Q}_l$*-vector spaces and the category of continuous representations of the fundamental group* $\pi_1(X, s)$ *on finite-dimensional vector spaces over* $\mathbb{Q}_l$.

Further properties of the fundamental group:
Let

$$f\colon X \to Y$$

be a purely inseparable morphism, and

$$s\colon \operatorname{Spec}(\Omega) \to X$$

a geometric point. From Theorem I, 3.16 follows

A I.9 Proposition. $\pi_1(X, s) = \pi_1(Y, f \circ s)$. *In particular,*

$$\pi_1(X_{\mathrm{red}}, s) = \pi_1(X, s).$$

Now we consider the category of finitely generated schemes X over the field $\mathbb{C}$ of complex numbers and the natural functor

$$X \mapsto X_{\mathrm{an}}$$

that assigns a complex analytic space to every such scheme. The set of points of X_{an} can be identified with the set $\operatorname{Hom}_{\operatorname{Spec}(\mathbb{C})}(\operatorname{Spec}(\mathbb{C}), X)$ of geometric points of X over $\operatorname{Spec}(\mathbb{C})$ with values in $\operatorname{Spec}(\mathbb{C})$. Let X be such a scheme over $\operatorname{Spec}(\mathbb{C})$, and $s\colon \operatorname{Spec}(\mathbb{C}) \to X$ a geometric point in $\operatorname{Hom}_{\operatorname{Spec}(\mathbb{C})}(\operatorname{Spec}(\mathbb{C}), X)$. The generalization of the Riemann existence theorem by Grauert and Remmert [52] implies:

The functor $Y \mapsto Y_{\mathrm{an}}$ *establishes an equivalence between the category of covering spaces of* X *and the category of (topological) covering spaces of* X_{an} *with finite fibres.* If X is complete over $\mathbb{C}$, this simply follows from the GAGA-theorems of Serre [131].

Let $\pi_1(X_{\mathrm{an}}, s)$ be the "topological" fundamental group of X_{an} with base point s. The subgroups of finite index in $\pi_1(X_{\mathrm{an}}, s)$ classify the finite connected covering spaces of X_{an}.

Let $\hat{\pi}_1(X_{\mathrm{an}}, s)$ be the profinite completion of $\pi_1(X_{\mathrm{an}}, s)$, i.e. the projective limit of all finite factor groups. We get:

A I.10 Proposition. *For a finitely generated scheme* X *over* $\operatorname{Spec}(\mathbb{C})$ *and a geometric point* $s \in \operatorname{Hom}_{\operatorname{Spec}(\mathbb{C})}(\operatorname{Spec}(\mathbb{C}), X)$, $\pi_1(X, s)$ *is canonically isomorphic to the profinite completion* $\hat{\pi}_1(X_{\mathrm{an}}, s)$ *of the fundamental group* $\pi_1(X_{\mathrm{an}}, s)$.

In Chapter III, on monodromy, we need some further results.

For a scheme over a field of positive characteristic, often only a certain "tame" part of the fundamental group can be determined.

A I.11 Abhyankar's Lemma. *Let A be a regular local ring with quotient field K, L a finite Galois field extension of K with Galois group G, and B the integral closure of A in L. Let x be a regular parameter, i.e. an element of a regular system of parameters $x_1, \ldots, x_{n-1}, x$ of A, and suppose that the principal ideal $A \cdot x$ is the only prime ideal of height 1 in A ramified in B. We consider a geometric point*

$$\alpha\colon \operatorname{Spec}(\Omega) \to \operatorname{Spec}(B)$$

that is localized at a prime ideal over $A \cdot x$. Suppose the order $e = \#(G_\alpha)$ of the ramification group $G_\alpha = \{\sigma \in G, \sigma \circ \alpha = \alpha\}$ is relatively prime to the characteristic of Ω. Then B is regular. More precisely: Let $\mathfrak{m}$ be a maximal ideal of B and let $\tilde{A}$ resp. $\tilde{B}_\mathfrak{m}$ be strict Henselizations of A and $B_\mathfrak{m}$. Then

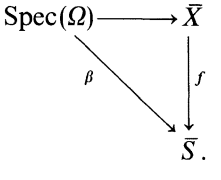
$$\tilde{B}_\mathfrak{m} = \tilde{A}[\sqrt[e]{x}].$$

The elements $x_1, \ldots, x_{n-1}, \sqrt[e]{x}$ form a regular system of parameters for $\tilde{B}_\mathfrak{m}$. The ramification degree e is relatively prime to the characteristic of the residue field of A.

For the Proof. One can easily see that it suffices to assume that A and B are strictly Henselian local rings. In a sufficiently large extension field of L we consider the regular ring $A[\sqrt[e]{x}]$ and the integral closure C of $B \cdot A[\sqrt[e]{x}]$ in its quotient field. First say $\dim A = 1$, with t a generating element of the maximal ideal of A. Since A is strictly Henselian, and because of our relative primality condition on $e = \# G_\alpha$, we have $B x = B t^e$. The ring $B[(\sqrt[e]{x})/t]$ is unramified over B. Since B is strictly Henselian, this implies $B[(\sqrt[e]{x})/t] = B$. For degree reasons this implies $B = A[\sqrt[e]{x}]$. This implies now in the general case that all prime ideals of height 1 in C are unramified over $A[\sqrt[e]{x}]$. Because of the purity of the ramification locus, C is thus unramified; hence it is etale over $A[\sqrt[e]{x}]$, and thus it coincides with $A[\sqrt[e]{x}]$. For degree reasons again we get

$$A[\sqrt[e]{x}] = C = B. \quad \square$$

Now we consider a finite surjective morphism of a normal connected scheme $\bar{X}$ to a normal connected scheme $\bar{S}$, a geometric point α of $\bar{X}$, and its image point β of $\bar{S}$,



Let a be the ordinary point of $\bar{X}$ associated with α, and $b=f(a)$ its image point. Suppose the local rings of the structure sheaf at a and b satisfy

$$\dim \mathcal{O}_{\bar{X},a}=\dim \mathcal{O}_{\bar{S},b}=1.$$

Let $\bar{X}$ be Galois over $\bar{S}$; that is, if $G=G(\bar{X}/\bar{S})$ is the automorphism group of $\bar{X}$ over $\bar{S}$, acting on $\bar{X}$ (on the right), and $F(\bar{X})$, $F(\bar{S})$ are the fields of rational functions, then $F(\bar{X})^G=F(\bar{S})$. If $\bar{X}$ is etale over $\bar{S}$, this means that $\bar{X}$ is a Galois covering space in the sense introduced at the start. We shall assume also that the order e of the ramification group

$$G_\alpha=\{\sigma\in G,\ \sigma\circ\alpha=\alpha\}$$

is invertible in $\mathcal{O}_{\bar{X},a}$. We say then: *$\bar{X}$ is tamely ramified at α.*

Let $\mu_e(\Omega)$ be the group of e-th roots of unity in Ω. It is easy to prove:

A I.12 Lemma. *There is a natural isomorphism*

$$\phi^\alpha:\ \mu_e(\Omega)\xrightarrow{\sim} G_\alpha\subseteq G.$$

It is characterized by

$$\phi^\alpha(\xi)\,x=t(\xi)\,x,\qquad d(t(\xi))=\xi$$

for a generating element x of the maximal ideal $\mathfrak{m}$ of $\mathcal{O}_{\bar{X},a}$.

Here $t(\xi)\in\mathcal{O}_{\bar{X},a}$ is an inverse image of $\xi\in\Omega$ under the mapping $\mathcal{O}_{\bar{X},a}\to\Omega$ induced by α. If α': $\mathrm{Spec}(\Omega)\to\bar{X}$ is another geometric point of $\bar{X}$ over β, then there is an element $\sigma\in G$ with

$$\alpha'\circ\sigma=\alpha,$$

and

$$\phi^{\alpha'}(\xi)=\sigma\,\phi^\alpha(\xi)\,\sigma^{-1}.$$

Thus ϕ^α and $\phi^{\alpha'}$ are conjugate.

Now let A be a closed subscheme of pure codimension 1 in $\bar{S}$, S the complement of A in $\bar{S}$, and

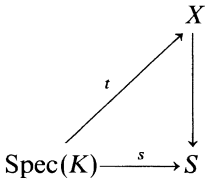
$$X\to S$$

a Galois covering space of S with Galois group G. Then X is an open subscheme of a uniquely determined normal irreducible scheme

$$\bar{X}\to\bar{S}$$

finite over $\bar{S}$ on which G likewise acts. The scheme $\bar{X}$ is the normalization of $\bar{S}$ in the function field $F(X)$ of X. We call the covering space $X\to S$ *tamely ramified* over S (with respect to the embedding $S\hookrightarrow\bar{S}$) when $\bar{X}$ is tamely ramified at all geometric points that lie over a generic point of an irreducible component of A.

Let s: $\mathrm{Spec}(K)\to S$ be a geometric point of S. We consider the category of those pointed Galois covering spaces



over (S, s) with Galois group $G(X/S)$ that are tamely ramified. One can show, for instance using Abhyankar's lemma, that this category is filtered.

A I.13 Definition. The projective limit

$$\varprojlim_{(X,t)} G(X/S) = \pi_1^t(S, s),$$

running over all tamely ramified pointed Galois covering spaces of (S, s), is called the *tame* fundamental group of (S, s) with respect to the embedding $S \hookrightarrow \bar{S}$. It is a factor group of the full fundamental group $\pi_1(S, s)$.

Let n be a natural number such that all primes $p \nmid n$ are invertible in $\mathcal{O}_{\bar{S},b}$. Let

$$\hat{\mathbb{Z}}^{(n)}(1) = \hat{\mathbb{Z}}^{(n)}(1)(\Omega) = \varprojlim_{(e,n)=1} \mu_e(\Omega).$$

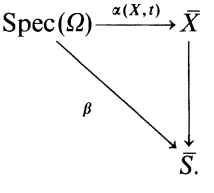
Here the system of all the $\mu_e(\Omega)$ is considered as a projective system with the mappings

$$\mu_e(\Omega) \to \mu_{e'}(\Omega)$$
$$\xi \mapsto \xi^{e/e'} \quad \text{for } e' \mid e.$$

We choose some element α in

$$\varprojlim_{(X,t)} \mathrm{Hom}_{\bar{S}}(\mathrm{Spec}(\Omega), \bar{X}) \text{ over a generic geometric point } \beta\colon \mathrm{Spec}(\Omega) \to A \subseteq \bar{S},$$

that is, a coherent system of geometric points



When we pass to the projective limit, the homomorphisms $\phi^{\alpha(X,t)}$ of Lemma 12 give us a homomorphism

$$\phi^{\alpha}\colon \hat{\mathbb{Z}}^{(n)}(1) \to \pi_1^t(S, s).$$

If we choose some other element α' from the projective limit instead of α, then $\phi^{\alpha'}$ arises by conjugation by an element in $\pi_1^t(S, s)$. *The conjugacy class thus depends only on* $\beta\colon \mathrm{Spec}(\Omega) \to \bar{S}$.

A I.14 Definition. *We denote by* γ_β: $\hat{\mathbb{Z}}^{(n)}(1) \to \pi_1^t(S, s)$ *any element from the conjugacy class just constructed of homomorphisms*

$$\phi^\alpha\colon\ \hat{\mathbb{Z}}^{(n)}(1) \to \pi_1^t(S, s).$$

Now let $\mathbb{P}^1$ be the projective line over the separably closed base field k, and K a separably closed extension field. We consider geometric points

$$s\colon\ \mathrm{Spec}(K) \xrightarrow[k]{} \mathbb{P}^1$$

$$\beta_v\colon\ \mathrm{Spec}(K) \xrightarrow[k]{} \mathbb{P}^1 \qquad v = 0, \ldots, r.$$

Suppose the underlying ordinary points a of s and b_v of β_v are pairwise distinct and the set $A = \{b_0, \ldots, b_r\}$ is closed. Set $p = 1$ when k has characteristic 0, otherwise set $p =$ characteristic of k. We want to determine the tame fundamental group

$$\pi_1^t(\mathbb{P}^1 - A, s)$$

of the pointed projective line.

If $k = K$ is the field of complex numbers, we can determine this group by transcendental methods using the comparison theorem A I.10: $\pi_1(\mathbb{P}^1 - A, s)$ is a free group. Suitable paths around the points $b_1, \ldots, b_r$ can be taken as the free generators. By base field extension and descent and specialization, one gets from this:

A I.15 Proposition. *For suitable choice of the homomorphisms*

$$\gamma_{\beta_v}\colon\ \hat{\mathbb{Z}}^{(p)}(1)(k) \to \pi_1^t(\mathbb{P}^1 - A, s), \qquad v = 1, \ldots, r,$$

in their conjugacy classes, the images generate a dense subgroup.

Remark. For our purposes we need only a simpler result: The images of the homomorphisms γ_{β_v} $(v = 1, \ldots, r)$ and their conjugates generate a dense subgroup of $\pi_1^t(\mathbb{P}^1 - A, s)$.

Proof. Otherwise there would be a nontrivial (unramified) irreducible covering space X of the full projective line. The genus of the curve X would have to be negative! □

Now we consider n-dimensional projective space $\mathbb{P}^n$ $(n \geqq 2)$ over the algebraically closed base field k, an *irreducible reduced hypersurface* $F \subset \mathbb{P}^n$, and a line $D \subset \mathbb{P}^n$ that meets F only transversally at smooth points.

$$F \cap D = A = \{b_0, \ldots, b_r\}.$$

Let

$$\beta_v\colon\ \mathrm{Spec}(k) \xrightarrow[k]{} D, \qquad v = 0, \ldots, r$$

be geometric points with the underlying ordinary points b_v, and let

$$s\colon\ \mathrm{Spec}(K) \xrightarrow[k]{} D - A \subset \mathbb{P}^n - F$$

be another geometric point of $D-A$, and thus also of $\mathbb{P}^n-F$. Here K is a separably closed extension field of k. Let

$$p=\begin{cases}1 & \operatorname{char} k=0\\ \operatorname{char} k & \text{otherwise.}\end{cases}$$

A I.16 Proposition. *The homomorphism*

$$q\colon\ \pi_1^t(D-A,\,s)\to\pi_1^t(\mathbb{P}^n-F,\,s)$$

is surjective. The homomorphisms

$$q\circ\gamma_{\beta_v}\colon\ \hat{\mathbb{Z}}^{(p)}(1)\to\pi_1^t(\mathbb{P}^n-F,\,s)$$

are conjugate in $\pi_1(\mathbb{P}^n-F,\,s)$.

The proof comes from the following considerations on Galois covering spaces of $\mathbb{P}^n$ "tamely ramified" over F. Let

$$f\colon\ \bar{X}\to\mathbb{P}^n$$

be an irreducible normal Galois scheme finite over $\mathbb{P}^n$, with automorphism group G. Suppose the scheme $\bar{X}$ is etale over $\mathbb{P}^n-F$ and *tamely ramified* over F. It follows from Abhyankar's lemma that $\bar{X}$ is nonsingular over all smooth points of F, and that the subscheme $\bar{X}_D=D\times_{\mathbb{P}^n}\bar{X}\subset\bar{X}$ is smooth over k and tamely ramified over $A\subset D$. It follows from Bertini's theorem and the Zariski connectedness theorem that X_D is connected and hence irreducible. On X_D we have G acting again as automorphism group over D. Let

$$\tilde{\beta},\,\tilde{\tilde{\beta}}\colon\ \operatorname{Spec}(k)\to\bar{X}_D\subset\bar{X}$$

be two geometric points over two of the geometric points

$$\beta_v\colon\ \operatorname{Spec}(k)\to\mathbb{P}^n.$$

We choose also two geometric points

$$a_1,\,a_2\colon\ \operatorname{Spec}(\Omega)\xrightarrow[k]{}\bar{X}$$

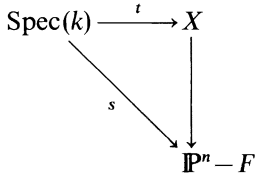
with the following property:

a_1 is a "generic" geometric point of the irreducible component of $f^{-1}(F)$ containing $\tilde{\beta}$, and a_2 is a "generic" geometric point of the component containing $\tilde{\tilde{\beta}}$. Since F is irreducible, these points are conjugate. Using Abhyankar's lemma, it is easy to see that

$$\phi^{(\tilde{\beta})}=\phi^{(a_1)}\qquad G_{\tilde{\beta}}=G_{a_1},$$
$$\phi^{(\tilde{\tilde{\beta}})}=\phi^{(a_2)}\qquad G_{\tilde{\tilde{\beta}}}=G_{a_2}.$$

As $\phi^{(a_1)}$ and $\phi^{(a_2)}$ are conjugate in G, so also are the homomorphisms $\phi^{(\tilde{\beta})}$ and $\phi^{(\tilde{\tilde{\beta}})}$!

Now we consider the tamely ramified pointed Galois covering spaces



$$\pi_1^t(\mathbb{P}^n - F, s) = \lim G(X/(\mathbb{P}^n - F)).$$

Each scheme X can be extended uniquely to a Galois normal irreducible scheme $\bar{X}$ over $\mathbb{P}^n$ on which $G(X/(\mathbb{P}^n - F))$ acts as automorphism group. We apply the preceding considerations to this scheme $\bar{X}$ over $\mathbb{P}^n$. □

A II. Derived Categories

(see: R. Hartshorne, Residues and Duality, Chap. I; Springer Lecture Notes in Mathematics No. 20).

For many purposes it is useful to handle homological algebra on the level of complexes. For this it is necessary to use the derived category of an abelian category, as introduced by Verdier. We shall assemble the most important ideas.

Let A be an abelian category. We denote by $K(A)$ the category of complexes in A modulo homotopy: The objects are the complexes $X^\bullet$ (with differential d_X of degree $+1$). The set $\mathrm{Hom}_{K(A)}(X^\bullet, Y^\bullet)$ for two complexes $X^\bullet$, $Y^\bullet$ is the set of ordinary homomorphisms of complexes $X^\bullet \to Y^\bullet$ (of degree zero) modulo those homomorphisms $X^\bullet \to Y^\bullet$ that are homotopic to zero. $K(A)$ is an additive category. We denote by

$$T\colon\ K(A) \to K(A)$$

the shift operator,

$$T(X^\bullet)^n = X^{n+1}, \qquad d_{T(X)} = -d_X,$$

and use T^i for the i-fold iterate

$$T^i(X^\bullet)^n = X^{n+i}, \qquad d_{T^i(X)} = (-1)^i d_X.$$

We also write $X^\bullet[i]$ for $T^i(X^\bullet)$. A morphism $X^\bullet \to T^i(Y^\bullet)$ is also called a homomorphism $X^\bullet \to Y^\bullet$ of degree i. Let

$$u\colon\ X^\bullet \to Y^\bullet$$

be a homomorphism in $K(A)$ (of degree 0). We construct the mapping cone $C^\bullet(u)$:

$$C^n(u) = X^{n+1} \oplus Y^n.$$

The differential operator is

$$(x, y) \mapsto (-d_X \cdot (x),\ u(x) + d_Y \cdot (y)).$$

We have natural homomorphisms

$$Y^\bullet \to C^\bullet(u), \qquad C^\bullet(u) \to T(X^\bullet).$$

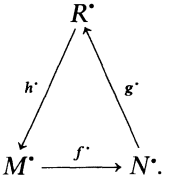
We call a system of mappings

$$M^{\cdot} \xrightarrow{f^{\cdot}} N^{\cdot}, \quad N^{\cdot} \xrightarrow{g^{\cdot}} R^{\cdot}, \quad R^{\cdot} \xrightarrow{h^{\cdot}} T(M^{\cdot})$$

an admissible triangle in $K(A)$ if it is (in the obvious way) isomorphic to a system

$$X^{\cdot} \xrightarrow{u} Y^{\cdot}, \quad Y^{\cdot} \to C^{\cdot}(u), \quad C^{\cdot}(u) \to T(X^{\cdot}).$$

Notation: $(M^{\cdot}, N^{\cdot}, R^{\cdot}; f, g, h)$, or



Thus $h: R^{\cdot} \to M^{\cdot}$ is a homomorphism of degree 1.

The following symmetry property holds:

$$(X^{\cdot}, Y^{\cdot}, Z^{\cdot}; u^{\cdot}; v^{\cdot}; w^{\cdot})$$

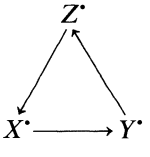
is an admissible triangle if and only if

$$(Y^{\cdot}, Z^{\cdot}, T(X^{\cdot}); v, w, -T(u))$$

is one. A covariant additive functor

$$F: K(A) \to B$$

into another abelian category B is called a cohomological functor, if for every admissible triangle



the induced sequence $F(X^{\cdot}) \to F(Y^{\cdot}) \to F(Z^{\cdot})$ is exact. Because of the symmetry property, we then get a long exact sequence

$$\ldots \to F(X^{\cdot}) \to F(Y^{\cdot}) \to F(Z^{\cdot}) \to F(T(X^{\cdot})) \to F(T(Y^{\cdot})) \to \ldots .$$

The corresponding things hold for a contravariant functor.

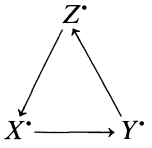
Examples of cohomological functors are:

(1) $\quad \mathrm{Hom}_{K(A)}(-, X^{\cdot})$

(2) $\quad \mathrm{Hom}_{K(A)}(X^{\cdot}, -)$

(3) $\quad H^0(-).$

Since $H^0(T^i(X^{\cdot}))$ equals $H^i(X^{\cdot})$, the i-th cohomology object of the complex $X^{\cdot}$, the long exact sequence for an admissible triangle



runs

$$\ldots \to H^n(X^\bullet) \to H^n(Y^\bullet) \to H^n(Z^\bullet) \to H^{n+1}(X^\bullet) \to H^{n+1}(Y^\bullet) \to \ldots .$$

A II.1 Definition. A homomorphism of complexes $f^\bullet: X^\bullet \to Y^\bullet$ is called a *quasi-isomorphism* when the induced mappings

$$H^n(X^\bullet) \to H^n(Y^\bullet)$$

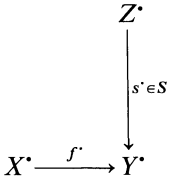
on the cohomology objects of these complexes are isomorphisms.

Obviously this property depends only on the class of $f^\bullet$ in $\mathrm{Hom}_{K(A)}(X^\bullet, Y^\bullet)$.

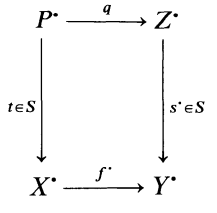
We list the most important properties of the class S of quasi-isomorphisms in $K(A)$.

(1) S is multiplicatively closed and contains the identity mapping on each complex $X^\bullet$.

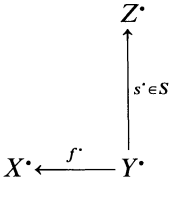
(2) A diagram



can be extended in $K(A)$ to a commutative diagram



The same holds for a diagram



(3) Let $f^\bullet, g^\bullet: X^\bullet \to Y^\bullet$ be two mappings in $K(A)$. There is an $s: P^\bullet \to X^\bullet$ in S with $f^\bullet \circ s^\bullet = g^\bullet \circ s^\bullet$ if and only if there is a $t^\bullet: Y^\bullet \to Q^\bullet$ in S with $t^\bullet \circ f^\bullet = t^\bullet \circ g^\bullet$.

(4) S is preserved by the shift operator T (and hence by its powers T^i). If three mappings $f^\bullet$, $g^\bullet$, $h^\bullet$ define a mapping of an admissible triangle into

another, and two of these mappings are quasi-isomorphisms, then so is the third one.

Properties (1)–(3) make it possible to construct a *fraction category* $D(A)$ of $K(A)$ with respect to S, in which all elements of S become *isomorphisms* (at least if A is contained in some universe):

The objects of $D(A)$ are again the complexes in A. The set $\mathrm{Hom}_{D(A)}(X^\cdot, Y^\cdot)$ of morphisms between two complexes $X^\cdot$, $Y^\cdot$ in $D(A)$ is defined as

$$\varinjlim_{I_{X^\cdot}} \mathrm{Hom}_{K(A)}(X^\cdot, Y^\cdot) \quad (\cong \varinjlim_{Y^\cdot \to \tilde{Y}^\cdot} \mathrm{Hom}(X^\cdot, \tilde{Y}^\cdot)).$$

Here $I_{X^\cdot}$ is the category of complexes $s\colon \tilde{X}^\cdot \to X^\cdot$ over $X^\cdot$ with $s^\cdot \in S$. The composition of morphisms in $D(A)$ can be defined in a natural way because of (1)–(3); $D(A)$ is an additive category. There is a natural functor

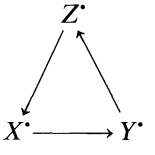
$$K(A) \to D(A)$$

that precisely takes quasi-isomorphisms to isomorphisms.

Because of (4), the shift operator T carries over to the category $D(A)$. Moreover, we again have a system of admissible triangles in $D(A)$ with properties corresponding to those in $K(A)$: A system of mappings in $D(A)$,

$$(X^\cdot \to Y^\cdot, Y^\cdot \to Z^\cdot, Z^\cdot \to T(X^\cdot)),$$

or



for short, is called an admissible triangle in $D(A)$ if it is (in the obvious way) isomorphic in $D(A)$ to the image in $D(A)$ of an admissible triangle in $K(A)$.

Each mapping in $D(A)$ is then again the base of an admissible triangle. The symmetry property holds. Every admissible triangle induces long exact sequences for the functors $\mathrm{Hom}_{D(A)}(-, X^\cdot)$ and $\mathrm{Hom}_{D(A)}(X^\cdot, -)$. As the functor $X^\cdot \mapsto H^n(X^\cdot)$ that assigns to a complex $X^\cdot$ its n-th cohomology group takes quasi-isomorphisms to isomorphisms, $H^n(X^\cdot)$ is in a natural way a functor on $D(A)$. Again every admissible triangle gives rise to a long exact sequence of cohomology objects. The long exact sequence for a short exact sequence

$$0 \to X^\cdot \xrightarrow{u^\cdot} Y^\cdot \xrightarrow{v^\cdot} Z^\cdot \to 0$$

of homomorphisms of complexes is also a special case of the exact sequence of a triangle:

Indeed, there is a natural mapping of the mapping cone of u into $Z^\cdot$,

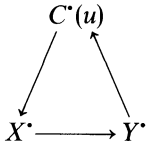
$$C^\cdot(u) \to Z^\cdot$$

$$X^{n+1} \otimes Y^n \ni (x, y) \mapsto v(y).$$

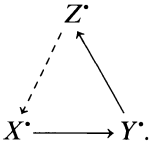
This is a *quasi-isomorphism* and induces an isomorphism from the long exact cohomology sequence

$$\ldots \to H^n(X^\cdot) \to H^n(Y^\cdot) \to H^n(C^\cdot(u)) \to H^{n+1}(X^\cdot) \to \ldots$$

of the triangle



to the long exact cohomology sequence of the original short exact sequence of complexes. There is a natural admissible triangle in $D(A)$,



A II.2 Definition. The category $D(A)$ thus constructed, with the additional structure of the shift operator T and the system of admissible triangles, is called the *derived category* of the abelian category A.

Certain *full* subcategories of $D(A)$ have a role to play: The full subcategory $D_+(A)$ of complexes that are isomorphic (in $D(A)$) to complexes bounded below, the full subcategory $D_-(A)$ of complexes isomorphic to complexes bounded above, and finally $D_b(A)$, the subcategory of complexes that are isomorphic to bounded complexes. We want to assume now that A has "enough" injective objects. Then the category $D_+(A)$ can be described (up to equivalence) more simply, as follows:

If $I^\cdot$ is a complex of injective objects, bounded below, then

$$\mathrm{Hom}_{K(A)}(X^\cdot, I^\cdot) \to \mathrm{Hom}_{D(A)}(X^\cdot, I^\cdot)$$

is an *isomorphism* for every complex $X^\cdot$. Moreover, for every complex $X^\cdot$ in $D_+(A)$ there is *an injective resolution*, i.e. a quasi-isomorphism

$$X^\cdot \to I(X^\cdot) \quad \text{(in the category of complexes)}$$

to an injective complex $I(X^\cdot)$ bounded below. Because of the previously stated isomorphism property for injective complexes bounded below, $I(X^\cdot)$ is uniquely determined up to homotopy.

$$X^\cdot \to I(X^\cdot)$$

can be extended in a natural way to a functor $D_+(A) \to K(A)$.

A II.3 Proposition. *This functor $D_+(A) \to K(A)$ represents an equivalence between the category $D_+(A)$ and the full subcategory of $K(A)$ formed of injective complexes bounded below.*

Now let $F\colon A \to B$ be an additive functor (covariant or contravariant) from A to another abelian category B. Then F can be extended componentwise to a functor from the category of complexes in A to the category of complexes in B, and it then takes homotopies to homotopies. Thus in a natural way we obtain a functor (covariant or contravariant)

$$F\colon K(A) \to K(B).$$

If F is exact, then it takes quasi-isomorphisms to quasi-isomorphisms, and thus it can be extended to a functor

$$D(A) \to D(B).$$

If F is only left exact or right exact, this is not possible. In place of it, one carries out the following construction.

We shall as an example treat the case of a *covariant left exact functor* F. We shall assume that A has enough injective objects. (The existence of "enough" F-acyclic objects will suffice.) Then we can compose the three functors

$$\begin{aligned} & D_+(A) \to K(A) \\ & \quad X^\cdot \mapsto I(X^\cdot) \quad \text{(injective resolution)}, \\ F\colon\ & K(A) \to K(B) \\ & K(B) \to D(B) \end{aligned}$$

to get a functor

$$\begin{aligned} \mathsf{R}F\colon\ & D_+(A) \to D(B) \\ & X^\cdot \mapsto F(I(X^\cdot)). \end{aligned}$$

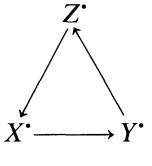
A II.4 Definition. The functor

$$\mathsf{R}F\colon D_+(A) \to D(B)$$

thus constructed is called the derived functor of F (in the sense of Verdier). It is exact in the following sense: It is compatible with the shift operator and takes admissible triangles to admissible triangles.

Remark. The functor $\mathsf{R}F$ can also be characterized by a universal mapping property.

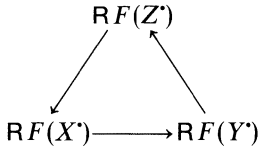
If we denote by $R^\nu F(X^\cdot)$ the ν-th cohomology object of the complex $\mathsf{R}F(X^\cdot)$, then each admissible triangle



in $D_+(A)$ gives a long exact sequence for the *derived functors* $R^\nu F$:

$$\ldots \to R^\nu F(X^\cdot) \to R^\nu F(Y^\cdot) \to R^\nu F(Z^\cdot) \to R^{\nu+1} F(X^\cdot) \to R^{\nu+1} F(Y^\cdot) \to \ldots,$$

namely, the long exact cohomology sequence for the triangle



In particular, since every short exact sequence

$$0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$$

of complexes bounded below in A can be fitted in a natural way into an admissible triangle in $D_+(A)$, it gives rise to a long exact sequence for the functors $R^i F$.

The category A can be embedded in $D_+(A)$:

$$M \mapsto 0 \to 0 \to \ldots \to M \to 0 \to 0 \to \ldots$$
$$\uparrow$$
$$\text{0-th place}$$

By construction then

$$H^\nu(\mathrm{R}F(M)) = R^\nu F(M)$$

is *the v-th derived functor of* $F(M)$ in the sense of homological algebra. The long exact sequence constructed above for a short exact sequence of complexes yields as a special case the long exact sequence familiar from homological algebra for the derived functors $R^n F$ and a short exact sequence $0 \to M \to N \to P \to 0$ of objects in A.

Further Remarks

Let $X^\bullet$ be a complex, bounded below, of F-acyclic objects X^n (i.e., $R^\nu F(X^n) = 0$ for all $\nu > 0$ and all n). Then

$$F(X^\bullet) = \mathrm{R}F(X^\bullet).$$

Let F and G be two covariant left exact functors,

$$A \xrightarrow{F} B \xrightarrow{G} C.$$

Assume $R^\nu F(M)$ is G-acyclic for every injective object M in A. Then

A II.5 Proposition.

$$\boxed{\mathrm{R}(G \circ F) = (\mathrm{R}G) \circ (\mathrm{R}F).}$$

This formula replaces the Leray spectral sequence for a composite functor $G \circ F$.

A II.6 *Remark.* Many results on the derived functor

$$\mathrm{R}F\colon\ D_+(X) \to D(Y)$$

can be reduced to results on the derived functors $R^\nu F(M)$ of objects M in A, using the long exact sequences for the functors $R^\nu F$. Let $X^\bullet$ be a complex

whose cohomology objects ($H^v(X^\cdot)$) vanish for $v<m$. Let L^m then be the image of X^{m-1} in X^m, and $\tilde{X}^\cdot$ the complex

$$0\to X^m/L^m\to X^{m+1}\to\dots$$

Then $\tilde{X}^\cdot$ is quasi-isomorphic to $X^\cdot$. There is an injection of $H^m(X^\cdot)$,

$$M=H^m(X^\cdot)\hookrightarrow X^\cdot.$$

The cokernel $\tilde{\tilde{X}}^\cdot$ of this injection is a complex with

$$H^v(\tilde{\tilde{X}}^\cdot)=\begin{cases}0 & v<m+1\\ H^v(X^\cdot) & v\geqq m+1.\end{cases}$$

We get a long exact sequence

$$\dots\to R^{v-m}F(H^m(X^\cdot))\to R^vF(X^\cdot)\to R^vF(\tilde{\tilde{X}}^\cdot)\to\dots.$$

Similarly one can truncate the top of a complex and work down from above.

Now let X be a scheme and Λ a ring. Then we denote by $D(X)$ the derived category of the category of etale sheaves of abelian groups on X, by $D(X,\Lambda)$ the derived category of the category of all etale sheaves of Λ-modules, and by $D(X,\mathrm{tor})\subset D(X)$ the derived category of the category of all torsion sheaves. $D_+(X)$, $D_+(X,\Lambda)$, ... have corresponding meanings.

Example of Derived Functors

$$\mathsf{R}\Gamma(X,M^\cdot)\colon\ D_+(X)\to D(\mathrm{Ab})$$
$$\uparrow$$
$$\text{Category of abelian groups}$$

is derived functor in the sense of Verdier of the functor of sections

$$\mathscr{G}\mapsto\Gamma(X,G)=\mathscr{G}(X).$$

Let $f\colon X\to Y$ be a morphism of schemes. Then we can form the derived functor of the direct image functor f_*,

$$\mathsf{R}f_*\colon\ D_+(X)\to D(Y).$$

The inverse image functor f^* is exact, and so it carries over to an exact mapping

$$D(Y)\to D(X).$$

Warning: The exact functor

$$\mathsf{R}f_!\colon\ D_+(X,\mathrm{tor})\to D(Y,\mathrm{tor})$$

is *not a derived functor* of the sort we have been describing. It lacks the universal mapping property.

As in topological sheaf theory, so also in etale sheaf theory the derived functors of certain functors of two variables play an important role.

Let Λ be a fixed (commutative) ring. In the following we consider only etale sheaves of Λ-modules on a scheme X.

The Functors $\mathsf{R}\,\mathrm{Hom}_\Lambda(F^\cdot, G^\cdot)$ *and* $\mathsf{R}\,\mathrm{Hom}_\Lambda(F^\cdot, G^\cdot)$

Let F and G be two sheaves on a scheme X. We denote by

$$\mathrm{Hom}_\Lambda(F, G)$$

the group of Λ-homomorphisms and by

the sheaf
$$\mathsf{Hom}_\Lambda(F, G)$$
$$(U \to X) \mapsto \mathrm{Hom}_\Lambda(F|U, G|U).$$

These define two left exact functors in the two variables F and G. Derived functors in the second variable, G, can be formed by using Λ-injective resolutions in the sense of homological algebra. We get the derived functors

$$\mathrm{Ext}^\nu_\Lambda(F, G) \quad \text{and} \quad \mathsf{Ext}^\nu_\Lambda(F, G).$$

They are ∂-functors in *both variables.*

Now let $F^\cdot$ and $G^\cdot$ be two complexes of sheaves. We define the complex

$$\mathrm{Hom}^\cdot_\Lambda(F^\cdot, G^\cdot) = K^\cdot$$

by
$$K^n = \prod_{p-q=n} \mathrm{Hom}_\Lambda(F^q, G^p).$$

The boundary operator is defined componentwise by

$$\mathrm{Hom}_\Lambda(F^q, G^p) \ni f \mapsto d_{K^\cdot}(f) = (-1)^{p-q} f \circ d_{F^\cdot} + d_{G^\cdot} \circ f.$$

Similarly we define the complex of sheaves

$$\mathsf{Hom}^\cdot_\Lambda(F^\cdot, G^\cdot).$$

$\mathrm{Hom}^\cdot_\Lambda(F^\cdot, G^\cdot)$ and $\mathsf{Hom}^\cdot_\Lambda(F^\cdot, G^\cdot)$ are in a natural way functors of two variables on the category of complexes.

One proves: If $G^\cdot$ is a complex of injective sheaves bounded below, then both functors take a quasi-isomorphism in the first variable to a quasi-isomorphism. Homotopies in both variables are taken to homotopies. Therefore, the following construction of derived functors in two variables makes sense:

$$\mathsf{R}\,\mathrm{Hom}_\Lambda(F^\cdot, G^\cdot)\colon\ D(X, \Lambda) \times D_+(X, \Lambda) \to D(\Lambda)$$
$$(F^\cdot, G^\cdot) \mapsto \mathrm{Hom}^\cdot_\Lambda(F^\cdot, J^\cdot(G^\cdot)),$$
$$\mathsf{R}\,\mathrm{Hom}_\Lambda(F^\cdot, G^\cdot)\colon\ D(X, \Lambda) \times D_+(X, \Lambda) \to D(\Lambda)$$
$$(F^\cdot, G^\cdot) \mapsto \mathsf{Hom}^\cdot_\Lambda(F^\cdot, J^\cdot(G^\cdot)).$$

Here $J^\cdot(G^\cdot)$ is again the functor "modulo homotopy" that assigns to the complex $G^\cdot \in D_+(X, \Lambda)$ an injective resolution $J^\cdot(G^\cdot)$ that is also bounded below.

$D(\Lambda)$ is the derived category of the category of Λ-modules.

Both derived functors are in the following sense exact in both variables: They are in the obvious way compatible with the shift operator in both variables and carry admissible triangles in both variables to admissible triangles. We also have

$$\mathsf{R}\Gamma(X, \mathsf{R}\,\mathrm{Hom}_\Lambda(F^\cdot, G^\cdot) = \mathsf{R}\,\mathrm{Hom}_\Lambda(F^\cdot, G^\cdot).$$

For two sheaves F and G,

$$H^\nu(\mathsf{R}\,\mathrm{Hom}_\Lambda(F, G)) = \mathrm{Ext}^\nu_\Lambda(F, G),$$
$$H^\nu(\mathsf{R}\,\mathrm{Hom}_\Lambda(F, G)) = \mathrm{Ext}^\nu_\Lambda(F, G).$$

The Derived Functors of Tensor Product

The tensor product

$$F \otimes_\Lambda G = F \otimes G$$

of two sheaves F and G of Λ-modules on the scheme X is the sheaf associated with the presheaf

$$(U \to X) \mapsto F(U) \otimes_\Lambda G(U).$$

One can easily show that the stalk of $F \otimes G$ at a geometric point

$$\alpha\colon\ \mathrm{Spec}(\Omega) \to X$$

is canonically isomorphic to the tensor product of the stalks,

$$(F \otimes G)_\alpha = F_\alpha \otimes_\Lambda G_\alpha .$$

From this follows:

(1) The tensor product is right exact in both variables.

(2) The tensor product is compatible with filtered direct limits in both variables.

(3) For a morphism $f\colon Y \to X$,

$$f^*(F \otimes G) = f^*(F) \otimes f^*(G).$$

The tensor product has enough acyclic objects in the sense of homological algebra, namely the Λ-flat sheaves:

A sheaf F is called *Λ-flat* when all stalks are Λ-flat. For example, the sheaf M^Λ of Λ-modules generated by a sheaf M of sets (4.9) is Λ-flat. For every sheaf G there is a flat sheaf F and a surjective homomorphism

$$F \to G.$$

The flat sheaf F can actually be constructed canonically, i.e. in functorial dependence on G. Just consider G as a sheaf of sets; then the identity mapping $G \to G$ induces a natural surjective Λ-homomorphism

$$F = G^\Lambda \to G.$$

If G is constructible, then F can be chosen to be constructible.

It follows that every sheaf G has a resolution (even a canonical one) by flat sheaves. (Left) derived functors of the tensor product can then be formed in both of the variables using such flat resolutions. In both cases one gets "the same" derived functors,

$$\mathrm{Tor}^\Lambda_n(F, G).$$

Let $F^\cdot$ and $G^\cdot$ be two complexes of sheaves. The tensor product

$$F^\cdot \otimes_\Lambda G^\cdot = F^\cdot \otimes G^\cdot = K^\cdot$$

of these complexes is defined as follows.

$$K^n = \bigoplus_{p+q=n} F^p \otimes G^q.$$

$$d_{K^\cdot} | F^p \otimes G^q = d_{F^\cdot} \otimes \mathrm{Id} + (-1)^p \, \mathrm{Id} \otimes d_{G^\cdot}.$$

One can prove: Suppose

$$F^\cdot \to G^\cdot$$

is a quasi-isomorphism, and $K^\cdot$ is another complex. If both of the complexes $F^\cdot$ and $G^\cdot$ are bounded above and flat, or if the complex $K^\cdot$ is bounded above and flat, then the morphism

$$F^\cdot \otimes K^\cdot \to G^\cdot \otimes K^\cdot$$

is also a quasi-isomorphism. The same holds with the two variables interchanged. These considerations lead to the construction of a left derived functor in thee sense of Verdier,

$$F^\cdot \underset{=}{\otimes} G^\cdot : \; D_-(X, \Lambda) \times D(X, \Lambda) \to D(X, \Lambda).$$

One constructs a flat resolution of $F^\cdot$, i.e. a complex of flat sheaves $P^\cdot(F^\cdot)$ bounded above and a quasi-isomorphism

$$P^\cdot(F^\cdot) \to F^\cdot.$$

Then

$$F^\cdot \underset{=}{\otimes} G^\cdot = P^\cdot(F^\cdot) \otimes G^\cdot.$$

Functorial dependence on the variable F can be obtained for instance by choosing a "canonical" resolution $P^\cdot(F^\cdot)$. In the same way, using a flat resolution of G, one gets a derived functor

$$F^\cdot \underset{=}{\otimes} G^\cdot : \; D(X, \Lambda) \times D_-(X, \Lambda) \to D(X, \Lambda).$$

The two constructions lead to "the same" derived functor on $D_-(X, \Lambda) \times D_-(X, \Lambda)$. $F^\cdot \underset{=}{\otimes} G^\cdot$ is exact in both variables, thus taking admissible triangles in either variable to admissible triangles. For sheaves F and G,

$$\mathrm{Tor}_\nu^\Lambda(F, G) = H^{-\nu}(F^\cdot \underset{=}{\otimes} G^\cdot).$$

The tensor product of complexes can be used for the construction of cup products.

Let X be a finitely generated scheme over a separably closed field, and let F, G be two sheaves on X. Let

$$P^\cdot \to F$$

be a flat resolution, bounded above, of one of the two sheaves, say F. Since the section functor Γ has finite homological dimension, $P^\cdot$ and G have Γ-acyclic resolutions bounded above,

$$P^\cdot \to J^\cdot(P^\cdot), \qquad G \to J^\cdot(G).$$

If Λ is $\mathbb{Z}/n\mathbb{Z}$, or more generally any noetherian ring, then we can even manage, e.g. using the Godement construction (Chap. I, Proof of 12.15), to get $J^\cdot(P^\cdot)$ also flat. Then $J^\cdot(P^\cdot) \otimes J^\cdot(G)$ is quasi-isomorphic to $P^\cdot \otimes G$. Let

$$J^\cdot(P^\cdot) \otimes J^\cdot(G) \to J^\cdot$$

be a Γ-acyclic resolution of $J^\cdot(P^\cdot)\otimes J^\cdot(G)$, bounded above. Then there are natural mappings of cohomology groups resp. hypercohomology groups

$$H^p(X,F)\otimes H^q(X,G)=H^p(\Gamma(X,J^\cdot(P^\cdot))\otimes H^q(\Gamma(X,J^\cdot(G))$$
$$\to H^{p+q}(\Gamma(X,J^\cdot))=H^{p+q}(X,P^\cdot\otimes G)\to H^{p+q}(X,F\otimes G).$$

Composition yields a cup product

$$H^p(X,F)\otimes H^q(X,G)\to H^{p+q}(X,F\otimes G)$$
$$\alpha\otimes\beta\mapsto\alpha\smile\beta.$$

A similar construction over a compactification of X gives further cup products,

$$H_c^p(X,F)\otimes H_c^q(X,G)\to H_c^{p+q}(X,F\otimes_\Lambda G),$$
$$H^p(X,F)\otimes H_c^q(X,G)\to H_c^{p+q}(X,F\otimes_\Lambda G),$$

and further

$$\mathrm{Ext}_\Lambda^p(F,G)\otimes_\Lambda H_c^q(X,F)\to H_c^{p+q}(X,F\otimes_\Lambda G).$$

In these last three formulas we assume that $\Lambda=\mathbb{Z}/n\mathbb{Z}$ with n relatively prime to the characteristic of the separably closed base field. The last pairing plays an essential role in the theory of Poincaré duality (Chap. II, §1). It can be derived from a natural transformation in the derived category,

$$\mathsf{R}\,\mathrm{Hom}_\Lambda(F,G)\to\mathsf{R}\,\mathrm{Hom}_\Lambda(\mathsf{R}\Gamma_c(X,F),\mathsf{R}\Gamma_c(X,G)).$$

This transformations is constructed as follows.

We consider a compactification

$$j\colon X\hookrightarrow\bar{X}$$

and Λ-injective resolutions

$$j_!(F)\to I^\cdot,\qquad j_!(G)\to J^\cdot.$$

The restrictions to X,

$$F\to j^*(I^\cdot),\qquad G\to j^*(J^\cdot),$$

are then injective resolutions of F and G. The complexes

$$\Gamma(\bar{X},I^\cdot)\cong\mathsf{R}\Gamma_c(X,F)$$
$$\Gamma(\bar{X},J^\cdot)\cong\mathsf{R}\Gamma_c(X,G)\quad(\text{in } D(\Lambda))$$

are complexes of injective Λ-modules.

There are natural transformations resp. isomorphisms in $D(\Lambda)$,

$$\begin{aligned}
\mathsf{R}\,\mathrm{Hom}_\Lambda(F,G)&\cong\mathrm{Hom}_\Lambda(F,j^*(J^\cdot))\\
&\quad\Big\downarrow\ \text{adjunction formula}\\
&\cong\mathrm{Hom}_\Lambda(j_!(F),J^\cdot)\xleftarrow{\cong}\mathrm{Hom}_\Lambda(I^\cdot,J^\cdot)\to\\
&\to\mathrm{Hom}_\Lambda(\Gamma(\bar{X},I^\cdot),\Gamma(\bar{X},J^\cdot))\\
&\cong\mathsf{R}\,\mathrm{Hom}_\Lambda(\mathsf{R}\Gamma_c(X,F),\mathsf{R}\Gamma_c(X,G)).
\end{aligned}$$

A III. Descent (see [56])

In the following, let

$$f\colon T \to S$$

be a faithfully flat morphism of schemes. In this appendix we shall allow non-noetherian schemes, and we require only that the morphism is quasi-compact, i.e. that the inverse image of an open subscheme of S is a finite union of open affine subschemes of T. Let p_i, $i=1, 2$, be the projections of $T\times_S T$ onto the i-th factor,

$$T\times_S T \underset{p_2}{\overset{p_1}{\rightrightarrows}} T \to S \qquad (f\circ p_1 = f\circ p_2)$$

(incidentally, $T\times_S T$ does not have to be noetherian even when T and S are noetherian).

If X is a scheme over S, we denote by

$$X_T = X\times_S T, \qquad X_{T\times_S T} = X\times_S (T\times_S T)$$

the schemes lifted to T resp. $T\times_S T$.

The projection mapping

$$X_T \to X$$

induces a diagram

$$X_T \times_X X_T \rightrightarrows X_T \to X.$$

There exist natural isomorphisms

$$\begin{aligned} X_T \times_X X_T \cong X_{T\times_S T} &= X_T \times_T T\times_S T \quad \text{(with respect to } p_1\text{)} \\ &= X_T \times_T T\times_S T \quad \text{(with respect to } p_2\text{)}. \end{aligned}$$

Let X' be another S-scheme. Every S-morphism $X\to X'$ induces a T-morphism $X_T \to X'_T$, and every T-morphism $X_T \to X'_T$ induces two $T\times_S T$-morphisms $X_{T\times_S T} \to X'_{T\times_S T}$, depending on whether X_T is lifted to $T\times_S T$ via p_1 or p_2 (and then identified with $X_{T\times_S T}$). We get a diagram

$$(*) \qquad \mathrm{Hom}_S(X, X') \to \mathrm{Hom}_T(X_T, X'_T) \rightrightarrows \mathrm{Hom}_{T\times_S T}(X_{T\times_S T}, X'_{T\times_S T}).$$

A III.1 Lemma. *If $T\to S$ is a faithfully flat quasi-compact morphism of schemes, then the sequence (*) is exact for every pair X, X' of S-schemes, i.e. the left arrow is injective and its image is the set where the two right arrows coincide.*

Corollary. *Let X be a noetherian scheme. Every representable functor*

$$F\colon \mathrm{Et}(X) \to \mathrm{Ens}$$

is an etale sheaf.

Before we sketch the simple proof of the lemma, we formulate an analogue for sheaves over S. Let $\mathscr{M}$ be a quasi-coherent $\mathscr{O}_S$-module. We denote by $\mathscr{M}_T$ and $\mathscr{M}_{T\times_S T}$ the inverse images of $\mathscr{M}$ on T and $T\times_S T$. If $\mathscr{M}'$ is another quasi-coherent sheaf on S, then as in the case of schemes one constructs a sequence

$$(**) \qquad \mathrm{Hom}(\mathscr{M}, \mathscr{M}') \to \mathrm{Hom}(\mathscr{M}_T, \mathscr{M}'_T) \rightrightarrows \mathrm{Hom}(\mathscr{M}_{T\times_S T}, \mathscr{M}'_{T\times_S T}).$$

Here $\mathrm{Hom}(\mathscr{M}, \mathscr{M}')$ denotes the set of all $\mathcal{O}_S$-linear sheaf homomorphisms, $\mathrm{Hom}(\mathscr{M}_T, \mathscr{M}'_T)$ the set of all $\mathcal{O}_T$-linear sheaf homomorphisms.

A III.2 Lemma. *If $T \to S$ is a faithfully flat quasi-compact morphism of schemes, then the sequence (**) is exact for all pairs of quasi-coherent $\mathcal{O}_S$-modules $\mathscr{M}$, $\mathscr{M}'$.*

From the special case $\mathscr{M} = \mathcal{O}_S$, we deduce:

Corollary. *Let X be a noetherian scheme, $\mathscr{M}$ a quasi-coherent $\mathcal{O}_X$-module. The functor*

$$\mathrm{Et} \to \mathrm{Ab}$$

$$(Z \xrightarrow{p} X) \mapsto (p^* \mathscr{M})(Z)$$

is an etale sheaf.

For the Proof of A III.1 and A III.2

Both lemmas can easily be reduced to the following "descent lemma" for modules over rings:

Descent Lemma. *Let $A \to B$ be a faithfully flat (in particular, injective) ring extension, M an A-module. The canonical mapping $M \to M \otimes_A B$, $m \mapsto m \otimes 1$, is then injective, and the image of M consists of the set where the two morphisms*

$$M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B$$

$$m \otimes b \mapsto \begin{cases} m \otimes 1 \otimes b \\ m \otimes b \otimes 1 \end{cases}$$

coincide. In other words, the sequence

$$0 \to M \to M \otimes_A B \to M \otimes_A B \otimes_A B$$

$$m \otimes b \mapsto m \otimes 1 \otimes b - m \otimes b \otimes 1$$

is exact.

Proof. If the extension $A \xrightarrow{j} B$ has a section $B \xrightarrow{s} A$ (i.e., an A-linear ring homomorphism with $s \circ j = \mathrm{id}_A$), then the statement is trivial even without the hypothesis of faithful flatness. In the general case, because of the faithful flatness it suffices to prove the exactness after tensoring with B. But the extension $B \to B \otimes_A B$ has a section, $s(b \otimes b') = b\,b'$. □

If M' is another A-module, then the descent lemma implies the exactness of

$$\mathrm{Hom}_A(M, M') \to \mathrm{Hom}_B(M \otimes_A B, M' \otimes_A B)$$

$$\rightrightarrows \mathrm{Hom}_{B \otimes_A B}(M \otimes_A B \otimes_A B, M' \otimes_A B \otimes_A B)$$

and thus gives A III.2 in the affine case. As a corollary, we get a corresponding exact sequence for A-algebras (instead of M, M') and algebra homomorphisms,

thus getting A III.1 in the affine case. The global case can easily be reduced to the affine case using quasi-compactness.

The Descent Lemma has a generalization proved in exactly the same way, the exactness of the Amitsur complex:

A III.3 *Remark. Let $A \to B$ be a faithfully flat ring extension, M an A-module. The sequence*

$$0 \to M \to M \otimes_A B \xrightarrow{d_0} M \otimes_A B \otimes_A B \xrightarrow{d_1} M \otimes_A B \otimes_A B \otimes_A B \xrightarrow{d_2} \dots$$

$$d_n(m \otimes b_0 \otimes \dots \otimes b_n) = \sum_{i=0}^{n} (-1)^i \, m \otimes b \otimes \dots \otimes b_{i-1} \otimes 1 \otimes b_i \otimes \dots \otimes b_n)$$

is exact.

Special Case

Let

$$A \to A_i, \quad i \in I,$$

be an etale covering of a noetherian ring A. The exact sequence above for $B = \prod_{i \in I} A_i$ is nothing but the Čech complex for the etale sheaf $\tilde{M}$ with respect to this covering.

A III.4 Corollary. *Let M be a module over a noetherian ring A. The Čech cohomology of the etale sheaf $\tilde{M}$ vanishes with respect to any etale covering of A.*

Descent Data

The question we now ask is, under what conditions can a sheaf (resp. a scheme) over T be pushed down to a sheaf (resp. a scheme) over S? For this we consider the extended diagram

$$T \times_S T \times_S T \underset{\xrightarrow{p_{23}}}{\overset{\xrightarrow{p_{12}}}{\xrightarrow{p_{13}}}} T \times_S T \rightrightarrows T \xrightarrow{f} S,$$

where p_{ij} $(i<j)$ denotes the projection on the i-th and j-th components.

Descent Data for Sheaves

Let $\mathcal{N}$ be a quasi-coherent sheaf on T. By descent data (with respect to $T \to S$) we mean an $\mathcal{O}_T$-linear isomorphism

$$\sigma\colon\ p_1^*(\mathcal{N}) \to p_2^*(\mathcal{N})$$

that satisfies the condition

$$p_{13}^*(\sigma) = p_{23}^*(\sigma) \circ p_{12}^*(\sigma).$$

Example. Let $\mathcal{M}$ be a quasi-coherent sheaf on S, $\mathcal{N} = f^*(\mathcal{M})$. Because $f \circ p_1 = f \circ p_2$, the sheaves $p_1^*(\mathcal{N})$ and $p_2^*(\mathcal{N})$ are isomorphic in a natural way. Thus we get

descent data for $\mathcal{N}$. In general, we call descent data $(\mathcal{N}, p_1^*\mathcal{N} \xrightarrow{\sigma} p_2^*\mathcal{N})$ *effective* if there is a quasi-coherent sheaf $\mathcal{M}$ on S and an isomorphism

$$f^*\mathcal{M} \xrightarrow{\sim} \mathcal{N}$$

for which the diagram

$$\begin{array}{ccc} p_1^*\mathcal{N} & \xrightarrow{\sigma} & p_2^*\mathcal{N} \\ \downarrow & & \downarrow \\ p_1^*f^*\mathcal{M} & \xrightarrow{\sim} & p_2^*f^*\mathcal{M} \\ \| & & \| \\ (f\circ p_1)^*\mathcal{M} & = & (f\circ p_2)^*\mathcal{M} \end{array}$$

is commutative.

A III.5 Proposition. *Let $T \to S$ be a faithfully flat quasi-compact morphism of schemes. Every set of descent data $(\mathcal{N}, p_1^*\mathcal{N} \to p_2^*\mathcal{N})$ ($\mathcal{N}$ a quasi-coherent sheaf on T) is effective.*

Because of A III.2, the descended sheaf $\mathcal{M}$ is uniquely determined up to isomorphism. Proposition A III.3 thus says that the category of quasi-coherent $\mathcal{O}_S$-modules is equivalent to the category of quasi-coherent $\mathcal{O}_T$-modules with descent data.

Descent Data for Schemes

Let Y be a scheme over T. By descent data with respect to $T \to S$ we mean a $(T \times_S T)$-isomorphism

$$\sigma\colon\ p_1^*Y \to p_2^*Y$$

with the property

$$p_{13}^*(\sigma) = p_{12}^*(\sigma) \circ p_{23}^*(\sigma).$$

We call the descent data (Y, σ) effective when there is a scheme X over S such that (Y, σ) is isomorphic to $X \times_T S$ furnished with the natural descent data $X \times_S T \times_S T \to X \times_S T \times_S T$ coming from the two projections p_1, p_2. Only under special assumptions will descent data (Y, σ) be effective. For example, it follows easily from A IIII.2 and A III.5 that (Y, σ) is effective when $Y \to T$ is affine, that is, when the inverse image of each open affine set in T is affine.

More generally, a morphism $Y \to T$ of schemes is called quasi-affine if it is quasi-compact and the inverse image of each open affine subscheme of T is isomorphic to a open subscheme of an affine scheme.

A III.6 Proposition. *Let $T \xrightarrow{f} S$ be a faithfully flat quasi-affine morphism of schemes, X a quasi-affine scheme over T. All descent data (X, σ) (with respect to f) are effective.*

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